



Tanzu Operations Manager

Tanzu Operations Manager 3.2

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Monitoring VM status in Tanzu Operations Manager

Monitoring VMs using the Tanzu Operations Manager interface

Tanzu Operations Manager VM disk space

Security and compliance in Tanzu Operations Manager

Security processes used in developing Tanzu Operations Manager

Tanzu Operations Manager security overview and policy

How VMware monitors for security vulnerabilities

How to report a vulnerability

Notification policy

Classes of vulnerabilities

High

Medium

Low

Release policy

Alerts and actions archive

Testing, release, and security lifecycle for Tanzu Operations Manager

Release mechanics

Versioning

Support

Patch releases

Upgrading

Release testing, integration, and validation

Test-Driven Development

Additional pre-release gates: Internal, PWS, and Compliance

Patch releases: security and bug fixes

Identifying security vulnerabilities

Fix, test, and release lifecycle

IaaS security considerations for Tanzu Operations Manager

Encrypting Tanzu Operations Manager VM disks

Encrypt disks or rotate keys

Azure

vSphere

AWS

Recreate BOSH-deployed disks

Security guidelines for your IaaS provider

Amazon Web Services (AWS)

Enable IMDSv2 in Tanzu Operations Manager

Google Cloud Platform (GCP)

Microsoft Azure

VMware vSphere

Stemcell security

Linux Stemcell Hardening

Windows stemcell hardening

Microsoft baseline security standard

Audit policies

Firewall settings

Floating stemcells in Tanzu Operations Manager

Floating stemcells: Benefits and limitations

Product tile upgrades

Uploading and managing stemcells

Tanzu Operations Manager assessment against NIST SP 800-53(r4) controls

Security-related Tanzu Operations Manager tiles and add-ons

Tiles

Add-ons

Troubleshooting and diagnostics for Tanzu Operations Manager

Troubleshooting problems in your Tanzu Operations Manager deployment

UAA token expires during product upload

Retry the deployment

Common issues

BOSH does not reinstall

Creating bound missing VMs times out

Install exits with a creates/updates/deletes app failure or with a 403 error

Install fails when Gateway instances exceed zero

Out of disk space error

Installing BOSH Director fails in vSphere

Deleting a deployment fails

Installing Elastic Application Runtime fails

Tanzu Operations Manager hangs during MicroBOSH install or HAProxy states "IP

Address Already Taken"

Poor performance

Common issues caused by firewalls

DNS resolution fails

Logging into Tanzu Operations Manager VMs with SSH

Log in to Tanzu Operations Manager VM with SSH

AWS

Azure

GCP

vSphere

Log in to the BOSH Director VM with SSH

Advanced Tanzu Operations Manager troubleshooting with the BOSH CLI

Gather credential and IP address information

Log in to the Tanzu Operations Manager VM with SSH

AWS

Azure

GCP

vSphere

Authenticate with the BOSH Director VM

Set the BOSH environment variables on the Tanzu Operations Manager VM

Create a local BOSH Director alias

Log in to the BOSH Director VM with UAA

Log in to the BOSH Director VM with SAML

SSH into the BOSH Director VM

Use the BOSH CLI for troubleshooting

BOSH VMs

BOSH Cloud check

Example scenarios

BOSH Create Recovery Plan and Recover

BOSH SSH

Troubleshooting BOSH Backup and Restore

Troubleshooting during a restore

Problem: Restore fails with a MySQL Monit start timeout

Problem: Deployment does not match the structure of the backup

General troubleshooting

Problem: Connection error

Problem: Error running metadata script

Troubleshooting Tanzu Operations Manager on Azure

Installation issues

Problem: Cannot copy the Tanzu Operations Manager image

Problem: Deployment fails at "create-env";

Insufficient external database permissions

Operational issues

Problem: Slow performance or timeouts

Problem: Service instance creation times out

Troubleshooting Tanzu Operations Manager for VMware vSphere

Tanzu Operations Manager installation fails

BOSH automated installation fails

Tanzu Operations Manager loses IP address after HA or reboot

Cannot connect to the OVF in a browser

Installation fails with failed network connection

Insufficient external database permissions

Additional Resources

[Documentation for other products related to Tanzu Operations Manager](#)

[Documentation Legal Notice](#)

Tanzu Operations Manager

This topic introduces VMware Tanzu Operations Manager and describes how you can get started.

Tanzu Operations Manager is a software tool to help you expand and manage products that you can install on your deployment. Each of these products is represented as a *tile* in the UI.

Tanzu Operations Manager deploys with the BOSH Director tile by default, which is a single point of control for VMs. The BOSH Director supports various IaaSes, including AWS, Azure, GCP, and vSphere.

With Tanzu Operations Manager, you can install the following types of tiles:

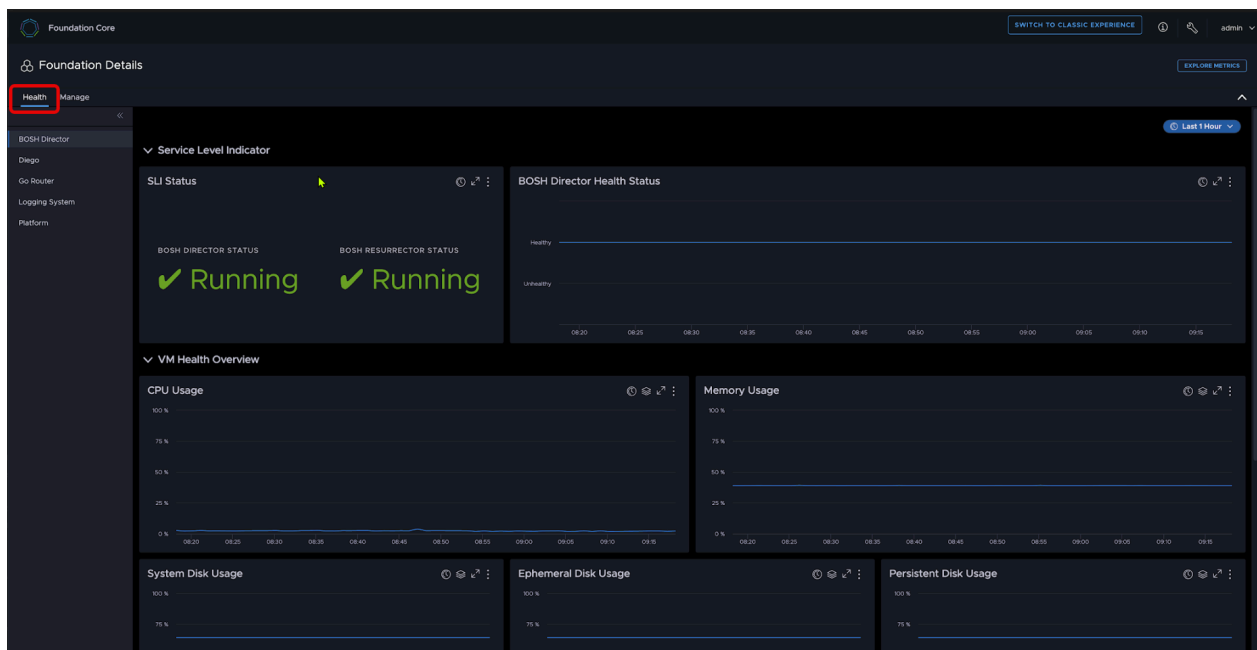
- **Runtime Tiles:** You can install VM-based or Kubernetes-based runtimes. These dynamically run apps and other processes to balance demand throughout app lifecycles. Example runtimes managed through Tanzu Operations Manager include Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI).
- **Service Tiles:** You can install various services that help support your deployment. Example services you can manage through Tanzu Operations Manager include Tanzu for MySQL, Tanzu for Valkey, and Tanzu Gemfire.

For more information about the tiles you can install, see [Broadcom Support portal](#).

Introducing Foundation Core

The full Operations Manager interface, plus new health metrics, is now available as Foundation Core (Beta). You can use Foundation Core just as you use Tanzu Operations Manager to install and configure your tiles. Tanzu Operations Manager functions are available on the Manage tab, and the observability metrics appear on the Health tab, where you can view health statistics for your deployment.

In the following screenshot, the Health tab is shown. On the Manage tab, you will find the familiar organization of the Tanzu Operations Manager configuration screens.



Beta Limitation: The BOSH Director cannot be configured from the Foundation Core in the Hub UI.

You must configure BOSH Director in Classic Operations Manager.

Other limitations of the Beta Foundation Core:

- Managing stemcells
- Configuring Tanzu Operations Manager settings
- Downloading logs
- Generating support bundles

Tanzu Operations Manager responsibilities

Tanzu Operations Manager is managed by an *operator*. The operator is responsible for installing and maintaining runtime and service tiles.

Operator responsibilities

The Tanzu Operations Manager operator is responsible for:

- Configuring Tanzu Operations Manager capabilities
- Integrating Tanzu Operations Manager with external systems
- Updating Tanzu Operations Manager and installed products
- Monitoring Tanzu Operations Manager health and performance
- Adjusting Tanzu Operations Manager resources and options to fix health or performance issues
- Diagnosing and troubleshooting Tanzu Operations Manager problems
- Managing Elastic Application Runtime users, resources, and infrastructure
- Installing software services and otherwise assisting developers
- Creating and managing Tanzu Operations Manager user and client accounts

Limited-access operator responsibilities

Operators can give limited-access view or write Tanzu Operations Manager responsibilities to others within an organization.

For more information about privileges, see [Configuring role-based access control in Tanzu Operations Manager](#).

Tanzu Operations Manager documentation

The following sections describe where to get started in the Tanzu Operations Manager documentation.

Release notes and upgrading

For information on new features, security patches, and upgrading Tanzu Operations Manager, see the following topics:

- [Tanzu Operations Manager v3.0 release notes](#)
- [Upgrading your Tanzu Operations Manager deployment](#)

Installing and configuring by IaaS

For information about managing each IaaS on Tanzu Operations Manager, see the following documentation sets:

- [Installing and configuring Tanzu Operations Manager on AWS](#)
- [Installing and configuring Tanzu Operations Manager on Azure](#)
- [Installing and configuring Tanzu Operations Manager on GCP](#)
- [Installing and configuring Tanzu Operations Manager on vSphere](#)

Using the UI

For information about how to fulfill common operator duties with the UI, see the following topics:

- [Using the Tanzu Operations Manager interface](#)
- [Adding and deleting products from Tanzu Operations Manager](#)
- [Importing and managing stemcells in Tanzu Operations Manager](#)
- [Managing errands in your Tanzu Operations Manager deployment](#)

Using the API or CLI

For information about how to use the Tanzu Operations Manager API or CLI and to configure deployments programmatically, see the following topics:

- [Using the Tanzu Operations Manager CLI](#)
- [Using the Tanzu Operations Manager API](#)
- [Using Tanzu Operations Manager programmatically and from the command line](#)

Backing up and troubleshooting

For information about how to back up, restore, and troubleshoot Tanzu Operations Manager deployments, see the following topics:

- [Backing up and restoring your Tanzu Operations Manager deployment](#)
- [Troubleshooting problems in your Tanzu Operations Manager deployment](#)

Monitoring and logging

For information about accessing logs and metrics for VMs and app containers, see the following topics:

- [Accessing logs and metrics for your Tanzu Operations Manager deployment](#)
- [Monitoring VM status in Tanzu Operations Manager](#)

VMware Tanzu Operations Manager v3.2 release notes

This topic contains release notes for VMware Tanzu Operations Manager v3.2.

Tanzu Platform for Cloud Foundry is now called Elastic Application Runtime. The current version of Elastic Application Runtime is 10.3.

You can upgrade directly from previous versions of Tanzu Operations Manager v3.0 or v3.1 to Tanzu Operations Manager v3.2. For more information, see [Jump Upgrading to Tanzu Operations Manager v3.2](#).

Tanzu Operations Manager is certified by the Cloud Foundry Foundation for 2023.

For more information about the Cloud Foundry Certified Provider Program, see [How Do I Become a Certified Provider?](#) on the Cloud Foundry website.

Releases

v3.2.0

Release Date: October 28, 2025

- [Feature] Adjust Avi loadbalancer verifier to better represent required permissions.
- [Feature] Expose tile name and version in informational memo section of BOSH manifest
- [Feature] vSphere VMs deployed by BOSH are now created with hardware version 17, will use Paravirtualized SCSI instead of LSI Logic SCSI, and the disk UUID extension is enabled by default.
- [Bug Fix] Improve OpenAPI documentation for Tanzu Operations Manager API

Component	Version
Tanzu Operations Manager	3.2.0-build.231
Stemcell (Bosh Director and Ops Manager)	1.915
BBR SDK	1.19.46
BOSH Director	282.1.0
BOSH DNS	1.39.18
System Metrics	3.0.15
CredHub	2.14.16
CredHub Maestro	11.0.1
om CLI	7.18.2
Syslog	12.3.12

Component	Version
Windows Syslog	1.3.10
Tanzu UAA	0.5.0
BPM	1.4.20
OS Conf	23.0.0
AWS CPI	106.0.3
Azure CPI	53.0.2
Google CPI	50.0.18
vSphere CPI	98.0.0
BOSH CLI	7.9.12
Credhub CLI	2.9.50
BBR CLI	1.9.75
Telemetry	2.2.0
Core Consumption	0.3.24
Tanzu License Verifier	2.5.7
Stitching Service	1.0.49
Observability Store	1.0.53
Prometheus	10.3.178
Foundation Core UI	1.0.19

New features in Tanzu Operations Manager v3.2

Tanzu Operations Manager v3.2 includes the following major features:

Tanzu Foundation Core UI

Tanzu Operations Manager is now deployed using Tanzu Foundation Core, which allows you to manage your foundation and view its health in one place.

In Foundation Core, you can see health metrics and beta Tanzu Operations Manager features.

You can switch back to the classic, full-function Tanzu Operations Manager UI experience using the “Switch to Classic Experience” button on the application header.

Breaking changes in Tanzu Operations Manager v3.2

Increased size requirements for Tanzu Operations Manager VM

Tanzu Operations Manager v3.2 increases the size requirements for the Tanzu Operations Manager VM. The minimum VM size requirements are now:

Machine Spec	Minimum Value
CPU	4 vCPUs

Machine Spec	Minimum Value
Memory	16 GB
Boot disk	100 GB

Customers should adjust any automation used to deploy the Tanzu Operations Manager VM when upgrading to Tanzu Operations Manager v3.2 to use an instance type that matches the new requirements.

OpenStack no longer supported

Tanzu Operations Manager v3.2 removes support for OpenStack.

Security fixes for Tanzu Operations Manager

The following tables list the CVEs that were fixed in each Tanzu Operations Manager / Foundation Core for VMware Tanzu Platform component since the previous patch version.

Release 3.2.0 CVE fixes for Azure

Component	Vulnerabilities Resolved
ops-manager-azure	• CVE-2022-2068 (Critical) • CVE-2022-1292 (Critical) • CVE-2025-4517 (Critical) • CVE-2024-8088 (High) • CVE-2024-9287 (High) • CVE-2023-6597 (High) • CVE-2023-4807 (High) • CVE-2025-8194 (High) • CVE-2025-4435 (High) • CVE-2025-4330 (High) • CVE-2025-4138 (High) • CVE-2024-7592 (High) • CVE-2024-6232 (High) • CVE-2024-4032 (High) • CVE-2023-36632 (High) • CVE-2023-0401 (High) • CVE-2023-0217 (High) • CVE-2023-0216 (High) • CVE-2023-0215 (High) • CVE-2022-4450 (High) • CVE-2022-3996 (High) • CVE-2022-3786 (High) • CVE-2022-3602 (High) • CVE-2022-3358 (High) • CVE-2022-1473 (High) • CVE-2024-0397 (High) • CVE-2023-0286 (High) • CVE-2025-30722 (Medium) • CVE-2023-2650 (Medium) • CVE-2025-0938 (Medium) • CVE-2024-0450 (Medium) • CVE-2025-4516 (Medium) • CVE-2024-50602 (Medium) • CVE-2022-4304 (Medium) • CVE-2022-1434 (Medium) • CVE-2025-30693 (Medium) • CVE-2024-6923 (Medium) • CVE-2024-12718 (Medium) • CVE-2023-40217 (Medium) • CVE-2023-27043 (Medium) • CVE-2022-2097 (Medium)

Component	Vulnerabilities Resolved
	• CVE-2022-1343 (Medium) • CVE-2022-4203 (Medium) • CVE-2025-6069 (Medium) • CVE-2024-11168 (Low) • CVE-2025-1795 (Low)

Release 3.2.0 CVE fixes for GCP

Component	Vulnerabilities Resolved
ops-manager-gcp	• CVE-2022-2068 (Critical) • CVE-2022-1292 (Critical) • CVE-2025-4517 (Critical) • CVE-2024-8088 (High) • CVE-2024-9287 (High) • CVE-2023-6597 (High) • CVE-2023-4807 (High) • CVE-2025-8194 (High) • CVE-2025-4435 (High) • CVE-2025-4330 (High) • CVE-2025-4138 (High) • CVE-2024-7592 (High) • CVE-2024-6232 (High) • CVE-2024-4032 (High) • CVE-2023-36632 (High) • CVE-2023-0401 (High) • CVE-2023-0217 (High) • CVE-2023-0216 (High) • CVE-2023-0215 (High) • CVE-2022-4450 (High) • CVE-2022-3996 (High) • CVE-2022-3786 (High) • CVE-2022-3602 (High) • CVE-2022-3358 (High) • CVE-2022-1473 (High) • CVE-2024-0397 (High) • CVE-2023-0286 (High) • CVE-2025-30722 (Medium) • CVE-2023-2650 (Medium) • CVE-2025-0938 (Medium) • CVE-2024-0450 (Medium) • CVE-2025-4516 (Medium) • CVE-2024-50602 (Medium) • CVE-2022-4304 (Medium) • CVE-2022-1434 (Medium) • CVE-2025-30693 (Medium) • CVE-2024-6923 (Medium) • CVE-2024-12718 (Medium) • CVE-2023-40217 (Medium) • CVE-2023-27043 (Medium) • CVE-2022-2097 (Medium)

Component	Vulnerabilities Resolved
	• CVE-2022-1343 (Medium) • CVE-2022-4203 (Medium) • CVE-2025-6069 (Medium) • CVE-2024-11168 (Low) • CVE-2025-1795 (Low)

Release 3.2.0 CVE fixes for vSphere

Component	Vulnerabilities Resolved
ops-manager-vmware	• CVE-2022-2068 (Critical) • CVE-2022-1292 (Critical) • CVE-2025-4517 (Critical) • CVE-2024-8088 (High) • CVE-2024-9287 (High) • CVE-2023-6597 (High) • CVE-2023-4807 (High) • CVE-2025-8194 (High) • CVE-2025-4435 (High) • CVE-2025-4330 (High) • CVE-2025-4138 (High) • CVE-2024-7592 (High) • CVE-2024-6232 (High) • CVE-2024-4032 (High) • CVE-2023-36632 (High) • CVE-2023-0401 (High) • CVE-2023-0217 (High) • CVE-2023-0216 (High) • CVE-2023-0215 (High) • CVE-2022-4450 (High) • CVE-2022-3996 (High) • CVE-2022-3786 (High) • CVE-2022-3602 (High) • CVE-2022-3358 (High) • CVE-2022-1473 (High) • CVE-2024-0397 (High) • CVE-2023-0286 (High) • CVE-2025-30722 (Medium) • CVE-2023-2650 (Medium) • CVE-2025-0938 (Medium) • CVE-2024-0450 (Medium) • CVE-2025-4516 (Medium) • CVE-2024-50602 (Medium) • CVE-2022-4304 (Medium) • CVE-2022-1434 (Medium) • CVE-2025-30693 (Medium) • CVE-2024-6923 (Medium) • CVE-2024-12718 (Medium) • CVE-2023-40217 (Medium) • CVE-2023-27043 (Medium) • CVE-2022-2097 (Medium)

Component	Vulnerabilities Resolved
	• CVE-2022-1343 (Medium) • CVE-2022-4203 (Medium) • CVE-2025-6069 (Medium) • CVE-2024-11168 (Low) • CVE-2025-1795 (Low)

Stemcell release notes

For information about the Stemcell versions used by Tanzu Operations Manager, see the [Stemcell Release Notes](#).

Architecture and planning for Tanzu Operations Manager and runtimes

See the following topics for information about platform architecture and how to plan your deployment of Tanzu Operations Manager, along with any runtime products:

- [Reference architectures for Tanzu Operations Manager and runtimes](#)
- [AVS reference architecture for Tanzu Operations Manager](#)
- [AWS reference architecture for Tanzu Operations Manager](#)
- [Azure reference architecture for Tanzu Operations Manager](#)
- [GCP reference architecture for Tanzu Operations Manager](#)
- [VMC reference architecture for Tanzu Operations Manager](#)
- [vSphere reference architecture for Tanzu Operations Manager](#)
 - [Using Edge Services Gateway on VMware NSX](#)
 - [Upgrading vSphere without causing downtime for your runtime](#)
 - [Migrating Tanzu Operations Manager to a new datastore in vSphere](#)
- [Global DNS load balancers for multi-foundation environments](#)

Reference architectures for Tanzu Operations Manager and runtimes

This topic describes reference architectures that you can use when installing Tanzu Operations Manager on any infrastructure to support the Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI) runtime environments. These reference architectures describe proven approaches for deploying Tanzu Operations Manager and runtimes on a specific IaaS, such as AWS, Azure, GCP, or vSphere.

These reference architectures meet the following requirements:

- They are secure.
- They are publicly accessible.
- They include common Tanzu Operations Manager-managed services such as Tanzu for MySQL, Tanzu RabbitMQ, and Spring Cloud Services.
- They can host at least 100 app instances.
- They have been deployed and validated by VMware to support Tanzu Operations Manager, Elastic Application Runtime, and Container Orchestration Engine.

You can use Tanzu Operations Manager reference architectures to help plan the best configuration for deploying your Tanzu Operations Manager on your IaaS.

Reference architecture and planning topics

All Tanzu Operations Manager reference architectures start with the base Elastic Application Runtime architecture and the base Container Orchestration Engine architecture.

These IaaS-specific topics build on these two common base architectures:

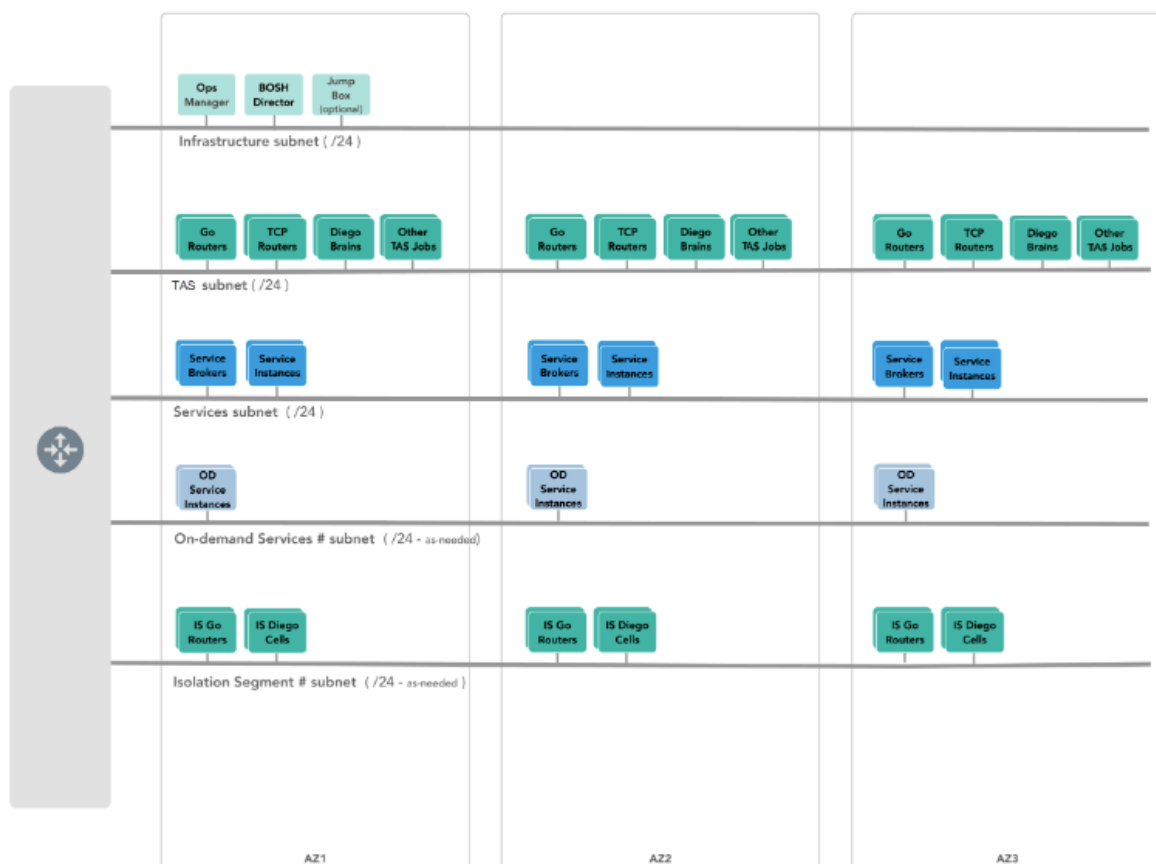
- [AWS reference architecture for Tanzu Operations Manager](#)
- [Azure reference architecture for Tanzu Operations Manager](#)
- [GCP reference architecture for Tanzu Operations Manager](#)
- [vSphere reference architecture for Tanzu Operations Manager](#)

The following topics address aspects of platform architecture and planning that the Tanzu Operations Manager reference architectures do not cover:

- [Implementing a multi-foundation Container Orchestration Engine deployment](#)
- [Global DNS load balancers for multi-foundation environments](#)

Elastic Application Runtime architecture

The following diagram illustrates a base architecture for Elastic Application Runtime and how its network topology places and replicates Elastic Application Runtime components across subnets and availability zones (AZs).



[View a larger version of this diagram.](#)

Internal components

The following table describes the internal component placements shown in the previous diagram:

Component	Placement and access notes
Tanzu Operations Manager VM	Deployed on one of the three public subnets. Accessible by fully-qualified domain name (FQDN) or through an optional jumpbox.
BOSH Director	Deployed on the infrastructure subnet.
Jumpbox	Optional. Deployed on the infrastructure subnet for accessing Elastic Application Runtime management components such as Tanzu Operations Manager and the Tanzu cf CLI.
Gorouters (HTTP routers in Tanzu Operations Manager)	Deployed on all three Elastic Application Runtime subnets, one per AZ. Accessed through the HTTP, HTTPS, and SSL load balancers.
Diego Brain	Deployed on all three Elastic Application Runtime subnets, one per AZ. The Diego Brain component is required, but SSH container access support through the Diego Brain is optional, and enabled through the SSH load balancers.
TCP routers	Optional. Deployed on all three Elastic Application Runtime subnets, one per AZ, to support TCP routing.
Service tiles	Service brokers and shared services instances are deployed to the Services subnet. Dedicated on-demand service instances are deployed to an on-demand services subnet.
Isolation segments	Deployed on an isolation segment subnet. Includes Diego Cells and Gorouters for running and accessing apps hosted within isolation segments.

Networks

The following sections describe Elastic Application Runtime's recommendations for defining your networks and load-balancing their incoming requests.

* Required subnets: Elastic Application Runtime requires these statically-defined networks to host its main component systems:

* Infrastructure subnet - `/24` segment
This subnet contains VMs that require access only for Platform Administrators, such as the Tanzu Operations Manager VM, the BOSH Director VM, and an optional jumpbox.

* Elastic Application Runtime subnet - `/24` segment
This subnet contains Elastic Application Runtime runtime VMs, such as Gorouters, Diego Cells, and Cloud Controllers.

* Services subnet - `/24` segment
The services and on-demand services networks support Tanzu Operations Manager tiles that you might add in addition to Elastic Application Runtime. These are the networks for everything that is not Elastic Application Runtime. Some services tiles can call for additional network capacity to grow into on-demand. If you use services with this capability, VMware recommends that you add an on-demand services network for each on-demand service.

* On-demand services subnets - `/24` segments
This is for services that can allocate network capacity on-demand from BOSH for their worker VMs. VMware recommends allocating a dedicated subnet to each on-demand service. For example, you can configure the Tanzu for Valkey tile as follows:

* **Network:** Enter the existing `Services` network to host the service broker.

* **Services network:** Deploy a new network `OD-Services1` to host the Redis worker VMs.

Another on-demand service tile can then also use `Services` for its broker and a new `OD-Services2` network for its workers, and so on.

* Isolation segment subnets - `/24` segments
You can add one or more isolation segment tiles to a Elastic Application Runtime installation to compartmentalize hosting and routing resources. For each isolation segment you deploy, you should designate a `/24` network for its range of address space.

Load balancing: Any Elastic Application Runtime installation needs a suitable load balancer to send incoming HTTP, HTTPS, SSH, and SSL traffic to its Gorouters and app containers. All installations approaching production-level use rely on external load balancing from hardware appliance vendors or other network-layer solutions.

The load balancer can also perform Layer 4 or Layer 7 load balancing functions. SSL can be terminated at the load balancer or used as a pass-through to the Gorouter.

Common deployments of load balancing in Elastic Application Runtime are:

- HTTP/HTTPS traffic to and from Gorouters
- TCP traffic to and from TCP routers
- Traffic from the Diego Brain, when developers access app containers through SSH

To load balance across multiple Elastic Application Runtime foundations, use an IaaS-specific or vendor-specific Global Traffic Manager or Global DNS load balancer.

For more information, see [Global DNS load balancers for multi-foundation environments](#).

High availability

Elastic Application Runtime is not considered high availability (HA) until it runs across at least two AZs. VMware recommends defining three AZs.

On an IaaS with its own HA capabilities, using the IaaS HA with a Elastic Application Runtime HA topology provides two benefits. Multiple AZs give Elastic Application Runtime redundancy, so that losing an AZ is not catastrophic. The BOSH Resurrector can then replace lost VMs as needed to repair a foundation.

To back up and restore a foundation externally, use BOSH backup and restore (BBR). For more information, see the [BOSH backup and restore](#) documentation.

Storage

Elastic Application Runtime requires disk storage for each component, both for persistent data and to allocate to ephemeral data. You size these disks in the Resource Config pane of the Elastic Application Runtime tile. For more information about storage configuration and capacity planning, see the corresponding section in the [reference architecture for your IaaS](#).

The platform also requires you to configure file storage for large shared objects. These blobstores can be external or internal. For details, see [Configuring File Storage for Elastic Application Runtime](#).

Security

For information about how Elastic Application Runtime implements security, see:

- [Tanzu Operations Manager security infrastructure](#)
- [Security concepts in Tanzu Operations Manager](#)
- [Certificates and TLS in your Tanzu Operations Manager deployment](#)
- [Routing Network Communications](#)

Domain names

Elastic Application Runtime requires these domain names to be registered:

- System domain for Elastic Application Runtime and other tiles: `sys.domain.name`
- App domain for your apps: `app.domain.name`

You must also define these wildcard domain names and include them when creating certificates that access Elastic Application Runtime and its hosted apps:

- `/*.SYSTEM-DOMAIN`

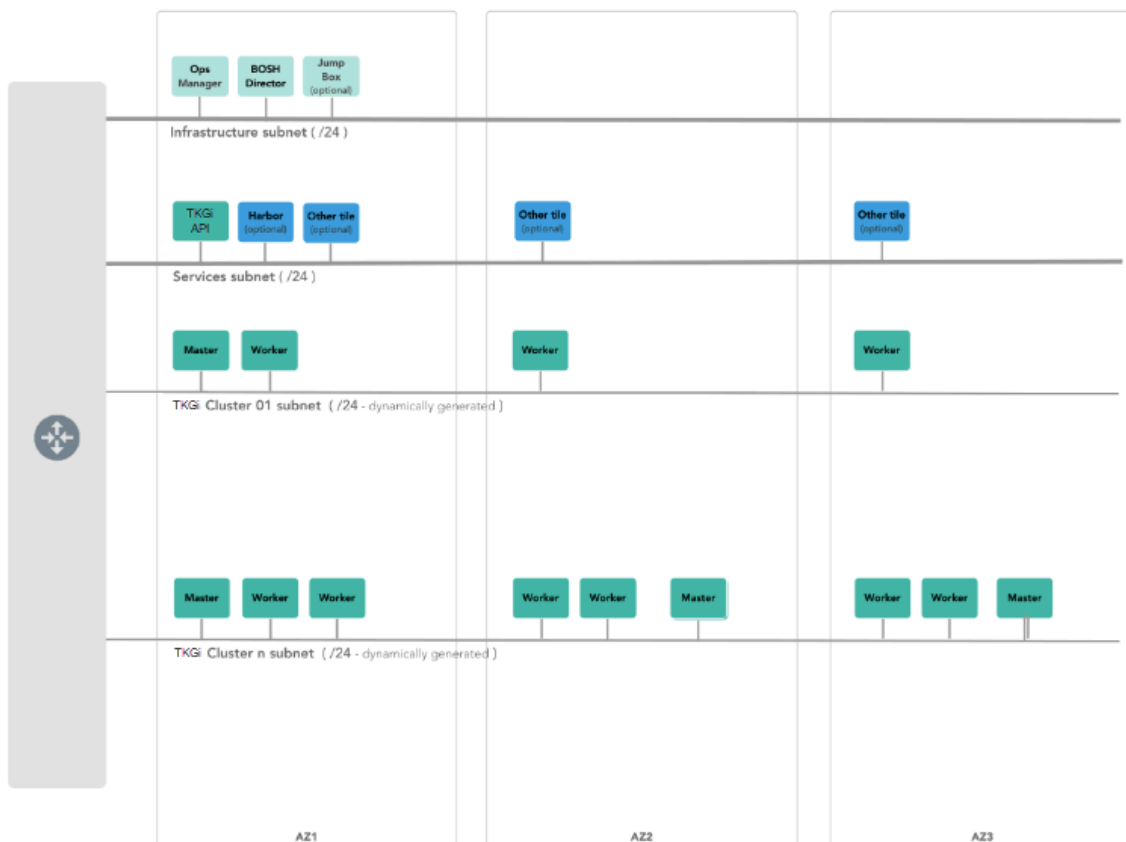
- `\*.APPS-DOMAIN`
- `\*.login.SYSTEM-DOMAIN`
- `\*.uaa.SYSTEM-DOMAIN`

Component scaling

For recommendations about scaling Elastic Application Runtime for different deployment scenarios, see [Scaling Elastic Application Runtime](#).

Container Orchestration Engine Architecture

The following diagram illustrates a base architecture for VMware Tanzu Container Orchestration Engine (formerly TKGI) and how its network topology places and replicates Container Orchestration Engine components across subnets and AZs:



[View a larger version of this diagram.](#)

Internal components

The table below describes the internal component placements shown in the diagram above:

Component	Placement and Access Notes
Tanzu Operations Manager VM	Deployed on one of the subnets. Accessible by fully-qualified domain name (FQDN) or through an optional jumpbox.
BOSH Director	Deployed on the infrastructure subnet.
Jumpbox	Optional. Deployed on the infrastructure subnet for accessing Container Orchestration Engine management components such as Tanzu Operations Manager and the <code>kubectl</code> command line interface.

Component	Placement and Access Notes
Container Orchestration Engine API	Deployed as a service broker VM on the Container Orchestration Engine services subnet. Handles Container Orchestration Engine API and service adapter requests, and manages Container Orchestration Engine clusters. For more information, see Container Orchestration Engine Architecture .
Harbor tile	Optional container images registry, typically deployed to the services subnet.
Container Orchestration Engine Cluster	Deployed to a dynamically-created, dedicated Container Orchestration Engine cluster subnet. Each cluster consists of worker nodes that run the workloads, or apps, and one or more master nodes.

Networks

These sections describe VMware's recommendations for defining your networks and load-balancing incoming requests.

- Subnets requirements: Container Orchestration Engine requires these defined networks to host the main elements:
 - Infrastructure subnet - [/24](#)
This subnet contains VMs that require access only for Platform Administrators, such as the Tanzu Operations Manager VM, the BOSH Director VM, and an optional jumpbox.
 - Container Orchestration Engine services subnet - [/24](#)
This subnet hosts Container Orchestration Engine API VM and other optional service tiles such as VMware Harbor Registry.
 - Container Orchestration Engine cluster subnets - each one a [/24](#) from a pool of pre-allocated IPs
These subnets host Container Orchestration Engine clusters.
- Load balancing: Load balancers can be used to manage traffic across primary nodes of a Container Orchestration Engine cluster or for deployed workloads. For information about how to configure load balancers for Container Orchestration Engine, see the corresponding section in the [reference architecture for your IaaS](#).

High availability

Container Orchestration Engine has no inherent HA capabilities to design for. Make the best efforts to design HA at the IaaS, storage, power, and access layers to support Container Orchestration Engine.

Storage

Container Orchestration Engine requires shared storage across all AZs for the deployed workloads to appropriately allocate their required storage.

Security

For information about how Container Orchestration Engine implements security, see [Enterprise Container Orchestration Engine Security](#) and [Firewall Ports](#).

Domain names

Container Orchestration Engine requires the [*.pks.domain.name](#) domain name to be registered when creating a wildcard certificate and Container Orchestration Engine tile configurations.

The wildcard certificate covers both the Container Orchestration Engine API domain, such as [api.pks.domain.name](#), and the Container Orchestration Engine clusters domains, such as

`cluster.pks.domain.name.`

Cluster management

For information about managing Container Orchestration Engine clusters, see [Managing clusters](#).

AVS reference architecture for Tanzu Operations Manager

This topic describes the Azure VMware Solution (AVS) reference architecture for Tanzu Operations Manager and any runtime products, including Tanzu Platform for Cloud Foundry. This architecture builds on the common base architectures described in [Platform architecture and planning](#).

For additional requirements and installation instructions for Tanzu Operations Manager on AVS, see the [Elastic Application Runtime documentation](#).

The AVS reference architecture for the Elastic Application Runtime runtime tiles is based on a software-defined networking (SDN) infrastructure. AVS offers NSX-T to support SDN infrastructure.

Elastic Application Runtime on AVS with NSX-T

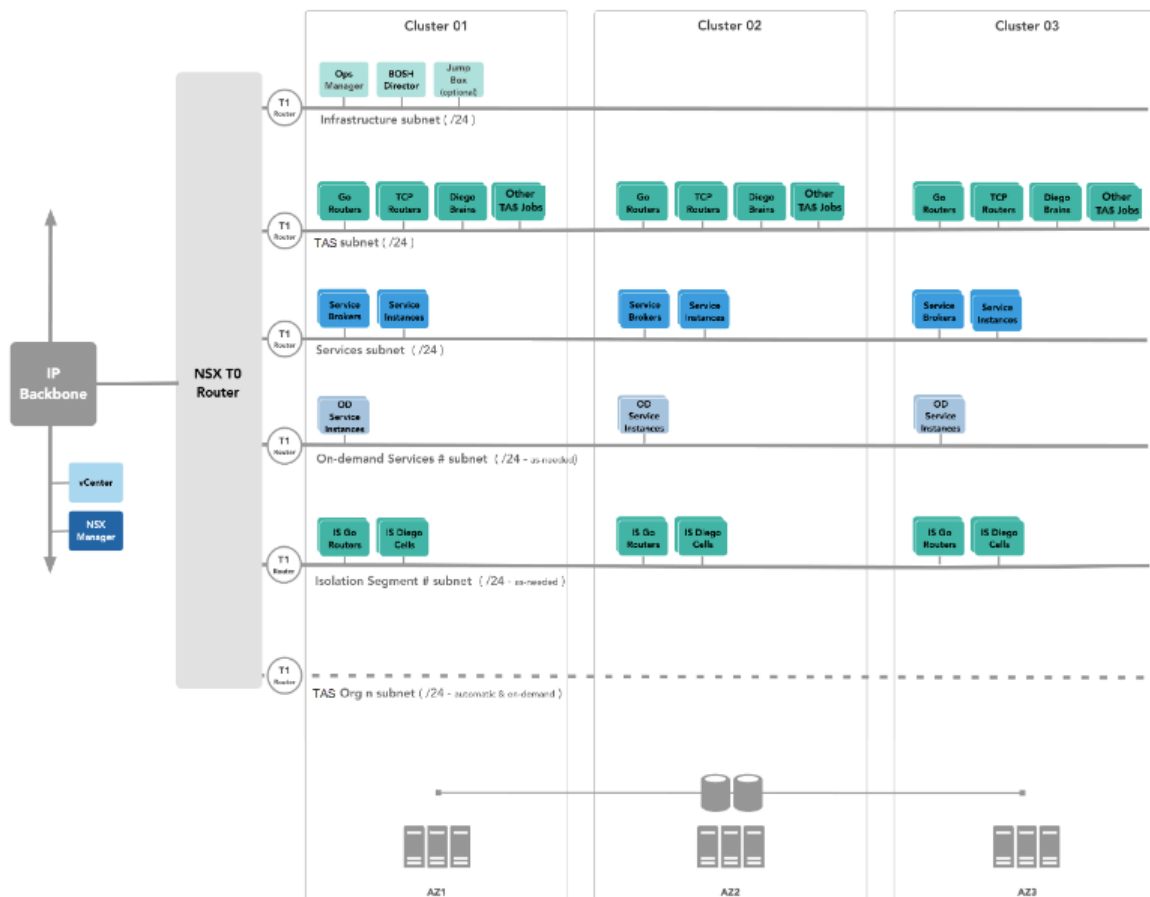
The following sections describe the reference architecture for Tanzu Operations Manager with Elastic Application Runtime on AVS with NSX-T deployments. They also provide requirements and recommendations for deploying Tanzu Operations Manager with Elastic Application Runtime on AVS with NSX-T, such as network, load balancing, and storage capacity requirements and recommendations.

Elastic Application Runtime on AVS with NSX-T supports these following SDN features:

- Virtualized, encapsulated networks and encapsulated broadcast domains
- VLAN exhaustion avoidance with the use of virtualized Logical Networks
- DNAT/SNAT services to create separate, non-routable network spaces for the Elastic Application Runtime installation
- Load balancing services to pass traffic through Layer 4 to pools of platform routers at Layer 7
- SSL termination at the load balancer at Layer 7 with the option to forward on at Layer 4 or 7 with unique certificates
- Virtual, distributed routing and firewall services native to the hypervisor

Architecture

The following diagram illustrates reference architecture for Elastic Application Runtime on AVS with NSX-T deployments:



[View a larger version of this diagram..](#)

Elastic Application Runtime deployments with NSX-T are deployed with three clusters and three availability zones (AZs).

An NSX-T Tier-0 router is on the front end of the Elastic Application Runtime deployment. This router is a central logical router into the Elastic Application Runtime platform. You can configure static or dynamic routing using Border Gateway Protocol (BGP) from the routed IP address backbone through the Tier-0 router with the edge gateway.

Several Tier-1 routers, such as the router for the Elastic Application Runtime and infrastructure subnets, connect to the Tier-0 router.

NSX-T Container Plug-in requirement

Elastic Application Runtime deployments require the VMware NSX-T Container Plug-in to enable the SDN features available through NSX-T.

The NSX-T Container Plug-in enables a container networking stack, and integrates with NSX-T.

To use NSX-T with Elastic Application Runtime, the NSX-T Container Plug-in must be installed, configured, and deployed at the same time as the Elastic Application Runtime tile. To download the NSX-T Container Plug-in, go to the [VMware NSX-T Container Plug-in](#) page on Broadcom Support portal.

Networking

These sections describe networking requirements and recommendations for Elastic Application Runtime on AVS with NSX-T deployments.

The Tier-0 router must have external route IP address space to advertise on the BGP network with its peers. Select a network range for the Tier-0 router with enough space so that you can separate the network into these two jobs:

- Routing incoming and outgoing traffic
- DNATs and SNATs, load balancer VIPs, and other platform components

Firewall

Tanzu Operations Manager requires that the NSX firewall routes are not blocked. By default, all communication between Tanzu Operations Manager VMs and vCenter or ESXi hosts route through the NSX firewall are blocked. For more information on which ports to enable, see [VMware ports and protocols](#) for vSphere.

DNS

Elastic Application Runtime requires a system domain, app domain, and several wildcard domains.

For more information about DNS requirements for Elastic Application Runtime, see [Domain names](#) in *Platform Planning and Architecture*.

Load balancing

The load balancing requirements and recommendations for Elastic Application Runtime on AVS with NSX-T deployments are:

- You must configure NSX-T load balancers for the Gorouters.
 - The domains for the Elastic Application Runtime system and apps must resolve to the load balancer VIP.
 - You must assign either a private or a public IP address assigned to the domains for the Elastic Application Runtime system and apps.
- Any TCP Gorouters and SSH Proxies within the platform also require NSX-T load balancers.
- Layer 4 and Layer 7 NSX-T load balancers are created automatically during app deployment.

Networking, subnets, and IP address spacing

The requirements and recommendations related to networks, subnets, and IP address spacing for Elastic Application Runtime on AVS with NSX-T deployments are:

- Elastic Application Runtime requires statically-defined networks to host its component VMs.
- The client side of an NSX-T deployment uses a series of non-routable address blocks when using DNAT/SNAT at the Tier-0 interface.
- The reference architecture for Elastic Application Runtime on AVS with NSX-T deployments uses a pattern in which all networks are calculated on the /24 8-bit network boundary. The network octet is numerically sequential.
- NSX-T dynamically assigns Elastic Application Runtime org networks and adds a Tier-1 router. These org networks are automatically instantiated based on a non-overlapping block of address space. You can configure the block of address space in the NCP Configuration section of the NSX-T tile in Tanzu Operations Manager. The default is /24. This means that every org in Elastic Application Runtime is assigned a new /24 network.

For more information about Elastic Application Runtime subnets, see [Required subnets](#) in *Platform Architecture and Planning Overview*.

High availability

AVS deploys a NSX-T with Active/Active HA.

For information about high availability (HA) requirements and recommendations for Elastic Application Runtime on AVS, see [High Availability](#) in *Platform Architecture and Planning Overview*.

Shared storage

Elastic Application Runtime requires shared storage. By default, AVS deploys with vSAN configured.

For information about storage on AVS, see the [Microsoft AVS documentation](#).

SQL Server

An internal MySQL database is sufficient for use in production environments.

However, an external database provides more control over database management for large environments that require multiple data centers.

For information about configuring system databases on Elastic Application Runtime, see the [Elastic Application Runtime documentation](#).

Security

For information about security requirements and recommendations for Elastic Application Runtime deployments, see [Security](#) in *Platform Architecture and Planning Overview*.

Blobstore storage

Use Azure Blob Storage as the external file storage option for Tanzu Operations Manager. This provides redundancy for high-availability deployments of Tanzu Operations Manager and unlimited scaling. Azure Blob Storage provides fully-redundant hot, cold, or archival storage in either local, regional, or global offerings.

Tanzu Operations Manager requires a bucket for the BOSH blobstore.

Elastic Application Runtime requires the following buckets:

- Buildpacks
- Droplets
- Packages
- Resources

These buckets require an associated role for read-write access.

HCX migrations

HCX Migration of Tanzu Operations Manager and any runtime products, including Elastic Application Runtime, on AVS is not supported at this time.

AWS reference architecture for Tanzu Operations Manager

This topic describes the Amazon Web Services (AWS) reference architecture for Tanzu Operations Manager, including Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI). This architecture builds on the common base architectures described in [Platform Architecture and Planning Overview](#).

For general requirements for running Tanzu Operations Manager and specific requirements for running Tanzu Operations Manager on AWS, see [Tanzu Operations Manager on AWS Requirements](#).

Elastic Application Runtime

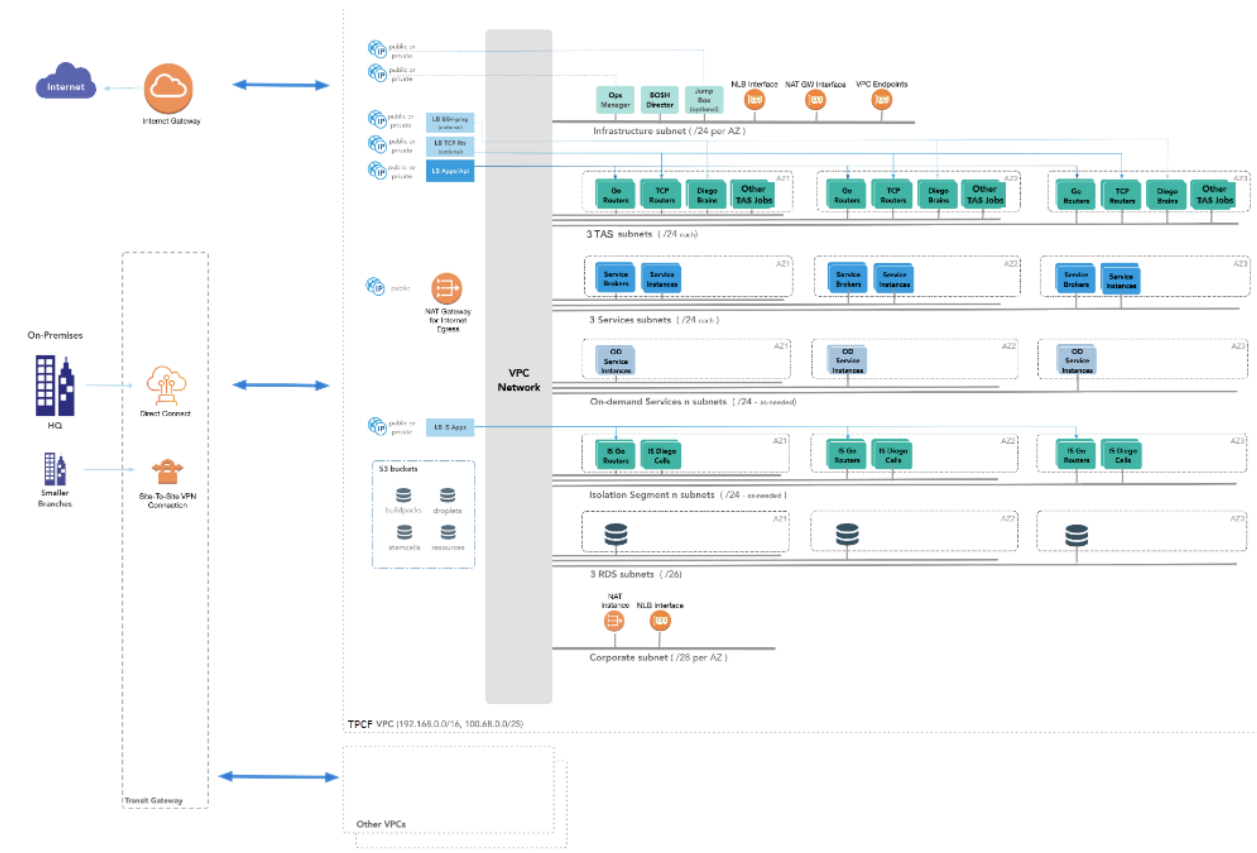
This section describes the reference architecture for a Elastic Application Runtime installation on AWS.

Reference architecture diagram

The following diagram illustrates the reference architecture for Elastic Application Runtime on AWS. The diagram shows the flow and components from the HQ, branch offices, and broader internet into the VPC

Network. Categories of elements in the VPC Network include:

- Infrastructure subnet (with Tanzu Operations Manager, BOSH Director, Jumpbox, NLB interface, NAT GW Interface, and VPC Endpoints)
- 3 Elastic Application Runtime subnets
- 3 Services subnets
- On-demand Services subnets
- Isolation Segment subnets
- 3 RDS subnets
- Corporate subnet (with a NAT instance and an NLB interface)



[View a larger version of this diagram..](#)

Networking

These sections provide guidance about networking resources.

DNS

You can use AWS Route 53 for DNS resolution to host your Elastic Application Runtime domains.

Load balancing

AWS offers Application Load Balancers (ALBs) and Network Load Balancers (NLBs). VMware recommends using NLBs because of the simplicity and security of leaving all encrypted traffic in place until it reaches the Gorouters. If you require Amazon Certificate Management or more complex Layer 7 routing rules, you can front or replace the NLB with an ALB.

Networks, subnets, and IP spacing planning

When planning networks, subnets, and IP spacing, some things to consider are:

- In AWS, each availability zone (AZ) requires its own subnet; the first five IPs must be reserved.
- Your organization might not have the IP space necessary to deploy Elastic Application Runtime in a consistent manner. To alleviate this, VMware recommends using the `100.64` address space to deploy a NAT instance and router so that the Tanzu Operations Manager deployment can use fewer IP addresses.
- If you want the front end of Elastic Application Runtime to be accessible from your corporate network, or the services running on Elastic Application Runtime to be able to access corporate resources, you must either:
 - provide routable IPs to your VPC
 - use NAT
- If you want a VPC that is only public-facing, no special consideration is necessary for the use of IPs.
- To understand the action required, depending on the type of traffic you want to allow for your Tanzu Operations Manager deployment, see the following table:

If you want...	then...
internet ingress	Create the load balancers with Elastic IPs.
internet egress	Create the NAT Gateway toward the internet.
corporate ingress	Create load balancers on your corporate subnet.
corporate egress (for accessing corporate resources)	Create the NAT instance on your corporate subnet, or if business requirements dictate, make the entire VPC corporate-routable.

- If you plan to install [VMware Tanzu Service Broker for AWS](https://techdocs.broadcom.com/tnz-aws-broker-cf), you might want to create another set of networks in addition to those outlined in the base reference architecture. This set of networks would be used for RDS and other AWS-managed instances.

Alternative network layouts

These options are available for choosing your network topology:

Type of traffic	Options
internet ingress	• Allow internet ingress by associating load balancers with Elastic IPs. • Do not allow internet ingress.
internet egress	• Allow Elastic Application Runtime to access the internet through a NAT gateway. • Do not allow Elastic Application Runtime to access the internet. Use only a corporate network.
peering	• Isolated. This is normally internet-only facing. • Peered with corporate or other VPCs through Transit Gateway • Peered directly with other VPCs and corporate connections
corporate peering	• SNAT for corporate resources. This requires three corporate IPs. • Routable IP addresses for VPC using /20 or a larger CIDR block.

RDS

Use a single RDS instance for BOSH and Elastic Application Runtime. This instance requires several databases. For more information, see:

- [Step 3: Director Config Page](#) in *Configuring BOSH Director on AWS*
- [Creating databases for Elastic Application Runtime](#)

Blobstore storage accounts

Tanzu Operations Manager requires a bucket for the BOSH blobstore.

Elastic Application Runtime requires these buckets:

- Buildpacks
- Droplets
- Packages
- Resources

These buckets require an associated role for read/write access.

Identity management

For identity management, use Instance Profiles whenever possible. For example, the AWS Config Page of the BOSH Director tile provides a Use AWS Instance Profile option. For more information, see [Step 2: AWS Config Page](#) in *Configuring BOSH Director on AWS*.

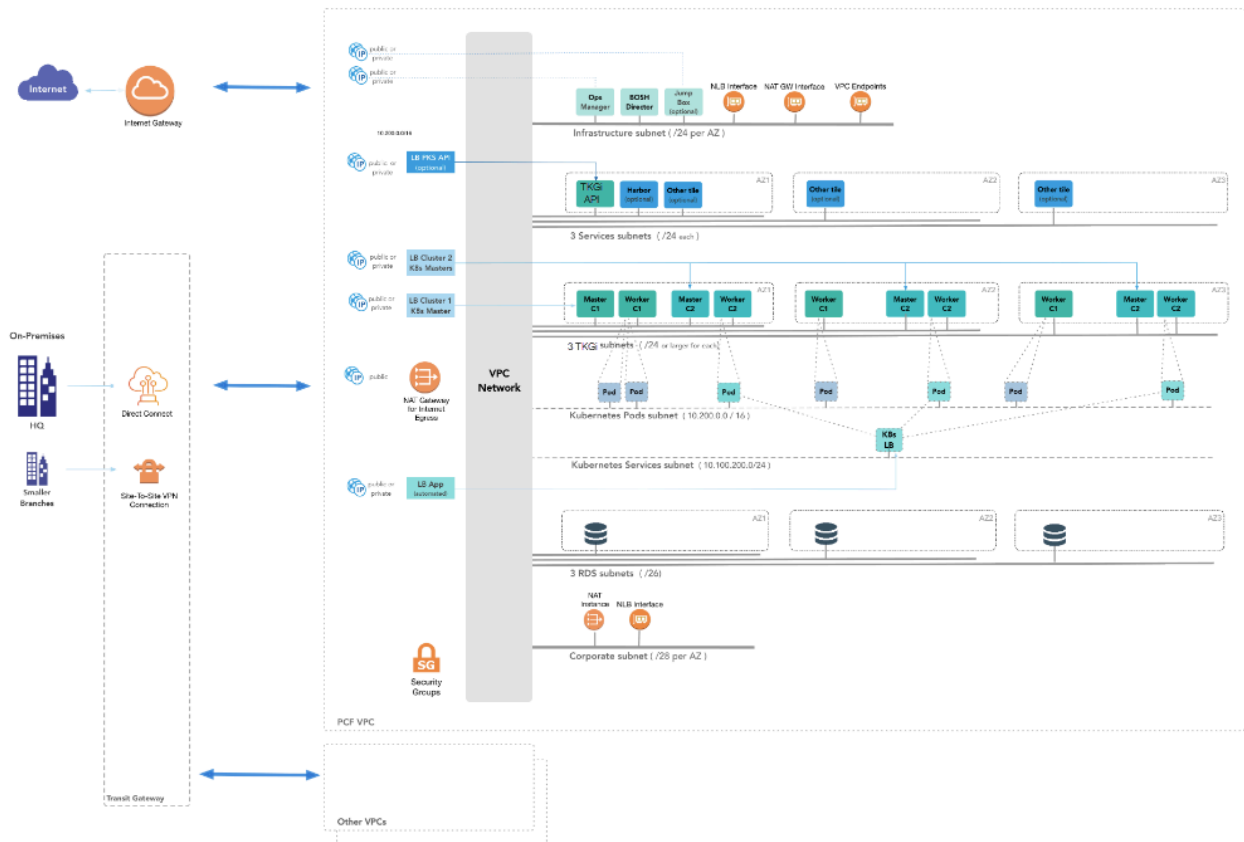
Container Orchestration Engine

The following sections describe the reference architecture for a Container Orchestration Engine installation on AWS.

Architecture diagram

The following diagram illustrates the reference architecture for Container Orchestration Engine on AWS. The diagram shows the flow and components from the HQ, branch offices, and the broader internet into the VPC Network. Categories of elements in the VPC Network include:

- Infrastructure subnet (with Tanzu Operations Manager, BOSH Director, Jumpbox, NLB interface, NAT GW Interface, and VPC Endpoints)
- 3 Services subnets
- 3 PKS subnets
- Kubernetes Pods subnet
- Kubernetes Services subnet
- 3 RDS subnets
- Corporate subnet (with a NAT instance and an NLB interface)



[View a larger version of this diagram..](#)

Networking

These sections provide guidance about networking resources.

DNS

You can use AWS Route 53 for DNS resolution to host your Container Orchestration Engine domains.

Load balancing

You can configure a front end load balancer to enable public access to the Container Orchestration Engine API VM.

For each Container Orchestration Engine cluster, VMware recommends creating a load balancer for the corresponding primary node or nodes. For more information, see [Creating and Configuring an AWS Load Balancer for Container Orchestration Engine Clusters](#).

For deployed workloads, Kubernetes can automatically create load balancers when applying the service definition to the app instances. For more information, see [Deploying and Exposing Basic Workloads](#).

Networks, subnets, and IP spacing planning

When planning networks, subnets, and IP spacing, some things to consider are:

- In AWS, each AZ requires its own subnet; the first five IPs must be reserved.
- * If you want the front end of Container Orchestration Engine to be accessible from your corporate network, or the services running on Container Orchestration Engine to be able to access corporate resources, you must either:
 - provide routable IPs to your VPC
 - use NAT
- Create the Container Orchestration Engine subnets that host the Kubernetes cluster VMs with enough IP address capacity to accommodate the number of VMs and clusters you expect to deploy. The reference architecture diagram above shows a /24 CIDR for the Container Orchestration Engine subnets, which allows for about 250 VMs per AZ. If you expect a larger number of clusters for your organization, consider defining larger subnets, such as /20, to make room for future growth.

Type of traffic	Options
internet ingress	- Allow internet ingress by associating load balancers with Elastic IPs. - Do not allow internet ingress.
internet egress	- Allow nodes and pods to access the internet through a NAT gateway. - Do not allow nodes or pods to access the internet. Use only a corporate network.
Peering	- Isolated. This is normally internet-only facing. - Peered with corporate or other VPCs through Transit Gateway. - Peered directly with other VPCs and corporate connections.
Corporate peering	- SNAT for corporate resources. This requires three corporate IPs. - Routable IP Addresses for VPC using /20 or a larger CIDR block.

RDS instance

Use a single RDS instance for BOSH. This instance requires a database. For more information, see [Step 3: Director Config page](#) in *Configuring BOSH Director on AWS*.

Identity management

To create and manage Kubernetes primary and worker nodes, use Instance Profiles. For more information, see [Kubernetes cloud provider](#) in *Installing Enterprise Container Orchestration Engine on AWS*.

For information about installing the required AWS constructs for Container Orchestration Engine, see [Installing Tanzu Operations Manager on AWS](#).

Azure reference architecture for Tanzu Operations Manager

This topic describes the Azure reference architecture for Tanzu Operations Manager and any runtime products, including Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI). This architecture builds on the common base architectures described in [Platform Architecture and Planning Overview](#).

For additional requirements and installation instructions for running Tanzu Operations Manager on Azure, see [Installing Tanzu Operations Manager on Azure](#).

Azure Virtual Data Center (VDC) uses a hub and spoke model for extending on-premise data centers. For more information about VDC, see the [Azure Virtual Datacenter e-book](#).

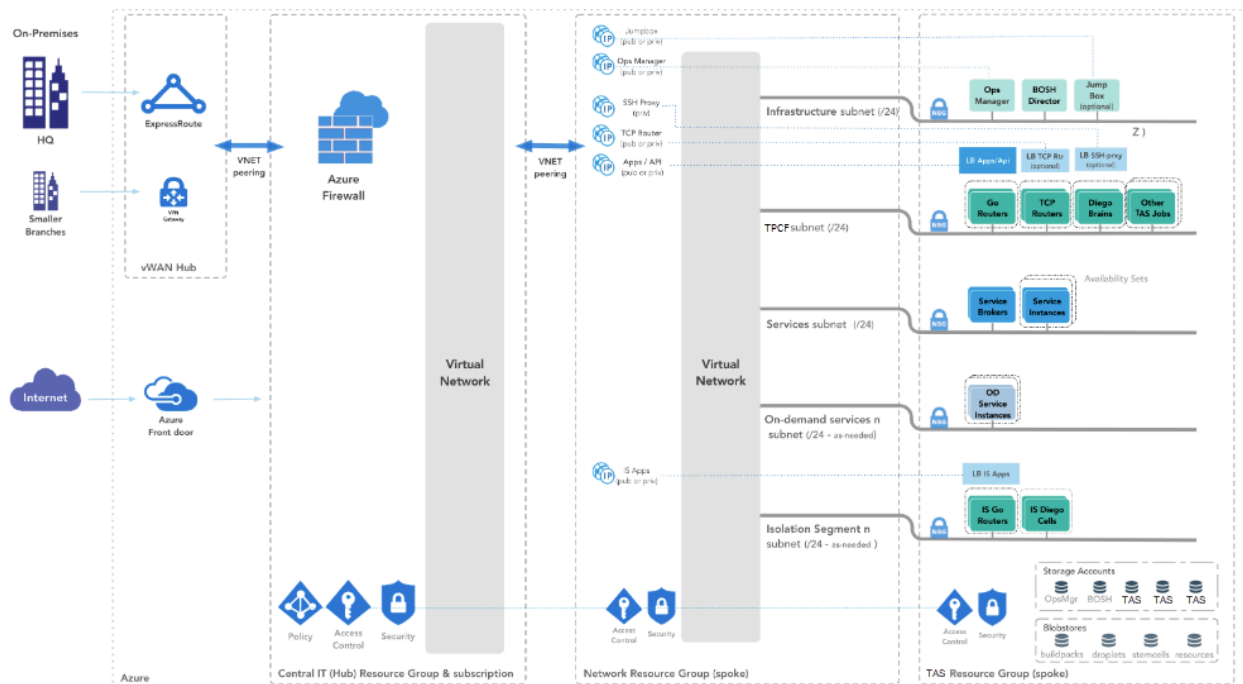
When you install Tanzu Operations Manager, you can use this model to place resources in resource groups. For example, a Tanzu Operations Manager installation contains:

- Hub:
 - A central IT resource group included in Azure VDC setup. This is where you define firewall, access control, and security policies.
- Spokes:
 - A Network resource group for all network resources.

- Elastic Application Runtime or Container Orchestration Engine resource groups for all Elastic Application Runtime or Container Orchestration Engine resources.

Reference architecture diagram

The following diagram illustrates the reference architecture for Tanzu Operations Manager on Azure using the hub and spoke model.



[View a larger version of this diagram..](#)

The following sections provide guidance about the resources that you use for running Tanzu Operations Manager on Azure.

Networking

The following sections provide guidance about networking resources.

General requirements and recommendations

The following are general requirements and recommendations related to networking when running Tanzu Operations Manager on Azure:

- You must enable virtual network peering between hub and spoke resource groups. For more information, see the [Microsoft Azure documentation](#).
- You must have separate subscriptions for each resource group. This includes:
 - The central network resource group. This is the hub.
 - Each workspace where workloads run. These are the spokes.
- Use ExpressRoutes for a dedicated connection from an on-premises datacenter to VDC. For more information, see the [ExpressRoute documentation](#). As a backup for when ExpressRoutes stops, you can use Azure VPN Gateways, which pass encrypted data into VDC through the internet.
- Use a central firewall in the hub resource group and a network appliance to prevent data infiltration and exfiltration.

- You can place network resources, such as DNS and NTP servers, either on-premise or on Azure Cloud.

DNS

For your domains, use a sub-zone of an Azure DNS zone.

Azure DNS supports DNS delegation, which allows sub-level domains to be hosted within Azure. For example, a company domain of `example.com` has a Tanzu Operations Manager zone in Azure DNS of `pcf.example.com`.

Azure DNS does not support recursion. To properly configure Azure DNS:

1. Create an `NS` record with your registrar that points to the four name servers supplied by your Azure DNS Zone configuration.
2. Create the required wildcard `A` records for the Elastic Application Runtime app and system domains, and any other records desired for your Tanzu Operations Manager deployments.

You do not need to make any configuration changes in Tanzu Operations Manager to support Azure DNS.

Load balancing

Use Standard Azure Load Balancers (ALBs). To allow internet ingress, you must create the ALBs with elastic IP addresses.

During installation, you configure ALBs for Elastic Application Runtime Gorouters. For more information, see [Create Load Balancers](#) in *Deploying Tanzu Operations Manager on Azure*. When applicable, TCP routers and SSH Proxies also require load balancers.

The Elastic Application Runtime system and app domains must resolve to your ALBs and have either a private or a public IP address assigned. You set these domains in the Domains pane of the Elastic Application Runtime tile.

Networks

Tanzu Operations Manager requires the networks defined in the base reference architectures for Elastic Application Runtime and Container Orchestration Engine. For more information, see the *Networks* sections in [Platform Architecture and Planning Overview](#).

Elastic Application Runtime on a single resource group

If shared network resources do not exist in an Azure subscription, you can use a single resource group to deploy Tanzu Operations Manager and define network constructs.

High Availability (HA)

Tanzu Operations Manager on Azure supports both availability zones (AZs) and Availability Sets.

Use AZs instead of Availability Sets in all Azure regions where they are available. AZs solve the placement group pinning issue associated with Availability Sets and allow greater control when defining high-availability configuration when using multiple Azure services.

For additional guidance depending on your use case, see the following table:

Foundation type	Guidance
New foundation	If you do not have strict requirements, select AZ-enabled regions. Examples of strict requirements include existing ExpressRoute connections from non-AZ enabled regions to customer data centers and app-specific latency requirements.
Existing foundation in AZ-enabled region	Migrate to a new Tanzu Operations Manager foundation, or to a Tanzu Operations Manager foundation so that you can use AZs. These existing foundations, especially those using older families of VMs such as DS_v2 , risk running out of capacity as they scale.
Existing foundation not in AZ-enabled region	Evaluate your requirements to determine the cost/benefit of migrating to an AZ-enabled region.

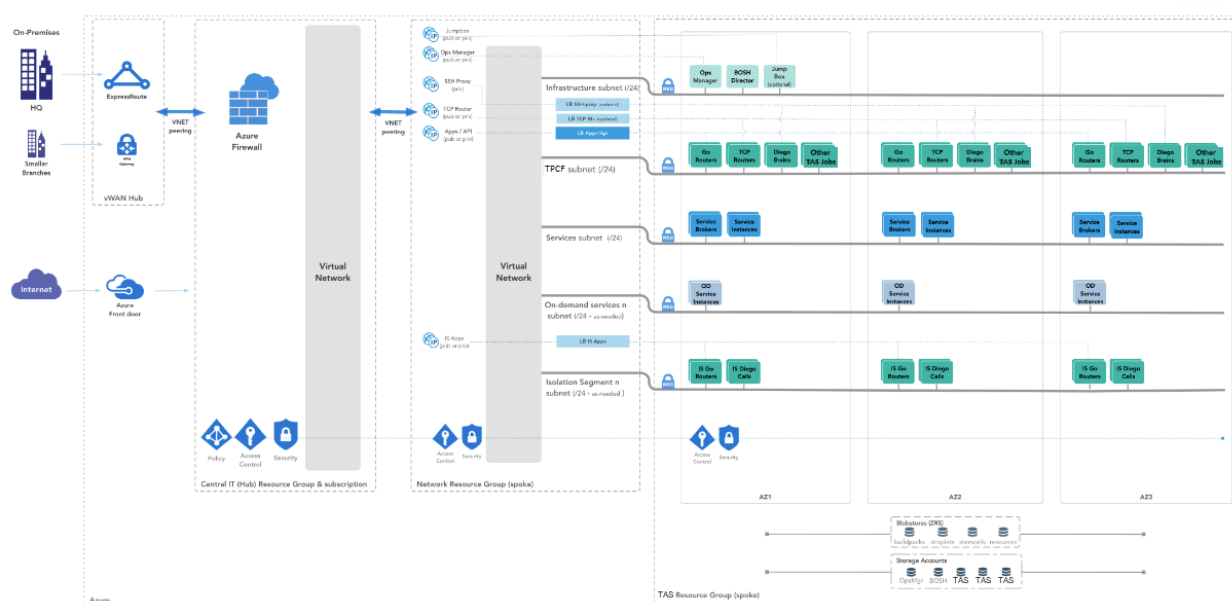
For more information about availability modes in Azure, see the following:

- [Availability Zones](#)
- [Availability Sets](#)
- The [Microsoft Azure documentation](#)

Availability zones

AZs are an Azure primitive that supports HA within a region. They are physically separate data centers in the same region, each with separate power, cooling, and network.

The following diagram illustrates Elastic Application Runtime deployed on Azure using AZs.



[View a larger version of this diagram..](#)

AZs have the following advantages:

- VMs and other zonal services deployed to independent zones provide greater physical separation than Fault Domains and avoid the placement group pinning issue associated with Availability sets.
- Zonal services are effectively in separate Update Domains as updates are rolled out to one zone at a time.
- AZs are visible and configurable. Two zonal services placed in the same logical zone are also placed in the same physical zone. This allows services of different types to take advantage of the highly available qualities that a zone provides.

Availability sets

Availability sets were the first Azure primitive to support high availability within a region.

Availability Sets are implemented at the hardware rack level, which can cause issues when deploying Tanzu Operations Manager on Azure. For more information, see [Placement group pinning issues](#).

Availability Sets have the following properties:

- They are a logical concept that ensures multiple VMs in the same set are spread evenly across three Fault Domains and either 5 or 20 Update Domains.
- When a hardware issue occurs, VMs on independent Fault Domains are guaranteed not to share power supply or networking and therefore maintain availability of a subset of VMs in the set.
- When updates are rolled out to a Cluster, VMs in independent Update Domains are not updated at the same time, ensuring that downtime caused by an update does not affect the availability of the set.

Placement group pinning issues

In Availability Set HA mode, VMs for a particular job are pinned to a placement group determined when the first VM in the Availability Set is provisioned. This is because BOSH places VMs for each job in the same Availability Set.

Cluster-pinning causes issues when:

- migrating jobs from one family of VMs to another, if they are backed by different hardware, such as `DS_v2` and `DS_v3`
- scaling up if there is no capacity remaining in the placement group

Storage

For storage, VMware recommends:

- For the storage account type, use the premium performance tier.
- Configure storage accounts for Zone Redundant Storage (ZRS) in AZ-enabled Regions. In non-AZ enabled regions, locally Redundant Storage (LRS) is sufficient.
- Configure VMs to use managed disks. Managed disks are an Azure feature that handles the correct distribution of VHDs to maintain high IOPS and high availability for the VMs.

SQL Server

The Internal MySQL database provided by Elastic Application Runtime is sufficient for production use.

Blobstore storage account

Use Azure Blob Storage as the external file storage option for Elastic Application Runtime. This provides redundancy for high-availability deployments of Elastic Application Runtime and unlimited scaling. Azure Blob Storage provides fully-redundant hot, cold, or archival storage in local, regional, or global offerings.

Elastic Application Runtime requires the following buckets:

- Buildpacks
- Droplets

- Packages
- Resources

These buckets require an associated role for read-write access.

Identity management

Use unique managed identities to deploy Tanzu Operations Manager, BOSH Director, [Cloud Service Broker for Microsoft Azure](#), and any other tiles that require independent credentials. Ensure that each managed identity is scoped to the least privilege necessary to operate.

Azure uses managed identities to handle service authorization. For more information, see the [Microsoft Azure documentation](#).

GCP reference architecture for Tanzu Operations Manager

This topic describes two Google Cloud Platform (GCP) reference architectures for Tanzu Operations Manager and any runtime products, including Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI) (on a shared virtual private cloud (VPC) and on a single-project VPC). This topic also outlines multiple networking variants for VPC deployment.

These architectures are validated for production-grade Tanzu Operations Manager deployments using multiple availability zones (AZs).

For general requirements for running Tanzu Operations Manager and specific requirements for running Tanzu Operations Manager on GCP, see [Tanzu Operations Manager on GCP Requirements](#).

A Tanzu Operations Manager reference architecture describes a proven approach for deploying Tanzu Operations Manager on a specific IaaS, such as GCP.

A Tanzu Operations Manager reference architecture must meet these requirements:

- Be secure
- Be publicly accessible
- Include common Tanzu Operations Manager-managed services such as Tanzu for MySQL, Tanzu RabbitMQ, and Spring Cloud Services
- Be able to host at least 100 app instances

VMware provides reference architectures to help you determine the best configuration for your Tanzu Operations Manager deployment.

Shared vs. single-project VPCs

A shared VPC installation is more difficult to configure than a Tanzu Operations Manager deployment on a single-project VPC, because the required account privileges and resource allocations are more granular and complex. But the shared VPC architecture allows network assets to be centrally located, which simplifies auditing and security. VMware recommends the shared VPC model for:

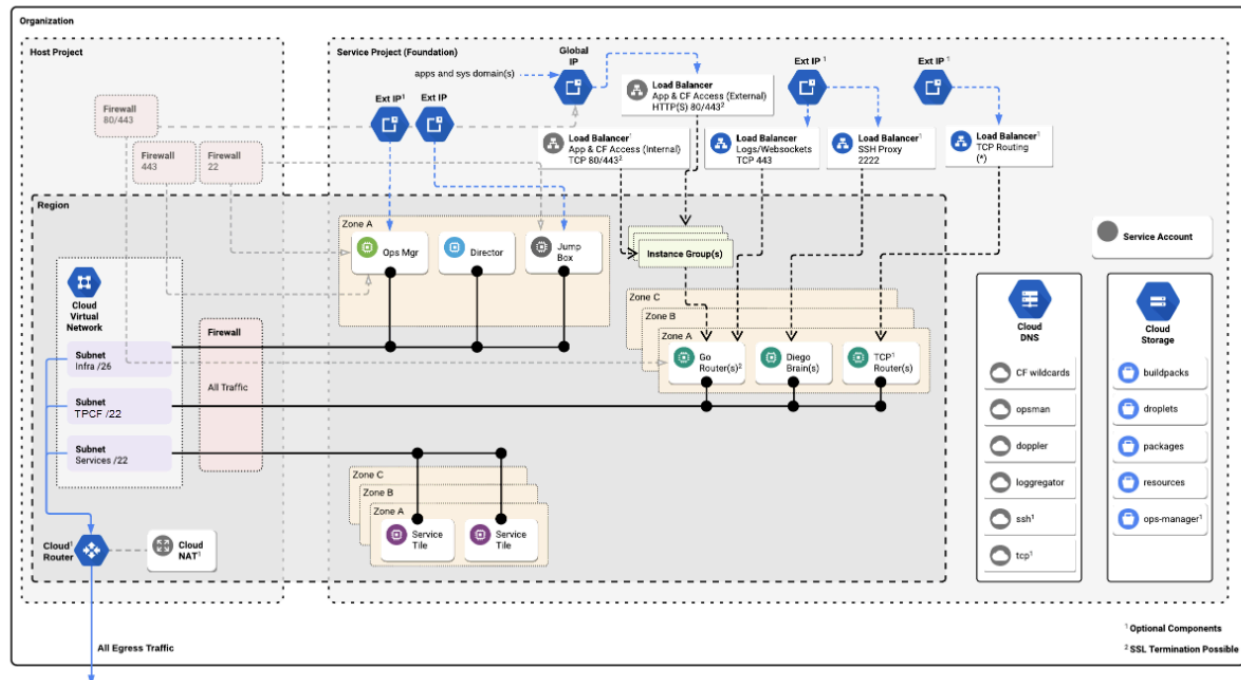
- Deployments with deep auditing and security requirements
- When networks hosting the foundation need to connect back to an internal network through a VPN or interconnect

A single-project VPC lets the platform architect give Tanzu Operations Manager full access to the VPC and its resources, which makes configuration easier. VMware recommends single-project VPC architecture for:

- Standalone deployments that do not connect to an internal network.
- Test and experimental deployments, and for projects that do not belong to an organization.

Shared VPC GCP reference architecture

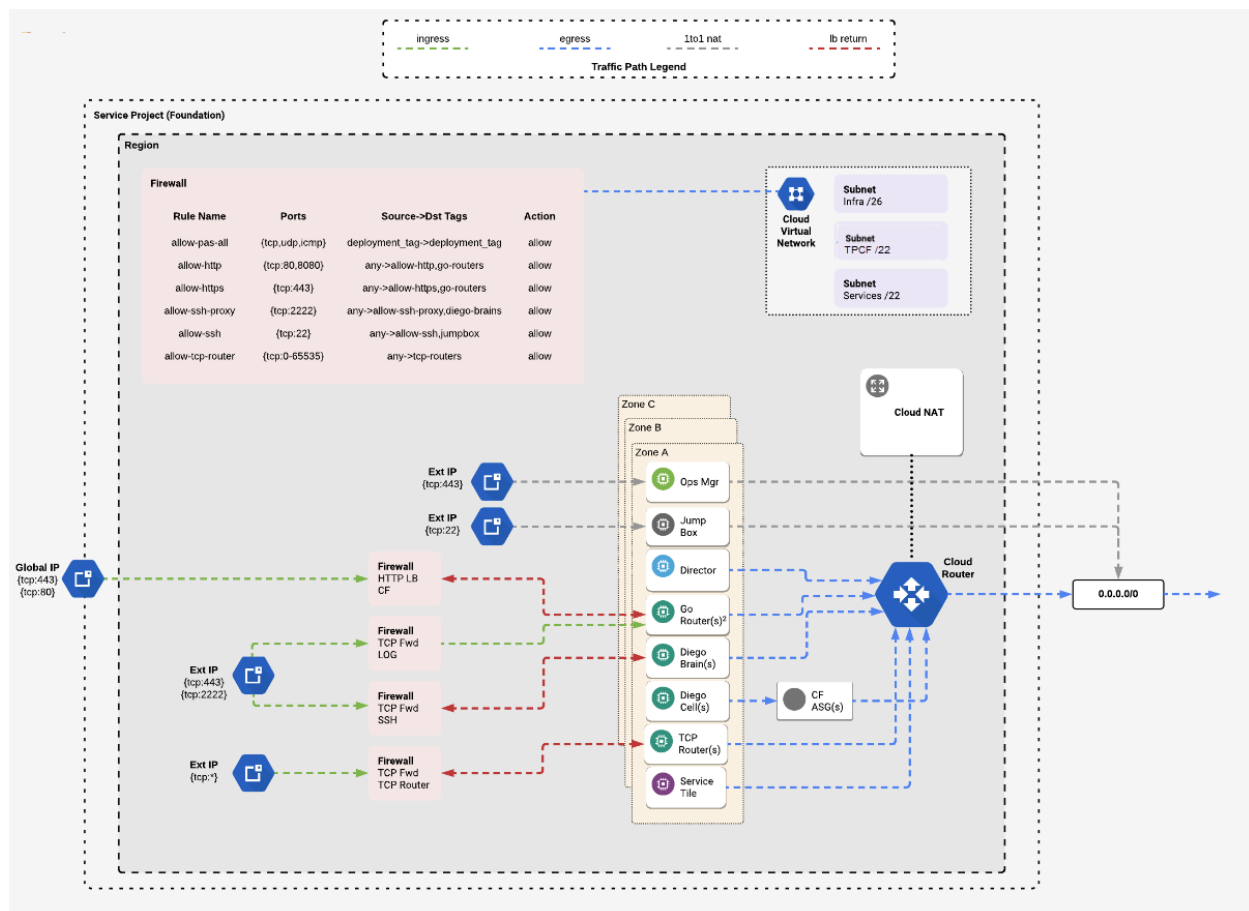
The following diagram illustrates a reference architecture for a deployment of Tanzu Operations Manager on a shared VPC on GCP. This architecture requires an organization on the VPC that contains a host project and a service project.



[View a larger version of this diagram.](#)

NAT network topology

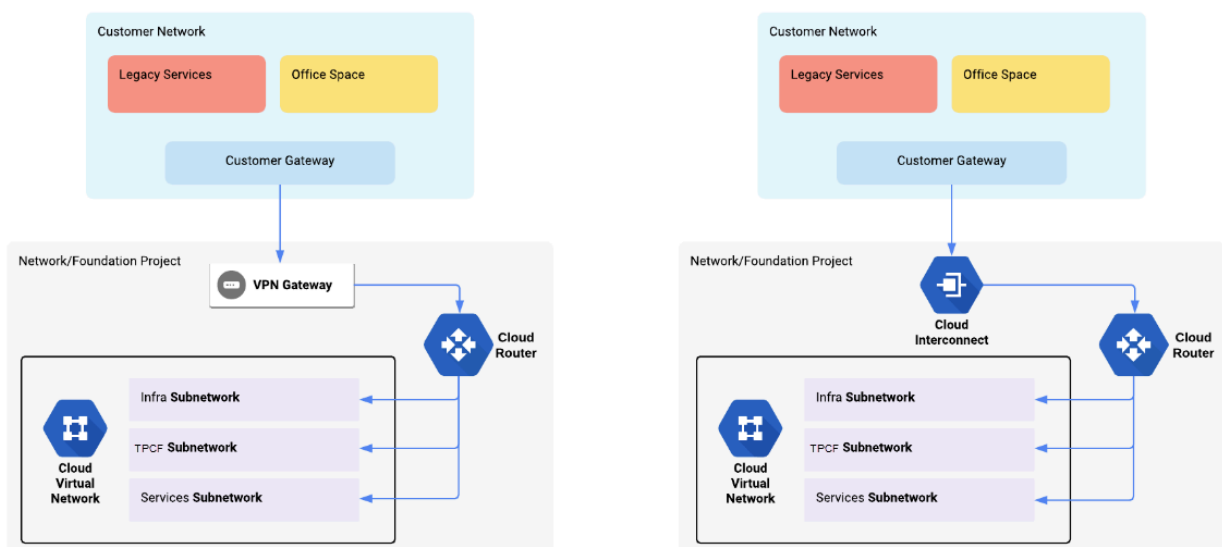
To expose a minimal number of public IP addresses, set up your NAT as shown in the following diagram:



[View a larger version of this diagram.](#)

Cloud Interconnect

To speed communication between data centers, use Google Cloud Interconnect, as shown in the following diagram:



[View a larger version of this diagram.](#)

The diagram shows two customer networks in GCP. One shows the traditional setup of linking to the VPN Gateway through the Customer Gateway. The other shows Customer Gateway connecting to Cloud Interconnect instead of VPN Gateway. In both cases, the gateway then connects to the Cloud Router, which goes to each of the three subnetworks.

Host and service architecture

GCP allocates resources using a hierarchy that centers around projects. To create a VPC, architects define a *host project* that allocates network resources for the VPC, such as address space and firewall rules. Then they can define one or more *service projects* to run within the VPC, which share the network resources allocated by the host project and include their own non-network resources, such as VMs and storage buckets.

To install Tanzu Operations Manager, Elastic Application Runtime, and services in a shared VPC on GCP, you create a host project for the VPC and a service project dedicated to running Tanzu Operations Manager, Elastic Application Runtime, and services.

Host project resources: The host project centrally manages these shared VPC network resources:

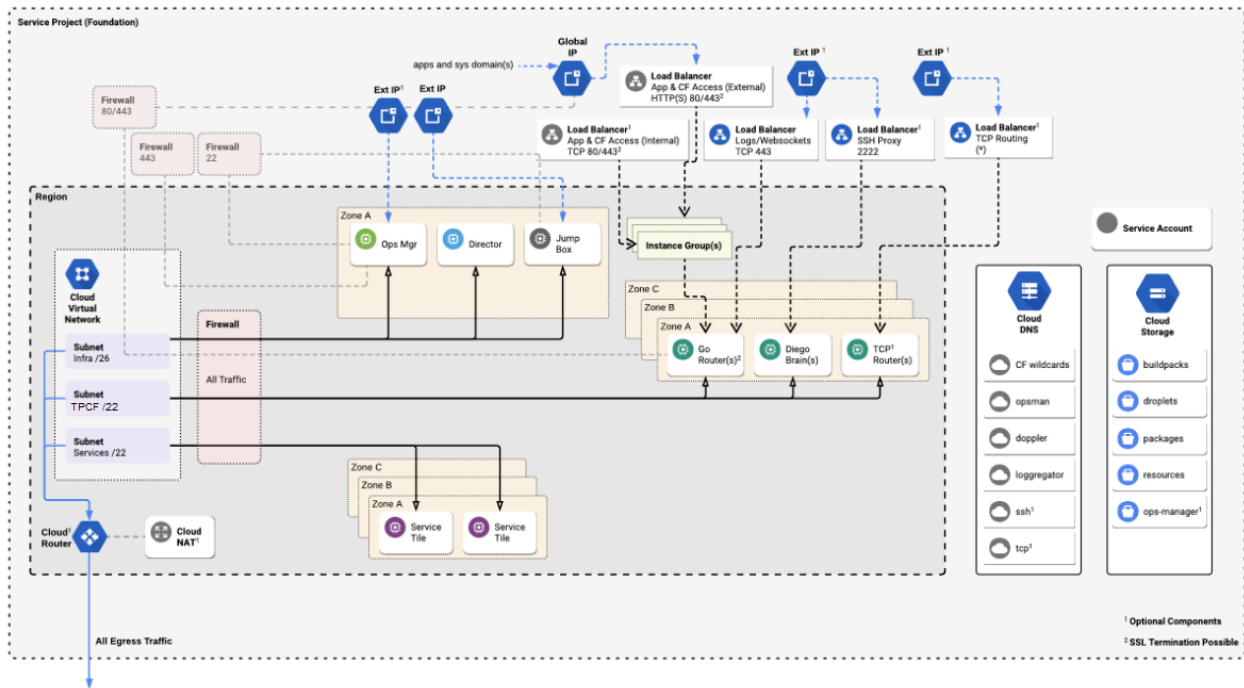
- Infra subnet (Tanzu Operations Manager VM and BOSH Director)
- Elastic Application Runtime subnet
- Services subnet
- Isolation segments
- Firewall rules
- NAT instances and gateway
- VPN or Google Cloud Interconnect
- Routes, such as egress internet through NAT or egress on-premise through a VPN

Service project resources: The service project manages these resources:

- Google Cloud Compute instances (VMs)
 - BOSH Director
 - Tanzu Operations Manager VM
 - VMs deployed by BOSH, such as runtime and service components
- Google Cloud Storage buckets for blobstore
 - BOSH Director
 - Resources
 - Buildpacks
 - Droplets
 - Packages
- Service account and a service account key for Elastic Application Runtime to access the storage buckets
- A service account for Tanzu Operations Manager
- Load balancers
- Google Cloud SQL instances, if using external databases

Single-project VPC base GCP reference architecture

The following diagram illustrates a reference architecture for a deployment of Tanzu Operations Manager on a single-project [VPC](#) on GCP:



[View a larger version of this diagram.](#)

Base reference architecture components

The following table lists the components that are part of a reference architecture deployment with three AZs.

Component	Reference Architecture Notes
Domains and DNS	Domain zones and routes in use by the reference architecture include: domains for *.apps and *.system (required), a route for Tanzu Operations Manager (required), a route for Doppler (required), a route for Loggregator (required), a route for SSH access to app containers (optional), and a route for TCP routing to apps (optional). The reference architecture uses GCP Cloud DNS as the DNS provider.
Tanzu Operations Manager VM	Deployed on the infrastructure subnet and accessible by fully-qualified domain name (FQDN) or through an optional jumpbox
BOSH Director	Deployed on the infrastructure subnet
Gorouters	Accessed through the HTTP and TCP WebSockets load balancers. Deployed on the Elastic Application Runtime subnet, one job per AZ
Diego Brains	Required. However, the SSH container access functionality is optional and enabled through the SSH Proxy load balancer. Deployed on the Elastic Application Runtime subnet, one job per AZ
TCP routers	Optional feature for TCP routing. Deployed on the Elastic Application Runtime subnet, one job per AZ
Database	Reference architecture uses GCP Cloud SQL rather than internal databases. Configure your database with a strong password and limit access only to components that require database access.
Blob Storage and Buckets	For buildpacks, droplets, packages, and resources. The reference architecture uses Google Cloud Storage rather than internal file storage.
Services	Deployed on the managed services subnet. Each service is deployed to each AZ.

Alternative GCP network layouts

This section describes the possible network layouts for Elastic Application Runtime deployments as covered by the reference architecture of Tanzu Operations Manager on GCP.

At a high level, there are currently two possible ways of granting public internet access to Elastic Application Runtime as described by the reference architecture:

- NAT provides connectivity from Elastic Application Runtime internals to the public internet.
 - The procedures in [Installing Tanzu Operations Manager on GCP](#) use this method.
- Every VM receives its own public IP address (no NAT).

Public IP addresses solution: If you prefer not to use a NAT solution, you can configure Elastic Application Runtime on GCP to assign public IP addresses for all components. This type of deployment might be more performant since most of the network traffic between Elastic Application Runtime components is routed through the front-end load balancer and the Gorouter.

Network objects: The following table lists the network objects expected for each type of reference architecture deployment with three AZs. This assumes you are using NAT.

Network Object	Notes	Minimum Number: NAT-Based	Minimum Number: Public IP Addresses
External IPs	For a NAT solution, use global IP address for apps and system access, and Tanzu Operations Manager or an optional jumpbox.	2	30+
NAT	One NAT per AZ.	3	0
Network	One per deployment. GCP Network objects allow multiple subnets with multiple CIDRs, so a typical deployment probably only ever requires one GCP Network object.	1	1
Subnets	Separate subnets for infrastructure (Tanzu Operations Manager VM, BOSH Director, jumpbox), Elastic Application Runtime, and services. Using separate subnets allows you to configure different firewall rules to meet your needs.	3	3
Routes	Routes are typically created by GCP dynamically when subnets are created, but you might need to create additional routes to force outbound communication to dedicated SNAT nodes. These objects are required to deploy Elastic Application Runtime without public IP addresses.	3+	3
Firewall Rules	GCP firewall rules are bound to a Network object and can be created to use IP ranges, subnets, or instance tags to match for source and destination fields in a rule. The preferred method used in the reference architecture deployment is instance tags.	6+	6+
Load balancers	Used to handle requests to Gorouters and infrastructure components. GCP uses two or more load balancers. The HTTP load balancer and TCP WebSockets load balancer are both required. The TCP router load balancer used for TCP routing and the SSH load balancer that allows SSH access to Diego apps are both optional. The HTTP load balancer provides SSL termination.	2+	2+
Jumpbox	Optional. Provides a way of accessing different network components. For example, you can configure it with your own permissions and then set it up to access the Broadcom Support portal to download tiles. Using a jumpbox is particularly useful in IaaS where Tanzu Operations Manager does not have a public IP address. In these cases, you can SSH into the Tanzu Operations Manager VM or any other component through the jumpbox.	(1)	(1)

Network communication in GCP deployments

These sections provide more background on the reasons behind certain network configuration decisions, specifically for the Gorouter.

Load balancer to Gorouter communications and TLS termination

In a Elastic Application Runtime on GCP deployment, the Gorouter receives two types of traffic:

- Unencrypted HTTP traffic on port 80 that is decrypted by the HTTP(S) load balancer
- Encrypted secure web socket traffic on port 443 that is passed through the TCP WebSockets load balancer

TLS is terminated for HTTPS on the HTTP load balancer, and for WebSockets (WSS) traffic on the Gorouter.

Elastic Application Runtime deployments on GCP use two load balancers to handle Gorouter traffic because HTTP load balancers currently do not support WebSockets.

ICMP

GCP routers do not respond to ICMP. VMware recommends deactivating ICMP checks in your BOSH Director network configuration. For more information, see [Step 5: Create Networks page](#) in *Configuring BOSH Director on GCP*.

VMC reference architecture for Tanzu Operations Manager

This topic describes the VMware Managed Cloud for AWS (VMC) reference architecture for Tanzu Operations Manager, including Tanzu Platform for Cloud Foundry. This architecture builds on the common base architectures described in [Platform architecture and Planning overview](#).

For specific installation instructions for running Elastic Application Runtime on VMC, see [Deploying Elastic Application Runtime to VMC](#).

Networking

This section provides guidance about networking resources.

- VMC SDDCs initially provision an NSX-T segment that provides a single class C subnet for workload VMs. Depending on your IP address requirements, you can use this subnet for your entire Elastic Application Runtime deployment, or add additional segments from the [VMC Console](#).
- If you want the front end of Elastic Application Runtime to be accessible from your corporate network or you need the services running on Elastic Application Runtime to be able to access corporate resources, you must do one of the following:
 - Provision public IP addresses from the SDDC console. For more information, see [Request or Release a Public IP Address](#).
 - Connect your SDDC to your on-premises data center with Direct Connect. For more information, see [Configure AWS Direct Connect Between Your SDDC and On-Premises Data Center](#).
 - Connect your SDDC to your on-premises data center with a VPN. For more information, see [Configure a VPN Connection Between Your SDDC and On-Premises Data Center](#).

AWS VPCs

When you deploy Tanzu Operations Manager and Elastic Application Runtime to VMC, VMware recommends creating a dedicated Virtual Private Cloud (VPC) on AWS to connect to your VMC SDDC.

Placing AWS services such as RDS in a dedicated connected VPC allows you to connect them to your SDDC while disallowing access from other networks. The AWS VPC is associated to your SDDC when you connect your AWS account.

For more information, see [Connected AWS account](#) in *Deploying and Managing a Software-Defined Data Center*.

RDS

Provision a single RDS instance to use as the external database for Elastic Application Runtime. For compatibility, VMware recommends using MySQL.

New database instances require several databases to be created. For more information, see [External system database configuration](#) in *Configuring Elastic Application Runtime*.

VMware recommends provisioning your RDS instance in the connected VPC. This allows you to connect to the endpoint for your RDS instance without making your database publicly accessible.

Blobstore storage accounts

Elastic Application Runtime requires S3 buckets for each of the following:

- Buildpacks
- Droplets
- Packages
- Resources

For better security, use a gateway endpoint so that only requests from your SDDC can access your S3 buckets. To achieve this configuration:

1. Create a gateway endpoint for S3. Follow Step 1 in [Access an S3 Bucket using an S3 endpoint](#).
2. Create the S3 buckets required by Elastic Application Runtime.
3. Add a security rule that disallows requests to the S3 bucket unless they come through the VPC endpoint. For more information, see [How can I restrict access to my Amazon S3 bucket using specific VPC endpoints or IP addresses?](#) in the AWS documentation. This can look like the following:

```
{
  "Id": "VPCe",
  "Version": "2012-10-17",
  "Statement": [{
    "Sid": "VPCe",
    "Effect": "Deny",
    "Principal": "*",
    "Action": "s3:*",
    "Resource": "arn:aws:s3::BUCKET-NAME/*",
    "Condition": {
      "StringNotEquals": {
        "aws:SourceVpce": "vpce-0123456abcdef0123"
      }
    }
  }]
}
```

Where:

- `BUCKET-NAME` is your S3 bucket's Amazon Resource Name (ARN)
- `vpce-0123456abcdef0123` is your gateway endpoint ID

vSphere reference architecture for Tanzu Operations Manager

This topic describes the vSphere reference architecture for Tanzu Operations Manager and any runtime products, including Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI). This architecture builds on the common base architectures described in [Platform architecture and planning](#).

For additional requirements and installation instructions for Tanzu Operations Manager on vSphere, see [Installing Tanzu Operations Manager on vSphere](#).

The vSphere reference architecture for the Elastic Application Runtime and Container Orchestration Engine runtime tiles is based on software-defined networking (SDN) infrastructure. vSphere offers NSX-T to support SDN infrastructure.

Tanzu Operations Manager supports these configurations for vSphere deployments:

- [Elastic Application Runtime on vSphere with NSX-T](#)
- [Elastic Application Runtime on vSphere without NSX](#)
- [Container Orchestration Engine on vSphere with NSX-T](#)
- [Container Orchestration Engine on vSphere without NSX-T](#)

Elastic Application Runtime on vSphere with NSX-T

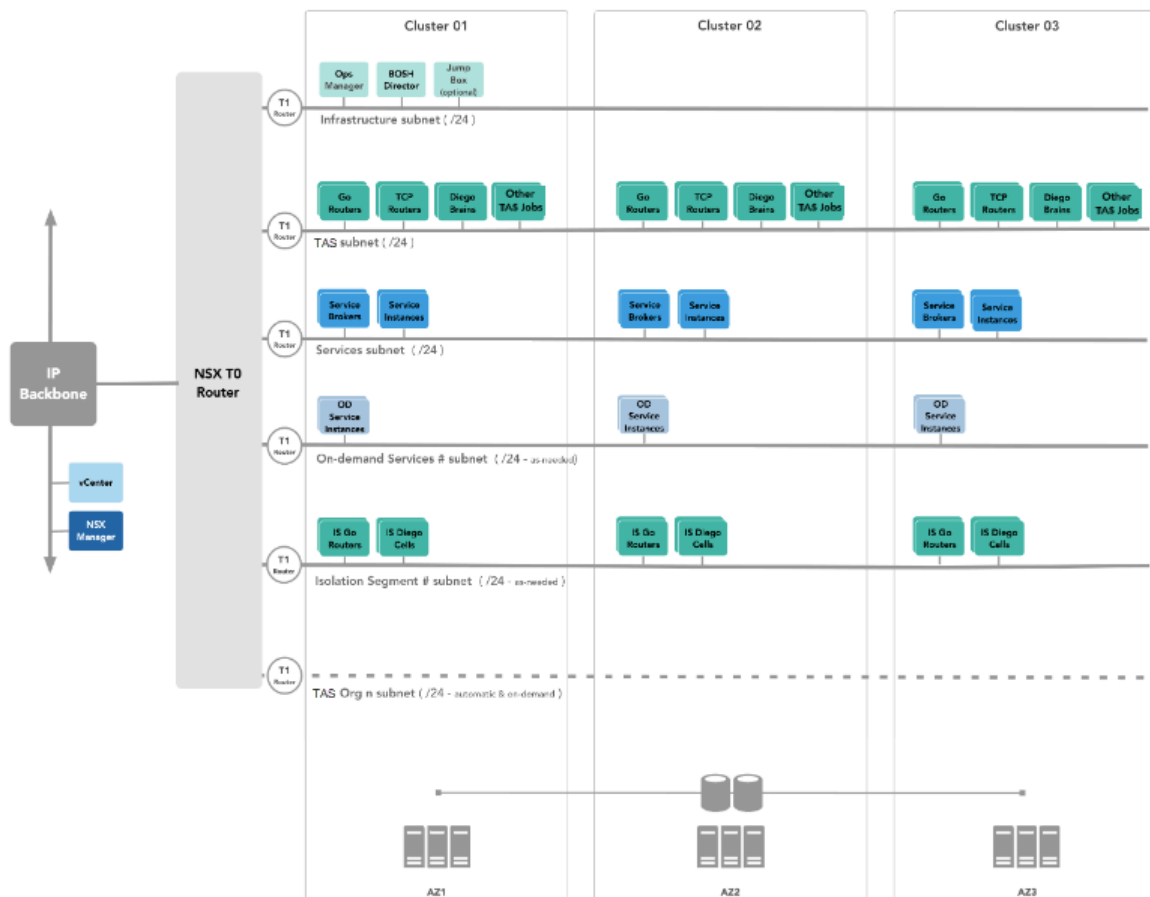
The following sections describe the reference architecture for Tanzu Operations Manager with Elastic Application Runtime on vSphere with NSX-T deployments, including vSphere deployments managed by VMware Cloud Foundation (VCF). They also provide requirements and recommendations for deploying Tanzu Operations Manager with Elastic Application Runtime on vSphere with NSX-T, such as network, load balancing, and storage capacity requirements and recommendations.

Elastic Application Runtime on vSphere with NSX-T supports the following SDN features:

- Virtualized, encapsulated networks and encapsulated broadcast domains
- VLAN exhaustion avoidance with the use of virtualized Logical Networks
- DNAT/SNAT services to create separate, non-routable network spaces for the Elastic Application Runtime installation
- Load balancing services to pass traffic through Layer 4 to pools of platform routers at Layer 7
- SSL termination at the load balancer at Layer 7 with the option to forward on at Layer 4 or 7 with unique certificates
- Virtual, distributed routing and firewall services native to the hypervisor

Architecture

The following diagram illustrates reference architecture for Elastic Application Runtime on vSphere with NSX-T deployments:



[View a larger version of this diagram..](#)

Elastic Application Runtime deployments with NSX-T are deployed with three clusters and three availability zones (AZs).

An NSX-T Tier-0 router is on the front end of the Elastic Application Runtime deployment. This router is a central logical router into the Elastic Application Runtime platform. You can configure static or dynamic routing using Border Gateway Protocol (BGP) from the routed IP address backbone through the Tier-0 router with the edge gateway.

Several Tier-1 routers, such as the router for the Elastic Application Runtime and infrastructure subnets, connect to the Tier-0 router.

NSX-T Container Plug-in requirement

Elastic Application Runtime deployments require the VMware NSX-T Container Plug-in to enable the SDN features available through NSX-T.

The NSX-T Container Plug-in integrates with NSX-T and enables a container networking stack.

To use NSX-T with Elastic Application Runtime, the NSX-T Container Plug-in must be installed, configured, and deployed at the same time as the Elastic Application Runtime tile. To download the NSX-T Container Plug-in, go to the [VMware NSX-T Container Plug-in](#) page on Broadcom Support portal.

Networking

These sections describe networking requirements and recommendations for Elastic Application Runtime on vSphere with NSX-T deployments.

The Tier-0 router must have external route IP address space to advertise on the BGP network with its peers. Select a network range for the Tier-0 router with enough space so that you can separate the network into these two jobs:

- Routing incoming and outgoing traffic
- DNATs and SNATs, load balancer VIPs, and other platform components

Firewall

Tanzu Operations Manager requires that the NSX firewall routes are not blocked. By default, all communication between Tanzu Operations Manager VMs and vCenter or ESXi hosts route through the NSX firewall are blocked. For more information on which ports to enable, see [VMware ports and protocols](#) for vSphere.

DNS

Elastic Application Runtime requires a system domain, app domain, and several wildcard domains.

For more information about DNS requirements for Elastic Application Runtime, see [Domain names](#) in *Platform Planning and Architecture*.

Load balancing

The load balancing requirements and recommendations for Elastic Application Runtime on vSphere with NSX-T deployments are:

- You must configure NSX-T load balancers for the Gorouters.
 - The domains for the Elastic Application Runtime system and apps must resolve to the load balancer IP.
 - You must assign either a private or a public IP address assigned to the domains for the Elastic Application Runtime system and apps.
You can use Layer 7 load balancers and stop TSL at the load balancers, but VMware discourages this approach because it results in additional overhead processing.
- Any TCP Gorouters and SSH Proxies within the platform also require NSX-T load balancers.
- Layer 4 and Layer 7 NSX-T load balancers are created automatically during app deployment.

Networking, subnets, and IP address spacing

The requirements and recommendations related to networks, subnets, and IP address spacing for Elastic Application Runtime on AVS with NSX-T deployments are:

- Elastic Application Runtime requires statically-defined networks to host its component VMs.
- The client side of an NSX-T deployment uses a series of non-routable address blocks when using DNAT/SNAT at the Tier-0 interface.
- The reference architecture for Elastic Application Runtime on vSphere with NSX-T deployments uses a pattern in which all networks are calculated on the /24 8-bit network boundary. The network octet is numerically sequential.
- NSX-T dynamically assigns Elastic Application Runtime org networks and adds a Tier-1 router. These org networks are automatically instantiated based on a non-overlapping block of address space. You can configure the block of address space in the NCP Configuration section of the NSX-T tile in Tanzu Operations Manager. The default is /24. This means that every org in Elastic Application Runtime is assigned a new /24 network.

For more information about Elastic Application Runtime subnets, see [Required subnets](#) in *Platform Architecture and Planning Overview*.

High availability

For information about high availability (HA) requirements and recommendations for Elastic Application Runtime on vSphere, see [High Availability](#) in *Platform Architecture and Planning Overview*.

- Shared storage

Elastic Application Runtime requires shared storage. You can allocate networked storage to the host clusters using one of two common approaches: horizontal or vertical. The approach you follow reflects how your data center arranges its storage and host blocks in its layout.

- Horizontal shared storage

With the horizontal shared storage approach, you grant all hosts access to all datastores and assign a subset to each Elastic Application Runtime installation.

For example, with six datastores `ds01` through `ds06`, you grant all nine hosts access to all six datastores. You then provision your first Elastic Application Runtime installation to use stores `ds01` through `ds03` and your second Elastic Application Runtime installation to use `ds04` through `ds06`.

- Vertical shared storage

With the vertical shared storage approach, you grant each cluster its own datastores, creating a cluster-aligned storage strategy. vSphere VSAN is an example of this architecture.

For example, with six datastores `ds01` through `ds06`, you assign datastores `ds01` and `ds02` to a cluster, `ds03` and `ds04` to a second cluster, and `ds05` and `ds06` to a third cluster. You then provision your first Elastic Application Runtime installation to use `ds01`, `ds03`, and `ds05`, and your second Elastic Application Runtime installation to use `ds02`, `ds04`, and `ds06`.

With this arrangement, all VMs in the same installation and cluster share a dedicated datastore.

Storage capacity

VMware recommends these storage capacity allocations for production and non-production Elastic Application Runtime environments:

- Production environments: Configure at least 8 TB of data storage. You can configure this as either one 8 TB store or a number of smaller volumes that add up to 8 TB. Frequently-used developments might require significantly more storage to accommodate new code and buildpacks.
- Non-production environments: Configure 4 to 6 TB of data storage.

The latest versions of Tanzu Operations Manager validated for the reference architecture do not support using vSphere Storage Clusters. Datastores can be listed in the vSphere tile by their native name, not the cluster name created by vCenter for the storage cluster.

If a datastore is part of a vSphere Storage Cluster using DRS storage (SDRS), you must deactivate the Storage vMotion feature on any datastores used by Tanzu Operations Manager. Otherwise, Storage vMotion activity can rename independent disks and cause BOSH to malfunction. For more information, see [Migrating Tanzu Operations Manager to a New Datastore in vSphere](#).

For more information about general storage requirements and recommendations for Elastic Application Runtime, see [Storage](#) in *Platform Architecture and Planning Overview*.

SQL Server

An internal MySQL database is sufficient for use in production environments. However, an external database provides more control over database management for large environments that require multiple data centers.

For information about configuring system databases on Elastic Application Runtime, see [Databases](#) in the Elastic Application Runtime documentation.

Security

For information about security requirements and recommendations for Elastic Application Runtime deployments, see [Security](#) in *Platform Architecture and Planning Overview*.

Blobstore storage

VMware recommends that you use these blobstore storage types for production and non-production Elastic Application Runtime environments:

- Production environments: Use an S3 storage appliance as the blobstore.
- Non-production environments: Use an NFS/WebDAV blobstore.
For non-production environments, the NFS/WebDAV blobstore can be the primary consumer of storage, as the NFS/WebDAV blobstore must be actively maintained. Elastic Application Runtime deployments experience downtime during events such as storage upgrades or migrations to new disks.

For more information about blobstore storage requirements and recommendations, see [Configure file storage](#) in the Elastic Application Runtime documentation.

Elastic Application Runtime on vSphere without NSX

These sections describe the differences in architecture for Elastic Application Runtime on vSphere without software-defined networking deployments.

Networking

Without an SDN, IP address allocations all come from routed network space. Discussions and planning within your organization are essential to acquiring the necessary amount of IP address space for a Elastic Application Runtime deployment with future growth considerations. This is because routed IP address space is a premium resource, and adding more later is difficult, costly, and time-consuming.

This is a best-guess layout for IP address space utilization in a single Elastic Application Runtime deployment:

- Infrastructure - /28
- Elastic Application Runtime deployment - /23
This size is almost completely dependent on the estimated desired capacity for containers and you might need less space than this. VMware discourages using a larger size in a single deployment.
- Services - /23
This size is almost completely dependent on the estimated desired capacity for services. Resize as necessary.

Isolation segments

Isolation segments can help with satisfying IP address space needs in a routed network design. You can build smaller groups of Gorouters and Diego Cells aligned to a particular service. Smaller groups use less IP address space.

Container Orchestration Engine on vSphere with NSX-T

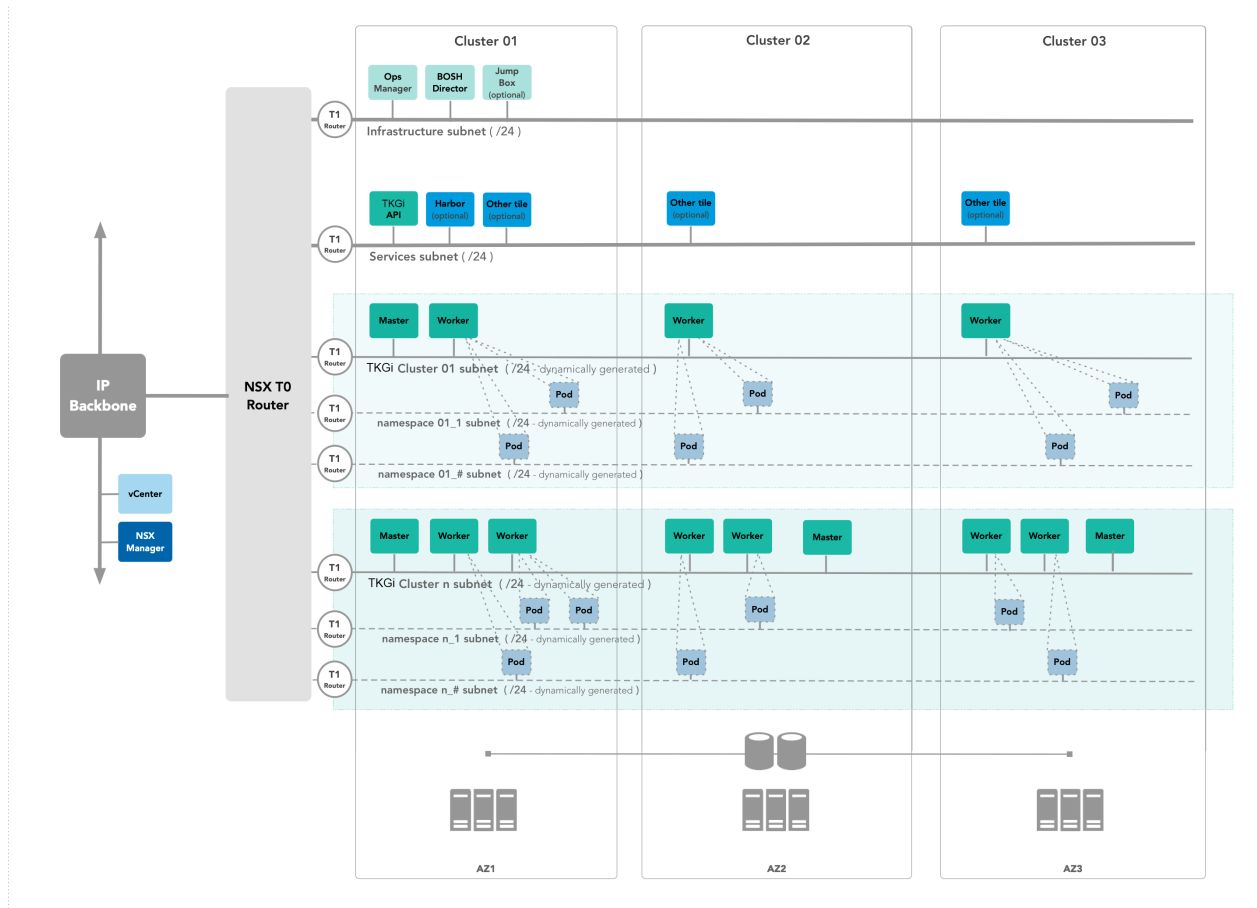
The following sections describe the reference architecture for Container Orchestration Engine on vSphere with NSX-T deployments. They also provide requirements and recommendations for deploying Container

Orchestration Engine on vSphere with NSX-T, such as network, load balancing, and storage capacity requirements and recommendations.

VMware recommends deploying Container Orchestration Engine on its own dedicated Tanzu Operations Manager instance, rather than on a shared Tanzu Operations Manager that also hosts other runtimes such as Elastic Application Runtime.

Architecture

The following diagram illustrates the reference architecture for Container Orchestration Engine on vSphere with NSX-T deployments.



[View a larger version of this diagram.](#)

Container Orchestration Engine deployments with NSX-T are deployed with three clusters and three AZs.

An NSX-T Tier-0 router is on the front end of the Container Orchestration Engine deployment. This router is a central logical router into the Container Orchestration Engine platform. You can configure static or dynamic routing using Border Gateway Protocol (BGP) from the routed IP address backbone through the Tier-0 router.

Several Tier-1 routers, such as the router for the infrastructure subnet, connect to the Tier-0 router. New Tier-1 routers are created on-demand as new clusters and namespaces are added to Container Orchestration Engine.

The Container Orchestration Engine on vSphere with NSX-T architecture supports multiple primary nodes.

Networking

These sections describe networking requirements and recommendations for Container Orchestration Engine on vSphere with NSX-T deployments.

Load balancing

The load balancing requirements and recommendations for Container Orchestration Engine on vSphere with NSX-T deployments are:

- Use standard NSX-T load balancers. Layer 4 and Layer 7 NSX-T load balancers are created automatically during app deployment.
- Use both Layer 4 and Layer 7 load balancers:
 - Use Layer 7 load balancers for ingress routing.
 - Use Layer 4 load balancers for **LoadBalancer** services. This allows you to terminate SSL at the load balancers, which reduces overhead processing.
- NSX-T provides ingress routing natively. You can also use a third-party service for ingress routing, such as Istio or NGINX. You run the third-party ingress routing service as a container in the cluster.
- If you use a third-party ingress routing service, you must:
 - Create wildcard DNS entries to point to the service.
 - Define domain information for the ingress routing service in the manifest of the Container Orchestration Engine on vSphere deployment. For example:

```
apiVersion: extensions/v1beta1
kind: Ingress
metadata:
  name: music-ingress
  namespace: music1
spec:
  rules:
    - host: music1.pks.domain.com
      http:
        paths:
          - path: /*
            backend:
              serviceName: music-service
              servicePort: 8080
```

- When you push a Container Orchestration Engine on vSphere deployment with a service **type** set to **LoadBalancer**, NSX-T automatically creates a new WIP for the deployment on the existing load balancer for that namespace. You must specify a listening and translation port in the service, a name for tagging, and a protocol. For example:

```
apiVersion: v1
kind: Service
metadata:
  ...
spec:
  type: LoadBalancer
  ports:
    - port: 80
      targetPort: 8080
      protocol: TCP
      name: web
```

Routable IPs

The routable IP address requirements and recommendations for Container Orchestration Engine with NSX-T deployments are:

- Deployments with Container Orchestration Engine NSX-T ingress: VMware recommends a /25 network for deployments with Container Orchestration Engine NSX-T ingress.
The Tier-0 router must have routable external IP address space to advertise on the BGP network with its peers. Select a network range for the Tier-0 router with enough space so that you can separate the network into these two jobs:
 - Routing incoming and outgoing traffic
 - DNATs and SNATs, load balancer WIPs, and other platform components
- Deployments with several load balancers: VMware recommends a /23 network for deployments that use several load balancers. Deployments with several load balancers have much higher address space consumption for load balancer WIPs. This is because Kubernetes service types allocate IP addresses very frequently. To accommodate the higher address space, allow for 4 times the address space.

Firewall

Tanzu Operations Manager requires that the NSX firewall routes are not blocked. By default, all communication between Tanzu Operations Manager VMs and vCenter or ESXi hosts route through the NSX firewall are blocked. For more information on which ports to enable, see [VMware ports and protocols](#) for vSphere.

Networks, subnets, and IP address spacing

These considerations and recommendations apply to networks, subnets, and IP address spacing for Container Orchestration Engine on vSphere with NSX-T deployments:

- Allocate a large network block for Container Orchestration Engine clusters and Pods:
 - Container Orchestration Engine clusters: Configure a 172.24.0.0/14 network block.
 - Container Orchestration Engine pods: Configure a 172.28.0.0/14 network block.

NSX-T creates IP address blocks of /24 from these /14 networks by default each time a new cluster or Pod is created. You can configure this CIDR range for Container Orchestration Engine in Tanzu Operations Manager.

- When you deploy Container Orchestration Engine with Tanzu Operations Manager, you must allow for a block of address space for dynamic networks that Container Orchestration Engine deploys for each namespace. The recommended address space allows you to view a queue of which jobs relate to each service.
- When a new Container Orchestration Engine cluster is created, Container Orchestration Engine creates a new /24 network from Container Orchestration Engine cluster address space.
- When a new app is deployed, new NSX-T Tier-1 routers are generated and Container Orchestration Engine creates a /24 network from the Container Orchestration Engine Pods network.
- Allocate a large IP address block in NSX-T for Kubernetes Pods. For example, a /14 network. NSX-T creates address blocks of /24 by default. This CIDR range for Kubernetes services network ranges is configurable in Tanzu Operations Manager.

For more information, see [Networks](#) in *Platform Architecture and Planning Overview*.

Multi-tenancy

For Container Orchestration Engine on vSphere with NSX-T deployments, networks are created dynamically for both Container Orchestration Engine clusters and Pods.

To accommodate these dynamically-created networks, VMware recommends that you use multiple clusters, rather than a single cluster with multiple namespaces.

Multiple clusters provide additional features such as security, customization on a per-cluster basis, privileged containers, failure domains, and version choice. Namespaces can be used as a naming construct and not as a tenancy construct.

Primary nodes

The Container Orchestration Engine on vSphere with NSX-T architecture supports multiple primary nodes.

You can define the number of primary nodes per plan in the Container Orchestration Engine tile in Tanzu Operations Manager. The number of primary nodes can be an odd number to allow `etcd` to form a quorum.

VMware recommends that you have at least one primary node per AZ for HA and disaster recovery.

High availability

For information about HA requirements and recommendations, see [High Availability](#) in *Platform Architecture and Planning Overview*.

Storage capacity

VMware recommends the following storage capacity allocation for production and non-production Container Orchestration Engine environments:

Container Orchestration Engine on vSphere supports static persistent volume provisioning and dynamic persistent volume provisioning.

For more information about storage requirements and recommendations, see [Persistent volume storage options on vSphere](#).

Security

For information about security requirements and recommendations, see [Security](#) in *Platform Architecture and Planning Overview*.

Container Orchestration Engine on vSphere without NSX-T

You can deploy Container Orchestration Engine without NSX-T. To deploy Container Orchestration Engine without NSX-T, click Flannel as your container network interface in the Networking pane of the Container Orchestration Engine tile.

Select from networks already identified in Tanzu Operations Manager to deploy the Container Orchestration Engine API and Container Orchestration Engine-provisioned Kubernetes clusters.

VMware recommends deploying Container Orchestration Engine on its own dedicated Tanzu Operations Manager instance, rather than on a shared Tanzu Operations Manager that also hosts other runtimes such as Elastic Application Runtime.

Upgrading vSphere without causing downtime for your runtime

You can upgrade the vSphere components that host your runtime deployment, either Tanzu Platform for Cloud Foundry or VMware Tanzu Container Orchestration Engine (formerly TKGI), without disrupting service.

Minimum requirements

At minimum, vSphere contains a vCenter Server and one or more ESXi hosts.

You cannot perform an in-place upgrade of vSphere without at least two ESXi hosts in your cluster.

If you do not meet this requirement, you have insufficient resources to evacuate an entire host, and you will experience runtime downtime during the upgrade.

To upgrade vSphere with only one ESXi host, or without sufficient headroom capacity, you must reduce your runtime deployment size. To do this, you can:

- Reduce the number of Diego Cells in your deployment

- Pause VMs to make more capacity available.

These actions might result in downtime for Elastic Application Runtime or Container Orchestration Engine.

Recommended starting configuration

If you run a Elastic Application Runtime or Container Orchestration Engine deployment following the base reference architecture described in [vSphere Reference architecture](#), your vSphere deployment has these components:

- One vCenter Server
- Three ESXi hosts per cluster
- Three or more clusters
- One NSX Edge appliance (or an HA pair)

VMware recommends having at least three ESXi hosts in your cluster to maintain high availability (HA) during your upgrade.

Upgrade vSphere

The following sections describe the procedure to upgrade the vSphere management layer that underlies Tanzu Operations Manager.

Step 1: Upgrade vCenter

To upgrade vCenter, see the instructions in [Overview of the vCenter server upgrade process](#) in the vSphere documentation.

Step 2: Upgrade ESXi hosts

After a successful vCenter upgrade, you must upgrade your ESXi hosts one at a time, starting with the first ESXi host.

To upgrade your ESXi hosts:

1. Verify that your ESXi hosts have sufficient resources and headroom to evacuate the VM workload of a single ESXi host to the two remaining hosts.

If you have enabled vSphere HA on your ESXi host, ensure that each ESXi host has sufficient headroom capacity; vSphere HA reserves 66% of available memory.
2. Use vMotion to move all the VMs on the host you want to upgrade to the other ESXi hosts. vMotion places the VMs on the other hosts based on available capacity. For more information, see [Migration with vMotion](#) in the vSphere documentation.
3. Upgrade the evacuated ESXi host. See [Upgrading ESXi hosts](#) in the vSphere documentation.
4. Repeat the previous steps for each remaining host, one at a time. vSphere automatically re-balances all VMs back onto the upgraded hosts through DRS after you have upgraded all hosts.

Migrating Tanzu Operations Manager to a new datastore in vSphere

This topic describes how to migrate your Tanzu Operations Manager installation to a new vSphere datastore.

Prerequisites

- The new and existing vSphere datastores must reside in the same data center.
- To avoid service disruption, VMware recommends that you configure your overall Tanzu Operations Manager deployment for high availability (HA). In addition, check for configurations necessary to achieve HA in each of your installed product tiles.

Before you begin

The following sections describe the steps to do prior to the migration.

Step 1: Back up your environment

Ensure that your Tanzu Operations Manager environment is fully backed up.

For more information about how to back up Tanzu Operations Manager, see [Backing up Deployments](#).

Step 2: Document current environment settings

Document your current environment settings. Record which VMs are running and in which datastore each one resides. If you experience any issues during or after the migration, you must have this information to restore your environment.

To obtain this information:

1. Run one of these commands:
 - If you use BOSH CLI v2, run:

```
bosh -e BOSH-ENVIRONMENT instances --details > instances.txt
```

Where **BOSH-ENVIRONMENT** is the alias you assigned to your BOSH Director.

- If you use the original version of the BOSH CLI, run:

```
bosh-old instances --details > instances.txt
```

2. Save the resulting **instances.txt** file to a safe location.
3. Note the datastore where each VM resides in vSphere.

Step 3: Check system health

Check the Status tab in your runtime product tiles, such as Tanzu Platform for Cloud Foundry or VMware Tanzu Container Orchestration Engine (formerly TKGI), and make sure there are no errors or reported issues.

Step 4: Check installed products health

In each tile installed in your Tanzu Operations Manager deployment, check the Status tab and verify that there are no errors or reported issues.

Step 5: Check BOSH Director status

To check the status of your BOSH Director:

1. Verify that there are no pending changes and that the status of all tiles is green.

2. Verify that the last installation log does not contain any errors.
3. Before you proceed with the migration, click Review Pending Changes and then click Apply Changes to make sure there are no errors in the installation log.

Migrate Tanzu Operations Manager to a new datastore

To migrate your Tanzu Operations Manager deployment to a new datastore:

1. Go to the Tanzu Operations Manager Installation Dashboard.
2. Click the BOSH Director tile.
3. Click vCenter Config.
4. Update the Ephemeral Datastore Names and Persistent Datastore Names text boxes to reflect the new datastore names.
If you use the datastore clustering feature in vSphere, provide only the datastore cluster name in the Persistent Datastore Cluster Names // Ephemeral Datastore Cluster Names fields.
5. Click Save.
6. Click Review Pending Changes.
7. Deselect the Select All Products check box so that only the BOSH Director tile is still selected.
8. Click Apply Changes.
9. Confirm that the BOSH Director VM has persistent disk on the new datastore.
 1. From vCenter, find the BOSH Director VM.
 2. In the Related Objects section, verify that the storage has been updated to the new datastore.
10. In the BOSH Director tile, click Director Config.
11. Select Recreate VMs deployed by the BOSH Director.
12. Click Review Pending Changes.
13. If you are using any service tiles, select the Recreate All On-Demand Service Instances errand for each tile you migrate.
14. Click Apply Changes.

Global DNS load balancers for multi-foundation environments

This topic describes how you can use Global DNS load balancers (GLBs) for multi-foundation environments. This topic also describes concepts such as foundation affinity and health checks.

If you want to configure a load balancer dedicated to one foundation and you are using an F5 LTM, see [Configuring an F5 Load Balancer for Elastic Application Runtime](#).

About multi-foundation environments

Multi-foundation environments are multiple instances of Tanzu Operations Manager and runtime products such as Tanzu Platform for Cloud Foundry or VMware Tanzu Container Orchestration Engine (formerly TKGI) that can communicate with each other in some ways. Each foundation can use different infrastructures to fit your preferences.

Multi-foundation environments are commonly deployed in an active-active or active-passive pattern. An *active-active pattern* is when instances of apps run on two foundations and they are both in use. An *active-passive pattern* is when instances of apps run on two foundations, but the instances might only be active on one foundation, and the other foundation becomes active only in the event of a failover.

Configure your GLB

The typical setup for foundation failover requires the GLB be authoritative for a wildcard app domain. The wildcard app domain is not the same domain as the foundation-default app domain.

To configure your GLB:

1. Find and record your wildcard app domain. For Elastic Application Runtime, your wildcard app domain is typically the Apps domain you configure in the Domains pane of the Elastic Application Runtime tile. For more information, see the [Elastic Application Runtime documentation](#).
2. Add the domain you recorded to both Elastic Application Runtime foundations using a shared domain. You can create a shared domain using the Tanzu cf CLI. For more information about the Tanzu cf CLI command to create shared domains, see the [Cloud Foundry CLI Reference Guide](#).
3. To support failover, set the time-to-live (TTL) in the wildcard DNS record to about 30 to 180 seconds. When you determine your TTL, consider the tradeoff between the app performance impact and the resulting time for failover to occur.

About foundation affinity

Foundation affinity occurs when the GLB favors one foundation over another during a route request. For example, users experience less latency if they are routed to a foundation that is geographically closer, so the GLB can favor that foundation.

Different GLBs have their own mechanisms to achieve this. For more information, see the following table for common foundation affinity concepts:

Term	Description
Topology / geographically-based affinity	The GLB attempts to direct traffic to the graphically nearest foundation based on IP geolocation or provided topology for private networks of the LDNS server performing the lookup.
Static-persist / member-hashing affinity	The Static Persist load balancing method uses the persist mask with the source IP address of the Local DNS in a deterministic algorithm to send requests to a specific pool member (virtual server). Using this method, the traffic manager sends DNS name resolution requests to the first available pool member based on a hash algorithm that determines the order of the pool members.
Active-passive foundations	If one foundation is usually idle, you can always pick the active foundation IP as long as it remains available.
Microservice-to-microservice affinity	For microservice apps, you typically use a Services Registry to manage traffic between microservices. There are two ways to do this: • If the microservices are using the domain that maps to the GLB, their traffic is routed through the GLB. • If the microservices are communicating using IP address or internal domains, such as when using an Envoy proxy, the traffic does not pass through the GLB.

About health checks

In Elastic Application Runtime, *health checks* determine whether a foundation for an app is healthy or not.

For the levels at which you can check the health of your foundation, see the following table:

Level	Description
Foundation	The GLB relies on a local load balancer in front of the Gorouters to determine the overall health of the foundation. The GLB can perform health checks on TCP port 443 or 80 or on the local load balancer. The local load balancer checks the health of the back end pool of Gorouters on port 8080.
App	Health checks are set only for apps that have instances on both foundations. Each app instance has canary DNS records. The canary DNS records are the same for the app instances on each foundation. You would also need to add more VIPs dedicated to these canary apps that would be used to check their health. (Not recommended)

VMware does not recommend setting health checks at the app level. Configuring health checks in this way can cause more frequent failover and delays while pushing apps. Complete failover of a foundation affects all apps on the platform. Health checks on a per-app basis can require additional overhead beyond the control of an app developer.

Installing Tanzu Operations Manager

These topics provide information about installing Tanzu Operations Manager, including IaaS-specific instructions.

- [Overview of installing and configuring Tanzu Operations Manager](#)
- [Preparing your firewall](#)
- [IaaS permissions guidelines](#)
- [Tanzu Operations Manager Runtimes](#)
- [Installing Tanzu Operations Manager in your air-gapped environment](#)
- [Installing and configuring Tanzu Operations Manager on AWS](#)
 - [Requirements and prerequisites for Tanzu Operations Manager on AWS](#)
 - [AWS permissions guidelines and best practices](#)
 - [Installing Tanzu Operations Manager on AWS manually](#)
 - [AWS objects required for deploying Tanzu Operations Manager](#)
 - [Preparing to deploy Tanzu Operations Manager on AWS](#)
 - [Deploying Tanzu Operations Manager on AWS](#)
 - [Configuring BOSH Director on AWS](#)
 - [Upgrading BOSH Director for Tanzu Operations Manager on AWS](#)
 - [Deleting your Tanzu Operations Manager deployment from AWS](#)
- [Installing and configuring Tanzu Operations Manager on Azure](#)
 - [Requirements and prerequisites for Tanzu Operations Manager on Azure](#)
 - [Installing Tanzu Operations Manager on Azure](#)
 - [Preparing to deploy Tanzu Operations Manager on Azure](#)
 - [Deploying Tanzu Operations Manager on Azure manually](#)
 - [Configuring BOSH Director on Azure](#)
 - [Troubleshooting Tanzu Operations Manager on Azure](#)
 - [Upgrading BOSH Director for Tanzu Operations Manager on Azure](#)
 - [Deleting your Tanzu Operations Manager deployment from Azure](#)
- [Installing and configuring Tanzu Operations Manager on GCP](#)
 - [Requirements and prerequisites for Tanzu Operations Manager on GCP](#)
 - [Recommended GCP resource quotas for Tanzu Operations Manager](#)
 - [Installing Tanzu Operations Manager on GCP manually](#)
 - [Preparing to deploy Tanzu Operations Manager on GCP](#)
 - [Deploying Tanzu Operations Manager on GCP](#)

- [Configuring BOSH Director on GCP](#)
- [Configuring a shared VPC on GCP with Tanzu Operations Manager](#)
- [Upgrading BOSH Director for Tanzu Operations Manager on GCP](#)
- [Deleting your Tanzu Operations Manager deployment from GCP](#)
- [Installing and configuring Tanzu Operations Manager on vSphere](#)
 - [Installing Tanzu Operations Manager on vSphere](#)
 - [vSphere service account requirements](#)
 - [Deploying Tanzu Operations Manager on vSphere](#)
 - [Configuring BOSH Director on vSphere](#)
 - [Using the Cisco Nexus 1000v Switch with Tanzu Operations Manager](#)
 - [Using the BOSH Resurrector](#)
 - [Understanding and using availability zones in vSphere](#)
 - [Using vSphere host groups with your Tanzu Operations Manager deployment](#)

Overview of installing and configuring Tanzu Operations Manager

This topic gives you an overview of how to install and configure Tanzu Operations Manager and app runtimes.

The Tanzu Operations Manager installation process takes place in the following steps:

- **Prepare your Tanzu Operations Manager Deployment:** To prepare your deployment, you must create the required components within your IaaS, which can include load balancers, blobstores, databases, TCP routers, DNS records, and more.
- **Deploy Tanzu Operations Manager:** To deploy Tanzu Operations Manager, you must download Tanzu Operations Manager from Broadcom Support portal, create a VM image, create a VM instance, and configure the VM instance to run Tanzu Operations Manager.
- **Configure Tanzu Operations Manager:** To configure Tanzu Operations Manager, you must log into the Tanzu Operations Manager interface to configure the settings.
- **Install Runtimes:** To install runtime tiles, you must download the tile from Broadcom Support portal and follow that runtime's procedure to configure and deploy the tile.
- **Install Services:** To install service tiles, you must download the tile from Broadcom Support portal and follow that service tile's procedure to configure and deploy the tile.

After you successfully install and configure Tanzu Operations Manager, you can learn how to run and operate Tanzu Operations Manager as an operator. For more information, see [Operating](#).

Install Tanzu Operations Manager

Tanzu Operations Manager, app runtimes, and associated products can run on multiple IaaSes.

Planning and installing Tanzu Operations Manager and app runtimes means building layers from the bottom up, starting with the details of your IaaS and progressing to configurations in the Tanzu Operations Manager interface.

The following sections includes lists of topics containing instructions for preparing, installing, and deploying Tanzu Operations Manager, by IaaS.

Part 1: Prepare your Tanzu Operations Manager deployment

For more information about planning your deployment, select the relevant topic:

- [Preparing to deploy Tanzu Operations Manager on AWS](#)
- [Preparing to deploy Tanzu Operations Manager on Azure](#)
- [Preparing to deploy Tanzu Operations Manager on GCP](#)
- [Installing Tanzu Operations Manager on vSphere](#)

Part 2: Deploy Tanzu Operations Manager

For more information about deploying Tanzu Operations Manager, select the relevant topic:

- [Deploying Tanzu Operations Manager on AWS](#)
- [Deploying Tanzu Operations Manager on Azure](#)
- [Deploying Tanzu Operations Manager on GCP](#)
- [Deploying Tanzu Operations Manager on vSphere](#)

Part 3: Configure Tanzu Operations Manager

For more information about configuring the BOSH Director tile by IaaS, select the relevant topic:

- [Configuring BOSH Director on AWS](#)
- [Configuring BOSH Director on Azure](#)
- [Configuring BOSH Director on GCP](#)
- [Configuring BOSH Director on vSphere](#)

Part 4: Install runtimes

For more information about the runtimes, see the following documentation sets:

- [VMware Elastic Application Runtime](#)
- [VMware Tanzu Container Orchestration Engine \(formerly TKGI\)](#)
- [Isolation Segment](#)
- [VMware Elastic Application Runtime \[Windows\]](#)
- [Installing Small Footprint Elastic Application Runtime](#)

Part 5: Install services

For more information about various services, see the documentation sets listed on the [VMware Tanzu Platform Documentation](#) home page.

Deploying BOSH and Tanzu Operations Manager

This topic explains how you can deploy BOSH and Tanzu Operations Manager.

[BOSH](#) and [Tanzu Operations Manager](#) provide the foundation for Tanzu Platform for Cloud Foundry and its associated products. When you install Elastic Application Runtime, you install the following components first.

- BOSH is an open-source tool that lets you run software systems in the cloud.

- BOSH and its IaaS-specific Cloud Provider Interfaces (CPIs) are what let Elastic Application Runtime run on multiple IaaSEs.
- See [Deploying Software with BOSH](#) for a mini overview of how you use BOSH to run software in the cloud.
- Tanzu Operations Manager is a GUI application that streamlines deploying subsequent software to the cloud via BOSH. Tanzu Operations Manager features are also accessible by API endpoints. For more information, see [Using the Tanzu Operations Manager API](#).
 - For routine tasks, you can use the Tanzu Operations Manager interface instead of the BOSH command-line interface (CLI), Tanzu Operations Manager API, or BOSH API.
 - Tanzu Operations Manager represents products as *tiles* with multiple configuration panes that let you input or select configuration values needed for the product.
 - Tanzu Operations Manager generates BOSH manifests containing the user-supplied configuration values, and sends them to the Director.
 - After you install Tanzu Operations Manager and BOSH, you use Tanzu Operations Manager to deploy almost all products.
- Deploying Tanzu Operations Manager deploys both BOSH and Tanzu Operations Manager with a single procedure.
 - You deploy BOSH and Tanzu Operations Manager by logging into your IaaS and allocating the network addresses, VMs, user accounts, and other resources needed to support Elastic Application Runtime and other products.

For all of the mentioned information, the specifics depend on the IaaS on which you are deploying Tanzu Operations Manager. The contents of this documentation are organized by IaaS.

IaaS-specific deployment guidelines

For information on deploying Tanzu Operations Manager to your IaaS, choose one of the following:

- Deploying to AWS: [Installing Tanzu Operations Manager on AWS](#)
- Deploying to GCP: [Installing Tanzu Operations Manager on GCP](#)
- Deploying to vSphere: [Installing Tanzu Operations Manager on vSphere](#)

Deploying software with BOSH

The following steps describe how you can use BOSH to run software in the cloud:

1. You create one or more *BOSH releases*, a directory that contains your software and anything that it depends on or needs to run on generic, empty VMs. For more information about using multiple BOSH releases, see [Including Multiple BOSH Releases in a deployment](#).
2. You create a *manifest .yml* file that identifies the component processes in the BOSH release, specifies the VMs they run on, how they communicate, how to start them up and shut them down, and any other configuration specifics that they need.
3. Using the BOSH command-line interface (CLI) or API, you upload the BOSH release and send the manifest to the *Director*, BOSH's executive process.
4. The BOSH Director provisions what it needs from the IaaS, *deploys* your software to run in the cloud, and heals automatically when VMs go down.

5. BOSH CLI and API commands let you control BOSH-managed processes and allocate or release IaaS resources.

Including multiple BOSH releases in a deployment

You can include additional features in the BOSH Director through targeted customization of the `manifest.yml`. The only software you can include in the BOSH Director is additional BOSH releases. All additional releases must be available at an HTTP address that Tanzu Operations Manager can reach. You cannot host or store additional BOSH releases on Tanzu Operations Manager or the BOSH Director.

To include custom BOSH releases in the BOSH Director, create as many BOSH releases as you need and use the Tanzu Operations Manager API to the location and necessary configuration properties each release.

For more information, see [Adding a job to the BOSH Director](#) in the Tanzu Operations Manager API documentation.

Customizing the Director manifest incorrectly can cause deployment downtime or failure. Tanzu Operations Manager does not perform any validations on software introduced to a deployment through manifest customizations. This feature is for advanced Tanzu Operations Manager operators only.

For more information about writing the `manifest.yml` file, see the [BOSH documentation](#).

Preparing your firewall

This topic describes how to configure your firewall for Tanzu Operations Manager and Tanzu Platform for Cloud Foundry, and how to verify that component VMs correctly resolve DNS entries behind your firewall.

Configure your firewall

Tanzu Operations Manager and Tanzu Platform for Cloud Foundry require the following open TCP ports:

- 25555: Routes from Tanzu Operations Manager VM to the BOSH Director
- 443: If configured, routes to your own load balancer
- 80: If configured, routes to your own load balancer
- 8844: Routes from Tanzu Operations Manager VM to BOSH CredHub
- 8443: Routes from Tanzu Operations Manager VM to BOSH Director UAA
- 6868: Routes to the BOSH Agent
- 2222: Necessary for using App SSH
- 25595: Routes from the Traffic Controller to the BOSH Director, to enable sending BOSH health metrics to the Firehose
- For vSphere Only: Tanzu Operations Manager requires that the NSX firewall routes are not blocked. By default, all communication between Tanzu Operations Manager VMs and vCenter or ESXi hosts route through the NSX firewall are blocked. For more information on which ports to enable, see [VMware ports and protocols](#) for vSphere.

If you want to use an external NTP server, UDP port 123 must be open.

For more information about required ports for additional installed products, see [Routing Network Communications](#).

Example: Configure firewall with iptables

The following example uses `iptables` commands to configure a firewall. `GATEWAY_EXTERNAL_IP` is a placeholder. Replace this value with your `PUBLIC_IP`.

1. Open `/etc/sysctl.conf`, a file that contains configurations for Linux kernel settings, by running:

```
sudo vi /etc/sysctl.conf
```

2. Add the line `net.ipv4.ip_forward=1` to `/etc/sysctl.conf` and save the file.
3. To remove all existing filtering or Network Address Translation (NAT) rules, run:

```
iptables --flush
iptables --flush -t nat
```

4. Add environment variables to use when creating the IP rules. Run:

```
export INTERNAL_NETWORK_RANGE=10.0.0.0/8
export GATEWAY_INTERNAL_IP=10.0.0.1
export GATEWAY_EXTERNAL_IP=203.0.113.242
export VMWARETANZU_IP=10.0.0.2
export LOAD_BALANCER_IP=10.0.0.254
```

5. To configure IP rules for the specified chains, run:

- FORWARD:

```
iptables -A FORWARD -i eth1 -j ACCEPT
iptables -A FORWARD -o eth1 -j ACCEPT
```

- POSTROUTING:

```
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables -t nat -A POSTROUTING -d $LOAD_BALANCER_IP -s $INTERNAL_NETWORK_RANGE \
-p tcp --dport 80 -j SNAT --to $GATEWAY_INTERNAL_IP
iptables -t nat -A POSTROUTING -d $LOAD_BALANCER_IP -s $INTERNAL_NETWORK_RANGE \
-p tcp --dport 443 -j SNAT --to $GATEWAY_INTERNAL_IP
```

- PREROUTING:

```
iptables -t nat -A PREROUTING -d $GATEWAY_EXTERNAL_IP -p tcp --dport \
25555 -j DNAT --to $VMWARETANZU_IP
iptables -t nat -A PREROUTING -d $GATEWAY_EXTERNAL_IP -p tcp --dport \
443 -j DNAT --to $LOAD_BALANCER_IP
iptables -t nat -A PREROUTING -d $GATEWAY_EXTERNAL_IP -p tcp --dport \
80 -j DNAT --to $LOAD_BALANCER_IP
iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 8443 -j DNAT \
--to $VMWARETANZU_IP:443
iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 80 -j DNAT \
--to $LOAD_BALANCER_IP:80
iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 8022 -j DNAT \
--to $VMWARETANZU_IP:22
iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 8080 -j DNAT \
--to $VMWARETANZU_IP:80
```

6. Save the `iptables` by running:


```
service iptables save
```

For more information about administering IP tables with [iptables](#), see the [iptables documentation](#).

Verify that component VMs resolve DNS entries behind your firewall

When you install Tanzu Operations Manager in an environment that uses a strong firewall, the firewall might block DNS resolution. For example, if you use [xip.io](#) to test your DNS configuration, the tests fail without warning if the firewall prevents Elastic Application Runtime from accessing [*.xip.io](#).

To verify that component VMs can correctly resolve DNS entries:

1. SSH into the Tanzu Operations Manager VM by following the procedure in [Log Into the Tanzu Operations Manager VM with SSH](#).
2. Run any of the following network administration commands with the IP address of the VM:
 - [nslookup](#)
 - [dig](#)
 - [host](#)
 - The appropriate [traceroute](#) command for your OS.
3. Review the output of the command and fix any blocked routes. If the output displays an error message, review the firewall logs to determine which blocked route or routes you need to clear.
4. Repeat steps 1 through 3 with the BOSH Director VM.

IaaS permissions guidelines

VMware recommends that you follow these practices for creating secure IaaS user roles before installing Tanzu Operations Manager.

The connection between Tanzu Operations Manager and IaaS providers requires IaaS accounts with appropriate permissions. These accounts act on behalf of the operator to access IaaS functionality, such as creating VMs, managing networks and storage, and other related services.

Tanzu Operations Manager and Tanzu Platform for Cloud Foundry can be configured with IaaS users in different ways, depending on your IaaS. Other product tiles and services might also use their own IaaS credentials. Refer to the documentation for those product tiles or services to configure them securely.

Least Privileged Users (LPUs)

VMware recommends following the principle of least privilege by scoping privileges to the most restrictive permissions possible for a given role. In the event that someone gains access to credentials by mistake or through malicious intent, LPUs limit the scope of the breach. VMware recommends following best practices for the particular IaaS you are deploying.

IaaS-specific guidelines

IaaS	Guidelines
AWS	AWS permissions guidelines

laaS	Guidelines
Azure	Preparing to deploy Tanzu Operations Manager on Azure . Use the minimum permissions necessary when creating your service principal.
GCP	Use one account with the minimum permissions required to create desired GCP resources in your GCP project, then create a separate service account with the minimum permissions required to deploy Elastic Application Runtime and other tiles. For more information about creating the service account, see Step 1: Set up IAM Service Accounts in <i>Preparing to Deploy Tanzu Operations Manager on GCP</i> .
vSphere	See Installing Tanzu Operations Manager on vSphere .

Tanzu Operations Manager Runtimes

This topic tells you about the runtime environments that you can install on Tanzu Operations Manager: Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI).

Tanzu Operations Manager includes runtime environment options that support apps and containers running on Linux and Windows VMs at different levels of resource usage and isolation. You install your chosen runtime or runtimes using the Tanzu Operations Manager Installation Dashboard.

Tanzu Operations Manager includes support for the following primary runtime environments:

- [Elastic Application Runtime](#)
- [Tanzu Hub](#)
- [Container Orchestration Engine](#)

Elastic Application Runtime

Elastic Application Runtime lets developers develop and manage cloud-native apps and software services running on Linux. Elastic Application Runtime is based on the open-source Cloud Foundry Application Runtime project.

To install and configure Elastic Application Runtime, see the [Elastic Application Runtime documentation](#).

As an alternative to Elastic Application Runtime, you can install [Elastic Application Runtime \[Small Footprint\]](#) (Elastic Application Runtime [Small Footprint]) to deploy your app using fewer resources.

For more information, see [Elastic Application Runtime \[Small Footprint\]](#).

After you install the Elastic Application Runtime runtime tile, you can also install the following runtime tiles that extend the functionality of Elastic Application Runtime:

- [Elastic Application Runtime Windows add-on](#)
- [Isolation Segment](#)

Elastic Application Runtime [Small Footprint]

Elastic Application Runtime [Small Footprint] is an alternative to Elastic Application Runtime that uses significantly fewer VMs than Elastic Application Runtime, but limits system capacity and availability.

For more information about Elastic Application Runtime [Small Footprint], see the [Elastic Application Runtime documentation](#).

Elastic Application Runtime Windows add-on

Elastic Application Runtime Windows add-on lets Tanzu Operations Manager host apps on Windows VMs and enables Tanzu Operations Manager developers to push .NET apps to these Windows Diego Cells. You must install Elastic Application Runtime before you can install Elastic Application Runtime Windows add-on.

For more information about Elastic Application Runtime Windows add-on, see the [Elastic Application Runtime Windows add-on documentation](#).

Isolation Segment

Isolation Segment extends Elastic Application Runtime to run apps in separate segments with their own isolated computing, routing, and logging resource pools. Operators replicate and configure an Isolation Segment tile for each new resource pool they want to create. You must install Elastic Application Runtime before you can install Isolation Segment.

To install Isolation Segment, see the [Elastic Application Runtime documentation](#).

Tanzu Hub

The Tanzu Hub tile installs Tanzu Hub on a Tanzu Operations Manager and BOSH foundation. To install the Tanzu Hub tile, see the [Tanzu Hub documentation](#).

By installing the Tanzu Connector tile in addition to the Tanzu Hub tile, you can use Tanzu Hub to view and analyze metrics and other data from a Cloud Foundry foundation. For more information, see [Connect, discover, and manage your foundations](#).

Container Orchestration Engine

Container Orchestration Engine lets operators provision, operate, and manage enterprise-grade Kubernetes clusters on Tanzu Operations Manager. Container Orchestration Engine is based on the open-source Cloud Foundry Container Runtime project.

To install and configure Container Orchestration Engine, see the [Container Orchestration Engine documentation](#).

Installing Tanzu Operations Manager in your air-gapped environment

This topic provides an overview of the components and resources you need to install Tanzu Operations Manager in air-gapped environments, including the typical corresponding automation resources.

Offline components

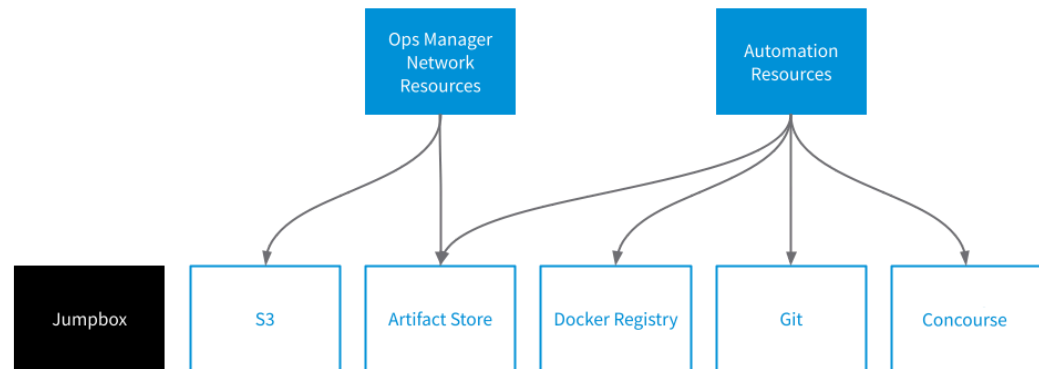
To run offline, you must obtain resources from the internet and move them into offline components that store and use them. The method you use to move a resource from the [Broadcom Support portal](#) or [GitHub](#) into an offline environment can vary from setting up a designated proxy to burning a DVD.

The following image displays types of resources, and the components you must move them to, in your offline environment. It also includes a jumpbox. The jumpbox is a Linux host for running commands such as `bosh`, `uaac`, and `fly`. You can use the Tanzu Operations Manager VM for this purpose.

Blocks shown in this configuration are:

- Tanzu Operations Manager Network Resources, feeding into:
 - S3

- Artifact Store
- Automation Resources, feeding into:
 - Artifact Store
 - Docker Registry
 - Git
 - Concourse



The following table provides more detail about resources and the component you must move them to:

Resource	Move to...
VMware Tanzu products such as tiles, Stemcells, BOSH releases, and Tanzu Operations Manager	S3 and artifact store
Pipelines and scripts	Git
Configuration	Git
Container images	Docker Registry, S3, or artifact store
Third-party resources such as NSX-T and OSS BOSH releases	Artifact store
Backup artifacts	S3

Architectural patterns

The following sections describe three architectural patterns for running in air-gapped environments. The pattern you use depends on how your environment is set up.

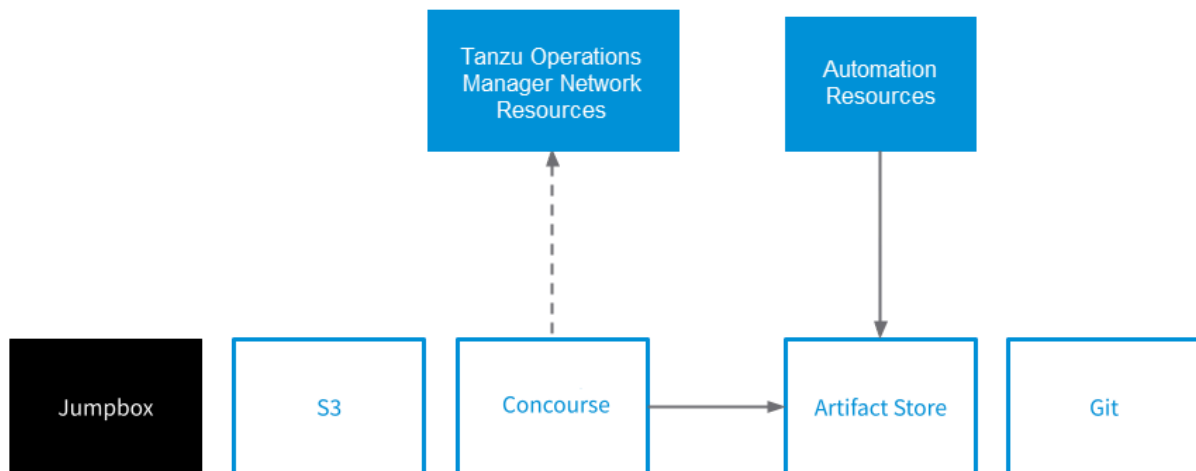
Pattern One: Artifact store with internet access

It is common for organizations to deploy an artifact store such as Artifactory or Nexus as a gateway to the Internet. In this configuration, the artifact store is typically configured as a Docker mirror.

This pattern includes an S3 component that is used only for backups of the installation. Blocks in this configuration are:

- Jumpbox
- S3
- Git
- Runway / Concourse, feeding into:
 - Tanzu Operations Manager Network Resources

- Artifact Store
- Automation Resources, feeding into:
 - Artifact Store



The following table provides an example of how you might handle resources in this scenario:

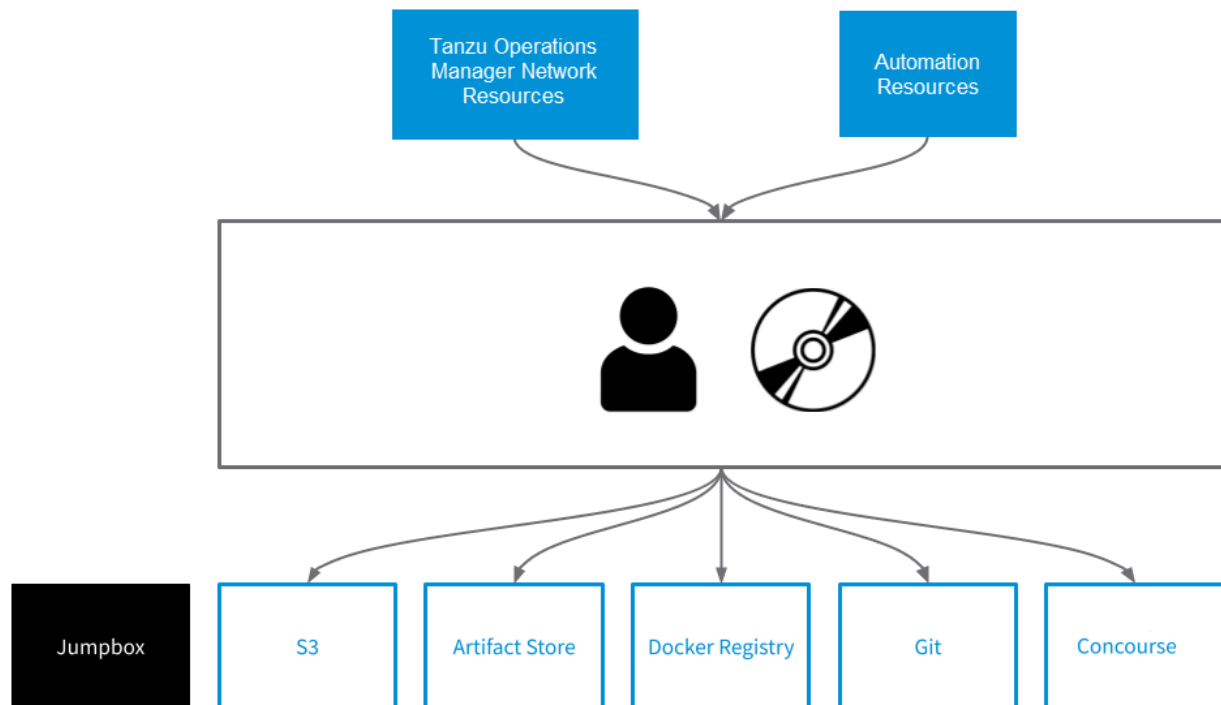
Resource	Component
VMware Tanzu products such as tiles, Stemcells, BOSH releases, and Tanzu Operations Manager	This architectural pattern presents the challenge of getting VMware Tanzu resources from the Internet into the artifact store. For example, Artifactory and Nexus cannot interact directly with Broadcom Support portal. You could place VMware Tanzu resources in an artifact store through a remote Concourse worker with access to the internet, or by downloading them from the internet and placing them in the artifact store manually, for example.
Pipelines and scripts	Store in Git. Use manual clone and push. Modify pipelines to use <code>registry_mirror</code> .
Configuration	Store in Git. Use manual clone and push.
Container images	Store in the artifact store that you configured as a mirror.
Third-party resources such as NSX-T and OSS BOSH releases	Store in a generic artifact store repository.
Backup artifacts	Store in a S3 blobstore.

Pattern Two: Completely offline

In completely offline environments, you must bring in all resources manually and store them in the available local components. Blocks shown in this configuration are:

- Jumpbox
- Tanzu Operations Manager Network Resources, feeding into:
 - Operator and media
- Automation Resources, feeding into:
 - Operator and media
- Operator and media, feeding into:
 - S3
 - Artifact Store
 - Docker Registry

- Git
- Concourse



In this architectural pattern, you must configure pipelines to watch components for changes and manually apply updates when they are available. The following table provides an example of how you might handle resources in this scenario:

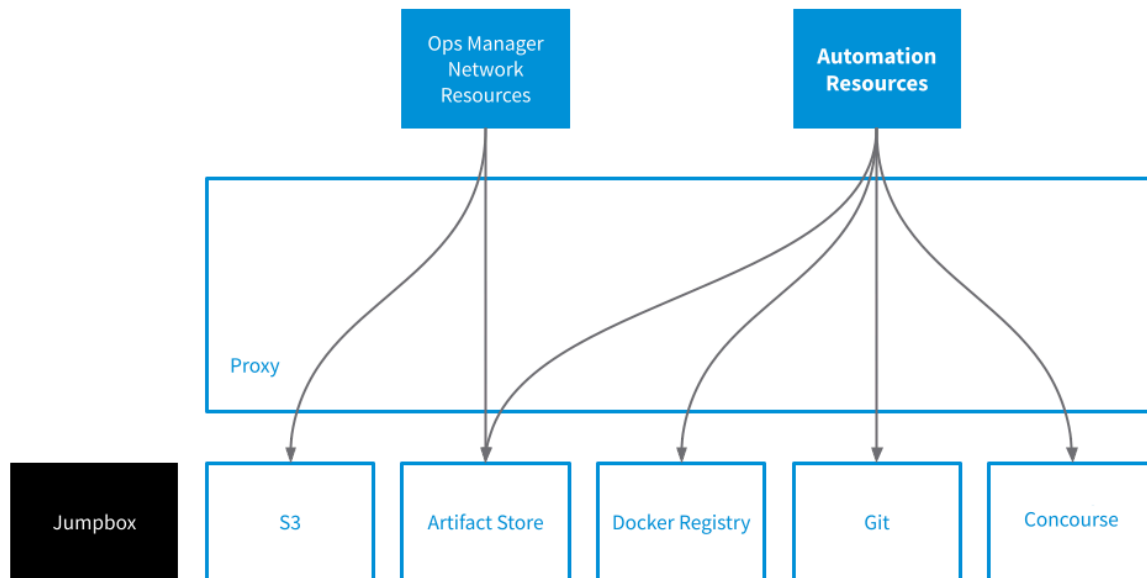
Resource	Component
VMware Tanzu products such as tiles, stemcells, BOSH releases, and Tanzu Operations Manager	Manually upload to Nexus.
Pipelines and scripts	Modify pipelines to use a local Harbor Docker registry. Manually clone an online environment, bring it to the offline environment on DVD, and push it to offline Git.
Configuration	Store in your offline Git.
Container images	Get from the Internet, transfer to USB, and push to offline Harbor.
Third-party resources such as NSX-T and OSS BOSH releases	Store in a generic artifact store repository.
Backup artifacts	Store in a S3 blobstore.

Pattern Three: TLS intercepting proxy

This pattern allows Internet access, but only through a proxy that decrypts and rewrites TLS certificates. In general, this resembles an online install, however it requires that you add your corporate certificate to all BOSH-deployed VMs. Blocks shown in this configuration are:

- Jumpbox
- Tanzu Operations Manager Network Resources block and Automation Resources block, feed (through a proxy) into:
 - S3
 - Artifact Store

- Docker Registry
- Git
- Concourse



Some of the challenges in this pattern include the following:

- Any automation that calls to the Internet fails if it does not have the corporate certificate in its trust store.
- Concourse tasks that do not use resources do not receive updated BOSH root certificates. This means you must configure tasks to ignore TLS errors or update the root certificates as part of the tasks.

Expanding your network with additional subnets

You can add more than one subnet to each network using the BOSH Director.

After you deploy Tanzu Operations Manager, you can expand your network with multiple, overlapping availability zones (AZs). You might want to create subnets with overlapping AZs if your current subnet is full and you determine that your network needs expansion.

Add a subnet

To add an additional subnet to your network:

1. Go to the Tanzu Operations Manager Installation Dashboard.
2. Click the BOSH Director tile.
3. Click Create Networks.
4. Click the name of your network to expand the network details.
5. Click Add Subnet. This action populates a new set of subnet fields. You might need to scroll down to see the fields.
6. Complete each of the subnet fields. For more information about configuring your subnet, see the Tanzu Operations Manager [configuration topic](#) for your IaaS installation.
7. Click Save.
8. Return to the Tanzu Operations Manager Installation Dashboard.
9. Click Review Pending Changes.
10. Click Apply Changes.

Installing and configuring Tanzu Operations Manager on AWS

The following topics describe how you can install and configure Tanzu Operations Manager on AWS:

- [Requirements and prerequisites for Tanzu Operations Manager on AWS](#)
- [AWS permissions guidelines and best practices](#)
- [Installing Tanzu Operations Manager on AWS manually](#)
- [AWS objects required for deploying Tanzu Operations Manager](#)
- [Preparing to deploy Tanzu Operations Manager on AWS](#)
 - [Tanzu Operations Manager for AWS policy document](#)
- [Deploying Tanzu Operations Manager on AWS](#)
- [Configuring BOSH Director on AWS](#)
- [Upgrading BOSH Director for Tanzu Operations Manager on AWS](#)
- [Deleting your Tanzu Operations Manager deployment from AWS](#)

Requirements and prerequisites for Tanzu Operations Manager on AWS

Ensure that you meet the following requirements and perform the AWS set up tasks before installing [Tanzu Operations Manager](#) on Amazon Web Services (AWS).

You can install Tanzu Operations Manager on AWS with either the Tanzu Platform for Cloud Foundry or VMware Tanzu Container Orchestration Engine (formerly TKGI) runtime. There are resource requirements specific to each runtime. Ensure that you meet the requirements for your runtime and the requirements specific to AWS before installing Tanzu Operations Manager on AWS.

Requirements

This section lists the resource requirements for installing Tanzu Operations Manager on AWS, both the general resource requirements, and the AWS-specific resource requirements.

Tanzu Operations Manager resource requirements

When launching Tanzu Operations Manager on AWS, you must use an instance type that meets the following minimum specifications: | Machine Spec | Minimum Value | | CPU | 4 vCPUs | | Memory | 16 GB | | Boot disk | 100 GB |

VMware recommends using a [m5.xlarge](#) instance type for the `<%= vars.ops_manager >` VM.

AWS resource requirements

These are the AWS-specific resource requirements for installing Tanzu Operations Manager on AWS with an external database and external file storage:

- Installing Tanzu Operations Manager on AWS requires a minimum of the following VM instance limits in your AWS account. The number of VMs required depends on the number of tiles and availability zones (AZs) you plan to deploy. Refer to one of the following topics, depending on the runtime you plan to install using Tanzu Operations Manager:
 - For Elastic Application Runtime-specific resource requirements, see [Elastic Application Runtime Resource Requirements](#).
 - For Container Orchestration Engine-specific resource requirements, see [AWS Prerequisites and Resource Requirements](#).

Before installing Tanzu Operations Manager

Before installing Tanzu Operations Manager on AWS, you must:

- Increase or remove the VM instance limits in your AWS account. Installing Tanzu Operations Manager requires more than the default 20 concurrent instances. For more information about VM resource requirements, see [Requirements](#).
- Configure your AWS account with the appropriate AWS region. For more information about selecting the correct region for your deployment, see [Region and Availability Zone Concepts](#) in the AWS documentation.
- Install the AWS CLI. Configure the AWS CLI with the user credentials that have admin access to your AWS account. To download the AWS CLI, see [AWS CLI](#).
- Configure an AWS EC2 key pair to use with your Tanzu Operations Manager deployment. For more information, see [Creating an EC2 Key Pair](#) in the AWS documentation.
- Register a wildcard domain for your Tanzu Operations Manager installation. For more information, see [Create an HTTPS listener for your Application Load Balancer](#) in the AWS documentation.
- Create an SSL certificate for your Tanzu Operations Manager domain. For more information, see the [AWS documentation about SSL certificates](#).

To deploy Tanzu Operations Manager to a production environment, you must obtain a certificate from a certificate authority. VMware recommends using a self-signed certificate generated by Tanzu Operations Manager for development and testing purposes only.

- (Elastic Application Runtime only): Configure sufficient IP allocation. For more information about IP allocation requirements, see [Elastic Application Runtime Resource Requirements](#).
- (Optional) (Elastic Application Runtime only): Configure external storage. VMware recommends using external storage if possible. For more information about how file storage location affects platform performance and stability during upgrades, see [Configure file storage](#) in *Configuring Elastic Application Runtime for Upgrades*.

- (Optional) (Elastic Application Runtime and Tanzu Operations Manager only): Configure external databases. VMware recommends using external databases in production deployments for BOSH Director and Elastic Application Runtime. An external database must be configured to use the UTC timezone.
- (Optional) Configure external user stores. When you deploy Tanzu Operations Manager, you can select SAML, LDAP, or OIDC to integrate existing user accounts.

Install Tanzu Operations Manager on AWS

To install Tanzu Operations Manager on AWS, see [Installing Tanzu Operations Manager on AWS](#).

Additional resources

For more information, see:

- [AWS Permissions Guidelines](#)

And in the AWS documentation:

- AWS identity and access management: [What is IAM?](#).
- User management: [Identities \(Users, Groups, and Roles\)](#).
- Best practices: [Temporary Security Credentials](#).

AWS permissions guidelines and best practices

VMware recommends:

- Minimizing use of primary account credentials by creating an IAM role and instance profile with the minimum required EC2, VPC, and EBS credentials.
- Following AWS account security best practices such as disabling root keys, using multi-factor authentication on the root account, and using CloudTrail for auditing API actions.

For more Amazon-specific best practices, see the following topics in the AWS documentation:

- [IAM Roles Best Practices](#)
- [AWS Security Best Practices Whitepaper](#)
- [AWS Well-Architected Framework](#)

Installing Tanzu Operations Manager on AWS manually

You can manually install Tanzu Operations Manager on Amazon Web Services (AWS) for use with the Tanzu Platform for Cloud Foundry runtime.

Do not use this topic if you intend to install the VMware Tanzu Container Orchestration Engine (formerly TKGI) runtime on AWS.

Installation procedure

To install Tanzu Operations Manager manually on AWS for use with Elastic Application Runtime:

1. (Optional) Review the resources you must deploy to AWS. See [Required AWS objects](#).

2. Prepare your AWS environment. See [Preparing to deploy Tanzu Operations Manager on AWS](#).
3. Deploy Tanzu Operations Manager. See [Deploying Tanzu Operations Manager on AWS](#).
4. (Optional) Verify that you deployed all of the required resources to AWS. See [Required AWS objects](#).
5. Configure BOSH Director. See [Configuring BOSH Director on AWS](#).

Next steps

After you complete the previous procedures, configure a runtime.

For information about installing and configuring a runtime, see the following:

- [Configuring Elastic Application Runtime](#)
- [Installing Container Orchestration Engine on AWS](#)
- [Installing and configuring Elastic Application Runtime Windows add-on](#)
- [Installing Small Footprint Elastic Application Runtime](#)

AWS objects required for deploying Tanzu Operations Manager

This topic tells you about the objects that can be created in Amazon Web Services (AWS) to deploy Tanzu Operations Manager.

Use the information in this topic to determine the resource requirements of Tanzu Operations Manager on AWS, or to verify that you have created the correct resources after completing the procedures in [Preparing to Deploy Tanzu Operations Manager on AWS](#) and [Deploying Tanzu Operations Manager on AWS](#).

S3 buckets

As part of the preparing to deploy process, you must create the following S3 buckets from the S3 Dashboard:

- `pcf-ops-manager-bucket`
- `pcf-buildpacks-bucket`
- `pcf-packages-bucket`
- `pcf-resources-bucket`
- `pcf-droplets-bucket`

These buckets must be empty when you install or reinstall Tanzu Operations Manager.

See [Create S3 Buckets](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

IAM role or user for Tanzu Operations Manager

You must create either an IAM role or an IAM user for Tanzu Operations Manager named `pcf-user` from the Identity and Access Management Dashboard, using the policy document included in [Tanzu Operations Manager for AWS Policy Document](#).

See [Create an IAM Role or User for Tanzu Operations Manager](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

Key pair

You must generate a key pair named `pcf-ops-manager-key`. For more information about setting up a key pair, see [Amazon EC2 Key Pairs](#) in the AWS documentation.

VPC (public and private subnets)

You must create a VPC with public and private subnets from the VPC Dashboard.

See [Create a VPC](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

The following table lists the subnets in CIDR block `10.0.0.0/16`.

Name	AZ	IPv4 CIDR block
<code>pcf-public-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.0.0/24</code>
<code>pcf-public-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.1.0/24</code>
<code>pcf-public-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.2.0/24</code>
<code>pcf-management-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.16.0/28</code>
<code>pcf-management-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.16.16/28</code>
<code>pcf-management-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.16.32/28</code>
<code>pcf-ert-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.4.0/24</code>
<code>pcf-ert-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.5.0/24</code>
<code>pcf-ert-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.6.0/24</code>
<code>pcf-services-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.8.0/24</code>
<code>pcf-services-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.9.0/24</code>
<code>pcf-services-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.10.0/24</code>
<code>pcf-rds-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.12.0/24</code>
<code>pcf-rds-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.13.0/24</code>
<code>pcf-rds-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.14.0/24</code>

NAT Gateway

You must create a NAT Gateway when creating a VPC.

See [Create a NAT Gateway](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

Security groups

The following sections describe the security groups you must create from the EC2 Dashboard.

Tanzu Operations Manager

The Tanzu Operations Manager Security Group must be named `pcf-ops-manager-security-group` and have the following inbound rules.

See [Configure a Security Group for Tanzu Operations Manager](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

Type	Protocol	Port range	Source
HTTP	TCP	80	My IP
HTTPS	TCP	443	My IP
SSH	TCP	22	My IP
BOSH Agent	TCP	6868	10.0.0.0/16
BOSH Director	TCP	25555	10.0.0.0/16

BOSH-deployed VMs

The BOSH-deployed VMs Security Group must be named `pcf-vms-security-group` and have the following inbound rules.

See [Configure a Security Group for BOSH-Deployed VMs](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

Type	Protocol	Port range	Source
All traffic	All	0 - 65535	Custom IP 10.0.0.0/16
Custom TCP rule	TCP	2222	Anywhere 0.0.0.0/0

Web ELB

The Web ELB Security Group must be named `pcf-web-elb-security-group` and have the following inbound rules.

See [Configure a Security Group for the Web ELB](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

Type	Protocol	Port range	Source
Custom TCP rule	TCP	4443	Anywhere 0.0.0.0/0
HTTP	TCP	80	Anywhere 0.0.0.0/0
HTTPS	TCP	443	Anywhere 0.0.0.0/0

TCP ELB

The TCP ELB Security Group must be named `pcf-tcp-elb-security-group` and have the following inbound rule.

See [Configure a Security Group for the TCP ELB](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

Type	Protocol	Port range	Source
Custom TCP rule	TCP	1024 - 1123	Anywhere 0.0.0.0/0

The TCP ELB Security Group must have the following outbound rule:

Type	Protocol	Port range	Source
All traffic	All	All	Anywhere 0.0.0.0/0

MySQL

The MySQL Security Group must be named `pcf-mysql-security-group` and have the following inbound rule:

See [Configure a Security Group for MySQL](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

Type	Protocol	Port range	Source	
MySQL	TCP	3306	Custom IP	10.0.0.0/16

The MySQL Security Group must have the following outbound rule.

Type	Protocol	Port range	Destination	
All traffic	All	All	Custom IP	10.0.0.0/16

Tanzu Operations Manager AMI

1. On the [Broadcom Support portal](#), click Tanzu Operations Manager for AWS.
2. In the PDF that is downloaded, find the AMI ID.
3. Using the AMI ID, locate the Tanzu Operations Manager AMI.

See [Step 1: Launch a Tanzu Operations Manager AMI](#) in *Deploying Tanzu Operations Manager on AWS*.

ELBs

The following sections describe the ELBs you must create from the EC2 Dashboard.

Web ELB

You must create a web ELB with the following configuration.

See [Step 2: Create Web Load Balancer](#) in *Deploying Tanzu Operations Manager on AWS*.

- Name: `pcf-web-elb`
- LB Inside: `pcf-vpc`
- Selected Subnet: `pcf-public-subnet-az0`, `pcf-public-subnet-az1`, `pcf-public-subnet-az2`
- Security Group: `pcf-elb-security-group`

SSH ELB

See [Step 3: Create SSH Load Balancer](#) in *Deploying Tanzu Operations Manager on AWS*.

- Name: `pcf-ssh-elb`
- LB Inside: `pcf-vpc`
- Selected Subnet: `pcf-public-subnet-az0`, `pcf-public-subnet-az1`, `pcf-public-subnet-az2`
- Security Group: `pcf-ssh-security-group`
- Health Check: TCP Port 2222

TCP ELB

See [Step 4: Create TCP Load Balancer](#) in *Deploying Tanzu Operations Manager on AWS*.

- Name: `pcf-tcp-elb`

- LB Inside: `pcf-vpc`
- Selected Subnet: `pcf-public-subnet-az0`, `pcf-public-subnet-az1`, `pcf-public-subnet-az2`
- Security Group: `pcf-tcp-security-group`
- Health Check: TCP Port 80

DNS configuration

You must go to your DNS provider and create CNAME and A records for all three of your load balancers.

See [Step 5: Configure DNS Records](#) in *Deploying Tanzu Operations Manager on AWS*.

RDS subnet group

You must create a subnet group for RDS named `pcf-rds-subnet-group` from the RDS Dashboard.

See [Step 6: Create RDS Subnet Group](#) in *Deploying Tanzu Operations Manager on AWS*.

MySQL database

You must create a MySQL database from the RDS Dashboard.

See [Step 7: Create a MySQL Database using AWS RDS](#) in *Deploying Tanzu Operations Manager on AWS*.

Preparing to deploy Tanzu Operations Manager on AWS

To deploy Tanzu Operations Manager on AWS, you must follow the procedures in this topic to create the objects in the AWS Management Console that Tanzu Operations Manager requires. You can manually configure the Amazon Web Services (AWS) components that you need to deploy Tanzu Operations Manager on AWS.

To view the list of AWS objects created by the procedures in this topic, see [Required AWS Objects](#).

File an AWS ticket/request

Log in to the AWS Management Console, and file a ticket with Amazon to ensure that your account can launch more than the default 20 instances. In the ticket, ask for these limits:

- 50 `t3.micro` instances
- 20 `c5.large` instances

You can verify the limits on your account by visiting the EC2 Dashboard on the AWS Management Console and clicking Limits on the left navigation menu.

For specialized deployments:

To deploy to...	Instructions
AWS for US Gov	Log in to the AWS GovCloud (US) Console instead of the standard AWS Management Console.
AWS China	Set up an AWS China account and contact the Platform Architect assigned for your account.

Create S3 buckets

S3 bucket names must be globally unique. When naming buckets, VMware recommends that you prefix the generic names below with a unique and easily identifiable string (For example, `ID-STRING-pcf-ops-manager-bucket`, `MY-IDENTIFIER-pcf-buildpacks-bucket`, and so on). Then you should use the same prefix when naming other associated resources, such as IAM policies.

Follow these steps to create five S3 buckets:

1. On the S3 Dashboard, click Create Bucket.
2. In Bucket name, enter `ID-STRING-pcf-ops-manager-bucket`.
3. In AWS Region, select your region.
4. Click Create bucket.
5. Repeat these steps to create 4 more S3 buckets:
 - `ID-STRING-pcf-buildpacks-bucket`
 - `ID-STRING-pcf-packages-bucket`
 - `ID-STRING-pcf-resources-bucket`
 - `ID-STRING-pcf-droplets-bucket`

Create a policy for Tanzu Operations Manager

Follow these steps to create an Amazon Identity and Access Management (IAM) policy with the minimal permissions necessary to run and install Tanzu Operations Manager:

1. Click IAM to access the IAM Dashboard.
2. Click Policies and then click Create Policy.
3. Copy the policy document included in [Tanzu Operations Manager for AWS Policy Document](#). You must edit the policy document so the names of the S3 buckets match the ones you created in [Create S3 Buckets](#) above.
4. Paste the policy document into the JSON tab on the Create policy page.
5. Click Next: Tags and click Next: Review.
6. In the Name text box, enter `pcf-iam-policy`.
7. Click Create policy.

Create an IAM role or user for Tanzu Operations Manager

Follow these steps to create an Amazon Identity and Access Management (IAM) user or role with the minimal permissions necessary to run and install Tanzu Operations Manager:

VMware recommends that you use IAM roles instead of users for better security.

Click IAM to access the IAM Dashboard.

- If you are configuring AWS access using an AWS instance profile, create an IAM role:
 1. Click Roles and then click Create role.
 2. On the Create role page, set:
 - Role type: AWS service
 - Use case: EC2
 3. Click Next.

4. On the Add permissions page, enter the name of the policy you created in the Permission policies search box. Select the check box next to the policy in the search results.
 5. Click Next.
 6. On the Name, review, and create page, enter a role name. For example, enter `pcf-role`.
 7. Click Create role.
- If you are configuring AWS access using AWS keys, create an IAM user:
 1. Click Users and then click Add users.
 2. Enter a user name. For example, enter `pcf-user`.
 3. In AWS access type, check the Access key - Programmatic access check box.

If you prefer to create your keys locally and import them into AWS, see the [Amazon documentation](#).
 4. Click Next: Permissions.
 5. Click the Attach existing policies directly tab.
 6. In the Filter policies search box, enter the name of the policy you created. Then, select the check box next to the policy in the search results.
 7. Click Next: Tags, and then click Next: Review.
 8. Click Create user.
 9. Click Download .csv to download the user security credentials.

The `credentials.csv` contains the IDs for your user security access key and secret access key. Keep the `credentials.csv` file for your currently active key pairs in a secure directory. You cannot recover a lost key pair.
 10. Click Close.

Create a VPC

1. Follow the AWS tutorial to create an elastic IP address, a VPC, and additional subnets. To access the tutorial, see the [AWS documentation](#). Refer to the following table when configuring the VPC wizard fields and subnets.

VPC wizard field	Instructions
Resources to create	Select <code>VPC and more</code>
Name tag	Enter <code>pcf-vpc</code>
IPv4 CIDR block	Select <code>IPv4 CIDR manual input</code>
IPv4 CIDR	Enter <code>10.0.0.0/16</code>
IPv6 CIDR block	Select No IPv6 CIDR block
Number of public subnets	Select 0
Number of private subnets	Select 0

For all other fields, accept the defaults.

2. Click Create VPC.
3. On the VPC dashboard, click Subnets, and then Create subnet.

4. In VPC ID, click the VPC that includes `pcf-vpc`.
5. Add the following subnets. For each subnet, enter the Subnet name, Availability Zone, and IPv4 CIDR block. Then click Add new subnet to add the next subnet in the table:

Subnet name	Availability zone	IPv4 CIDR block
<code>pcf-public-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.0.0/24</code>
<code>pcf-public-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.1.0/24</code>
<code>pcf-public-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.2.0/24</code>
<code>pcf-management-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.16.0/28</code>
<code>pcf-management-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.16.16/28</code>
<code>pcf-management-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.16.32/28</code>
<code>pcf-tas-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.4.0/24</code>
<code>pcf-tas-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.5.0/24</code>
<code>pcf-tas-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.6.0/24</code>
<code>pcf-services-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.8.0/24</code>
<code>pcf-services-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.9.0/24</code>
<code>pcf-services-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.10.0/24</code>
<code>pcf-rds-subnet-az0</code>	REGION-#a (for example, <code>us-west-2a</code>)	<code>10.0.12.0/24</code>
<code>pcf-rds-subnet-az1</code>	REGION-#b (for example, <code>us-west-2b</code>)	<code>10.0.13.0/24</code>
<code>pcf-rds-subnet-az2</code>	REGION-#c (for example, <code>us-west-2c</code>)	<code>10.0.14.0/24</code>

6. Click Add new subnet.

Create an internet gateway

1. On the VPC dashboard, click Internet gateways.
2. Click Create internet gateway.
3. In Name, enter `pcf-internet-gateway`.
4. Click Create internet gateway.
5. In Actions, click Attach to VPC.
6. In Available VPCs, click the VPC that includes `pcf-vpc`.
7. Click Attach internet gateway.

Create a NAT gateway

1. On the VPC dashboard, click NAT gateways.
2. Click Create NAT gateway.
3. In Name, enter `pcf-nat-gateway`.
4. In Subnet, select the subnet that includes `pcf-public-subnet-az0`.
5. Click Allocate Elastic IP.

6. Click Create NAT gateway.

Create route table for public subnets

1. On the VPC dashboard, click Route tables.
2. Click Create route table.
3. In Name, enter `pcf-public-route-table`.
4. In VPC, click the VPC that includes (`pcf-vpc`).
5. Click Create route table.
 1. In the Routes section, click Edit routes.
 2. Click Add route.
 3. In Destination, enter `0.0.0.0/0`.
 4. In Target, click Internet Gateway, then click the internet gateway that includes (`pcf-internet-gateway`).
 5. Click Save changes.
6. Click the Subnet associations tab.
 1. In the Explicit subnet associations section, click Edit subnet associations.
 2. Select `pcf-public-subnet-az0`, `pcf-public-subnet-az1`, and `pcf-public-subnet-az2`.
 3. Click Save associations.

Create route table for remaining subnets

1. On the VPC dashboard, click Route tables.
2. Click Create route table.
3. In Name, enter `pcf-management-route-table`.
 1. In VPC, click the VPC that includes (`pcf-vpc`).
 2. Click Create route table.
4. In the Routes section, click Edit routes.
 1. Click Add route.n Destination, enter `0.0.0.0/0`.
 2. In Target, click NAT Gateway, then click the NAT gateway that includes (`pcf-nat-gateway`).
 3. Click Save changes.
5. Click the Subnet associations tab.
 1. In the Explicit subnet associations section, click Edit subnet associations.
 2. Click `pcf-management-subnet-az0`, `pcf-management-subnet-az1`, `pcf-management-subnet-az2`, `pcf-tas-subnet-az0`, `pcf-tas-subnet-az1`, `pcf-tas-subnet-az2`, `pcf-services-subnet-az0`, `pcf-services-subnet-az1`, and `pcf-services-subnet-az2`.
 3. Click Save associations.

Configure a security group for Tanzu Operations Manager

1. Return to the EC2 Dashboard.

2. Click Security Groups, then Create Security Group.
3. In Security group name, enter `pcf-ops-manager-security-group`.
4. In Description, enter a description to identify this security group.
5. In VPC, click the VPC in which you want to deploy Tanzu Operations Manager.
6. In the Inbound rules section, add rules using the information in the following table.

VMware recommends limiting access to Tanzu Operations Manager to IP ranges within your organization, but you might reduce the IP restrictions after configuring authentication for Tanzu Operations Manager.

Type	Protocol	Port range	Source	
HTTP	TCP	80	My IP	
HTTPS	TCP	443	My IP	
SSH	TCP	22	My IP	
Custom TCP (BOSH Agent)	TCP	6868	Custom	10.0.0.0/16
Custom TCP (BOSH Director)	TCP	25555	Custom	10.0.0.0/16

7. Click Create security group.

Configure a security group for BOSH-deployed VMs

1. On the Security Groups page, click Create Security Group to create another security group.
2. In Security group name, enter `pcf-vms-security-group`.
3. In Description, enter a description to identify this security group.
4. In VPC, click the VPC where you want to deploy the BOSH-deployed VMs.
5. In the Inbound rules section, add rules for all traffic from your public and private subnets to your private subnet, as shown in the table. This rule configuration does the following:
 - Enables BOSH to deploy Tanzu Platform for Cloud Foundry and other services
 - Enables app VMs to communicate through the router
 - Allows the load balancer to send traffic to Elastic Application Runtime

Type	Protocol	Port range	Source	
All traffic	All	All	Custom	10.0.0.0/16
Custom TCP rule	TCP	2222	Anywhere-IPv4	0.0.0.0/0

6. Click Create security group.

Configure a security group for the Web ELB

1. On the Security Groups page, click Create Security Group to create another security group.
2. In Security group name, enter `pcf-web-elb-security-group`.
3. In the Description text box, enter a description to identify this security group.
4. In VPC, click the VPC where you want to deploy this Elastic Load Balancer (ELB).

5. In the Inbound rules section, add rules to allow traffic to ports 80, 443, and 4443 from 0.0.0.0/0, as shown in the table.

Allow traffic to port 4443 only if you are in an AWS cloud region that does not support AWS ALBs. For example, the GovCloud region. For more information about AWS regions and availability zones, see [AWS Global Infrastructure](#).

For finer control over what can reach Elastic Application Runtime, change 0.0.0.0/0 to be more restrictive. This security group governs external access to Elastic Application Runtime from apps such as the Tanzu cf CLI and app URLs.

Type	Protocol	Port range	Source	
Custom TCP	TCP	4443	Anywhere-IPv4	0.0.0.0/0
HTTP	TCP	80	Anywhere-IPv4	0.0.0.0/0
HTTPS	TCP	443	Anywhere-IPv4	0.0.0.0/0

6. Click Create security group.

Configure a security group for the TCP ELB

1. On the Security Groups page, click Create Security Group to create another security group.
2. In Security group name, enter `pcf-tcp-elb-security-group`.
3. In Description text box, enter a description to identify this security group.
4. In VPC, click the VPC in which you want to deploy this ELB.
5. In the Inbound rules section, add the following rule:

Type	Protocol	Port range	Source	
Custom TCP rule	TCP	1024 - 1123	Anywhere-IPv4	0.0.0.0/0

6. Click Create security group.

Configure a security group for MySQL

If you plan to use an internal database, skip this step. If you are using RDS, you must configure a security group that enables the Tanzu Operations Manager VM and BOSH Director VM to access the database.

1. On the Security Groups page, click Create Security Group to create another security group.
2. In Security group name, enter `pcf-mysql-security-group`.
3. In Description text box, enter a description to identify this security group.
4. In VPC, click the VPC where you want to deploy MySQL.
5. In the Inbound rules section, add a rule of type MySQL and specify the subnet of your VPC in Source, as shown in the table.

Type	Protocol	Port range	Source	
MYSQL/Aurora	TCP	3306	Custom	10.0.0.0/16

6. In the Outbound rules section, add a rule of type **All traffic** and specify the subnet of your VPC in Destination, as shown in the table.

Type	Protocol	Port range	Source	
All traffic	All	All	Custom	10.0.0.0/16

7. Click Create security group.

Next step

Proceed to the next step, [Deploying Tanzu Operations Manager on AWS](#).

Tanzu Operations Manager for AWS policy document

You can use this policy document to complete the procedure in [Create an IAM User for Tanzu Operations Manager](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

Policy document base code

For the S3 Bucket policy sections, make sure to update the generic bucket naming with your custom/unique bucket names.

From the AWS Management Console, copy and paste the following text into the Policy Document text box.

You can add more policies to enable additional features. For more information, see [Add more AWS policies](#).

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Deny",
      "Action": [
        "iam:Add*",
        "iam:Attach*",
        "iam:ChangePassword",
        "iam:Create*",
        "iam:DeactivateMFADevice",
        "iam>Delete*",
        "iam:Detach*",
        "iam:EnableMFADevice",
        "iam:GenerateCredentialReport",
        "iam:GenerateServiceLastAccessedDetails",
        "iam:GetAccessKeyLastUsed",
        "iam:GetAccountAuthorizationDetails",
        "iam:GetAccountPasswordPolicy",
        "iam:GetAccountSummary",
        "iam:GetContextKeysForCustomPolicy",
        "iam:GetContextKeysForPrincipalPolicy",
        "iam:GetCredentialReport",
        "iam:GetGroup",
        "iam:GetGroupPolicy",
        "iam:GetLoginProfile",
        "iam:GetOpenIDConnectProvider",
        "iam:GetPolicy",
        "iam:GetPolicyVersion",
        "iam:GetRole",
        "iam:GetRolePolicy",
        "iam:GetSAMLProvider",
        "iam:GetSSHPublicKey",
        "iam:GetServerCertificate",
        "iam:GetServiceLastAccessedDetails",
        "iam:GetUser",
        "iam:GetUserPolicy",
        "iam:List*",
        "iam:Put*",
        "iam:RemoveClientIDFromOpenIDConnectProvider",
        "iam:RemoveRoleFromInstanceProfile",
        "iam:RemoveUserFromGroup",
        "iam:ResyncMFADevice",
        "iam:SetDefaultPolicyVersion",
        "iam:SimulateCustomPolicy",
        "iam:SimulatePrincipalPolicy",
        "iam:Update*"
      ],
      "Resource": [
        "*"
      ]
    }
  ],
  {

```

```

    "Sid": "OpsMgrInfrastructureIaasConfiguration",
    "Effect": "Allow",
    "Action": [
        "ec2:DescribeKeypairs",
        "ec2:DescribeVpcs",
        "ec2:DescribeSecurityGroups",
        "ec2:DescribeAvailabilityZones",
        "ec2:DescribeAccountAttributes"
    ],
    "Resource": "*"
},
{
    "Sid": "OpsMgrInfrastructureDirectorConfiguration",
    "Effect": "Allow",
    "Action": [
        "s3:*"
    ],
    "Resource": [
        "arn:aws:s3::pcf-ops-manager-bucket",
        "arn:aws:s3::pcf-ops-manager-bucket/*",
        "arn:aws:s3::pcf-buildpacks-bucket",
        "arn:aws:s3::pcf-buildpacks-bucket/*",
        "arn:aws:s3::pcf-packages-bucket",
        "arn:aws:s3::pcf-packages-bucket/*",
        "arn:aws:s3::pcf-resources-bucket",
        "arn:aws:s3::pcf-resources-bucket/*",
        "arn:aws:s3::pcf-droplets-bucket",
        "arn:aws:s3::pcf-droplets-bucket/*"
    ]
},
{
    "Sid": "OpsMgrInfrastructureAvailabilityZones",
    "Effect": "Allow",
    "Action": [
        "ec2:DescribeAvailabilityZones"
    ],
    "Resource": "*"
},
{
    "Sid": "OpsMgrInfrastructureNetworks",
    "Effect": "Allow",
    "Action": [
        "ec2:DescribeSubnets"
    ],
    "Resource": "*"
},
{
    "Sid": "DeployMicroBosh",
    "Effect": "Allow",
    "Action": [
        "ec2:DescribeImages",
        "ec2:RunInstances",
        "ec2:DescribeInstances",
        "ec2:TerminateInstances",
        "ec2:RebootInstances",

```



```

        "elasticloadbalancing:DescribeLoadBalancers",
        "elasticloadbalancing:DeregisterInstancesFromLoadBalancer",
        "elasticloadbalancing:RegisterInstancesWithLoadBalancer",
        "ec2:DescribeAddresses",
        "ec2:DisassociateAddress",
        "ec2:AssociateAddress",
        "ec2:CreateTags",
        "ec2:DescribeVolumes",
        "ec2:CreateVolume",
        "ec2:AttachVolume",
        "ec2>DeleteVolume",
        "ec2:DetachVolume",
        "ec2:CreateSnapshot",
        "ec2>DeleteSnapshot",
        "ec2:DescribeSnapshots",
        "ec2:DescribeRegions"
    ],
    "Resource": "*"
}
]
}

```

Add more AWS policies

To perform specific tasks, you might need to add more policies to your Policy Document text box in the AWS Management Console.

Refer to the following table to determine what additional policies you might want to include. Then paste the code from the corresponding row. Enter the code in your existing policy after any `},`.

For more information about AWS policies, see [Example policies](#) in the AWS documentation.

I need to...

So I paste this code.

Use an AWS Instance Profile instead of AWS Keys.

```
{
  "Sid": "AllowToGetInfoAboutCurrentInstanceProfile",
  "Effect": "Allow",
  "Action": [
    "iam:GetInstanceProfile"
  ],
  "Resource": [
    "IAM-INSTANCE-ARN-PROFILE"
  ]
},
{
  "Sid": "AllowToCreateInstanceWithCurrentInstanceProfile",
  "Effect": "Allow",
  "Action": [
    "iam:PassRole"
  ],
  "Resource": [
    "IAM-OPS-MANAGER-ARN-ROLE"
  ]
},
}
```

Where:

- **IAM-INSTANCE-ARN-PROFILE** is the IAM instance ARN profile that you configure in AWS.
- **IAM-OPS-MANAGER-ARN-ROLE** is the IAM Tanzu Operations Manager ARN role that you configure in AWS.

Encrypt EBS volumes with AWS Key Management Service (KMS).

```
{
  "Sid": "RequiredIfUsingHeavyStemcells",
  "Effect": "Allow",
  "Action": [
    "ec2:RegisterImage",
    "ec2:DeregisterImage"
  ],
  "Resource": "*"
},
{
  "Sid": "RequiredIfEncryptingStemcells",
  "Effect": "Allow",
  "Action": [
    "ec2:CopyImage"
  ],
  "Resource": "*"
},
}
```

I need to...

So I paste this code.

Use a custom KMS encryption key that I created in AWS.

```
{
  "Sid": "RequiredIfUsingCustomKMSKeys",
  "Effect": "Allow",
  "Action": [
    "kms:ReEncrypt*",
    "kms:GenerateDataKey*",
    "kms:CreateGrant",
    "kms:DescribeKey*"
  ],
  "Resource": [
    "((kms_key_arn))"
  ]
},
```

Use the load balancer target group.

```
{
  "Sid": "RequiredIfUsingLBTargetGroupCloudProperties",
  "Effect": "Allow",
  "Action": [
    "elasticloadbalancing:DescribeLoadBalancers",
    "elasticloadbalancing:DescribeTargetGroups",
    "elasticloadbalancing:DescribeTargetHealth",
    "elasticloadbalancing:RegisterTargets"
  ],
  "Resource": "*"
},
```

Use the AWS Spot instance advisor.

```
{
  "Sid": "RequiredIfUsingSpotBidPriceCloudProperties",
  "Effect": "Allow",
  "Action": [
    "ec2:CancelSpotInstanceRequests",
    "ec2:DescribeSpotInstanceRequests",
    "ec2:RequestSpotInstances"
  ],
  "Resource": "*"
},
```

Create and replace AWS routes.

```
{
  "Sid": "RequiredIfUsingAdvertisedRoutesCloudProperties",
  "Effect": "Allow",
  "Action": [
    "ec2:CreateRoute",
    "ec2:DescribeRouteTables",
    "ec2:ReplaceRoute"
  ],
  "Resource": "*"
},
```

Deploying Tanzu Operations Manager on AWS

You can deploy Tanzu Operations Manager on Amazon Web Services (AWS).

Before you deploy Tanzu Operations Manager using the procedure in this topic, complete the preparation steps in [Preparing to Deploy Tanzu Operations Manager on AWS](#).

Step 1: Launch a Tanzu Operations Manager AMI

To launch an Amazon Machine Image (AMI) for Tanzu Operations Manager:

1. Go to the Tanzu Operations Manager section of the [Broadcom Support portal](#).
2. Click the version of Tanzu Operations Manager you want to install from the Release list.
3. In the Release Download Files, click the file named Tanzu Operations Manager for AWS to download a PDF.
4. Open the PDF and identify the AMI ID for your region.
5. Return to the EC2 Dashboard.
6. Click AMIs from the Images menu.
7. Using the Owned by me drop-down filter, click Public images .
8. Paste the AMI ID for your region into the search bar and press enter.

There is a different AMI for each region. If you cannot locate the AMI for your region, verify that you have set your AWS Management Console to your desired region. Additionally, AWS automatically deprecates public AMIs 2 years after they are published, including the AMIs for Tanzu Operations Manager and stemcells. These Tanzu Operations Manager AMIs do not appear in AMI searches, but can still be launched using the [aws](#) CLI or Platform Automation. Deploying a deprecated light stemcell using AWS CPI v100 or below results in an error.

9. (Optional) If you want to encrypt the VM that runs Tanzu Operations Manager with AWS Key Management Service (KMS), perform the following additional steps:
 1. Right-click the row that lists your AMI and click Copy AMI.
 2. Click your Destination region.
 3. Click Encryption. For more information about AMI encryption, see [Encryption and AMI Copy](#) from the *Copying an AMI* topic in the AWS documentation.
 4. Click your KMS Key. To create a new custom key, see [Creating Keys](#) in the AWS documentation.
 5. Click Copy AMI. You can use the new AMI you copied for the following steps.
10. Click the row that lists your Tanzu Operations Manager AMI and click Launch instance from AMI.
11. On the Launch an instance page, for Name, enter a name for the Tanzu Operations Manager VM. For example, enter [pcf-ops-manager](#).
12. Click m5.large for your instance type.
13. Click the [pcf-ops-manager-key](#) key pair, and confirm that you have access to the private key file. You use this key pair to access the Tanzu Operations Manager VM.
14. In the Network settings section, click Edit and configure the following for your instance:
 - Network: Click the VPC that you created.
 - Subnet: Click [pcf-public-subnet-az0](#) to allow traffic from public IP addresses, or [pcf-management-subnet-az0](#) to allow traffic only from private IP addresses.
 - Auto-assign for Public IP: Click Enable to allow traffic from public IP addresses, or Disable to allow traffic only from private IP addresses.
 - Firewall (security groups): Click Select an existing security group and click the [pcf-ops-manager-security-group](#) that you created in [Configure a Security Group for Tanzu](#)

[Operations Manager](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.

- For all other fields, accept the default values.
- 15. In the Configure storage section, adjust the Size (GiB) value. The default persistent disk value is 8 GB. VMware recommends increasing this value to a minimum of 100 GB.
- 16. (Optional) If you are using IAM roles, perform the following additional steps:
 - 1. Click Advanced details to expand the section.
 - 2. Click the IAM role associated with the role created in [Create an IAM Role or User for Tanzu Operations Manager](#) in *Preparing to Deploy Tanzu Operations Manager*.
 - 3. For all other fields, accept the default values.
- 17. In the Summary section, ensure that the number of instances is 1.
- 18. Click Launch instance.
- 19. Click View all instances to access the Instances page on the EC2 Dashboard.

Step 2: Create web load balancer

- 1. On the EC2 Dashboard, click Load Balancers.
- 2. Click Create Load Balancer.
- 3. Under Application Load Balancer, click Create.
- 4. For Step 1: Configure Load Balancer:
 - 1. Under Basic Configuration:
 - For Name, enter `pcf-web-elb`.
 - For Scheme, click internet-facing to allow traffic from public IP addresses, or internal to allow traffic only from private IP addresses.
 - For IP address type, click the type of IP addresses that you want to allow.
 - 2. Under Listeners, click Add listener. For Load Balancer Protocol, click HTTPS.
 - 3. Under Availability Zones, click your VPC.
 - 4. Check all availability zones. For each availability zone, click `pcf-public-subnet-az0` to allow traffic from public IP addresses, or `pcf-management-subnet-az0` to allow traffic only from private IP addresses.
 - 5. Click Next: Configure Security Settings.
- 5. For Step 2: Configure Security Settings, do the following:
 - 1. Under Select default certificate, do one of the following:
 - If you already have a certificate from AWS Certificate Manager (ACM), click Choose a certificate from ACM.
 - If you do not have a certificate from ACM, click Upload a certificate to ACM. For more information, see [Importing Certificates into AWS Certificate Manager](#) in the AWS documentation.

For a production or production-like environment, use a certificate from a Certificate Authority (CA). This can be an internal certificate or a purchased certificate. For a sandbox environment, you can use a self-signed certificate, leaving the Certificate chain entry blank.

2. Click Next: Configure Security Groups.
6. For Step 3: Configure Security Groups, do the following:
 1. Under Assign a security group, click Select an existing security group.
 2. From the list of security groups, click the `pcf-web-elb-security-group` security group that you configured in [Configure a Security Group for the Web ELB](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.
 3. Click Next: Configure Routing.
7. For Step 4: Configure Routing, do the following:
 1. Under Target Group, enter the following values:
 - Name: Enter `pcf-web-elb-target-group`.
 - Target type: Click Instance.
 - Protocol: Click HTTP.
 - Port: Enter `80`.
 2. Under Health checks, enter the following values:
 - Protocol: Click HTTP.
 - Path: Enter `/health`.
 3. Under Advanced health check settings, enter the following values:
 - Port: Click override and enter `8080`.
 - Healthy threshold: Enter `6`.
 - Unhealthy threshold: Enter `3`.
 - Timeout: Enter `3`.
 - Interval: Enter `5`.
 - Success codes: Enter `200`.
 4. Click Next: Register Targets.
8. For Step 5: Register Targets, accept the default values and click Next: Review.
9. For Step 6: Review, review the load balancer details and then click Create. A message appears to confirm that AWS has successfully created the load balancer.

Step 3: Create SSH load balancer

1. From the Load Balancers page, click Create Load Balancer.
2. Click Network Load Balancer.
3. For Step 1: Configure Load Balancer, do the following:
 1. Under Basic Configuration, do the following:
 - For Name, enter `pcf-ssh-elb`.
 - For Scheme, click internet-facing to allow traffic from public IP addresses, or internal to allow traffic only from private IP addresses.
 - For IP address type, click the type of IP addresses that you want to allow.

2. Under Listeners, edit the existing listener. For Load Balancer Protocol, click TCP, and for Load Balancer Port, enter **2222**.
3. Under Availability Zones, click your VPC.
4. Check all availability zones. For each availability zone, click **pcf-public-subnet-az0** to allow traffic from public IP addresses, or **pcf-management-subnet-az0** to allow traffic only from private IP addresses.
5. Click Next: Configure Security Settings.
4. On the Configure Security Settings page, ignore the Improve your load balancer's security error message.
5. Click Next: Configure Routing.
6. For Step 3: Configure Routing, do the following:
 1. Under Target Group, enter the following values:
 - Name: Enter **pcf-ssh-elb-target-group**.
 - Target type: Click Instance.
 - Protocol: Click TCP.
 - Port: Enter **2222**.
 2. Under Health checks, enter the following values:
 - Protocol: Click TCP.
 3. Under Advanced health check settings, enter the following values:
 - Port: Click traffic port.
 - Healthy threshold: Enter **6**.
 - Interval: Click **10 seconds**.
 4. Click Next: Register Targets.
7. For Step 4: Register Targets, accept the default values and click Next: Review.
8. For Step 5: Review, review the load balancer details and then click Create. A message appears to confirm that AWS has successfully created the load balancer.

Step 4: Create TCP load balancer

1. On the Load Balancers page, click Create Load Balancer.
2. Click Classic Load Balancer.
3. Define the load balancer with the following information:
 - Load Balancer name: Enter **pcf-tcp-elb**.
 - Create LB Inside: Click the **pcf-vpc** VPC that you created in [Create a VPC in Preparing to Deploy Tanzu Operations Manager on AWS](#).
 - If you want to allow traffic from public IP addresses, ensure that the Create an internal load balancer check box is not selected. If you want to allow traffic only from private IP addresses, you must click this check box.
4. Under Listener Configuration, add the following rules:

Load balancer protocol	Load balancer port	Instance protocol	Instance port
TCP	1024	TCP	1024
TCP	1025	TCP	1025
TCP	1026	TCP	1026
...	...	...	...
TCP	1123	TCP	1123

The ... entry indicates that you must add listening rules for each port between 1026 and 1123.

- Under Select Subnets, click either the public or private subnets you configured in [Create a VPC](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*, and click Next: Assign Security Groups.
- On the Assign Security Groups page, click the security group `pcf-tcp-elb-security-group` that you configured in [Configure a Security Group for the TCP ELB](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*, and click Next: Configure Security Settings.
- On the Configure Security Settings page, ignore the Improve your load balancer's security error message.
- Click Next: Configure Health Check.
- On the Configure Health Check page, enter the following values:
 - Ping Protocol: Click `TCP`.
 - Ping Port: Set to `80`.
 - Response Timeout: Set to `3` seconds.
 - Interval: Set to `5` seconds.
 - Unhealthy threshold: Set to `3`.
 - Health threshold: Set to `6`.
- Click Next: Add EC2 Instances.
- Accept the defaults on the Add EC2 Instances page, and click Next: Add Tags.
- Accept the defaults on the Add Tags page, and click Review and Create.
- Review and confirm the load balancer details, and click Create.

Step 5: Configure DNS records

- Perform the following steps for all three of the load balancers you created in previous steps, named `pcf-web-elb`, `pcf-ssh-elb`, and `pcf-tcp-elb`:
 - From the Load Balancers page, click the load balancer.
 - On the Description tab, locate the Basic Configuration section and record the DNS name of the load balancer.
- Click Instances on the navigational menu to view your EC2 instances.
- Click the `pcf-ops-manager` instance created in [Step 1: Launch an Tanzu Operations Manager AMI](#).
- On the Description tab, record the value for IPv4 Public IP.
- Go to your DNS provider and create the following `CNAME` and `A` records:

- CNAME: *.apps.DOMAIN.com and *.system.DOMAIN.com point to the DNS name of the pcf-web-elb load balancer.
- CNAME: ssh.system.DOMAIN.com points to the DNS name of the pcf-ssh-elb load balancer.
- CNAME: tcp.system.DOMAIN.com points to the DNS name of the pcf-tcp-elb load balancer.
- A: pcf.DOMAIN.com points to the public IP address of the pcf-ops-manager EC2 instance.

Where DOMAIN is a domain name. VMware recommends that you use the same domain name for each record.

6. Click Assign Security Groups.

Step 6: Create RDS subnet group

Perform the following steps to create a RDS Subnet Group for the two RDS subnets.

1. Go to the RDS Dashboard.
2. Click Subnet Groups, then Create DB Subnet Group.
3. Enter the following values:
 - Name: Enter pcf-rds-subnet-group.
 - Description: Enter a description to identify this subnet group.
 - VPC ID: Click pcf-vpc.
 - Availability Zone and Subnet ID: Choose the AZ and subnet for pcf-rds-subnet-az0 and click Add.
4. Repeat the preceding steps to add pcf-rds-subnet-az1 and pcf-rds-subnet-az2 to the group.
5. Click Create. On the Subnet Group page, you might need to refresh the page to view the new group.

Step 7: Create a MySQL database using AWS RDS

You must have an empty MySQL database when you install or reinstall Tanzu Operations Manager on AWS.

1. Go to the RDS Dashboard.
2. Click Create database to launch the wizard.
3. Under Engine type, click MySQL.
4. Under Templates, click Production to create a database for production environments.
5. Specify the following database details:
 - Multi-AZ deployment: Click Multi-AZ DB instance.
 - DB cluster identifier: Enter pcf-ops-manager-director.
 - Enter a secure Master username and Master password. Record the user name and password. You will use these credentials later when configuring the Director Config page in the BOSH Director tile.
 - DB instance class: Click db.m5.large - 2 vCPUs, 8 GiB RAM.
 - Storage type: Click Provisioned IOPS SSD (io1).

- Allocated storage: Enter 100 GiB.
6. In the Connectivity section, enter the following values:
 - Virtual private cloud (VPC): Click [pcf-vpc](#).
 - DB subnet group: Click the [pcf-rds-subnet-group](#) that you created in [Step 6: Create RDS Subnet Group](#).
 - Public access: Click No.
 - VPC security group (firewall): Click the [pcf-mysql-security-group](#) that you created in [Configure a Security Group for MySQL in Preparing to Deploy Tanzu Operations Manager on AWS](#).
 - Accept the default values for the remaining fields.
 7. In the Additional configuration section, enter the following values:
 - Initial database name: Enter [bosh](#).
 - Accept the default values for the remaining fields.
 8. Click Create database. Creating the database might take several minutes.

Next steps

When the instance has run, do the following:

- Review [Required AWS Objects](#) to verify that you created the correct resources in AWS.
- Proceed to [Configuring BOSH Director on AWS](#) to continue deploying Tanzu Operations Manager.

Configuring BOSH Director on AWS

You can configure BOSH Director to deploy Tanzu Operations Manager on Amazon Web Services (AWS).

Before you begin this procedure, ensure that you have followed all of the steps in [Deploying Tanzu Operations Manager on AWS](#).

Step 1: Log in to Tanzu Operations Manager

1. In a web browser, go to the fully qualified domain that you created in the [Configure DNS Records](#) step of *Deploying Tanzu Operations Manager on AWS*.
2. Log in to Tanzu Operations Manager. To log in, see [Log In to Tanzu Operations Manager for the First Time](#).

Step 2: Configure AWS

1. Click the BOSH Director tile.
2. Click AWS Config. The AWS Management Console Config page appears.

AWS Management Console Config

☐ Use AWS Keys

Access Key ID*

AWS Secret Key*

☒ Use AWS Instance Profile

AWS IAM Instance Profile*

Security Group ID*

Key Pair Name*

SSH Private Key*

Region*

VM Type Catalog

AWS EBS disk type*

Changing this value will force BOSH to recreate all VMs

☐ Encrypt Linux EBS Volumes

Custom Encryption Key

Save

3. Click Use AWS Keys or Use AWS Instance Profile.

- Access Key ID and AWS Secret Key: To retrieve your AWS key information, use the AWS Identity and Access Management (IAM) credentials that you generated in [Step 3: Create an IAM User for Tanzu Operations Manager in Preparing to Deploy Tanzu Operations Manager on AWS](#).
- AWS IAM Instance Profile: Enter the name of the IAM profile you created in [Step 3: Create an IAM User for Tanzu Operations Manager in Preparing to Deploy Tanzu Operations Manager on AWS](#).

4. Complete the rest of the AWS Management Console Config page:

- Security Group ID: Enter the Group ID of the `pcf-vms-security-group` you created for your Tanzu Operations Manager VMs in the [Configure a Security Group for Tanzu Operations Manager VMs](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*. Locate the Group ID in the Security Groups tab of your EC2 Dashboard.
- Key Pair Name: Enter `pcf-ops-manager-key`.
- SSH Private Key: Open the AWS key pair `pcf-ops-manager-keys.pem` file you generated in [Step 3: Create an IAM User for Tanzu Operations Manager in Preparing to Deploy Tanzu Operations Manager on AWS](#). Copy the contents of the `.pem` file and paste it into the SSH Private Key text box.
- Region: Click the region in which you deployed Tanzu Operations Manager.
- VM Type Catalog Click the VM catalog that you want to use for your BOSH-deployed VMs.
- AWS EBS disk type: Click the type of EBS volume for BOSH deployed VMs to use.

If you change this value, BOSH recreates all VMs when you Apply Changes.

- Encrypt Linux EBS Volumes: Click this check box to enable full encryption on persistent disks of all BOSH deployed VMs except the Tanzu Operations Manager VM and BOSH Director VM.

For more information about using EBS encryption, see [Configuring Amazon EBS Encryption for Tanzu Operations Manager on AWS](#).

Enabling EBS encryption only encrypts Linux VMs. The Windows VMs deployed with Elastic Application Runtime Windows add-on or Elastic Application Runtime Windows add-on 2012 R2 are not encrypted.

5. Click Save.

Step 3: Complete the Director Config page

To configure the Director Config pane:

1. Click Director Config.

Director Config

Go to
[Certificate Duration Overrides](#)
[Health Monitor](#)
[CredHub Encryption Provider](#)
[Blobstore Location](#)
[Database Location](#)

NTP Servers (comma delimited)*

One or more NTP server addresses for consistent and valid time stamps used in syslog, etc.

Bosh HM Forwarder IP Address

Should only be used if OS Bosh HM Forwarder job is deployed

☐ Enable VM Resurrector Plugin

☒ Enable System Metrics

☐ Enable Post Deploy Scripts

☐ Enable Short Lived NATS Bootstrap Credentials
This will make the Bosh Director rotate the NATs bootstrap credentials of a VM. This increases security by invalidating the NATS credentials that are stored in the IAAS metadata service.

☐ Recreate VMs deployed by the BOSH Director
This will force BOSH to recreate all Bosh Deployed VMs on the next deploy. This does not recreate the BOSH Director. Persistent disk will be preserved.

☐ Recreate BOSH Director VM
This will force BOSH to recreate the Bosh Director VM on the next deploy. Persistent disk will be preserved.

☐ Recreate All Persistent Disks
Check this box to recreate all Persistent Disks for the Director and all other Tiles.

☐ Enable bosh deploy retries
This will attempt to re-deploy a failed deployment up to 5 times.

- The NTP server configuration is updated only after VM recreation. If you modify the value of this text box, select the Recreate VMs deployed by the BOSH Director check box to re-create your BOSH Director-deployed VMs and update the NTP server configuration. If you have any service tiles installed, ensure that the Recreate All Service Instances errand runs for each service tile.
2. Leave the Bosh HM Forwarder IP Address text box blank. BOSH-reported component metrics are available in Loggregator Firehose by default. If you continue to use the BOSH HM Forwarder to consume these component metrics, you might receive duplicate data.
3. Select the Enable VM Resurrector Plugin check box to enable the BOSH Resurrector functionality and increase your runtime availability.

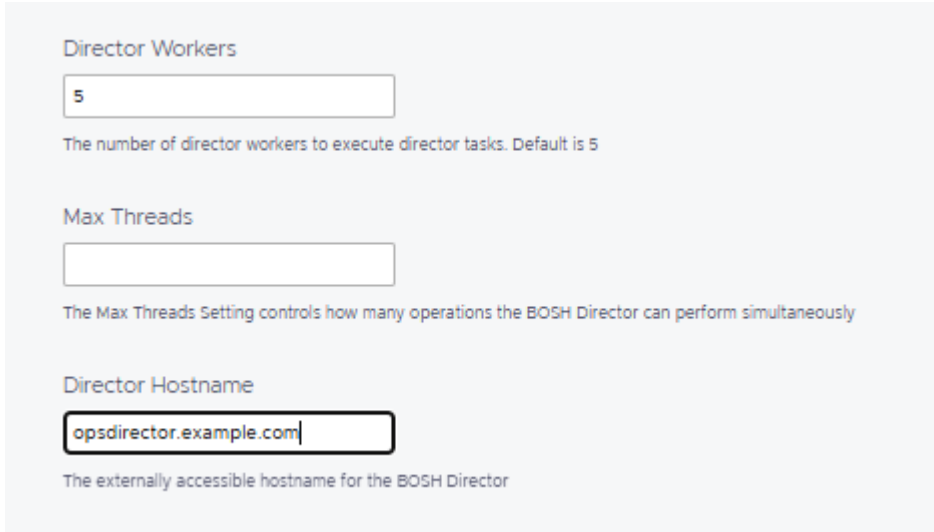
4. If you want to collect detailed metrics from BOSH-managed VMs, select the Enable System Metrics check box. For a list of metrics that the System Metrics Agent collects from BOSH-managed VMs, see [System Metrics Agent](#) on GitHub. When you select this check box, ensure that the System Metrics Scraper is able to connect on port **53035** to all BOSH-managed VMs.
5. Select Enable Post Deploy Scripts to run a post-deployment script after deployment. This script allows the job to execute additional commands against a deployment.
If you intend to install VMware Tanzu Container Orchestration Engine (formerly TKGI), you must enable post-deploy scripts.
6. To configure your VMs to use short-lived NATS credentials during the bootstrap process, select the Enable Short Lived NATS Bootstrap Credentials check box. This increases the security of your Tanzu Operations Manager deployment by rotating the TLS certificate that the BOSH Agents in your VMs use to communicate with the BOSH Director through BOSH NATS.
For more information, see [Using Short-Lived NATS Bootstrap credentials](#).

Configuring your VMs to use short-lived NATS bootstrap credentials restarts the BOSH Agent in all VMs as they are being created, so VMs are created somewhat more slowly.

This configuration affects all new deployments of Tanzu Operations Manager. If you want to configure VMs in existing Tanzu Operations Manager deployments to use short-lived bootstrap credentials, you must recreate them.

7. Select Recreate VMs deployed by the BOSH Director to force BOSH to recreate BOSH-deployed VMs on the next deployment. This process does not recreate the BOSH Director VM or destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
8. Select Recreate BOSH Director VM to force the BOSH Director VM to be recreated on the next deploy. This process does not destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
9. Select Recreate All Persistent Disks to force BOSH to migrate and recreate persistent disks for the BOSH Director and all tiles. This process does not destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
10. Select Enable bosh deploy retries to instruct Tanzu Operations Manager to retry failed BOSH operations up to five times.
11. Select Skip Director Drain Lifecycle to prevent drain scripts from running when the BOSH Director is recreated.
12. Select Store BOSH Job Credentials on tmpfs (beta) to store credentials for BOSH jobs on temporary file storage (tmpfs) memory, rather than on disk. You must re-create all BOSH-deployed VMs for this setting to take effect.
13. Select Keep Unreachable Director VMs to preserve BOSH Director VMs after a failed deployment for troubleshooting purposes.
14. (Optional) Select Enable automatic rotation of the BOSH DNS CA certificate (experimental) if you want Tanzu Operations Manager to automatically rotate the BOSH DNS CA certificate and associated leaf certificates when you click Apply Changes. For more information about this feature, see [Automatic rotation of BOSH DNS certificates](#).
15. (Optional) Modify the Director Workers value, which sets the number of workers available to run BOSH Director tasks. The default value is **5**.

16. (Optional) Max Threads sets the maximum number of threads that the BOSH Director can run simultaneously. VMware recommends that you leave the text box blank to use the default value, unless doing so results in rate limiting or errors on your IaaS.
17. (Optional) To add a custom URL for your BOSH Director, enter a valid hostname in the Director Hostname text box. You can also use this text box to configure a load balancer in front of your BOSH Director. For more information, see [How to set up a Load Balancer in front of Tanzu Operations Manager](#).



The screenshot shows a configuration panel for the BOSH Director. It contains three sections, each with a title, a text input field, and a descriptive subtitle.

- Director Workers:** The input field contains the number '5'. The subtitle reads: 'The number of director workers to execute director tasks. Default is 5'.
- Max Threads:** The input field is empty. The subtitle reads: 'The Max Threads Setting controls how many operations the BOSH Director can perform simultaneously'.
- Director Hostname:** The input field contains the text 'opsdirector.example.com'. The subtitle reads: 'The externally accessible hostname for the BOSH Director'.

18. (Optional) To set a custom banner that users see when you log in to the BOSH Director using SSH, enter text in the Custom SSH Banner text box.
19. (Optional) Enter your comma-separated custom Identification Tags. For example, `iaas:foundation1,hello:world`. You can use the tags to identify your foundation when viewing VMs or disks from your IaaS.
20. For Certificate Duration Overrides, you can choose whether certificates generated within Tanzu Operations Manager and CredHub use the default duration specified for the certificate, or a custom duration value that you specify.
 - Off: Use this option to use default duration for certificates created by all products.
 - On: Use this option to set a value to override the duration for certificates created by all products. If a product creates a certificate with a longer duration than the value you set, the longer duration is used.
 1. CA Certificate Duration (days): Enter the number of days for which CA certificates are valid.
 2. Leaf Certificate Duration (days): Enter the number of days for which leaf certificates are valid. This value must be less than or equal to the CA certificate duration.After you set a certificate duration override, you must take additional steps to apply the setting to all certificates. For more information, see [Overriding Duration for Certificates](#).

Certificate Duration Overrides

Set overrides for CA and leaf certificate durations.
For detailed information about which certificates are affected by this setting and which are not, see [Override Certificate Duration in the TAS for VMs or TKGi documentation](#).

☐ Off
☒ On

Enter values to set the duration for CA and leaf certificates in the fields below.
These values will be applied when new certificates are generated via new product deployment or certificate rotation.

CA Certificate Duration (days)*

1460

Enter a value between 365 and 3650 days

Leaf Certificate Duration (days)*

730

Leaf Certificate Duration must be less than or equal to the CA Certificate Duration.

21. Select HM Pager Duty Plugin to enable Health Monitor integration with PagerDuty.

- Service Key: Enter your API service key from PagerDuty.
- HTTP Proxy: Enter an HTTP proxy for use with PagerDuty.

☒ HM Pager Duty Plugin

Service Key*

YOUR_PAGERDUTY-SERVICE-KEY

HTTP Proxy

YOUR-HTTP-PROXY

22. Select HM Email Plugin to enable Health Monitor integration with email.

- Host: Enter your email host name.
- Port: Enter your email port number.
- Domain: Enter your domain.
- From: Enter the sender's address.
- Recipients: Enter a comma-separated list of addresses of intended recipients.
- Username: Enter the user name for your email server.
- Password: Enter the password for your email server.
- Enable TLS: Select this check box to enable Transport Layer Security to the email host.

☒ HM Email Plugin

Host*
smtp.example.com

Port*
25

Domain*
cloudfoundry.example.com

From*
user2.example.com

Recipients*
user2@example.com, user1.exempl

Username
user

Password
.....

☒ Enable TLS

23. For CredHub Encryption Provider, you can choose whether BOSH CredHub stores its encryption key internally on the BOSH Director and CredHub VM, or in an external hardware security module (HSM). The HSM option is more secure.

Before configuring an HSM encryption provider in the Director Config pane, you must follow the procedures and collect information described in [Preparing CredHub HSMs for configuration](#). After you deploy Tanzu Operations Manager with an HSM encryption provider, you cannot change BOSH CredHub to store encryption keys internally.

- Internal: Use this option for internal CredHub key storage. This option is selected by default and requires no additional configuration.
- Luna HSM: Use this option to use a SafeNet Luna HSM as your permanent CredHub encryption provider, and fill in the following fields:

If you use multiple HSM hosts, you must use the Tanzu Operations Manager API to set or update the HSM configuration. For more information, see [Updating a staged director's properties](#) in the Tanzu Operations Manager API documentation.

1. Encryption Key Name: Any name to identify the key that the HSM uses to encrypt and decrypt the CredHub data. Changing this key name after you deploy Tanzu Operations Manager can cause service downtime.
2. Provider Partition: The partition that stores your encryption key. Changing this partition after you deploy Tanzu Operations Manager can cause service downtime. For this value and the ones that follow, use values gathered in [Preparing CredHub HSMs for configuration](#).

3. Provider Partition Password
4. Partition Serial Number
5. Provider Client Certificate: The certificate that validates the identity of the HSM when CredHub connects as a client.
6. Provider Client Certificate Private Key
7. HSM Host Address
8. HSM Port Address: If you do not know your port address, enter 1792.
9. HSM Certificate: The certificate that the HSM presents to CredHub to establish a two-way mTLS connection.

CredHub Encryption Provider

- ☐ Internal
- ☒ Luna HSM

Encryption Key Name*

Provider Partition*

Provider Partition Password*

Partition Serial Number*

Provider Client Certificate*

Provider Client Certificate Private Key*

HSM Host Address*

HSM Port Address*



24. Click Blobstore Location to configure the blobstore as either an internal server or an external endpoint.

After you deploy the BOSH Director, only Tanzu Operations Manager users in Advanced Mode can change the blobstore location. For more information, see [Platform operator roles and permissions in Tanzu Operations Manager](#).

- Internal: Use this option for an internal blobstore. No additional configuration is required.
- S3 Compatible Blobstore: Use this option for an external S3-compatible endpoint. To create an S3 bucket, follow the procedures in [Sign up for Amazon S3](#) and [Creating a Bucket](#) in the AWS documentation. When you have created an S3 bucket, configure the following fields:

1. S3 Endpoint:

1. If you are using a public S3 endpoint:

1. Locate the endpoint for your region. To find the endpoint, see the Amazon Simple Storage Service (S3) table in [Regions and Endpoints](#) in the AWS documentation.
 2. Construct a URL using your region's endpoint. For example, if you are using the [us-west-2](#) region, the URL you create would be <https://s3.us-west-2.amazonaws.com>. Enter this URL in the S3 Endpoint field.

2. If you are using a non-public S3-compatible endpoint:

1. Enter the URL for the non-public endpoint.
 2. Add the custom public CA certificate you used to sign the S3 endpoint to the Trusted Certificates field on the Director Config page.

2. Bucket Name: Enter the name of the S3 bucket.

3. Access Key and Secret Key: Enter the keys you generated when creating your S3 bucket.

4. Enable signed URLs: Select this check box to configure BOSH VMs to generate short lived, pre-signed URLs for communication between the BOSH Agent and the blobstore. If you enable this feature, BOSH Agents do not use credentials to communicate with the blobstore, and VM disks do not store blobstore credentials. This feature is available only for BOSH VMs that use Xenial stemcell line 621 or later.

5. Select V2 Signature or V4 Signature. If you select V4 Signature, enter your Region. AWS recommends using Signature Version 4 (V4). For more information

about AWS S3 Signatures, see [Authenticating Requests](#) in the AWS documentation.

6. S3 Backup Strategy: To configure whether and how to back up the BOSH Director's S3 blobstore, select one of the following options:

- No backups
- Use a versioned bucket: This option uses blobstore bucket versioning to save backups. It requires an S3 blobstore that supports versioning.
- Copy into an additional bucket: This option saves blobstore backups to a separate bucket, which is useful with blobstores that do not support versioning. This option requires you to configure the Backup Bucket Region and Backup Bucket Name fields.

To backup and restore with BBR, if you are using S3 Compatible Blobstore and you have set up an [iam_instance_policy](#), additional steps are needed:

1. Attach the [iam_instance_policy](#) to the [backup_restore](#) job using the Tanzu Operations Manager API. See [Custom VM Extensions](#).
2. In addition to creating backup buckets, update the Elastic Application Runtime tile (Settings > File Storage) to use S3-compatible storage. See [External S3-Compatible filestore](#).
3. Ensure that your IAM Instance Policy is set up to access the backup bucket. For more information about the IAM Instance Policy, see the [Tanzu Operations Manager for AWS policy document](#).

- GCS Blobstore: Use this option for an external Google Cloud Storage (GCS) endpoint.

To create a GCS bucket, follow the procedures in [Creating Storage Buckets](#) in the GCS documentation. To create a GCS bucket, you must have a GCS account. When you have created a GCS bucket, configure the following fields:

1. Bucket Name: Enter the name of your GCS bucket.
2. Storage Class: Use this storage class for your GCS bucket. For more information, see [Storage Classes](#) in the GCP documentation.
3. Service Account Key: Follow the steps in [Set up IAM Service Accounts](#) in *Preparing to Deploy Tanzu Operations Manager on GCP* to download a JSON file with a private key. Enter the contents of the JSON file into the text box.
4. GCS Backup Strategy: To configure whether and how to back up the BOSH Director's GCS blobstore, click one of the following options:
 - No backups
 - Copy into an additional bucket: This option saves blobstore backups to a bucket separate from the blobstore itself, and requires you to configure the Backup Bucket Name text box.

Blobstore Location

- ☐ Internal
- ☐ S3 Compatible Blobstore
- ☒ GCS Blobstore

Bucket Name*

Storage Class*

Regional

Service Account Key*

Paste the JSON file containing the private key for your GCP service account.

GCS Backup Strategy

- ☒ No backups
- ☐ Copy into an additional bucket

Backup Bucket Name*

25. Click Save.

Step 4: Create Availability Zones

VMware recommends at least three availability zones (AZs) for a highly available installation of Tanzu Platform for Cloud Foundry. The procedures in [Preparing to Deploy Tanzu Operations Manager on AWS](#) use 3 AZs.

1. Click Create Availability Zones.
2. To add the three AZs you specified in the [Create a VPC](#) section of *Preparing to Deploy Tanzu Operations Manager on AWS*:
 1. Click Add.
 2. For Amazon Availability Zone, enter the name of the AZ.
 3. Repeat until you have entered all three AZs, in the format **REGION-#a**, **REGION-#b**, and **REGION-#c**. For example, **us-west-2a**, **us-west-2b**, and **us-west-2c**.
3. Click Save.

Create Availability Zones

Availability Zones*

+ us-central-1-a

Amazon Availability Zone*

SAVE

Step 5: Create Networks

1. Click Create Networks.

Create Networks

Warning: Tanzu Ops Manager recommends keeping the IP settings throughout the life of your installation. Tanzu Ops Manager may prevent you from changing them in the future. Contact VMware support for help completing such a change.

Verification Settings

☐ Enable ICMP checks

Networks

One or many IP ranges upon which your products will be deployed

+ first-network

management

Name*

management

A unique name for this network

Subnets*

Add Network

Add Subnet

2. (Optional) Click Enable ICMP checks to enable ICMP on your networks. Tanzu Operations Manager uses ICMP checks to confirm that components within your network are reachable.
3. Perform the following steps to add the network configuration that you created for your VPC in the [Create a VPC](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*. Record your VPC CIDR if you set a CIDR other than the recommended one.
 1. Click Add Network.
 2. For Name, enter **management**.
 3. Create a subnet for each availability zone by clicking Add Subnet. See the following table for the information required to create all three subnets. When entering the VPC Subnet ID for each subnet, you must enter the corresponding Subnet ID associated with that subnet in the AWS VPC dashboard:

First Subnet	VPC subnet ID	Subnet ID for pcf-management-subnet-az0
	CIDR	10.0.16.0/28
	Reserved IP ranges	10.0.16.0-10.0.16.4
	DNS	10.0.0.2*
	Gateway	10.0.16.1
	Availability zones	REGION-#a . Example: us-west-2a

Second Subnet	VPC subnet ID	Subnet ID for <code>pcf-management-subnet-az1</code>
	CIDR	<code>10.0.16.16/28</code>
	Reserved IP ranges	<code>10.0.16.16-10.0.16.20</code>
	DNS	<code>10.0.0.2</code> *
	Gateway	<code>10.0.16.17</code>
	Availability zones	<code>REGION-#b</code> . Example: <code>us-west-2b</code>
Third Subnet	VPC subnet ID	Subnet ID for <code>pcf-management-subnet-az2</code>
	CIDR	<code>10.0.16.32/28</code>
	Reserved IP ranges	<code>10.0.16.32-10.0.16.36</code>
	DNS	<code>10.0.0.2</code> *
	Gateway	<code>10.0.16.33</code>
	Availability zones	<code>REGION-#c</code> . Example: <code>us-west-2c</code>

If you set a VPC CIDR other than recommended, enter the second IP in your VPC CIDR. For example, for a `10.0.0.0/24` VPC CIDR, enter `10.0.0.2` in each subnet.

- Click Add Network.
- For Name, enter a unique name.
- Create a subnet for each availability zone by clicking Add Subnet. For more information, see the following table for the information required to create all three subnets:

First Subnet	VPC subnet ID	Subnet ID for <code>pcf-tas-subnet-az0</code> or <code>pcf-pks-subnet-az0</code>
	CIDR	<code>10.0.4.0/24</code>
	Reserved IP ranges	<code>10.0.4.0-10.0.4.4</code>
	DNS	<code>10.0.0.2</code> *
	Gateway	<code>10.0.4.1</code>
	Availability zones	<code>REGION-#a</code> . Example: <code>us-west-2a</code>
Second Subnet	VPC subnet ID	Subnet ID for <code>pcf-tas-subnet-az1</code> or <code>pcf-pks-subnet-az1</code>
	CIDR	<code>10.0.5.0/24</code>
	Reserved IP ranges	<code>10.0.5.0-10.0.5.4</code>
	DNS	<code>10.0.0.2</code> *
	Gateway	<code>10.0.5.1</code>
	Availability zones	<code>REGION-#b</code> . Example: <code>us-west-2b</code>

Third Subnet	VPC subnet ID	Subnet ID for <code>pcf-tas-subnet-az2</code> or <code>pcf-pks-subnet-az2</code>
	CIDR	<code>10.0.6.0/24</code>
	Reserved IP ranges	<code>10.0.6.0-10.0.6.4</code>
	DNS	<code>10.0.0.2*</code>
	Gateway	<code>10.0.6.1</code>
	Availability zones	<code>REGION-#c</code> . Example: <code>us-west-2c</code>

If you set a VPC CIDR other than recommended, enter the second IP in your VPC CIDR. For example, for a `10.0.0.0/24` VPC CIDR, enter `10.0.0.2` in each subnet.

- Click Add Network.
- For Name, enter `services`.
- Create a subnet for each availability zone by clicking Add Subnet. See the following table for the information required to create all three subnets:

First Subnet	VPC subnet ID	Subnet ID for <code>pcf-services-subnet-az0</code>
	CIDR	<code>10.0.8.0/24</code>
	Reserved IP ranges	<code>10.0.8.0-10.0.8.3</code>
	DNS	<code>10.0.0.2*</code>
	Gateway	<code>10.0.8.1</code>
	Availability zones	<code>REGION-#a</code> . Example: <code>us-west-2a</code>
Second Subnet	VPC subnet ID	Subnet ID for <code>pcf-services-subnet-az1</code>
	CIDR	<code>10.0.9.0/24</code>
	Reserved IP ranges	<code>10.0.9.0-10.0.9.3</code>
	DNS	<code>10.0.0.2*</code>
	Gateway	<code>10.0.9.1</code>
	Availability zones	<code>REGION-#b</code> . Example: <code>us-west-2b</code>
Third Subnet	VPC subnet ID	Subnet ID for <code>pcf-services-subnet-az2</code>
	CIDR	<code>10.0.10.0/24</code>
	Reserved IP ranges	<code>10.0.10.0-10.0.10.3</code>
	DNS	<code>10.0.0.2*</code>
	Gateway	<code>10.0.10.1</code>
	Availability zones	<code>REGION-#c</code> . Example: <code>us-west-2c</code>

If you set a VPC CIDR other than recommended, enter the second IP in your VPC CIDR. For example, for a `10.0.0.0/24` VPC CIDR, enter `10.0.0.2` in each subnet.

4. Click Save.

If you are deploying VMware Tanzu Container Orchestration Engine (formerly TKGI) with a Container Orchestration Engine workload load balancer, you must tag each AWS subnet with your Container Orchestration Engine Kubernetes cluster unique identifier before you create the load balancer. For more information about tagging subnets with a Container Orchestration Engine cluster unique identifier, see [AWS Prerequisites](#).

After you deploy Tanzu Operations Manager, you add subnets with overlapping AZs to expand your network. For more information about configuring additional subnets, see [Expanding your Network with Additional Subnets](#).

Step 6: Assign AZs and Networks

1. Click Assign AZs and Networks.
2. Click a Singleton Availability Zone from the drop-down menu. The BOSH Director is deployed into this AZ.
3. Click **management** under Network from the drop-down menu. The BOSH Director is deployed into this network.
4. Click Save.

Assign AZs and Networks

The BOSH Director is a single instance.

Choose the availability zone in which to place that instance. It is highly recommended that you backup this VM on a regular basis to preserve settings.

Singleton Availability Zone*

us-central1-a

Network*

management

Save

Step 7: Configure Security

1. Click Security.

Security

Trusted Certificates

These certificates enable BOSH-deployed components to trust a custom root certificate.

☐ Include Tanzu Ops Manager Root CA in Trusted Certs
Checking this box will place the Tanzu Ops Manager Root CA into the trusted certificate field. The Bosh Director will then place this CA into the trust store of every VM that it deploys.

☐ Clear the Default Trusted Certificates Store
Deletes all the certificates in the system certificate trust store every time BOSH deploys a VM. Certificates added in the 'Trusted Certificates' field will still be trusted and you should add the certificates required to access your IaaS Provider services.

Generate VM passwords or use single password for all VMs

☒ Generate passwords

☐ Use default BOSH password

SAVE

2. In Trusted Certificates, paste in your custom certificate authority (CA) certificates to insert into your organization's certificate trust chain. This feature enables all BOSH-deployed components in your deployment to trust custom root certificates. If you want to use Docker registries to run Elastic Application Runtime app instances in Docker containers, enter the certificate for your private Docker registry in this text box. For more information about running app instances in Elastic Application Runtime using Docker registries, see [Using Docker Registries](#).

To enter multiple certificates, paste in your certificates one after the other. For example, format your certificates as shown in the following example:

```
-----BEGIN CERTIFICATE-----
ABCDEF12345678ABCDEF12345678ABCDEF12345678AB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
BCDEF12345678ABCDEF12345678ABCDEF12345678ABB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
CDEF12345678ABCDEF12345678ABCDEF12345678ABBB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
```

3. To include both the Tanzu Operations Manager root CA and the certificates pasted into Trusted Certificates in the `trusted_certs` field in the BOSH director manifest, select the Include Tanzu Ops Manager Root CA in Trusted Certs check box. BOSH Director includes this CA in the trust store of every VM that it deploys.
- You must select this check box if you are using Tanzu Operations Manager to generate certificates for a load balancer or a router.
4. To clear the default trusted certificates from all BOSH-deployed Linux VMs, select Clear the Default Trusted Certificates Store.

If you select this check box, you must provide your own trusted certificates. All TLS communication fails if trusted certificates are not available.

5. Select Generate passwords or Use default BOSH password. VMware recommends using the Generate passwords option for greater security.
6. Click Save. To view your saved BOSH Director password, click the Credentials tab.

Step 8: Configure DNS for BOSH

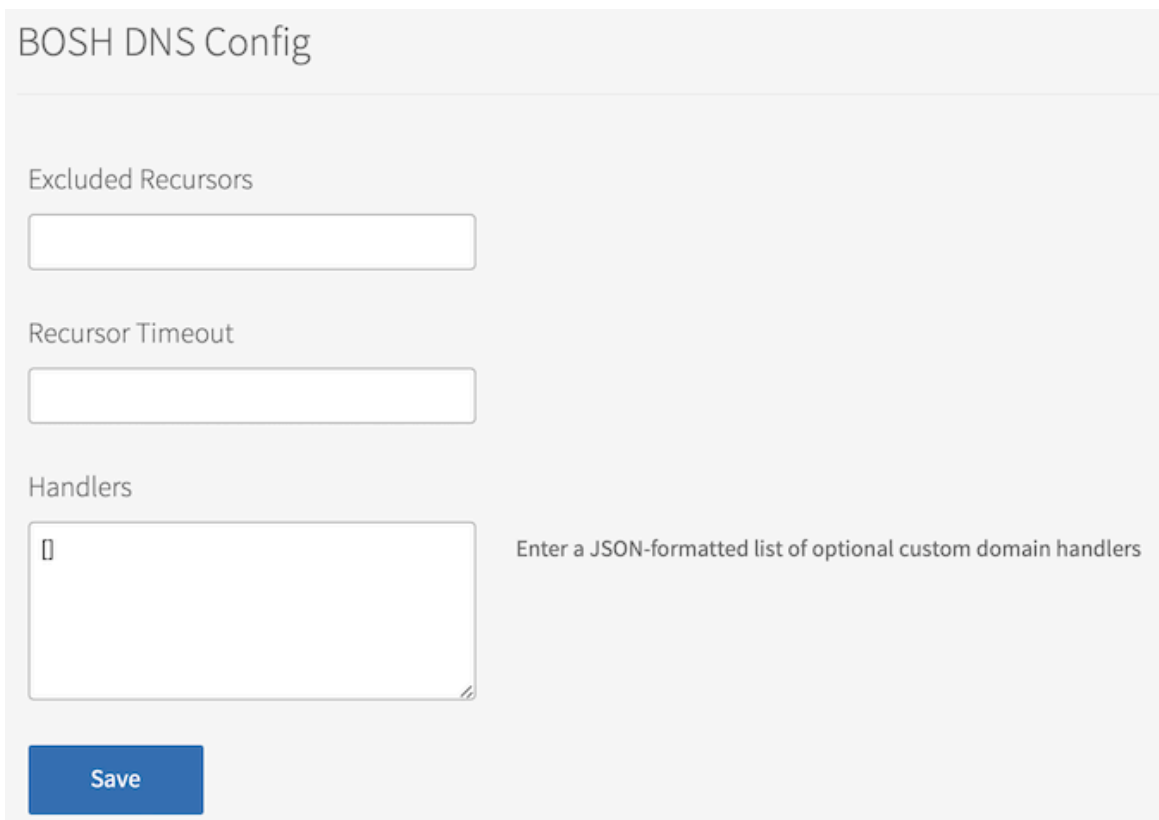
In the BOSH DNS Config pane, you can configure DNS for BOSH Director by adding excluded recursors, a recursor timeout, and handlers.

The text boxes in the BOSH DNS Config pane are:

- Excluded Recursors: Exclude recursor addresses, which are URL redirects, so that they are not contacted by the BOSH DNS server. For more information about how the BOSH DNS release selects recursors, see [Recursors](#) in *Native DNS Support* in the BOSH documentation.
- Recursor Timeout: Specify a timeout for the BOSH DNS server when contacting any connected recursor addresses. The time limit includes dialing, writing, and reading from the recursor. If any of these actions exceeds the time limit, the action fails.
- Handlers: Specify recursor addresses that apply to specific domains. For example, you can use handlers to forward all requests to a domain to a private DNS for resolution. For more information about using handlers, see [Additional handlers](#) in *Native DNS Support* in the BOSH documentation.

To add excluded recursors, a recursor timeout, or handlers to the BOSH DNS release:

1. Click BOSH DNS Config.



The screenshot shows the 'BOSH DNS Config' pane. It contains three input fields: 'Excluded Recursors' (a single-line text box), 'Recursor Timeout' (a single-line text box), and 'Handlers' (a multi-line text box containing an empty JSON array '[]'). To the right of the 'Handlers' field is the instruction 'Enter a JSON-formatted list of optional custom domain handlers'. At the bottom left of the pane is a blue 'Save' button.

2. (Optional) In Excluded Recursors, enter a list of prohibited recursor addresses.

3. (Optional) In Recursor Timeout, enter a time limit for contacting the connected recursors. This time limit must include one of the Go parse duration time units. For example, entering **5s** sets the timeout limit to five seconds. For more information about supported time units, see [func ParseDuration](#) in the Go Programming Language documentation.
4. (Optional) In Handlers, enter an array of custom domain handlers in JSON format. For example:

```
[
  {
    "cache": {
      "enabled": true
    },
    "domain": "example.com",
    "source": {
      "type": "http",
      "url": "http://example.endpoint.local"
    }
  }
]
```

5. Click Save.

Step 9: Configure Syslog

1. Click Syslog.
2. (Optional) To send BOSH Director system logs to a remote server, select Yes.
3. In Address, enter the IP address or DNS name for the remote server.
4. In Port, enter the port number that the remote server listens on.
5. Using the Transport Protocol drop-down menu, select TCP or UDP to specify the transport protocol to use to send the logs to the remote server.
6. (Optional) Select the Enable TLS check box to send encrypted logs to remote server with TLS. After you select the check box:
 1. In Permitted Peer, enter either the name or SHA1 fingerprint of the remote peer.
 2. In SSL Certificate, enter the SSL certificate for the remote server.

VMware recommends that you enable TLS encryption when you are forwarding logs. Logs can contain sensitive information, such as cloud provider credentials.

7. (Optional) In Queue Size, enter an integer that specifies the number of log entries held in the buffer. The default value is 100,000.
8. (Optional) Select the Forward Debug Logs check box to forward the logs to an external source. This option is deselected by default. If you select it, you might generate a large amount of log data.
9. (Optional) In the Custom rsyslog Configuration text box, enter the configuration details for **rsyslog**. This text box requires the rainerscript syntax.
10. Click Save.

Step 10: Configure Resources for BOSH Director

1. Click Resource Config.

JOB	INSTANCES	VM TYPE	PERSISTENT DISK TYPE
> BOSH Director	Automatic: 1	Automatic: e2-standard-2 disk (machine_type: e2-standard-2, cpu: 2, ram: 8 GB, disk: ...)	Automatic: 50 GB
> Main Compilation Job	1	e2-highmem-2 (machine_type: e2-highmem-2, cpu: 2, ram: 16 GB, disk: 64 GB)	None

2. Adjust any values as necessary for your deployment. Under the Instances, VM Type, and Persistent Disk Type fields, and select Automatic from the drop-down menu to allocate the recommended resources for the job. If the Persistent Disk Type field reads None, the job does not require persistent disk space.
 - For the BOSH Director job, click a VM type with at least 8 GB memory.
 - If you set a field to Automatic and the recommended resource allocation changes in a future version, Tanzu Operations Manager automatically uses the new recommended allocation.
 - If you install Elastic Application Runtime Windows add-on, provision your Main Compilation Job with at least 100 GB of disk space.
3. (Optional) To configure an Application Load Balancer (ALB) for your BOSH Director, click the icon next to the BOSH Director job name to see more fields. In the Load Balancers field for the BOSH Director job, enter your AWS target group name. Prepend the name with **alb:**. For example, enter **alb:target-group-name**.

To create an ALB and target group, follow the procedures in [Getting Started with Application Load Balancers](#) in the AWS documentation. Then go to Target Groups in the EC2 Dashboard menu to find your target group Name.

To configure an ALB, you must have the following AWS IAM permissions.

```
"elasticloadbalancing:DescribeLoadBalancers",
"elasticloadbalancing:DeregisterInstancesFromLoadBalancer",
"elasticloadbalancing:RegisterInstancesWithLoadBalancer",
"elasticloadbalancing:DescribeTargetGroups",
"elasticloadbalancing:RegisterTargets"
```

4. Click Save.

Step 11: Complete the BOSH Director installation

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes.
3. Click Apply Changes.
4. The BOSH Director installation begins. The Changes Applied dialog box appears when the installation is complete.

Next steps

After you have configured the BOSH director on AWS, you can install one of the runtime environments. For more information, see [Tanzu Operations Manager runtimes](#).

Upgrading BOSH Director for Tanzu Operations Manager on AWS

This topic describes how you can upgrade BOSH Director for Tanzu Operations Manager on Amazon Web Services (AWS). Perform these tasks as part of the Tanzu Operations Manager upgrade process.

For information about upgrading Tanzu Operations Manager, see [Task 2: Upgrade Tanzu Operations Manager and installed products to v3.2](#) in *Upgrading Tanzu Operations Manager*.

In this procedure, you create a Tanzu Operations Manager VM instance to host the upgraded version of Tanzu Operations Manager. Then, to complete the Tanzu Operations Manager upgrade, you export your existing Tanzu Operations Manager installation onto this new VM.

For Tanzu Operations Manager installations on AWS, the VM that hosts the new version of Tanzu Operations Manager uses an Amazon Machine Image (AMI) specific to your region.

To create a Tanzu Operations Manager VM instance:

1. Retrieve the Tanzu Operations Manager AMI ID. See [Retrieve Tanzu Operations Manager AMI ID](#).
2. Launch the AMI. See [Launch Tanzu Operations Manager AMI](#).
3. Edit the Tanzu Operations Manager DNS A Record. See [Edit Tanzu Operations Manager DNS A Record](#).

Prerequisites

To complete the Tanzu Operations Manager upgrade, you must have your Tanzu Operations Manager decryption passphrase. You defined this decryption passphrase during the initial installation of Tanzu Operations Manager.

Retrieve Tanzu Operations Manager AMI ID

Tanzu Operations Manager has a different AMI ID for each region. Retrieve and record the AMI ID for your region. You must use this AMI ID when launching the Tanzu Operations Manager AMI instance.

To retrieve the AMI ID for Tanzu Operations Manager:

1. Go to the [Tanzu Operations Manager](#) page on the Broadcom Support portal.
2. Select the version of Tanzu Operations Manager you want to install from the Release list.
3. In the Release Download Files, click the file named Tanzu Operations Manager for AWS to download a PDF.
4. Open the PDF and record the AMI ID for your region.

Launch Tanzu Operations Manager AMI

Use the AMI ID for your region to launch an AMI instance that hosts the new version of Tanzu Operations Manager.

To launch the AMI for Tanzu Operations Manager:

1. Go to your EC2 Dashboard.
2. In the Images menu, click AMIs.
3. In the Owned by me drop-down filter, click Public images.
4. Paste the AMI ID for your region into the search bar and press enter.

There is a different AMI for each region. If you cannot locate the AMI for your region, verify that you have set your AWS Management Console to your desired region. If you still cannot locate the AMI, contact [Support](#).

5. (Optional) If you want to encrypt the VM that runs Tanzu Operations Manager with AWS Key Management Service (KMS):
 1. Right-click the row that lists your AMI and click Copy AMI.
 2. Select your Destination region.
 3. Enable Encryption. For more information about AMI encryption, see [Encryption and AMI Copy](#) in *Copying an AMI* in the AWS documentation.
 4. Select your KMS Key. To create a new custom key, see [Creating Keys](#) in the AWS documentation.
 5. Click Copy AMI. You can use the new AMI you copied for the next steps.
6. Select the row that lists your Tanzu Operations Manager AMI and click Launch.
7. Select m5.large for your instance type.
8. Click Next: Configure Instance Details.
9. Configure the following fields for your instance:
 - Network: Select the VPC that you created.
 - Subnet: Click `pcf-public-subnet-az0`.
 - Auto-assign for Public IP: Click Enable.
 - IAM role: Select the IAM role associated with your pcf-user profile. If you have not created one, click Create new IAM role and follow the procedures in [AWS Permissions Guidelines](#).
 - For all other fields, accept the default values.
10. Click Next: Add Storage.
11. Adjust the Size (GiB) value. The default persistent disk value is 50 GB. VMware recommends increasing this value to a minimum of 100 GB.
12. Click Next: Tag Instance
13. On the Add Tags pane, add a tag with the key `Name` and value `pcf-ops-manager`.
14. Click Next: Configure Security Group.
15. Select the `pcf-ops-manager-security-group` that you created in [Configure a Security Group for Tanzu Operations Manager](#) in *Preparing to Deploy Tanzu Operations Manager on AWS*.
16. Click Review and Launch and confirm the instance launch details.
17. Click Launch.
18. Click the `pcf-ops-manager-key` key pair, confirm that you have access to the private key file, and click Launch Instances. You use this key pair to access the Tanzu Operations Manager VM.

Edit Tanzu Operations Manager DNS A record

After you deploy the new Tanzu Operations Manager VM, edit the Tanzu Operations Manager DNS record in the EC2 dashboard to point to the IP address for the new VM.

To edit the Tanzu Operations Manager DNS A record:

1. Click View Instances to access the Instances page on the EC2 Dashboard.
2. Select the VM that you created in the previous section.
3. Locate the IPv4 Public IP value in the instance Description tab, and record this value for use in the next step.
4. In your DNS provider, edit the A record for `pcf.SYSTEM-DOMAIN` to point to the IP address recorded in the previous step, where `SYSTEM-DOMAIN` is the system domain of your Tanzu Operations Manager deployment.

Next steps

After you complete this procedure, continue to the upgrade instructions in [Upgrading Tanzu Operations Manager](#).

Deleting your Tanzu Operations Manager deployment from AWS

This topic describes how you can delete the AWS resources associated with a Tanzu Operations Manager deployment. You can use the AWS console to remove an installation of all components, but retain the objects in your bucket for a future deployment.

Prerequisites

Before you can delete resources from AWS, you must have an account with access to your AWS Management Console.

For more information, see [AWS Management Console](#) in the AWS documentation.

Delete resources from AWS

To delete the resources associated with Tanzu Operations Manager from AWS:

1. Log into the AWS Management Console.
2. Go to the EC2 Dashboard.
 1. Click Instances on the navigation menu.
 1. Select the Tanzu Operations Manager instance from the list.
 2. Click Actions to open the drop-down menu.
 3. In Instance State, click Terminate.
 4. A dialog box appears that prompts you to confirm that you want to stop the instances. Click Yes, then Terminate to confirm.
 2. Click Load Balancers on the navigation menu.
 1. Select the web, SSH, and TCP load balancers from the list.
 2. Click Actions to open the drop-down menu.
 3. To delete all load balancers, click Delete.
 4. A dialog box appears that prompts you to confirm the deletion of the load balancers. Click Yes, then Delete to confirm.
3. From the AWS Console, click RDS.
 1. Click Instances from the navigation menu.

1. Click Instance Actions to open the drop-down menu.
 2. To delete all RDS instances, click Delete.
 3. Click Create final Snapshot from the drop-down menu.
 4. To delete the database instance, click Delete.
4. On the AWS Console, click S3.
 1. Click Buckets from the navigation menu.
 2. For each bucket created:
 1. Select the bucket.
 2. Click Delete.
 3. On the page that appears, type the name of the bucket in the confirmation text box, then click Delete bucket.
5. On the AWS Console, click VPC.
 1. Click NAT gateways from the navigation menu.
 1. Select the NAT gateway you created for your installation.
 2. Click Actions to open the drop-down menu.
 3. To delete the NAT gateway, click Delete NAT gateway.
 4. A dialog appears that prompts you to confirm the deletion of the NAT gateway. Enter **delete** into the confirmation text box, then click Delete.
 2. Click Your VPCs on the navigation menu.
 1. Select the VPC you created for your installation.
 2. Click Actions to open the drop-down menu.
 3. To delete the VPCs, click Delete VPC.
 4. A dialog appears prompting you to confirm the deletion of the VPC. Enter “delete” in the confirmation text box, then click Delete.

Installing and configuring Tanzu Operations Manager on Azure

The following topics tell you how to install and configure Tanzu Operations Manager on Microsoft Azure:

- [Requirements and prerequisites for Tanzu Operations Manager on Azure](#)
- [Installing Tanzu Operations Manager on Azure](#)
- [Preparing to deploy Tanzu Operations Manager on Azure](#)
- [Deploying Tanzu Operations Manager on Azure manually](#)
- [Configuring BOSH Director on Azure](#)
- [Troubleshooting Tanzu Operations Manager on Azure](#)
- [Upgrading BOSH Director for Tanzu Operations Manager on Azure](#)
- [Deleting your Tanzu Operations Manager deployment from Azure](#)
- Additional information:
 - [Migrating from Storage Accounts to Managed Disks on Azure](#)
 - [Configuring BOSH Director on Azure Government Cloud](#)

Requirements and prerequisites for Tanzu Operations Manager on Azure

Ensure that you meet the following requirements and prerequisites before installing Tanzu Operations Manager on Microsoft Azure.

You can install Tanzu Operations Manager on either Azure or Azure Government Cloud. For information about Azure regions, see the [Azure documentation](#). For more information about Azure Government Cloud, see [Configuring BOSH Director on Azure Government Cloud](#).

You can install Tanzu Operations Manager on Azure with the Tanzu Platform for Cloud Foundry runtime. There are resource requirements specific to both Elastic Application Runtime and Azure. Ensure that you meet the requirements for Elastic Application Runtime and the requirements specific to Azure before installing Tanzu Operations Manager on Azure. For more information, see [Requirements](#).

Requirements

This section lists the resource requirements for installing Tanzu Operations Manager on Azure.

Tanzu Operations Manager resource requirements

When launching Tanzu Operations Manager on Azure, you must use an instance type that meets the following minimum specifications: | Machine Spec | Minimum Value | | CPU | 4 vCPUs | | Memory | 16 GB | | Boot disk | 120 GB |

VMware recommends using a [Standard_D4s_v3](#) instance type for the `<%= vars.ops_manager > VM`.

Azure resource requirements

The following are the minimum resource requirements for deploying a Tanzu Operations Manager deployment with Elastic Application Runtime on Azure:

- An OS disk of 120 GB for the Tanzu Operations Manager VM.
Tanzu Operations Manager v3.2 requires a BOSH Director VM with at least 8 GB memory.
- A minimum of the following VM instance limits in your Azure account. The number of VMs required depends on the number of tiles and availability zones (AZs) you plan to deploy.
 - For Elastic Application Runtime-specific resource requirements, see [Elastic Application Runtime Resource Requirements](#).
 - For Container Orchestration Engine-specific resource requirements, see [Azure Prerequisites and Resource Requirements](#).

Some VM instance types are only supported in certain Azure regions. For more information, see [Products available by region](#) in the Microsoft Azure documentation. To deploy Tanzu Operations Manager on an Azure region that does not support the default VM instance types, you must override the default VM sizes.

For more information, see [Overriding defaults with custom VM types](#) in the Tanzu Operations Manager API documentation. Changing the default VM sizes might increase the cost of your deployment.

Prerequisites

Before you install Tanzu Operations Manager on Azure, you must:

- Install the Azure CLI. To install the Azure CLI for your operating system, see [Preparing to Deploy Tanzu Operations Manager on Azure](#).

- Configure sufficient IP allocation for Elastic Application Runtime. For more information about IP allocation requirements, see [Elastic Application Runtime Resource Requirements](#).
- Create an SSL certificate for your Tanzu Operations Manager domain. To deploy Tanzu Operations Manager to a production environment, you must obtain a certificate from a certificate authority (CA). VMware recommends using a self-signed certificate generated by Tanzu Operations Manager for development and testing purposes only.
- Assign administrative rights to a domain for Tanzu Operations Manager. You must be able to add wildcard records to this domain.
- Create a wildcard DNS record that points to your Gorouter or load balancer. Alternatively, you can use a service such as xip.io. For example, `203.0.113.0.xip.io`. Then, create at least one wildcard TLS certificate that matches the DNS record you configured. With a wildcard DNS record, every hostname in your domain resolves to the IP address of your Gorouter or load balancer. For example, if you create a DNS record `*.example.com` pointing to your Gorouter, every app deployed to the `example.com` domain resolves to the IP address of your Gorouter.
- Create one or more NTP servers, if the NTP servers are not already provided by your Azure project.
- Install the most recent version of the Tanzu cf CLI. For more information, see the [Cloud Foundry Command Line Interface](#) in the GitHub repository.
- (Optional) Configure external storage. VMware recommends using external storage if possible. For more information about how file storage location affects platform performance and stability during upgrades, see [Configure file storage](#) in *Configuring Elastic Application Runtime for Upgrades*.
- (Optional) Configure external databases. VMware recommends using external databases for BOSH Director and Elastic Application Runtime in production deployments. An external database must be configured to use the UTC timezone.
- (Optional) Configure external user stores. When you deploy Tanzu Operations Manager, you can select SAML, LDAP, or OIDC to integrate existing user accounts.

Install Tanzu Operations Manager on Azure

To install Tanzu Operations Manager on Azure, see [Installing Tanzu Operations Manager on Azure](#).

Install Tanzu Operations Manager on Azure Government Cloud

To install Tanzu Operations Manager on Azure Government Cloud, see [Configuring BOSH Director on Azure Government Cloud](#).

Additional resources

The following are additional resources related to installing Tanzu Operations Manager on Microsoft Azure:

- [Troubleshooting Tanzu Operations Manager on Azure](#)
- [Reference architecture for Tanzu Operations Manager on Azure](#)
- [Azure security documentation](#)
- [IaaS Permissions Guidelines](#)
- [Deleting Tanzu Operations Manager from Azure](#)

- [Upgrading BOSH Director on Azure](#)

Installing Tanzu Operations Manager on Azure

This topic explains how you can manually install Tanzu Operations Manager on Microsoft Azure.

Complete the following steps:

1. Deploy Tanzu Operations Manager. See [Deploying Tanzu Operations Manager on Azure](#).
2. Configure BOSH Director. See [Configuring BOSH Director on Azure](#).

Next steps

After completing the procedures, configure a runtime.

For information about installing and configuring a runtime, see the following:

- [Configuring Elastic Application Runtime](#)
- [Installing and configuring Elastic Application Runtime Windows add-on](#)
- [Installing Small Footprint Elastic Application Runtime](#)
- [Installing Container Orchestration Engine on Azure](#)

Preparing to deploy Tanzu Operations Manager on Azure

This topic describes how you can prepare Azure to deploy Tanzu Operations Manager. You must deploy Tanzu Operations Manager before you can deploy Tanzu Platform for Cloud Foundry or VMware Tanzu Container Orchestration Engine (formerly TKGI).

Step 1: Install and configure the Azure CLI

1. Install the Azure CLI by following the instructions for your operating system in [Install the Azure CLI](#) in the Microsoft Azure documentation.
2. On the command line, set your cloud with a `name` value by running the following command:

```
az cloud set --name CLOUD-NAME
```

Where `CLOUD-NAME` is one of the following values. Use the value that corresponds to the Azure environment on which you are installing Tanzu Operations Manager:

- For Azure: `AzureCloud`
- For Azure China: `AzureChinaCloud`. If log in to `AzureChinaCloud` fails with `CERT_UNTRUSTED`, use the latest version of node, 4.x or later. For more information about this error, see [Failed to login AzureChinaCloud](#) in the Azure/azure-xplat-cli GitHub repository.
- For Azure Government Cloud: `AzureUSGovernment`

For example:

```
$ az cloud set --name AzureCloud
```

3. To log in, run the following command:

```
az login
```

4. Authenticate by going to the URL in the output of the command, entering the provided code, and clicking your account.

Step 2: Set your default subscription

1. To list your Azure subscriptions, run the following command:

```
az account list
```

For example:

```
$ az account list
[
  {
    "id": "12345678-1234-5678-1234-567891234567",
    "name": "Sample Subscription",
    "user": {
      "name": "Sample Account",
      "type": "user"
    },
    "tenantId": "11111111-1234-5678-1234-678912345678",
    "state": "Enabled",
    "isDefault": true,
    "registeredProviders": [],
    "environmentName": "AzureCloud"
  },
  {
    "id": "87654321-1234-5678-1234-678912345678",
    "name": "Sample Subscription1",
    "user": {
      "name": "Sample Account1",
      "type": "user"
    },
    "tenantId": "22222222-1234-5678-1234-678912345678",
    "state": "Enabled",
    "isDefault": false,
    "registeredProviders": [],
    "environmentName": "AzureCloud"
  }
]
```

2. Identify your default subscription in the output of this command by locating the subscription that has `isDefault` set to `true`. To deploy Tanzu Operations Manager to a different subscription, you must set the other subscription as the default. To set a new default subscription, run the following command:

```
az account set --subscription SUBSCRIPTION-ID
```

Where `SUBSCRIPTION-ID` is the value of the `id` field of the subscription that you want to set as the default.

Example: The following command sets the default to a subscription that has an `id` with the value of `"87654321-1234-5678-1234-567891234567"`.

```
$ az account set --subscription "87654321-1234-5678-1234-567891234567"
```

3. Record the value of the `id` of the default subscription. You need this value in future configuration steps.
4. Record the value of `tenantID` of the default subscription. This is your `TENANT_ID` for creating a service principal. If your `tenantID` value is not defined, you might be using a personal account to log in to your Azure subscription.

Step 3: Create an AAD application

1. To create an Azure Active Directory (AAD) application, run the following command:

```
az ad app create --display-name "Service Principal for BOSH" \  
--web-home-page-url "http://BOSHAzureCPI" \  
--identifier-uris "http://BOSHAzureCPI"
```

For example:

```
$ az ad app create --display-name "Service Principal for BOSH" \  
--web-home-page-url "http://BOSHAzureCPI" \  
--identifier-uris "http://BOSHAzureCPI"
```

You can provide any string for the `web-home-page-url` and `identifier-uris` flags, but the value of `identifier-uris` must be unique within the organization associated with your Azure subscription. For the `web-home-page-url`, VMware recommends using `http://BOSHAzureCPI`, as shown in the preceding example.

2. From the output of the `az ad app create` command that you ran in the previous step, locate and record the value of `appId`. This is your `APPLICATION_ID` for creating a service principal.

The following example output shows an `appId` value of `"5c552e8f-b977-45f5-a50b-981cfe17cb9d"`.

```
{  
  "appId": "5c552e8f-b977-45f5-a50b-981cfe17cb9d",  
  "appPermissions": null,  
  "availableToOtherTenants": false,  
  "displayName": "Service Principal for BOSH",  
  "web": {  
    "homePageUrl": "http://BOSHAzureCIP",  
  },  
  "identifierUris": [  
    "http://BOSHAzureCPI"  
  ],  
  "objectId": "f3884df4-7d1d-4894-a78c-c1fe75750436",  
  "objectType": "Application",  
  "replyUrls": []  
}
```

Step 4: Create and configure a service principal

1. To create a service principal, run the following command:

```
az ad sp create-for-rbac --display-name "Service Principal for BOSH" \
-n YOUR-APPLICATION-ID \
--role Owner \
--scopes /subscriptions/SUBSCRIPTION-ID
```

Where **YOUR-APPLICATION-ID** is the **APPLICATION_ID** that you recorded in [Create an AAD Application](#).

For example:

```
$ az ad sp create-for-rbac --display-name "Service Principal for BOSH" \
-n 5c552e8f-b977-45f5-a50b-981cfe17cb9d \
--role Owner \
--scopes /subscriptions/1e9249d9-c682-410b-8d50-31fda4736c8b
{
  "appId": "5c552e8f-b977-45f5-a50b-981cfe17cb9d",
  "displayName": "Service Principal for BOSH",
  "password": "9Hf8Q~4kYUPJ7wDqWoBVM5DctJZ~8xAxg8842axC",
  "tenant": "b39165ca-3cee-4b8a-a4d6-cd1239dd62f0"
}
```

2. From the output of `az ad sp create-for-rbac` command in the previous step, locate and record the value of **password**. This is your **CLIENT_SECRET** for creating a service principal.

You must have the **Owner** role on your service principal to deploy Tanzu Operations Manager to Azure. If you need to use multiple resource groups for your deployment on Azure, you can define custom roles for your service principal. These roles enable BOSH to deploy to pre-existing network resources outside of the resource group.

For more information about Azure Role-Based Access Control, see [Built-in roles for Azure resource](#) the Microsoft Azure documentation.

3. To verify the role assignment, run the following command:

```
az role assignment list --assignee "SERVICE-PRINCIPAL-NAME"
```

Where **"SERVICE-PRINCIPAL-NAME"** is any value of **servicePrincipalNames** from the preceding output, such as **YOUR-APPLICATION-ID**.

For example:


```
$ az role assignment list --assignee "5c552e8f-b977-45f5-a50b-981cfe17cb9d"
[
  {
    "id": "/subscriptions/995b7eed-77ef-45ff-a5c9-1a405ffb8243/providers/Microsoft.Authorization/roleAssignments/32e644cf-ba1a-4f43-bf7c-68bf4583e463",
    "name": "32e644cf-ba1a-4f43-bf7c-68bf4583e463",
    "properties": {
      "principalId": "cc13c685-4c3b-461e-ae96-7a0563960b83",
      "principalName": "http://BOSHAzureCPI",
      "roleDefinitionId": "/subscriptions/995b7eed-77ef-45ff-a5c9-1a405ffb8243/providers/Microsoft.Authorization/roleDefinitions/b24988ac-6180-42a0-ab88-20f7382dd24c",
      "roleDefinitionName": "Owner",
      "scope": "/subscriptions/995b7eed-77ef-45ff-a5c9-1a405ffb8243"
    },
    "type": "Microsoft.Authorization/roleAssignments"
  }
]
```

Step 5: Verify your service principal

To verify your service principal, run the following command to log in:

```
az login --username APPLICATION_ID --password CLIENT_SECRET \
--service-principal --tenant TENANT_ID
```

Where:

- **APPLICATION_ID** is the **APPLICATION_ID** that you recorded in [Create an AAD Application](#).
- **CLIENT_SECRET** is the password that you recorded in [Create a service principal](#).
- **TENANT_ID** is the value of **tenantID** of the default subscription that you recorded in [Set your Default Subscription](#).

For example:

```
$ az login --username 5c552e8f-b977-45f5-a50b-981cfe17cb9d \
--password "Swordfish" \
--service-principal --tenant 22222222-1234-5678-1234-678912345678
[
  {
    "cloudName": "AzureCloud",
    "id": "995b7eed-77ef-45ff-a5c9-1a405ffb8243",
    "isDefault": true,
    "name": "my-user-name",
    "state": "Enabled",
    "tenantId": "22222222-1234-5678-1234-678912345678",
    "user": {
      "name": "5c552e8f-b977-45f5-a50b-981cfe17cb9d",
      "type": "servicePrincipal"
    }
  }
]
```

If you cannot log in, the service principal is invalid. Create a new service principal and log in again.

Step 6: Complete registrations

1. To register your subscription with Microsoft.Storage, run:

```
az provider register --namespace Microsoft.Storage
```

2. To register your subscription with Microsoft.Network, run:

```
az provider register --namespace Microsoft.Network
```

3. To register your subscription with Microsoft.Compute, run:

```
az provider register --namespace Microsoft.Compute
```

Next steps

After you complete this procedure, follow the instructions in [Deploying Tanzu Operations Manager on Azure](#).

Deploying Tanzu Operations Manager on Azure manually

You can use these instructions to deploy BOSH and Tanzu Operations Manager on Azure by using individual commands to create resources. VMware recommends this manual procedure for deploying to Azure China and Azure Government Cloud.

Before you follow the procedures in this topic, you must complete the procedures in [Preparing to Deploy Tanzu Operations Manager on Azure](#).

Sometimes Azure portal displays the names of resources with incorrect capitalization. Always use the Azure CLI to retrieve the correctly capitalized name of a resource.

Step 1: Create network resources

To create network resources for your deployment:

1. Go to the Azure portal, click Resource groups, then click Add to create a new resource group for your deployment.
2. Enter a Resource group name.
3. Click your Subscription, click a Resource group location, then click Create.
4. Export the name of your resource group as the environment variable `$RESOURCE_GROUP`.

```
export RESOURCE_GROUP="YOUR-RESOURCE-GROUP-NAME"
```

If you are on a Microsoft Windows machine, use `set` instead of `export`.

```
set RESOURCE_GROUP="YOUR-RESOURCE-GROUP-NAME"
```

5. Export your location. For example, `westus`.

```
export LOCATION=westus
```

For a list of available locations, run `az account list-locations`.

6. Create a network security group named `pcf-nsg`.

```
az network nsg create --name pcf-nsg \  
--resource-group $RESOURCE_GROUP \  
--location $LOCATION
```

7. Add network security group rules to the `pcf-nsg` group to enable traffic to known ports from the public Internet.

```
az network nsg rule create --name ssh \  
--nsg-name pcf-nsg --resource-group $RESOURCE_GROUP \  
--protocol Tcp --priority 100 \  
--destination-port-range '22'
```

```
az network nsg rule create --name http \  
--nsg-name pcf-nsg --resource-group $RESOURCE_GROUP \  
--protocol Tcp --priority 200 \  
--destination-port-range '80'
```

```
az network nsg rule create --name https \  
--nsg-name pcf-nsg --resource-group $RESOURCE_GROUP \  
--protocol Tcp --priority 300 \  
--destination-port-range '443'
```

```
az network nsg rule create --name diego-ssh \  
--nsg-name pcf-nsg --resource-group $RESOURCE_GROUP \  
--protocol Tcp --priority 400 \  
--destination-port-range '2222'
```

- To block traffic from the public Internet, append `--source-address-prefixes AzureLoadBalancer` to enable traffic from only the Azure load balancer or `--source-address-prefixes VirtualNetwork` to only enable traffic from the virtual network.
- To enable traffic from both the Azure load balancer and the virtual network, to create duplicates of each rule, one specifying `--source-address-prefixes AzureLoadBalancer` and one specifying `--source-address-prefixes VirtualNetwork`.

8. Create a network security group named `opsmgr-nsg`.

```
az network nsg create --name opsmgr-nsg \  
--resource-group $RESOURCE_GROUP \  
--location $LOCATION
```

9. Add a network security group rule to the `opsmgr-nsg` group to enable HTTP traffic to the Tanzu Operations Manager VM.

```
az network nsg rule create --name http \  
--nsg-name opsmgr-nsg --resource-group $RESOURCE_GROUP \  
--protocol Tcp --priority 100 \  
--destination-port-range 80
```

10. Add a network security group rule to the `opsmgr-nsg` group to enable HTTPS traffic to the Tanzu Operations Manager VM.

```
az network nsg rule create --name https \
--nsg-name opsmgr-nsg --resource-group $RESOURCE_GROUP \
--protocol Tcp --priority 200 \
--destination-port-range 443
```

11. Add a network security group rule to the `opsmgr-nsg` group to enable SSH traffic to the Tanzu Operations Manager VM.

```
az network nsg rule create --name ssh \
--nsg-name opsmgr-nsg --resource-group $RESOURCE_GROUP \
--protocol Tcp --priority 300 \
--destination-port-range 22
```

- To block traffic from the public Internet, append `--source-address-prefixes AzureLoadBalancer` to enable traffic from only the Azure load balancer or `--source-address-prefixes VirtualNetwork` to only enable traffic from the virtual network.
- To enable traffic from both the Azure load balancer and the virtual network, create duplicates of each rule, one specifying `--source-address-prefixes AzureLoadBalancer` and one specifying `--source-address-prefixes VirtualNetwork`.
- Optionally, if you want to use private IP ranges with Tanzu Operations Manager and enable all internal traffic, you can create the Network Security Groups to enable all internal traffic.

```
az network nsg rule create --name internal-virtual-network \
--nsg-name internal-traffic --resource-group $RESOURCE_GROUP \
--protocol Tcp --priority 100 \
--destination-port-range * \
--source-address-prefixes VirtualNetwork
az network nsg rule create --name internal-from-lb \
--nsg-name internal-traffic --resource-group $RESOURCE_GROUP \
--protocol Tcp --priority 110 \
--destination-port-range * \
--source-address-prefixes AzureLoadBalancer
```

12. Create a virtual network named `pcf-virtual-network`.

```
az network vnet create --name pcf-virtual-network \
--resource-group $RESOURCE_GROUP --location $LOCATION \
--address-prefixes 10.0.0.0/16
```

13. Add subnets to the network for Tanzu Operations Manager, BOSH director, and the VMs for your runtime, and attach the Network Security Group.

```

az network vnet subnet create --name pcf-infrastructure-subnet \
--vnet-name pcf-virtual-network \
--resource-group $RESOURCE_GROUP \
--address-prefix 10.0.4.0/26 \
--network-security-group pcf-nsg
az network vnet subnet create --name pcf-pas-subnet \
--vnet-name pcf-virtual-network \
--resource-group $RESOURCE_GROUP \
--address-prefix 10.0.12.0/22 \
--network-security-group pcf-nsg
az network vnet subnet create --name pcf-services-subnet \
--vnet-name pcf-virtual-network \
--resource-group $RESOURCE_GROUP \
--address-prefix 10.0.8.0/22 \
--network-security-group pcf-nsg

```

Step 2: Create Tanzu Operations Manager storage account

Tanzu Operations Manager is distributed as a VHD disk image on Azure. In order to launch the Tanzu Operations Manager VM, you will need to copy the VHD to a storage account in your resource group.

To create the storage account:

1. Choose a name for your Tanzu Operations Manager storage account, and export it as the environment variable `$STORAGE_NAME`. Storage account names must be globally unique across Azure, between 3 and 24 characters in length, and contain only lowercase letters and numbers.

```
export STORAGE_NAME="YOUR-BOSH-STORAGE-ACCOUNT-NAME"
```

2. Create a standard storage account with the following command. This account is used to copy the VHD required to launch the Tanzu Operations Manager VM, but does not have to be used for running any VMs.

```

az storage account create --name $STORAGE_NAME \
--resource-group $RESOURCE_GROUP \
--sku Standard_LRS \
--location $LOCATION

```

If the command fails, verify that you followed the rules for naming your storage account. Export another new storage account name if necessary.

3. Configure the Azure CLI to use the Tanzu Operations Manager storage account as the default setting.
 1. Retrieve the connection string for the account.

```

az storage account show-connection-string \
--name $STORAGE_NAME --resource-group $RESOURCE_GROUP

```

The command returns output similar to the following:

```

{
  "connectionString": "DefaultEndpointsProtocol=https;EndpointSuffix=core.windows.net;AccountName=cfdocsboshstorage;AccountKey=EXAMPLEaaaaabbb
rnc5igFxyWsgq016Tu9uGwse0l8bqNBEL/2tp7wX92QMUM19Pz9BYTXt8aq4A=="
}

```

2. Record the full value of `connectionString` from the preceding output, starting with and including `DefaultEndpointsProtocol=`.
3. Export the value of `connectionString` as the environment variable `$CONNECTION_STRING`.

```
export CONNECTION_STRING="YOUR-ACCOUNT-KEY-STRING"
```

4. Create a blob container in the Tanzu Operations Manager storage account, named `opsmanager`.

```
az storage container create --name opsmanager \  
--connection-string $CONNECTION_STRING
```

Step 3: Create load balancers

Your load balancer configuration depends on whether you want apps to be available from public IP addresses, private IP addresses, or both.

To create load balancers:

1. Required: Elastic Application Runtime Load Balancer

1. Create a load balancer named `pcf-lb`. A static IP address is automatically created for Standard SKU load balancers unless specified otherwise with `--public-ip-address-allocation Dynamic`.

```
az network lb create --name pcf-lb \  
--resource-group $RESOURCE_GROUP --location $LOCATION \  
--backend-pool-name pcf-lb-be-pool --frontend-ip-name pcf-lb-fe-ip \  
--public-ip-address pcf-lb-ip --public-ip-address-allocation Static \  
--sku Standard
```

This back end pool is empty when you create it.

2. Add a probe to the load balancer.

```
az network lb probe create --lb-name pcf-lb \  
--name http8080 --resource-group $RESOURCE_GROUP \  
--protocol Http --port 8080 --path health
```

3. Add a load balancing rule for HTTP.

```
az network lb rule create --lb-name pcf-lb \  
--name http --resource-group $RESOURCE_GROUP \  
--protocol Tcp --frontend-port 80 \  
--backend-port 80 --frontend-ip-name pcf-lb-fe-ip \  
--backend-pool-name pcf-lb-be-pool \  
--probe-name http8080
```

4. Add a load balancing rule for HTTPS.

```
az network lb rule create --lb-name pcf-lb \  
--name https --resource-group $RESOURCE_GROUP \  
--protocol Tcp --frontend-port 443 \  
--backend-port 443 --frontend-ip-name pcf-lb-fe-ip \  
--backend-pool-name pcf-lb-be-pool \  
--probe-name http8080
```

5. (Optional) For private IPs:

1. Create a load balancer named **pcf-lb**.

```
az network lb create --name pcf-lb \
--resource-group $RESOURCE_GROUP --location $LOCATION \
--backend-pool-name pcf-lb-be-pool --frontend-ip-name pcf-lb-fe-ip \
--private-ip-address 10.0.12.6 --sku Standard \
--vnet-name pcf-virtual-network \
--subnet pcf-pas-subnet
```

2. Add a probe to the load balancer.

```
az network lb probe create --lb-name pcf-lb \
--name http8080 --resource-group $RESOURCE_GROUP \
--protocol Http --port 8080 --path health
```

3. Add a load balancing rule for HTTP.

```
az network lb rule create --lb-name pcf-lb \
--name http --resource-group $RESOURCE_GROUP \
--protocol Tcp --frontend-port 80 \
--backend-port 80 --frontend-ip-name pcf-lb-fe-ip \
--backend-pool-name pcf-lb-be-pool \
--probe-name http8080
```

4. Add a load balancing rule for HTTPS.

```
az network lb rule create --lb-name pcf-lb \
--name https --resource-group $RESOURCE_GROUP \
--protocol Tcp --frontend-port 443 \
--backend-port 443 --frontend-ip-name pcf-lb-fe-ip \
--backend-pool-name pcf-lb-be-pool \
--probe-name http8080
```

This back end pool is empty when you create it.

6. Go to your DNS provider and create an entry that points ***.YOUR-SUBDOMAIN** to the public IP address of your load balancer. For example, create an entry that points **azure.example.com** to **198.51.100.1**. You can retrieve it by running:

```
az network public-ip show --name pcf-lb-ip --resource-group $RESOURCE_GROUP
```

2. Optional: Diego SSH Load Balancer

1. Create a load balancer named **pcf-ssh-lb**. This back end pool is empty when you create it.

```
az network lb create --name pcf-ssh-lb \
--resource-group $RESOURCE_GROUP --location $LOCATION \
--backend-pool-name pcf-ssh-lb-be-pool --frontend-ip-name pcf-ssh-lb-fe-ip \
--public-ip-address pcf-ssh-lb-ip --public-ip-address-allocation Static \
--sku Standard
```

2. (Optional) For private IPs, create a load balancer named **pcf-ssh-lb**. This back end pool is empty when you create it.

```
az network lb create --name pcf-ssh-lb \
--resource-group $RESOURCE_GROUP --location $LOCATION \
--backend-pool-name pcf-ssh-lb-be-pool --frontend-ip-name pcf-ssh-lb-fe-ip \
--private-ip-address 10.0.12.7 \
--sku Standard \
--vnet-name pcf-virtual-network \
--subnet pcf-pas-subnet
```

3. Add a probe to the load balancer.

```
az network lb probe create --lb-name pcf-ssh-lb \
--name tcp2222 --resource-group $RESOURCE_GROUP \
--protocol Tcp --port 2222
```

4. Add a load balancing rule for SSH.

```
az network lb rule create --lb-name pcf-ssh-lb \
--name diego-ssh --resource-group $RESOURCE_GROUP \
--protocol Tcp --frontend-port 2222 \
--backend-port 2222 --frontend-ip-name pcf-ssh-lb-fe-ip \
--backend-pool-name pcf-ssh-lb-be-pool \
--probe-name tcp2222
```

5. Go to your DNS provider, and create an entry that points `ssh.sys.YOUR-SUBDOMAIN` to the public IP address of your load balancer. For example, create an entry that points `azure.example.com` to `198.51.100.1`. You can retrieve it by running:

```
az network public-ip show --name pcf-ssh-lb-ip --resource-group $RESOURCE_GROUP
```

Step 4: Boot Tanzu Operations Manager

To boot Tanzu Operations Manager:

1. Go to the [Broadcom Support portal](#) and download the latest release of Tanzu Operations Manager for Azure.
2. View the downloaded PDF and locate the Tanzu Operations Manager image URL appropriate for your region.
3. Export the Tanzu Operations Manager image URL as an environment variable.

```
export OPS_MAN_IMAGE_URL="YOUR-OPS-MAN-IMAGE-URL"
```

4. Download the Tanzu Operations Manager image. For compatibility when upgrading to future versions of Tanzu Operations Manager, choose a unique name for the image that includes the Tanzu Operations Manager version number. For example, replace `opsman-image-3.0.x` in the following examples with a string containing this release; e.g. `opsman-image-3.0.0`.

1. Copy the Tanzu Operations Manager image into your storage account using the Azure CLI:

```
az storage blob copy start --source-uri $OPS_MAN_IMAGE_URL \
--connection-string $CONNECTION_STRING \
--destination-container opsmanager \
--destination-blob opsman-image-3.0.x.vhd
```

2. Copying the image might take several minutes. Run the following command and examine the output under `"copy"`:


```

az storage blob show --name opsman-image-3.0.x.vhd \
--container-name opsmanager \
--connection-string $CONNECTION_STRING
...
"copy": {
  "completionTime": "2017-06-26T22:24:11+00:00",
  "id": "b9c8b272-a562-4574-baa6-f1a04afcefd",
  "progress": "53687091712/53687091712",
  "source": "https://opsmanagerwestus.blob.core.windows.net/images/opsman-image-3.0.x.vhd",
  "status": "success",
  "statusDescription": null
},

```

3. Wait a few moments and run the preceding command again if **status** is **pending**. When **status** reads **success**, continue to the next step.
5. Create a public IP address named **ops-manager-ip**. If you do not want to use a public IP for Tanzu Operations Manager, skip this step.
Record the **ipAddress** from the preceding output. This is the public IP address of Tanzu Operations Manager.

```

az network public-ip create --name ops-manager-ip \
--resource-group $RESOURCE_GROUP --location $LOCATION \
--allocation-method Static
{
  "publicip": {
    "dnsSettings": null,
    "etag": "W/\"4450ebe2-9e97-4b17-9cf2-44838339c661\"",
    "id": "/subscriptions/995b7eed-77ef-45ff-a5c9-1a405ffb8243/resourceGroups/cf-docs/providers/Microsoft.Network/publicIPAddresses/ops-manager-ip",
    "idleTimeoutInMinutes": 4,
    "ipAddress": "40.83.148.183",
    "ipConfiguration": null,
    "location": "westus",
    "name": "ops-manager-ip",
    "provisioningState": "Succeeded",
    "publicIpAddressVersion": "IPv4",
    "publicIpAllocationMethod": "Static",
    "resourceGroup": "cf-docs",
    "resourceGuid": "950d4831-1bec-42da-8a79-959bcddea9dd",
    "tags": null,
    "type": "Microsoft.Network/publicIPAddresses"
  }
}

```

6. Create a network interface for Tanzu Operations Manager. If you do not want use a public IP address for Tanzu Operations Manager, remove the **--public-ip-address ops-manager-ip** flag and value.

```

az network nic create --vnet-name pcf-virtual-network \
--subnet pcf-infrastructure-subnet --network-security-group opsmgr-nsg \
--private-ip-address 10.0.4.4 \
--public-ip-address ops-manager-ip \
--resource-group $RESOURCE_GROUP \
--name opsman-nic --location $LOCATION

```

7. Create a keypair on your local machine with the user name **ubuntu**. For example, enter the following command:

```
ssh-keygen -t rsa -f opsman -C ubuntu
```

When you are prompted for a passphrase, press the **enter** key to provide an empty passphrase.

8. Create the Tanzu Operations Manager VM.

1. Create a managed image from the Tanzu Operations Manager VHD file:

```
az image create --resource-group $RESOURCE_GROUP \  
--name opsman-image-3.0.x \  
--source https://$STORAGE_NAME.blob.core.windows.net/opsmanager/image-3.0.x.vhd \  
--location $LOCATION \  
--os-type Linux
```

If you are using Azure China or Azure Government Cloud, replace **blob.core.windows.net** with the following:

- For Azure China, use **blob.core.chinacloudapi.cn**. For more information, see the [Azure documentation](#) for more information.
- For Azure Government Cloud, use **blob.core.usgovcloudapi.net**. For more information, see the [Azure documentation](#) for more information.

2. Create your Tanzu Operations Manager VM, replacing **PATH-TO-PUBLIC-KEY** with the path to your public key **.pub** file.

```
az vm create --name opsman-3.0.x --resource-group $RESOURCE_GROUP \  
--location $LOCATION \  
--nics opsman-nic \  
--image opsman-image-3.0.x \  
--os-disk-size-gb 128 \  
--os-disk-name opsman-3.0.x-osdisk \  
--admin-username ubuntu \  
--size Standard_D2s_v3 \  
--storage-sku Standard_LRS \  
--ssh-key-value PATH-TO-PUBLIC-KEY
```

9. If you plan to install more than one tile in this Tanzu Operations Manager installation, you can increase the size of the Tanzu Operations Manager VM disk. You can repeat this process and increase the disk again at a later time if necessary.

1. To stop the VM and detach the disk, run:

```
az vm deallocate --name opsman-3.0.x \  
--resource-group $RESOURCE_GROUP
```

2. To resize the disk to 128 GB, run:

```
az disk update --size-gb 128 --name opsman-3.0.x-osdisk \  
--resource-group $RESOURCE_GROUP
```

3. To start the VM, run:

```
az vm start --name opsman-3.0.x --resource-group $RESOURCE_GROUP
```

Step 5: Complete BOSH Director configuration

To finish configuring BOSH Director:

1. Go to your DNS provider, and create an entry that points a fully qualified domain name (FQDN) to the public IP address of Tanzu Operations Manager. As a best practice, always use the FQDN to access Tanzu Operations Manager.
2. Continue to [Configuring BOSH Director on Azure](#).

Configuring BOSH Director on Azure

After you deploy Tanzu Operations Manager on Azure, follow the steps in this topic to configure the BOSH Director tile within Tanzu Operations Manager on Azure.

Prerequisites

Before you follow the procedures in this topic, you must have completed the procedures in these topics:

- [Preparing to deploy Tanzu Operations Manager on Azure](#)
- [Deploying Tanzu Operations Manager on Azure](#)

You can also follow the procedures in this topic using the Tanzu Operations Manager API. For more information, see [Using the Tanzu Operations Manager API](#).

Prerequisites for Azure Government Cloud

If you use Azure Government Cloud, you must set Azure Generation 2 VM types for a successful deployment. You can set custom VM types with the Tanzu Operations Manager API. For more information, see [Using the Tanzu Operations Manager API](#).

Azure Generation 2 VM types

Tanzu Operations Manager configures VMs to be deployed using Azure Generation 2 VM types by default. If necessary, you can reconfigure the VMs in your deployment to use Azure Generation 1 VM types.

The following table shows the mapping between Generation 1 and Generation 2 Azure VM types:

Generation 1 VM Type	Generation 2 VM Type
Standard_DS1_v2	Standard_B2s
Standard_DS2_v2	Standard_D2s_v3
Standard_DS3_v2	Standard_D4s_v3
Standard_DS4_v2	Standard_D8s_v3
Standard_DS5_v2	Standard_D16s_v3
	Standard_D32s_v3
	Standard_D48s_v3
	Standard_D64s_v3
Standard_DS11_v2	Standard_E2s_v3
Standard_DS12_v2	Standard_E4s_v3
Standard_DS13_v2	Standard_E8s_v3
Standard_DS14_v2	Standard_E16s_v3

Generation 1 VM Type	Generation 2 VM Type
	Standard_E20s_v3
	Standard_E32s_v3
	Standard_E48s_v3
	Standard_E64s_v3
Standard_F1s	Standard_B1ms
Standard_F2s	Standard_F2s_v2
Standard_F4s	Standard_F4s_v2
Standard_F8s	Standard_F8s_v2
	Standard_F16s_v2
	Standard_F32s_v2
	Standard_F48s_v2
	Standard_F64s_v2

Step 1: Log into Tanzu Operations Manager

1. In a web browser, go to the fully qualified domain name (FQDN) of Tanzu Operations Manager that you created in [Deploying Tanzu Operations Manager on Azure](#).
2. Log in to Tanzu Operations Manager. See [Log in to Tanzu Operations Manager for the First Time](#).

Step 2: Azure Config page

1. Click the BOSH Director tile.
2. Click Azure Config.
3. Complete the following fields with information you obtained in [Preparing to Deploy Tanzu Operations Manager on Azure](#).
 - Subscription ID: Enter the ID of your Azure subscription.
 - Tenant ID: Enter your **TENANT_ID**.
 - Application ID: Enter the **APPLICATION_ID** that you created in the [Create an AAD Application](#) step of *Preparing to Deploy Tanzu Operations Manager on Azure*.
 - Client Secret: Enter your **CLIENT_SECRET**.
4. Complete the following fields:
 - Resource Group Name: Enter the name of your resource group, which you exported as the **\$RESOURCE_GROUP** environment variable.
5. For Cloud Storage Type, select one of the following options based on your Azure VM storage settings.
 - Use Managed Disks: Select this option if you use Azure Managed Disks. For more information, see [Azure Managed Disks Overview](#) in the Microsoft documentation.
 1. (Optional) Custom Disk Encryption Set Name: Enter the name of the Disk Encryption Set that will be used to encrypt managed disks using the

corresponding customer-managed key. You must redeploy all VMs for this setting to re-encrypt existing disks. When left blank, managed disks will be encrypted using a platform-managed key.

- Use Storage Accounts (Deprecated): Select this option if you use storage accounts to store your Azure VMs. Azure has announced that using storage accounts for VM disks will no longer be supported starting on September 30th, 2025. To migrate to managed disks, see [Migrating from Storage Accounts to Managed Disks on Azure](#).

You can update your deployment from using storage accounts to using managed disks. However, after you click Use Managed Disks and deploy Tanzu Operations Manager, you cannot change your deployment back to use storage accounts.

6. BOSH Storage Account Name: Enter the name of your storage account, which you exported as the `$STORAGE_NAME` environment variable. This field is required when “Use Storage Accounts” is selected.
7. Deployments Storage Account Name (Deprecated): Enter the base storage name you used to create your deployment storage accounts, prepended and appended with the wildcard character `*`. For example, if you created accounts named `xyzdeploymentstorage1`, `xyzdeploymentstorage2`, and `xyzdeploymentstorage3`, enter `*deploymentstorage*`. Tanzu Operations Manager requires that you specify an asterisk at both the beginning and the end of the base storage account name. This field is required when “Use Storage Accounts” is selected.
8. From the Storage Tier ephemeral / persistent disks drop-down menu, click the storage option that corresponds with your Azure subscription:
 - For HDD-based storage, click `Standard_LRS`.
 - For SSD-based storage, click `Premium_LRS`.
9. If you want all BOSH-created disks to use the same storage tier for the root and OS disks, select the Use the same Storage Tier for root / OS disks check box. This overrides the default behavior, which is to use a storage tier based on the instance size. For more information, see the [Azure documentation](#).
10. (Optional) Enter your Default Security Group. For more information about Network Security Groups, see [Filter network traffic with network security groups](#) in the Microsoft Azure documentation. The Azure portal sometimes displays the names of resources with incorrect capitalization. Always use the Azure CLI to retrieve the correctly capitalized name of a resource.
11. For SSH Public Key, copy and paste the contents of your public key in the `opsman.pub` file. You created this file in [Deploying Tanzu Operations Manager on Azure](#).
12. For SSH Private Key, copy and paste the contents of your private key in the `opsman` file.
13. For VM Type Catalog, click the VM Catalog that you want to use for your BOSH Managed VMs.
14. For Availability Mode, choose whether to use Availability Sets or Availability Zones. You can further configure this option in the Assign AZs and Networks pane. For more information, see [Assign AZs and Networks](#). Availability Zones is selected by default, but you can choose Availability Sets.

You can only choose between Availability Zones and Availability Sets in a staged BOSH Director tile that has not yet been deployed. After you click Apply Changes

and the BOSH Director deploys successfully, you cannot switch between Availability Zones and Availability Sets.

15. For Azure Environment, select the Azure cloud you want to use. VMware recommends Azure Commercial Cloud for most Azure environments. If you selected Azure Government Cloud as your Azure environment, you must configure custom VM types. For more information, see [Prerequisites for Azure Government Cloud](#) in *Configuring BOSH Director on Azure*.

To deploy Tanzu Operations Manager on Azure using Active Directory Federation Services (ADFS) see [../opsguide/adfs-ssso-configuration.html](#).

16. Click Save.

Step 3: Director Config page

To configure the Director Config pane:

1. Click Director Config.

The screenshot shows the 'Director Config' page. At the top, there's a 'Go to' section with links: 'Certificate Duration Overrides', 'Health Monitor', 'CredHub Encryption Provider', 'Blobstore Location', and 'Database Location'. Below this is a section for 'NTP Servers (comma delimited)*' with a text box containing 'time.google.com' and a note: 'One or more NTP server addresses for consistent and valid time stamps used in syslog, etc.'. Next is 'Bosh HM Forwarder IP Address' with an empty text box and a note: 'Should only be used if OS Bosh HM Forwarder job is deployed'. A list of checkboxes follows: 'Enable VM Resurrector Plugin' (unchecked), 'Enable System Metrics' (checked), 'Enable Post Deploy Scripts' (unchecked), 'Enable Short Lived NATS Bootstrap Credentials' (unchecked) with a note 'This will make the Bosh Director rotate the NATs bootstrap credentials of a VM. This increases security by invalidating the NATS credentials that are stored in the IAAS metadata service.', 'Recreate VMs deployed by the BOSH Director' (unchecked) with a note 'This will force BOSH to recreate all Bosh Deployed VMs on the next deploy. This does not recreate the BOSH Director. Persistent disk will be preserved.', 'Recreate BOSH Director VM' (unchecked) with a note 'This will force BOSH to recreate the Bosh Director VM on the next deploy. Persistent disk will be preserved.', 'Recreate All Persistent Disks' (unchecked) with a note 'Check this box to recreate all Persistent Disks for the Director and all other Tiles.', and 'Enable bosh deploy retries' (unchecked) with a note 'This will attempt to re-deploy a failed deployment up to 5 times.'

- The NTP server configuration is updated only after VM recreation. If you modify the value of this text box, select the Recreate VMs deployed by the BOSH Director check box to re-create your BOSH Director-deployed VMs and update the NTP server configuration. If you have any service tiles installed, ensure that the Recreate All Service Instances errand runs for each service tile.

2. Leave the Bosh HM Forwarder IP Address text box blank. BOSH-reported component metrics are available in Loggregator Firehose by default. If you continue to use the BOSH HM Forwarder to consume these component metrics, you might receive duplicate data.
3. Select the Enable VM Resurrector Plugin check box to enable the BOSH Resurrector functionality and increase your runtime availability.
4. If you want to collect detailed metrics from BOSH-managed VMs, select the Enable System Metrics check box. For a list of metrics that the System Metrics Agent collects from BOSH-managed VMs, see [System Metrics Agent](#) on GitHub. When you select this check box, ensure that the System Metrics Scraper is able to connect on port 53035 to all BOSH-managed VMs.
5. Select Enable Post Deploy Scripts to run a post-deployment script after deployment. This script allows the job to execute additional commands against a deployment. If you intend to install VMware Tanzu Container Orchestration Engine (formerly TKGI), you must enable post-deploy scripts.
6. To configure your VMs to use short-lived NATS credentials during the bootstrap process, select the Enable Short Lived NATS Bootstrap Credentials check box. This increases the security of your Tanzu Operations Manager deployment by rotating the TLS certificate that the BOSH Agents in your VMs use to communicate with the BOSH Director through BOSH NATS. For more information, see [Using Short-Lived NATS Bootstrap credentials](#).

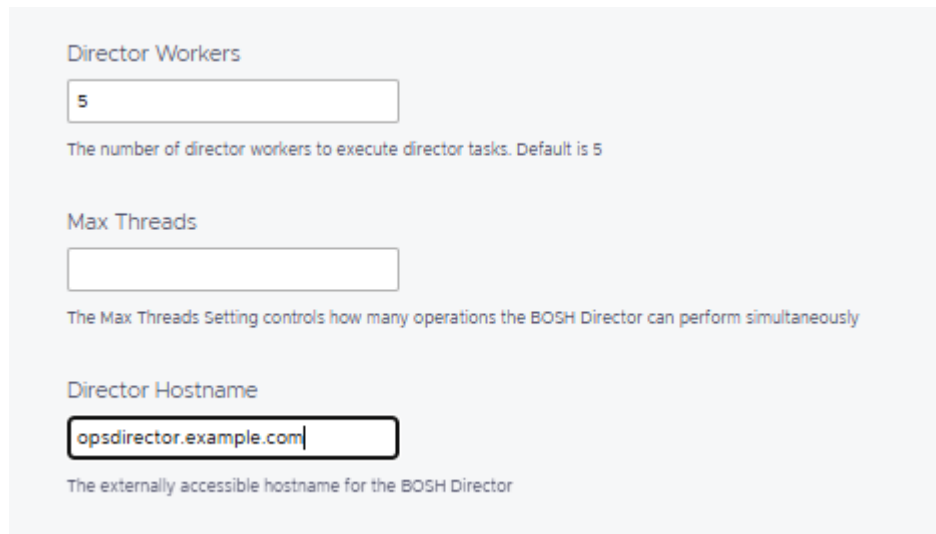
Configuring your VMs to use short-lived NATS bootstrap credentials restarts the BOSH Agent in all VMs as they are being created, so VMs are created somewhat more slowly.

This configuration affects all new deployments of Tanzu Operations Manager. If you want to configure VMs in existing Tanzu Operations Manager deployments to use short-lived bootstrap credentials, you must recreate them.

7. Select Recreate VMs deployed by the BOSH Director to force BOSH to recreate BOSH-deployed VMs on the next deployment. This process does not recreate the BOSH Director VM or destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
8. Select Recreate BOSH Director VM to force the BOSH Director VM to be recreated on the next deploy. This process does not destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
9. Select Recreate All Persistent Disks to force BOSH to migrate and recreate persistent disks for the BOSH Director and all tiles. This process does not destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
10. Select Enable bosh deploy retries to instruct Tanzu Operations Manager to retry failed BOSH operations up to five times.
11. Select Skip Director Drain Lifecycle to prevent drain scripts from running when the BOSH Director is recreated.
12. Select Store BOSH Job Credentials on tmpfs (beta) to store credentials for BOSH jobs on temporary file storage (tmpfs) memory, rather than on disk. You must re-create all BOSH-deployed VMs for this setting to take effect.
13. Select Keep Unreachable Director VMs to preserve BOSH Director VMs after a failed deployment for troubleshooting purposes.
14. (Optional) Select Enable automatic rotation of the BOSH DNS CA certificate (experimental) if you want Tanzu Operations Manager to automatically rotate the BOSH DNS CA certificate and

associated leaf certificates when you click Apply Changes. For more information about this feature, see [Automatic rotation of BOSH DNS certificates](#).

15. (Optional) Modify the Director Workers value, which sets the number of workers available to run BOSH Director tasks. The default value is 5.
16. (Optional) Max Threads sets the maximum number of threads that the BOSH Director can run simultaneously. VMware recommends that you leave the text box blank to use the default value, unless doing so results in rate limiting or errors on your IaaS.
17. (Optional) To add a custom URL for your BOSH Director, enter a valid hostname in the Director Hostname text box. You can also use this text box to configure a load balancer in front of your BOSH Director. For more information, see [How to set up a Load Balancer in front of Tanzu Operations Manager](#).



The screenshot shows a configuration interface for the BOSH Director. It contains three sections, each with a title, a text input field, and a descriptive subtitle.

- Director Workers:** The input field contains the number '5'. The subtitle reads: 'The number of director workers to execute director tasks. Default is 5'.
- Max Threads:** The input field is empty. The subtitle reads: 'The Max Threads Setting controls how many operations the BOSH Director can perform simultaneously'.
- Director Hostname:** The input field contains the text 'opsdirector.example.com'. The subtitle reads: 'The externally accessible hostname for the BOSH Director'.

18. (Optional) To set a custom banner that users see when you log in to the BOSH Director using SSH, enter text in the Custom SSH Banner text box.
19. (Optional) Enter your comma-separated custom Identification Tags. For example, `iaas:foundation1,hello:world`. You can use the tags to identify your foundation when viewing VMs or disks from your IaaS.
20. For Certificate Duration Overrides, you can choose whether certificates generated within Tanzu Operations Manager and CredHub use the default duration specified for the certificate, or a custom duration value that you specify.
 - Off: Use this option to use default duration for certificates created by all products.
 - On: Use this option to set a value to override the duration for certificates created by all products. If a product creates a certificate with a longer duration than the value you set, the longer duration is used.
 1. CA Certificate Duration (days): Enter the number of days for which CA certificates are valid.
 2. Leaf Certificate Duration (days): Enter the number of days for which leaf certificates are valid. This value must be less than or equal to the CA certificate duration.After you set a certificate duration override, you must take additional steps to apply the setting to all certificates. For more information, see [Overriding Duration for Certificates](#).

Certificate Duration Overrides

Set overrides for CA and leaf certificate durations.
For detailed information about which certificates are affected by this setting and which are not, see [Override Certificate Duration in the TAS for VMs or TKGi documentation](#).

☐ Off
☒ On

Enter values to set the duration for CA and leaf certificates in the fields below.
These values will be applied when new certificates are generated via new product deployment or certificate rotation.

CA Certificate Duration (days)*

1460

Enter a value between 365 and 3650 days

Leaf Certificate Duration (days)*

730

Leaf Certificate Duration must be less than or equal to the CA Certificate Duration.

21. Select HM Pager Duty Plugin to enable Health Monitor integration with PagerDuty.

- Service Key: Enter your API service key from PagerDuty.
- HTTP Proxy: Enter an HTTP proxy for use with PagerDuty.

☒ HM Pager Duty Plugin

Service Key*

YOUR_PAGERDUTY-SERVICE-KEY

HTTP Proxy

YOUR-HTTP-PROXY

22. Select HM Email Plugin to enable Health Monitor integration with email.

- Host: Enter your email host name.
- Port: Enter your email port number.
- Domain: Enter your domain.
- From: Enter the sender's address.
- Recipients: Enter a comma-separated list of addresses of intended recipients.
- Username: Enter the user name for your email server.
- Password: Enter the password for your email server.
- Enable TLS: Select this check box to enable Transport Layer Security to the email host.

☒ HM Email Plugin

Host*
smtp.example.com

Port*
25

Domain*
cloudfoundry.example.com

From*
user2.example.com

Recipients*
user2@example.com, user1.exempl

Username
user

Password
.....

☒ Enable TLS

23. For CredHub Encryption Provider, you can choose whether BOSH CredHub stores its encryption key internally on the BOSH Director and CredHub VM, or in an external hardware security module (HSM). The HSM option is more secure.

Before configuring an HSM encryption provider in the Director Config pane, you must follow the procedures and collect information described in [Preparing CredHub HSMs for configuration](#). After you deploy Tanzu Operations Manager with an HSM encryption provider, you cannot change BOSH CredHub to store encryption keys internally.

- Internal: Use this option for internal CredHub key storage. This option is selected by default and requires no additional configuration.
- Luna HSM: Use this option to use a SafeNet Luna HSM as your permanent CredHub encryption provider, and fill in the following fields:

If you use multiple HSM hosts, you must use the Tanzu Operations Manager API to set or update the HSM configuration. For more information, see [Updating a staged director's properties](#) in the Tanzu Operations Manager API documentation.

1. Encryption Key Name: Any name to identify the key that the HSM uses to encrypt and decrypt the CredHub data. Changing this key name after you deploy Tanzu Operations Manager can cause service downtime.
2. Provider Partition: The partition that stores your encryption key. Changing this partition after you deploy Tanzu Operations Manager can cause service downtime. For this value and the ones that follow, use values gathered in [Preparing CredHub HSMs for configuration](#).

3. Provider Partition Password
4. Partition Serial Number
5. Provider Client Certificate: The certificate that validates the identity of the HSM when CredHub connects as a client.
6. Provider Client Certificate Private Key
7. HSM Host Address
8. HSM Port Address: If you do not know your port address, enter 1792.
9. HSM Certificate: The certificate that the HSM presents to CredHub to establish a two-way mTLS connection.

CredHub Encryption Provider

- ☐ Internal
- ☒ Luna HSM

Encryption Key Name*

Provider Partition*

Provider Partition Password*

Partition Serial Number*

Provider Client Certificate*

Provider Client Certificate Private Key*

HSM Host Address*

HSM Port Address*

24. Click Blobstore Location to configure the blobstore as either an internal server or an external endpoint.

After you deploy the BOSH Director, only Tanzu Operations Manager users in Advanced Mode can change the blobstore location. For more information, see [Platform operator roles and permissions in Tanzu Operations Manager](#).

- Internal: Use this option for an internal blobstore. No additional configuration is required.
- S3 Compatible Blobstore: Use this option for an external S3-compatible endpoint. To create an S3 bucket, follow the procedures in [Sign up for Amazon S3](#) and [Creating a Bucket](#) in the AWS documentation. When you have created an S3 bucket, configure the following fields:

1. S3 Endpoint:

1. If you are using a public S3 endpoint:

1. Locate the endpoint for your region. To find the endpoint, see the Amazon Simple Storage Service (S3) table in [Regions and Endpoints](#) in the AWS documentation.
 2. Construct a URL using your region's endpoint. For example, if you are using the [us-west-2](#) region, the URL you create would be <https://s3.us-west-2.amazonaws.com>. Enter this URL in the S3 Endpoint field.

2. If you are using a non-public S3-compatible endpoint:

1. Enter the URL for the non-public endpoint.
 2. Add the custom public CA certificate you used to sign the S3 endpoint to the Trusted Certificates field on the Director Config page.

2. Bucket Name: Enter the name of the S3 bucket.

3. Access Key and Secret Key: Enter the keys you generated when creating your S3 bucket.

4. Enable signed URLs: Select this check box to configure BOSH VMs to generate short lived, pre-signed URLs for communication between the BOSH Agent and the blobstore. If you enable this feature, BOSH Agents do not use credentials to communicate with the blobstore, and VM disks do not store blobstore credentials. This feature is available only for BOSH VMs that use Xenial stemcell line 621 or later.

5. Select V2 Signature or V4 Signature. If you select V4 Signature, enter your Region. AWS recommends using Signature Version 4 (V4). For more information

about AWS S3 Signatures, see [Authenticating Requests](#) in the AWS documentation.

6. S3 Backup Strategy: To configure whether and how to back up the BOSH Director's S3 blobstore, select one of the following options:

- No backups
- Use a versioned bucket: This option uses blobstore bucket versioning to save backups. It requires an S3 blobstore that supports versioning.
- Copy into an additional bucket: This option saves blobstore backups to a separate bucket, which is useful with blobstores that do not support versioning. This option requires you to configure the Backup Bucket Region and Backup Bucket Name fields.

To backup and restore with BBR, if you are using S3 Compatible Blobstore and you have set up an [iam_instance_policy](#), additional steps are needed:

1. Attach the [iam_instance_policy](#) to the [backup_restore](#) job using the Tanzu Operations Manager API. See [Custom VM Extensions](#).
2. In addition to creating backup buckets, update the Elastic Application Runtime tile (Settings > File Storage) to use S3-compatible storage. See [External S3-Compatible filestore](#).
3. Ensure that your IAM Instance Policy is set up to access the backup bucket. For more information about the IAM Instance Policy, see the [Tanzu Operations Manager for AWS policy document](#).

- GCS Blobstore: Use this option for an external Google Cloud Storage (GCS) endpoint.

To create a GCS bucket, follow the procedures in [Creating Storage Buckets](#) in the GCS documentation. To create a GCS bucket, you must have a GCS account. When you have created a GCS bucket, configure the following fields:

1. Bucket Name: Enter the name of your GCS bucket.
2. Storage Class: Use this storage class for your GCS bucket. For more information, see [Storage Classes](#) in the GCP documentation.
3. Service Account Key: Follow the steps in [Set up IAM Service Accounts](#) in *Preparing to Deploy Tanzu Operations Manager on GCP* to download a JSON file with a private key. Enter the contents of the JSON file into the text box.
4. GCS Backup Strategy: To configure whether and how to back up the BOSH Director's GCS blobstore, click one of the following options:
 - No backups
 - Copy into an additional bucket: This option saves blobstore backups to a bucket separate from the blobstore itself, and requires you to configure the Backup Bucket Name text box.

Blobstore Location

- ☐ Internal
- ☐ S3 Compatible Blobstore
- ☒ GCS Blobstore

Bucket Name*

Storage Class*

Regional

Service Account Key*

Paste the JSON file containing the private key for your GCP service account.

GCS Backup Strategy

- ☒ No backups
- ☐ Copy into an additional bucket

Backup Bucket Name*

25. Click Save.

Step 4: Create Networks page

Create the following networks:

- [Infrastructure Network](#)
- [Runtime Network](#)
- [Services Network](#)

The Azure portal sometimes displays the names of resources with incorrect capitalization. Always use the Azure CLI to retrieve the correctly capitalized name of a resource.

Infrastructure network

To create the infrastructure network:

1. Click Create Networks.
2. (Optional) Click Enable ICMP checks if you want to enable ICMP on your networks. Tanzu Operations Manager uses ICMP checks to confirm that components in your network are reachable.
3. Click Add Network. For name, enter **infrastructure**.
4. Under Subnets, create a subnet using the information in the following table:

Name	infrastructure
Azure network name	Enter your Azure network name in the format provided on the UI. You can use either the NETWORK-NAME/SUBNET-NAME format or the RESOURCE-GROUP/NETWORK-NAME/SUBNET-NAME format. If you specify a resource group, it must exist under the same subscription ID that you provided in the Azure Config Page .
CIDR	Enter 10.0.4.0/26 .
Reserved IP ranges	Enter the first nine IP addresses of the subnet. For example, 10.0.4.1-10.0.4.9 .
DNS	Enter 168.63.129.16 .
Gateway	Enter the first IP address of the subnet. For example, 10.0.4.1 .

infrastructure

Name*

infrastructure

A unique name for this network

Subnets*

Add Subnet

Azure Network Name*

tcf-virtual-network/tcf-infrastructure

Should be of the format "<network-name>/<subnet-name>/<region-name>" or "<shared-vpc-host-project-name>/<network-name>/<subnet-name>/<region-name>"

CIDR*

10.0.8.0/26

A valid CIDR block in which to deploy VMs, e.g. "10.9.9.0/24"

Reserved IP Ranges*

10.0.8.1-10.0.8.22

Tanzu Ops Manager will not deploy VMs to any IP in this range, e.g. "10.9.9.0-10.9.9.100, 10.9.9.200-10.9.9.255"

DNS*

8.8.8.8

One or more Domain Name Servers used by VMs

Gateway*

10.0.8.1

The network gateway IP to be used by VMs

Availability Zones *

☐ us-central1-a

SAVE

5. Click Save.

After you deploy Tanzu Operations Manager, you add subnets with overlapping AZs to expand your network. For more information about configuring additional subnets, see [Expanding your Network with Additional Subnets](#).

Runtime network

To create the runtime network:

1. Click Add Network. For Name, enter a unique name for this network. Use a name that indicates the runtime you are using.
2. Under Subnets, create a subnet using the information in the following table:

Name	Enter a unique name for this network.
Azure network name	Enter your Azure network name in the format provided on the UI. You can use either the NETWORK-NAME/SUBNET-NAME format or the RESOURCE-GROUP/NETWORK-NAME/SUBNET-NAME format. If you specify a resource group, it must exist under the same subscription ID that you provided in the Azure Config Page .
CIDR	Enter 10.0.12.0/22 .
Reserved IP ranges	Enter the first nine IP addresses of the subnet. For example, 10.0.12.1-10.0.12.9 .
DNS	Enter 168.63.129.16 .
Gateway	Enter the first IP address of the subnet. For example, 10.0.12.1 .

TCF

Name*

TCF

A unique name for this network

Subnets*

Azure Network Name*

tcf-virtual-network/tcf-subnet

Should be of the format "<network-name>/<subnet-name>/<region-name>" or "<shared-vpc-host-project-name>/<network-name>/<subnet-name>/<region-name>"

CIDR*

10.0.12.0/22

A valid CIDR block in which to deploy VMs, e.g. '10.9.9.0/24'

Reserved IP Ranges*

10.0.12.1-10.0.12.9

Tanzu Ops Manager will not deploy VMs to any IP in this range, e.g. '10.9.9.0-10.9.9.100, 10.9.9.200-10.9.9.255'

DNS*

168.129.16

One or more Domain Name Servers used by VMs

Gateway*

10.0.12.1

The network gateway IP to be used by VMs

Availability Zones *

☐ us-central1-a

SAVE

3. Click Save.

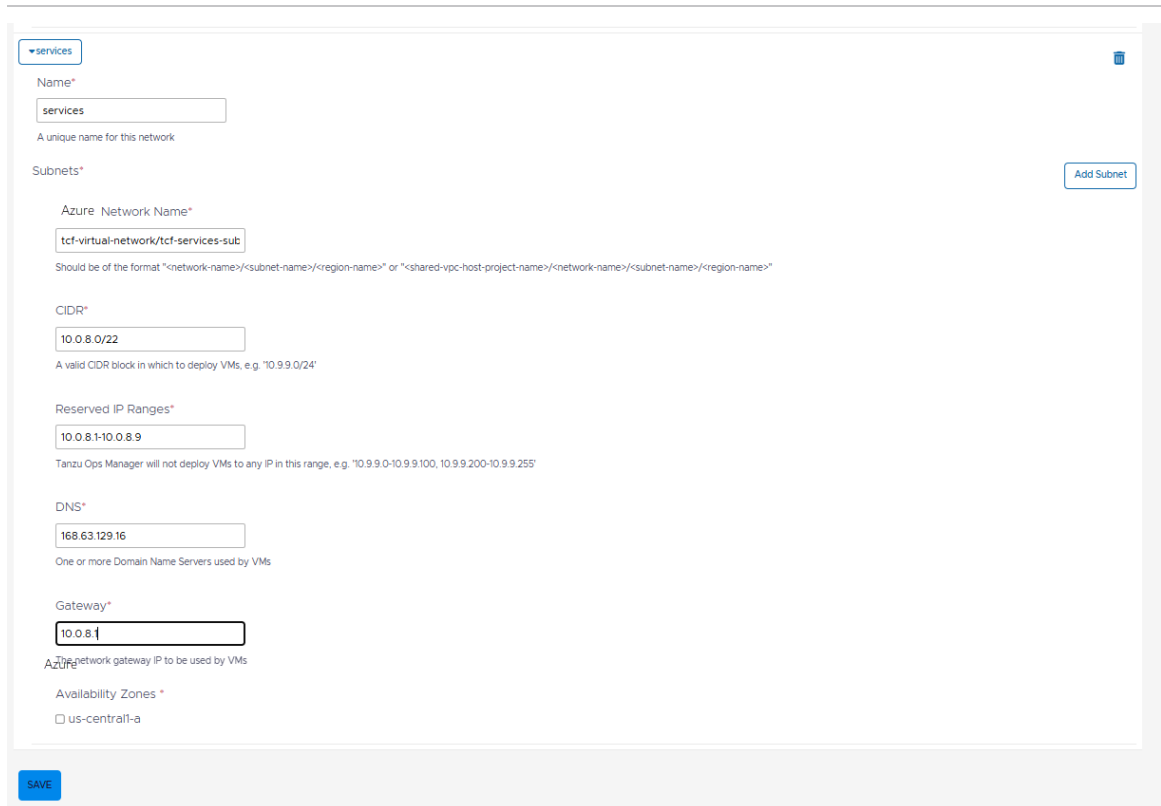
Services network

To create the services network:

1. Click Add Network and in the Name text box, enter **services**.
2. Under Subnets, create a subnet using the information in the following table:

Name	services
Azure network name	Enter your Azure network name in the format provided on the UI. You can use either the NETWORK-NAME/SUBNET-NAME format or the RESOURCE-GROUP/NETWORK-NAME/SUBNET-NAME format. If you specify a resource group, it must exist under the same subscription ID that you provided in the Azure Config Page .

CIDR	Enter 10.0.8.0/22 .
Reserved IP ranges	Enter the first nine IP addresses of the subnet. For example, 10.0.8.1-10.0.8.9 .
DNS	Enter 168.63.129.16 .
Gateway	Enter the first IP address of the subnet. For example, 10.0.8.1 .



3. Click Save.

You might see red warnings. You can ignore them *only if* Enable ICMP checks is not selected.

Step 5: Assign AZs and Networks

- Click Assign AZs and networks.
The appearance of this page changes based on whether you are using Availability Zones or Availability Sets. For more information, see [Azure Config Page](#).
 - If you use Availability Zones, click one of the three available zones from the Singleton Availability Zone drop-down menu, then click a network from the Networks drop-down menu. The networks that appear in this list are networks you created earlier in this procedure.
 - If you use Availability Sets, click a network from the Networks drop-down menu. The networks that appear in this list are networks you created earlier in this procedure.
- Click Save.

Step 6: Security page

1. Click Security.

Security

Trusted Certificates

These certificates enable BOSH-deployed components to trust a custom root certificate.

☐ Include Tanzu Ops Manager Root CA in Trusted Certs
Checking this box will place the Tanzu Ops Manager Root CA into the trusted certificate field. The Bosh Director will then place this CA into the trust store of every VM that it deploys.

☐ Clear the Default Trusted Certificates Store
Deletes all the certificates in the system certificate trust store every time BOSH deploys a VM. Certificates added in the "Trusted Certificates" field will still be trusted and you should add the certificates required to access your IaaS Provider services.

Generate VM passwords or use single password for all VMs

☒ Generate passwords

☐ Use default BOSH password

SAVE

2. In Trusted Certificates, paste in your custom certificate authority (CA) certificates to insert into your organization's certificate trust chain. This feature enables all BOSH-deployed components in your deployment to trust custom root certificates. If you want to use Docker registries to run Elastic Application Runtime app instances in Docker containers, enter the certificate for your private Docker registry in this text box. For more information about running app instances in Elastic Application Runtime using Docker registries, see [Using Docker Registries](#).

To enter multiple certificates, paste in your certificates one after the other. For example, format your certificates as shown in the following example:

```
-----BEGIN CERTIFICATE-----
ABCDEF12345678ABCDEF12345678ABCDEF12345678AB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
BCDEF12345678ABCDEF12345678ABCDEF12345678ABB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
CDEF12345678ABCDEF12345678ABCDEF12345678ABBB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
```

3. To include both the Tanzu Operations Manager root CA and the certificates pasted into Trusted Certificates in the `trusted_certs` field in the BOSH director manifest, select the Include Tanzu Ops Manager Root CA in Trusted Certs check box. BOSH Director includes this CA in the trust store of every VM that it deploys.
You must select this check box if you are using Tanzu Operations Manager to generate certificates for a load balancer or a router.
4. To clear the default trusted certificates from all BOSH-deployed Linux VMs, select Clear the Default Trusted Certificates Store.

If you select this check box, you must provide your own trusted certificates. All TLS communication fails if trusted certificates are not available.

5. Select Generate passwords or Use default BOSH password. VMware recommends using the Generate passwords option for greater security.
6. Click Save. To view your saved BOSH Director password, click the Credentials tab.

Step 7: BOSH DNS Config page

In the BOSH DNS Config pane, you can configure DNS for BOSH Director by adding excluded recursors, a recursor timeout, and handlers.

The text boxes in the BOSH DNS Config pane are:

- **Excluded Recursors:** Exclude recursor addresses, which are URL redirects, so that they are not contacted by the BOSH DNS server. For more information about how the BOSH DNS release selects recursors, see [Recursors](#) in *Native DNS Support* in the BOSH documentation.
- **Recursor Timeout:** Specify a timeout for the BOSH DNS server when contacting any connected recursor addresses. The time limit includes dialing, writing, and reading from the recursor. If any of these actions exceeds the time limit, the action fails.
- **Handlers:** Specify recursor addresses that apply to specific domains. For example, you can use handlers to forward all requests to a domain to a private DNS for resolution. For more information about using handlers, see [Additional handlers](#) in *Native DNS Support* in the BOSH documentation.

To add excluded recursors, a recursor timeout, or handlers to the BOSH DNS release:

1. Click BOSH DNS Config.

The screenshot shows the 'BOSH DNS Config' interface. It has a title bar 'BOSH DNS Config'. Below it are three sections: 'Excluded Recursors' with a single-line text input field; 'Recursor Timeout' with a single-line text input field; and 'Handlers' with a multi-line text input field containing '[]'. To the right of the Handlers input field is the text 'Enter a JSON-formatted list of optional custom domain handlers'. At the bottom left is a blue 'Save' button.

2. (Optional) In Excluded Recursors, enter a list of prohibited recursor addresses.

3. (Optional) In Recursor Timeout, enter a time limit for contacting the connected recursors. This time limit must include one of the Go parse duration time units. For example, entering **5s** sets the timeout limit to five seconds. For more information about supported time units, see [func ParseDuration](#) in the Go Programming Language documentation.
4. (Optional) In Handlers, enter an array of custom domain handlers in JSON format. For example:

```
[
  {
    "cache": {
      "enabled": true
    },
    "domain": "example.com",
    "source": {
      "type": "http",
      "url": "http://example.endpoint.local"
    }
  }
]
```

5. Click Save.

Step 8: Syslog page

1. Click Syslog.
2. (Optional) To send BOSH Director system logs to a remote server, select Yes.
3. In Address, enter the IP address or DNS name for the remote server.
4. In Port, enter the port number that the remote server listens on.
5. Using the Transport Protocol drop-down menu, select TCP or UDP to specify the transport protocol to use to send the logs to the remote server.
6. (Optional) Select the Enable TLS check box to send encrypted logs to remote server with TLS. After you select the check box:
 1. In Permitted Peer, enter either the name or SHA1 fingerprint of the remote peer.
 2. In SSL Certificate, enter the SSL certificate for the remote server.

VMware recommends that you enable TLS encryption when you are forwarding logs. Logs can contain sensitive information, such as cloud provider credentials.

7. (Optional) In Queue Size, enter an integer that specifies the number of log entries held in the buffer. The default value is 100,000.
8. (Optional) Select the Forward Debug Logs check box to forward the logs to an external source. This option is deselected by default. If you select it, you might generate a large amount of log data.
9. (Optional) In the Custom rsyslog Configuration text box, enter the configuration details for **rsyslog**. This text box requires the rainerscript syntax.
10. Click Save.

Step 9: Resource Config page

1. Click Resource Config.

The screenshot shows the 'Resource Config' interface with a table of configurations. The table has four columns: JOB, INSTANCES, VM TYPE, and PERSISTENT DISK TYPE. There is a 'SAVE' button in the top right corner.

JOB	INSTANCES	VM TYPE	PERSISTENT DISK TYPE
> BOSH Director	Automatic: 1	Automatic: e2-standard-2.disk (machine_type: e2-standard-2, cpu: 2, ram: 8 GB, di...	Automatic: 50 GB
> Main Compilation Job	1	e2-highmem-2 (machine_type: e2-highmem-2, cpu: 2, ram: 16 GB, disk: 64 GB)	None

- Under the Instances, VM Type, and Persistent Disk Type fields, click Automatic from the drop-down menu to allocate the recommended resources for the job. If the Persistent Disk Type field displays None, the job does not require persistent disk space. For the BOSH Director job, click a VM type with at least 8 GB memory.

If you set a field to Automatic and the recommended resource allocation changes in a future version, Tanzu Operations Manager automatically uses the updated recommended allocation.

If you install Elastic Application Runtime Windows add-on, provision your Main Compilation Job with at least 100 GB of disk space.

- (Optional) To add a load balancer to the BOSH Director:
 - Click the icon next to the BOSH Director job name to see the Load Balancers field and Internet Connected check box.
 - Ensure that the Internet Connected check box is not selected.
 - For Load Balancers, enter your Azure application gateway name for each associated job. To create an application gateway, follow the procedures in [Configure an application gateway for SSL offload by using Azure Resource Manager](#) from the Azure documentation. When you create the application gateway, associate the gateway's IP address with your system domain.

This feature is not recommended for production use. The Azure load balancer does not support an override port in the healthcheck configuration.

- Click Save.

Step 10: Add custom VM extensions (optional)

Use the Tanzu Operations Manager API to add custom properties to your VMs; associated security groups and load balancers, for example. For more information, see [Managing custom VM extensions](#).

Step 11: Complete the BOSH Director installation

- Click Review Pending Changes, and then Apply Changes. If a red ICMP error message appears and you have deactivated ICMP, click Ignore errors and start the install.
- BOSH Director installation begins. This might take a few moments. When the installation process completes successfully, the Changes Applied window appears.
- Click the Tanzu Operations Manager Installation Dashboard link to return to the Installation Dashboard.

Next steps

- (Optional) When Tanzu Operations Manager finishes deploying, deploy BOSH Add-ons to your system. For more information, see [Deploying BOSH Add-Ons](#).

- Install one or more Tanzu Operations Manager runtime environments to complete your installation. For more information, see [Tanzu Operations Manager Runtimes](#).

Troubleshooting Tanzu Operations Manager on Azure

This topic describes how you can troubleshoot known issues when deploying or running Tanzu Operations Manager on Azure.

Installation issues

This section describes how to troubleshoot known installation issues.

Problem: Cannot copy the Tanzu Operations Manager image

Symptom	You cannot copy the Tanzu Operations Manager image into your storage account as part of Step 4: Boot Tanzu Operations Manager in <i>Deploying Tanzu Operations Manager on Azure</i> .
Explanation	You have an outdated version of the Azure CLI. You need Azure CLI v2.0.0 or later. To display your current Azure CLI version, run: <pre>az --version</pre>
Solution	Install the Azure CLI v2.0 by following the instructions for your operating system in Install the Azure CLI in the Microsoft Azure documentation.

Problem: Deployment fails at “create-env”

Symptom	After you click Review Pending Changes, then Apply Changes to install Tanzu Operations Manager and Tanzu Platform for Cloud Foundry, the deployment fails at <code>create-env</code> with an error message similar to the following: <pre>Command 'deploy' failed: Deploying: Creating instance 'bosh/0': Waiting until instance is ready: Starting SSH tunnel: Parsing private key file '/tmp/bosh_ec2_private_key.pem': asn1: structure error: tags don't match (16 vs {class:3 tag:28 length:127 isCompound:false}) {optional:false explicit:false application:false defaultValue: tag: stringType:0 timeType:0 set:false omitEmpty:false} pkcs1PrivateKe y @2 ===== 2016-09-29 16:28:22 UTC Finished "bosh create-env" /var/tempest/workspaces/default/deployments/bosh.yml"; Duration: 328s; Exit Status: 1 Exited with 1.</pre>
Explanation	You provided a passphrase when you created your key pair in Step 4: Boot Tanzu Operations Manager in <i>Deploying Tanzu Operations Manager on Azure</i> .
Solution	Create a new key pair with no passphrase and do the installation again. For more information, see Step 4: Boot Tanzu Operations Manager in <i>Deploying Tanzu Operations Manager on Azure</i> .

Insufficient external database permissions

Upgrade issues can arise when the external database user that was used for the network policy database is given insufficient permissions. To avoid this upgrade issue, ensure that the Network Policy Server database user has the **ALL PRIVILEGES** permission.

Operational issues

This section describes how to troubleshoot known operational issues.

Problem: Slow performance or timeouts

Symptom	Developers suffer from slow performance or timeouts when pushing or managing apps, and end users suffer from slow performance or timeouts when accessing apps.
Explanation	The Azure Load Balancer (ALB) disconnects active TCP connections when they have been idle for more than four minutes.
Solution	To mitigate slow performance or timeouts, the default value of the Back end request and idle timeout field is set to 900 seconds, but operators can tune this parameter to fit their infrastructure. To edit the Back end request and idle timeout field: 1. Go to Tanzu Operations Manager Installation Dashboard. 2. Click the Elastic Application Runtime tile. 3. Click Networking. 4. Enter your desired time, in seconds, in the Back end request and idle timeout text box. 5. Click Save.

Problem: Service instance creation times out

Symptom	You are unable to provision a service instance of a Java or Go service. Running cf create-service fails with error Failure provisioning service instance... Timed out after 8 minutes... or similar.
Explanation	HTTP libraries for Java and Go, running with default settings, prune connections that have been idle for 240 seconds from their connection pool without sending a TCP reset message back to the client service broker. This removes the ability of the broker to provision new service instances.
Solution	In the Networking pane of the Elastic Application Runtime tile, enter a value of 240 or less in the Front end idle timeout field, so that Tanzu Operations Manager regenerates the front end connection before it times out.

Upgrading BOSH Director for Tanzu Operations Manager on Azure

You can upgrade BOSH Director for Tanzu Operations Manager on Azure. Complete these tasks as part of the Tanzu Operations Manager upgrade process.

For information about upgrading Tanzu Operations Manager, see [Upgrading Tanzu Operations Manager](#).

In this procedure, you create a Tanzu Operations Manager VM instance to host the upgraded version of Tanzu Operations Manager. Then, to complete the Tanzu Operations Manager upgrade, you export your existing Tanzu Operations Manager installation onto this new VM.

The Azure portal sometimes displays the names of resources with incorrect capitalization. Always use the Azure CLI to retrieve the correctly capitalized name of a resource.

To create a Tanzu Operations Manager VM instance:

1. Export environment variables. For more information, see [Export Environment Variables](#).
2. Copy the Tanzu Operations Manager image into your storage account. For more information, see [Copy the Tanzu Operations Manager image](#).
3. Configure the Tanzu Operations Manager IP Address. For more information, see [Configure Tanzu Operations Manager IP Address](#).
4. Create the Tanzu Operations Manager VM instance. For more information, see [Create Tanzu Operations Manager VM Instance](#).

Prerequisites

To complete the Tanzu Operations Manager upgrade, you must have your Tanzu Operations Manager decryption passphrase. You defined this decryption passphrase during the initial installation of Tanzu Operations Manager.

Export environment variables

Export the `connectionString` variable for your Azure account and assign it to a new environment variable. In the following procedure, you use the `connectionString` environment variable when you copy the Tanzu Operations Manager image.

To export environment variables:

1. Install the Azure CLI v2.0 by following the instructions for your operating system in [Install the Azure CLI](#), in the Microsoft Azure documentation.
2. Set your cloud by running:

```
az cloud set --name AzureCloud
```

If you deployed Tanzu Operations Manager in an environment other than Azure Cloud, consult the following list:

- For Azure China, replace `AzureCloud` with `AzureChinaCloud`. If logging in to `AzureChinaCloud` fails with a `CERT_UNTRUSTED` error, use Node v4 or later.
- For Azure Government Cloud, replace `AzureCloud` with `AzureUSGovernment`.

3. Log in:

```
az login
```

Authenticate by going to the URL in the output, entering the provided code, and clicking your account.

4. Ensure that the following environment variables are set to the names of the resources you created when deploying Tanzu Operations Manager. This is included in the procedures in [Deploying Tanzu Operations Manager on Azure](#):
 - `$RESOURCE_GROUP`: This must be set to the name of your resource group. To list the resource groups for your subscription, run:

```
az group list
```

- **\$LOCATION**: This must be set to your location, such as **westus**. To view a list of available locations, run:

```
az account list-locations
```

- **\$STORAGE_NAME**: This must be set to your Tanzu Operations Manager storage account name. To list your storage accounts, run:

```
az storage account list
```

5. Retrieve the connection string for the account by running:

```
az storage account show-connection-string \
--name $STORAGE_NAME --resource-group $RESOURCE_GROUP
```

The command returns output similar to the following:

```
{
  "connectionString": "DefaultEndpointsProtocol=https;EndpointSuffix=core.windows.net;AccountName=cfdocsboshstorage;AccountKey=EXAMPLEaleiIy/Lprnc5igFXYWsgq016Tu9uGwse0l8bqNBEL/2tp7wX92QMUM19Pz9BYTXt8aq4A=="
}
```

6. From the **data:** field in the preceding output, record the full value of **connectionString**, starting with and including **DefaultEndpointsProtocol=**.
7. Export the value of **connectionString** as the environment variable **\$AZURE_STORAGE_CONNECTION_STRING** by running:

```
export AZURE_STORAGE_CONNECTION_STRING="ACCOUNT-KEY-STRING"
```

Where **ACCOUNT-KEY-STRING** is the value of **connectionString**.

Copy the Tanzu Operations Manager Image

Copy the Tanzu Operations Manager image into your storage account. You use the Tanzu Operations Manager image to create the VM that hosts the new version of Tanzu Operations Manager.

To copy the Tanzu Operations Manager image into your storage account:

1. Go to the [Tanzu Operations Manager](#) page on the Broadcom Support portal and download the release of Tanzu Operations Manager for Azure to which you want to upgrade.
2. View the downloaded PDF and locate the Tanzu Operations Manager image URL appropriate for your region.
3. Export the Tanzu Operations Manager image URL as an environment variable by running:

```
export OPS_MAN_IMAGE_URL="OPS-MANAGER-IMAGE-URL"
```

Where **OPS-MANAGER-IMAGE-URL** is the URL of your Tanzu Operations Manager image.

This command overrides the old Tanzu Operations Manager image URL with the new Tanzu Operations Manager image URL.

4. Copy the Tanzu Operations Manager image into your storage account. For compatibility when upgrading to future versions of Tanzu Operations Manager, choose a unique name for the image

that includes the Tanzu Operations Manager version number. For example, replace `opsman-image-version` in the following examples with `opsman-image-3.0.1`. Run:

```
az storage blob copy start --source-uri $OPS_MAN_IMAGE_URL \
--connection-string $AZURE_STORAGE_CONNECTION_STRING \
--destination-container opsmanager \
--destination-blob opsman-image-version.vhd
```

5. Copying the image might take several minutes. Run:

```
az storage blob show --name opsman-image-version.vhd \
--container-name opsmanager \
--account-name $STORAGE_NAME
```

6. Examine the output under `"copy"`, as in the following example:

```
"copy": {
  "completionTime": "2017-06-26T22:24:11+00:00",
  "id": "b9c8b272-a562-4574-baa6-f1a04afcefd",
  "progress": "53687091712/53687091712",
  "source": "https://opsmanagerwestus.blob.core.windows.net/images/ops-manager-3.0.x.vhd",
  "status": "success",
  "statusDescription": null
},
```

When `status` reads `success`, continue to the next step.

Configure Tanzu Operations Manager IP Address

Remove the old Tanzu Operations Manager VM and record the IP address for the Tanzu Operations Manager network interface. You use this IP address when creating the new Tanzu Operations Manager VM that hosts the upgraded version of Tanzu Operations Manager.

To configure the Tanzu Operations Manager IP address, follow one of these procedures:

- [Reuse existing Dynamic Public IP Address](#)
- [Use a new Dynamic Public IP Address](#)

Reuse existing Dynamic Public IP Address

To reuse an existing dynamic public IP address:

1. List your VMs and record the name of your Tanzu Operations Manager VM by running:

```
az vm list
```

2. Delete your old Tanzu Operations Manager VM by running:

```
az vm delete --name OPS-MANAGER-VM --resource-group $RESOURCE_GROUP
```

Where `OPS-MANAGER-VM` is your old Tanzu Operations Manager VM.

3. List your network interfaces and record the name of the Tanzu Operations Manager network interface. Run:

```
az resource list --resource-group $RESOURCE_GROUP \
--resource-type Microsoft.Network/networkInterfaces
```

Use a new Dynamic Public IP Address

To use a new dynamic public IP address:

1. Create a new public IP address named `ops-manager-ip-new` by running:

```
az network public-ip create --name ops-manager-ip-new \
--resource-group $RESOURCE_GROUP --location $LOCATION \
--allocation-method Static
{
  "publicIp": {
    "dnsSettings": null,
    "etag": "W/\"4450ebe2-9e97-4b17-9cf2-44838339c661\"",
    "id": "/subscriptions/995b7eed-77ef-45ff-a5c9-1a405ffb8243/resourceGroups/cf-docs/providers/Microsoft.Network/publicIPAddresses/ops-manager-ip-new",
    "idleTimeoutInMinutes": 4,
    "ipAddress": "40.83.148.183",
    "ipConfiguration": null,
    "location": "westus",
    "name": "ops-manager-ip-new",
    "provisioningState": "Succeeded",
    "publicIpAddressVersion": "IPv4",
    "publicIpAllocationMethod": "Static",
    "resourceGroup": "cf-docs",
    "resourceGuid": "950d4831-1bec-42da-8a79-959bcddea9dd",
    "tags": null,
    "type": "Microsoft.Network/publicIPAddresses"
  }
}
```

2. Record the value for `ipAddress` from the output of the preceding step. This is the public IP address of Tanzu Operations Manager.
3. Create a network interface for Tanzu Operations Manager by running:

```
az network nic create --vnet-name pcf-net \
--subnet pcf --network-security-group opsmgr-nsg \
--private-ip-address 10.0.0.5 \
--public-ip-address ops-manager-ip-new \
--resource-group $RESOURCE_GROUP \
--name ops-manager-nic-new --location $LOCATION
```

4. Shut down your old Tanzu Operations Manager VM, if it still exists, and run:

```
az vm deallocate --name ops-manager --resource-group $RESOURCE_GROUP
```

If your Tanzu Operations Manager VM is not named `ops-manager`, provide the correct name. To list all VMs in your account, run:

```
az vm list
```

5. Update your DNS record to point to the new public IP address of Tanzu Operations Manager.

Create Tanzu Operations Manager VM Instance

Create a Tanzu Operations Manager VM instance to host the new version of Tanzu Operations Manager.

To create a Tanzu Operations Manager VM instance:

1. To use the key pair from your previous Tanzu Operations Manager, locate the path to the file on your local machine. To create a new key pair, enter:

```
ssh-keygen -t rsa -f opsman -C ubuntu
```

When prompted for a passphrase, press the **Enter** key to provide an empty passphrase.

2. Create the Tanzu Operations Manager VM.

1. Create a managed image from the Tanzu Operations Manager VHD file by running:

```
az image create --resource-group $RESOURCE_GROUP \
--name OPS-MANAGER-VERSION \
--source https://$STORAGE_NAME.blob.core.windows.net/opsmanager/OPS-MANAGER-VERSION.vhd \
--location $LOCATION \
--os-type Linux
```

Where **OPS-MANAGER-VERSION** is the version of Tanzu Operations Manager you are deploying.

If you are using Azure China or Azure Government Cloud, replace **blob.core.windows.net** with the following:

- For Azure China, use **blob.core.chinacloudapi.cn**. For more information, see [Azure China developer guide](#) in the Microsoft Azure documentation.
- For Azure Government Cloud, use **blob.core.usgovcloudapi.net**. For more information, see [Azure Government storage](#) in the Microsoft Azure documentation.

2. Create your Tanzu Operations Manager VM by running:

```
az vm create --name opsman-version --resource-group $RESOURCE_GROUP \
--location $LOCATION \
--nics opsman-nic \
--image OPS-MANAGER-IMAGE-VERSION \
--os-disk-name OPS-MANAGER-VERSION-osdisk \
--admin-username ubuntu \
--size Standard_D2s_v3 \
--storage-sku Standard_LRS \
--ssh-key-value PATH-TO-PUBLIC-KEY
```

Where:

- **opsman-version** is the Tanzu Operations Manager you are deploying. For example, **opsman-version-3.0.1**.
 - **OPS-MANAGER-IMAGE-VERSION** is the image version of Tanzu Operations Manager you are deploying. For example, **opsman-image-3.0.1**.
 - **OPS-MANAGER-VERSION** is the version of Tanzu Operations Manager you are deploying. For example, **opsman-3.0.1**.
 - **PATH-TO-PUBLIC-KEY** is the path to your public key **.pub** file.
3. If you deploy more than one tile in this Tanzu Operations Manager installation, increase the size of the Tanzu Operations Manager VM disk. You can repeat this process and increase the disk again at a later time if necessary.

1. Stop the VM and detach the disk by running:

```
az vm deallocate --name OPS-MANAGER-VERSION \  
--resource-group $RESOURCE_GROUP
```

Where **OPS-MANAGER-VERSION** is the version of Tanzu Operations Manager you are deploying. For example, **opsman-3.0.1**.

2. Resize the disk to 128 GB by running:

```
az disk update --size-gb 128 --name OPS-MANAGER-VERSION-osdisk \  
--resource-group $RESOURCE_GROUP
```

Where **OPS-MANAGER-VERSION** is the version of Tanzu Operations Manager you are deploying. For example, **opsman-3.0.1**.

3. Start the VM by running:

```
az vm start --name OPS-MANAGER-VERSION --resource-group $RESOURCE_GROUP
```

Where **OPS-MANAGER-VERSION** is the version of Tanzu Operations Manager you are deploying. For example, **opsman-3.0.1**.

Next steps

After you complete this procedure, continue to the upgrade instructions in [Upgrading Tanzu Operations Manager](#).

Deleting your Tanzu Operations Manager deployment from Azure

This topic describes how you can delete the Azure resources associated with a Tanzu Operations Manager deployment. The fastest way to remove resources is to delete the resource group (or groups) associated with your Tanzu Operations Manager on Azure installation.

Delete the resource group

To delete a resource group:

1. Go to the [Azure Portal](#) and log in to your subscription.
2. From inside your subscription, click Resource Groups.
3. Click the resource group you want to delete.
4. In the details pane for the resource group, click the trash can icon. Review the information in the confirmation pane before you proceed.
5. To confirm deletion, type the resource group name and click Delete.

For more information about managing resource groups in Azure, see the [Azure documentation](#).

Migrating from Storage Accounts to Managed Disks on Azure

This topic describes how to migrate from using Storage Accounts (unmanaged disks) to Managed Disks on Azure with Tanzu Operations Manager.

Tanzu Operations Manager allows users on Azure to configure the BOSH Director tile between two Cloud Storage Type options: Use Storage Accounts and Use Managed Disks. When Use Storage Accounts is selected, BOSH places all disks associated with VMs in the configured Azure storage account created by

the user. This is referred to as “unmanaged disks” in Azure. When Use Managed Disks is selected, BOSH allows the Azure platform to manage the storage of disks.

Azure has announced that unmanaged disks will no longer be supported starting on September 30th, 2025. After that date, any VMs that have unmanaged disks attached to them will be automatically stopped and will be unlaunchable. Users that have configured the BOSH Director to use storage accounts must migrate to using managed disks before that date.

For more information, see [Migrate your Azure unmanaged disks by September 30, 2025](#).

The migration procedure requires redeploying all virtual machines managed by Tanzu Operations Manager and BOSH. All ephemeral and persistent disks will be recreated during the redeploy. It is advised that you set a maintenance window that is appropriate for the number of VMs.

The migration procedure requires manual cleanup after finishing the redeploy. It is strongly advised that you run the cleanup steps immediately after completing the migration. (See [Post-migration cleanup](#), after the migration procedure on this page.)

Prerequisites

- You are running Tanzu Operations Manager v3.0.38 or later.
- In the BOSH Director tile, on the Azure Config page, you have previously selected Use Storage Accounts for the Cloud Storage Type field.

Migrate procedure

Migrate the Tanzu Operations Manager VM

To migrate the Tanzu Operations Manager VM, follow the instructions in [Upgrading BOSH Director for Tanzu Operations Manager on Azure](#) using managed disks instead of storage accounts.

Migrate BOSH-deployed VMs

1. On the Dashboard page, click the BOSH Director tile.
2. In Cloud Storage Type, select Use Managed Disks.
3. (Optional) If you want to use a customer-managed encryption key, enter the name of the Disk Encryption Set resource in Azure into the Custom Disk Encryption Set Name field. By default, Azure encrypts managed disks using platform-managed encryption keys.
4. Click Save.
5. On the Dashboard page, click Review Pending Changes.
6. Select all tiles. For each service tile, select its corresponding upgrade errands to ensure that service instances are redeployed.
7. Click Apply Changes.
8. If you have manually deployed manifests outside of Tanzu Operations Manager using [bosh deploy](#), then redeploy each of those deployments.
9. On the Dashboard page, click the BOSH Director tile.
10. Clear the Deployments Storage Account Name field. This field is no longer needed when using managed disks.
11. Click Save.

12. On the Dashboard page, click Review Pending Changes.
13. Unselect all tiles except for BOSH Director.
14. Click Apply Changes.

After the Apply Changes has completed, all VMs have been migrated to use managed disks. You can verify that the migration is complete in the Azure Portal UI by checking each of the disks attached to BOSH-deployed VMs to see if they are managed.

Post-migration cleanup

Before cleaning up old disks, ensure that you have redeployed all VMs associated with tiles, service instances, and deployments made using `bosh deploy` outside of Tanzu Operations Manager. Deleting a disk that is still in use causes downtime.

The migration procedure does not delete the original disks stored in storage accounts. After you confirm that all disks have been migrated to use managed disks, the remaining disks in any previously configured storage accounts can be deleted.

If you chose to use a customer-managed encryption key, then BOSH initially converts each disk in the storage accounts to a managed disk encrypted by a platform-managed key, then create the final managed disk, encrypted using the customer-managed key. You can view a list of these disks by running `bosh disks -orphaned`. These disks can be deleted after 5 days by running `bosh clean-up`, or immediately by running `bosh clean-up --all`.

Installing and configuring Tanzu Operations Manager on GCP

The following topics tell you how to install and configure Tanzu Operations Manager on GCP:

- [Requirements and prerequisites for Tanzu Operations Manager on GCP](#)
- [Recommended GCP resource quotas for Tanzu Operations Manager](#)
- [Installing Tanzu Operations Manager on GCP manually](#)
- [Preparing to deploy Tanzu Operations Manager on GCP](#)
- [Deploying Tanzu Operations Manager on GCP](#)
- [Configuring BOSH Director on GCP](#)
- [Configuring a shared VPC on GCP with Tanzu Operations Manager](#)
- [Upgrading BOSH Director for Tanzu Operations Manager on GCP](#)
- [Deleting your Tanzu Operations Manager deployment from GCP](#)

Requirements and prerequisites for Tanzu Operations Manager on GCP

Ensure that you meet the following requirements and prerequisites before installing Tanzu Operations Manager on Google Cloud Platform (GCP).

You can install Tanzu Operations Manager on GCP with either the Tanzu Platform for Cloud Foundry or VMware Tanzu Container Orchestration Engine (formerly TKGI) runtime. There are resource requirements specific to each runtime. Ensure that you meet the requirements for your runtime and the requirements specific to GCP before installing Tanzu Operations Manager on GCP.

Requirements

This section lists the following resource requirements for installing Tanzu Operations Manager on GCP:

- General Tanzu Operations Manager resource requirements. For more information, see [Tanzu Operations Manager Resource requirements](#).
- GCP-specific resource requirements. For more information, see [GCP resource requirements](#).

Tanzu Operations Manager Resource requirements

This section lists resource requirements for installing Tanzu Operations Manager on GCP. It includes links to general resource requirements for both the Elastic Application Runtime and Container Orchestration Engine runtimes.

See one of the following topics, depending on the runtime you plan to install using Tanzu Operations Manager:

- For Elastic Application Runtime-specific resource requirements, see [Elastic Application Runtime Resource Requirements](#).
- For Container Orchestration Engine-specific resource requirements, see [AWS prerequisites and resource requirements](#).

When launching Tanzu Operations Manager on GCP, you must use an instance type that meets the following minimum specifications: | Machine Spec | Minimum Value | | CPU | 4 vCPUs | | Memory | 16 GB | | Boot disk | 100 GB |

VMware recommends using a `n2-standard-4` instance type for the `<%= vars.ops_manager >` VM.

GCP resource requirements

The following are GCP-specific resource requirements for installing Tanzu Operations Manager on GCP:

Installing Tanzu Operations Manager on GCP requires a minimum of the following VM instance limits in your GCP account. The number of VMs required depends on the number of tiles and availability zones you plan to deploy. For Elastic Application Runtime-specific resource requirements, see [Elastic Application Runtime Resource Requirements](#).

- (Elastic Application Runtime-only) Your GCP project must have sufficient quota to deploy all the VMs needed to install Tanzu Operations Manager with Elastic Application Runtime. For a list of suggested quotas, see [Recommended GCP quotas](#).

Prerequisites

To install Tanzu Operations Manager on GCP, you must do the following:

- Install the Google Cloud SDK on your machine and authenticate it to your GCP account. To download the Google Cloud SDK, see [Google Cloud SDK](#).
- Increase or remove the VM instance limits in your GCP account. For VM instance requirements, see [GCP Resource Requirements](#), above.
- Update your GCP account with the following required permissions:
 - The following permissions are required to create firewalls, networks, load balancers, and other resources:
 - Compute Engine, then Compute Instances Admin (beta)
 - Compute Engine, then Compute Network Admin

- Compute Engine, then Compute Security Admin
- To use Google Cloud Storage (GCS) for Cloud Controller file storage, the following permission is required to create buckets:
 - Storage, then Storage Admin
- To use Cloud DNS, the following permission is required to add and modify DNS entries:
 - Project, then Owner
- Create an SSL certificate for your Tanzu Operations Manager domain.
To deploy Tanzu Operations Manager to a production environment, you must obtain a certificate from a certificate authority. VMware recommends using a self-signed certificate generated by Tanzu Operations Manager for development and testing purposes only.
- Assign administrative rights to a domain for Tanzu Operations Manager. You need to be able to add wildcard records to this domain.
- Create a wildcard DNS record that points to your router or load balancer. Alternatively, you can use a service such as xip.io; for example, `203.0.113.0.xip.io`. Then create at least one wildcard TLS certificate that matches the DNS record you configured.
With a wildcard DNS record, every hostname in your domain resolves to the IP address of your router or load balancer. For example, if you create a DNS record `*.example.com` pointing to your router, every app deployed to the `example.com` domain resolves to the IP address of your router.
- Create one or more NTP servers, if the NTP servers are not already provided by your GCP project.
- Install the most recent version of one of the following CLIs, depending on your runtime:
 - Elastic Application Runtime: The Tanzu cf CLI. For more information, see [Cloud Foundry Command Line Interface](#) in GitHub.
 - Container Orchestration Engine: The Container Orchestration Engine CLI. For more information, see [Installing the Container Orchestration Engine CLI](#).
- Elastic Application Runtime-only:
 - Request a quota increase for your GCP project. For GCP quota requirements, see [GCP Resource Requirements](#). To request an increase, see [Quotas](#) in the GCP console.
 - Configure sufficient IP allocation. For more information about IP allocation requirements, see [Elastic Application Runtime Resource Requirements](#) above.
 - (Optional) Configure external storage. VMware recommends using external storage if possible. For more information about how file storage location affects platform performance and stability during upgrades, see [Configure file storage](#).
 - (Optional) (Elastic Application Runtime and Tanzu Operations Manager-only) Configure external databases. VMware recommends using external databases in production deployments for BOSH Director and Elastic Application Runtime. An external database must be configured to use the UTC timezone.
- (Optional) Configure external user stores. When you deploy Tanzu Operations Manager, you can select SAML, LDAP, or OIDC to integrate existing user accounts.

Install Tanzu Operations Manager on GCP

You can install Tanzu Operations Manager on GCP.

To install Tanzu Operations Manager on GCP, see [Installing Tanzu Operations Manager on GCP](#)

Additional resources

The following are additional resources related to installing Tanzu Operations Manager on GCP:

- For general authentication guidelines for GCP, see [GCP authentication documentation](#).
- For more information about troubleshooting the Tanzu Operations Manager on GCP infrastructure, see [Troubleshooting Tanzu Operations Manager on GCP](#).
- For production-level deployment options for Tanzu Operations Manager on GCP, see [Reference Architecture for Tanzu Operations Manager on GCP](#).

Recommended GCP resource quotas for Tanzu Operations Manager

The default quotas on a new GCP subscription are not sufficient for a typical production level Tanzu Operations Manager deployment. This topic lists the quotas that you need for a typical Tanzu Operations Manager deployment.

For a single Tanzu Operations Manager deployment, ensure that you have the following quotas.

Metric	Resource	Suggested minimum quota
CPUs	Regional	150*
Firewall rules	Global	15
Forwarding rules	Global	15
Backend services	Global	5
Health checks	Global	5
Images	Global	10
Static IP addresses**	Regional	5
In-use IP addresses	Global	5
In-use IP addresses**	Regional	5
Networks	Global	5
Subnetworks	Global	5
Routes	Global	20
Target pools	Global	5
Target HTTP proxies	Global	5
Target HTTPS proxies	Global	5
Persistent disk standard (GB)	Regional	15,000

* Assuming a deployment with 100 app instances.

** Assuming a SNAT topology.

To view production-level deployment options for Tanzu Operations Manager on GCP, see the [Reference architecture for Tanzu Operations Manager on GCP](#).

For instructions about setting up the GCP resources required to deploy Tanzu Operations Manager, see [Preparing to deploy Tanzu Operations Manager on GCP](#).

Installing Tanzu Operations Manager on GCP manually

You can install Tanzu Operations Manager on GCP manually, with either the Tanzu Platform for Cloud Foundry or VMware Tanzu Container Orchestration Engine (formerly TKGI) runtime.

Install with Elastic Application Runtime

To install manually with Elastic Application Runtime:

1. Prepare to deploy Tanzu Operations Manager. See [Preparing to deploy Tanzu Operations Manager on GCP](#).
2. Deploy Tanzu Operations Manager. See [Deploying Tanzu Operations Manager on GCP](#).
3. Configure BOSH Director. See [Configuring BOSH Director on GCP](#).
4. (Optional). Configure a shared VPC. See [Configuring a Shared VPC on GCP](#).

Install with Container Orchestration Engine

To install manually with Container Orchestration Engine:

1. Prepare to deploy Tanzu Operations Manager. See [Preparing to deploy Tanzu Operations Manager on GCP](#).
2. Deploy Tanzu Operations Manager. See [Deploying Tanzu Operations Manager on GCP](#).
3. Configure BOSH Director. See [Configuring BOSH Director on GCP](#).
4. (Optional). Configure a shared VPC. See [Configuring a shared VPC on GCP](#).

Next steps

After you complete the previous procedures, configure a runtime.

For information about installing and configuring a runtime, see the following:

- Configure Elastic Application Runtime. See [Configuring Elastic Application Runtime](#)
 - [Installing and configuring Elastic Application Runtime Windows add-on](#)
 - [Installing Small Footprint Elastic Application Runtime](#)
- Configure Container Orchestration Engine. See [Installing Container Orchestration Engine on GCP](#)

Preparing to deploy Tanzu Operations Manager on GCP

You must follow these steps before you install Tanzu Operations Manager on Google Cloud Platform (GCP).

Prerequisites

Before you prepare your Tanzu Operations Manager installation:

- If you are deploying Tanzu Platform for Cloud Foundry, see [Tanzu Operations Manager on GCP Requirements](#).
- If you are deploying Enterprise VMware Tanzu Container Orchestration Engine (formerly TKGI), see [GCP Prerequisites and Resource Requirements](#).

Configuration and components

This section outlines high-level infrastructure options for Tanzu Operations Manager on GCP. A Tanzu Operations Manager deployment includes Tanzu Operations Manager and your chosen runtime; for example, both Tanzu Operations Manager with Elastic Application Runtime and Tanzu Operations Manager with Container Orchestration Engine are Tanzu Operations Manager deployments. For more information, review the deployment options and recommendations in [Reference Architecture for Tanzu Operations Manager on GCP](#).

You can deploy Tanzu Operations Manager using one of two main configurations on a GCP virtual private cloud (VPC):

- A *single-project* configuration that gives Tanzu Operations Manager full access to VPC resources
- A *shared VPC* configuration in which Tanzu Operations Manager shares VPC resources

See [Shared vs. Single-Project VPCs](#) in *Reference Architecture for Tanzu Operations Manager on GCP* for a full discussion and recommendations.

When deploying Tanzu Operations Manager on GCP, VMware recommends using the following GCP components:

- [Google Cloud SQL](#) for external database services
- [NAT Gateway Instances](#) to limit the number of VMs with public IP addresses
- [Google Cloud Storage](#) for external file storage

Step 1: Set up IAM service accounts

Tanzu Operations Manager uses IAM service accounts to access GCP resources.

For a single-project installation: Complete the following steps to create a service account for Tanzu Operations Manager.

For a shared-VPC installation: Complete the following steps twice, first, to create a host account, and second, to create a service account for Tanzu Operations Manager.

1. From the GCP console, click IAM & Admin, then Roles.
2. Click Create New Role:
 - Title: Enter a title; for example, “BOSH Director”
 - ID: Enter a unique id; for example, “bosh.director”
 - Click Add Permissions. Select each of the following permissions, then click Add.
 - `compute.addresses.get`
 - `compute.addresses.list`
 - `compute.backendServices.get`
 - `compute.backendServices.list`
 - `compute.diskTypes.get`
 - `compute.disks.delete`
 - `compute.disks.list`
 - `compute.disks.get`
 - `compute.disks.createSnapshot`

- `compute.snapshots.create`
- `compute.disks.create`
- `compute.disks.resize`
- `compute.images.useReadOnly`
- `compute.globalOperations.get`
- `compute.images.delete`
- `compute.images.get`
- `compute.images.create`
- `compute.instanceGroups.get`
- `compute.instanceGroups.list`
- `compute.instanceGroups.update`
- `compute.instances.setMetadata`
- `compute.instances.setLabels`
- `compute.instances.setTags`
- `compute.instances.reset`
- `compute.instances.start`
- `compute.instances.list`
- `compute.instances.get`
- `compute.instances.delete`
- `compute.instances.create`
- `compute.subnetworks.use`
- `compute.subnetworks.useExternalIp`
- `compute.instances.detachDisk`
- `compute.instances.attachDisk`
- `compute.disks.use`
- `compute.instances.deleteAccessConfig`
- `compute.instances.addAccessConfig`
- `compute.addresses.use`
- `compute.addresses.useInternal`
- `compute.machineTypes.get`
- `compute.regionOperations.get`
- `compute.zoneOperations.get`
- `compute.networks.get`
- `compute.subnetworks.get`
- `compute.snapshots.delete`
- `compute.snapshots.get`

- `compute.targetPools.list`
 - `compute.targetPools.get`
 - `compute.targetPools.addInstance`
 - `compute.targetPools.removeInstance`
 - `compute.instances.use`
 - `storage.buckets.create`
 - `storage.objects.create`
 - `compute.zones.list`
 - `resourceManager.projects.get`
 - `compute.subnetworks.list`
 - `compute.networks.list`
 - If you intend to use the Tanzu Ops Manager VM Service Account option for authentication instead of AuthJSON, then the following permissions are also required:
 - `compute.instances.setServiceAccount`
 - Add each permission associated with the `Service Account User` role
 - Click Create
3. From the GCP console, click IAM & Admin, then Service accounts.
 4. Click Create Service Account:
 - Service account name: Enter a name; for example, `bosh`.
 - Role: Use the drop-down menu, select the following roles:
 - Service Accounts, then Service Account User
 - Service Accounts, then Service Account Token Creator
 - Custom, then the name of the role you previously created; for example, BOSH Director

You must scroll down in the pop-up windows to select all required roles.

The Service Account User role is required only if you plan to use the Tanzu Operations Manager VM Service Account to deploy Tanzu Operations Manager. For more information about the Tanzu Operations Manager VM Service Account, see [Step 2: Google Cloud Platform Config](#) in *Configuring BOSH Director on GCP*.
 - Service account ID: The text box automatically generates a unique ID based on the user name.
 - Furnish a new private key: Select this check box and JSON as the Key type.
 5. Click Create. Your browser automatically downloads a JSON file with a private key for this account. Save this file in a secure location.

You can use this service account to configure file storage for Elastic Application Runtime. For more information, see [GCP](#) in *Configuring File Storage for Elastic Application Runtime*.

Step 2: Enable Google Cloud APIs

Tanzu Operations Manager manages GCP resources using the Google Compute Engine and Cloud Resource Manager APIs. To enable these APIs:

1. Log in to the Google Developers Console at <https://console.developers.google.com>.
2. In the console, go to the GCP projects where you want to install Tanzu Operations Manager.
 - For a single-project installation, complete the following steps for the Tanzu Operations Manager project.
 - For a shared-VPC installation, complete the following steps for both host and service projects, to enable them to access the Google Cloud API.
3. Click API Manager, then Library.
4. Under Google Cloud APIs, click Compute Engine API.
5. On the Google Compute Engine API pane, click Enable.
6. In the search text box, type **Google Cloud Resource Manager API**.
7. On the Google Cloud Resource Manager API pane, click Enable.
8. To verify that the APIs have been enabled, complete the following steps:

1. Log in to GCP using the IAM service account you created in [Set up IAM Service Accounts](#):

```
$ gcloud auth activate-service-account --key-file JSON_KEY_FILENAME
```

2. List your projects:

```
$ gcloud projects list
PROJECT_ID          NAME                      PROJECT_NUMBER
my-host-project-id  my-host-project-name     #####
my-service-project-id my-service-project-name  #####
```

This command lists the projects in which you enabled Google Cloud APIs.

Step 3: Create a GCP network with subnets

1. Log in to the [GCP console](#).
2. Go to the GCP project where you want to install Tanzu Operations Manager. For a shared VPC installation, go to the host project.
3. Click VPC network, then CREATE VPC NETWORK.
4. In the Name text box, enter a name of your choice for the VPC network. This name helps you identify resources for this deployment in the GCP console. Network names must be lowercase; for example, **pcf-virt-net**.
 1. Under Subnets, create an infrastructure subnet for Tanzu Operations Manager and NAT instances. Complete the form as follows:

Name	pcf-infrastructure-subnet-GCP-REGION Example: pcf-infrastructure-subnet-us-west1
Region	A region that supports three availability zones. For help selecting the correct region for your deployment, see the Google documentation about regions and zones .

IP address range	A CIDR ending in /26 Example: 192.168.101.0/26
------------------	---

For deployments that do not use external IP addresses, enable Private Google access to allow your runtime to make API calls to Google services.

- Click Add subnet to add a second subnet for the BOSH Director and components specific to your runtime. Complete the form as follows:

Name	pcf-RUNTIME-subnet-GCP-REGION Example: pcf-pas-subnet-us-west1
Region	The same region you selected for the infrastructure subnet
IP address range	A CIDR ending in /22 Example: 192.168.16.0/22

- Click Add subnet to add a third Subnet with the following details:

Name	pcf-services-subnet-GCP-REGION Example: pcf-services-subnet-us-west1
Region	The same region you selected for the previous subnets
IP address range	A CIDR in /22 Example: 192.168.20.0/22

- Under Dynamic routing mode, leave Regional selected.
- Click Create.

Step 4: Create NAT instances

Use NAT instances when you want to expose only a minimal number of public IP addresses.

Creating NAT instances permits internet access from cluster VMs. You might, for example, need this internet access for pulling Docker images or enabling internet access for your workloads.

For more information, see [Reference Architecture for Tanzu Operations Manager on GCP](#) and the [GCP documentation](#).

- In the GCP console, with your single project or shared-VPC host project selected, go to Compute Engine, then VM instances.
- Click CREATE INSTANCE.
- Complete the following text boxes:
 - Name: Enter pcf-nat-gateway-pri.
This is the first, or primary, of three NAT instances you need. If you use a single AZ, you need only one NAT instance.
 - Zone: Click the first zone from your region.
Example: For region us-west1, click us-west1-a zone.
 - Machine type: Click n1-standard-4.
 - Boot disk: Click Change and click Ubuntu 14.04 LTS.
- Expand the additional configuration text boxes by clicking Management, disks, networking, SSH keys.
- In the Startup script text box under Automation, enter the following text:

```
#!/bin/bash
sudo sysctl -w net.ipv4.ip_forward=1
sudo sh -c 'echo net.ipv4.ip_forward=1 >> /etc/sysctl.conf'
sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
```

6. Click Networking to open additional network configuration text boxes:

1. In the Network tags text box, add the following:
 - `nat-traverse`
 - `pcf-nat-instance`.
2. Click the Networking tab and the pencil icon to edit the Network interface.
3. For Network, click `pcf-virt-net`. You created this network in [Step 1: Create a GCP Network with Subnets](#).
4. For Subnetwork, click `pcf-infrastructure-subnet-GCP-REGION`.
5. For Primary internal IP, click `Ephemeral (Custom)`. Enter an IP address; for example, `192.168.101.2`, in the Custom ephemeral IP address text box. The IP address must meet the following requirements:
 - The IP address must exist in the CIDR range you set for the `pcf-infrastructure-subnet-GCP-REGION` subnet.
 - The IP address must exist in a reserved IP range set later in BOSH Director. The reserved range is typically the first `.1` through `.9` addresses in the CIDR range you set for the `pcf-infrastructure-subnet-GCP-REGION` subnet.
 - The IP address cannot be the same as the Gateway IP address set later in Tanzu Operations Manager. The Gateway IP address is typically the first `.1` address in the CIDR range you set for the `pcf-infrastructure-subnet-GCP-REGION` subnet.
6. For External IP, click `Ephemeral`.
If you select a static external IP address for the NAT instance, then you can use the static IP to further secure access to your CloudSQL instances.
7. Set IP forwarding to `On`.
8. Click Done.

7. Click Create to finish creating the NAT instance.

8. Repeat steps 2 through 6 to create two additional NAT instances with the names and zones specified in the following table. The rest of the configuration remains the same.

Instance 2	Name	<code>pcf-nat-gateway-sec</code>
	Zone	Select the second zone from your region. Example: For region <code>us-west1</code> , select zone <code>us-west1-b</code> .
	Internal IP	Select <code>Custom</code> and enter an IP address in the Internal IP address text box. Example: <code>192.168.101.3</code> . As described previously, this address must in the CIDR range you set for the <code>pcf-infrastructure-subnet-GCP-REGION</code> subnet, must exist in a reserved IP range set later in BOSH Director, and cannot be the same as the Gateway IP address set later in Tanzu Operations Manager.

Rule 1	This rule allows SSH from public networks. Name: <code>pcf-allow-ssh</code> Network: <code>pcf-virt-net</code> Allowed protocols and ports: <code>tcp:22</code> Source filter: IP ranges Source IP ranges: <code>0.0.0.0/0</code> Target tags: <code>allow-ssh</code>
Rule 2	This rule allows HTTP from public networks. Name: <code>pcf-allow-http</code> Network: <code>pcf-virt-net</code> Allowed protocols and ports: <code>tcp:80</code> Source filter: IP ranges Source IP ranges: <code>0.0.0.0/0</code> Target tags: <code>allow-http, router</code>
Rule 3	This rule allows HTTPS from public networks. Name: <code>pcf-allow-https</code> Network: <code>pcf-virt-net</code> Allowed protocols and ports: <code>tcp:443</code> Source filter: IP ranges Source IP ranges: <code>0.0.0.0/0</code> Target tags: <code>allow-https, router</code>
Rule 4	This rule allows Gorouter health checks. Name: <code>pcf-allow-http-8080</code> Network: <code>pcf-virt-net</code> Allowed protocols and ports: <code>tcp:8080</code> Source filter: IP ranges Source IP Ranges: <code>0.0.0.0/0</code> Target tags: <code>router</code>
Rule 5	This rule allows communication between BOSH-deployed jobs. Name: <code>pcf-allow-pas-all</code> Network: <code>pcf-virt-net</code> Allowed protocols and ports: <code>tcp;udp;icmp</code> Source filter: Source tags Target tags: <code>pcf, pcf-opsman, nat-traverse</code> Source tags: <code>pcf, pcf-opsman, nat-traverse</code>
Rule 6 (Optional)	This rule allows access to the TCP router. Name: <code>pcf-allow-cf-tcp</code> Network: <code>pcf-virt-net</code> Source filter: IP ranges Source IP ranges: <code>0.0.0.0/0</code> Allowed protocols and ports: <code>tcp:1024-65535</code> Target tags: <code>pcf-cf-tcp</code>
Rule 7 (Optional)	This rule allows access to the SSH proxy. Name: <code>pcf-allow-ssh-proxy</code> Network: <code>pcf-virt-net</code> Source filter: IP ranges Source IP ranges: <code>0.0.0.0/0</code> Allowed protocols and ports: <code>tcp:2222</code> Target tags: <code>pcf-ssh-proxy, diego-brain</code>

If you want your firewall rules to permit traffic only within your private network, modify the Source IP Ranges from the table accordingly.

3. If you are using your GCP project only to deploy Tanzu Operations Manager, then you can delete the following default firewall rules:
 - `default-allow-http`
 - `default-allow-https`
 - `default-allow-icmp`
 - `default-allow-internal`
 - `default-allow-rdp`
 - `default-allow-ssh`

If you are deploying Container Orchestration Engine only, continue to [Next steps](#).

If you are deploying Elastic Application Runtime or other runtimes, continue to Create database instance and databases.

Step 6: Create database instance and databases

Create database instance

1. For a shared-VPC installation, click the service project in the GCP console. This step and the following steps allocate resources to the service project, not the host project.
2. From the GCP console, click SQL and then click CREATE INSTANCE.
3. Ensure MySQL is selected and click Next.
4. Under MySQL, click Second Generation instance type.
5. Click Configure MySQL under your choice for instance type: Development, Staging, or Production.
6. Configure the instance as follows:
 - Instance ID: `pcf-pas-sql`
 - Root password: Set a password for the root user.
 - Region: Select the region you specified when creating networks.
 - Zone: Select Any.
 - Configure machine type and storage:
 - Click Change and then select db-n1-standard-2.
 - Ensure that Enable automatic storage increases is selected. This allows DB storage to grow automatically when space is required.
 - Enable auto backups and high availability: Make the following selections:
 - Leave Automate backups and Enable binary logging selected.
 - Under High availability, select the Create failover replica check box.
 - Authorize Networks: Click Add network and create a network named `all` that allows traffic from `0.0.0.0/0`.

If you assigned static IP addresses to your NAT instances, you can instead limit access to the database instances by specifying the NAT IP addresses.

7. Click Create.

Create databases

1. Go to the Instances page and select the database instance you just created.
2. Select the Databases tab.
3. Click Create database to create the following databases:
 - `account`
 - `app_usage_service`
 - `autoscale`
 - `ccdb`
 - `console`
 - `diego`
 - `locket`
 - `networkpolicyserver`
 - `nfsvolume`
 - `notifications`
 - `routing`
 - `silk`
 - `uaa`
 - `credhub`
4. Select the USERS tab.
5. Click Create user account to create a unique user name and password for each database you previously created. For Host name, select Allow any host. You must create a total of fourteen user accounts.

Ensure that the `networkpolicyserver` database user has the `ALL PRIVILEGES` permission.

Step 7: Create storage buckets

1. With your single project or shared-VPC service project selected in the GCP console, click Storage, then Browser.
2. Click CREATE BUCKET and create buckets with the following names. For Default storage class, click Multi-Regional:
 - `PREFIX-pcf-buildpacks`
 - `PREFIX-pcf-droplets`
 - `PREFIX-pcf-packages`
 - `PREFIX-pcf-resources`
 - `PREFIX-pcf-backup`

PREFIX is a prefix of your choice. It is required to make the bucket name unique.

Step 8: Create HTTP load balancer

For load balancing, you can use a global HTTP load balancer or an internal, regional load balancer with a private IP address.

Single project, standalone installations typically use a global HTTP load balancer. See [Create HTTP Load Balancer](#) for how to set this up.

Shared-VPC installation typically use an internal TCP/UDP load balancer to minimize public IP addresses. See [Create Internal Load Balancer](#) for how to set this up.

Create internal load balancer

To create an internal load balancer for Tanzu Operations Manager on GCP, do the following.

1. Create an internal-facing TCP/UDP load balancer for each region of your Tanzu Operations Manager deployment.

GCP Internal Load Balancer (iLB) is a regional product. Within the same VPC/network, client VMs in a different region from the iLB cannot access the iLB. For more information, see the [GCP documentation](#).

2. Assign private IP addresses to the load balancers.
3. After you have deployed Tanzu Operations Manager, follow instructions in [Create or Update a VM Extension](#) to add a custom VM extension that applies internal load balancing to all VMs deployed by BOSH.

For example, the following manifest code adds a VM extension **backend-pool** to Tanzu Operations Manager VMs:

```
vm_extensions:  
- name: backend-pool  
  cloud_properties:  
    ephemeral_external_ip: true  
  backend_service:  
    name: name-of-backend-service  
    scheme: INTERNAL
```

Create HTTP load balancer

To create a global HTTP load balancer for Tanzu Operations Manager on GCP:

1. [Create Instance Group](#).
2. [Create Health Check](#).
3. [Configure Back End](#).
4. [Configure Front End](#).

To create an instance group:

1. Go to Compute Engine, then Instance groups.
2. Click CREATE INSTANCE GROUP.
3. Complete the form as follows:
 - For Name, enter **pcf-http-lb**

- For Location, click Single-zone.
 - For Zone, click the first zone from your region; for example: For region `us-west1`, click zone `us-west1-a`.
 - Under Group type, click Unmanaged instance group.
 - For Network, click `pcf-virt-net`.
 - For Subnetwork, click the `pcf-pas-subnet-my-gcp-region` subnet that you created previously.
 - Click Create.
4. Create a second instance group with the following details:
- Name: `pcf-http-lb`
 - Location: Single-zone
 - Zone: Click the second zone from your region; for example: For region `us-west1`, click zone `us-west1-b`.
 - Group type: Click Unmanaged instance group.
 - Network: Click `pcf-virt-net`.
 - Subnetwork: Click the `pcf-pas-subnet-my-gcp-region` subnet that you created previously.
5. Create a third instance group with the following details:
- Name: `pcf-http-lb`
 - Location: Single-zone
 - Zone: Click the third zone from your region; for example: For region `us-west1`, click zone `us-west1-c`.
 - Group type: Click Unmanaged instance group.
 - Network: Click `pcf-virt-net`.
 - Subnetwork: Click the `pcf-pas-subnet-my-gcp-region` subnet that you created previously.

To create the health check:

1. Go to Compute Engine, then Health checks.
2. Click CREATE HEALTH CHECK.
3. Complete the form as follows:
 - Name: `pcf-cf-public`
 - Port: `8080`
 - Request path: `/health`
 - Check interval: `30`
 - Timeout: `5`
 - Healthy threshold: `10`
 - Unhealthy threshold: `2`
4. Click Create.

To configure the back end:

1. Go to Network services, then Load balancing.

2. Click CREATE LOAD BALANCER.
3. Under HTTP(S) Load Balancing, click Start configuration.
4. For the Name, enter `pcf-global-pcf`.
5. Select Backend configuration.
6. From the drop-down menu, click Backend services, then Create a backend service.
7. Complete the form as follows:
 - Name: `pcf-http-lb-backend`
 - Protocol: `HTTP`
 - Named port: `http`
 - Timeout: `10 seconds`
 - Under Backends, then New backend, click the Instance group that corresponds to the first zone of the multi-zone instance group you created; for example: `pcf-http-lb (us-west1-a)`. Click Done.
 - Click Add backend, click the Instance group that corresponds to the second zone of the multi-zone instance group you created; for example: `pcf-http-lb (us-west1-b)`. Click Done.
 - Click Add backend, click the Instance group that corresponds to the third zone of the multi-zone instance group you created; for example: `pcf-http-lb (us-west1-c)`. Click Done.
 - Health check: Click the `pcf-cf-public` health check that you created.
 - Cloud CDN: Ensure that Cloud CDN is deactivated.
 - From the drop-down menu, click Backend services, then Create a backend service.
 - Complete the form as follows:
 - Name: `pcf-http-lb-backend`
 - Protocol: `HTTP`
 - Named port: `http`
 - Timeout: `10 seconds`
 - Under Backends, then New backend, click the Instance group that corresponds to the first zone of the multi-zone instance group you created; for example: `pcf-http-lb (us-west1-a)`. Click Done.
 - Click Add backend, click the Instance group that corresponds to the second zone of the multi-zone instance group you created; for example: `pcf-http-lb (us-west1-b)`. Click Done.
 - Click Add backend, click the Instance group that corresponds to the third zone of the multi-zone instance group you created; for example: `pcf-http-lb (us-west1-c)`. Click Done.
 - Health check: Click the `pcf-cf-public` health check that you created.
 - Cloud CDN: Make sure Cloud CDN is inactive.
8. Click Create.

To configure the front end:

1. Click Host and path rules to populate the default text boxes and a green check mark.

2. Click Frontend configuration, and add the following:
 - Name: `pcf-cf-lb-http`
 - Protocol: `HTTP`
 - IP: Perform the following steps:
 1. Click Create IP address.
 2. Enter a Name for the new static IP address and an optional description; for example, `pcf-global-pcf`.
 3. Click Reserve.
 - Port: `80`
3. Click Add Frontend IP and port and add the following:

Skip this step if you do not have either a self-signed or trusted SSL certificate.

When you configure the tile for your chosen runtime, you are given the opportunity to create a new self-signed certificate. Upon creating a certificate, you can complete the Add Frontend IP and port section.

- Name: `pcf-cf-lb-https`
 - Protocol: `HTTPS`
 - IP address: Click the `pcf-global-pcf` address you create for the previous Frontend IP and Port.
 - Port: `443`
 - Select Create a new certificate. The Create a New Certificate dialog is displayed.
 - In the Name text box, enter a name for the certificate.
 - In the Public key certificate text box, copy in the contents of your public certificate, or upload your certificate as a .pem file. If the certificate is runtime-generated, copy and paste the generated contents from the runtime's Certificate text box into the BOSH Director Public key certificate text box.
 - In the Certificate chain text box, enter or upload your certificate chain in the .pem format. If you are using a self-signed certificate, such as a Elastic Application Runtime or Container Orchestration Engine-generated certificate, do not enter a value in the Certificate Chain text box.
 - In the Private key text box, copy in the contents or upload the .pem file of the private key for the certificate. If the certificate is runtime-generated, copy and paste the generated contents from the runtime's Private Key text box into the BOSH Director Private key text box.
4. Review the completed frontend configuration.
 5. Click Review and finalize to verify your configuration.
 6. Click Create.

Step 9: Create TCP WebSockets load balancer

The load balancer for tailing logs with WebSockets for Tanzu Operations Manager on GCP operates on TCP port `443`.

1. From the GCP console, click Network services, then Load balancing.
2. Click Create load balancer.
3. Under TCP Load Balancing, click Start configuration.
4. On the Create a load balancer configuration UI, make the following selections:
 - Under Internet facing or internal only, click From Internet to my VMs.
 - Under Multiple regions or single region, click Single region only.
 - Under Connection termination, click No (TCP).
5. Click Continue.
6. In the New TCP load balancer window, enter `pcf-wss-logs` in the Name text box.
7. Click Backend configuration to configure the Backend service:
 - Region: Click the region you used to create the network in [Create a GCP Network with Subnets](#).
 - From the Health check drop-down menu, create a health check with the following details:
 - Name: `pcf-gorouter`
 - Port: `8080`
 - Request path: `/health`
 - Check interval: `30`
 - Timeout: `5`
 - Healthy threshold: `10`
 - Unhealthy threshold: `2`

The Backend configuration section shows a green check mark.
8. Click Frontend configuration. Fill in the text boxes:
 - Protocol: `TCP`
 - IP: Perform the following steps:
 1. Click Create IP address.
 2. For name Name for the new static IP address and an optional description; for example, `pcf-gorouter-wss`.
 3. Click Reserve.
 - Port: `443`
9. Click Review and finalize to verify your configuration.
10. Click Create.

Step 10: Create SSH proxy load balancer

1. From the GCP console, click Network services, then Load balancing.
2. Click Create load balancer.
3. Under TCP Load Balancing, click Start configuration.
4. Under Internet facing or internal only, click From Internet to my VMs.

5. Under Connection termination, click No (TCP).
6. Click Continue.
7. In the New TCP load balancer window, enter `pcf-ssh-proxy` in the Name text box.
8. Click Backend configuration, and enter the following values:
 - Region: Click the region you used to create the network in [Create a GCP Network with Subnet](#).
 - Backup pool: `None`
 - Failover ratio: `10%`
 - Health check: `No health check`
9. Click Frontend configuration, and add the following:
 - Protocol: `TCP`
 - IP: Perform the following steps:
 1. Click Create IP address.
 2. Enter a Name for the new static IP address and an optional description; for example, `pcf-ssh-proxy`.
 3. Click Reserve.
 - Port: `2222`
10. (Optional) Review and finalize your load balancer.
11. Click Create.

Step 11: (Optional) Create load balancer for TCP router

This step required only if you enable TCP routing in your deployment.

To create a load balancer for TCP routing in GCP:

1. From the GCP console, click Network services, then Load balancing.
2. Click Create load balancer.
3. Under TCP Load Balancing, click Start configuration.
4. Under Connection termination, click No (TCP).
5. Click Continue.
6. On the New TCP load balancer pane, enter a unique name for the load balancer in the Name text box; for example, `pcf-cf-tcp-lb`.
7. Click Backend configuration, and enter the following values:
 - Region: Click the region you used to create the network in [Create a GCP Network with Subnet](#).
 - From the Health check drop-down menu, create a health check with the following details:
 - Name: `pcf-tcp-lb`
 - Port: `80`
 - Request path: `/health`
 - Check interval: `30`

- Timeout: 5
 - Healthy threshold: 10
 - Unhealthy threshold: 2
 - Click Save and continue.
8. Click Frontend configuration, and add the front end IP and port entry as follows:
 - Protocol: TCP
 - IP: Perform the following steps:
 1. Click Create IP address.
 2. Enter a Name for the new static IP address and an optional description; for example, pcf-cf-tcp-lb.
 3. Click Reserve.
 - Port: 1024-65535
 9. Click Review and finalize to verify your configuration.
 10. Click Create.

Step 12: Add DNS records for your load balancers

In this step, you redirect queries for your domain to the IP addresses of your load balancers.

1. Locate the static IP addresses of the load balancers you created in [Preparing to deploy Tanzu Operations Manager on GCP](#):
 - An HTTP(S) load balancer named pcf-global-pcf
 - A TCP load balancer for WebSockets named pcf-wss-logs
 - A TCP load balancer named pcf-ssh-proxy
 - A TCP load balancer named pcf-cf-tcp-lb

You can locate the static IP address of each load balancer by clicking its name under Network services, then Load balancing in the GCP console.

2. Log in to the DNS registrar that hosts your domain. Examples of DNS registrars include Network Solutions, GoDaddy, and Register.com.
3. Create A records with your DNS registrar that map domain names to the public static IP addresses of the load balancers located previously:

Create and map this record...	To the IP of this load balancer	Required
*.sys.MY-DOMAIN Example: *.sys.example.com	pcf-global-pcf	Yes
*.apps.MY-DOMAIN Example: *.apps.example.com	pcf-global-pcf	Yes
doppler.sys.MY-DOMAIN Example: doppler.sys.example.com	pcf-wss-logs	Yes
loggregator.sys.MY-DOMAIN Example: loggregator.sys.example.com	pcf-wss-logs	Yes

Create and map this record...	To the IP of this load balancer	Required
<code>ssh.sys.MY-DOMAIN</code> Example: <code>ssh.sys.example.com</code>	<code>pcf-ssh-proxy</code>	Yes, to allow SSH access to apps
<code>tcp.MY-DOMAIN</code> Example: <code>tcp.example.com</code>	<code>pcf-cf-tcp-lb</code>	No, only set up if you have enabled the TCP routing feature

4. Save your changes within the web interface of your DNS registrar.
5. Run the following `dig` command to confirm that you created your A record successfully:

```
dig SUBDOMAIN.EXAMPLE-URL.com
```

Where `SUBDOMAIN.EXAMPLE-URL` is the subdomain for your load balancer.

You should see the A record that you just created:

```
;; ANSWER SECTION:
xyz.EXAMPLE.COM.      1767      IN      A      203.0.113.1
```

Next steps

(Optional) To prepare for deploying either a Elastic Application Runtime or Container Orchestration Engine tile on GCP, you can download the required runtime tile in advance:

- To download Elastic Application Runtime, log in to the [Broadcom Support portal](#), select your desired release version, and download Elastic Application Runtime.
- To download Container Orchestration Engine, log in to the [Broadcom Support portal](#), select your desired release version, and download VMware Tanzu Container Orchestration Engine (formerly TKGI).

After initiating the tile download, proceed to the next step, [Deploying Tanzu Operations Manager on GCP](#).

Deploying Tanzu Operations Manager on GCP

This topic describes how you can deploy Tanzu Operations Manager on Google Cloud Platform (GCP).

Before you deploy Tanzu Operations Manager, see the preparation steps in [Preparing to deploy Tanzu Operations Manager on GCP](#).

After you complete this procedure, follow the instructions in [Configuring BOSH Director on GCP](#).

Step 1: Locate the Tanzu Operations Manager installation file

1. Log in to the [Broadcom Support portal](#).
2. Go to the [VMware Tanzu Operations Manager](#) page.
3. From the Release list, select the release for your installation.
4. Select one of the following download files:
 - Tanzu Operations Manager for GCP
 - Tanzu Operations Manager YAML for GCP

When you click the download link, your browser downloads or opens the `OpsManager_VERSION_onGCP.pdf` or `OpsManager_VERSION_onGCP.yml` file.

These documents provide the GCP location of the Tanzu Operations Manager `.tar.gz` installation file based on the geographic location of your installation.

5. Copy the filepath string of the Tanzu Operations Manager image based on your deployment location.

Step 2: Create a private VM image

1. Log in to the [GCP console](#).
2. In the navigation panel, click Compute Engine, then click Images.
3. Click Create Image.
4. Complete the following text boxes:
 - Name: Enter a name; for example, `om-pcf`.
 - Encryption: Leave Automatic (recommended) selected.
 - Source: Select Cloud Storage file.
 - Cloud Storage file: Paste in the Google Cloud Storage filepath you copied from the PDF file in the [previous step](#).
5. Click Create. The file might take a few minutes to import.

Step 3: Create the Tanzu Operations Manager VM instance

1. Select the check box for the image that you previously created.
2. Click Create Instance.
3. In the Create an instance form, complete the following text boxes:
 - Name: Enter a name that matches the naming conventions of your deployment.
 - Zone: Select a zone from the region in which you created your network.
 - Machine type: Select `n2-standard-2`.
 - Boot disk: Click Change, then perform the following steps:
 - Click Custom images if it is not already selected.
 - Click Boot disk type. If your environment has high performance needs, click SSD. An environment used to develop Tanzu Operations Manager tiles, for example, might benefit from a higher performing Tanzu Operations Manager VM boot disk. (See the [Tile Developer Guide](#).) For most environments, you can select Standard.
 - Set the Size (GB) of the boot disk to the minimum or higher.

Machine Spec	Minimum Value
Boot disk	100 GB

- Select the Tanzu Operations Manager image you created in the previous step if it is not already selected.

- Click Select to save.
- Under Identity and API access, select the Service account you created when preparing your environment during the step [Set up IAM Service Accounts](#).
- Allow HTTP traffic: Leave this check box deselected.
- Allow HTTPS traffic: Leave this check box deselected.
- Networking: Select the Networking tab, and perform the following steps:
 - Under Network interfaces, perform the following steps:
 - Remove the `default` network interface if this interface still exists.
 - Select the network you created when preparing your environment in [Create a GCP Network with Subnet](#) in *Preparing to Deploy Tanzu Operations Manager on GCP*. For example, `pcf-virt-net`.
 - Under Subnetwork, select the `pcf-infrastructure-subnet-GCP-REGION` subnet that you created when preparing your environment in [Create a GCP Network with Subnet](#) in *Preparing to Deploy Tanzu Operations Manager on GCP*.
 - For Primary internal IP, select Ephemeral (Custom). Enter an IP address (for example, `192.168.101.5`) in the Custom ephemeral IP address field. Specify the next available internal IP address located within the reserved IP range that you will [configure later in BOSH Director](#). Do not use the Gateway IP (for example `192.168.101.1`).
 - For External IP, select Create IP address. In the next form, enter a name for the static IP. For example, `om-public-ip`. Click Reserve. In the External IP drop-down menu, select the static IP address you just reserved.
 - For Network tags, enter `pcf-opsman` and `allow-https`. These tags apply the firewall rules you created in [Create Firewall Rules for the Network](#) to the Tanzu Operations Manager VM.
- 4. Click Create to deploy the new Tanzu Operations Manager VM. This might take a few moments to finish.
- 5. Go to your DNS provider and create an entry that points a fully qualified domain name (FQDN) `opsman.MY-DOMAIN` to the `pcf-opsman` static IP address of Tanzu Operations Manager that you created in a previous step.
 To set up Tanzu Operations Manager authentication correctly, VMware recommends using an FQDN to access Tanzu Operations Manager. Using an ephemeral IP address to access Tanzu Operations Manager can cause authentication errors upon subsequent access.

Next steps

After you complete this procedure, follow the instructions in [Configuring BOSH Director on GCP](#).

Later, if you need to SSH into the Tanzu Operations Manager VM to perform diagnostic troubleshooting, see [Log Into the Tanzu Operations Manager VM](#).

Configuring BOSH Director on GCP

You can use the steps in this topic to configure BOSH Director in Tanzu Operations Manager on Google Cloud Platform (GCP).

You can also perform the procedures in this topic using the Tanzu Operations Manager API. For more information, see [Using the Tanzu Operations Manager API](#).

Step 1: Log into Tanzu Operations Manager

To configure BOSH Director, you must first access your Tanzu Operations Manager installation and determine how your user database is maintained.

To access Tanzu Operations Manager:

1. In a web browser, go to the fully qualified domain name (FQDN) of Tanzu Operations Manager that you set up in [Deploying Tanzu Operations Manager on GCP](#).
To set up Tanzu Operations Manager authentication correctly, VMware recommends using an FQDN to access Tanzu Operations Manager. Using an ephemeral IP address to access Tanzu Operations Manager can cause authentication errors upon subsequent access.
2. Log in to Tanzu Operations Manager. For more information about logging in, see [Log in to Tanzu Operations Manager for the First Time](#).

Step 2: Configure GCP tile

To configure your GCP tile:

1. In the Tanzu Operations Manager Installation Dashboard, click the BOSH Director tile.
2. Click Google Config and fill in the following text boxes:
 - Project ID: Enter your GCP project ID in lowercase characters; for example, `your-gcp-project-id`.
 - Default Deployment Tag: Enter the prefix that you used when creating the GCP resources for this Tanzu Operations Manager installation.
 - Click AuthJSON: Paste in the contents of the JSON file that you downloaded in [Set up IAM Service Accounts](#) in *Preparing to Deploy Tanzu Operations Manager on GCP*.
As an alternative, you can click the Tanzu Ops Manager VM Service Account to use the service account that you set up in [Set up IAM Service Accounts](#) in *Preparing to Deploy Tanzu Operations Manager on GCP*. To use this option, the service account must be assigned the Service Account Actor role.
 - VM Type Catalog: Click the VM catalog that you want to use for your BOSH-deployed VMs.
 - Google Disk Type: Click the Google Disk Type that you want to use for your BOSH-deployed VMs.

Google Cloud Platform Config

Project ID*

ops-manager-ci

Project ID on GCP

Default Deployment Tag

parmesan-vms

The tag that will be added to all VMs deployed by this OpsManager

☐ The Tanzu Ops Manager VM Service Account

☒ AuthJSON

AuthJSON*

Change

JSON downloaded from Google that can access your project.

VM Type Catalog

E2_SERIES

Google Disk Type*

pd-standard (cost-effective)

The type determines the performance characteristics of the disk

Changing this value will force BOSH to recreate all VMs and (depending on configuration) may recreate persistent disks

SAVE

3. Click Save.

Step 3: Configure Director

To configure the Director Config pane in the BOSH Director tile, you must have the correct permissions. To configure these permissions, see [Required permissions](#) in the BOSH documentation and [Creating storage buckets](#) in the GCP documentation.

To configure the Director Config pane:

1. Click Director Config.

Director Config

[Go to](#)
[Certificate Duration Overrides](#)
[Health Monitor](#)
[CredHub Encryption Provider](#)
[Blobstore Location](#)
[Database Location](#)

NTP Servers (comma delimited)*

One or more NTP server addresses for consistent and valid time stamps used in syslog, etc.

Bosh HM Forwarder IP Address

Should only be used if OS Bosh HM Forwarder job is deployed

☐ Enable VM Resurrector Plugin
☒ Enable System Metrics
☐ Enable Post Deploy Scripts
☐ Enable Short Lived NATS Bootstrap Credentials
This will make the Bosh Director rotate the NATs bootstrap credentials of a VM. This increases security by invalidating the NATS credentials that are stored in the IAAS metadata service.
☐ Recreate VMs deployed by the BOSH Director
This will force BOSH to recreate all Bosh Deployed VMs on the next deploy. This does not recreate the BOSH Director. Persistent disk will be preserved.
☐ Recreate BOSH Director VM
This will force BOSH to recreate the Bosh Director VM on the next deploy. Persistent disk will be preserved.
☐ Recreate All Persistent Disks
Check this box to recreate all Persistent Disks for the Director and all other Tiles.
☐ Enable bosh deploy retries
This will attempt to re-deploy a failed deployment up to 5 times.

- The NTP server configuration is updated only after VM recreation. If you modify the value of this text box, select the Recreate VMs deployed by the BOSH Director check box to re-create your BOSH Director-deployed VMs and update the NTP server configuration. If you have any service tiles installed, ensure that the Recreate All Service Instances errand runs for each service tile.
- 2. Leave the Bosh HM Forwarder IP Address text box blank. BOSH-reported component metrics are available in Loggregator Firehose by default. If you continue to use the BOSH HM Forwarder to consume these component metrics, you might receive duplicate data.
- 3. Select the Enable VM Resurrector Plugin check box to enable the BOSH Resurrector functionality and increase your runtime availability.
- 4. If you want to collect detailed metrics from BOSH-managed VMs, select the Enable System Metrics check box. For a list of metrics that the System Metrics Agent collects from BOSH-managed VMs, see [System Metrics Agent](#) on GitHub. When you select this check box, ensure that the System Metrics Scraper is able to connect on port 53035 to all BOSH-managed VMs.
- 5. Select Enable Post Deploy Scripts to run a post-deployment script after deployment. This script allows the job to execute additional commands against a deployment. If you intend to install VMware Tanzu Container Orchestration Engine (formerly TKGI), you must enable post-deploy scripts.
- 6. To configure your VMs to use short-lived NATS credentials during the bootstrap process, select the Enable Short Lived NATS Bootstrap Credentials check box. This increases the security of your Tanzu Operations Manager deployment by rotating the TLS certificate that the BOSH Agents

in your VMs use to communicate with the BOSH Director through BOSH NATS.
For more information, see [Using Short-Lived NATS Bootstrap credentials](#).

Configuring your VMs to use short-lived NATS bootstrap credentials restarts the BOSH Agent in all VMs as they are being created, so VMs are created somewhat more slowly.

This configuration affects all new deployments of Tanzu Operations Manager. If you want to configure VMs in existing Tanzu Operations Manager deployments to use short-lived bootstrap credentials, you must recreate them.

7. Select **Recreate VMs** deployed by the BOSH Director to force BOSH to recreate BOSH-deployed VMs on the next deployment. This process does not recreate the BOSH Director VM or destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
8. Select **Recreate BOSH Director VM** to force the BOSH Director VM to be recreated on the next deploy. This process does not destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
9. Select **Recreate All Persistent Disks** to force BOSH to migrate and recreate persistent disks for the BOSH Director and all tiles. This process does not destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
10. Select **Enable bosh deploy retries** to instruct Tanzu Operations Manager to retry failed BOSH operations up to five times.
11. Select **Skip Director Drain Lifecycle** to prevent drain scripts from running when the BOSH Director is recreated.
12. Select **Store BOSH Job Credentials on tmpfs (beta)** to store credentials for BOSH jobs on temporary file storage (tmpfs) memory, rather than on disk. You must re-create all BOSH-deployed VMs for this setting to take effect.
13. Select **Keep Unreachable Director VMs** to preserve BOSH Director VMs after a failed deployment for troubleshooting purposes.
14. (Optional) Select **Enable automatic rotation of the BOSH DNS CA certificate (experimental)** if you want Tanzu Operations Manager to automatically rotate the BOSH DNS CA certificate and associated leaf certificates when you click **Apply Changes**. For more information about this feature, see [Automatic rotation of BOSH DNS certificates](#).
15. (Optional) Modify the **Director Workers** value, which sets the number of workers available to run BOSH Director tasks. The default value is **5**.
16. (Optional) **Max Threads** sets the maximum number of threads that the BOSH Director can run simultaneously. VMware recommends that you leave the text box blank to use the default value, unless doing so results in rate limiting or errors on your IaaS.
17. (Optional) To add a custom URL for your BOSH Director, enter a valid hostname in the **Director Hostname** text box. You can also use this text box to configure a load balancer in front of your BOSH Director. For more information, see [How to set up a Load Balancer in front of Tanzu Operations Manager](#).

Director Workers

5

The number of director workers to execute director tasks. Default is 5

Max Threads

The Max Threads Setting controls how many operations the BOSH Director can perform simultaneously

Director Hostname

opsdirector.example.com

The externally accessible hostname for the BOSH Director

18. (Optional) To set a custom banner that users see when you log in to the BOSH Director using SSH, enter text in the Custom SSH Banner text box.
19. (Optional) Enter your comma-separated custom Identification Tags. For example, `iaas:foundation1,hello:world`. You can use the tags to identify your foundation when viewing VMs or disks from your IaaS.
20. For Certificate Duration Overrides, you can choose whether certificates generated within Tanzu Operations Manager and CredHub use the default duration specified for the certificate, or a custom duration value that you specify.
 - Off: Use this option to use default duration for certificates created by all products.
 - On: Use this option to set a value to override the duration for certificates created by all products. If a product creates a certificate with a longer duration than the value you set, the longer duration is used.
 1. CA Certificate Duration (days): Enter the number of days for which CA certificates are valid.
 2. Leaf Certificate Duration (days): Enter the number of days for which leaf certificates are valid. This value must be less than or equal to the CA certificate duration.

After you set a certificate duration override, you must take additional steps to apply the setting to all certificates. For more information, see [Overriding Duration for Certificates](#).

Certificate Duration Overrides

Set overrides for CA and leaf certificate durations.
For detailed information about which certificates are affected by this setting and which are not, see [Override Certificate Duration in the TAS for VMs or TKGI documentation](#).

☐ Off
☒ On

Enter values to set the duration for CA and leaf certificates in the fields below.
These values will be applied when new certificates are generated via new product deployment or certificate rotation.

CA Certificate Duration (days)*

1460

Enter a value between 365 and 3650 days

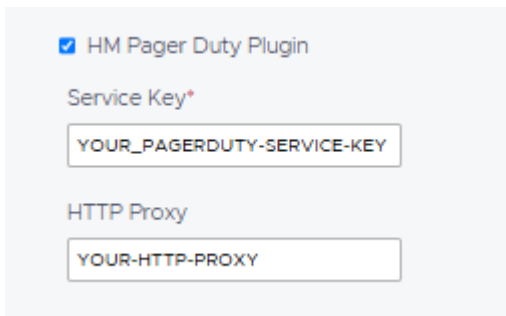
Leaf Certificate Duration (days)*

730

Leaf Certificate Duration must be less than or equal to the CA Certificate Duration.

21. Select HM Pager Duty Plugin to enable Health Monitor integration with PagerDuty.
 - Service Key: Enter your API service key from PagerDuty.

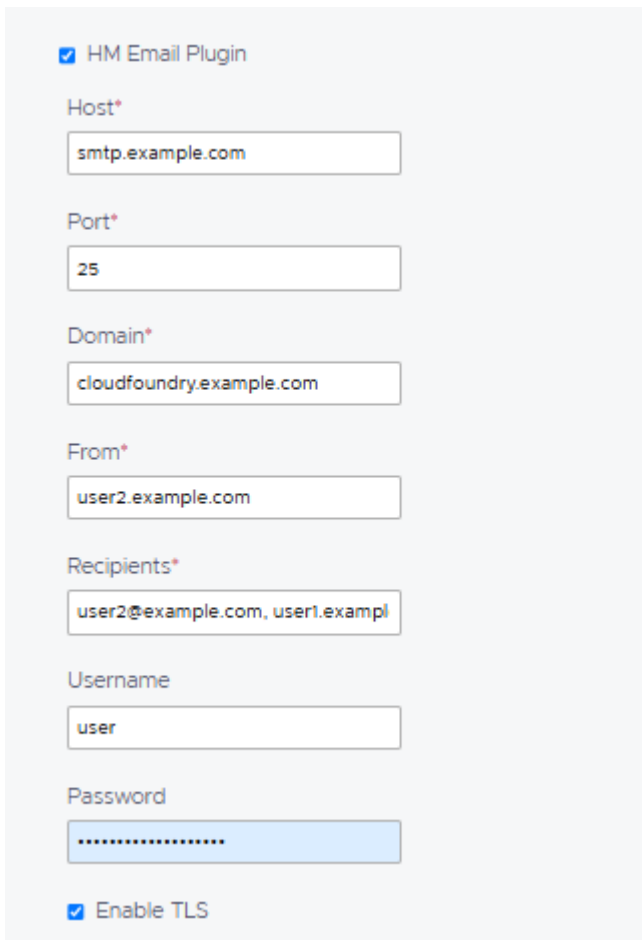
- HTTP Proxy: Enter an HTTP proxy for use with PagerDuty.



A screenshot of a configuration form for the 'HM Pager Duty Plugin'. The form is set against a light gray background. At the top, there is a checked checkbox labeled 'HM Pager Duty Plugin'. Below it, the label 'Service Key*' is followed by a text input field containing the placeholder text 'YOUR_PAGERDUTY-SERVICE-KEY'. Further down, the label 'HTTP Proxy' is followed by another text input field containing the placeholder text 'YOUR-HTTP-PROXY'.

22. Select HM Email Plugin to enable Health Monitor integration with email.

- Host: Enter your email host name.
- Port: Enter your email port number.
- Domain: Enter your domain.
- From: Enter the sender's address.
- Recipients: Enter a comma-separated list of addresses of intended recipients.
- Username: Enter the user name for your email server.
- Password: Enter the password for your email server.
- Enable TLS: Select this check box to enable Transport Layer Security to the email host.



A screenshot of a configuration form for the 'HM Email Plugin'. The form is set against a light gray background. At the top, there is a checked checkbox labeled 'HM Email Plugin'. Below it, the label 'Host*' is followed by a text input field containing 'smtp.example.com'. The label 'Port*' is followed by a text input field containing '25'. The label 'Domain*' is followed by a text input field containing 'cloudfoundry.example.com'. The label 'From*' is followed by a text input field containing 'user2.example.com'. The label 'Recipients*' is followed by a text input field containing 'user2@example.com, user1.exempl'. The label 'Username' is followed by a text input field containing 'user'. The label 'Password' is followed by a text input field filled with dots. At the bottom, there is a checked checkbox labeled 'Enable TLS'.

23. For CredHub Encryption Provider, you can choose whether BOSH CredHub stores its encryption key internally on the BOSH Director and CredHub VM, or in an external hardware security module (HSM). The HSM option is more secure.

Before configuring an HSM encryption provider in the Director Config pane, you must follow the procedures and collect information described in [Preparing CredHub HSMs for configuration](#). After you deploy Tanzu Operations Manager with an HSM encryption provider, you cannot change BOSH CredHub to store encryption keys internally.

- Internal: Use this option for internal CredHub key storage. This option is selected by default and requires no additional configuration.
- Luna HSM: Use this option to use a SafeNet Luna HSM as your permanent CredHub encryption provider, and fill in the following fields:

If you use multiple HSM hosts, you must use the Tanzu Operations Manager API to set or update the HSM configuration. For more information, see [Updating a staged director's properties](#) in the Tanzu Operations Manager API documentation.

1. Encryption Key Name: Any name to identify the key that the HSM uses to encrypt and decrypt the CredHub data. Changing this key name after you deploy Tanzu Operations Manager can cause service downtime.
2. Provider Partition: The partition that stores your encryption key. Changing this partition after you deploy Tanzu Operations Manager can cause service downtime. For this value and the ones that follow, use values gathered in [Preparing CredHub HSMs for configuration](#).
3. Provider Partition Password
4. Partition Serial Number
5. Provider Client Certificate: The certificate that validates the identity of the HSM when CredHub connects as a client.
6. Provider Client Certificate Private Key
7. HSM Host Address
8. HSM Port Address: If you do not know your port address, enter **1792**.
9. HSM Certificate: The certificate that the HSM presents to CredHub to establish a two-way mTLS connection.

CredHub Encryption Provider

- ☐ Internal
- ☒ Luna HSM

Encryption Key Name*

Provider Partition*

Provider Partition Password*

Partition Serial Number*

Provider Client Certificate*

Provider Client Certificate Private Key*

HSM Host Address*

HSM Port Address*



24. Click Blobstore Location to configure the blobstore as either an internal server or an external endpoint.

After you deploy the BOSH Director, only Tanzu Operations Manager users in Advanced Mode can change the blobstore location. For more information, see [Platform operator roles and permissions in Tanzu Operations Manager](#).

- Internal: Use this option for an internal blobstore. No additional configuration is required.
- S3 Compatible Blobstore: Use this option for an external S3-compatible endpoint. To create an S3 bucket, follow the procedures in [Sign up for Amazon S3](#) and [Creating a Bucket](#) in the AWS documentation. When you have created an S3 bucket, configure the following fields:

1. S3 Endpoint:

1. If you are using a public S3 endpoint:

1. Locate the endpoint for your region. To find the endpoint, see the Amazon Simple Storage Service (S3) table in [Regions and Endpoints](#) in the AWS documentation.
2. Construct a URL using your region's endpoint. For example, if you are using the [us-west-2](#) region, the URL you create would be <https://s3.us-west-2.amazonaws.com>. Enter this URL in the S3 Endpoint field.

2. If you are using a non-public S3-compatible endpoint:

1. Enter the URL for the non-public endpoint.
2. Add the custom public CA certificate you used to sign the S3 endpoint to the Trusted Certificates field on the Director Config page.

2. Bucket Name: Enter the name of the S3 bucket.

3. Access Key and Secret Key: Enter the keys you generated when creating your S3 bucket.

4. Enable signed URLs: Select this check box to configure BOSH VMs to generate short lived, pre-signed URLs for communication between the BOSH Agent and the blobstore. If you enable this feature, BOSH Agents do not use credentials to communicate with the blobstore, and VM disks do not store blobstore credentials. This feature is available only for BOSH VMs that use Xenial stemcell line 621 or later.

5. Select V2 Signature or V4 Signature. If you select V4 Signature, enter your Region. AWS recommends using Signature Version 4 (V4). For more information

about AWS S3 Signatures, see [Authenticating Requests](#) in the AWS documentation.

6. S3 Backup Strategy: To configure whether and how to back up the BOSH Director's S3 blobstore, select one of the following options:

- No backups
- Use a versioned bucket: This option uses blobstore bucket versioning to save backups. It requires an S3 blobstore that supports versioning.
- Copy into an additional bucket: This option saves blobstore backups to a separate bucket, which is useful with blobstores that do not support versioning. This option requires you to configure the Backup Bucket Region and Backup Bucket Name fields.

To backup and restore with BBR, if you are using S3 Compatible Blobstore and you have set up an [iam_instance_policy](#), additional steps are needed:

1. Attach the [iam_instance_policy](#) to the [backup_restore](#) job using the Tanzu Operations Manager API. See [Custom VM Extensions](#).
2. In addition to creating backup buckets, update the Elastic Application Runtime tile (Settings > File Storage) to use S3-compatible storage. See [External S3-Compatible filestore](#).
3. Ensure that your IAM Instance Policy is set up to access the backup bucket. For more information about the IAM Instance Policy, see the [Tanzu Operations Manager for AWS policy document](#).

- GCS Blobstore: Use this option for an external Google Cloud Storage (GCS) endpoint.

To create a GCS bucket, follow the procedures in [Creating Storage Buckets](#) in the GCS documentation. To create a GCS bucket, you must have a GCS account. When you have created a GCS bucket, configure the following fields:

1. Bucket Name: Enter the name of your GCS bucket.
2. Storage Class: Use this storage class for your GCS bucket. For more information, see [Storage Classes](#) in the GCP documentation.
3. Service Account Key: Follow the steps in [Set up IAM Service Accounts](#) in *Preparing to Deploy Tanzu Operations Manager on GCP* to download a JSON file with a private key. Enter the contents of the JSON file into the text box.
4. GCS Backup Strategy: To configure whether and how to back up the BOSH Director's GCS blobstore, click one of the following options:
 - No backups
 - Copy into an additional bucket: This option saves blobstore backups to a bucket separate from the blobstore itself, and requires you to configure the Backup Bucket Name text box.

Blobstore Location

☐ Internal

☐ S3 Compatible Blobstore

☒ GCS Blobstore

Bucket Name*

Storage Class*

Regional

Service Account Key*

Paste the JSON file containing the private key for your GCP service account.

GCS Backup Strategy

☒ No backups

☐ Copy into an additional bucket

Backup Bucket Name*

25. Click Save.

Step 4: Configure Create Availability Zones

VMware recommends at least three availability zones (AZs) for a highly available installation.

- For configuring a Elastic Application Runtime deployment with multiple AZs, see [Reference Architecture for Tanzu Operations Manager on GCP](#).
- For Container Orchestration Engine best practices for distributing workloads across multiple AZs and clusters, see [Maintaining Workload Uptime](#).

To configure the Availability Zones:

1. Click Create Availability Zones.
2. Click Add.
3. For Google Availability Zone:

1. Enter one of the zones that you associated to the back end service instance groups of the HTTP(S) load balancer. For example, if you are using the `us-central1` region and selected `us-central1-a` as one of the zones for your HTTP(S) load balancer instance groups, enter `us-central1-a`.
2. Click Add.
3. Repeat the preceding step for all the AZs that you associated to instance groups in [Preparing to deploy Tanzu Operations Manager on GCP](#).

The screenshot shows a 'Create Availability Zones' dialog box. At the top, it says 'Create Availability Zones'. Below that, there's a section 'Availability Zones*' with a list containing 'us-central1-a'. To the right of this list is an 'Add' button. Below the list is a 'Google Availability Zone*' input field with a placeholder text 'The Google Availability Zone name'. At the bottom left is a 'Save' button.

4. Click Save.
4. Repeat the preceding step for all the AZs you are using in your deployment.
5. When you are done, click Save.

Step 5: Configure Create Networks

To create networks:

1. Click Create Networks.
2. Verify that Enable ICMP checks is not selected. GCP routers do not respond to ICMP pings.
3. Click Add Network. Create the three networks described in the following tables.
To use a shared VPC network, enter the shared VPC host project name before the network name in the format `VPC-PROJECT-NAME/NETWORK-NAME/SUBNET-NAME/REGION-NAME`. For example, `vpc-project/opsmgr/central/us-central1`. For more information, see [Configuring a Shared VPC on GCP](#).

Infrastructure Network

Network name	<code>infrastructure</code>
Google network name	Enter the name of the infrastructure network that you created in Create a GCP Network with Subnets in <i>Preparing to Deploy Tanzu Operations Manager on GCP</i> . The format is: <code>pcf-virt-net/pcf-infrastructure-subnet-MY-REGION/MY-GCP-REGION</code>
CIDR	Enter the name of the CIDR ending in <code>/26</code> that you used when you created the infrastructure subnet in GCP. For example: <code>192.168.101.0/26</code>
Reserved IP ranges	Enter the first <code>.1</code> through <code>.9</code> addresses from the CIDR. For example, if the CIDR is <code>192.168.101.0/26</code> , enter the range <code>192.168.101.1-192.168.101.9</code> .
DNS	<code>169.254.169.254</code>
Gateway	Enter the first <code>.1</code> address from the CIDR. For example, if the CIDR is <code>192.168.101.0/26</code> , enter <code>192.168.101.1</code> .

Availability zones	Select all three AZs.
--------------------	-----------------------

Runtime Network

Name	Enter the name of the runtime that you intend to deploy in this environment. For example, <code>pas</code> or <code>pks</code> .
Google network name	Enter the name of the runtime network that you created when you prepared your GCP environment. The format is: <code>pcf-virt-net/pcf-RUNTIME-subnet-MY-REGION/MY-GCP-REGION</code>
CIDR	Enter the name of the CIDR ending in <code>/22</code> that you used when you created the runtime subnet in GCP. For example: <code>192.168.16.0/22</code>
Reserved IP ranges	Enter the first <code>.1</code> through <code>.9</code> addresses from the CIDR. For example, if the CIDR is <code>192.168.16.0/22</code> , enter the range <code>192.168.16.1-192.168.16.9</code> .
DNS	<code>169.254.169.254</code>
Gateway	Enter the first <code>.1</code> address from the CIDR. For example, if the CIDR is <code>192.168.16.0/22</code> , enter <code>192.168.16.1</code> .
Availability zones	Select all three AZs.

Services network

Network name	<code>services</code>
--------------	-----------------------

Google network Name	Enter the name of the services network you created when you prepared your GCP environment. The format is: <code>pcf-virt-net/pcf-services-subnet-MY-REGION/MY-GCP-REGION</code>
CIDR	Enter the name of the CIDR ending in <code>/22</code> that you used when you created the services subnet in GCP. For example: <code>192.168.20.0/22</code>
Reserved IP ranges	Enter the first <code>.1</code> through <code>.9</code> addresses from the CIDR. For example, if the CIDR is <code>192.168.20.0/22</code> , enter the range <code>192.168.20.1-192.168.20.9</code> .
DNS	<code>169.254.169.254</code>
Gateway	Enter the first <code>.1</code> address from the CIDR. For example, if the CIDR is <code>192.168.20.0/22</code> , enter <code>192.168.20.1</code> .
Availability zones	Select all three AZs.

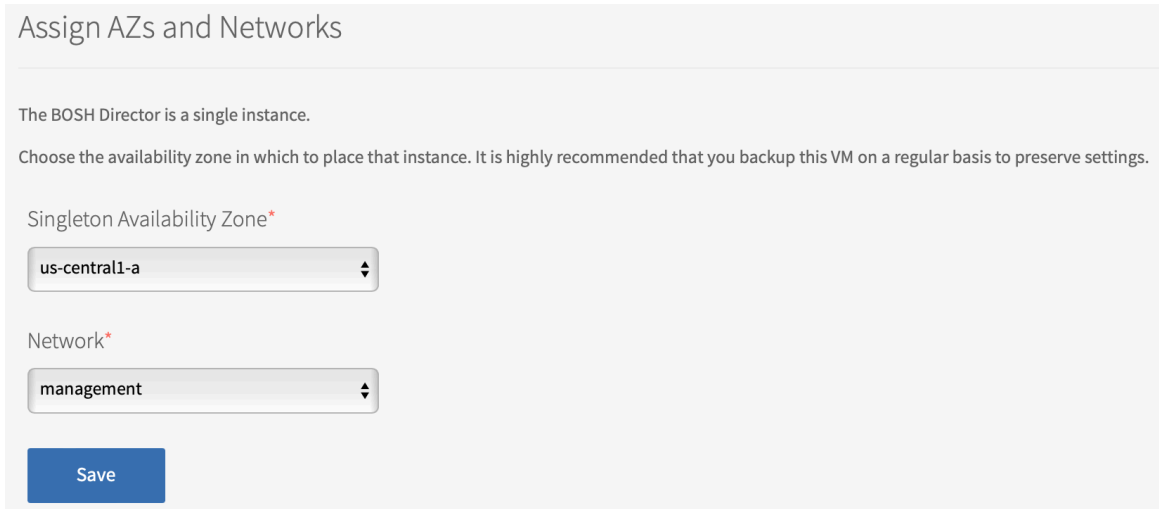
After you deploy Tanzu Operations Manager, you add subnets with overlapping AZs to expand your network. For more information about configuring additional subnets, see [Expanding your Network with Additional Subnets](#).

Ensure that `169.254.169.254`, the IP address of the `metadata.google.internal` metadata server, is always specified at the beginning of this step's DNS resolver lists.

Step 6: Configure Assign AZs and Networks

To configure the Assign AZs and Networks pane:

1. Click Assign AZs and Networks.

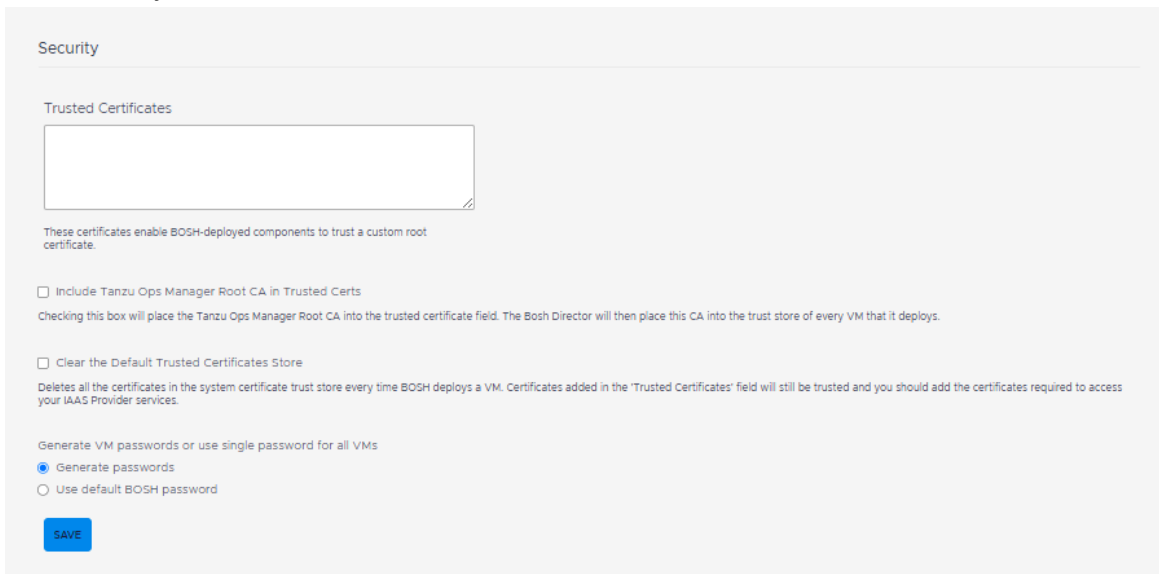


2. Use the drop-down menu to select a Singleton Availability Zone. The BOSH Director installs in this AZ.
3. Under Network, click the **infrastructure** network for your BOSH Director.
4. Click Save.

Step 7: Configure security

To configure security:

1. Click Security.



2. In Trusted Certificates, paste in your custom certificate authority (CA) certificates to insert into your organization's certificate trust chain. This feature enables all BOSH-deployed components in your deployment to trust custom root certificates. If you want to use Docker registries to run Elastic Application Runtime app instances in Docker containers, enter the certificate for your private Docker registry in this text box. For more information about running app instances in Elastic Application Runtime using Docker registries, see [Using Docker Registries](#).

To enter multiple certificates, paste in your certificates one after the other. For example, format your certificates as shown in the following example:

```

-----BEGIN CERTIFICATE-----
ABCDEF12345678ABCDEF12345678ABCDEF12345678AB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
BCDEF12345678ABCDEF12345678ABCDEF12345678ABB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
CDEF12345678ABCDEF12345678ABCDEF12345678ABBB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----

```

3. To include both the Tanzu Operations Manager root CA and the certificates pasted into Trusted Certificates in the `trusted_certs` field in the BOSH director manifest, select the Include Tanzu Ops Manager Root CA in Trusted Certs check box. BOSH Director includes this CA in the trust store of every VM that it deploys.
You must select this check box if you are using Tanzu Operations Manager to generate certificates for a load balancer or a router.
4. To clear the default trusted certificates from all BOSH-deployed Linux VMs, select Clear the Default Trusted Certificates Store.

If you select this check box, you must provide your own trusted certificates. All TLS communication fails if trusted certificates are not available.

5. Select Generate passwords or Use default BOSH password. VMware recommends using the Generate passwords option for greater security.
6. Click Save. To view your saved BOSH Director password, click the Credentials tab.

Step 8: Configure BOSH DNS config

In the BOSH DNS Config pane, you can configure DNS for BOSH Director by adding excluded recursors, a recursor timeout, and handlers.

The text boxes in the BOSH DNS Config pane are:

- **Excluded Recursors:** Exclude recursor addresses, which are URL redirects, so that they are not contacted by the BOSH DNS server. For more information about how the BOSH DNS release selects recursors, see [Recursors](#) in *Native DNS Support* in the BOSH documentation.
- **Recursor Timeout:** Specify a timeout for the BOSH DNS server when contacting any connected recursor addresses. The time limit includes dialing, writing, and reading from the recursor. If any of these actions exceeds the time limit, the action fails.
- **Handlers:** Specify recursor addresses that apply to specific domains. For example, you can use handlers to forward all requests to a domain to a private DNS for resolution. For more information about using handlers, see [Additional handlers](#) in *Native DNS Support* in the BOSH documentation.

To add excluded recursors, a recursor timeout, or handlers to the BOSH DNS release:

1. Click BOSH DNS Config.

BOSH DNS Config

Excluded Recursors

Recursor Timeout

Handlers

Enter a JSON-formatted list of optional custom domain handlers

Save

2. (Optional) In Excluded Recursors, enter a list of prohibited recursor addresses.
3. (Optional) In Recursor Timeout, enter a time limit for contacting the connected recursors. This time limit must include one of the Go parse duration time units. For example, entering `5s` sets the timeout limit to five seconds. For more information about supported time units, see [func ParseDuration](#) in the Go Programming Language documentation.
4. (Optional) In Handlers, enter an array of custom domain handlers in JSON format. For example:

```
[
  {
    "cache": {
      "enabled": true
    },
    "domain": "example.com",
    "source": {
      "type": "http",
      "url": "http://example.endpoint.local"
    }
  }
]
```

5. Click Save.

Step 9: Configure Syslog

1. Click Syslog.
2. (Optional) To send BOSH Director system logs to a remote server, select Yes.
3. In Address, enter the IP address or DNS name for the remote server.
4. In Port, enter the port number that the remote server listens on.

5. Using the Transport Protocol drop-down menu, select TCP or UDP to specify the transport protocol to use to send the logs to the remote server.
6. (Optional) Select the Enable TLS check box to send encrypted logs to remote server with TLS. After you select the check box:
 1. In Permitted Peer, enter either the name or SHA1 fingerprint of the remote peer.
 2. In SSL Certificate, enter the SSL certificate for the remote server.

VMware recommends that you enable TLS encryption when you are forwarding logs. Logs can contain sensitive information, such as cloud provider credentials.

7. (Optional) In Queue Size, enter an integer that specifies the number of log entries held in the buffer. The default value is 100,000.
8. (Optional) Select the Forward Debug Logs check box to forward the logs to an external source. This option is deselected by default. If you select it, you might generate a large amount of log data.
9. (Optional) In the Custom rsyslog Configuration text box, enter the configuration details for [rsyslog](#). This text box requires the rainerscript syntax.
10. Click Save.

Step 10: Configure resources

To enable traffic, you must associate load balancers with the VMs in your deployment. You do this in the Resource Config pane.

To configure the Resource Config pane:

1. Click Resource Config.

The screenshot shows the 'Resource Config' pane. It has a 'SAVE' button in the top right. Below it is a table with four columns: 'JOB', 'INSTANCES', 'VM TYPE', and 'PERSISTENT DISK TYPE'. The first row is for the 'BOSH Director' job, with 'Automatic: 1' instances, 'Automatic: e2-standard-2 disk (machine_type: e2-standard-2, cpu: 2, ram: 8 GB, di: ...)' VM type, and 'Automatic: 50 GB' persistent disk type. Below the table, there is a checkbox labeled 'Internet Connected' which is checked, and a text box labeled 'Load Balancers'. The second row of the table is for the 'Main Compilation Job', with '1' instance, 'e2-highmem-2 (machine_type: e2-highmem-2, cpu: 2, ram: 16 GB, disk: 64 GB)' VM type, and 'None' persistent disk type.

2. To add a load balancer to the BOSH Director:
 - Click the icon next to the BOSH Director job name to see the Load Balancers text box and the Internet Connected check box.
 - Ensure that the Internet Connected check box is not selected for any job. The check box gives VMs a public IP address that enables outbound internet access.
 - In [Preparing to deploy Tanzu Operations Manager on GCP](#), you provisioned a Network Address Translation (NAT) box to provide internet connectivity to your VMs. For more information about using NAT in GCP, see [VPC network overview](#) in the GCP documentation.

If you install Elastic Application Runtime Windows add-on, provision your Main Compilation Job with at least 100 GB of disk space.

Step 11: Add custom VM extensions (optional)

Use the Tanzu Operations Manager API to add custom properties to your VMs; associated security groups and load balancers, for example. For more information, see [Managing custom VM extensions](#).

Step 12: Complete the BOSH Director installation

To complete your BOSH Director installation:

1. Click Installation Dashboard to return to the Installation Dashboard.
2. Click Review Pending Changes, then Apply Changes.

If you encounter an error message saying “Cannot reach gateway with IP [IP address] (ignorable if ICMP is disabled)”, return to the Network Config pane and ensure that you have deselected the Enable ICMP Checks check box. Click Review Pending Changes, then click Apply Changes again.

3. BOSH Director installation begins. This might take a few moments. When the installation process is complete, the Changes Applied dialog box appears.

Next steps

After you complete this procedure, follow the instructions for the runtime you intend to install.

- To deploy Elastic Application Runtime, see [Configuring Elastic Application Runtime](#).
- To prepare to deploy Container Orchestration Engine, see [Creating Service Accounts in GCP for Container Orchestration Engine](#).

Configuring a shared VPC on GCP with Tanzu Operations Manager

This topic describes how you can configure and integrate a shared Virtual Private Cloud (VPC) on Google Cloud Platform (GCP) with Tanzu Operations Manager.

GCP Shared VPC, formerly known as Google Cross-Project Networking (XPN), allows you to assign GCP resources to individual projects within an organization, and allows communication and shared services between projects. For more information about shared VPCs, see the [GCP documentation](#).

Prerequisites

To configure a shared VPC, you must assign your project to a Cloud Organization. Confirm that you have a Cloud Organization associated with your GCP account using one of these methods:

- GCP Console: From the [GCP console](#), click the Organization drop-down menu to display all organizations you belong to.
- gcloud Command Line Interface (CLI): From the command line, run `gcloud organizations list` to display all organizations you belong to. To install the gcloud CLI, see [gcloud Overview](#) in the Google documentation.

For more information, see [Creating and Managing Organizations](#) in the GCP documentation. If you do not have a Cloud Organization, contact GCP support.

Step 1: Provision the shared VPC

Follow the procedure in [Enabling a shared VPC host project](#) in the GCP documentation. This procedure requires shared VPC admin permissions.

Step 2: Create a shared VPC network

Follow the procedures in [Preparing to Deploy Tanzu Operations Manager on GCP](#) to create a new network with firewall rules. Then follow the procedures in:

- [Step 3: Create a GCP Network with Subnet](#)
- [Step 5: Create Firewall Rules for the Network](#)

Step 3: Connect the shared VPC to Tanzu Operations Manager

You can use the [GCP console](#) or the gcloud CLI to connect the shared VPC host project with Tanzu Operations Manager.

For more information, see [VPC Network Peering](#) in the GCP documentation.

VPC Network Peering is currently in beta and intended for evaluation and test purposes only.

Set up VPC network peering with the GCP console

To set up VPC network peering with the GCP console:

1. From the GCP console, click Networking, then VPC network peering.
2. Click Create Connection.
3. Enter a name for the network connection from the Tanzu Operations Manager project to the new shared network, such as `opsmanager-to-xpn`.
4. Click Save.
5. Click Create Connection.
6. Enter a name for the network connection from the new shared network to the Tanzu Operations Manager project, such as `xpn-to-opsmanager`.
7. Click Save.

Set up VPC network peering with the gcloud CLI

To set up VPC network peering with the gcloud CLI:

1. Run:

```
gcloud config set project OPSMANAGER-PROJECT
```

Where `OPSMANAGER-PROJECT` is the name of the project that contains your Tanzu Operations Manager installation.

2. To create a connection from the Tanzu Operations Manager project to the new shared VPC project, run:

```
gcloud beta compute networks peerings create OPSMANAGER-TO-VPC \  
  --network OPSMANAGER-NETWORK \  
  --peer-project VPC-HOST-PROJECT \  
  --peer-network VPC-NETWORK \  
  --auto-create-routes
```

Where:

- **OPSMANAGER-TO-VPC** is a name you choose for the connection, such as **om-to-vpc**.
- **OPSMANAGER-NETWORK** is the name of the network assigned to the Tanzu Operations Manager project in GCP, such as **my-om-project**.
- **VPC-HOST-PROJECT** is the name you gave the shared VPC project in [Step 1: Provision the Shared VPC](#).
- **VPC-NETWORK** is the name of the network you gave the shared VPC project in [Step 2: Create shared VPC networks](#).

3. Run:

```
gcloud config set project VPC-HOST-PROJECT
```

Where **VPC-HOST-PROJECT** is the new shared VPC project you created in [Step 1: Provision the Shared VPC](#).

4. To create a connection from the new shared VPC project to the Tanzu Operations Manager project, run:

```
gcloud beta compute networks peerings create VPC-TO-OPSMANAGER \  
  --network VPC-NETWORK \  
  --peer-project OPSMANAGER-PROJECT \  
  --peer-network OPSMANAGER-NETWORK \  
  --auto-create-routes
```

Where:

- **VPC-TO-OPSMANAGER** is a name you choose for the connection, such as **vpc-to-om**.
- **VPC-NETWORK** is the name of the network you gave the shared VPC project in [Step 2: Create shared VPC network](#).
- **OPSMANAGER-PROJECT** is the name of the project that contains your Tanzu Operations Manager installation.
- **OPSMANAGER-NETWORK** is the name of the network assigned to the Tanzu Operations Manager project in GCP.

Step 4: Verify the shared VPC configuration

After you configure a shared VPC, use the following procedure to verify that the shared VPC host project VM appears in the Tanzu Operations Manager project.

1. From the GCP console, select the Tanzu Operations Manager project from the drop-down menu at the top of the page.
2. Click Networking, then VPC networks.

3. Confirm that the shared VPC network name appears in the Subnets list.
4. Confirm that the shared VPC network IP address ranges match what you set for the new VPC project in [Step 2: Create a shared VPC network](#).

Upgrading BOSH Director for Tanzu Operations Manager on GCP

This topic describes how you can upgrade BOSH Director for Tanzu Operations Manager on Google Cloud Platform (GCP). Perform these tasks as part of the Tanzu Operations Manager upgrade process.

For information about upgrading Tanzu Operations Manager, see [Task 2: Upgrade Tanzu Operations Manager and Installed Products to v3.2](#) in *Upgrading Tanzu Operations Manager*.

In this procedure, you create a Tanzu Operations Manager VM instance to host the upgraded version of Tanzu Operations Manager. Then, to complete the Tanzu Operations Manager upgrade, you export your existing Tanzu Operations Manager installation onto this new VM.

To create a Tanzu Operations Manager VM instance:

1. Locate the Tanzu Operations Manager installation file. For more information, see [Locate the Tanzu Operations Manager installation file](#).
2. Create a private VM image. For more information, see [Create a private VM image](#).
3. Create the Tanzu Operations Manager VM. For more information, see [Create the Tanzu Operations Manager VM instance](#).

Prerequisites

To complete the Tanzu Operations Manager upgrade, you must have your Tanzu Operations Manager decryption passphrase. You defined this decryption passphrase during the initial installation of Tanzu Operations Manager.

Locate the Tanzu Operations Manager installation file

The Tanzu Operations Manager installation file includes a filepath to the Tanzu Operations Manager image. You use this image file to create a private VM image for the new Tanzu Operations Manager VM.

To locate the Tanzu Operations Manager installation file:

1. Log in to the [Broadcom Support portal](#) and search for “Tanzu Operations Manager.” Click the name to see the list of active releases.
2. From the Release list, select the release for your upgrade.
3. Select one of the following download files:
 - Tanzu Operations Manager for GCP
 - Tanzu Operations Manager YAML for GCPWhen you click the download link, your browser downloads or opens the [OpsManager_version_onGCP.pdf](#) or [OpsManager_version_onGCP.yml](#) file. These documents provide the GCP location of the Tanzu Operations Manager [.tar.gz](#) installation file based on the geographic location of your installation.
4. Copy the filepath string of the Tanzu Operations Manager image based on your existing deployment location.

Create a private VM image

Create a private VM image to use when you create the new Tanzu Operations Manager VM.

To create a private VM image:

1. Log in to the [GCP console](#).
2. In the navigation panel, click Compute Engine, then Images.
3. Click Create Image.
4. Complete the following fields:
 - Name: Enter a name that matches the naming convention of your existing Tanzu Operations Manager image files.
 - Encryption: Leave Automatic (recommended) selected.
 - Source: Choose Cloud Storage file.
 - Cloud Storage file: Paste in the Google Cloud Storage filepath you copied from the PDF or YAML file in the previous task.
5. Click Create. The file might take a few minutes to import.

Create the Tanzu Operations Manager VM instance

Create a Tanzu Operations Manager VM instance to host the new version of Tanzu Operations Manager.

To create a Tanzu Operations Manager VM instance:

1. Select the check box for the image you previously created.
2. Click Create Instance.
3. In the Create an instance form, fill in the following text boxes:
 - Name: Enter a name that matches the naming conventions of your existing deployment.
 - Zone: Select a zone from the region of your existing deployment.
 - Machine type: Click Customize to manually configure the vCPU and memory. A Tanzu Operations Manager VM instance requires the following minimum specifications:

Machine Spec	Minimum Value
CPU	4 vCPUs
Memory	16 GB

- Boot disk: Click Change, then perform the following steps:
 - Click Custom images, if it is not already selected.
 - Click the Boot disk type. If you have a Tanzu Operations Manager environment with high performance needs, select SSD. As an example, environments used to develop Tanzu Operations Manager tiles may benefit from a higher performing Tanzu Operations Manager VM boot disk. For most environments, however, you can select Standard.
 - Set the Size (GB) of the boot disk to the minimum or higher.

Machine Spec	Minimum Value
Boot disk	100 GB

- Click the Tanzu Operations Manager image you created in the previous step if it is not already selected.
 - Click Select to save.
- In Identity and API access, click the Service account you created when you initially installed Tanzu Operations Manager. For more information, see [Step 1: Set up IAM Service Accounts](#) in *Preparing to Deploy Tanzu Operations Manager on GCP*.
- Allow HTTP traffic: Select this check box only if you selected it in your original Tanzu Operations Manager VM configuration.
- Allow HTTPS traffic: Select this check box only if you selected it in your original Tanzu Operations Manager VM configuration.
- Networking: Click the Networking tab and do the following:
 - For Network and Subnetwork, select the network and subnetwork you created when you initially deployed Tanzu Operations Manager. For more information, see [Step 3: Create a GCP Network with Subnets](#) in *Preparing to Deploy Tanzu Operations Manager on GCP*.
 - For Network tags, enter any tags that you applied to your original Tanzu Operations Manager. For example, if you used the `pcf-opsmanager` tag to apply the firewall rule you created in [Step 5: Create Firewall Rules for the Network](#) in *Preparing to Deploy Tanzu Operations Manager on GCP*, then apply the same tag to this Tanzu Operations Manager VM.
 - For Internal IP, click **Custom**. In the Internal IP address field, enter a spare address located within the reserved IP range you configured in your existing BOSH Director as part of [Step 5: Create Networks Page](#) in *Configuring BOSH Director on GCP*. Do not use `10.0.0.1`, which is configured for the Gateway.
 - For External IP, click New static IP address. In the next form, enter a name for the static IP. For example, `om-public-ip`. Click Reserve. In the External IP drop-down menu, select the static IP address you just reserved.
- 4. Click Create to deploy the new Tanzu Operations Manager VM. This might take a few moments.
- 5. Go to your DNS provider, and modify the entry that points a fully-qualified domain name (FQDN) to the Tanzu Operations Manager VM. Replace the original Tanzu Operations Manager static IP address with the public IP address of the new Tanzu Operations Manager VM you created in a previous step.

To set up Tanzu Operations Manager authentication correctly, VMware recommends using a FQDN to access Tanzu Operations Manager. Using an ephemeral IP address to access Tanzu Operations Manager can cause authentication errors upon subsequent access. If you are importing older Tanzu Operations Manager settings, you must use the FQDN.

Next steps

After you complete this procedure, continue the upgrade instructions in [Upgrading Tanzu Operations Manager](#).

Later, if you need to SSH into the Tanzu Operations Manager VM to perform diagnostic troubleshooting, see [Log in to Tanzu Operations Manager VM with SSH](#).

Deleting your Tanzu Operations Manager deployment from GCP

This topic describes how you can delete the Google Cloud Platform (GCP) resources associated with a Tanzu Operations Manager deployment.

You can delete the resources in one of the following ways:

- If you created a separate project for your Tanzu Operations Manager deployment, follow the procedure in [Delete the project](#).
- If the project that contains your Tanzu Operations Manager deployment also contains other resources that you want to preserve, follow the procedure in [Delete Tanzu Operations Manager resources](#).

Delete the project

To delete the project for your Tanzu Operations Manager deployment:

1. Go to the GCP Console Dashboard.
2. Go to Project, and click Manage project settings.
3. Click DELETE PROJECT.
4. Enter your project ID and click SHUT DOWN to confirm.

Delete Tanzu Operations Manager resources

To delete the resources associated with your Tanzu Operations Manager deployment:

1. Go to the GCP Console Dashboard.
2. Click the upper left icon and click Networking.
3. Click Load balancing.
4. Complete the following steps for all load balancers associated with your Tanzu Operations Manager deployment:
 1. Click the trash can icon next to the load balancer.
 2. In the next dialog, click any health checks and backend services associated with the load balancer.
 3. Click DELETE LOAD BALANCER AND THE SELECTED RESOURCES.
5. Click the upper left icon and click Compute Engine.
6. Complete the following steps for VM instances, Instance groups, and Disks:
 1. Select the check box next to the Tanzu Operations Manager resource.
 2. When all Tanzu Operations Manager resources are selected, click DELETE.
 3. Click DELETE again to confirm.
7. Click the upper left icon and click Networking.
8. Click External IP addresses.
9. Select all external IP addresses associated with your Tanzu Operations Manager deployment, and click RELEASE STATIC ADDRESS.
10. Click Networks, and complete the steps for any networks you created for Tanzu Operations Manager:

1. Click the name of the network.
 2. Click DELETE NETWORK.
 3. Click DELETE to confirm.
11. Click the upper left icon and click IAM & Admin.
 12. Click the trash can icon next to the `bosh` service account you created for Tanzu Operations Manager and click REMOVE.
 13. Go to Compute Engine, then Metadata, followed by SSH Keys. Delete the `vcap` SSH key that you created for the project.

Installing and configuring Tanzu Operations Manager on vSphere

The following topics describe how you can install and configure Tanzu Operations Manager on vSphere:

- [Installing Tanzu Operations Manager on vSphere](#)
- [vSphere service account requirements](#)
- [Deploying Tanzu Operations Manager on vSphere](#)
- [Configuring BOSH Director on vSphere](#)
- [Using the Cisco Nexus 1000v Switch with Tanzu Operations Manager](#)
- [Using the BOSH Resurrector](#)
- [Understanding and using availability zones in vSphere](#)
- [Using vSphere host groups with your Tanzu Operations Manager deployment](#)

Installing Tanzu Operations Manager on vSphere

This guide describes how you can install Tanzu Operations Manager on vSphere.

If you experience a problem while using the following steps, see the [Elastic Application Runtime documentation](#) or the *Known Issues* sections of the following topics:

- [Tanzu Operations Manager v3.2 Release Notes](#)
- [Elastic Application Runtime v10.3 Release Notes](#) in the Elastic Application Runtime documentation
- [Isolation Segment v10.3 Release Notes](#) in the Elastic Application Runtime documentation
- [Elastic Application Runtime Windows add-on v10.3 Release Notes](#) in the Elastic Application Runtime Windows add-on documentation
- [Release Notes](#) in the Container Orchestration Engine documentation

Requirements

This section describes the resource requirements for installing Tanzu Operations Manager.

If you are using the Cisco Nexus 1000v Switch, see the [Using the Cisco Nexus 1000v Switch with Tanzu Operations Manager](#).

When you install Tanzu Operations Manager on a vSphere environment with multiple ESXi hosts, you must use network-attached or shared storage devices. Local storage

devices do not support sharing across multiple ESXi hosts.

Version requirements

Tanzu Operations Manager can be installed on vSphere, and the BOSH Director tile in Tanzu Operations Manager can be configured on vSphere.

The following versions of vSphere are supported.

The BOSH Director tile within Tanzu Operations Manager can optionally be configured with the following versions of NSX-T.

	Supported versions
VMware Cloud Foundation (VCF)	9
vSphere	v8.0 v7.0 v6.7 v6.5
NSX-T (optional)	v4.0 v3.1 v3.0

Resource requirements

When launching Tanzu Operations Manager on vSphere, you must use an instance type that meets the following minimum specifications:

Machine Spec	Minimum Value
CPU	4 vCPUs
Memory	16 GB
Boot disk	100 GB

You can install Tanzu Operations Manager with one of two runtimes. For more information about the requirements for each runtime, see the following:

- Tanzu Platform for Cloud Foundry: [Elastic Application Runtime on vSphere requirements](#).
For requirements when deploying with NSX-T, see [Prerequisites](#) in *Deploying Elastic Application Runtime with NSX-T Networking*.
- VMware Tanzu Container Orchestration Engine (formerly TKGI):
 - Without NSX-T: [vSphere Prerequisites and Resource Requirements](#)
 - With NSX-T: [Preparing to Install Container Orchestration Engine on vSphere with NSX](#).

Firewall port requirements

Tanzu Operations Manager requires that the NSX firewall routes are not blocked. By default, all communication between Tanzu Operations Manager VMs and vCenter or ESXi hosts route through the NSX firewall are blocked. For more information on which ports to enable, see [VMware ports and protocols](#) for vSphere.

vSphere service account requirements

Tanzu Operations Manager requires read/write permissions to the data center level of the [vSphere Inventory Hierarchy](#) to successfully install. VMware recommends defining a custom role for the service account that has all privileges for all objects in the data center, including propagating privileges to children.

Because Tanzu Operations Manager passes all required credentials through to BOSH, you only need one service account with the required vSphere privileges to complete the installation. Setting up separate service accounts for Tanzu Operations Manager and BOSH is not necessary or recommended.

For more information about service account requirements, see the following topics:

- Elastic Application Runtime: [vSphere service account requirements](#)
- Container Orchestration Engine: [Create the BOSH/Tanzu Operations Manager service account in Preparing vSphere Before Deploying Container Orchestration Engine](#)

You can also apply the default [VMware Administrator System Role](#) to the service account to achieve the appropriate permission level.

vSphere security documents

The following links contain best practices for securing and managing a vSphere installation:

- [vSphere v6.5 Security Guide \(PDF\)](#)
- [vSphere v6.7 Release Notes](#)
- [vSphere v7.0 Security Guide \(PDF\)](#)

Installation procedure

Complete the following procedures to install Tanzu Operations Manager on vSphere:

1. [Deploying Tanzu Operations Manager to vSphere](#)
2. [Configuring BOSH Director on vSphere](#)

Additional configuration

See the following topics for additional configuration options:

- [vSphere Virtual Disk Types](#)
- [Using the BOSH Resurrector](#)
- [Availability Zones in vSphere](#)

Next steps

After you complete the previous procedures, configure a runtime.

For information about installing and configuring a runtime, see the following:

- Elastic Application Runtime:
 - Without NSX-T: [Configuring Elastic Application Runtime](#)
 - With NSX-T: [Deploying Elastic Application Runtime with NSX-T Networking](#)

You must install the NSX-T tile before you install or upgrade the Elastic Application Runtime tile.

- Elastic Application Runtime [Windows]: [Installing and configuring Elastic Application Runtime Windows add-on](#)
- Small Footprint Elastic Application Runtime: [Getting Started with Small Footprint Elastic Application Runtime](#)
- Container Orchestration Engine:
 - Without NSX-T: [Installing Container Orchestration Engine](#)
 - With NSX-T: [Installing NSX-T for Container Orchestration Engine](#)

vSphere service account requirements

This topic describes the minimum privileges required by the vSphere BOSH CPI.

Overview of vSphere service account requirements

A vSphere admin must grant minimum privileges to the vSphere service account that Tanzu Operations Manager uses to manage vSphere resources.

The Tanzu Operations Manager account needs privileges at both the vCenter server level and at the Datacenter level.

The recommended permissions in this topic are configured using the API. UI permissions are not included because they vary among vSphere versions. API permissions are consistent across vSphere versions.

For more information about how permission levels and inheritance work in vSphere, see [Hierarchical inheritance of permissions](#).

For more information about vSphere permissions, see [vSphere permissions and user management tasks](#).

vCenter-level privileges

Tanzu Operations Manager assigns custom attributes to the virtual machines (VMs) it deploys to identify BOSH releases and job index information about each VM. vCenter APIs require vCenter server level access to manage these custom attributes.

The following table summarizes the privileges that a Tanzu Operations Manager account requires at the vCenter Server instance level. Some of these privileges are inherited, and others must be granted by a vCenter admin.

Object	Privilege (API)
Role	System.Anonymous
	System.Read
	System.View
Global	Global.ManageCustomFields
	Global.SetCustomField
	Extension.Register
Profile-Driven Storage	StorageProfile.Update
	StorageProfile.View

Datacenter-level privileges

The following privileges must be set at the data center level.

Object	Privilege (API)
Datastore	Datastore.FileManagement
Network	Network.Assign

Folder and datastore-level privileges

You must grant the following privileges on any entities in a datacenter where you deploy Tanzu Operations Manager.

Datastore object

Privilege (API)
Datastore.AllocateSpace
Datastore.Browse
Datastore.DeleteFile
Datastore.FileManagement

Folder object

Tanzu Operations Manager creates a folder for VMs, stemcells, and persistent disks during installation. The folder contents change frequently as Tanzu Operations Manager applies changes.

Privilege (API)
Folder.Create
Folder.Delete
Folder.Move
Folder.Rename

Host object

Privilege (API)
Host.Inventory.EditCluster
Host.Config.SystemManagement (Required if using a vSAN with encryption enabled)

Inventory service object

Privilege (API)
InventoryService.Tagging.CreateTag
InventoryService.Tagging.EditTag
InventoryService.Tagging.DeleteTag

Resource object

When using **vAppImport** to clone a VM, BOSH requires the resource migration privileges to create a new, powered-off VM based on a given stemcell. BOSH migrates the VM to the destination datastore, where Tanzu Operations Manager deploys the VM and powers it on.

Privilege (API)
Resource.AssignVMToPool
Resource.ColdMigrate
Resource.HotMigrate

Profile-driven storage object

Privilege (API)
StorageProfile.Update
StorageProfile.View

Virtual Machine object

See the following tables. They describe Virtual Machines (VMs) in the following categories:

- Configuration
- Guest Operations
- Interaction
- Inventory
- Provisioning
- Snapshot Management

Configuration

Privilege (API)
VirtualMachine.Config.AddExistingDisk
VirtualMachine.Config.AddNewDisk
VirtualMachine.Config.AddRemoveDevice
VirtualMachine.Config.AdvancedConfig
VirtualMachine.Config.CPUCount
VirtualMachine.Config.Resource
VirtualMachine.Config.ManagedBy
VirtualMachine.Config.ChangeTracking
VirtualMachine.Config.DiskLease
VirtualMachine.Config.MksControl
VirtualMachine.Config.DiskExtend
VirtualMachine.Config.Memory

Privilege (API)

VirtualMachine.Config.EditDevice

VirtualMachine.Config.RawDevice

VirtualMachine.Config.ReloadFromPath

VirtualMachine.Config.RemoveDisk

VirtualMachine.Config.Rename

VirtualMachine.Config.ResetGuestInfo

VirtualMachine.Config.Annotation

VirtualMachine.Config.Settings

VirtualMachine.Config.SwapPlacement

VirtualMachine.Config.UpgradeVirtualHardware

Guest operations

Privilege (API)

VirtualMachine.GuestOperations.Execute

VirtualMachine.GuestOperations.Modify

VirtualMachine.GuestOperations.Query

Interaction

Privilege (API)

VirtualMachine.Interact.AnswerQuestion

VirtualMachine.Interact.SetCDMedia

VirtualMachine.Interact.ConsoleInteract

VirtualMachine.Interact.DefragmentAllDisks

VirtualMachine.Interact.DeviceConnection

VirtualMachine.Interact.GuestControl

VirtualMachine.Interact.PowerOff

VirtualMachine.Interact.PowerOn

VirtualMachine.Interact.Reset

VirtualMachine.Interact.Suspend

VirtualMachine.Interact.ToolsInstall

Inventory

Privilege (API)

VirtualMachine.Inventory.CreateFromExisting

VirtualMachine.Inventory.Create

VirtualMachine.Inventory.Move

Privilege (API)

VirtualMachine.Inventory.Register

VirtualMachine.Inventory.Delete

VirtualMachine.Inventory.Unregister

Provisioning

When cloning a stemcell, BOSH sets custom specifications, such as host names and network configurations, based on the stemcell operating system.

The VM download privilege allows BOSH to modify files within a VM, including links between VMs and persistent disks. When vMotion migrates disks in vSphere, BOSH uses these links to maintain the connections between VMs and their persistent disks.

Privilege (API)

VirtualMachine.Provisioning.DiskRandomAccess

VirtualMachine.Provisioning.DiskRandomRead

VirtualMachine.Provisioning.GetVmFiles

VirtualMachine.Provisioning.PutVmFiles

VirtualMachine.Provisioning.CloneTemplate

VirtualMachine.Provisioning.Clone

VirtualMachine.Provisioning.Customize

VirtualMachine.Provisioning.DeployTemplate

VirtualMachine.Provisioning.MarkAsTemplate

VirtualMachine.Provisioning.MarkAsVM

VirtualMachine.Provisioning.ModifyCustSpecs

VirtualMachine.Provisioning.PromoteDisks

VirtualMachine.Provisioning.ReadCustSpecs

Snapshot management

Before Tanzu Operations Manager deploys a new VM, it uses a snapshot to clone the stemcell image to the destination.

Privilege (API)

VirtualMachine.State.CreateSnapshot

VirtualMachine.State.RemoveSnapshot

VirtualMachine.State.RenameSnapshot

VirtualMachine.State.RevertToSnapshot

vApp object

These privileges must be set at the resource pool level. [VApp.ApplicationConfig](#) is required when attaching or detaching persistent disks.

Deploying Tanzu Operations Manager on vSphere

You can use these instructions to deploy Tanzu Operations Manager to VMware vSphere. The instructions in this topic are based on vSphere 6.5 using the vSphere Client (HTML5) interface.

Runtime considerations

For security reasons, VMware requires a dedicated instance of Tanzu Operations Manager for use with VMware Tanzu Container Orchestration Engine (formerly TKGI). If you are installing Container Orchestration Engine on vSphere with NSX-T integration, follow the instructions in [Deploying Tanzu Operations Manager with NSX-T for Container Orchestration Engine](#) instead of performing the steps described in this topic.

Prepare vSphere

1. Log in to vCenter using the vSphere Client (HTML5).
2. Select the vSphere Datacenter where you are deploying Tanzu Operations Manager.
3. Create one or more vSphere clusters. A vSphere cluster is a collection of ESXi hosts and associated virtual machines (VMs) with shared resources and a shared management interface.

Follow the next steps to create a vSphere cluster:

1. Right-click the vSphere Datacenter object.
2. Click New Cluster.
3. Name the cluster; for example, **Cluster1**.
4. Click Turn ON vSphere DRS to enable DRS. Set DRS Automation to Partially Automated or Fully Automated.
5. Click Turn ON vSphere HA to enable vSphere HA.

For highly available, production-ready environments, you must create more than one cluster. For more information, see [Compute and HA considerations](#) and [Scaling and capacity management](#) in *vSphere Reference Architecture*.

4. If you created more than one vSphere cluster, configure shared storage for the clusters. You can allocate shared storage to the clusters using one of two common approaches: horizontal or vertical. For more information about configuring shared storage for clusters, see [Storage Design](#) in *vSphere Reference Architecture*.

Failure to properly configure shared storage for vSphere clusters could result in downtime and unrecoverable loss of data.

5. Create a resource pool for the management components in your deployment. Resource pools are vSphere objects used to organize resources.

The management resource pool separates non-HA components such as BOSH Director. For example, if you deploy Tanzu Operations Manager for use with Container Orchestration Engine, the Container Orchestration Engine management plane includes Tanzu Operations Manager, BOSH Director, the Container Orchestration Engine control plane, and Harbor (optional).

Follow the next steps to create a vSphere resource pool:

1. Right-click the vSphere cluster object that you created and click New Resource Pool.
2. Name the resource pool; for example, **RP-MGMT**.
3. Review the default CPU and Memory settings and adjust if necessary. Adjustments are not commonly needed.
4. Click OK.
6. (Optional) If desired, create host groups within the clusters. For information about using host groups in vSphere, see [Host groups in vSphere](#).
 1. Click the Cluster where you want to create a host group.
 2. Click the Configure tab.
 3. In Configuration, click VM/Host Groups.
 4. Click Add and configure the host group as follows:
 - Name: Enter a name for the host group.
 - Type: Select Host Group from the drop-down menu.
 - Click Add and select the ESXi host(s) to include in the host group.
 - Click OK.

Deploy Tanzu Operations Manager

Tanzu Operations Manager for vSphere is provided as an OVA file. An OVA file is a template for a VM. To deploy Tanzu Operations Manager on vSphere, you import the OVA file into your vSphere environment.

1. Download the Tanzu Operations Manager for vSphere OVA file from the [Broadcom Support portal](#). Select a Tanzu Operations Manager release from the Release list.
2. Right-click the management resource pool that you created and select Deploy OVF Template. Alternatively, from the vCenter menu, select Actions, then Deploy OVF Template.
3. On the Select an OVF template pane, click Browse.
4. Click the Tanzu Operations Manager OVA file that you downloaded, and click Open.
5. Review the template selection and click Next.
6. On the Select a name and folder pane:
 - Enter a name for the Tanzu Operations Manager virtual machine, or use the default name.
 - Select the object labeled Datacenter.
 - Click Next.
7. On the Select a compute resource pane, select the compute resource where the Tanzu Operations Manager VM runs, and click Next. Select the Resource Pool for the management components.

Hardware virtualization must be turned off if your vSphere host does not support VT-X/EPT. Refer to [vSphere Requirements](#) in *Tanzu Operations Manager on vSphere Requirements* for more information.

8. On the Review Details pane, confirm the configuration up to this point, and click Next.
9. At the Select Storage pane, select the virtual disk format.
 1. Choose a Datastore.
 - If you use vSAN, select the vSAN datastore and apply the default storage policy.
 - If you use a dedicated disk, select the appropriate disk format for your deployment needs. The default type is Thin Provision, but you can choose to use a thick provisioned disk such as Thick Provisioned Lazy Zeroed. For more information, see [vSphere virtual disk types](#).
 2. Click Next.

Tanzu Operations Manager requires a VM with at least 8 GB memory.

10. On the Select Networks pane, either select the default VM Network, or ideally, create an Infrastructure Network, and click Next.
11. On the Customize template pane, enter the following information, and click Next.
 - IP Address: The IP address of the Tanzu Operations Manager network interface; for example, [10.0.0.2](#).
 - Netmask: The network mask for Tanzu Operations Manager; for example, [255.255.255.0](#).
 - Default Gateway: The default gateway for Tanzu Operations Manager to use; for example, [10.0.0.1](#).
 - DNS: One or more DNS servers for the Tanzu Operations Manager VM to use; for example, [10.20.20.1](#).
 - NTP Servers: The IP address of one or more NTP servers for Tanzu Operations Manager; for example, [10.113.60.176](#).
 - Public SSH Key: Enter the pre-existing public SSH key value to enable SSH access to the Tanzu Operations Manager VM. Enter the full key in the [ssh-rsa AAA...](#) format. You cannot deploy Tanzu Operations Manager successfully without adding a public key. If you do not add a key, Tanzu Operations Manager shuts down automatically because it cannot find a key, and it might enter a reboot loop.
 - Custom hostname: The hostname for the Tanzu Operations Manager VM; for example, [ops-manager](#).
12. On the Ready to complete pane, review the configuration settings and click Finish.

This action begins the OVA import and deployment process.
13. Use the Recent Tasks pane at the bottom of the vCenter dashboard to check the progress of the OVA import and deployment.

If the import or deployment is unsuccessful, check the configuration for errors. Fix any errors and restart the deployment process.

Start Tanzu Operations Manager

1. Create a DNS entry for the IP address that you used for Tanzu Operations Manager. You must use this fully qualified domain name when you log into Tanzu Operations Manager.

Tanzu Operations Manager security features require you to create a fully qualified domain name to access Tanzu Operations Manager during the initial configuration. For more information, see [Tanzu Operations Manager on vSphere Requirements](#).

2. Select the Tanzu Operations Manager VM and click Actions, then Power, followed by Power On.

After the VM boots, the Tanzu Operations Manager web interface is available at the IP address that you specified.

It is normal to experience a brief delay while the web server and VM start.

3. Navigate to the fully qualified domain of your Tanzu Operations Manager in a web browser.

Next steps

After you complete this procedure, follow the instructions in [Configuring BOSH Director on vSphere](#).

Configuring BOSH Director on vSphere

This topic describes how you can configure the BOSH Director tile for VMware vSphere.

You can also perform the procedures in this topic using the Tanzu Operations Manager API. For more information, see [Using the Tanzu Operations Manager API](#).

If you are installing VMware Tanzu Container Orchestration Engine (formerly TKGI) on vSphere with NSX-T integration, follow the instructions in [Configuring BOSH Director with NSX-T for Container Orchestration Engine](#) instead of performing the procedure described in this topic.

Prerequisite

Before you begin this procedure, you must complete all steps in [Deploying Tanzu Operations Manager to vSphere](#).

Step 1: Log into Tanzu Operations Manager

1. In a web browser, go to the fully qualified domain of your Tanzu Operations Manager.
2. Log in to Tanzu Operations Manager. See [Log in to Tanzu Operations Manager for the first time](#).

Step 2: Configure vCenter

1. Click the BOSH Director for vSphere tile.
2. Click vCenter Config. Configure the one or more vCenters that host your Tanzu Operations Manager foundation, as described in the following steps.

vCenter Config
Add vCenter Config

Name*

vCenter Host*

vCenter Username*

vCenter Password*
[Change](#)

vCenter CA Certificate

Datacenter Name*

Virtual Disk Type*

thin

Ephemeral Datastore Names (comma delimited)*

NOTE: Removing an Ephemeral Datastore after an initial deploy can result in a system outage and/or data loss.

Ephemeral Datastore Cluster Names (comma delimited)

NOTE: Removing an Ephemeral Datastore Cluster after an initial deploy can result in a system outage and/or data loss.

Persistent Datastore Names (comma delimited)*

NOTE: Removing a Persistent Datastore after an initial deploy can result in a system outage and/or data loss.

Persistent Datastore Cluster Names (comma delimited)

3. Complete the following fields on the vCenter Config pane:

- **Name:** A name that you provide for your vCenter configuration. This field is used to identify the datacenter configuration in Tanzu Operations Manager if you are configuring multiple datacenters.
- **vCenter Host:** The hostname of the vCenter that manages ESXi/vSphere.
- **vCenter Username:** The user name of a vCenter user that has create and delete privileges for VMs and folders.

- vCenter Password: The password for the vCenter user specified earlier.
- vCenter CA Certificate: The CA certificate or certificate chain that signs the TLS certificate for the vCenter host. When a value is provided, TLS connections to the vCenter server validate that the server certificate is trusted. Otherwise, TLS connections are established but the certificate is not validated.
- Datacenter Name: The name of the datacenter as it appears in vCenter.
- Virtual Disk Type: The Virtual Disk Type to provision for all VMs. For guidance on selecting a virtual disk type, see [vSphere virtual disk types](#).

Datastores do not support whitespace characters in their names. Including whitespace characters in datastore names causes an error.

- Ephemeral Datastore Names (comma-delimited): The names of the datastores that store ephemeral VM disks deployed by Tanzu Operations Manager.
- Ephemeral Datastore Cluster Names (comma-delimited): The names of the datastore clusters that store ephemeral VM disks deployed by Tanzu Operations Manager. If Datastore Names and Datastore Clusters are specified, both are considered for disk placement. If you deactivate Storage DRS for a datastore cluster, the BOSH CPI ignores the cluster. For more information, see [Activate and deactivate storage DRS](#). As of Tanzu Operations Manager v3.0.11+LTS-T, datastore clusters located in folders can be referenced using the full path to the datastore cluster. Folder names must be followed by a forward slash character.
- Persistent Datastore Names (comma-delimited): The names of the datastores that store persistent VM disks deployed by Tanzu Operations Manager.
- Persistent Datastore Cluster Names (comma-delimited): The names of the datastore clusters that store persistent VM disks deployed by Tanzu Operations Manager. If Datastore Names and Datastore Clusters are specified, both are considered for disk placement. If you deactivate Storage DRS for a datastore cluster, the BOSH CPI ignores the cluster. For more information, see [Activate and deactivate storage DRS](#). As of Tanzu Operations Manager v3.0.11+LTS-T, datastore clusters located in folders can be referenced using the full path to the datastore cluster. Folder names must be followed by a forward slash character.

A datastore cluster is a collection of datastores with shared resources and a shared management interface.

4. To configure networking, complete the vCenter Config fields described:

- Standard vCenter Networking: This is the default option when upgrading Tanzu Operations Manager. If you select this option, you do not need to configure the NSX-T Networking fields.

If you intend to deploy Container Orchestration Engine using Flannel as the CNI, select this option to configure the BOSH Director. Do not choose NSX-T Networking. When you configure the Container Orchestration Engine tile later on, you configure Flannel as the container network interface (CNI).

If you intend to deploy Container Orchestration Engine with NSX-T, see the [Container Orchestration Engine documentation](#).

When deploying on VMC (vSphere on AWS), you must select Standard vCenter Networking. NSX-T Networking does not work with VMC.

- NSX-T Networking: Select this option to allow NSX-T network virtualization for non-Container Orchestration Engine products such as Elastic Application Runtime. This virtualization allows you to configure component load balancing and security in the Resource Config pane of the BOSH Director tile. For more information, see [Step 10: Resource Config pane](#).

Configure NSX networking by entering the following information:

- Use NSX-T Policy API: Select whether to use the NSX-T Policy API instead of the Manager API when placing VMs in NSX-T groups and server pools. You can use NSX-T groups or server pools to designate VMs as load balancer back ends. This feature requires NSX-T Data Center v3.0 or later.
- Use NSX-T Policy API Migration Mode: Activate this check box to activate migration mode on the CPI. This allows you to transition infrastructure created with the Management API over to the Policy API.
- NSX Address: The address of the NSX Manager.
- NSX-T Authentication: Select how BOSH Director authenticates to the NSX Manager:
 - Local User Authentication authenticates with a user name and password.
 - For this option, fill in the NSX Username and NSX Password fields. Specify an NSX user with [Enterprise Administrator](#) privileges.
 - Certificate Authentication authenticates with a certificate and private key.
 - For this option, fill in the NSX Manager Principal Identity Certificate and NSX Manager Principal Identity Private Key fields.
- NSX CA Cert: A CA certificate in PEM format that establishes a secure connection to the NSX server before the BOSH Director authenticates to the NSX Manager. If the NSX Manager generated a self-signed certificate, use the following command to retrieve the CA certificate using OpenSSL:

```
openssl s_client -showcerts -connect NSX-MANAGER-ADDRESS:443 < /dev/null 2> /dev/null | openssl x509
```

Where [NSX-MANAGER-ADDRESS](#) is the address of the NSX manager.

- Use NSX-T Policy API Migration Mode: This is a special mode to migrate an existing deployment from earlier NSX-T Manager API objects such as NS Groups to the newer NSX-T Policy API objects such as Groups, and must only be used with NSX-T migration scripts. Leave unselected if you are unsure.
- AVI Load Balancer: Select this check box to allow Load Balancing provided by the VMware AVI Load Balancer Appliance for non-Container Orchestration Engine products such as Elastic Application Runtime. This Appliance allows you to configure component load balancing in the Resource Config pane of the BOSH Director tile. For more information, see [Step 10: Resource Config pane](#).

Ensure that you change the AVI appliance configuration to use Basic Auth before setting

up the AVI Appliance in Tanzu Operations Manager.

Configure the AVI Appliance by entering the following information:

- AVI Appliance Address: The address of the AVI Appliance
 - AVI Appliance CA Cert: (Optional) A CA certificate in PEM format that establishes a secure connection to the AVI Appliance before the BOSH Director authenticates to the AVI Appliance
 - AVI Appliance Authentication: Select how BOSH Director authenticates to the AVI Appliance:
 - Token Authentication: Authenticates to the AVI Appliance with a pre-created authentication token
 - AVI Appliance Auth Token: The pre-created AVI Appliance authentication token
 - Username and Password Authentication: Authenticates to the AVI Appliance with HTTP Basic Authentication
 - AVI Appliance Username: The username of the authenticating user
 - AVI Appliance Password: The password of the authenticating user
5. Configure the following folder names. After initial deployment, you cannot edit folder names.
- VM Folder: The vSphere datacenter folder where Tanzu Operations Manager places VMs. Enter `[YOUR-DEPLOYMENT]_vms` where `YOUR-DEPLOYMENT` corresponds to a descriptive name for your deployment. For example, `my_pcf_vms`.
 - Template Folder: The vSphere datacenter folder where Tanzu Operations Manager places templates. Enter `[YOUR-DEPLOYMENT]_templates`. For example, `my_pcf_templates`.
 - Disk path Folder: The vSphere datastore folder where Tanzu Operations Manager creates attached disk images. Do not nest this folder. Enter `[YOUR-DEPLOYMENT]_disk`. For example, `my_pcf_disk`.
6. Click Save.
7. (Optional) Click Add vCenter Config to configure additional vCenters. After you click Save, your multiple vCenter Configs are listed in the vCenter Configs pane. For more information about multiple vCenter configs, see [Managing multiple data centers](#).

Step 3: Director Config pane

To configure the Director Config pane:

1. Click Director Config.

Director Config

[Go to](#)
[Certificate Duration Overrides](#)
[Health Monitor](#)
[CredHub Encryption Provider](#)
[Blobstore Location](#)
[Database Location](#)

NTP Servers (comma delimited)*

One or more NTP server addresses for consistent and valid time stamps used in syslog, etc.

Bosh HM Forwarder IP Address

Should only be used if OS Bosh HM Forwarder job is deployed

☐ Enable VM Resurrector Plugin
☒ Enable System Metrics
☐ Enable Post Deploy Scripts
☐ Enable Short Lived NATS Bootstrap Credentials
This will make the Bosh Director rotate the NATs bootstrap credentials of a VM. This increases security by invalidating the NATS credentials that are stored in the IAAS metadata service.
☐ Recreate VMs deployed by the BOSH Director
This will force BOSH to recreate all Bosh Deployed VMs on the next deploy. This does not recreate the BOSH Director. Persistent disk will be preserved.
☐ Recreate BOSH Director VM
This will force BOSH to recreate the Bosh Director VM on the next deploy. Persistent disk will be preserved.
☐ Recreate All Persistent Disks
Check this box to recreate all Persistent Disks for the Director and all other Tiles.
☐ Enable bosh deploy retries
This will attempt to re-deploy a failed deployment up to 5 times.

- The NTP server configuration is updated only after VM recreation. If you modify the value of this text box, select the Recreate VMs deployed by the BOSH Director check box to re-create your BOSH Director-deployed VMs and update the NTP server configuration. If you have any service tiles installed, ensure that the Recreate All Service Instances errand runs for each service tile.
- 2. Leave the Bosh HM Forwarder IP Address text box blank. BOSH-reported component metrics are available in Loggregator Firehose by default. If you continue to use the BOSH HM Forwarder to consume these component metrics, you might receive duplicate data.
- 3. Select the Enable VM Resurrector Plugin check box to enable the BOSH Resurrector functionality and increase your runtime availability.
- 4. If you want to collect detailed metrics from BOSH-managed VMs, select the Enable System Metrics check box. For a list of metrics that the System Metrics Agent collects from BOSH-managed VMs, see [System Metrics Agent](#) on GitHub. When you select this check box, ensure that the System Metrics Scraper is able to connect on port 53035 to all BOSH-managed VMs.
- 5. Select Enable Post Deploy Scripts to run a post-deployment script after deployment. This script allows the job to execute additional commands against a deployment. If you intend to install VMware Tanzu Container Orchestration Engine (formerly TKGI), you must enable post-deploy scripts.
- 6. To configure your VMs to use short-lived NATS credentials during the bootstrap process, select the Enable Short Lived NATS Bootstrap Credentials check box. This increases the security of your Tanzu Operations Manager deployment by rotating the TLS certificate that the BOSH Agents

in your VMs use to communicate with the BOSH Director through BOSH NATS.
For more information, see [Using Short-Lived NATS Bootstrap credentials](#).

Configuring your VMs to use short-lived NATS bootstrap credentials restarts the BOSH Agent in all VMs as they are being created, so VMs are created somewhat more slowly.

This configuration affects all new deployments of Tanzu Operations Manager. If you want to configure VMs in existing Tanzu Operations Manager deployments to use short-lived bootstrap credentials, you must recreate them.

7. Select **Recreate VMs** deployed by the BOSH Director to force BOSH to recreate BOSH-deployed VMs on the next deployment. This process does not recreate the BOSH Director VM or destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
8. Select **Recreate BOSH Director VM** to force the BOSH Director VM to be recreated on the next deploy. This process does not destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
9. Select **Recreate All Persistent Disks** to force BOSH to migrate and recreate persistent disks for the BOSH Director and all tiles. This process does not destroy any persistent disk data. This check box is cleared automatically after a successful redeployment.
10. Select **Enable bosh deploy retries** to instruct Tanzu Operations Manager to retry failed BOSH operations up to five times.
11. Select **Skip Director Drain Lifecycle** to prevent drain scripts from running when the BOSH Director is recreated.
12. Select **Store BOSH Job Credentials on tmpfs (beta)** to store credentials for BOSH jobs on temporary file storage (tmpfs) memory, rather than on disk. You must re-create all BOSH-deployed VMs for this setting to take effect.
13. Select **Keep Unreachable Director VMs** to preserve BOSH Director VMs after a failed deployment for troubleshooting purposes.
14. (Optional) Select **Enable automatic rotation of the BOSH DNS CA certificate (experimental)** if you want Tanzu Operations Manager to automatically rotate the BOSH DNS CA certificate and associated leaf certificates when you click **Apply Changes**. For more information about this feature, see [Automatic rotation of BOSH DNS certificates](#).
15. (Optional) Modify the **Director Workers** value, which sets the number of workers available to run BOSH Director tasks. The default value is **5**.
16. (Optional) **Max Threads** sets the maximum number of threads that the BOSH Director can run simultaneously. VMware recommends that you leave the text box blank to use the default value, unless doing so results in rate limiting or errors on your IaaS.
17. (Optional) To add a custom URL for your BOSH Director, enter a valid hostname in the **Director Hostname** text box. You can also use this text box to configure a load balancer in front of your BOSH Director. For more information, see [How to set up a Load Balancer in front of Tanzu Operations Manager](#).

Director Workers

5

The number of director workers to execute director tasks. Default is 5

Max Threads

The Max Threads Setting controls how many operations the BOSH Director can perform simultaneously

Director Hostname

opsdirector.example.com

The externally accessible hostname for the BOSH Director

18. (Optional) To set a custom banner that users see when you log in to the BOSH Director using SSH, enter text in the Custom SSH Banner text box.
19. (Optional) Enter your comma-separated custom Identification Tags. For example, `iaas:foundation1,hello:world`. You can use the tags to identify your foundation when viewing VMs or disks from your IaaS.
20. For Certificate Duration Overrides, you can choose whether certificates generated within Tanzu Operations Manager and CredHub use the default duration specified for the certificate, or a custom duration value that you specify.
 - Off: Use this option to use default duration for certificates created by all products.
 - On: Use this option to set a value to override the duration for certificates created by all products. If a product creates a certificate with a longer duration than the value you set, the longer duration is used.
 1. CA Certificate Duration (days): Enter the number of days for which CA certificates are valid.
 2. Leaf Certificate Duration (days): Enter the number of days for which leaf certificates are valid. This value must be less than or equal to the CA certificate duration.

After you set a certificate duration override, you must take additional steps to apply the setting to all certificates. For more information, see [Overriding Duration for Certificates](#).

Certificate Duration Overrides

Set overrides for CA and leaf certificate durations.
For detailed information about which certificates are affected by this setting and which are not, see [Override Certificate Duration in the TAS for VMs or TKGI documentation](#).

☐ Off
☒ On

Enter values to set the duration for CA and leaf certificates in the fields below.
These values will be applied when new certificates are generated via new product deployment or certificate rotation.

CA Certificate Duration (days)*

1460

Enter a value between 365 and 3650 days

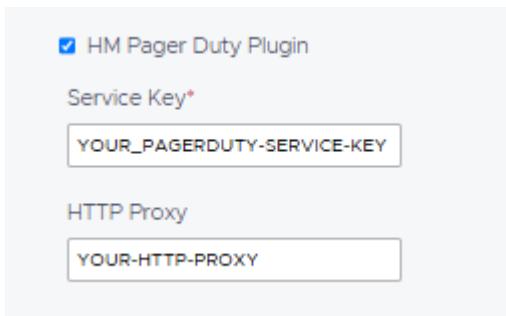
Leaf Certificate Duration (days)*

730

Leaf Certificate Duration must be less than or equal to the CA Certificate Duration.

21. Select HM Pager Duty Plugin to enable Health Monitor integration with PagerDuty.
 - Service Key: Enter your API service key from PagerDuty.

- HTTP Proxy: Enter an HTTP proxy for use with PagerDuty.

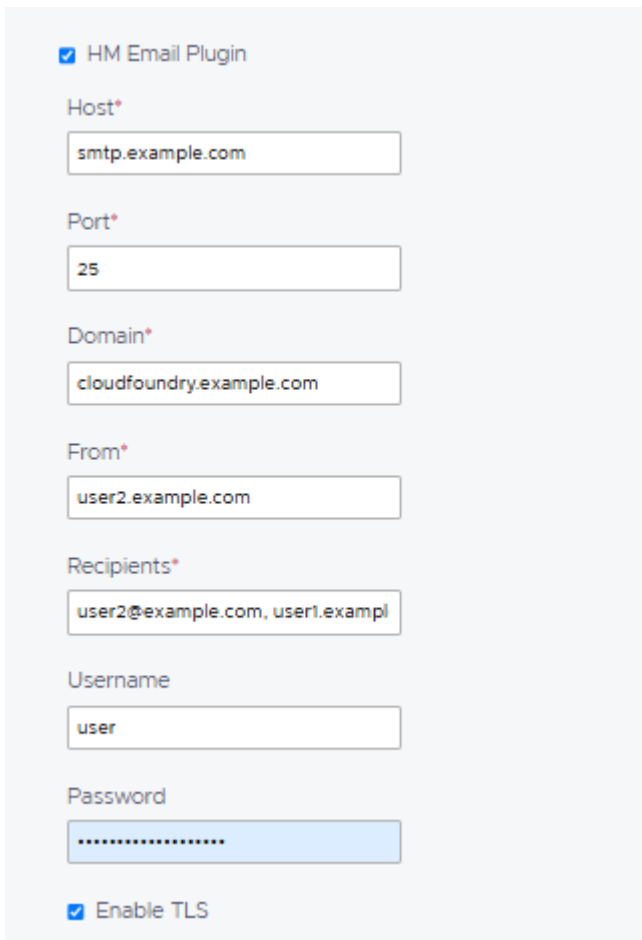


☒ HM Pager Duty Plugin
 Service Key*

 HTTP Proxy

22. Select HM Email Plugin to enable Health Monitor integration with email.

- Host: Enter your email host name.
- Port: Enter your email port number.
- Domain: Enter your domain.
- From: Enter the sender's address.
- Recipients: Enter a comma-separated list of addresses of intended recipients.
- Username: Enter the user name for your email server.
- Password: Enter the password for your email server.
- Enable TLS: Select this check box to enable Transport Layer Security to the email host.



☒ HM Email Plugin
 Host*

 Port*

 Domain*

 From*

 Recipients*

 Username

 Password

☒ Enable TLS

23. For CredHub Encryption Provider, you can choose whether BOSH CredHub stores its encryption key internally on the BOSH Director and CredHub VM, or in an external hardware security module (HSM). The HSM option is more secure.

Before configuring an HSM encryption provider in the Director Config pane, you must follow the procedures and collect information described in [Preparing CredHub HSMs for configuration](#). After you deploy Tanzu Operations Manager with an HSM encryption provider, you cannot change BOSH CredHub to store encryption keys internally.

- Internal: Use this option for internal CredHub key storage. This option is selected by default and requires no additional configuration.
- Luna HSM: Use this option to use a SafeNet Luna HSM as your permanent CredHub encryption provider, and fill in the following fields:

If you use multiple HSM hosts, you must use the Tanzu Operations Manager API to set or update the HSM configuration. For more information, see [Updating a staged director's properties](#) in the Tanzu Operations Manager API documentation.

1. Encryption Key Name: Any name to identify the key that the HSM uses to encrypt and decrypt the CredHub data. Changing this key name after you deploy Tanzu Operations Manager can cause service downtime.
2. Provider Partition: The partition that stores your encryption key. Changing this partition after you deploy Tanzu Operations Manager can cause service downtime. For this value and the ones that follow, use values gathered in [Preparing CredHub HSMs for configuration](#).
3. Provider Partition Password
4. Partition Serial Number
5. Provider Client Certificate: The certificate that validates the identity of the HSM when CredHub connects as a client.
6. Provider Client Certificate Private Key
7. HSM Host Address
8. HSM Port Address: If you do not know your port address, enter **1792**.
9. HSM Certificate: The certificate that the HSM presents to CredHub to establish a two-way mTLS connection.

CredHub Encryption Provider

- ☐ Internal
- ☒ Luna HSM

Encryption Key Name*

Provider Partition*

Provider Partition Password*

Partition Serial Number*

Provider Client Certificate*

Provider Client Certificate Private Key*

HSM Host Address*

HSM Port Address*



24. Click Blobstore Location to configure the blobstore as either an internal server or an external endpoint.

After you deploy the BOSH Director, only Tanzu Operations Manager users in Advanced Mode can change the blobstore location. For more information, see [Platform operator roles and permissions in Tanzu Operations Manager](#).

- Internal: Use this option for an internal blobstore. No additional configuration is required.
- S3 Compatible Blobstore: Use this option for an external S3-compatible endpoint. To create an S3 bucket, follow the procedures in [Sign up for Amazon S3](#) and [Creating a Bucket](#) in the AWS documentation. When you have created an S3 bucket, configure the following fields:

1. S3 Endpoint:

1. If you are using a public S3 endpoint:

1. Locate the endpoint for your region. To find the endpoint, see the Amazon Simple Storage Service (S3) table in [Regions and Endpoints](#) in the AWS documentation.
2. Construct a URL using your region's endpoint. For example, if you are using the [us-west-2](#) region, the URL you create would be <https://s3.us-west-2.amazonaws.com>. Enter this URL in the S3 Endpoint field.

2. If you are using a non-public S3-compatible endpoint:

1. Enter the URL for the non-public endpoint.
2. Add the custom public CA certificate you used to sign the S3 endpoint to the Trusted Certificates field on the Director Config page.

2. Bucket Name: Enter the name of the S3 bucket.

3. Access Key and Secret Key: Enter the keys you generated when creating your S3 bucket.

4. Enable signed URLs: Select this check box to configure BOSH VMs to generate short lived, pre-signed URLs for communication between the BOSH Agent and the blobstore. If you enable this feature, BOSH Agents do not use credentials to communicate with the blobstore, and VM disks do not store blobstore credentials. This feature is available only for BOSH VMs that use Xenial stemcell line 621 or later.

5. Select V2 Signature or V4 Signature. If you select V4 Signature, enter your Region. AWS recommends using Signature Version 4 (V4). For more information

about AWS S3 Signatures, see [Authenticating Requests](#) in the AWS documentation.

6. S3 Backup Strategy: To configure whether and how to back up the BOSH Director's S3 blobstore, select one of the following options:

- No backups
- Use a versioned bucket: This option uses blobstore bucket versioning to save backups. It requires an S3 blobstore that supports versioning.
- Copy into an additional bucket: This option saves blobstore backups to a separate bucket, which is useful with blobstores that do not support versioning. This option requires you to configure the Backup Bucket Region and Backup Bucket Name fields.

To backup and restore with BBR, if you are using S3 Compatible Blobstore and you have set up an [iam_instance_policy](#), additional steps are needed:

1. Attach the [iam_instance_policy](#) to the [backup_restore](#) job using the Tanzu Operations Manager API. See [Custom VM Extensions](#).
2. In addition to creating backup buckets, update the Elastic Application Runtime tile (Settings > File Storage) to use S3-compatible storage. See [External S3-Compatible filestore](#).
3. Ensure that your IAM Instance Policy is set up to access the backup bucket. For more information about the IAM Instance Policy, see the [Tanzu Operations Manager for AWS policy document](#).

- GCS Blobstore: Use this option for an external Google Cloud Storage (GCS) endpoint.

To create a GCS bucket, follow the procedures in [Creating Storage Buckets](#) in the GCS documentation. To create a GCS bucket, you must have a GCS account. When you have created a GCS bucket, configure the following fields:

1. Bucket Name: Enter the name of your GCS bucket.
2. Storage Class: Use this storage class for your GCS bucket. For more information, see [Storage Classes](#) in the GCP documentation.
3. Service Account Key: Follow the steps in [Set up IAM Service Accounts](#) in *Preparing to Deploy Tanzu Operations Manager on GCP* to download a JSON file with a private key. Enter the contents of the JSON file into the text box.
4. GCS Backup Strategy: To configure whether and how to back up the BOSH Director's GCS blobstore, click one of the following options:
 - No backups
 - Copy into an additional bucket: This option saves blobstore backups to a bucket separate from the blobstore itself, and requires you to configure the Backup Bucket Name text box.

Blobstore Location

☐ Internal

☐ S3 Compatible Blobstore

☒ GCS Blobstore

Bucket Name*

Storage Class*

Regional

Service Account Key*

Paste the JSON file containing the private key for your GCP service account.

GCS Backup Strategy

☒ No backups

☐ Copy into an additional bucket

Backup Bucket Name*

25. Click Save.

Step 4: Create Availability Zones pane

Tanzu Operations Manager Availability Zones (AZs) correspond to your vCenter clusters, resource pools, or host groups.

Multiple AZs allow you to provide high availability and load balancing to your applications. When you run more than one instance of an application, Tanzu Operations Manager balances those instances across all of the AZs assigned to the application.

VMware recommends that you use at least three AZs for a highly available installation of your chosen runtime.

For more information about AZs and high availability in vSphere, see [Compute and HA considerations](#) in *vSphere Reference architecture*.

1. Select Create Availability Zones.
2. Use the following steps to create the first AZ.

Click Add.

- Enter a unique Name for the AZ.
 - Select a vCenter config name from the IaaS Configuration drop-down menu to associate your AZ with a vCenter. If you have only one vCenter config, or if you are creating your first AZ, IaaS Configuration defaults to the first vCenter and cannot be configured.
 - Enter the name of the vCenter Cluster where you deployed Tanzu Operations Manager. You created this cluster in the [Prepare vSphere](#) step of *Deploying Tanzu Operations Manager to vSphere* (for example, [Cluster1](#)).
 - Enter the name of the Resource Pool that you created in the [Prepare vSphere](#) step of *Deploying Tanzu Operations Manager to vSphere* to organize management components like BOSH Director (for example, [RP-MGMT](#)). The jobs running in this AZ share the CPU and memory resources defined by the pool. You only need to specify this resource group in only one AZ.
 - (Optional) Enter the name of a Host Group for the AZ to use. The host group must be within the cluster you designated in the Clusters field. You created host groups in the [Prepare vSphere](#) step of *Deploying Tanzu Operations Manager to vSphere*. For more information about using host groups, see [Host groups in vSphere](#).
 - (Optional) In the VM-Host Affinity Rule drop-down menu, select SHOULD. Selecting SHOULD allows vSphere to restart VMs in another host group in the event of an AZ failure. For more information, see [DRS host-affinity rules and stretched clusters in Host Groups in vSphere](#).
The value for VM-Host Affinity Rule applies only if you are using Host Groups. If you do not specify a Host Group for the cluster, then any changes you make to VM-Host Affinity Rule are ignored.
 - (Optional) Click Add Cluster to add more clusters to the AZ as required by your installation size and high availability requirements.
3. (Optional) Use the following steps to create additional AZs.
- Click Add.
 - Specify Cluster and optionally, a Resource Pool or a Host Group with a VM-Host Affinity Rule. Typically, you specify either a resource pool or a host group for an AZ, but not both. For more information about the difference between host groups and resource pools, see [Host Groups in vSphere](#).
 - Click Add Cluster to assign more clusters to the AZ as necessary for high availability. Click the trash icon to delete a cluster. The first cluster cannot be deleted.
4. Click Save.

Step 5: Create Networks pane

1. Select Create Networks.
2. Select Enable ICMP checks to enable ICMP on your networks. Tanzu Operations Manager uses ICMP checks to confirm that components within your network are reachable.
3. Depending on the runtime that you are deploying, do one of the following:
 - For Elastic Application Runtime, click Add Network and create the following networks:

- **infrastructure**: This network is for Tanzu Operations Manager and the BOSH Director.
- **pas**: This network is for all the Elastic Application Runtime VMs including Gorouter, Diego cells, and Cloud Controller.
- **services**: This network is for any service tiles to be deployed alongside Elastic Application Runtime.

Use the values from the following table as a guide when you create each network, replacing the IP addresses with ranges that are available in your vSphere environment:

Infrastructure Network	Field	Configuration Example
	Name	infrastructure
	vSphere Network Name *	pcf-virt-net/infrastructure
	CIDR	192.168.101.0/24
	Reserved IP Ranges †	192.168.101.1-192.168.101.9
	DNS	192.168.101.2
	Gateway	192.168.101.1
Deployment Network	Field	Configuration Example
	Name	pas
	vSphere Network Name *	pcf-virt-net/pcf-pas-subnet
	CIDR	192.168.16.0/24
	Reserved IP Ranges †	192.168.16.1-192.168.16.9
	DNS	192.168.16.2
	Gateway	192.168.16.1
Services Network	Field	Configuration Example
	Name	services
	vSphere Network Name *	pcf-virt-net/pcf-services-subnet
	CIDR	192.168.20.0/24
	Reserved IP Ranges †	192.168.20.1-192.168.20.9
	DNS	192.168.20.2
	Gateway	192.168.20.1

* In vSphere Network Name, enter the full path of the network as it is displayed in vCenter; for example, enter **YOUR-DIRECTORY-NAME/YOUR-NETWORK-NAME**. If your vSphere Network Name contains a forward slash character, replace the forward slash with the URL-encoded forward slash character **%2f**.

† For Reserved IP Ranges, enter any IP addresses from the CIDR that you do not want BOSH to use in your installation. Tanzu Operations Manager does not deploy VMs to any address in this range.

- For Container Orchestration Engine, click Add Network and create the following networks:
 - infrastructure**: This network is for Tanzu Operations Manager, the BOSH Director, the Container Orchestration Engine broker, and the Container Orchestration Engine API.
 - pks**: If you have a large deployment with multiple tiles, you can choose to deploy the Container Orchestration Engine broker and Container Orchestration Engine API to a separate network named **pks**. For more information, see the following table.
 - services**: Network for creating the primary and worker VMs for Kubernetes clusters. The CIDR cannot conflict with the pod overlay network **10.200.0.0/16** or the reserved Kubernetes services CIDR of **10.100.200.0/24**.

Multiple networks allow you to place vCenter on a private network and the rest of your deployment on a public network. Isolating vCenter in this manner denies access to it from outside sources and reduces possible security vulnerabilities.

Use the values from the following table as a guide when you create each network, replacing the IP addresses with ranges that are available in your vSphere environment:

Infrastructure Network	Field	Configuration Example
	Name	infrastructure
	vSphere Network Name *	pcf-virt-net/pks-infrastructure-subnet
	CIDR	192.168.101.0/26
	Reserved IP Ranges †	192.168.101.1-192.168.101.9
	DNS	192.168.101.2
	Gateway	192.168.101.1
Main Network (Optional)	Field	Configuration Example
	Name	pks
	vSphere Network Name *	pcf-virt-net/pks-subnet
	CIDR	192.168.16.0/26
	Reserved IP Ranges †	192.168.16.1-192.168.16.9
	DNS	192.168.16.2
	Gateway	192.168.16.1
Service Network	Field	Configuration Example
	Name	services
	vSphere Network Name *	pcf-virt-net/pks-service-subnet
	CIDR	192.168.20.0/22
	Reserved IP Ranges †	192.168.20.1-192.168.20.9
	DNS	192.168.20.2
	Gateway	192.168.20.1

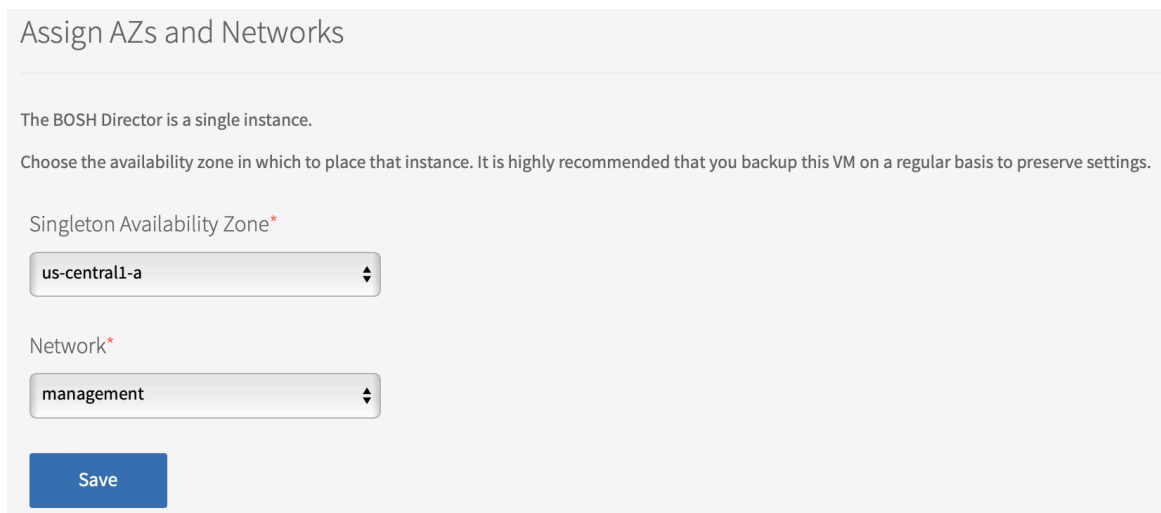
* In vSphere Network Name, enter the full path of the network as it is displayed in vCenter; for example, enter **YOUR-DIRECTORY-NAME/YOUR-NETWORK-NAME**. If your vSphere Network Name contains a forward slash character, replace the forward slash with the URL-encoded forward slash character **%2f**.

† For Reserved IP Ranges, enter any IP addresses from the CIDR that you do not want BOSH to use in your installation. Tanzu Operations Manager does not deploy VMs to any address in this range.

4. For each network you create, select the Availability Zones to use with the network. Assign as many AZs to your network as needed by your deployment. For more information, see [Compute and HA considerations](#) and [Scaling and Capacity Management](#) sections of *vSphere Reference Architecture*.
5. Click Save.

Step 6: Assign AZs and Networks

1. Select Assign AZs and Networks.



Assign AZs and Networks

The BOSH Director is a single instance.

Choose the availability zone in which to place that instance. It is highly recommended that you backup this VM on a regular basis to preserve settings.

Singleton Availability Zone*

us-central1-a

Network*

management

Save

2. Use the Singleton Availability Zone drop-down menu to select a singleton AZ. The BOSH Director is installed as a single instance in this Availability Zone.
3. Use the Network drop-down menu to select a network for your BOSH Director.
4. Click Save.

Step 7: Security pane

1. Click Security.

Security

Trusted Certificates

These certificates enable BOSH-deployed components to trust a custom root certificate.

☒ Include Tanzu Ops Manager Root CA in Trusted Certs
Checking this box will place the Tanzu Ops Manager Root CA into the trusted certificate field. The Bosh Director will then place this CA into the trust store of every VM that it deploys.

☐ Clear the Default Trusted Certificates Store
Deletes all the certificates in the system certificate trust store every time BOSH deploys a VM. Certificates added in the 'Trusted Certificates' field will still be trusted and you should add the certificates required to access your IaaS Provider services.

Generate VM passwords or use single password for all VMs

☒ Generate passwords

☐ Use default BOSH password

SAVE

2. In Trusted Certificates, paste in your custom certificate authority (CA) certificates to insert into your organization's certificate trust chain. This feature enables all BOSH-deployed components in your deployment to trust custom root certificates. If you want to use Docker registries to run Elastic Application Runtime app instances in Docker containers, enter the certificate for your private Docker registry in this text box. For more information about running app instances in Elastic Application Runtime using Docker registries, see [Using Docker Registries](#).

To enter multiple certificates, paste in your certificates one after the other. For example, format your certificates as shown in the following example:

```
-----BEGIN CERTIFICATE-----
ABCDEF12345678ABCDEF12345678ABCDEF12345678AB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
BCDEF12345678ABCDEF12345678ABCDEF12345678ABB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
CDEF12345678ABCDEF12345678ABCDEF12345678ABBB
EFGH12345678ABCDEF12345678ABCDEF12345678ABCDEF
GH12345678ABCDEF12345678ABCDEF12345678...
-----END CERTIFICATE-----
```

3. To include both the Tanzu Operations Manager root CA and the certificates pasted into Trusted Certificates in the `trusted_certs` field in the BOSH director manifest, select the Include Tanzu Ops Manager Root CA in Trusted Certs check box. BOSH Director includes this CA in the trust store of every VM that it deploys.
- You must select this check box if you are using Tanzu Operations Manager to generate certificates for a load balancer or a router.
4. To clear the default trusted certificates from all BOSH-deployed Linux VMs, select Clear the Default Trusted Certificates Store.

If you select this check box, you must provide your own trusted certificates. All TLS communication fails if trusted certificates are not available.

5. Select Generate passwords or Use default BOSH password. VMware recommends using the Generate passwords option for greater security.
6. Click Save. To view your saved BOSH Director password, click the Credentials tab.

Step 8: BOSH DNS Config pane

In the BOSH DNS Config pane, you can configure DNS for BOSH Director by adding excluded recursors, a recursor timeout, and handlers.

The text boxes in the BOSH DNS Config pane are:

- Excluded Recursors: Exclude recursor addresses, which are URL redirects, so that they are not contacted by the BOSH DNS server. For more information about how the BOSH DNS release selects recursors, see [Recursors](#) in *Native DNS Support* in the BOSH documentation.
- Recursor Timeout: Specify a timeout for the BOSH DNS server when contacting any connected recursor addresses. The time limit includes dialing, writing, and reading from the recursor. If any of these actions exceeds the time limit, the action fails.
- Handlers: Specify recursor addresses that apply to specific domains. For example, you can use handlers to forward all requests to a domain to a private DNS for resolution. For more information about using handlers, see [Additional handlers](#) in *Native DNS Support* in the BOSH documentation.

To add excluded recursors, a recursor timeout, or handlers to the BOSH DNS release:

1. Click BOSH DNS Config.

The screenshot shows the 'BOSH DNS Config' pane. It contains three input fields: 'Excluded Recursors' (a single-line text box), 'Recursor Timeout' (a single-line text box), and 'Handlers' (a multi-line text box containing an empty JSON array '[]'). To the right of the Handlers text box is the instruction 'Enter a JSON-formatted list of optional custom domain handlers'. At the bottom left of the pane is a blue 'Save' button.

2. (Optional) In Excluded Recursors, enter a list of prohibited recursor addresses.

3. (Optional) In Recursor Timeout, enter a time limit for contacting the connected recursors. This time limit must include one of the Go parse duration time units. For example, entering **5s** sets the timeout limit to five seconds. For more information about supported time units, see [func ParseDuration](#) in the Go Programming Language documentation.
4. (Optional) In Handlers, enter an array of custom domain handlers in JSON format. For example:

```
[
  {
    "cache": {
      "enabled": true
    },
    "domain": "example.com",
    "source": {
      "type": "http",
      "url": "http://example.endpoint.local"
    }
  }
]
```

5. Click Save.

Step 9: Syslog pane

BOSH Director logs contain sensitive information that can be considered privileged. For example, these logs might contain cloud provider credentials. If you choose to forward logs to an external syslog endpoint, use TLS encryption to prevent information from being intercepted by a third party.

1. Click Syslog.
2. (Optional) To send BOSH Director system logs to a remote server, select Yes.
3. In Address, enter the IP address or DNS name for the remote server.
4. In Port, enter the port number that the remote server listens on.
5. Using the Transport Protocol drop-down menu, select TCP or UDP to specify the transport protocol to use to send the logs to the remote server.
6. (Optional) Select the Enable TLS check box to send encrypted logs to remote server with TLS. After you select the check box:
 1. In Permitted Peer, enter either the name or SHA1 fingerprint of the remote peer.
 2. In SSL Certificate, enter the SSL certificate for the remote server.

VMware recommends that you enable TLS encryption when you are forwarding logs. Logs can contain sensitive information, such as cloud provider credentials.

7. (Optional) In Queue Size, enter an integer that specifies the number of log entries held in the buffer. The default value is 100,000.
8. (Optional) Select the Forward Debug Logs check box to forward the logs to an external source. This option is deselected by default. If you select it, you might generate a large amount of log data.
9. (Optional) In the Custom rsyslog Configuration text box, enter the configuration details for **rsyslog**. This text box requires the rainerscript syntax.
10. Click Save.

Step 10: Resource Config pane

1. Click Resource Config.

Resource Config

SAVE

JOB	INSTANCES	VM TYPE	PERSISTENT DISK TYPE
▼ BOSH Director NSX-T CONFIGURATION NS Groups group1, group2 VIF Type PARENT or null Logical Load Balancer ⓘ { "server_pools": [] }	Automatic: 1	Automatic: large.disk (cpu)	Automatic: 50 GB
▶ Main Compilation Job	Automatic: 4	Automatic: xlarge.disk (cpu)	None

2. Under Instances, VM Type, and Persistent Disk Type, choose Automatic from the drop-down menus to allocate the recommended resources for the job.
3. If the Persistent Disk Type field displays the value None, the job does not require persistent disk space.

For the BOSH Director job, select a VM type with at least 8 GB memory. If you set a field to **Automatic** and the recommended resource allocation changes in a future version, Tanzu Operations Manager automatically uses the updated recommended allocation. If you install Elastic Application Runtime Windows add-on, provision your **Main Compilation Job** with at least 100 GB of disk space.

4. (Optional) If you configured NSX-T networking in the vCenter Config pane, and you want to configure security and load balancing for the BOSH Director job:
 1. Click the menu expansion icon next to the BOSH Director job name. The NSX-T Configuration configuration fields appear.
 2. Enter your NSX-T configuration values in these fields.
5. (Optional) If you configured AVI Load Balancing in the vCenter Config pane, and you want to configure load balancing for the BOSH Director job using the AVI Load Balancer:
 1. Click the menu expansion icon next to the BOSH Director job name. The AVI Load Balancer Configuration configuration fields appear.
 2. Enter your AVI Load Balancer configuration values in these fields.

6. Click Save.

Step 11: Add custom VM extensions (optional)

Use the Tanzu Operations Manager API to add custom properties to your VMs; associated security groups and load balancers, for example. For more information, see [Managing custom VM extensions](#).

Step 12: Complete the BOSH Director installation

1. Click the Tanzu Operations Manager Installation Dashboard link to return to the Installation Dashboard.
2. Click Review Pending Changes. Select the product that you intend to deploy and review the changes. For more information, see [Reviewing pending product changes](#).

Configuring proxy settings for the BOSH CPI

If you use a proxy to mediate traffic between your BOSH installation and your IaaS endpoint, you must enter your proxy information when Tanzu Operations Manager starts for the first time.

This section describes how to configure proxy settings for the BOSH Cloud Provider Interface (CPI), the API that the BOSH Director uses to interact with your IaaS endpoint.

For more information about BOSH components, see the [BOSH documentation](#).

The following procedure assumes that you have already installed Tanzu Operations Manager.

Configure the proxy

1. When Tanzu Operations Manager starts for the first time, you must select Use an Identity Provider or Internal Authentication.
2. Both the Use an Identity Provider and the Internal Authentication login pages include fields for HTTP proxy, HTTPS proxy, and No proxy.

For the Tanzu Platform for Cloud Foundry runtime, editing proxy settings in the BOSH Director propagates those settings to the Garden Diego cells in Elastic Application Runtime. Sharing these proxy settings between Tanzu Operations Manager and Elastic Application Runtime enables you to reach a Docker Hub through a proxy within an environment that lacks internet connectivity.

3. Complete the fields with the following information:
 - HTTP proxy: Enter the URL for the proxy that handles HTTPS requests between the BOSH CPI and the IaaS API.
 - HTTPS proxy: Enter the URL for the proxy that handles HTTP requests between the BOSH CPI and the IaaS API. If you are using the same proxy to handle both HTTPS and HTTP traffic, enter the same URL as HTTP proxy.
 - No proxy: Enter the following IP addresses:
 - 127.0.0.1
 - 169.254.169.254 - This is the metadata server for many IaaS.
 - The IP address of your BOSH Director. This is the first IP address not in your reserved range. For example, if you enter 10.0.16.0-10.0.16.10 in the Reserved IP Ranges field of the Create Networks page when configuring the BOSH Director, the BOSH Director IP is 10.0.16.11.

When the BOSH Director talks to other components on its VM, such as the database and the User Account and Authentication (UAA) module, it communicates with them as if they were external. You must enter the BOSH Director IP address in the no-proxy list to prevent traffic from the BOSH Director to other components on the same VM from passing through the proxy.

The values in the No Proxy field must be comma-delimited, with no spaces or quotation marks between values.

Example: `127.0.0.1,169.254.169.254,10.0.16.11`.

Next steps

After you complete this procedure, follow the instructions for the runtime that you intend to install:

- To install Elastic Application Runtime, see the [Elastic Application Runtime documentation](#).
- To install Container Orchestration Engine, see the [Container Orchestration Engine documentation](#).

For more information about Tanzu Operations Manager runtimes, see [Installing Runtimes](#).

Using the Cisco Nexus 1000v Switch with Tanzu Operations Manager

You can use the Cisco Nexus 1000v Switch with Tanzu Operations Manager.

Begin by configuring Tanzu Operations Manager through Step 4 in [Configuring BOSH Director on vSphere](#). Then configure your network using the following steps.

1. From your Tanzu Operations Manager Installation Dashboard, click the BOSH Director tile.
2. Click Create Networks.
3. Click the network name to configure the network settings. The name is `default` if you have not changed it.

Create Networks

Warning: Tanzu Ops Manager recommends keeping the IP settings throughout the life of your installation. Tanzu Ops Manager may prevent you from changing them in the future. Contact VMware support for help completing such a change.

Verification Settings

☒ Enable ICMP checks

Networks

One or many IP ranges upon which your products will be deployed

▼ default

Name*

default

Subnets*

▼ Add Subnet

vSphere Network Name*

CIDR*

Reserved IP Ranges*

DNS*

Gateway*

Availability Zones *

☐ first-az

Save

- Find the folder name and port group name for the switch, as you configured them in vCenter. For example, you might want to use the switch configured on the **beer-apple** port group, in the **drinks dc** folder.

Folder names can be strings of multiple words separated by spaces, or strings of words separated by other punctuation, including dashes.

- In the vSphere Network Name text box, instead of entering your network name, enter the folder name and port group name for the switch, as you configured them in vCenter. For the example vSphere environment described earlier, you would enter **drinks dc/beer-apple** to use the switch configured on the **beer-apple** port group.

Create Networks

Warning: Tanzu Ops Manager recommends keeping the IP settings throughout the life of your installation. Tanzu Ops Manager may prevent you from changing them in the future. Contact VMware support for help completing such a change.

Verification Settings

☒ Enable ICMP checks

Add Network

Networks

One or many IP ranges upon which your products will be deployed

▼ NETWORK_NAME

Name*

NETWORK_NAME

Subnets*

Add Subnet

vSphere Network Name*

drinks dc/beer-apple

The name of the network as it appears in vCenter. It can be of the format "<network-name>", or "<folder>/<network-name|distributed-port-name>" or "<folder>/<distributed-switch-name>/<distributed-port-name>"

CIDR*

Reserved IP Ranges*

DNS*

Gateway*

Availability Zones *

☐ first-az

Save

- Click Save.
- Return to [Configuring BOSH Director on vSphere](#) to complete the Tanzu Operations Manager installation.

Using the BOSH Resurrector

This topic tells you about the BOSH Resurrector, which monitors VMs and recreates VMs that become unavailable. You can enable the BOSH Resurrector to ensure high availability (HA) for your Tanzu Operations Manager deployment.

For vSphere users, this topic explains how to enable vSphere HA to work along with the BOSH Resurrector.

- vSphere HA restarts unreachable VMs at the IaaS level, below the BOSH layer where the BOSH Resurrector operates.
- VMware recommends enabling the vSphere HA, the BOSH Resurrector, and a highly-available storage solution to protect the VMs in your deployment.

For more information about the BOSH Resurrector, see [Auto-healing Capabilities](#) in the BOSH documentation.

How the BOSH Resurrector increases Tanzu Operations Manager availability

The BOSH Resurrector increases Tanzu Operations Manager availability by:

- Reacting to hardware failures and network disruptions by recreating virtual machines on active, stable hosts
- Detecting operating system failures by continuously monitoring virtual machines and recreating them as required
- Continuously monitoring the BOSH Agent running on each virtual machine and recreating the VMs as required

The BOSH Resurrector continuously monitors the status of all virtual machines in a Tanzu Operations Manager deployment. The Resurrector also monitors the BOSH Agent on each VM. If either the VM or the BOSH Agent fails, the Resurrector recreates the virtual machine on another active host.

BOSH Resurrector limitations

The following limitations apply when using the BOSH Resurrector:

- The Resurrector does not monitor or protect the Tanzu Operations Manager VM or the BOSH Director VM.
- The Resurrector might not be able to resolve issues caused by the loss of an entire host.
- The Resurrector does not monitor or protect data storage.

To mitigate these limitations, VMware recommends using vSphere HA. For more information, see [vSphere HA](#) below.

Enable the BOSH Resurrector

To enable the BOSH Resurrector:

1. Log in to the Tanzu Operations Manager web interface.
2. On the Product Dashboard, click BOSH Director.
3. In the navigation menu, click Director Config.
4. Click Enable VM Resurrector Plugin and then click Save.

Enable vSphere HA

In the event of a host failure, vSphere HA restarts but does not recreate VMs on alternate ESXi hosts in the same cluster. A host failure could result from a sudden power off, a crash due to hardware fault, or the vSphere host becoming disconnected from the network. vSphere HA is recommended because BOSH Resurrector cannot recover from a host failure.

For more information, see [vSphere High Availability](#) in the BOSH documentation.

To enable vSphere High Availability:

1. Run the vSphere Management Console.

2. In the left panel, right-click the cluster that contains the Tanzu Operations Manager deployment and click Edit Settings.
3. Select the Turn on vSphere High Availability check box.
4. Click OK to enable vSphere High Availability on the cluster.

Understanding and using availability zones in vSphere

This topic describes what an availability zone (AZ) is in vSphere, and gives an example of how you might scale an app across AZs.

VMware defines an AZ as an operator-assigned, functionally independent segment of network infrastructure. In cases of partial infrastructure failure, a tile distributes and balances all instances of running apps across remaining AZs. Strategic use of AZs contributes to the fault tolerance and high availability of a tile deployment.

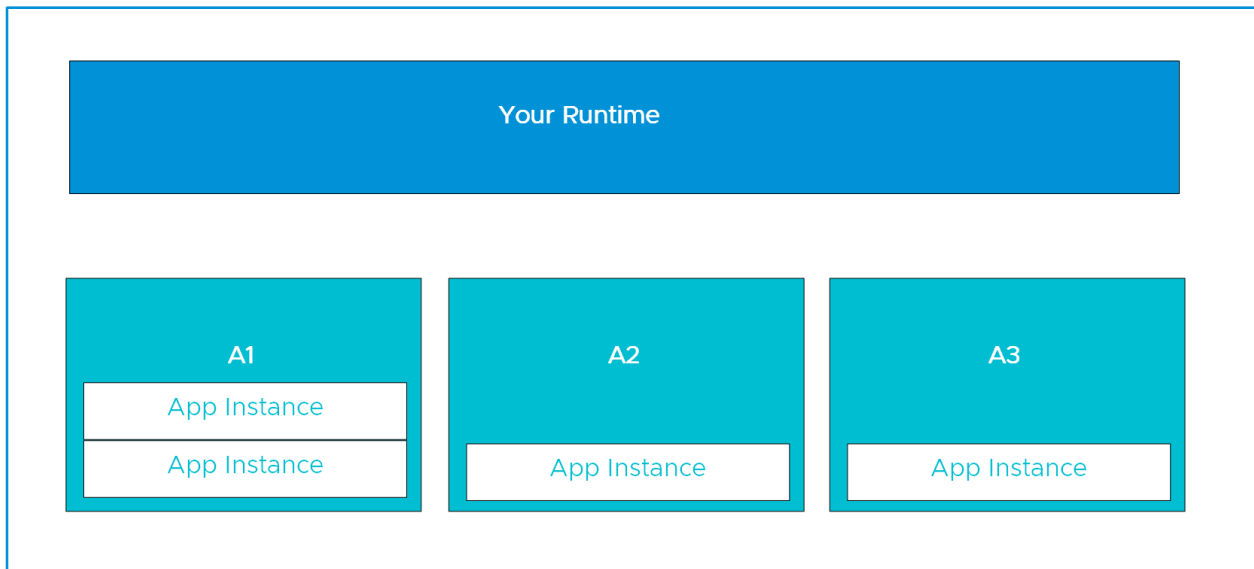
vSphere supports distributing deployments across multiple AZs. For more information, see [Step 4: Create Availability Zones](#) in *Configuring BOSH Director on vSphere*.

VMware recommends using three AZs to operate a highly available tile deployment.

Example scenario: Balancing across AZs during failure

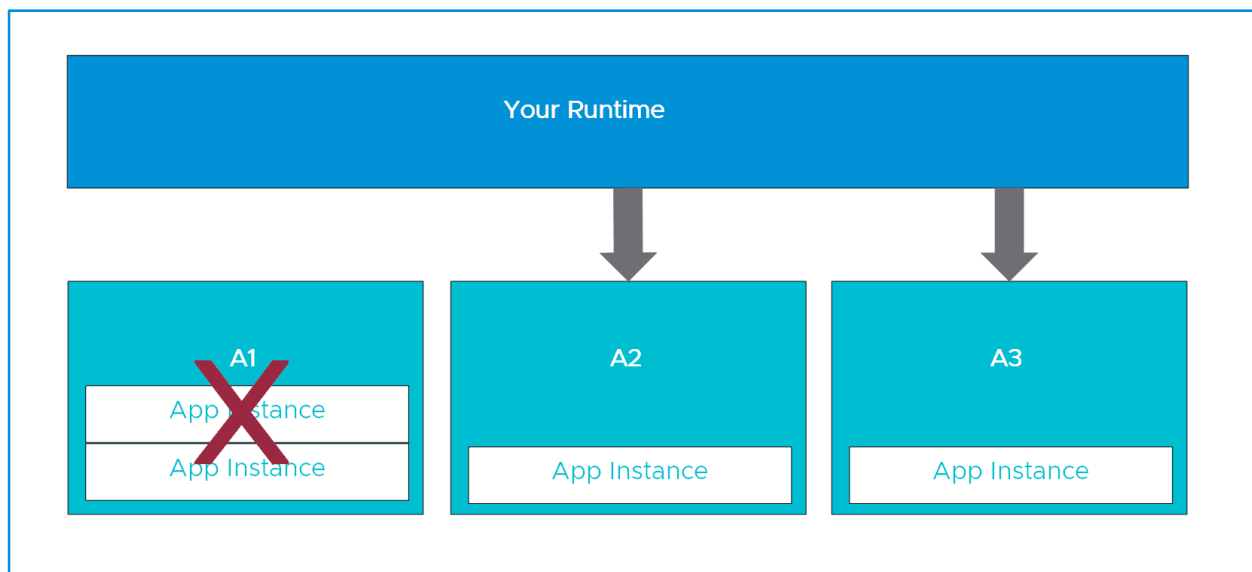
You can scale an app to four instances in a tile environment distributed across three AZs: A1, A2, and A3. The environment allocates the instances according to the Diego Auction. For more information about the Diego Auction, see [How Diego balances app processes in Cloud Foundry](#).

For this scenario, see the following diagram:



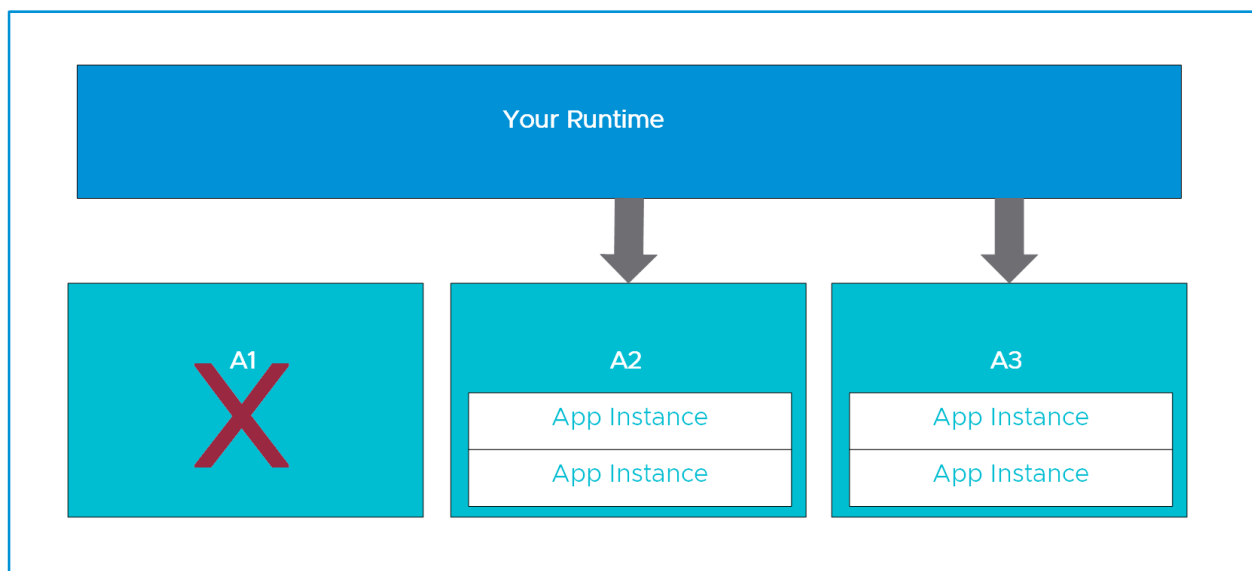
If A1 experiences a power outage or hardware failure, the two app instances running in A1 stop while the app instances in zones A2 and A3 continue to run.

See the following diagram:



If A1 remains unavailable, the tile balances new instances of the app across the remaining AZs.

For this scenario, see the following diagram:



Using vSphere host groups with your Tanzu Operations Manager deployment

You can use the vSphere host groups feature with Tanzu Operations Manager to build flexible scalability and high availability into your BOSH deployments. This topic describes the host groups feature and how you can configure the Distributed Resource Scheduler (DRS) Host-Affinity Rule for host groups.

About host groups in vSphere

vSphere host groups and resource pools are both methods for segmenting and organizing resources within a vSphere cluster. A vSphere cluster is a collection of ESXi servers that run VMs. A resource pool is a collection of vSphere resources in the cluster, while a host group is a subset of ESXi servers within the cluster.

For BOSH deployments on vSphere infrastructure, you associate vSphere clusters with availability zones (AZs). You use Tanzu Operations Manager to define these AZs for your BOSH Director deployments. In BOSH, AZs provide a mechanism for balancing apps and jobs across resources for resiliency and fault tolerance. Optionally, when defining an AZ, you can specify a resource pool or host group that is

associated with the cluster. The association of vSphere resource pools or host groups with AZs allows BOSH to distribute deployments across predefined vSphere resources.

For more information about configuring AZs in BOSH Director, see [Create Availability Zones](#).

When to use host groups

This section describes the types of deployments that benefit most from using host groups. For multi-use vSphere clusters, it's better to use host groups instead of resource pools.

In some cases, configuring host groups does not provide an advantage. In deployments where the underlying clusters are dedicated to a single use, such as running only app containers, it's simpler to not use host groups, and rather, to use AZs to balance associated jobs and apps.

Stricter resource boundaries in multi-use clusters

Host groups are the most useful for deployments where the underlying clusters are used by multiple BOSH-managed deployments and you want greater control of cluster resource usage than is provided by resource pools. This greater control of resources is possible because you can use host groups to restrict BOSH VM deployments to a specific set of compute resources.

A resource pool allows both deployments to have access to the total compute capacity of the cluster, but also puts up a protective resource boundary contention between the deployments. The deployments can share the resources, but in the event that resource consumption greatly increases, a resource management engine enforces resource usage limits. In this situation, the engine might deny pool resources to one of the deployments. Depending on the nature of the deployments, denying resources to one of the deployments might not be ideal.

Vertical scaling

Host groups also allow you to scale up or down the number of ESXi hosts used by your AZ. They also allow you to scale resources without forcing BOSH to redeploy its AZs.

For example, with host groups, you can begin by associating a single server from a cluster with an AZ. When one of the deployments requires more resources, instead of updating AZs or tuning resource pools, you can move additional servers from the existing cluster into the host group. This prevents disruption to the BOSH deployments.

For more information about using host groups in scaling platforms, see the [VMware Tanzu Architecture for Dell EMC VxRail documentation](#).

DRS Host-Affinity Rules and stretched clusters

When you configure host groups, configure the host group's Distributed Resource Scheduler (DRS) Host-Affinity Rule to **SHOULD**. This configuration allows VMs to start up in a new host group and AZ in the event that an AZ fails, or in case you want to scale up the number of VM instances for a job.

To illustrate the reason behind this configuration, consider a deployment where the Host-Affinity Rule is set to **MUST**. The **MUST** rule requires that each VM is deployed to its associated host group. If you have a three-instance job, then BOSH deploys one instance to each host group and AZ. If you add a fourth instance and the rule is set to **MUST**, vSphere does not allow that fourth VM to turn on to achieve DRS separation.

In deployments that use stretched clusters, configuring the DRS Host-Affinity Rule to **SHOULD** is especially important for ensuring high availability in the event that an AZ fails.

You configure this rule when setting the host group for an AZ in BOSH Director. For more information, see [Step 4: Create Availability Zones](#) in *Configuring BOSH Director on vSphere*.

For more information about DRS Host-Affinity Rules, see [VM-Host affinity rules](#) in the *VMware vSphere Product Documentation*.

Upgrading your Tanzu Operations Manager deployment

The following topics provide information about upgrading your Tanzu Operations Manager deployment:

- [Upgrading your Tanzu Operations Manager deployment](#)
- [Jump upgrading to Tanzu Operations Manager v3.2](#)
- [Upgrade preparation checklist for Tanzu Operations Manager v3.2](#)
- [Upgrading Elastic Application Runtime and other Tanzu Operations Manager products](#)

Upgrading your Tanzu Operations Manager deployment

You can upgrade Tanzu Operations Manager and Tanzu Platform for Cloud Foundry to v3.2, and then upgrade your product tiles.

Breaking Changes: Read the [Tanzu Operations Manager v3.2 Release Notes](#), [Elastic Application Runtime v10.3 Release Notes](#), [Isolation Segment v10.3 Release Notes](#), and [Elastic Application Runtime Windows add-on v10.3 Release Notes](#), including the *Breaking Changes* and *Known Issues* sections, before you start the upgrade process.

During an upgrade, the apps in your deployment continue to run. However, you cannot write to your deployment or make changes to apps during the upgrade. For details about how upgrading impacts individual Elastic Application Runtime components, see [What Happens During Elastic Application Runtime Upgrades](#).

VMware recommends upgrading directly to Tanzu Operations Manager v3.2 from Tanzu Operations Manager v2.5 through v2.9. To upgrade Tanzu Operations Manager to v3.2 from Tanzu Operations Manager v2.5 through v2.9, see [Jump Upgrade to Tanzu Operations Manager v3.2](#). To upgrade both Tanzu Operations Manager and Elastic Application Runtime, see [Task 1: Prepare to Upgrade](#).

Jump upgrade from Tanzu Operations Manager v2.5 through v2.9

Jump upgrading describes the process of skipping minor versions when upgrading Tanzu Operations Manager. Jump upgrading enables you to upgrade directly to the latest version of Tanzu Operations Manager, which is v3.2. For more information, see [Jump Upgrading to Tanzu Operations Manager v3.2](#).

To jump upgrade to Tanzu Operations Manager v3.2:

1. Review the prerequisites and breaking changes described in [Jump upgrading to Tanzu Operations Manager v3.2](#).
2. Complete the steps to prepare Tanzu Operations Manager, the BOSH Director, and BOSH-managed add-ons for upgrading in [Upgrade preparation checklist for Tanzu Operations Manager v3.2](#). Do not complete the steps to prepare Elastic Application Runtime for upgrading.
3. Follow the procedure in this topic to upgrade Tanzu Operations Manager, starting at [Task 1: Prepare to upgrade](#). Do not complete the steps to upgrade Elastic Application Runtime.

Upgrade Elastic Application Runtime

Skip this step if you are using the jump upgrade method described in the previous section.

If you are upgrading Elastic Application Runtime to the next minor release:

1. Locate the version of [Upgrade preparation checklist for Tanzu Operations Manager](#) that is specific to the minor release version to which you are upgrading and complete the steps that are specific to Elastic Application Runtime.
2. For more information, see the versions of [Configuring Elastic Application Runtime for upgrades](#) in the Elastic Application Runtime documentation and [Upgrading Elastic Application Runtime and Other Tanzu Operations Manager Products](#) that are specific to the minor release version to which you are upgrading.

Task 1: Prepare to upgrade

If you have not already, complete the steps in [Upgrade preparation checklist for Tanzu Operations Manager v3.2](#).

Task 2: Upgrade Tanzu Operations Manager and installed products to v3.2

The following sections describe how to upgrade Tanzu Operations Manager and installed products to Tanzu Operations Manager v3.2.

Import installation to Tanzu Operations Manager v3.2 VM

To import your Tanzu Operations Manager installation to the Tanzu Operations Manager v3.2 VM:

1. Download the Tanzu Operations Manager VM Template v3.2 from the [Tanzu Operations Manager](#) download page on Broadcom Support portal.
2. Record the fully qualified domain name (FQDN) address of the existing Tanzu Operations Manager VM.
3. To avoid conflicts, power off the existing Tanzu Operations Manager VM.
4. Deploy the new Tanzu Operations Manager VM by following the steps in one of these topics:
 - AWS: [Upgrading BOSH Director on AWS](#).
 - Azure: [Upgrading BOSH Director on Azure](#).
 - GCP: [Upgrading BOSH Director on GCP](#).
 - vSphere: [Deploying Tanzu Operations Manager on vSphere](#).
5. When redirected to the Welcome to Tanzu Operations Manager page, click Import Existing Installation.
6. When you are prompted, enter the Decryption Passphrase for this Tanzu Operations Manager installation. You set this passphrase during your initial installation of Tanzu Operations Manager. If lost, the Decryption Passphrase cannot be recovered.
7. Click Choose File and go to the installation ZIP file exported in the installation export step of your upgrade preparation. For more information, see [Export Your Installation](#) in *Upgrade Preparation*

Checklist for Tanzu Operations Manager v3.2.

8. Click Import.
Some browsers do not provide feedback on the status of the import process, and might appear to fail.
9. A Successfully imported installation message appears upon completion.

Tanzu Operations Manager automatically imports the Telemetry tile. This tile collects product usage data, which helps improve VMware products and services. Using Telemetry is optional and the tile does not share product usage data until you add and configure it. For more information, see the documentation for the [Telemetry](#) tile.

Import new Elastic Application Runtime and product tiles

After you upgrade to Tanzu Operations Manager v3.2, you must upgrade your product versions.

To upgrade your product versions:

1. Click IMPORT A PRODUCT to upload the product file.
2. In IMPORT A BUTTON, click + next to tile. This adds the tile to your staging area.
3. Click the newly-added tile to review and save any configurable options.
4. (Optional) If you use other service tiles, you can upgrade them following the same procedure. For more information, see [Upgrading Elastic Application Runtime and other Tanzu Operations Manager products](#).

Complete your upgrade

If the installation fails or returns errors, contact [Broadcom Support](#). Do not attempt to roll back the upgrade by restarting the previous version's Tanzu Operations Manager VM.

To complete your upgrade to Tanzu Operations Manager v3.2:

1. Go to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes, then click Apply Changes. This immediately imports and applies upgrades to all tiles in a single transaction.
3. Click each service tile, click the Status tab, and confirm that all VMs appear and are in good health.
4. After confirming that the new installation functions correctly, remove the previous version's Tanzu Operations Manager VM.

Task 3: Monitor the upgrade (optional)

The following sections describe how to monitor your Tanzu Operations Manager foundation during the upgrade and troubleshoot issues.

Check status and performance

You can monitor the progress of the upgrade by checking the status of the foundation at various locations.

VMware recommends live-monitoring your upgrade with [HealthWatch](#). HealthWatch captures, calculates, stores, visualizes, and alerts on platform metrics, including:

- BOSH-reported VM metrics
- Runtime component metrics
- [Key Performance Indicators](#) and [Key Scaling Indicators](#)
- Healthwatch-generated [super metrics](#)

For more information, see the [Healthwatch documentation](#).

If you are not using Healthwatch, you can do some or all of the following to monitor upgrade progress:

- Use the BOSH CLI to check the status of a task, VM, or instance. For more information, see [Check Status using BOSH CLI](#).
- Check app availability.
- Run Tanzu cf CLI commands.
- Check the availability of the Tanzu Operations Manager UI.
- If using a network-attached storage (NAS) server, check the server's performance.
- If on vSphere, check vSphere performance.

Check status using the BOSH CLI

To check the status of a task, run:

```
...
bosh -e ALIAS task TASK-NUMBER
...
```

Where:

- * `ALIAS` is your BOSH alias.
- * `TASK-NUMBER` is the number of the upgrade task you want to check.

To check the status of a VM, run:

```
```console
bosh -e ALIAS vms --vitals
```
```

Where `ALIAS` is your BOSH alias.

To check the status of an instance, run:

```
...
bosh -e ALIAS instances --ps
...
```

Where `ALIAS` is your BOSH alias.

Check Diego state

You can use the CF Diego Operator Toolkit (cfdot) to check Diego component instance count by current state. For more information, see the [cfdot](#) repository on GitHub.

Take snapshots of storage metrics

You can periodically take snapshots of storage metrics. VMware recommends this if you have a large foundation and have experienced storage issues in the past.

Collect logs

If you encounter problems during upgrade, collect the following information: * All job logs * Task debug logs for VM upgrade tasks * The installation log from Tanzu Operations Manager

This information helps determine the cause of upgrade issues.

Task 4: After upgrade

The following sections describe how to prepare for use of your new environment, check its health status, and clean up after upgrading.

Recreate BOSH alias

To log in to BOSH after upgrading Tanzu Operations Manager, you must re-create your alias.

To recreate your alias using BOSH:

1. Run:

```
bosh alias-env ALIAS -e DIRECTOR-IP
```

Where:

- **ALIAS** is the BOSH alias you are re-creating.
- **DIRECTOR-IP** is the IP address of your BOSH Director VM.

Install new Tanzu cf CLI

To install the version of the Tanzu Command Line Interface (Tanzu cf CLI) packaged with the Elastic Application Runtime tile on the Broadcom Support portal:

1. Go to the [Elastic Application Runtime](#) download page on the Broadcom Support portal.
2. Click your release number, and then download the CF CLI.
3. Unzip the cf CLI ZIP file containing compressed binaries for the cf CLI.
4. Follow the procedure in the [Elastic Application Runtime documentation](#) to install the Tanzu cf CLI.

Check the health of your deployment

Check the health of your deployment to ensure that all jobs and processes are running as expected.

To check the health of your deployment:

1. Check the health of each BOSH deployment in your installation by running:

```
bosh -e ALIAS -d DEPLOYMENT-NAME instances --ps
bosh -e ALIAS vms --vitals
bosh -e ALIAS -d DEPLOYMENT-NAME cck --report
```

Where:

- **ALIAS** is your BOSH alias.
 - **DEPLOYMENT-NAME** is the name of your BOSH deployment.
2. Push and horizontally scale a test app to test Elastic Application Runtime performance.
 3. If you are running Elastic Application Runtime MySQL as a cluster, run the **mysql-diag** tool to validate health of the cluster. For more information, see the BOSH CLI v2 instructions in [Running mysql-diag](#) in the Elastic Application Runtime documentation.

Check resource settings

If you added custom VM Type or Persistent Disk Type options, you must ensure that these values are set correctly and were not overwritten.

To check your custom resource settings:

1. Go to the Installation Dashboard and click the BOSH Director tile.
2. Click Resource Config.
3. Ensure that the values for VM Type and Persistent Disk Type are correct.

Run BOSH clean-up

To clean up old stemcells, releases, orphaned disks, and other unused resources, run:

```
...  
bosh -e ALIAS clean-up --all  
...  
Where `ALIAS` is the BOSH alias you are recreating.
```

Restage apps

VMware recommends telling developers to restage any apps that use buildpacks that were updated as a part of this upgrade.

For more information about how developers can restage apps, see [Restage your app](#) in *Starting, Restarting, and Restaging Apps* in the Elastic Application Runtime documentation.

Jump upgrading to Tanzu Operations Manager v3.0

This topic describes how you can upgrade to Tanzu Operations Manager v3.2 or later directly from v2.5 and later, skipping minor versions. This is called a jump upgrade.

Prerequisites

Before you jump upgrade to Tanzu Operations Manager v3.2 or later:

- If you are using Tanzu Operations Manager v2.10.8 or earlier, you must complete any in-progress CA rotation before you can jump upgrade. Failure to complete the CA rotation results in the inability to Apply Changes due to safety violations.
- If you are on Tanzu Operations Manager v2.4 or earlier, you must do standard upgrades for each version up to v2.5 before you can jump upgrade to v3.2.
- To upgrade to Tanzu Operations Manager v3.2, you must ensure that the products you have installed are compatible with Tanzu Operations Manager v3.2.
- You must have no staged changes for any product tile or service instance.
- If you are using vSphere with NSX-T, you must have NSX-T v2.3 or later installed.

BOSH DNS certificates regeneration

In Tanzu Operations Manager v2.10.9 and later, BOSH DNS leaf certificates are generated to include the subject alternative name (SAN) field. This migration process is required before you upgrade to Tanzu Operations Manager v2.10.33 and later. If you do not complete this migration before upgrading to

v2.10.33 or later, you might experience unexpected BOSH DNS resolution issues until the change is rolled out to all VMs.

When you upgrade to v2.10.9 or later, the BOSH DNS leaf certificates are automatically regenerated to include the SAN field on the next Apply Changes.

To allow DNS certificate regeneration and avoid communication issues between system components:

1. Run the Upgrade service instances errand on all service tiles.
2. Apply Changes to all tiles.

Jump upgrade

After you ensure your deployment meets the requirements described in [Prerequisites](#), you can upgrade to Tanzu Operations Manager to v3.2 or later.

To jump upgrade to Tanzu Operations Manager to v3.2 or later:

1. Review the table in [Breaking changes](#) below and address any breaking changes that apply to your current Tanzu Operations Manager version.
2. Follow the procedure in [Upgrading Tanzu Operations Manager](#).

Breaking changes

The following table describes the breaking changes that affect Tanzu Operations Manager v2.5 through v2.9 and how to address them before you upgrade:

| Upgrading from | Breaking change | Action before upgrading |
|-----------------------|---|--|
| Any prior to v2.10.51 | When upgrading to Tanzu Operations Manager v3.0.36 or later, you must upgrade through v2.10.51 before attempting to upgrade to v3.0.36 or later. | Versions of Tanzu Operations Manager prior to v2.10.51 include a version of the BOSH Director that uses Postgres v10, and the version of BOSH Director included with Tanzu Operations Manager v2.10.51 is the <code>_last_</code> version that is capable of upgrading the BOSH Director postgres job from version 10 of Postgres. |
| Any prior to v2.10.33 | For Tanzu Operations Manager v2.10.9 and later, BOSH DNS leaf certificates are generated to include the subject alternative name (SAN) field. If you do not distribute the BOSH DNS leaf certificates to all VMs, you can experience downtime caused by communication issues between system components. | When upgrading to Tanzu Operations Manager v2.10.9 or later, run the Upgrade service instances errand on all service tiles and Apply Changes to all tiles. For more information, see BOSH DNS certificates regeneration . |
| v2.5 | You can SSH onto the Tanzu Operations Manager VM in a vSphere deployment only with a private SSH key, not a password. You could get an error when authenticating that causes Tanzu Operations Manager to shut down and enter a reboot loop. | Add a public key to the OVF template for the Tanzu Operations Manager VM. Then, use the private key to SSH onto the Tanzu Operations Manager VM. For more information, see Deploy Tanzu Operations Manager in <i>Deploying Tanzu Operations Manager on vSphere</i> . |
| v2.5 | The <code>stemcell</code> property for the Tanzu Operations Manager API <code>/api/v0/diagnostic_report</code> endpoint is removed. This could cause an error when retrieving Tanzu Operations Manager diagnostic reports. | Update any automation scripts that use the <code>stemcell</code> property. For more information about "Stemcell Property for Diagnostic Report API Endpoint is Removed" in the Tanzu Operations Manager v2.6 Release Notes, contact Broadcom Support . |

| Upgrading from | Breaking change | Action before upgrading |
|----------------------------|--|--|
| v2.5 | The <code>stemcell</code> version key for pending changes is moved and renamed. This could cause an error when retrieving Tanzu Operations Manager API pending changes. | Update your automation scripts that use the <code>stemcell</code> information in the <code>/api/v0/pending_changes</code> endpoint. |
| v2.5 | Trusty stemcells 3541 and earlier are no longer supported. If your deployment is on AWS and using outdated stemcells, Tanzu Operations Manager fails to boot. | Install Xenial stemcells or Trusty stemcell 3586 or later. For more information about "AWS Deployments Use 5th Generation Instances" in the Tanzu Operations Manager v2.6 Release Notes, contact Broadcom Support . |
| v2.5 and v2.6 | The <code>/api/v0/stemcells_assignments</code> endpoint in the Tanzu Operations Manager API is removed. This could cause an error when retrieving Tanzu Operations Manager API stemcell assignments. | Update your automation scripts to use the <code>/api/v0/stemcell_associations</code> endpoint instead. For more information, see Stemcell Associations for Products APIs in the Tanzu Operations Manager API documentation. |
| v2.5 and v2.6 | Tanzu Operations Manager v2.10 installs a later version of the BOSH CLI. This can cause your BOSH CLI automation scripts to stop working. | Update your automation scripts to use BOSH CLI v2. For more information, see Installing the CLI and Commands in the BOSH documentation. |
| v2.5 and v2.6 | Certain properties within a collection or selector are no longer configurable in the Tanzu Operations Manager UI and API. | Update your automation scripts to remove these properties. For more information about the tile properties you can configure in Tanzu Operations Manager v2.10.2, see the release notes for that tile. |
| v2.5 and v2.6 | The <code>nsx_security_groups</code> property is renamed <code>security_groups</code> for NSX-V deployments and <code>ns_groups</code> for NSX-T deployments. Additionally, the Tanzu Operations Manager API does not recognize NSX-related properties that are not nested under an <code>nsx</code> or <code>nsxt</code> key. If your deployment is on vSphere, Tanzu Operations Manager API does not update parameters for NSX-V or NSX-T. | Update your automation scripts to use the appropriate properties for your deployment. For more information, see Configuring resources for a job in the Tanzu Operations Manager API documentation. |
| v2.5 and v2.6 | Manually set CredHub certificates are assumed to be generated by CredHub, and will be unintentionally rotated by the Tanzu Operations Manager API | You must reset any certificates that were manually set in CredHub on Tanzu Operations Manager v2.6 and earlier to prevent them from being rotated with other certificates generated by CredHub. To find and reset any manually set certificates in CredHub, see Reviewing manually set certificates in CredHub . |
| v2.5, v2.6, and v2.7 | If TLS is not enabled before upgrade and Windows stemcells are in use, Apply Changes can fail. Tanzu Operations Manager enables TLS by default when an internal blobstore is configured. | Enable the internal blobstore TLS on the Director Config pane under the BOSH Director tile. To re-create all Windows-based VMs, do one of the following: <ul style="list-style-type: none"> • Upload and deploy a newer Windows stemcell to all relevant tiles. • Enable the Recreate VMs deployed by the BOSH Director check box in the Director Config pane and deploy all relevant tiles. |
| v2.5, v2.6, v2.7, and v2.8 | Tanzu Operations Manager v2.9 is incompatible with vSphere Data Center NSX-T v2.2 and earlier. | Upgrade to NSX-T v2.3 or later. For more information about "Tanzu Operations Manager v2.9 Incompatible with vSphere Data Center NSX-T v2.2 and Earlier" in the Tanzu Operations Manager v2.9 Release Notes, contact Broadcom Support . |

| Upgrading from | Breaking change | Action before upgrading |
|----------------|---|--|
| v2.8 and v2.9 | Tanzu Operations Manager v2.10 includes CredHub Maestro v8.0. In this version of CredHub Maestro, the <code>maestro update-transitional latest</code> command is removed. | If you have scripts that rely on the <code>maestro update-transitional latest</code> command, remove references to the command before you upgrade to Tanzu Operations Manager v3.2. In CredHub Maestro v8.0, you run <code>maestro regenerate ca</code> to regenerate a certificate authority (CA) and mark the latest version of the CA as transitional. This command performs both actions, while previous versions of CredHub Maestro use a separate command for each task. |

For breaking changes between Tanzu Operations Manager v2.10 and v3.2, see [Breaking changes in Tanzu Operations Manager v3.2](#).

Upgrade preparation checklist for Tanzu Operations Manager v3.0

This checklist contains important preparation steps that you must follow before beginning to upgrade Tanzu Operations Manager and Tanzu Platform for Cloud Foundry from v2.9 to v3.2. If you do not follow these instructions, you may jeopardize your existing deployment data and cause the upgrade to fail.

VMware recommends upgrading directly to Tanzu Operations Manager v3.2 from Tanzu Operations Manager v2.5 through v2.9. To upgrade directly to Tanzu Operations Manager v3.2 from Tanzu Operations Manager v2.5 through v2.9, see [Jump Upgrade from Tanzu Operations Manager v2.5 Through v2.9](#) in *Upgrading Tanzu Operations Manager*.

Although you can skip minor versions when upgrading Tanzu Operations Manager, do not skip minor versions when upgrading Elastic Application Runtime. Skipping minor versions when upgrading Elastic Application Runtime might result in additional breaking changes. To avoid this, upgrade Elastic Application Runtime to the minor version that directly follows your current version of Elastic Application Runtime. For more information, see [Upgrade Elastic Application Runtime](#) in *Upgrading Tanzu Operations Manager*.

Any certificate rotation must be completed before upgrading Tanzu Operations Manager. Failure to complete the CA rotation results in the inability to Apply Changes due to safety violations.

Back up your Tanzu Operations Manager deployment

VMware recommends backing up your Tanzu Operations Manager deployment before upgrading, to restore in the case of failure. To do this, follow the instructions in [Backing up Deployments with BBR](#).

Find your decryption passphrase for Tanzu Operations Manager

To complete the Tanzu Operations Manager upgrade, you must have your Tanzu Operations Manager decryption passphrase. You defined this decryption passphrase during the initial installation of Tanzu Operations Manager.

Review changes in Tanzu Operations Manager and Elastic Application Runtime v10.3

Review each of the following links to understand the changes in the new release, such as new features, known issues, and breaking changes.

- Release Notes
 - [Tanzu Operations Manager v3.2 Release Notes](#)
 - [Elastic Application Runtime 10-3 Release Notes](#)
- Known Issues
 - [Tanzu Operations Manager v3.2 Known Issues](#)
 - [Elastic Application Runtime 10-3 Known Issues](#)
- Breaking Changes
 - [Tanzu Operations Manager v3.2 Breaking Changes](#)
 - [Elastic Application Runtime 10-3 Breaking Changes](#)
- KPI Changes
 - [KPI Changes](#)
- Diego Network Communications
 - [Diego Network Communications](#)

Check VMware NSX-T Data Center version

For vSphere deployments, the vSphere BOSH CPI that is included in Tanzu Operations Manager v3.2 no longer supports NSX-T v2.2 and earlier. If you are using NSX-T v2.2 or earlier, you must upgrade NSX-T to a supported version.

The specific version of NSX-T required for your deployment depends on which runtime you plan to deploy and integrate with NSX-T.

- For Elastic Application Runtime deployments, see the [Support matrix: Elastic Application Runtime, VMware NSX-T Container Plug-in](#).
- For VMware Tanzu Container Orchestration Engine (formerly TKGI) deployments, see the [VMware product interoperability matrices](#).

Update tiles and add-ons

These sections describe changes you must make to your product tiles and add-ons before upgrading Tanzu Operations Manager.

Review service tile compatibility

Before you upgrade, check whether the service tiles that you currently have are compatible with the new version of Tanzu Operations Manager.

To check all the service tiles in your current Tanzu Operations Manager deployment, and how to use Upgrade Planner, see [Upgrade Planner](#).

Alternatively, you can do the following:

1. Go to the tile's download page on the [Broadcom Support portal](#).
2. Search for the tile you want to check, and click the name of the tile.
3. Click the tile version on the Release list.
4. Select I agree to Terms and Conditions.

5. Download the compressed file. Review the documentation and release notes online.

If the currently-deployed version of a tile is not compatible with Tanzu Operations Manager v3.2, you must upgrade the tile to a compatible version before you upgrade Tanzu Operations Manager. You do not need to upgrade tiles that are compatible with both Tanzu Operations Manager v2.9 and v3.2.

Some Partner service tiles might be incompatible with Tanzu Operations Manager v3.2. For information about Partner service tile compatibility, see the tile release documentation in the [Broadcom Support portal](#), or contact the Partner organization that produces the service tile.

Environment details

You can use the following empty table as a template for recording and tracking the tile versions that you have deployed in all of your environments.

| | Sandbox | Non-Prod | Prod | Other |
|---|---|----------|------|-------|
| Tanzu Operations Manager | Tanzu Operations Manager | | | |
| | Elastic Application Runtime | | | |
| Tanzu Operations Manager Services | Tanzu for MySQL | | | |
| | Tanzu for Valkey | | | |
| | Tanzu RabbitMQ | | | |
| | Single Sign-On (SSO) | | | |
| | Spring Cloud Services | | | |
| | Concourse | | | |
| | ... | | | |
| Tanzu Operations Manager Partner Services | New Relic | | | |
| | ... | | | |

Upgrade services tiles

Upgrade all service tiles to versions that are compatible with Tanzu Operations Manager v3.2. Service tiles are add-on products you install alongside your runtime. For example, Tanzu for MySQL, Healthwatch, and Tanzu RabbitMQ [VMs] are service tiles.

Do not upgrade runtime tiles, such as Elastic Application Runtime, Elastic Application Runtime Windows add-on, or Container Orchestration Engine at this time.

To verify version compatibility, see [Upgrade Planner](#) and the service tile documentation.

Configure BOSH Director

With each release of a new Tanzu Operations Manager version, BOSH Director might require updates before upgrading to the new version. For actions to take before upgrading to Tanzu Operations Manager v3.2, see the following sections.

Check required machine specifications

Check the required machine specifications for Tanzu Operations Manager v3.2. These specifications are specific to your IaaS. If these specifications do not match your existing Tanzu Operations Manager deployment, modify the values of your Tanzu Operations Manager VM instance. For example, if the boot disk of your existing Tanzu Operations Manager deployment is 50 GB and the new Tanzu Operations

Manager deployment requires 100 GB, increase the size of your Tanzu Operations Manager boot disk to 100 GB.

Configure Elastic Application Runtime

With each release of a new Tanzu Operations Manager version, Elastic Application Runtime might require updates before upgrading to the new version. For more information, see the following sections for what action to take before upgrading to Tanzu Operations Manager v3.2.

To deactivate hostname validation for external databases on GCP and Azure:

This pre-upgrade step applies only to existing Elastic Application Runtime deployments where both of these conditions are met:

- In Elastic Application Runtime 3.0+, you want to use the same external GCP or Azure database configured in the Databases pane.
- You enabled TLS communication for the GCP or Azure external database by adding a certificate authority (CA) certificate to the Database CA certificate field in the Databases pane.

If you meet these conditions, you must deactivate hostname validation before you upgrade to Elastic Application Runtime 3.0 and beyond. Failure to deactivate hostname validation can cause the upgrade to fail for deployments that use external databases on GCP or Azure.

To deactivate hostname validation:

1. After you stage Elastic Application Runtime for upgrade, go to the Databases pane in the Elastic Application Runtime tile.
2. Unselect the Require hostname verification check box. By default, this check box is selected.

For more information about database configuration in Elastic Application Runtime v10.3, see the [Elastic Application Runtime documentation](#).

To deactivate unused errands (optional):

To save upgrade time, you can deactivate unused Elastic Application Runtime post-deploy errands. For more information, see [Post-Deploy Errands](#) in *Errands*. Only deactivate these errands if your environment does not need them.

In some cases, if you have previously deactivated lifecycle errands for any installed product to reduce deployment time, you might want to activate these errands again before upgrading. For more information, see [Add and import products](#) in *Adding and Deleting Products*.

Check OS compatibility of BOSH-managed add-ons and tiles

Before upgrading to Tanzu Operations Manager v3.2, operators who have deployed any BOSH-managed add-ons such as IPsec Encryption, Anti-Virus Scanning, or File Integrity Monitoring and who have deployed or are planning to deploy Elastic Application Runtime Windows add-on must modify the add-on manifest to specify a compatible OS stemcell. For more information, see the [Elastic Application Runtime Windows add-on documentation](#).

For example, File Integrity Monitor (FIM) is not supported on Windows. Therefore, the manifest must use an `include` directive to specify the target OS stemcell of `ubuntu-xenial` and `ubuntu-jammy`.

To upgrade to a Jammy stemcell, see the documentation for each add-on and follow the instructions.

To update an add-on manifest:

1. Locate your add-on manifest file. For example, for FIM, locate the `fim.yml` you uploaded to the Tanzu Operations Manager VM.
2. Modify the manifest to include the following `include` directive to your manifest:

```
include:
  stemcell:
    - os: ubuntu-xenial
    - os: ubuntu-jammy
```

3. Upload the modified manifest file to your Tanzu Operations Manager deployment. For instructions, see [Installing file integrity monitoring on BOSH Director](#) in the File Integrity Monitoring documentation.

If you use any other BOSH-managed add-ons in your deployment, verify OS compatibility for those components as well. For more information about configuring BOSH add-on manifests, see [Addons Block](#).

Check certificate authority expiration dates

Depending on the requirements of your deployment, you might need to rotate your certificate authority (CA) certificates. The non-configurable certificates in your deployment expire every two years. You must regenerate and rotate them so that critical components do not face an outage.

To retrieve information about all the RSA and CA certificates for the BOSH Director and other products in your deployment, you can run the following using the Tanzu Operations Manager API: `GET /api/v0/deployed/certificates?expires_within=TIME`

`expires_within` is optional, and valid values are:

- `d` for days
- `w` for weeks
- `m` for months
- `y` for years

For example, to search for certificates expiring within one month, run:

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/certificates?expires_within=1m" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where:

- `OPS-MANAGER-FQDN` is the fully-qualified domain name (FQDN) of your Tanzu Operations Manager deployment.
- `UAA-ACCESS-TOKEN` is your UAA access token.

For information about regenerating and rotating CA certificates, see [Overview of certificate rotation](#).

Check the capacity of your deployment

These sections describe steps for ensuring your deployment has adequate capacity to perform the upgrade.

Confirm adequate disk space

Confirm that the BOSH Director VM has adequate disk space for your upgrades. You need at least 20 GB of free disk space to upgrade Tanzu Operations Manager and Elastic Application Runtime. If you plan to upgrade other products, the amount of disk space required depends on how many tiles you plan to deploy to your upgraded Tanzu Operations Manager deployment.

To check current persistent disk usage:

1. Go to Tanzu Operations Manager Installation Dashboard.
2. Click the BOSH Director tile.
3. Click the Status tab.
4. Check the value of the PERSISTENT DISK TYPE column. If persistent disk usage is higher than 50%:
 1. Click the Settings tab.
 2. Click Resource Config.
 3. Increase your persistent disk space to handle the size of the resources. If you do not know how much disk space to allocate, set the value to at least **100 GB**.

Check Diego Cell RAM and disk

Check that Diego Cells have sufficient available RAM and disk capacity to support app containers.

The KPIs that monitor these these resources are:

- `rep.CapacityRemainingMemory`
- `rep.CapacityRemainingDisk`

Adjust Diego Cell limits

If needed, adjust the maximum number of Diego Cells that the platform can upgrade simultaneously, to avoid overloading the other Diego Cells. For more information, see [Limit component instance restarts](#).

The maximum number of Diego Cells that can update at once, `max_in_flight`, is 4%. This setting is configured in the BOSH manifest, in the Diego Cell definition. For more information, see [Prevent overload](#).

For more information about these KPIs, see [Diego Cell metrics](#).

Review file storage IOPS and other upgrade limiting factors

During the Tanzu Operations Manager upgrade process, a large quantity of data is moved around on disk.

To ensure a successful upgrade of your deployment, verify that your underlying Elastic Application Runtime file storage is performant enough to handle the upgrade. For more information about the configurations to evaluate, see [Configure file storage](#).

In addition to file storage IOPS, consider additional existing deployment factors that can impact overall upgrade duration and performance:

| Factor | Impact |
|-----------------|---|
| Network latency | Network latency can contribute to how long it takes to move app instance data to new containers. |
| Number of ASGs | A large number of App Security Groups (ASGs) in your deployment can contribute to an increase in app instance container startup time. For more information, see App Security Groups in the Elastic Application Runtime documentation. |

| Factor | Impact |
|--|---|
| Number of app instances and app growth | A large increase in the number of app instances and average droplet size since the initial deployment can increase the upgrade impact on your system. |

Run BOSH clean-up

To clean up old stemcells, releases, orphaned disks, and other resources before upgrade, run:

```
bosh -e ALIAS clean-up --all
```

Where **ALIAS** is your BOSH deployment alias.

This cleanup helps prevent the product and stemcell upload process from exceeding the BOSH Director's available persistent disk space.

Check the health of your deployment

These sections describe steps for ensuring your deployment is healthy before you perform the upgrade.

Collect foundation health status

For collecting foundation health status, VMware recommends using Healthwatch, which monitors the current health, performance, and capacity of your Tanzu Operations Manager deployment, and generates alerts. For more information, see the [Healthwatch documentation](#).

If you are not using Healthwatch, you can do some or all of the following to collect foundation health status:

- If your Tanzu Operations Manager deployment has external metrics monitoring set up, verify that VM CPU, RAM, and disk use levels are within reasonable levels.
- Check your system status.
 - To check the status of your BOSH instances, run:

```
bosh -e ALIAS -d DEPLOYMENT-NAME instances --ps
```

Where:

- **ALIAS** is your BOSH deployment alias.
- **DEPLOYMENT-NAME** is the name of the BOSH deployment with the instances you want to check.

Running **bosh instances** with the flags **--ps**, **--vitals**, or **--failing** reveals individual job failure.

- To check the status of your BOSH VMs, run:

```
bosh -e ALIAS vms --vitals
```

Where **ALIAS** is your BOSH deployment alias.

This command reveals VMs with high CPU, high memory, high disk utilization, and those with a **state** other than **running**.

- To check the status of your BOSH cloud config, run:

```
bosh -e ALIAS -d DEPLOYMENT-NAME cck --report
```

Where:

- **ALIAS** is your BOSH deployment alias.
 - **DEPLOYMENT-NAME** is the name of the BOSH deployment with the cloud config you want to check.
- Check the Status tab of each Elastic Application Runtime tile for VM CPU, RAM, and disk use levels.
 - Check that Tanzu Operations Manager persistent disk usage is less than 50%. If not, follow the procedure in [Confirm Adequate Disk Space](#).
 - (Optional) Check the logs for errors before proceeding with the upgrade. For more information, see the [Elastic Application Runtime documentation](#).

Validate MySQL cluster health

If you are running Elastic Application Runtime MySQL as a cluster, run the `mysql-diag` tool to validate health of the cluster.

For BOSH CLI v2 instructions, see the [Elastic Application Runtime documentation](#).

Review pending and recent changes

To review pending and recent changes:

1. Confirm that there are no outstanding changes in Tanzu Operations Manager or any other tile. All tiles must be green. If all tiles are not green, click Review Pending Changes, then click Apply Changes.
2. After applying changes, click Recent Install Logs to confirm that the changes completed cleanly. You must see the following output:

```
Cleanup complete
{"type": "step_finished", "id": "clean_up_bosh.cleaning_up"}
Exited with 0.
```

Export your installation

To export your installation:

1. In Tanzu Operations Manager Installation Dashboard, click the account drop-down and menu and select Settings.
2. On the Settings page, click Export Installation Settings, then click Export Installation Settings.

This exports the current Tanzu Operations Manager installation with all of its assets.

When you export an installation, the export contains the base VM images, necessary packages, and configuration settings, but does not include releases between upgrades if Tanzu Operations Manager has already uploaded to BOSH. When backing up your deployment, you must take this into account by backing up the BOSH blobstore that contains the uploaded releases. BOSH Backup and Restore (BBR) backs up the BOSH blobstore. For more information, see [Backing up Deployments with BBR](#).

- The export time depends on the size of the exported file.

- Some browsers do not provide feedback on the status of the export process and might appear to hang.

Some operating systems automatically unzip the exported installation. If this occurs, create a ZIP file of the unzipped export. Do not start compressing at the "installation" folder level. Instead, start compressing at the level containing the `rails_database_dump.postgres` file.

Next steps

After you have completed the Upgrade Preparation Checklist, continue to [Upgrading Tanzu Operations Manager](#).

Upgrading Elastic Application Runtime and other Operations Manager products

You can upgrade to a point release (for example, from v3.2.0 to v3.2.1) of Tanzu Platform for Cloud Foundry and other product tiles without upgrading Tanzu Operations Manager. You might need to perform this upgrade if a security update for Elastic Application Runtime is released, or if new features are introduced in a point release of a product tile.

For Elastic Application Runtime component and version information, see the [Elastic Application Runtime documentation](#).

If you cannot download products from Broadcom Support portal due to restricted network connectivity, see [Installing Tanzu Operations Manager in Air-Gapped Environments](#).

Prerequisites

Before you upgrade to point releases:

- You must have completed the procedure in [Upgrading Tanzu Operations Manager](#).
- For the latest compatibility information, see [Upgrade planner](#) before upgrading Elastic Application Runtime.
- Important: Read the *Known Issues* sections for the products that you plan to install before starting. For all available product release notes, see:
 - [Tanzu Operations Manager Release Notes](#) in the Tanzu Operations Manager documentation
 - [Elastic Application Runtime Release Notes](#) in the Elastic Application Runtime documentation
 - [Isolation Segment Release Notes](#) in the Elastic Application Runtime documentation
 - [Elastic Application Runtime Windows add-on Release Notes](#) in the Elastic Application Runtime Windows add-on documentation
 - [VMware Tanzu Container Orchestration Engine \(formerly TKGI\) Release Notes](#) in the VMware Tanzu Container Orchestration Engine (formerly TKGI) documentation

Check BOSH Director disk space

Confirm that the BOSH Director VM has adequate disk space for your upgrades. You need at least 20 GB of free disk space to upgrade Tanzu Operations Manager and Elastic Application Runtime. If you plan to upgrade other products, the amount of disk space required depends on how many tiles you plan to deploy to your upgraded Tanzu Operations Manager deployment.

To check current persistent disk usage:

1. Go to Tanzu Operations Manager Installation Dashboard.
2. Click the BOSH Director tile.
3. Click the Status tab.
4. Check the value of the PERSISTENT DISK TYPE column. If persistent disk usage is higher than 50%:
 1. Click the Settings tab.
 2. Click Resource Config.
 3. Increase your persistent disk space to handle the size of the resources. If you do not know how much disk space to allocate, set the value to at least **100 GB**.

Upgrading Elastic Application Runtime

If you use the Broadcom Support portal API, the latest product versions automatically appear in the Tanzu Operations Manager Installation Dashboard. For more information, see [Using Broadcom Support portal API to upgrade products](#) in *Adding and Deleting Products*.

To upgrade Elastic Application Runtime without upgrading Tanzu Operations Manager:

1. Download the product file from [Broadcom Support portal](#).
2. Import the product file to your Tanzu Operations Manager Installation Dashboard.
3. Click the + icon next to the uploaded product description to add this product to your staging area.
4. Click the newly added tile to review any configurable options.
5. Click Review Pending Changes.
6. Click Apply Changes to install the service.

Upgrading Tanzu Operations Manager products

If you use the Broadcom Support portal API, the latest product versions automatically appear in the Tanzu Operations Manager Installation Dashboard. For more information, see [Using Broadcom Support portal API to upgrade products](#) in *Adding and Deleting Products*.

This section describes how to upgrade individual products like [Single Sign-On](#), [Tanzu for MySQL](#), [Tanzu RabbitMQ](#), and [App Metrics](#) in your Tanzu Operations Manager deployment. Ensure you review the individual product upgrade procedure for each tile.

To upgrade your products:

1. Browse to [Broadcom Support portal](#) and sign in.
2. Download the latest release for the product or products that you want to upgrade. Every product is tied to exactly one stemcell. Download the stemcell that matches your product and version.
3. Go to the Tanzu Operations Manager Installation Dashboard.
4. Click Import a Product.

5. Select the `.pivotal` file that you downloaded from Broadcom Support portal or received from your software distributor, then click Open. If the product is successfully added, it appears in your product list. If the product that you selected is not the latest version, the most up-to-date version appears in your product list.
6. Click the + icon to add the product tile to the Tanzu Operations Manager Installation Dashboard.
7. Repeat the import, upload, and upgrade steps for each product you download.
8. If you are upgrading a product that uses a self-signed certificate from v1.1 to v1.2, you must configure the product to trust the self-signed certificate. To configure a product to trust the self-signed certificate:
 1. Click the product tile.
 2. Select the setting page containing the SSL certificate configuration. For example, for Elastic Application Runtime, select the HAProxy pane.
 3. Check the Trust Self-Signed Certificates box.
 4. Click Save.
9. Click Review Pending Changes.
10. Click Apply Changes.

Upgrading replicated tiles

To upgrade a replicated tile, you must:

- Generate a replica of the newer version of the tile using the Replicator.
- Give the new replica the same name as the existing replica.

Example workflow

This example workflow assumes the following:

- You used the Replicator to generate a replica of the Isolation Segment v1 tile.
- You used the `name` seg-one.
- You installed the tile in Tanzu Operations Manager.
- You used the following sample Replicator command for the initial installation:

```
./replicator-darwin\  
--name seg-one\  
--path /download/p-isolation-segment-v1.pivotal\  
--output /output/p-isolation-segment-v1-seg-one.pivotal
```

To upgrade your replicated tile to Isolation Segment v2:

1. Download and unzip the new Replicator from the [Isolation Segment](#) page on Broadcom Support portal. You must download the version of Replicator that corresponds with the version of the Isolation Segment tile you want to replicate.
2. Create the replica by running:

```
./replicator-darwin\  
--name NAME-OF-EXISTING-REPLICA \  
--path PATH-TO-NEW-TILE \  
--output /output/p-isolation-segment-v2-seg-one.pivotal
```

Where:

- **NAME-OF-EXISTING-REPLICA** must be the same as the name used for the existing replica. In this example, this name is seg-one.
 - **PATH-TO-NEW-TILE** is the path to the new Isolation Segment tile.
3. After you create the replica tile p-isolation-segment-v2-seg-one.pivotal, upload it to the Tanzu Operations Manager Installation Dashboard. This upgrades the Isolation Segment v1 replica tile in place.

Operating your Tanzu Operations Manager deployment

See the following topics for information about operating your Tanzu Operations Manager deployment:

- [Using the Tanzu Operations Manager UI](#)
 - [Using the Tanzu Operations Manager interface](#)
 - [Reviewing your pending product changes in Tanzu Operations Manager](#)
 - [Adding and deleting products from Tanzu Operations Manager](#)
 - [Importing and managing stemcells in Tanzu Operations Manager](#)
 - [Managing errands in your Tanzu Operations Manager deployment](#)
- [Using the API or CLI](#)
 - [Using Tanzu Operations Manager programmatically and from the command line](#)
 - [Using the Tanzu Operations Manager API](#)
 - [Using the Tanzu Operations Manager CLI](#)
- [Advanced features in Tanzu Operations Manager](#)
 - [Modifying installation and product template files](#)
 - [Managing verifiers in your Tanzu Operations Manager deployment](#)
 - [Managing custom VM extensions with the Tanzu Operations Manager API](#)
 - [Tanzu Operations Manager fields that lock on deploy](#)
 - [Overriding manifest properties in Advanced Mode](#)
 - [Property template references for Tanzu Operations Manager tiles](#)
- [User accounts and permissions in Tanzu Operations Manager](#)
 - [Platform operator roles and permissions in Tanzu Operations Manager](#)
 - [Creating UAA clients for the BOSH Director](#)
 - [Creating and managing Tanzu Operations Manager user and client accounts](#)
 - [Configuring Role-Based Access Control in Tanzu Operations Manager](#)
- [Certificates and credentials in Tanzu Operations Manager](#)
 - [Certificate types used by Tanzu Operations Manager](#)
 - [Managing certificates with the Tanzu Operations Manager API](#)
 - [CredHub for centralized Tanzu Operations Manager credential management](#)
 - [Overview of certificate rotation in Tanzu Operations Manager](#)
 - [Checking Tanzu Operations Manager certificate expiration dates and rotation procedures](#)
 - [Overriding duration for Tanzu Operations Manager and CredHub certificates](#)
 - [Rotating CAs and leaf certificates using the Tanzu Operations Manager API](#)

- [Rotating non-configurable leaf certificates in Tanzu Operations Manager](#)
- [Rotating configurable Tanzu Operations Manager leaf certificates](#)
- [Rotating identity provider SAML certificates in Tanzu Operations Manager](#)
- [Using custom certificate authorities in Tanzu Operations Manager](#)
- [Retrieving credentials from your Tanzu Operations Manager deployment](#)
- [Using short-lived NATS bootstrap credentials in Tanzu Operations Manager](#)
- [Certificates and credentials managed by CredHub](#)
 - [CredHub](#)
 - [Credential types supported by CredHub](#)
 - [Resetting manually-set certificates in BOSH CredHub](#)
 - [Getting started with CredHub Maestro](#)
 - [Advanced certificate rotation with CredHub Maestro](#)
 - [Tanzu Operations Manager tiles that are compatible with CredHub Maestro](#)
 - [Troubleshooting CredHub Maestro safety violations](#)

Using Tanzu Operations Manager

The following topics show you how to use the Tanzu Operations Manager UI:

- [Using the Tanzu Operations Manager interface](#)
- [Reviewing your pending product changes in Tanzu Operations Manager](#)
- [Adding and deleting products from Tanzu Operations Manager](#)
- [Importing and managing stemcells in Tanzu Operations Manager](#)
- [Managing errands in your Tanzu Operations Manager deployment](#)

Logging into Tanzu Operations Manager

This topic describes how you can log into Tanzu Operations Manager for the first time after deploying Tanzu Operations Manager. It also describes how you can configure the Tanzu Operations Manager default authentication with either SAML, LDAP, or internal authentication.

You can now also manage your deployment (foundation) in [Tanzu Hub](#).

You can install, configure, and update tiles (called Capabilities in Hub) from one interface.

Tanzu Operations Manager is now deployed using Tanzu Foundation Core, which allows you to manage your foundation and view its health in the Tanzu Hub UI. In Foundation Core, you can see health metrics and beta Tanzu Operations Manager features. The BOSH Director configuration is not available, so it may not be suitable for production environments.

You can switch back to the classic, full-function Tanzu Operations Manager UI experience using the Switch to Classic Experience button on the application header.

About login and authentication methods

When you have a new installation of Tanzu Operations Manager, you choose the default authentication and login method.

When you log in for the first time, you go to the fully qualified domain name (FQDN) in your web browser. You configure the FQDN when you first deploy Tanzu Operations Manager. To log in, see [Log In to Tanzu Operations Manager For the First Time](#).

Your login method and authentication choices are:

- [Internal Authentication](#): Tanzu Operations Manager maintains and manages your user database.
- [SAML Identity Provider](#): An external identity server maintains your user database.
- [LDAP Server](#): An external identity server maintains your user database.
- [OIDC Identity Provider](#): An external identity server maintains your user database.

Log in to Tanzu Operations Manager for the first time

To log in to Tanzu Operations Manager for the first time with a new Tanzu Operations Manager deployment:

1. In a web browser, go to Tanzu Operations Manager using your FQDN. You set your FQDN when you configure Tanzu Operations Manager before deployment.
2. When Tanzu Operations Manager starts for the first time, choose one of the following procedures:
 - [Internal Authentication](#)
 - [SAML Identity Provider](#)
 - [LDAP Server](#)

Log in to Tanzu Operations Manager with internal authentication

To set up internal authentication that Tanzu Operations Manager maintains:

1. When redirected to the Internal Authentication page:
 - Enter a Username, Password, and Password confirmation to create an Admin user.
 - Enter a Decryption passphrase and the Decryption passphrase confirmation. This passphrase encrypts the Tanzu Operations Manager datastore and is not recoverable.
 - If you are using an HTTP proxy or HTTPS proxy, follow the instructions in [Configuring Proxy Settings for the BOSH CPI](#).
 - Read the End User License Agreement, and select the check box to accept the terms.
 - Click Setup Authentication.
2. Log in to Tanzu Operations Manager with the Admin user name and password you created in the previous step.

Log in to Tanzu Operations Manager with SAML identity provider

To configure Tanzu Operations Manager to log in by default using a SAML identity provider for user authentication:

1. Log in to your identity provider console and download the identity provider metadata XML. Optionally, if your identity provider supports metadata URL, you can copy the metadata URL instead of the XML.
2. Do one of the following, depending on if you use a separate identity provider for BOSH:

- For the same identity provider: Copy the identity provider metadata XML or URL to the Tanzu Operations Manager SAML Identity Provider login page.
 - For a separate identity provider: Copy the metadata XML or URL from that identity provider and enter it into the BOSH identity provider metadata text box in the Tanzu Operations Manager login page.
3. Enter values for the following fields. Failure to provide values in these text boxes results in a 500 error.
 - SAML admin group: Enter the name of the SAML group that contains all Tanzu Operations Manager administrators. This text box is case-sensitive.
 - SAML groups attribute: Enter the groups attribute tag name with which you configured the SAML server. This text box is case sensitive.
 4. Enter a Decryption passphrase and the Decryption passphrase confirmation. This passphrase encrypts the Tanzu Operations Manager datastore, and is not recoverable.
 5. If you are using an HTTP proxy or HTTPS proxy, see [Configuring Proxy Settings for the BOSH CPI](#).
 6. Read the End User License Agreement and select the check box to accept the terms.
 7. Click Provision an admin client in the BOSH UAA. You can use this to enable BOSH automation with scripts and tooling. For more information about this option, see [Provision admin client in Creating UAA clients for the BOSH Director](#).
 8. Click Setup Authentication.
 9. Your Tanzu Operations Manager login page appears. Enter your user name and password and click Login.
 10. Download your SAML service provider metadata (SAML Relying Party metadata) by navigating to the following URLs:
 - Tanzu Operations Manager SAML service provider metadata: <https://OPS-MAN-FQDN:443/uaa/saml/metadata>
 - BOSH Director SAML service provider metadata: <https://BOSH-IP-ADDRESS:8443/saml/metadata>.
Where **BOSH-IP-ADDRESS** is in the Status pane of the BOSH Director tile.
 11. Import the Tanzu Operations Manager SAML provider metadata to your identity provider. If your identity provider does not support importing, provide the following values:
 - Single sign on URL: <https://OPS-MAN-FQDN:443/uaa/saml/SSO/alias/OPS-MAN-FQDN>
 - Audience URI (SP Entity ID): <https://OP-MAN-FQDN:443/uaa>
 - Name ID: Email Address
 - SAML authentication requests are always signed
 12. Import the BOSH Director SAML provider metadata to your identity provider. If the identity provider does not support an import, provide the following values:
 - Single sign on URL: <https://BOSH-IP:8443/saml/SSO/alias/BOSH-IP>
 - Audience URI (SP Entity ID): <https://BOSH-IP:8443>
 - Name ID: Email Address
 - SAML authentication requests are always signed

Log in to Tanzu Operations Manager with LDAP

To configure Tanzu Operations Manager to log in by default using an LDAP server for user authentication:

1. For Server URL, enter the URL that points to your LDAP server. With multiple LDAP servers, separate their URLs with spaces. Each URL must include one of the following protocols:
 - `ldap://`: This specifies that the LDAP server uses an unencrypted connection.
 - `ldaps://`: This specifies that the LDAP server uses SSL for an encrypted connection and requires that the LDAP server holds a trusted certificate or that you import a trusted certificate to the JVM truststore.

2. For LDAP Username and LDAP Password, enter the LDAP Distinguished Name (DN) and the password for binding to the LDAP Server. Example DN:
`cn=admin,ou=Users,dc=example,dc=com` `%= vars.company_name %>` recommends that you provide LDAP credentials that grant read-only permissions on the LDAP Search Base and the LDAP Group Search Base. In addition to this, if the bind user belongs to a different search base, you must use the full DN.

VMware recommends against reusing LDAP service accounts across environments. LDAP service accounts are not subject to manual lockouts. For example, lockouts that result from users using the same account. Also, LDAP service accounts are not subject to automated deletions, since disruption to these service accounts could prevent user logins.

3. For User Search Base, enter the location in the LDAP directory tree from which any LDAP User search begins. The typical LDAP Search Base matches your domain name. For example, a domain named “cloud.example.com” typically uses the following LDAP User Search Base:
`ou=Users,dc=example,dc=com`
4. For User Search Filter, enter a string that defines LDAP User search criteria. These search criteria allow LDAP to perform more effective and efficient searches. For example, the standard LDAP search filter `cn=Smith` returns all objects with a common name equal to `Smith`.

In the LDAP search filter string that you use to configure your runtime, use `{0}` instead of the user name. For example, use `cn={0}` to return all LDAP objects with the same common name as the user name.

In addition to `cn`, other attributes commonly searched for and returned are `mail`, `uid` and, in the case of Active Directory, `sAMAccountName`.

For instructions for testing and troubleshooting your LDAP search filters, see [Configuring LDAP Integration](#).

5. For Group Search Base, enter the location in the LDAP directory tree from which the LDAP Group search begins. For example, a domain named `cloud.example.com` typically uses the following LDAP Group Search Base: `ou=Groups,dc=example,dc=com`
6. For Group Search Filter, enter a string that defines LDAP Group search criteria. The standard value is `member={0}`.
7. For Email Attribute, enter the attribute name in your LDAP directory that corresponds to the email address in each user record. For example, `mail`.
8. For LDAP RBAC Admin Group Name, enter the DN of the LDAP group you want to have admin permissions in Tanzu Operations Manager.
9. From the drop-down menu, select how the UAA handles LDAP server referrals out to other external user stores. The UAA can:

- Automatically follow any referrals.
 - Ignore referrals and return partial result.
 - Throw exception for each referral and abort.
10. For Server SSL Cert, paste in the root certificate from your CA certificate or your self-signed certificate.
 11. Enter a Decryption passphrase and the Decryption passphrase confirmation. This passphrase encrypts the Tanzu Operations Manager datastore, and is not recoverable.
 12. If you are using an HTTP proxy or HTTPS proxy, follow the instructions in [Configuring Proxy Settings for the BOSH CPI](#).
 13. Read the End User License Agreement, and select the check box to accept the terms.
 14. Click Provision an admin client in the BOSH UAA. You can use this to enable BOSH automation with scripts and tooling. For more information, see [Provision Admin Client](#) in *Creating UAA Clients for BOSH Director*.
 15. Click Setup Authentication.

Log in to Tanzu Operations Manager with OIDC Identity Provider

To configure Tanzu Operations Manager to default to an OpenID Connect (OIDC) provider for user authentication:

1. In your OIDC provider, create a new client for Tanzu Operations Manager to use for authentication.
 - In Grant Type, select Authorization Code.
 - Register <https://OPS-MAN-FQDN/uaa/login/callback/oidc> as a valid `redirect_uri` for the client.
 - If you plan to use OIDC authentication for the BOSH CLI, register <https://BOSH-FQDN-OR-IP:8443/uaa/login/callback/oidc> as a valid `redirect_uri` for the client. If you intend to use the BOSH FQDN, you must configure that later in the BOSH Director configuration.
2. In Discovery URL, enter your OIDC service provider discovery URL.
3. In Client ID, enter the client ID created in Step 1.
4. In Client Secret, enter the client secret created in Step 1.
5. In Scopes, enter the scopes to request from the OIDC provider, as a comma-separated list. You must include the following scopes.
 - The `openid` scope
 - A scope that allows access to the group claim
 - Standard `email` and `profile` scopes, if you plan to use the claims listed in the next step to populate common fields in UAA
6. Enter the claims used to populate the UAA user store with data from the OIDC provider.
 - Enter the External Groups Claim to populate associated groups for the user in UAA. Enter the OIDC provider's token claim that contains the groups to which the user belongs. Only the provided OIDC Admin Group Name and the default group names of `opsman.full_control`, `opsman.restricted_control`, `opsman.full_view` and `opsman.restricted_view` are mapped to UAA groups.

- (Optional) Enter the Username Claim to populate the `username` field in UAA. Tanzu Operations Manager uses this to show the current logged-in user.
7. In OIDC Admin Group Name, enter the OIDC provider group name that corresponds to users who receive admin access. Users in this OIDC group are granted the `opsman.admin` scope in UAA.
 8. Enter a Decryption passphrase and the Decryption passphrase confirmation. This passphrase encrypts the Tanzu Operations Manager datastore, and is not recoverable.
 9. If you are using an HTTP proxy or HTTPS proxy, see [Configuring Proxy Settings for the BOSH CPI](#).
 10. Read the End User License Agreement and select the check box to accept the terms.
 11. Click Provision an admin client in the BOSH UAA. You can use this to enable BOSH automation with scripts and tooling. For more information about this option, see [Provision admin client in Creating UAA clients for the BOSH Director](#).
 12. Click Setup Authentication.

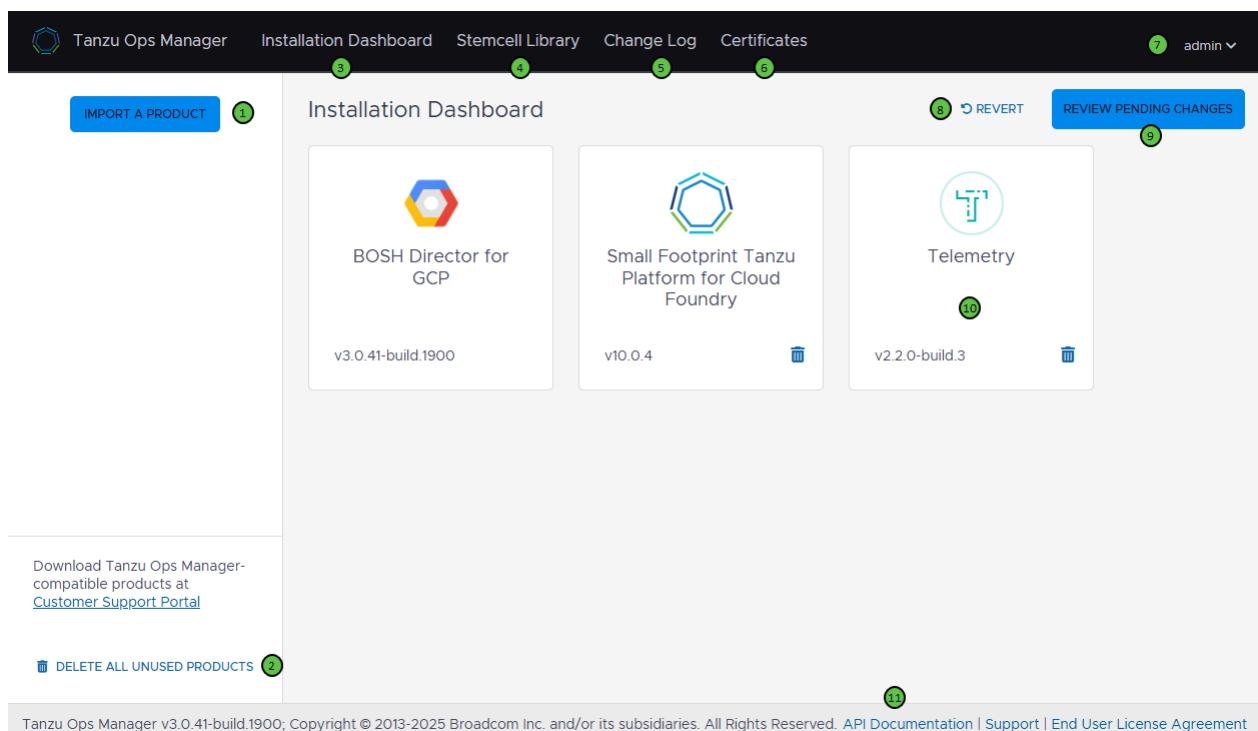
Using the Tanzu Operations Manager interface

This topic tells you about the key features of the Tanzu Operations Manager interface.

Installation Dashboard page

The Installation Dashboard is the central Tanzu Operations Manager interface for managing your deployment. It displays the BOSH Director configuration for your IaaS and product tiles for your deployment.

The following image shows the Tanzu Operations Manager Installation Dashboard. Each section is labeled with a green number. Click the image to see it at full size.



The following list describes each section that is labeled for the Installation Dashboard:

- 1 - Import a Product: Click this button to add a new product to Tanzu Operations Manager. You can download Tanzu Operations Manager-compatible product files from the [Broadcom Support portal](#).

- If you configure a [VMware Tanzu Network API](#) token in the Settings section of Tanzu Operations Manager, a list of the latest versions of already imported products appears automatically.
- 2 - Delete All Unused Products: Click this link to delete any unused products.
- 3 - Installation Dashboard: Click this link to return to the Tanzu Operations Manager Installation Dashboard page from other Tanzu Operations Manager pages.
- 4 - Stemcell Library: Click this link to open the Stemcell Library. In the Stemcell Library you can import stemcells, stage stemcells, and review your stemcell version numbers. For more information, see [Importing and Managing Stemcells](#).
- 5 - Change Log: Click this link to view and search a log of your previous installations. For more information, see the [Change Log Page](#) section of this page for details.
- 6 - Certificates: Click this link to view certificates in your installation. For more information, see the [Certificates Page](#) section of this page for details.
- 7 - User Account Menu: Click your user name to view the menu. Use this menu to go to your Settings page, view My Account to change your email and password, or log out of the Tanzu Operations Manager Installation Dashboard. For more details about accounts, see the [My Account page](#) section of this page for details about account information. For more information, see the [Settings page](#) section of this page for information about Tanzu Operations Manager settings.
- 8 - Revert: Click the Revert button to return to the previous successful installation. Any pending changes you have made to the current installation settings are lost.
- 9 - Review Pending Changes: Click the button to go to the Review Pending Changes page, which organizes pending changes by tile. You can activate or deactivate each tile to selectively deploy individual tiles and their dependencies. For more information, see [Reviewing pending product changes](#).
- 10 - Orange Bar: Indicates that additional configuration for the product tile is required before deployment. Click the product tile to complete its configuration.

If an imported product is missing a required stemcell, a Missing Stemcell link appears on the tile. Click this link to open the stemcell library. For more information about stemcells, see [Importing and managing stemcells](#).

- 11 - API Documentation: Click this link to go to the Tanzu Operations Manager API documentation, which details how you can manage Tanzu Operations Manager through the API rather than with the UI. For more information about the Tanzu Operations Manager API, see [Using the Tanzu Operations Manager API](#).

Change Log page

Go to the Change Log page by clicking the corresponding link in the dashboard header. This page shows the Tanzu Operations Manager's deployment history, and a record of all of its Apply Changes actions.

The following table lists the following top-level attributes for each deployment:

| Label | Type | Description |
|---------------|----------|---|
| STATUS | Icon | Whether the deployment succeeded or failed |
| DEPLOYMENT ID | Text | A sequentially-numbered identifier for the deployment |
| USER | Username | The user who initiated the deployment |

A LOGS button for each deployment opens the deployment's full Installation Log.

Within each deployment listing, a table shows details for the individual products deployed in, or deleted from, each installation.

The details listed for each product are as follows:

| Label | Type | Description |
|----------|--|--|
| PRODUCT | Text | The name of the product |
| ACTION | One of the following labels: ADDED, UPDATED, DELETED, NO CHANGES, FAILED | The action, if any, that the deployment took with the product. |
| STARTED | UTC timestamp | When Tanzu Operations Manager began deploying the product. |
| FINISHED | UTC timestamp | When Tanzu Operations Manager stopped deploying or attempting to deploy the product. |
| DURATION | Time, in minutes | How long Tanzu Operations Manager took to deploy or attempt to deploy the product. |

A Total row lists all product totals for the STARTED, FINISHED, AND DURATION text boxes.

The Change Log page shows up to 10 Apply Changes at a time, starting with the most recent. You can navigate additional Apply Changes records using the following buttons:

- First Page - Click First Page to return to the most recent 10 Apply Changes records.
- Previous Page / Next Page - Click Previous Page, or Next Page to load older or newer entries.

Certificates page

Go to the Certificates page by clicking the corresponding link in the dashboard header. This page shows the certificates in your Tanzu Operations Manager installation.

As shown in the following screenshot, leaf certificates are nested under the CA certificate that signs them. Certificates are listed only once instead of per deployment, and a rotation status is listed for each certificate.

| <div>  Tanzu Ops Manager Installation Dashboard Stemcell Library Change Log Certificates admin </div> | | | | | | |
|--|-----------------------------|---|-------------|--------------------------|-----------------|--|
| Certificates | | | | | | |
| <div>  FILTER TABLE </div> | | | | | | |
| Type | Certificate name | Deployments | Propagation | Rotation Status | Expiration Date | Rotation procedure |
| <div> <div>▼</div> CA </div> | /opsmgr/bosh_dns/tls_ca | 1  | 1 of 1 | Ready to regenerate CA | May 1, 2029 | Standard Non-Configurable Leaf Procedure |
| <div> <div>—</div> Leaf </div> | /bosh_dns_health_client_tls | 1  | 1 of 1 | Ready to regenerate leaf | May 2, 2026 | Standard Non-Configurable Leaf Procedure |
| <div> <div>—</div> Leaf </div> | /bosh_dns_health_server_tls | 1  | 1 of 1 | Ready to regenerate leaf | May 2, 2026 | Standard Non-Configurable Leaf Procedure |
| <div> <div>—</div> Leaf </div> | /dns_api_client_tls | 1  | 1 of 1 | Ready to regenerate leaf | May 2, 2026 | Standard Non-Configurable Leaf Procedure |
| <div> <div>—</div> Leaf </div> | /dns_api_server_tls | 1  | 1 of 1 | Ready to regenerate leaf | May 2, 2026 | Standard Non-Configurable Leaf Procedure |

The following information is listed on the Certificates page:

- Type: a certificate authority (CA) or leaf certificate
- Certificate name: the name of the certificate
- Deployments: the number of deployments that use the certificate. This links to a page listing the deployments associated with the certificate and whether they've been deployed or not
- Propagation: the propagation progress of a certificate after it has been regenerated. The first number is how many deployments are using the updated version of the certificate. The second number is the total number of deployments associated with the certificate. For instance, "1 of 5" would indicate that only 1 deployment is using the new certificate out of 5 deployments, total.
- Rotation Status: the next step necessary to rotate the certificate. The following table shows the possible values for Rotation Status:

| Column | Description |
|--------------------------|--|
| Needs propagation | New versions of the certificate are available and need to be deployed. To deploy the certificate, initiate Apply Changes on all tiles associated with the certificate, selecting the "Upgrade all service instances" errand for any service tiles. For any non-tile deployments associated with the certificate, use <code>bosh deploy</code> . For auto-rotated certificates, a stemcell upgrade is required to complete propagation of the new certificate versions. |
| Ready to regenerate CA | The CA certificate is ready to be regenerated with the recommended procedure. |
| Ready to activate new CA | The new version of the CA certificate is ready to be activated with the recommended procedure. |

| Column | Description |
|---------------------------------------|---|
| Ready to regenerate leaf | For CA certificates, the certificate is ready to regenerate all its child certificates to be signed by the new CA. For leaf certificates, the certificate is ready to be regenerated. |
| Ready to regenerate configurable leaf | The certificate is ready to be regenerated using its CA, or it is ready to be updated with a new external certificate. When using an externally created certificate, ensure that the certificate that signs the configurable leaf is trusted. For example, the leaf certificate is signed by a globally trusted CA, or the signing CA is added in the BOSH Director tile. (Security tab > Trusted Certificates field) |
| Ready to delete old CA | The new CA certificate and its leafs have been deployed, and the old certificate version can be deleted. |
| Rotate parent CA | The signing CA certificate is expiring soon. Rather than rotate the current certificate directly, you should rotate its CA certificate. Rotating the CA certificate rotates the current certificate as part of its rotation procedure. |
| Unknown state - contact support | Tanzu Operations Manager is unable to determine the next step of the rotation. This can mean that the certificate is in a state that generates safety violations when rotated. Contact Broadcom Support and include a support bundle that contains information such as the output of maestro topology . |

Certificate propagation status page

Each certificate on the Certificates page contains a link in the Deployments column that leads to a propagation status page for that certificate. This page lists information about which tiles and deployments need to be redeployed to distribute new versions of the certificate.

Individual BOSH deployments (such as service instances or Container Orchestration Engine clusters) are nested under the tile that creates those deployments. BOSH deployments that are created outside of Tanzu Operations Manager are nested under Non-Tile Deployments.


[Tanzu Ops Manager](#)
[Installation Dashboard](#)
[Stemcell Library](#)
[Change Log](#)
[Certificates](#)

Certificates

[FILTER TABLE](#)

| Type | Certificate name | Deployments  | Propagation  | Rotation Status  | Expiration Date |
|------|-----------------------------|---|---|---|-----------------|
| ▼ CA | /opsmgr/bosh_dns/tls_ca | 1  | 1 of 1 | Ready to regenerate CA | May 1, 2029 |
| Leaf | /bosh_dns_health_client_tls | 1  | 1 of 1 | Ready to regenerate leaf | May 2, 2026 |
| Leaf | /bosh_dns_health_server_tls | 1  | 1 of 1 | Ready to regenerate leaf | May 2, 2026 |

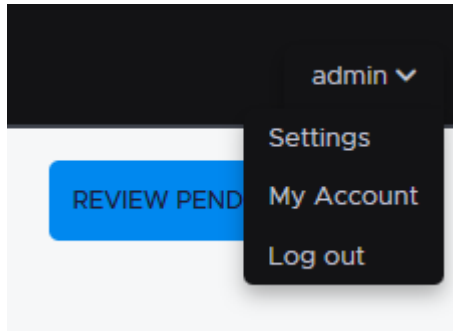
| Column | Description |
|-------------|---|
| Deployments | The name of the BOSH deployment that uses the certificate |
| Propagation | The status of the latest version of the certificate for the corresponding deployment; for example "1 of 4." |

- Expiration Date: the date that the certificate expires
- Rotation Procedure: a link to documentation about how to rotate the certificate

- Location: whether CredHub or Tanzu Operations Manager stores and manages the certificate
- Configurable: whether you can configure the certificate. If a certificate is configurable, you can generate your own and paste it into Tanzu Operations Manager configuration panes.
- Auto Rotation: whether the certificate is automatically rotated by Tanzu Operations Manager. Rotation of these certificates is advanced during Apply Changes without the need to make API calls to rotate them. For more information, see [Automatic rotation of BOSH DNS certificates](#).

Settings page

To go to the Settings page, click your user name and then click Settings. On the screen that appears, click Change Decryption Passphrase.



The following sections describe the configuration pages.

If you modify these settings, you do not have to return to the Installation Dashboard and click Apply Changes. These settings apply to the Tanzu Operations Manager VM. The BOSH Director does not apply them to your deployment.

Change decryption passphrase

To reset your decryption passphrase, enter the following details, and click Change Decryption Passphrase.

- Current Decryption Passphrase
- New Decryption Passphrase
- Confirm New Decryption Passphrase

Tanzu Ops Manager Installation Dashboard Stemcell Library Change Log Certificates

Settings

- Change Decryption Passphrase
- Internal Authentication Settings
- SAML Settings
- LDAP Settings
- OIDC Settings
- SSL Certificate

Change Decryption Passphrase

Current Decryption Passphrase*

New Decryption Passphrase*

Confirm New Decryption Passphrase*

CHANGE DECRYPTION PASSPHRASE

Internal authentication settings

You can use the Internal Authentication Settings page to view and update the settings for your internal authentication method.

This pane includes the following text boxes:

- Current Decryption Passphrase: Update the decryption passphrase for your internal authentication method.
- Admin Username: Update the user name for the admin user.
- Admin Password and Admin Password Confirmation: Update the password for the admin user.

SAML settings

To change your Identity Provider (IdP) to SAML, configure the following text boxes:

- Current Decryption Passphrase: Enter your decryption passphrase.
- SAML IDP Metadata: Enter the full URL or XML SAML IdP metadata.
- BOSH IDP Metadata: (Optional) Enter the full URL or XML BOSH IdP metadata. If you leave it blank, the default is the same metadata as used in the SAML IDP Metadata text box.
- SAML Admin Group: Enter the name of the SAML group that contains all of the Tanzu Operations Manager administrators. This text box is case-sensitive.
- Groups Attribute: Enter the groups attribute tag name with which you configured the SAML server. The groups attribute is case-sensitive.
- Provision an admin client in the Bosh UAA: Select this check box to enable provisioning an admin client in the BOSH UAA. For more information, see [Provision Admin Client](#).

LDAP settings

Use this page to change your IdP to LDAP.

For more information about changing your IdP to SAML or LDAP, view the following instructions for your IaaS configuration:

- [Configuring BOSH Director on AWS](#)
- [Configuring BOSH Director on GCP](#)
- [Configuring BOSH Director on Azure](#)
- [Configuring BOSH Director on vSphere](#)

OIDC settings

To change your Identity Provider (IdP) to OIDC, configure the following text boxes:

1. In your OIDC provider, create a new client for Tanzu Operations Manager to use for authentication.
 - In Grant Type, select Authorization Code.
 - Register <https://OPS-MAN-FQDN/uaa/login/callback/oidc> as a valid `redirect_uri` for the client.
 - If you plan to use OIDC authentication for the BOSH CLI, register <https://BOSH-FQDN-OR-IP:8443/uaa/login/callback/oidc> as a valid `redirect_uri` for the client. If you intend to use the BOSH FQDN, you must configure that later in the BOSH Director configuration.
2. In Discovery URL, enter your OIDC service provider discovery URL.
3. In Client ID, enter the client ID created in Step 1.
4. In Client Secret, enter the client secret created in Step 1.
5. In Scopes, enter the scopes to request from the OIDC provider, as a comma-separated list. You must include the following scopes.
 - The `openid` scope
 - A scope that allows access to the group claim
 - Standard `email` and `profile` scopes, if you plan to use the claims listed in the next step to populate common fields in UAA
6. Enter the claims used to populate the UAA user store with data from the OIDC provider.
 - Enter the External Groups Claim to populate associated groups for the user in UAA. Enter the OIDC provider's token claim that contains the groups to which the user belongs. Only the provided OIDC Admin Group Name and the default group names of `opsman.full_control`, `opsman.restricted_control`, `opsman.full_view` and `opsman.restricted_view` are mapped to UAA groups.
 - (Optional) Enter the Username Claim to populate the `username` field in UAA. Tanzu Operations Manager uses this to show the current logged-in user.
7. In OIDC Admin Group Name, enter the OIDC provider group name that corresponds to users who receive admin access. Users in this OIDC group are granted the `opsman.admin` scope in UAA.

SSL certificate

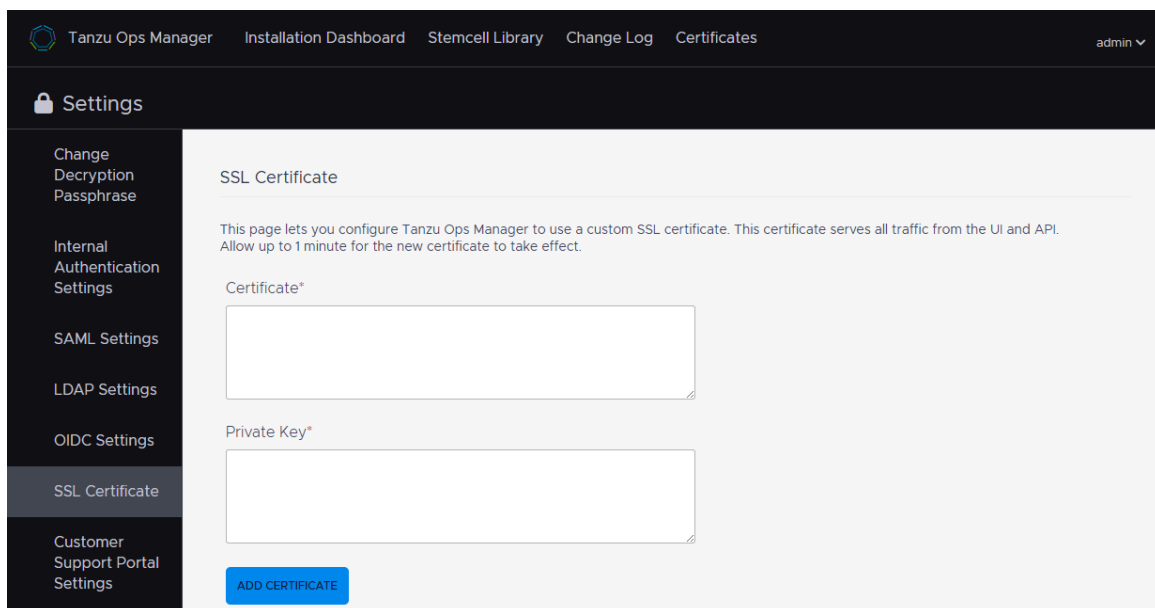
You can use the SSL Certificate pane to configure Tanzu Operations Manager to use a custom SSL certificate for all Tanzu Operations Manager traffic both through the UI and API.

This pane includes the following text boxes:

- Certificate: Enter a custom certificate.
- Private Key: Enter the private key for the certificate.

If you leave the text boxes blank, Tanzu Operations Manager uses an auto-generated self-signed certificate rather than your custom certificate and private key.

1. Click Add Certificate.



The screenshot shows the 'Settings' page in the Tanzu Ops Manager UI. The left sidebar contains a list of settings categories: Change Decryption Passphrase, Internal Authentication Settings, SAML Settings, LDAP Settings, OIDC Settings, SSL Certificate (which is highlighted), and Customer Support Portal Settings. The main content area is titled 'SSL Certificate' and includes a description: 'This page lets you configure Tanzu Ops Manager to use a custom SSL certificate. This certificate serves all traffic from the UI and API. Allow up to 1 minute for the new certificate to take effect.' Below the description are two text input fields: 'Certificate*' and 'Private Key*'. At the bottom of the form is a blue button labeled 'ADD CERTIFICATE'.

2. If you have previously added a custom certificate and want to replace it with a self-signed certificate, click Revert to self-signed certificate.

Broadcom Support Portal settings

Enter your [Tanzu API Token](#) from the Broadcom Support Portal and click Add Token to connect your Installation Dashboard to Broadcom Support Portal.

Proxy settings

If you are using a proxy to connect to Tanzu Operations Manager, update your Proxy Settings. Provide one of the following:

- HTTP proxy
- HTTPS proxy
- No proxy.

Custom banner

Create a custom text banner to communicate important messages to operators. For UI Banner, enter the text you want to show on each page of the Tanzu Operations Manager UI. For SSH Banner, enter the text that appears when an operator logs in to Tanzu Operations Manager.

Export installation settings

This feature exports the current installation with all of its assets. When you export an installation, the exported file contains references to the base VM images, necessary packages, and configuration settings.

Syslog

Viewable by administrators only. Configure a custom syslog server for Tanzu Operations Manager. When you click Yes and fill the following text boxes, Tanzu Operations Manager produces and sends all syslog entries from the Tanzu Operations Manager VM to the configured syslog endpoint. Tanzu Operations Manager also sends BOSH Director access events to the syslog endpoint.

Tanzu Operations Manager syslog entries are sent in RFC 3164 format.

To configure a custom syslog endpoint for Tanzu Operations Manager logs:

1. Select Syslog.
2. (Optional) Select Yes to send Tanzu Operations Manager system logs to a remote server.
3. In Address, enter the IP address or DNS name for the remote server.
4. In Port, enter the port number that the remote server listens on.
5. In the Transport Protocol drop-down menu, select TCP or UDP. This selection determines which transport protocol is used to send the logs to the remote server.
6. (Optional) Select the Enable TLS check box to send encrypted logs to remote server with TLS. After you select the check box:
 1. In Permitted Peer, enter either the name or SHA1 fingerprint of the remote peer.
 2. In SSL Certificate, enter the SSL certificate for the remote server.

VMware strongly recommends that you enable TLS encryption when you are forwarding logs. Logs can contain sensitive information, such as cloud provider credentials.

7. (Optional) In Queue Size, enter an integer. This value specifies the number of log messages held in the buffer. The default value is 100,000.
8. (Optional) Select the Forward Debug Logs check box to forward the logs to an external source. This option is deselected by default. If you select it, you may generate a large amount of log data.
9. (Optional) In Custom rsyslog Configuration, enter the configuration details for rsyslog. This text box requires the `rainerscript` syntax.
10. Click Save.

Pruning settings

Pruning settings are available in Tanzu Operations Manager 3.0.18+LTS-T and later.

You configure the pruning of Change Logs that are created each time you Apply Changes.

If pruning is activated, Change Logs that are older than the configured date are pruned once per day. The most recent Change Log and the most recent successful Change Log for a tile are not pruned even if they are older than the configured date.

Advanced options

The following settings can be applied for the account:

- Disable Foundation Core UI for Ops Manager: Turns off the ability to switch to the new Beta UI in Tanzu Hub. This setting affects all users of this instance.
- Download Activity Data: Downloads a directory that contains the configuration file for the installation, the deployment history, and version information.

- **Download Root CA Cert:** Downloads the root CA certificate of your deployment as an alternative to using cURL to access the Tanzu Operations Manager API.
- **Download Core Consumption Data:** Downloads the historical core consumption as CSV, instead of using cURL to access the Tanzu Operations Manager API. The CSV download contains an hourly reading of the chargeable cores that are consumed by each product, in the following form:

```
timestamp,product_identifier,physical_cores,logical_cores
```

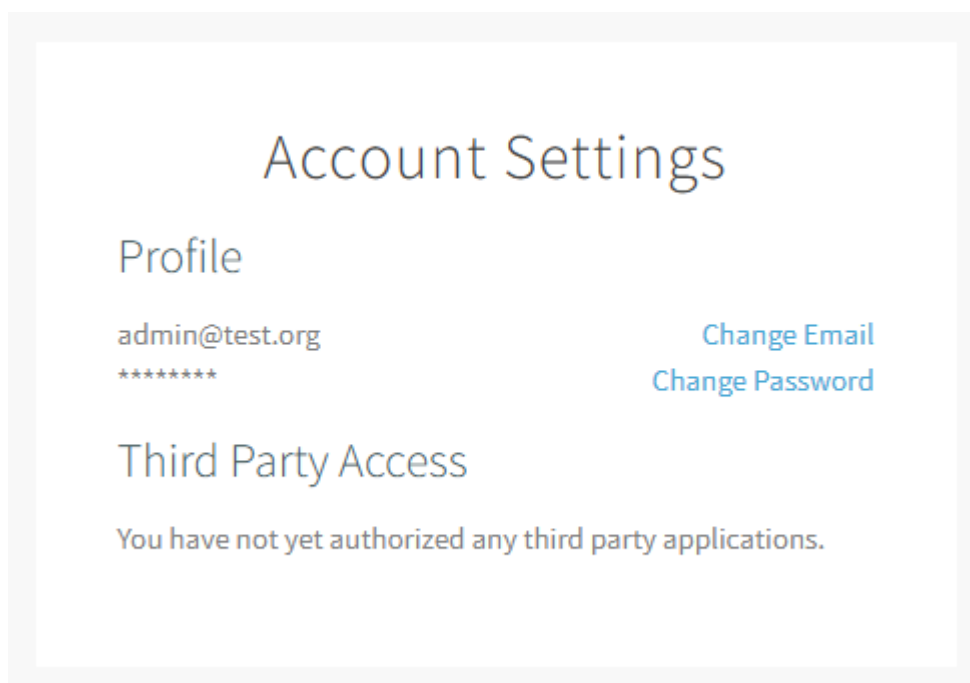
Products that do not support core counting do not appear in the CSV download.

- **View Diagnostic Report:** Displays various types of information about the configuration of your deployment.
- **Delete This Installation:** Deletes your Tanzu Operations Manager installation permanently.

My Account page

To change your email and password, go to the My Account page by clicking your user name, located at the upper right corner of the screen, and selecting My Account. The Account Settings page includes the following elements:

- **Profile:** The current email address and obscured password are shown.
- **Third Party Access:** Any third-party applications authorized for use are listed here.
- **Change Email button**
- **Change Password button**



Support page

Click Support in the footer of the Installation Dashboard in Tanzu Operations Manager to access the Support page.

The Support page allows you to download a ZIP file that includes Tanzu Operations Manager logs, deployed manifests and configurations, and BOSH deployment diagnostics.

The contents of the ZIP file help Support quickly address any issues in your deployment.

You can also access [Broadcom Support](#) to search for Knowledge base articles and other documentation.

Reviewing your pending product changes in Tanzu Operations Manager

This topic describes the Review Pending Changes page and how you can selectively deploy individual or multiple tiles in Tanzu Operations Manager. Deploying tiles selectively can dramatically reduce deployment time.

Review Pending Changes page

The following image shows the Review Pending Changes page. The features called out with green numbers are described in the following sections.

The screenshot displays the 'Review Pending Changes' page in the Tanzu Operations Manager interface. The page features a navigation bar at the top with links to 'Tanzu Ops Manager', 'Installation Dashboard', 'Stemcell Library', 'Change Log', and 'Certificates'. Below the navigation bar, the page title 'Review Pending Changes' is shown. The main content area contains a list of pending product changes for four tiles. Each tile entry includes a checkbox, a status indicator (Staged or Incomplete), and an 'APPLY CHANGES' button. The 'Small Footprint Tanzu Platform for Cloud Foundry' tile is highlighted with a green border, and the 'Tanzu for MySQL on Cloud Foundry' tile is highlighted with a red border.

| Tile Name | Version | Status | Dependencies | Changes |
|--|---------------------------|------------|--|--|
| BOSH Director for GCP | Version 3.0.41-build.1900 | Staged | No Dependencies | |
| Small Footprint Tanzu Platform for Cloud Foundry | Version 10.0.4 | Staged | No Dependencies | Update product to 10.0.4 Update stemcell to ubuntu-jammy 1.808 |
| Telemetry | Version 2.2.0-build.3 | Staged | Small Footprint Tanzu Platform for Cloud Foundry (cf) >=2.7 (optional) | |
| Tanzu for MySQL on Cloud Foundry | Version 10.0.0 | Incomplete | Small Footprint Tanzu Platform for Cloud Foundry (cf) >= 2.11.0 | Add stemcell ubuntu-jammy 1.808 |

Each labeled section of the Review Pending Changes page works as follows:

- **Select All Products:** When activated, the check box selects all available tiles. When deactivated, the check box deselects all available tiles.
- **Select Product:** The check boxes next to each tile listing allow you to select the tiles you want to deploy. The BOSH Director is always selected.
- **Tile Listing:** Each tile listing displays the version of the tile staged for deployment. The top of each listing shows one of the following three colors:
 - Green - The tile is configured and ready for deployment.
 - Orange - The tile is not fully configured and cannot be deployed yet.
 - Red - The tile is pending deletion.
- **Depends on:** This section lists tile dependencies and versions. If you choose to deploy a tile that has dependencies, you must also deploy all dependencies for that tile in the same deployment. Dependencies are shown in the following colors:
 - Green - The correct version of the dependency is installed.
 - Red - An incompatible version of the dependency is installed. If the red text includes "(optional)," you can still deploy the tile.
 - Gray - The dependency is not installed, but it is optional.
- **Warnings:** Red text in orange tile listings indicates changes you must make before you deploy the tile. You might see some of the following warnings:
 - Missing stemcell(s) - You must import and stage one or more stemcells in the Stemcell Library. The warning includes a link to the Stemcell Library page.
 - Stemcell(s) out of date - You must update one or more stemcells for the tile in the Stemcell Library. For more information, see [Importing and Managing Stemcells](#).
 - Configuration is incomplete - You must finish configuring the tile.
 - Configuration is invalid - You must resolve invalid configurations within the tile.
- **Changes:** This section lists high-level pending changes to the tile that Tanzu Operations Manager has not yet attempted to deploy. When Tanzu Operations Manager tries to deploy a tile but the deployment fails, it clears this list, and the pending changes are retained for the next deployment. To see if a tile has successfully deployed, see [Change Log Page](#) in *Using the Tanzu Operations Manager Interface*.
Listed changes include:
 - Tile deletion is staged.
 - Tile version is updated.
 - Stemcell is added.
 - Stemcell is updated.
- **Errands:** If you expand this section, you can activate or deactivate errands that trigger when that tile deploys. For example, you may activate an errand to log properties. For more information, see [Managing Errands in Tanzu Operations Manager](#).
- **Staged:** This section indicates the status of the tile.
- **See Changes:** When you click this button, it takes you to the Pending Changes page for that tile. For more information, see [About Pending Changes for a Specific Tile](#).
- **Apply Changes:** Clicking this button deploys all selected tiles.

- **Rotate All Non-Configurable Leaf Certificates:** When this check box is activated, all non-configurable leaf certificates are rotated as the first step of re-deployment. The page displays when the next non-configurable leaf certificate is to expire, allowing you to decide whether to activate the check box without needing to look at the Certificates page.

When you activate the Rotate All Non-Configurable Leaf Certificates check box, you must redeploy all tiles to ensure that all VMs receive the latest version of the certificates. You must also activate the Upgrade All Service instances errand for any tiles that have that errand. This errand re-deploys VMs associated with service instances, ensuring that they also receive the latest version of the certificates. For more information, see [Rotating Non-Configurable Leaf Certificates](#).

About pending changes for a specific tile

If you have changed the configuration of a tile in your deployment, use the Pending Changes page to review the changes to the existing manifest or configurations line-by-line. The Pending Changes page summarizes staged changes to the configuration of the existing tile. The changes displayed on this page have not yet been applied to the tile.

This allows you to verify changes to the manifest and configurations, and approve any staged changes before re-deploying tiles.

Using the Pending Changes page

The change summary fields display features that have been activated or deactivated through the Tanzu Operations Manager Installation Dashboard or API. These changes are staged, but have not yet been implemented. You must click Apply Changes to put these changes into effect.

To see changes for a specific tile:

1. On the Review Pending Changes page, click See Changes on the tile you want to review. The Pending Changes tab opens.
2. Review the changes that appear. Settings that have been *activated* are displayed in green with a + (plus) sign. Settings that have been *deactivated* are displayed in red with a - (minus) sign.
3. Click the Return to Review Pending Changes link to apply the changes or review other tiles.

When you make changes to a tile, change summaries are displayed for the following tile areas:

- Manifest
- Runtime configurations
- CPI configurations
- Cloud configurations

If an aspect of the tile has not changed, the change summary displays “No changes.”

Changes to errands are not displayed in the UI.

Adding and deleting products from Tanzu Operations Manager

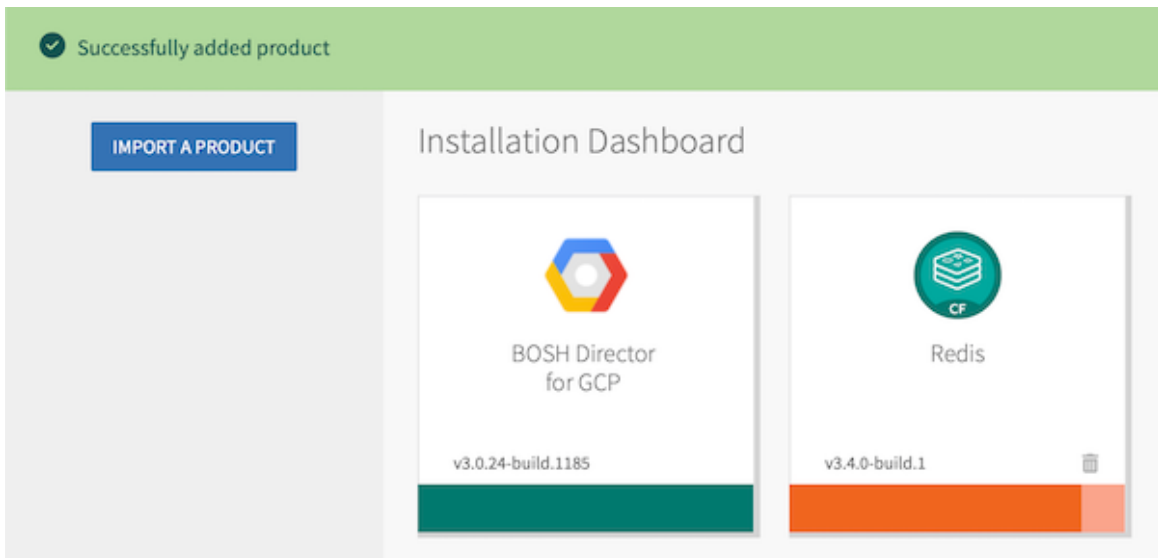
This topic describes how you can add and delete additional products, such as [Tanzu RabbitMQ](#), from Tanzu Operations Manager.

In Tanzu Operations Manager, all product tiles use floating stemcells by default. This increases the security of your deployment by enabling tiles to automatically use the latest patched version of a stemcell, but it might significantly increase the amount of time required by a tile upgrade. For more information, see [Floating Stemcells](#).

Add and import products

To add or import a product to your Tanzu Operations Manager installation:

1. Download products from [Broadcom Support portal](#). If you cannot download products from Broadcom Support portal due to restricted network connectivity, see [Installing Tanzu Operations Manager in Air-Gapped Environments](#).
2. Go to the Tanzu Operations Manager Installation Dashboard and log in.
3. Click Import a Product.
4. Select the `.pivotal` file that you downloaded from Broadcom Support portal or received from your software distributor, and click Open. If the product is successfully added, it appears in the product list beneath the Import a Product button. If the product that you selected is not the latest version, the most up-to-date version appears in your product list.
5. Add the product tile to the Tanzu Operations Manager Installation Dashboard by clicking the green + (plus) icon in the bottom-right corner of the product listing.
6. The product tile appears in the Tanzu Operations Manager Installation Dashboard. If the product requires configuration, the bar at the bottom of the tile appears orange. If the bottom bar is orange, configure the product.



7. (Optional) In the product configuration view, select the Errands pane to configure post-install errands or review the default settings. Post-install errands are scripts that automatically run after a product installs, before Tanzu Operations Manager makes the product available for use. For more information about post-install errands, see [Using BOSH errands with your tile](#) in the Tanzu Operations Manager Tile Developer Guide.

By default, Tanzu Operations Manager reruns errands even if they are not necessary, due to settings left from a previous install. Leaving errands checked at all times can cause updates and other processes to take longer. To prevent an errand from running, select the Off option from the drop-down menu for the errand in the Errands pane of the Settings tab in the product tile before installing the product.

Errands

Errands are scripts that run at designated points during an installation.

Post-Deploy Errands

Register On-demand MySQL Broker

Default (On) ▼

Smoke Tests

Default (On) ▼

Validate no IP-based bindings in use before upgrade-all-service-instances

Default (On) ▼

Upgrade all On-demand MySQL Service Instances

Default (On) ▼

Recreate all On-demand MySQL Service Instances

Default (Off) ▼

Pre-Delete Errands

Delete All Service Instances and Deregister Broker

Default (On) ▼

SAVE

The Broker Registrar drop-down menu is an example of an errand available for a product. When you select On, this errand registers service brokers with the Cloud Controller and updates any broker URL and credential values that have changed since the previous registration.

- Click Save.
- Click Review Pending Changes.
- Click Apply Changes to start the installation and run post-install lifecycle errands for the product.

Using the Broadcom Support portal API to upgrade products

Rather than manually downloading each new version of the products in your Tanzu Operations Manager installation, you can upgrade your products by connecting your Tanzu Operations Manager Installation

Dashboard to Broadcom Support portal using an API token. After you have uploaded a product, all subsequent product upgrades appear automatically in the Tanzu Operations Manager Installation Dashboard.

You can use the Broadcom Support portal API only if you have access to the Internet, because communication between Tanzu Operations Manager and Broadcom Support portal is necessary to import your products. If you are on an isolated network, do not save your API token.

Connect the Tanzu Operations Manager Installation Dashboard with Broadcom Support portal

To connect the Tanzu Operations Manager Installation Dashboard with Broadcom Support portal:

1. Log in to [Broadcom Support portal](#).
2. Click your user name.
3. From the drop-down menu, click My Profile.
4. Copy the Legacy API Token (Deprecated).
5. Go to the Tanzu Operations Manager Installation Dashboard and log in.
6. Click your user name, located in the upper top right side of the page.
7. From the drop-down menu, click Settings, then External API Access.
8. In the Set API token text box, paste the Legacy API Token you copied from Broadcom Support portal.
9. Click Save.

Update existing products

After you save the Broadcom Support portal API token to the Tanzu Operations Manager Installation Dashboard, the latest versions of your existing products appear in the Tanzu Operations Manager Installation Dashboard.

To upgrade an existing product to the latest version:

1. In the product list of the Tanzu Operations Manager Installation Dashboard, locate the product you want to upgrade.
2. Click the green download icon.
3. When the download is complete, refresh the page to use the product.
4. If necessary, configure the product.
5. Click Review Pending Changes.
6. Click Apply Changes.

Apply changes to BOSH Director

You can use the Tanzu Operations Manager API or the Review Pending Changes page in the Tanzu Operations Manager Installation Dashboard to apply pending changes only to the BOSH Director when you stage multiple products in a new Tanzu Operations Manager installation or as part of an upgrade.

For more information, see [Reviewing Pending Product Changes](#).

Delete products

To delete a product from your Tanzu Operations Manager installation:

From the Installation Dashboard:

1. Click the trash icon on a product tile to remove that product.
2. In the Delete Product pane that appears, click Confirm.

You cannot delete the BOSH Director product.

3. Click Review Pending Changes, then Apply Changes.

After you delete a product, the product tile is removed from the installation, and the Installation Dashboard, however, the product appears in the Available Products view.

Importing and managing stemcells in Tanzu Operations Manager

This topic shows you how to use the Stemcell Library in Tanzu Operations Manager to import and stage stemcells to products. For conceptual information about floating stemcells and stemcell upgrades, see [Floating Stemcells](#).

Some product releases require Xenial stemcells. If you are importing and using Xenial stemcells for the first time, see the [Release Notes](#). See also the [Stemcell release notes](#).

Import and stage a stemcell

To import and stage a stemcell:

1. Download the appropriate .tgz stemcell file from the [Broadcom Support portal](#).

Tanzu Operations Manager does not support heavy or full stemcells for GCP. For GCP, you must use a light stemcell. For more information about light stemcells, see [Light Stemcells](#) in the BOSH documentation.
2. Click Import Stemcell to permanently import a stemcell into Tanzu Operations Manager. The Import Stemcell dialog box appears.
3. Select the products you want to stage.
4. Click Apply Stemcell to Products.

Choose a stemcell version

If you uploaded multiple versions of a stemcell, you can use the drop-down menu in the Staged column to choose which version to use.

Tanzu Ops Manager
Installation Dashboard
Stemcell Library
Change Log
Certificates
admin

Stemcell Library

IMPORT STEMCELL
SAVE

| Product | Deployed | Required | Staged |
|--|--------------------|-----------------------|---|
| BOSH Director for GCP
Version 3.0.41-build.1900 | ubuntu-jammy 1.808 | >= ubuntu-jammy 1.808 | <div>ubuntu-jammy 1.808</div> <div>ubuntu-jammy 1.808</div> |
| Telemetry
Version 2.2.0-build.3 | ubuntu-jammy 1.808 | >= ubuntu-jammy 1 | <div>ubuntu-jammy 1.808</div> <div>Latest stemcell</div> |
| Small Footprint Tanzu Platform for Cloud Foundry
Version 10.0.4 | ubuntu-jammy 1.808 | >= ubuntu-jammy 1.8 | <div>ubuntu-jammy 1.808</div> <div>Latest stemcell</div> |
| Tanzu for MySQL on Cloud Foundry
Version 10.0.0 | - | >= ubuntu-jammy 1 | <div>ubuntu-jammy 1.808</div> <div>Latest stemcell</div> |

If the stemcell is displayed with a green checkmark and the words Latest stemcell below the Staged drop-down menu, the stemcell is the latest available version on your host. An outdated stemcell displays Stemcell out-of-date.

When you upgrade Tanzu Operations Manager, any older stemcell versions that were not in use by a product are removed from the Stemcell Library. If a stemcell was in use by a product prior to upgrading, it will continue to be available after the upgrade. If you want to use a stemcell that was removed as part of the upgrade, you must upload it again.

Managing errands in your Tanzu Operations Manager deployment

Errands are scripts that can run at the beginning and at the end of an installed product's availability time.

This topic describes the product errands used in Tanzu Operations Manager, and how you can use Tanzu Operations Manager to adjust whether and when these errands run.

Product tiles include two types of errands:

- Post-deploy errands run after the product installs but before Tanzu Operations Manager makes the product available for use. One example is an errand that publishes a newly-installed service to the Services Marketplace.
- Pre-delete errands run after an operator chooses to delete the product but before Tanzu Operations Manager actually deletes it. One example is a clean-up task that removes all data objects used by the errand.

When you click Review Pending Changes, then Apply Changes in Tanzu Operations Manager, BOSH either creates a VM for each errand that runs or co-locates errands on existing VMs. Tile developers determine where BOSH deploys the errands for their product.

Tanzu Platform for Cloud Foundry provides several post-deploy errands including smoke tests, Apps Manager, notification, Tanzu Operations Manager Account, and autoscaling errands. For more information about Elastic Application Runtime errands, see the [Elastic Application Runtime documentation](#).

For information about the errands associated with any other Tanzu Operations Manager product, see the documentation provided with the product tile.

Errand run rules

You can configure two different run rules for errands: On and Off. These rules control when Tanzu Operations Manager executes the errand.

When the errand is configured to be On, then it always runs, even if there are no changes to the product manifest. When the errand is configured to be Off, it never runs.

By default, Tanzu Operations Manager applies the On rule to all errands.

For any errand, you can override these Tanzu Operations Manager defaults with your own tile-specific defaults defined in the tile [property blueprints](#).

Configure errand run rules in Tanzu Operations Manager

You can configure the run rules for errands in two places in Tanzu Operations Manager. The [Errands](#) pane saves your configuration and applies the configuration to future installations. The [Pending Changes](#) view applies the rules only to the next time you run a Tanzu Operations Manager install, without saving them.

Errands pane: persistent rules

Product tiles for Elastic Application Runtime and other Tanzu Operations Manager products have an Errands pane that configures the run rules for the product's errands and saves the settings for later.

The Errands pane lists all errands for the product and lets you select On or Off. The Default option differs depending on the errand, and reflects the default setting used by Tanzu Operations Manager for the errand or any tile-specific default that overrides it.

Errands

Errands are scripts that run at designated points during an installation.

Post-Deploy Errands

Smoke Test Errand

Default (On) ▼

Runs Smoke Tests against your installation.

Usage Service Errand

Default (On) ▼

Pushes the Usage Service application to your installation. Apps Manager depends on this application.

Offline Docs Errand

Default (Off) ▼

Pushes the Offline Docs application to your installation.

Apps Manager Errand

Default (On) ▼

Pushes the Apps Manager application to your installation.

Notifications Errand

Default (On) ▼

Pushes the Notifications application to your installation.

Notifications UI Errand

Default (On) ▼

Pushes the Notifications UI component to your installation.

App Autoscaler Errand

Default (On) ▼

Pushes the App Autoscaler application to your installation.

App Autoscaler Smoke Test Errand

Default (On) ▼

Runs Smoke Tests against the App Autoscaling Service.

Metric Registrar Smoke Test Errand

Default (Off) ▼

Runs Smoke Tests against the Metric Registrar. Requires that "Enable Metric Registrar" is checked in the Metric Registrar tab.

NFS Broker Errand

Default (On) ▼

Pushes the NFS Broker application to your installation.

SMB Broker Errand

Default (On) ▼

Pushes the SMB Broker application to your installation.

Rotate CC Database Key

Default (Off) ▼

Re-encrypts credentials in Cloud Controller DB with selected encryption key.

There are no pre-delete errands for this product.

SAVE

To configure run rules for a tile:

1. In Tanzu Operations Manager, click the tile to open it.
2. Under the Settings tab, open the Errands pane.

3. Use the drop-down menus to configure the run rule choice for each errand: On or Off.
4. Click Save to save the configuration values and return to the Installation Dashboard.
5. Click Review Pending Changes, then Apply Changes to redeploy the tile with the new settings.

Review Pending Changes: one-time rules

Tanzu Operations Manager lets you quickly configure one-time errand run rules for any product queued up for installation:

1. In Tanzu Operations Manager, go to the Review Pending Changes section that shows products that Tanzu Operations Manager has yet to install or update.
2. Expand the ERRANDS link, for the product you want to configure. A list of errands associated with the product appears.
3. Select any errands you want to run during the Apply Changes action. Tanzu Operations Manager uses these settings when you Apply Changes to install the product, but does not save the settings for future installations.
4. Click Apply Changes to redeploy the tile.

Related links

If you are a product developer and want to learn more about adding errands to your product tile for Tanzu Operations Manager, see [Errands](#) in *Tanzu Operations Manager Tile Developer Guide*.

Using the API or CLI

This guide includes the following topics:

- [Using Tanzu Operations Manager programmatically and from the command line](#)
- [Using the Tanzu Operations Manager API](#)
- [Using the Tanzu Operations Manager CLI](#)

Using Tanzu Operations Manager programmatically and from the command line

You can do Tanzu Operations Manager operations programmatically and from the command line, rather than from the Tanzu Operations Manager UI.

Tanzu Operations Manager includes three related tools for performing operations programmatically and from the command line:

- The Tanzu Operations Manager API. For more information, see the [Tanzu Operations Manager API](#) documentation.
- The `om` command-line interface (CLI). For more information about `om`, see the [om repository](#).
- Platform Automation tasks. For more information, see the [Platform Automation documentation](#).

These three tools control the Tanzu Operations Manager VM at different levels of abstraction. For more information about tile development, see the [Tanzu Operations Manager Tile Developer Guide](#).

The Tanzu Operations Manager API

The Tanzu Operations Manager API offers the most granular control of the Tanzu Operations Manager VM. The API enables users to do the following:

- Automate component-level testing of Tanzu Operations Manager product tiles
- Manage Tanzu Operations Manager user accounts
- Develop and debug product tiles
- Access product property values that are not exposed in a product tile GUI
- Control the Tanzu Operations Manager VM directly, programmatically or manually, bypassing the Tanzu Operations Manager UI.

For information about getting started with the Tanzu Operations Manager API, see [Access the Tanzu Operations Manager API](#) in *Using the Tanzu Operations Manager API*.

Your Tanzu Operations Manager serves a local copy of the complete Tanzu Operations Manager API documentation at <https://YOUR-OPS-MANAGER-FQDN/docs>.

The Tanzu Operations Manager CLI (om)

The Tanzu Operations Manager CLI, **om**, calls the Tanzu Operations Manager API to do the following:

- Perform Tanzu Operations Manager operations from the command line or scripts
- Manage credentials from the command line or scripts
- Generate configuration files from, and apply saved configurations to, staged product tiles.

For information, including how to install **om** and a usage example, see [Using the Tanzu Operations Manager CLI](#).

Platform Automation

Platform Automation tasks call the **om** CLI to do the following:

- Serve as building blocks for Continuous Integration (CI) pipelines that automate installation and updating of Tanzu Operations Manager foundations
- Within pipelines, store and retrieve product configuration values from secrets vaults like CredHub and code repositories like GitHub.

For a complete list of Platform Automation tasks, see [Task Reference](#) in the Platform Automation documentation.

Examples of equivalent operations

The following table lists some common operations using the Tanzu Operations Manager UI, Tanzu Operations Manager API, **om** CLI, and Platform Automation tasks, as examples of how different Tanzu Operations Manager interfaces relate to one another.

For complete lists of operations, consult the references in the previous sections.

| UI action | API command | om Command | Platform automation task |
|---|---|------------------------------|---------------------------|
| Manually configure the BOSH Director tile | Multiple commands; see Configure the BOSH Director in <i>Using the Tanzu Operations Manager API</i> . | om configure-director | configure-director |

| UI action | API command | om Command | Platform automation task |
|---|---|-----------------------------------|--------------------------------|
| Click IMPORT A PRODUCT | <code>POST</code> product file to <code>/available_products</code> | <code>om upload-product</code> | <code>upload-product</code> |
| Click the + symbol next to a product listing left of Installation Dashboard | <code>POST</code> product name and version to <code>/staged/products</code> | <code>om stage-product</code> | <code>stage-product</code> |
| Manually configure a product tile | Multiple commands; see Configure a Product in Using the Tanzu Operations Manager API . | <code>om configure-product</code> | <code>configure-product</code> |
| Click STEMCELL LIBRARY, then IMPORT STEMCELL | <code>POST</code> stemcell file to <code>/stemcells</code> | <code>om upload-stemcell</code> | <code>upload-stemcell</code> |
| Click Apply Changes | <code>POST</code> specified products (or <code>all</code>) and errand lists to <code>/installations</code> | <code>om apply-changes</code> | <code>apply-changes</code> |

Using Tanzu Operations Manager API

This topic explains how you can access the Tanzu Operations Manager API, and gives examples for using the API to perform common operations.

The Tanzu Operations Manager API controls the Tanzu Operations Manager VM directly, bypassing the Tanzu Operations Manager UI. Platform operators use the Tanzu Operations Manager API to automate deployments, retrieve and manage credentials, and otherwise work with Tanzu Operations Manager. Tile Developers use the Tanzu Operations Manager API to test and debug Tanzu Operations Manager product tiles.

For more information about the Tanzu Operations Manager API, see the [Tanzu Operations Manager API](#) documentation. Your Tanzu Operations Manager serves a local copy of this documentation at <https://YOUR-OPS-MANAGER-FQDN/docs>.

Related tools

For running Tanzu Operations Manager operations from the command line or within shell scripts, the Tanzu Operations Manager command-line interface (CLI) `om` is usually a better choice than the Tanzu Operations Manager API.

For more information about `om`, see the [om repository](#).

For an overview of the Tanzu Operations Manager API and the tools based on it, see [Using Tanzu Operations Manager Programmatically and from the Command Line](#).

Access the Tanzu Operations Manager API

To access the Tanzu Operations Manager API, you must authenticate to the Tanzu Operations Manager User Account and Authentication (UAA) server and log in as described below.

For more information about UAA, see [User Accounts and Authentication overview](#).

Install the UAAC

If you have not already, install the UAA Command Line Interface (UAAC) by running the following command from a terminal window:

```
gem install cf-uaac
```

Log in to Tanzu Operations Manager

To log in to the Tanzu Operations Manager VM and retrieve your token:

1. Use the UAAC to target your Tanzu Operations Manager UAA server:

```
uaac target https://OPS-MAN-FQDN/uaa
```

Where **OPS-MANAGER-FQDN** is the fully qualified domain name of Tanzu Operations Manager.

For example:

```
$ uaac target https://my-opsmanager.example.com/uaa
```

2. Run a command to retrieve your UAA token. See the following options for the appropriate command to run, and the responses to provide for the authentication prompts.

- If Tanzu Operations Manager UAA uses an internal or LDAP identity provider, run:

```
uaac token owner get
Client ID: opsman
Client secret: [Leave Blank]
User name: OPS-MAN-USERNAME
Password: OPS-MAN-PASSWORD
```

Where **OPS-MAN-USERNAME** and **OPS-MAN-PASSWORD** are the credentials that you use to log in to the Tanzu Operations Manager web interface.

For example:

```
$ uaac token owner get
Client ID: opsman
Client secret:
User name: my-opsman-username
Password: my-opsman-password
```

- If Tanzu Operations Manager UAA uses a SAML or OIDC identity provider, run:

```
uaac token sso get
Client ID: opsman
Client secret: [Leave Blank]
Passcode: OPS-MAN-PASSCODE
```

Where **OPS-MAN-PASSCODE** is the value you retrieve from <https://OPS-MAN-FQDN/uaa/passcode>.

For example:

```
$ uaac token sso get
Client ID: opsman
Client secret:
Password: my-opsman-passcode
```

If authentication is successful, the UAAC displays the following message: **Successfully fetched token via owner password grant**.

Run API commands

To run any Tanzu Operations Manager API command, you pass your authorization token to the API command endpoint in a header that follows the format **Authorization: Bearer YOUR-ACCESS-TOKEN**.

If you are calling Tanzu Operations Manager API commands from the command line, the machine that you must be logged into depends on the location of your Tanzu Operations Manager UAA, as follows:

- Internal: You must run Tanzu Operations Manager API commands from the Tanzu Operations Manager VM.
- External: You can run commands from your local machine.

API access test

The following procedure tests whether you can access the Tanzu Operations Manager API from the command line by retrieving a list of deployed products:

1. List your tokens. Run:

```
uaac contexts
```

In the command output, locate the entry for your Tanzu Operations Manager FQDN. From **client_id: opsman**, record the value for **access_token**.

2. Use the **GET /api/v0/deployed/products** endpoint to retrieve a list of deployed products:

```
curl "https://OPS-MAN-FQDN/api/v0/deployed/products" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where **UAA-ACCESS-TOKEN** is the access token recorded in the previous step.

The command output looks like the following:

```
$ curl "https://my-opsmanager.example.com/api/v0/deployed/products" \  
-X GET \  
-H "Authorization: Bearer my-access-token" \  
[{"installation_name":"p-bosh","guid":"p-bosh-00000000000000000000","type":"p-bosh","product_version":"1.10.0.0"}, {"installation_name":"cf-00000000000000000000","guid":"cf-00000000000000000000","type":"cf","product_version":"1.10.0"}]
```

Example Tanzu Operations Manager API procedures

The following procedures illustrate how to test API calls from the command line. If you are a developer writing deployment or test routines in programming languages, these procedures show how you might test the routines line-by-line while building them.

The Tanzu Operations Manager command-line interface (CLI) **om** is usually a better tool than the Tanzu Operations Manager API for running Tanzu Operations Manager operations directly from the command-line or within shell scripts.

These methods are intended for advanced Tanzu Operations Manager operators and administrators.

Configure the BOSH Director

The following procedure configures the BOSH Director for your IaaS using the Tanzu Operations Manager API.

VMware recommends that beginning users configure the BOSH Director through the Tanzu Operations Manager UI, which provides context and explanations for each option. For information about configuring the BOSH Director tile in the Tanzu Operations Manager UI, see one of the following:

- [AWS](#)
- [Azure](#)
- [GCP](#)
- [vSphere](#)

To configure the BOSH Director with the Tanzu Operations Manager API:

1. Access the Tanzu Operations Manager API by following the procedure in [Access the Tanzu Operations Manager API](#), earlier on this page.
2. To perform initial setup, enter a **POST** request to the Ops Man API **setup** endpoint, passing metadata to the **-d** flag:

```
curl "https://YOUR-OPS-MANAGER-URL.com/api/v0/setup" \
-X POST \
-H "Content-Type: application/json" \
-d '{ "setup": {
  "decryption_passphrase": "PASSPHRASE",
  "decryption_passphrase_confirmation": "PASSPHRASE",
  "identity_provider": "IDP-LOCATION",
  "admin_user_name": "ADMIN-USERNAME",
  "admin_password": "ADMIN-PASSWORD",
  "admin_password_confirmation": "ADMIN-PASSWORD",
  "http_proxy": HTTP-PROXY,
  "https_proxy": HTTPS-PROXY,
  "no_proxy": "127.0.0.1"
}}
```

Where:

- **PASSPHRASE** is your deployment's decryption passphrase
- **IDP-LOCATION** is your identity provider location, **internal** or **external**
- **ADMIN-USERNAME** is your Tanzu Operations Manager admin user name
- **ADMIN-PASSWORD** is your Tanzu Operations Manager admin password
- **HTTP-PROXY** is your HTTP proxy, if applicable
- **HTTPS-PROXY** is your HTTPS proxy, if applicable

For example:

```
$ curl "https://my-opsman.example.com/api/v0/setup" \
-X POST \
-H "Content-Type: application/json" \
-d '{ "setup": {
  "decryption_passphrase": "my-passphrase",
  "decryption_passphrase_confirmation": "my-passphrase",
  "identity_provider": "internal",
  "admin_user_name": "my-username",
  "admin_password": "my-password",
  "admin_password_confirmation": "my-password",
  "http_proxy": "http://myproxy.example.com",
  "https_proxy": "https://myproxy.example.com",
  "no_proxy": "127.0.0.1"
} }'
```

You do not need to specify UAA or other authentication details on a first-time deploy. Making the `setup` request automatically creates a UAA client when it is complete. If you specify a UAA client for a first-time deployment, the deployment will fail.

3. A **200 OK** response appears.
4. After the successful response, a UAA client with the metadata you specified is created and launched. Tanzu Operations Manager users can authenticate with UAAC, and if they have a pre-created client, they can target UAAC with the client name and password. For more information on pre-created clients, see [Creating and Managing Tanzu Operations Manager User and Client Accounts](#).
5. Send an HTTP **PUT** request to the `api/v0/staged/director/properties` endpoint to configure your IaaS and BOSH Director. There are many configuration parameters available to customize your BOSH Director. All the commands are IaaS-agnostic except the IaaS configuration key. Missing required fields can cause an error. For more information about configuring the BOSH Director, see [Fetching Director, IaaS, and Security Properties](#).
6. Send an HTTP **POST** request to the `api/v0/staged/director/availability_zones` endpoint to create Availability Zones (AZs) for your product.

The `create-azs` endpoint is optional for Azure-based deployments, because you cannot manually configure Azure AZs.

7. Send an HTTP **PUT** request to the `api/v0/staged/director/networks` endpoint to create networking rules for the deployment. Specify whether or not to use ICMP checks by setting the `icmp_checks_enabled` parameter to `true` or `false`.
8. Send an HTTP **PUT** request to the `api/v0/staged/director/network_and_az` endpoint to assign a singleton AZ and a network where your BOSH Director will be located.
9. You must update the BOSH Director's resource configuration settings before deploying BOSH. To update the resource config:

1. Send a **GET** request to `api/v0/staged/products` to find your Director's GUID.

1. Use the following HTTP request to list all jobs on the Director:

```
GET api/v0/staged/products/:BOSH-DIRECTOR-GUID/jobs
```

2. Use the following HTTP request to update the resource config of each job on the BOSH Director:


```
PUT api/v0/staged/products/:BOSH-DIRECTOR-GUID/jobs/:JOB-GUID/resource_config
```

Configure a product

Use the following procedure to upload, stage, and configure a product tile using the Tanzu Operations Manager API:

1. Upload or import the product into Tanzu Operations Manager.
 1. Send an HTTP **POST** request to the `api/v0/available_products` endpoint to upload the product.
 2. Send an HTTP **GET** request to the `api/v0/available_products` endpoint to list the names of available products.
 3. In the output list from `available_products`, find the product you uploaded by referencing its name and version.
 4. Send an HTTP **POST** request to the `/api/v0/staged/products` endpoint to add the uploaded product to Tanzu Operations Manager.
2. Send an HTTP **GET** request to the `api/v0/staged_products` endpoint to confirm that the product is staged for deployment in Tanzu Operations Manager, and display the product GUID.
3. Send an HTTP **PUT** request to the `PUT /api/v0/staged/products/:PRODUCT-GUID/networks_and_azs` to assign availability zones (AZs) and networks to the product, where `PRODUCT-GUID` is the product GUID.
4. Send an HTTP **PUT** request to the `/api/v0/staged/products/:PRODUCT-GUID/properties` endpoint to update the product's properties, where `PRODUCT-GUID` is the product GUID.
5. Send an HTTP **PUT** request to the `/api/v0/staged/products/:PRODUCT-GUID/syslog_configuration` endpoint to configure syslog for the product, where `PRODUCT-GUID` is the product GUID.
6. Update the product's resource configuration by doing the following:
 1. Send an HTTP **GET** request to the `api/v0/staged/products/:PRODUCT-GUID/jobs` endpoint to list all jobs on a product and display the GUID for each job, where `PRODUCT-GUID` is the product GUID.
 2. Send an HTTP **PUT** request to the `api/v0/staged/products/:PRODUCT-GUID/jobs/:JOB-GUID/resource_config` endpoint to update the resource config for a particular job, where:
 - `PRODUCT-GUID` is the product GUID.
 - `JOB-GUID` is the job GUID.

Upgrade Tanzu Operations Manager

The following procedure upgrades Tanzu Operations Manager using the Tanzu Operations Manager API.

Before you upgrade Tanzu Operations Manager with the Tanzu Operations Manager API, you must have the following:

- Two versions of Tanzu Operations Manager: an existing deployment and a new deployment of a later version. The new deployment cannot be configured in any way.
- DNS information for the existing deployment of Tanzu Operations Manager

Follow this procedure to upgrade your Tanzu Operations Manager:

1. Send an HTTP **GET** request to the `/api/v0/installation_asset_collection` endpoint of your old Tanzu Operations Manager to export your existing installation data.

2. Send an HTTP **POST** request to the `/api/v0/installation_asset_collection` endpoint of your new Tanzu Operations Manager to import your installation data to the new deployment.
3. Send an HTTP **POST** request to the `/api/v0/installations` endpoint of your new Tanzu Operations Manager to trigger a new installation process.
4. Test the new deployment.
5. (Optional) Delete the old deployment.

Using Tanzu Operations Manager CLI

This topic explains how to perform common operations with the Tanzu Operations Manager CLI, a tool that helps you configure and deploy tiles to Tanzu Operations Manager without using the Tanzu Operations Manager UI. The Tanzu Operations Manager CLI communicates with the Tanzu Operations Manager API.

In addition to using the om CLI directly, you can use it to manage Tanzu Operations Manager programmatically through platform automation. For information about platform automation, see the [Platform Automation Toolkit](#).

The Tanzu Operations Manager CLI has the following benefits over the Tanzu Operations Manager API:

- You can **om curl** Tanzu Operations Manager API calls with easier authentication
- om CLI commands are easier to use and memorize
- om CLI output can be easier to parse

For more information about using the Tanzu Operations Manager CLI, see [om](#) in GitHub.

Install the Tanzu Operations Manager CLI

You can install the Tanzu Operations Manager CLI in one of the following ways:

- Mac or Linux Users: [Install om CLI using Homebrew](#)
- Windows, Mac, or Linux Users: [Install om CLI from direct download](#)

Install the Tanzu Operations Manager CLI using Homebrew

If you are on a Mac or Linux machine, you can use Homebrew to install the Tanzu Operations Manager CLI. You must first have Homebrew installed to use this method.

To install the Tanzu Operations Manager CLI using Homebrew:

1. Download the Tanzu Operations Manager CLI.

```
brew tap pivotal-cf/om https://github.com/pivotal-cf/om
```

2. Install the om CLI.

```
brew install om
```

3. Test your installation.

```
om version
```

Install the Tanzu Operations Manager CLI from direct download

You can install the Tanzu Operations Manager CLI directly from the om CLI GitHub repo.

To install the Tanzu Operations Manager CLI from direct download:

1. Go to [om CLI releases](#) in GitHub and choose your release.
2. Download the release from Assets. Choose the asset that corresponds with your operating system. For example, download [om-windows-x.x.x.zip](#) for Windows machines.
3. Unzip your file, if necessary, and run the executable file.
4. Wait for your machine to indicate that the installation is complete.
5. Test your installation by opening a Shell window and running [om version](#).

Authenticate with the Tanzu Operations Manager CLI

For most om CLI commands, you must authenticate with Tanzu Operations Manager by putting your Tanzu Operations Manager login information into a YAML file.

To authenticate with the Tanzu Operations Manager CLI:

1. In your workspace, create a YAML file that represents your Tanzu Operations Manager environment. Give the file a name. For example:

```
touch om-cli-env.yml
```

2. Add the following content to the file. For more information about configuring the file, see [Generating an Env File](#) in the Platform Automation Toolkit documentation.

```
---
target: https://OPS-MANAGER-FQDN
username: OPS-MANAGER-USERNAME
password: OPS-MANAGER-PASSWORD
```

Where:

- [OPS-MANAGER-FQDN](#) is the FQDN of your Tanzu Operations Manager deployment.
 - [OPS-MANAGER-USERNAME](#) is the user name that you use when you log into the Tanzu Operations Manager UI.
 - [OPS-MANAGER-PASSWORD](#) is the password that you use when you log into the Tanzu Operations Manager UI.
3. On your machine, set an environment variable to the absolute filepath of your environment file for easy access.

```
export OM_ENV=PATH-TO-WORKING-DIRECTORY/om-cli-env.yml
```

4. Test authentication in your Tanzu Operations Manager environment by running the [om products](#) command.

```
om --env=$OM_ENV products
```

If this is successful, you will see product output that looks similar to this example:

| NAME | AVAILABLE | STAGED | DEPLOYED |
|----------------------|---------------|------------------|------------------|
| cf | 2.10.12 | 2.10.12 | 2.10.12 |
| p-bosh | | 2.10.8-build.168 | 2.10.8-build.168 |
| pivotal-telemetry-om | 1.1.2-build.2 | | |

Tanzu Operations Manager CLI commands

For a list of all Tanzu Operations Manager CLI commands, see [Commands](#) in the om CLI GitHub documentation.

Advanced features in Tanzu Operations Manager

This guide describes advanced features in Tanzu Operations Manager. You can use these features to complete functions that are outside of the scope of typical Tanzu Operations Manager behavior.

Misuse of any advanced feature could deactivate or destroy deployments. These features are for advanced operators only. VMware recommends contacting [Support](#) before using these features.

This guide includes the following topics:

- [Modifying Your Tanzu Operations Manager installation and product template files](#)
- [Managing Tanzu Operations Manager Verifiers](#)
- [Managing custom VM Extensions](#)
- [Tanzu Operations Manager fields that lock on deploy](#)
- [Overriding Manifest properties in Advanced Mode](#)
- [Property and template references](#)

Modifying installation and product template files

This topic describes how you can modify your Tanzu Operations Manager installation by decrypting and editing the YAML files that Tanzu Operations Manager uses to store configuration data.

You can use these procedures to view and change values that you cannot access through the Tanzu Operations Manager web interface. You can also modify the product templates that Tanzu Operations Manager uses to create forms and obtain user input.

You might want to modify the Tanzu Operations Manager installation and product template files for a number of reasons, including the following:

- To change the User Account and Authentication (UAA) admin password of the deployment
- To retrieve key values
- To migrate content across different Tanzu Operations Manager releases

Be careful when you make changes to your Tanzu Operations Manager installation and product template files. Use spaces instead of tabs, and remember that YAML files use

whitespace as a delimiter. Finally, VMware does not officially support these procedures, so use them at your own risk.

About installation and product template files

During the installation process, Tanzu Operations Manager combines information from the installation and product template files to generate the manifests that define your deployment.

- **Installation files:** Tanzu Operations Manager stores user-entered data and automatically-generated values for Tanzu Operations Manager in installation YAML files on the Tanzu Operations Manager VM. Tanzu Operations Manager encrypts and stores these files in the directory `/var/tempest/workspaces/default`. To make changes, you must decrypt the files, edit them as necessary, and then re-encrypt them.
- **Product templates:** Tanzu Operations Manager uses product templates to create forms and obtain user input. The `job_types` and `property_blueprint` key-value pairs in a product template determine how the `jobs` and `properties` sections are displayed in the installation file. Tanzu Operations Manager stores product templates as YAML files in the directory `/var/tempest/workspaces/default/metadata` on the Tanzu Operations Manager VM. These files are not encrypted, so you can edit them without decrypting. User input does not alter these files.

When you upgrade Tanzu Operations Manager your changes might be removed from the installation and product template files.

Modify the installation files

Perform the following steps to locate, decrypt, and edit the Tanzu Operations Manager installation files `installation.yml` and `actual-installation.yml`.

1. SSH into the Tanzu Operations Manager VM by following the steps in [Log Into the Tanzu Operations Manager VM with SSH](#).

2. On the command line, go to the scripts directory:

```
$ cd /home/tempest-web/tempest/web/scripts/
```

3. Run the following command to decrypt the installation YAML file and make a temporary copy of the decrypted file. When prompted for a passphrase, enter the decryption passphrase you created when you launched Tanzu Operations Manager for the first time:

```
$ sudo -u tempest-web SECRET_KEY_BASE="s" ./decrypt /var/tempest/workspaces/default/installation.yml /tmp/installation.yml
```

4. Open `/tmp/installation.yml` to view or edit values.

5. If you plan to make changes, make a backup of the original installation YAML file:

```
$ cp /var/tempest/workspaces/default/installation.yml ~/installation-orig.yml
```

6. If you have made changes to your copy of the installation YAML file, you must encrypt it and overwrite the original with it:

```
$ sudo -u tempest-web SECRET_KEY_BASE="s" RAILS_ENV=production /home/tempest-web/tempest/web/scripts/encrypt /tmp/installation.yml /var/tempest/workspaces/default/installation.yml
```

When you are prompted, enter a passphrase.

7. Delete the temporary copy of the decrypted file:

```
$ rm /tmp/installation.yml
```

8. Repeat steps 2 through 7 for `/tmp/actual-installation.yml`. For each step in which you see `installation.yml`, replace it with `actual-installation.yml`. For example, for step 3 you run:

```
$ sudo -u tempest-web SECRET_KEY_BASE="s" ./decrypt /var/tempest/workspaces/default/actual-installation.yml /tmp/actual-installation.yml
```

9. Restart the Tanzu Operations Manager web interface:

```
$ sudo service tempest-web stop && sudo service tempest-web start
```

10. Go to Tanzu Operations Manager in a browser and enter your decryption passphrase.
11. Log in to Tanzu Operations Manager.
12. Click Review Pending Changes, then Apply Changes.
13. If Tanzu Operations Manager cannot load your changes, see [Revert to your backup](#) to restore your previous settings.

Modify product template files

Complete the following steps to locate and edit your Tanzu Operations Manager product template files:

1. SSH into the Tanzu Operations Manager VM by following the steps in [Log into the Tanzu Operations Manager VM with SSH](#).
2. On the Tanzu Operations Manager VM, go to the `/var/tempest/workspaces/default/metadata` directory.

```
$ cd /var/tempest/workspaces/default/metadata
```

3. The `/var/tempest/workspaces/default/metadata` directory contains the product templates as YAML files. If you plan to make changes, make a backup of the original product template YAML file:

```
$ cp /var/tempest/workspace/default/metadata/YOUR-PRODUCT-TEMPLATE.yml ~/YOUR-PRODUCT-TEMPLATE-orig.yml
```

4. Open and edit the product template YAML file as necessary. For more information about product templates, see [Product template references](#).
5. Go to Tanzu Operations Manager to see your changes.
6. If Tanzu Operations Manager cannot load your changes, see [Revert to your backup](#) to restore your previous settings.

Revert to your backup

Complete the following steps to revert to your backup of an installation or product template file:

1. SSH into the Tanzu Operations Manager VM by following the steps in [Log into the Tanzu Operations Manager VM with SSH](#).
2. Overwrite the modified file with the backup:
 - For the installation file, run the following command:

```
$ cp ~/installation-orig.yml /var/tempest/workspaces/default/installation.yml
```

- For a product template file, run the following command:

```
$ cp ~/YOUR-PRODUCT-TEMPLATE-orig.yml /var/tempest/workspaces/default/metadata/YOUR-PRODUCT-TEMPLATE.yml
```

3. Restart the Tanzu Operations Manager web interface:

```
$ sudo service tempest-web stop && sudo service tempest-web start
```

4. Go to Tanzu Operations Manager in a browser and enter your decryption passphrase.
5. Log in to Tanzu Operations Manager.
6. Click Review Pending Changes, then Apply Changes.

Managing verifiers in your Tanzu Operations Manager deployment

You can deactivate and reactivate verifiers for fields in the BOSH Director and other Tanzu Operations Manager product tiles.

Tanzu Operations Manager provides this option for troubleshooting purposes. For example, your deployment may have a unique configuration that the verifier cannot detect. In this case, you can unblock your deployment by disabling the verifier.

VMware recommends contacting [Support](#) before you deactivate Tanzu Operations Manager verifiers.

VMware recommends recording and routinely revisiting which verifiers you deactivate. The Tanzu Operations Manager UI does not indicate which verifiers you deactivated. Deactivated verifiers persist through upgrades.

About verifiers

Tanzu Operations Manager verifiers narrow the feedback loop for operators by checking that fields are configured correctly. Tanzu Operations Manager verifiers reach out and find objects in the world. For example, given an IP, a verifier can ping the IP to see that it responds.

Verifiers run early in a deployment. With verifiers turned off, deployments might fail much later.

Verifiers run on two occasions:

| Verifier label | Triggers when you... | Blocks deployment? |
|---------------------------|----------------------|--|
| <code>form</code> | Click Save in a pane | Does not block deployment.
Error message is a warning. |
| <code>install_time</code> | Click Apply Changes | Blocks you from deploying.
Disabling verifiers unblocks deployment. |

Deactivate verifiers by type

You can deactivate verifiers by verifier type. For example, you can deactivate all MySQL verifiers for a tile. To deactivate a verifier type for a tile, you must find and record the type, then use the Tanzu Operations Manager API to complete the change.

Step 1: Record the verifier type

To record the verifier type:

1. Log in to Tanzu Operations Manager and go to Review Pending Changes.
2. Click Apply Changes and see the verifier error message.
3. Record the **type** in the verifier error message. For example, **WildcardDomainVerifier**.

Step 2: Use the Tanzu Operations Manager API

To deactivate verifiers, you must use the Tanzu Operations Manager API. To get started with the Tanzu Operations Manager API, follow the procedures in [Using the Tanzu Operations Manager API](#).

The API endpoint varies by tile. Go to one of the following procedures:

- [Deactivate verifiers for the BOSH Director tile](#)
- [Deactivate verifiers for other tiles](#)

To deactivate a verifier type for the BOSH Director tile:

1. Run the following API command:

```
curl "https://OPS-MAN-FQDN/api/v0/staged/director/verifiers/install_time/VERIFIER-TYPE" \  
-X PUT \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \  
-H "Content-Type: application/json" \  
-d '{"enabled": false}'
```

Where:

- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
- **VERIFIER-TYPE** is the verifier type you recorded in [Step 1: Record the Verifier Type](#).
- **UAA_ACCESS_TOKEN** is your UAA access token. For more information, see [Access the API](#).

For more information about this endpoint, see [Endpoints](#) in the Tanzu Operations Manager API documentation.

2. To confirm that your verifier type is deactivated, run the following command:

```
curl "https://OPS-MAN-FQDN/api/v0/staged/director/verifiers/install_time" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Confirm that the Tanzu Operations Manager API shows **false** for the verifier type you deactivated. In the following example, **DirectorConfigurationVerifier** is set to **false**:


```
{
  "verifiers": [
    {
      "type": "AllocatedIpAddressVerifier",
      "enabled": true
    },
    {
      "type": "AvailabilityZonesVerifier",
      "enabled": true
    },
    {
      "type": "DirectorConfigurationVerifier",
      "enabled": false
    },
    {
      "type": "IaaSConfigurationVerifier",
      "enabled": true
    },
    {
      "type": "NetworksExistenceVerifier",
      "enabled": true
    },
    {
      "type": "NetworksPingableVerifier",
      "enabled": true
    },
    {
      "type": "PrivilegeVerifier",
      "enabled": true
    }
  ]
}
```

For more information about this endpoint, see [Retrieving a list of install time verifiers for a specific product](#) in the Tanzu Operations Manager API documentation.

3. Record the product and the verifier type that you deactivated in a secure location. VMware recommends recording and routinely revisiting which verifiers you deactivate.

To deactivate a verifier type for tiles other than the BOSH Director tile:

1. Run the following command and record the **guid** corresponding to your product.

```
curl "https://OPS-MAN-FQDN/api/v0/staged/products" \
-X GET \
-H "Authorization: Bearer UAA_ACCESS_TOKEN"
```

Where:

- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
- **UAA_ACCESS_TOKEN** is your UAA access token. For more information, see [Access the API](#).

See the following example output:

```
{[{
  "installation_name": "component-type1-installation-name",
  "guid": "component-type1-guid",
  "type": "component-type1"
}],
{
  "installation_name": "p-bosh-installation-name",
  "guid": "p-bosh-guid",
  "type": "p-bosh"
}]}
```

2. To deactivate your verifier by type, run the following command.

```
curl "https://OPS-MAN-FQDN/api/v0/staged/products/PRODUCT-GUID/verifiers/install_time/VERIFIER-TYPE" \
-X PUT \
-H "Authorization: Bearer UAA_ACCESS_TOKEN" \
-H "Content-Type: application/json" \
-d '{"enabled": false}'
```

Where:

- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
- **PRODUCT-GUID** is the GUID you recorded in the previous step.
- **VERIFIER-TYPE** is the verifier type you recorded in [Step 1: Record the Verifier type](#).

3. To confirm that your verifier type is deactivated, run the following command:

```
curl "https://OPS-MAN-FQDN/api/v0/staged/products/PRODUCT-GUID/verifiers/install_time" \
-X GET \
-H "Authorization: Bearer UAA_ACCESS_TOKEN"
```

Confirm that the Tanzu Operations Manager API shows **false** for the verifier type you deactivated. In the following example, **MysqlDatabaseVerifier** is set to **false**:

```
{
  "verifiers":[
    {
      "type":"BlobstoreVerifier",
      "enabled":true
    },
    {
      "type":"MysqlDatabaseVerifier",
      "enabled":false
    },
    {
      "type":"SsoUrlVerifier",
      "enabled":true
    },
    {
      "type":"WildcardDomainVerifier",
      "enabled":true
    }
  ]
}
```

4. Record the product and the verifier type that you deactivated in a secure location. VMware recommends recording and routinely revisiting which verifiers you deactivate.

Re-enable verifiers by type

VMware recommends recording and routinely revisiting which verifiers you deactivate. The Tanzu Operations Manager UI does not indicate which verifiers you deactivate, and deactivated verifiers persist through upgrades.

To enable verifiers again, follow the steps detailed in [Use the Tanzu Operations Manager API](#), and rather than setting `enabled` to `false`, set it to `true`.

If you do not know what verifiers are deactivated, `curl` the `verifiers` endpoint for each installed product.

Managing custom VM extensions with Tanzu Operations Manager API

You can use the Tanzu Operations Manager API to manage VM extensions. VM extensions allow you to specify IaaS-specific configurations for your VMs, such as associated security groups and load balancers. Using the Tanzu Operations Manager API, you can create custom VM extensions, give them names, and associate those names with BOSH jobs.

The Resource Config pane in a product tile lets you configure some IaaS-specific load balancing and security properties, and create VM extensions to support them. For examples from the Elastic Application Runtime product, see [Configuring Load Balancing for Elastic Application Runtime](#).

If you use custom VM extensions, you might override other settings. For example, if you use a custom VM extension to add tags, the default tags are removed from all instance groups. To use default tags with custom VM extensions, apply the tags manually.

Prerequisites

To manage custom VM extensions, you need the following:

- Access to the Tanzu Operations Manager API: For more information about the Tanzu Operations Manager API, see [Using the Tanzu Operations Manager API](#).
- Your UAA Access Token Value: Follow the procedure in [Access the API](#) in *Using the Tanzu Operations Manager API* to retrieve your access token. In each example, replace `UAA_ACCESS_TOKEN` with the value of your access token.
- Knowledge of your IaaS-specific configurations: To find configurations you can set for custom VM extensions, see the *VM Types* section of your IaaS CPI in the [BOSH documentation](#). For example, see [VM Types](#) in the AWS CPI docs for a list of AWS VM extension options.

Create or update a VM extension

From the `/api/v0/staged/vm_extensions/NAME` endpoint, use `PUT` to enter your custom VM extension. Replace `NAME` in the endpoint with the name of the VM extension.

Enter the following properties:

- `name`: Enter a memorable string value as the name of your new VM extension. You will associate this name to a job later.

If you are updating a custom VM extension rather than creating a new one, enter the name of the VM extension you want to override.

- **cloud_properties**: Enter your IaaS-specific VM extension properties.

See the following AWS example:

```
curl "https://example.com/api/v0/staged/vm_extensions/example_vm_extension_name" \
-X PUT \
-H "Authorization: Bearer UAA_ACCESS_TOKEN" \
-H "Content-Type: application/json" \
-d '{"name": "example_vm_extension_name", "cloud_properties": { "source_dest_check": false } }'
```

For more information about this command, see [VM extensions APIs](#) in the Tanzu Operations Manager API documentation.

Delete a VM extension

From the `/api/v0/staged/vm_extensions/NAME` endpoint, use **DELETE** to delete a custom VM extension by **name**. Replace **NAME** in the API endpoint with the name of the VM extension.

See the following example:

```
curl "https://example.com/api/v0/staged/vm_extensions/vm_ext1" \
-X DELETE \
-H "Authorization: Bearer UAA_ACCESS_TOKEN" \
-H "Content-Type: application/json" \
-d '{"name": "vm_ext1"}'
```

For more information about this command, see [VM extensions APIs](#) in the Tanzu Operations Manager API docs.

Apply VM extensions to a job

From the `/api/v0/staged/products/PRODUCT_GUID/jobs/JOB_GUID/resource_config` endpoint, use **PUT** to configure a job with the names of your VM extensions.

For the API endpoint, replace **PRODUCT_GUID** with the Globally Unique Identifier (GUID) of your job. You can find this value by requesting `GET /api/v0/deployed/products`. Also replace **JOB_GUID** with the GUID of your job. You can find this value by requesting `GET /api/v0/staged/products/PRODUCT_GUID/jobs`.

For the **additional_vm_extensions** property, list the names of each of the custom VM extensions you want to associate with that job.

You must apply changes in order for the new custom VM extension to take effect.

See the following GCP example:

```
curl "https://example.com/api/v0/staged/products/product-type1-guid/jobs/example-job-guid/resource_config" \
  -X PUT \
  -H "Authorization: Bearer UAA_ACCESS_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{
    "instances": 1,
    "instance_type": {
      "id": "automatic"
    },
    "persistent_disk": {
      "size_mb": "20480"
    },
    "internet_connected": true,
    "elb_names": ["tcp:load_balancer_1", "http:load_balancer_2"],
    "additional_vm_extensions": ["vm_ext_setting_automatic_restart", "vm_ext_setting_on_host_maintenance"]
  }'
```

For more information about this command, see [Configuring resources for a job](#) in the Tanzu Operations Manager API docs.

See deployed VM extensions

From the [/api/v0/deployed/vm_extensions](#) endpoint, use [GET](#) to see your applied custom VM extensions.

See the following example command:

```
curl "https://example.com/api/v0/deployed/vm_extensions" \
  -X GET \
  -H "Authorization: Bearer UAA_ACCESS_TOKEN"
```

The output lists all your VM extension names and its associated custom properties defined in [cloud_properties](#).

See the following example output:

```
{
  "vm_extensions": [
    {
      "name": "vm_ext1",
      "cloud_properties": {
        "source_dest_check": false
      }
    },
    {
      "name": "vm_ext2",
      "cloud_properties": {
        "key_name": "operations_keypair"
      }
    }
  ]
}
```

For more information about this command, see [View a list of deployed VM extension](#) in the Tanzu Operations Manager API documentation.

Tanzu Operations Manager fields that lock on deploy

This topic describes how locked fields work in Tanzu Operations Manager, and lists the unlockable and permanently locked fields in installation tiles.

Types of configurable fields in Tanzu Operations Manager

Some fields in Tanzu Operations Manager lock during deployment. After deployment, you can no longer configure a locked field through the Tanzu Operations Manager UI or API endpoints. Product tile authors configure fields as locked to prevent irrevocable VM malfunction, deployment corruption, and data loss. Most users do not need to edit locked fields after initial deployment.

Because deployment requirements might evolve in unanticipated ways over time, some locked fields are *unlockable*, which means that an advanced Tanzu Operations Manager operator can unlock them to allow reconfiguration. Other fields are *permanently locked*, and their values cannot be changed. Unlocking the unlockable fields requires you to use Tanzu Operations Manager's Advanced Mode. For more information about Advanced Mode, see [Unlockable Fields](#).

In summary, there are three types of configurable fields in Tanzu Operations Manager:

- Standard, which can be reconfigured after deployment as many times as you need.
- Unlockable, which lock on deployment, but can be unlocked and configured with Advanced Mode.
- Permanently locked, which lock on deployment and cannot be unlocked.

Unlockable fields

This section describes the fields that lock during deployment and can be unlocked and configured after deployment. To unlock these fields, you must use Tanzu Operations Manager's Advanced Mode. For more information, see [How to enable Advanced Mode in Tanzu Operations Manager](#).

After you have enabled Advanced Mode, you can use the Tanzu Operations Manager UI or the API to reconfigure fields that were locked previously. Those fields are listed in the following sections.

VMware recommends that only skilled operators use Tanzu Operations Manager's Advanced Mode. If you use Advanced Mode incorrectly, you can deactivate or destroy your deployment.

Unlockable fields in the BOSH Director tile

In the Director Config pane, you can unlock the following fields in Advanced Mode after deployment:

- In the CredHub Encryption Provider section:
 - Under Luna HSM:
 - HSM Host Address
 - HSM Port Address
 - Partition Serial Number
- In the Blobstore Location section:
 - Blobstore Location

- Under S3 Compatible Blobstore:
 - S3 Endpoint
 - Bucket Name
- In the Database Location section:
 - Under External MySQL Database:
 - Host
 - Port
 - Database

In the Assign AZs and Networks pane, you can unlock the following fields after deployment:

- The Singleton Availability Zone drop-down menu

Unlockable fields in product tiles

Product tiles also have unlockable fields.

In the Assign AZs and Networks pane, you can unlock the following fields after deployment:

- Place singleton jobs in
- Balance other jobs

Permanently locked fields

This section describes the fields that lock during deployment and cannot be unlocked even in Advanced Mode. These fields cannot be updated after deployment, either through the Tanzu Operations Manager UI or through API endpoints.

Permanently locked fields in the BOSH Director tile

In the Director Config pane, the following fields lock permanently after deployment:

- In the CredHub Encryption Provider section:
 - CredHub Encryption Provider
 - Under Luna HSM:
 - Encryption Key Name
 - Provider Partition
 - Provider Partition Password
- Database Location: You cannot change the database location after deploying the BOSH Director.

In the Create Availability Zones pane, the following field locks permanently after deployment:

- Availability Zone (AZ): You cannot delete an in-use Availability Zone (AZ).

Configurable fields in product tiles

No fields other than those previously listed lock in third-party product tiles. However, tile authors can specify which additional fields must lock on deploy by passing the `freeze-on-deploy` flag.

For more information about this flag, see [Common Property Blueprint Attributes](#) in *Property and Template References* in the Tanzu Operations Manager Tile Developer Guide.

Overriding manifest properties in Advanced Mode

You can use Advanced Mode to override some manifest properties that you cannot change through the Tanzu Operations Manager UI or API.

Tanzu Operations Manager is an interface to the BOSH Director manifest and the runtime configs that underlie VMware Tanzu. Tanzu Operations Manager does not allow you to edit the manifest and configs directly.

When you configure Tanzu Operations Manager through the UI or API, you set many of the properties contained in the manifest and configs. Tanzu Operations Manager sets some other properties for you, and other properties use the default values provided by VMware Tanzu or BOSH.

Use the manifest property override feature described in this topic to edit properties that cannot be modified in other ways.

Procedure

To override manifest properties:

1. If the property that you want to change is edited through the Tanzu Operations Manager UI or the Tanzu Operations Manager API, use the UI or API to change the property.
2. If the property cannot be modified through the Tanzu Operations Manager UI or the Tanzu Operations Manager API:
 1. Verify that the property can be overridden. See [Overridable properties](#).
 2. If the property is in the BOSH Director Manifest, see [Override BOSH Director manifest properties](#).
 3. If the property is in one of the runtime configs, see [Override Runtime Config properties](#).

You can also use the Tanzu Operations Manager API to view the BOSH Director manifest, view the list of overrides for the BOSH Director and runtime configs, and delete the overrides.

Overridable properties

You can edit properties in the BOSH Director manifest, another tile manifest, or two of the runtime configs.

BOSH Director manifest

In the BOSH Director manifest, you can override properties in the following sections:

- `network`
- `instance_group jobs`

For an example, see [Change the network type to dynamic](#).

You cannot:

- Edit properties in other sections of the BOSH Director manifest
- Add or delete a network
- Add or delete an instance group

Tile manifest

In the tile manifest, you can override properties in the following sections:

- [features](#)
- [update](#)

For example, see [Change number of Canary instances](#) or [Enable returning DNS addresses in links](#).

Runtime configs

You can edit properties in two add-ons in each of two of the runtime configs files. The following table lists these properties:

| Runtime config | Add-on |
|--|--|
| ops_manager_dns_runtime | <ul style="list-style-type: none"> • bosh_dns • bosh_dns_windows |
| ops_manager_system_metrics_runtime | <ul style="list-style-type: none"> • system_metrics_agent • system_metrics_agent_windows |

Override BOSH Director manifest properties

To override properties in the BOSH Director manifest:

1. Set Tanzu Operations Manager to Advanced Mode.

VMware recommends that only skilled operators use Advanced Mode in Tanzu Operations Manager. If you use Advanced Mode incorrectly, you can deactivate or destroy your deployment. Tanzu Operations Manager does not validate the properties that you change using Advanced Mode. For example, if you enter a string where an integer is required, you do not see an error message.

2. Use the [/api/v0/staged/director/overrides](#) endpoint.
3. After you complete your changes, deactivate Advanced Mode by following the procedure in [How to Enable Advanced Mode in the Tanzu Operations Manager](#).

Override tile manifest properties

To override properties in the tile manifest:

1. Set to Advanced Mode by following the procedure in [How to Enable Advanced Mode in the Tanzu Operations Manager](#). For more information about Advanced Mode, see [Advanced Mode APIs](#) in the Tanzu Operations Manager API documentation.

VMware recommends that only skilled operators use Advanced Mode. If you use Advanced Mode incorrectly, you can deactivate or destroy your deployment. and it does not validate the properties that you change using Advanced Mode. For example, if you enter a string where an integer is required, you do not see an error message.

2. Use the [/api/v0/staged/products](#) endpoint to get and record the product GUID from the output of the command. For more information, see [All staged products](#) in the Tanzu Operations Manager API documentation.

3. Use the `/api/v0/staged/products/PRODUCT-GUID/overrides` endpoint. `PRODUCT-GUID` is the GUID you recorded earlier in this procedure.
4. After you complete your changes, deactivate Advanced Mode by following the procedure in [How to Enable Advanced Mode in the Tanzu Operations Manager](#).

Override runtime config properties

To override properties in the runtime configs:

1. Set to Advanced Mode by following the procedure in [How to Enable Advanced Mode in Tanzu Operations Manager](#).

VMware recommends that only skilled operators use Advanced Mode. If you use Advanced Mode incorrectly, you can deactivate or destroy your deployment. It does not validate the properties that you change using Advanced Mode. For example, if you enter a string where an integer is required, you do not see an error message.

2. Use the `/api/v0/staged/director/runtime_configs/overrides` endpoint. For more information, see:
 - [BOSH DNS job spec](#) in GitHub. By default, this file shows the properties and defaults, for the `bosh_dns` add-on in the `ops_manager_dns_runtime`.
 - [Override BOSH Director manifest properties](#).
3. After you complete your changes, deactivate Advanced Mode by following the procedure in [How to Enable Advanced Mode in the Tanzu Operations Manager](#).

Examples of overridable properties

This section contains examples of properties that can be overridden.

BOSH Director manifest example: change the network type to dynamic

By default, the network type is set to `manual`. You cannot change this property through the UI or API.

You can override this default by changing the value of the `type` property in the `networks` section of the BOSH Director manifest.

Tile manifest example: change number of canary instances

By default, the number of canary instances is set to 1. You cannot change this property through the UI or API.

You can override this default by changing the number of canary instances in the `update` section of the tile manifest.

For a list of all the properties you can change in the `update` section, see [Update Block](#) in the BOSH documentation.

Tile manifest example: enable returning DNS addresses in links

By default, `use_dns_addresses` is set to `false`. You cannot change this property through the UI or API.

You can override this default by changing the value to `true` in the `features` section of the tile manifest.

For a list of all the properties you can change in the [features](#) section, see [Features block](#) in the BOSH documentation.

Runtime config example: change the timeout for the BOSH DNS network health check

By default, the network timeout for synchronous health checks is 1 second. You cannot change this property through the UI or API.

You can override this default by changing the value of the `health.health.synchronous_check_timeout` property in the `bosh-dns` add-on of the `ops_manager_dns_runtime` runtime config.

Property template references for Tanzu Operations Manager tiles

For information about the product template for Tanzu Operations Manager tiles and how you can modify it to include different properties, see the [Tile Developer documentation](#).

User accounts and permissions in Tanzu Operations Manager

The following topics cover user accounts and permissions in your Tanzu Operations Manager deployment:

- [Platform operator roles and permissions in Tanzu Operations Manager](#)
- [Creating UAA clients for the BOSH Director](#)
- [Creating and managing Tanzu Operations Manager user and client accounts](#)
- [Configuring Role-Based Access Control in Tanzu Operations Manager](#)

Platform operator roles and permissions in Tanzu Operations Manager

This topic describes the roles and permissions of the operator user type in a Tanzu Operations Manager and Tanzu Platform for Cloud Foundry deployment.

There are various user types in Tanzu Operations Manager and Elastic Application Runtime deployments. Roles are assigned categories that more specifically define functions that a user can perform. For more information about user types in Elastic Application Runtime, see [Elastic Application Runtime User Types](#).

Operators are users who run a deployment and have admin privileges. Operators are also referred to as “Tanzu Operations Manager admins” and “runtime admins” because they perform an admin role in these contexts.

Operator tools and tasks

Operators fulfill system admin roles covering the entire deployment. They work primarily with their IaaS and Tanzu Operations Manager to configure and maintain Elastic Application Runtime runtime component VMs. The component VMs support the VMs that host apps.

Typical operator tasks include:

- Deploying and configuring Tanzu Operations Manager, runtimes, and other product and service tiles
- Maintaining and upgrading Elastic Application Runtime deployments
- Creating user accounts for Elastic Application Runtime users and the orgs that the users work in

- Creating service plans that define the access granted to end users

Operator user accounts

When Tanzu Operations Manager starts up for the first time, the operator specifies one of the following authentication systems for operator user accounts:

- Internal authentication, using a new UAA database that Tanzu Operations Manager creates
- External authentication, through an existing identity provider accessed through SAML protocol

The operator can then use the UAA CLI (UAAC) to create more operator accounts. For more information, see [Creating and managing Tanzu Operations Manager user and client accounts](#).

Operator roles and permissions

The following table summarizes the operator user type, including their roles, the tools they use, the System of Record (SOR) that stores their accounts, and the accounts they can provision.

| User type | Available roles | Tools they use | Account SOR | Accounts they can provision |
|-----------|--|--|---|---|
| Operator | <ul style="list-style-type: none"> • UAA admin • SSO plan admin • Other system admins | <ul style="list-style-type: none"> • IaaS UI • Broadcom Support portal • Tanzu Operations Manager • Tanzu cf CLI • UAA CLI (UAAC) • SSO Dashboard • Marketplace | Tanzu Operations Manager user store through UAA
<i>or</i>
External store through SAML | Operators and Elastic Application Runtime runtime users |

Creating UAA clients for the BOSH Director

This topic describes how to create a UAA client for the BOSH Director. You must create an automation client to run BOSH from a script or set up a continuous integration pipeline.

This topic assumes you are using [BOSH CLI v2+](#).

To create an automation client, do one of the following:

- **Local authentication:** When using Internal Authentication, use the existing admin client to manually create an automation client with the correct privileges.
- **Provision admin client:** When setting up SAML, LDAP, or OIDC authentication, ensure that Tanzu Operations Manager provisions an admin client.

Local authentication

To complete this procedure, the UAAC client must be installed on the Tanzu Operations Manager virtual machine (VM).

1. Open a terminal and SSH into the Tanzu Operations Manager VM by following the instructions for your IaaS in [Log into the Tanzu Operations Manager VM with SSH](#).
2. Go to the Tanzu Operations Manager Installation Dashboard and click the BOSH Director tile. In BOSH Director, click the Status tab, and record the IP address.
3. Using the `uaac target` command, target BOSH Director UAA on port `8443` using the IP address you copied, and specify the location of the root certificate. The default location is `/var/tempest/workspaces/default/root_ca_certificate`.

```
$ uaac target https://BOSH-DIRECTOR-IP:8443 --ca-cert \
/var/tempest/workspaces/default/root_ca_certificate
```

```
Target: https://10.85.16.4:8443
```

Where `BOSH-DIRECTOR-IP` is the IP you recorded in the Status tab of the BOSH Director.

You can also curl or point your browser to the following endpoint to obtain the root certificate: `https://OPS-MANAGER-FQDN/api/v0/security/root_ca_certificate`

4. Log in to the BOSH Director UAA and retrieve the owner token. Complete the following steps to obtain the values for `UAA-LOGIN-CLIENT-PASSWORD` and `UAA-ADMIN-CLIENT-PASSWORD`:
 1. Select the BOSH Director tile on the Tanzu Operations Manager Installation Dashboard.
 2. Click the Credentials tab, and record the entries for Uaa Login Client Credentials and Uaa Admin User Credentials.
 3. For each entry, click Link to Credential to obtain the password.

```
$ uaac token owner get login -s UAA-LOGIN-CLIENT-PASSWORD
User name: admin
Password: UAA-ADMIN-CLIENT-PASSWORD
Successfully fetched token via owner password grant.
Target: https://10.85.16.4:8443
Context: admin, from client login
```

To obtain the password for the UAA log in and admin clients, you can also curl or point your browser to the following endpoints: `https://OPS-MANAGER-FQDN/api/v0/deployed/director/credentials/uaa_login_client_credentials` and `https://OPS-MANAGER-FQDN/api/v0/deployed/director/credentials/uaa_admin_user_credentials`.

5. Create a new UAA Client with `bosh.admin` privileges.

```
$ uaac client add ci --authorized_grant_types client_credentials \
--authorities bosh.admin --secret CI-SECRET

scope: uaa.none
client_id: ci
resource_ids: none
authorized_grant_types: client_credentials
autoapprove:
action: none
authorities: bosh.admin
name: ci
lastmodified: 1469727130702
id: ci
```

6. Set the client and secret as environment variables on the VM.

```
ubuntu@ip-10-0-0-12:~$ export BOSH_CLIENT=ci
ubuntu@ip-10-0-0-12:~$ export BOSH_CLIENT_SECRET=CI-SECRET
```

7. Set an alias for the BOSH Director environment.

```
bosh alias-env MY-ENVIRONMENT-NAME -e BOSH-DIRECTOR-IP \
--ca-cert /var/tempest/workspaces/default/root_ca_certificate
```

You can now use the UAA client you created to run BOSH in automated or scripted environments, such as continuous integration pipelines.

Provision admin client

Tanzu Operations Manager provides an option to create UAA clients during SAML, LDAP, or OIDC configuration so that operators can use scripts and tooling to automate BOSH.

1. When configuring Tanzu Operations Manager for SAML, LDAP, or OIDC, select Provision an admin client in the Bosh UAA.
2. After deploying BOSH Director (BOSH), click the Status tab and record the IP address.
3. In the BOSH Director tile, click the Credentials tab.
4. Click the link for the Uaa Bosh Client Credentials to record the client name and secret.
5. Open a terminal and SSH into the Tanzu Operations Manager VM. For more information, see [Log into the Tanzu Operations Manager VM with SSH](#).
6. Set the client and secret as environment variables on the Tanzu Operations Manager VM.

```
export BOSH_CLIENT=bosh_admin_client
export BOSH_CLIENT_SECRET=UAA-BOSH-CLIENT-SECRET
```

Where **UAA-BOSH-CLIENT-SECRET** is the client secret you recorded in an earlier step.

For example:

```
$ ubuntu@ip-10-0-0-12:~$ export BOSH_CLIENT=bosh_admin_client
$ ubuntu@ip-10-0-0-12:~$ export BOSH_CLIENT_SECRET=aBcDeFGhijKabdsadfdfdf
```

7. Set an alias for the BOSH Director environment.

```
bosh alias-env MY-ENVIRONMENT-NAME -e BOSH-DIRECTOR-IP \
--ca-cert /var/tempest/workspaces/default/root_ca_certificate
```

Where **BOSH-DIRECTOR-IP** is the IP address you recorded in the Status tab of the BOSH Director.

Creating and managing Tanzu Operations Manager user and client accounts

You can add and remove Tanzu Operations Manager users, and you can use an administrator Tanzu Operations Manager account to create client accounts that you can use for Tanzu Operations Manager automation.

Tanzu Operations Manager supports multiple user accounts in Tanzu Operations Manager. A User Account and Authentication (UAA) module co-located on the Tanzu Operations Manager VM manages access permissions to Tanzu Operations Manager.

When Tanzu Operations Manager boots for the first time, you create an admin user. However, you do not create additional users through the Tanzu Operations Manager web interface. If you want to create additional users who can log into Tanzu Operations Manager, you must use the UAA API, either through `curl` or the UAA Command Line Client (UAAC).

Users created this way do not persist if UAA authentication settings change.

In addition to creating user accounts, you can also create client accounts, which connect automation tools and scripts to Tanzu Operations Manager. VMware recommends using clients to handle automated tasks.

Client accounts are not bound to the same authentication protocols as user accounts. A user account that controls automated components can cause those components to fail if the account experiences inconsistent availability due to permission or authentication issues.

You create client accounts after you deploy Tanzu Operations Manager, or during configuration for an initial deployment. For more information about adding clients during initial configuration or after deployment, see [Add Pre-Created Client](#).

Add Tanzu Operations Manager users

This section describes how to add users with UAAC. If you do not already have the UAAC installed, run `gem install cf-uaac` on the command line.

You can manage users on the Tanzu Operations Manager UAA module *only* if you choose to use Internal Authentication instead of an external Identity Provider when configuring Tanzu Operations Manager.

To add Tanzu Operations Manager users:

1. Target your Tanzu Operations Manager UAA:

```
uaac target https://YOUR-OPSMANAGER-FQDN/uaa/
```

Where **YOUR-OPSMANAGER-FQDN** is the fully qualified domain name of your Tanzu Operations Manager installation.

2. Get your token:

```
uaac token owner get
Client ID: opsmn
Client Secret:
Username: OPSMANAGER-ADMIN-USERNAME
Password: OPSMANAGER-ADMIN-PASSWORD
Successfully fetched token via client credentials grant.
Target https://YOUR-OPSMANAGER-FQDN/uaa/
```

Where:

- **YOUR-OPSMANAGER-FQDN** is the fully qualified domain name of your Tanzu Operations Manager installation.
- **OPSMANAGER-ADMIN-USERNAME** and **OPSMANAGER-ADMIN-PASSWORD** are the user name and password for the Tanzu Operations Manager admin user.

The **Client Secret** field does not require a value.

3. Add a user.

```
uaac user add USER-NAME -p USER-PASSWORD --emails USER-EMAIL@EXAMPLE.COM
```

Where:

- **USER-NAME** is the user name of the user you are adding.
- **USER-PASSWORD** is the password with which this user authenticates.
- **USER-EMAIL** is the email address associated with this user.

4. (Optional) Set the Role-Based Access Control (RBAC) permissions for your user. For more information, see [Configuring Role-Based Access Control \(RBAC\) in Tanzu Operations Manager](#).

Remove Tanzu Operations Manager users

This section describes how to remove users with UAAC. If you do not already have the UAAC installed, run `gem install cf-uaac` on the command line.

To remove Tanzu Operations Manager users:

1. Target your Tanzu Operations Manager UAA:

```
uaac target https://YOUR-OPSMANAGER-FQDN/uaa/
```

2. Get your token:

```
uaac token owner get
Client ID: opsmn
Client Secret:
Username: OPSMANAGER-ADMIN-USERNAME
Password: OPSMANAGER-ADMIN-PASSWORD
&nbsp;
Successfully fetched token via client credentials grant.
Target https://YOUR-OPSMAN-FQDN/uaa/
```

Where:

- **YOUR-OPSMANAGER-FQDN** is the fully qualified domain name of your Tanzu Operations Manager installation.

- **OPSMANAGER-ADMIN-USERNAME** and **OPSMANAGER-ADMIN-PASSWORD** are the user name and password for the Tanzu Operations Manager admin user.

The **Client Secret** field does not require a value.

3. Delete a user:

```
uaac user delete USER-NAME
```

Where **USER-NAME** is the user name of the user you wish to delete.

Add Tanzu Operations Manager client accounts

The following sections describe how to create client accounts for Tanzu Operations Manager automation using an admin account.

Log in to UAAC as an admin

To configure a client, you must first log in to UAAC as an admin.

Use one of the following two methods to authenticate to UAAC:

- [Authenticate Using SAML or SSO](#)
- [Authenticate Using LDAP](#)

Authenticate using SAML or SSO

If you're using SAML or SSO, authenticate to UAAC as an admin before creating a client.

To authenticate to UAAC:

1. Target your UAA server.

```
uaac target https://YOUR-OPSMANAGER-FQDN/uaa
```

Where **YOUR-OPSMANAGER-FQDN** is the fully qualified domain name of your Tanzu Operations Manager installation.

2. Log in as an admin.

```
uaac token sso get
```

3. When prompted, type the Client ID and passcode. Leave the client secret blank.

```
Client ID: opsmn
Client secret:
Passcode (from http://YOUR-OPSMANAGER-FQDN/uaa/passcode): YOUR-UAA-PASSCODE
```

Where:

- **YOUR-OPSMANAGER-FQDN** is the fully qualified domain name of your Tanzu Operations Manager installation.
- **YOUR-UAA-PASSCODE** is the passcode with which you authenticate to UAA.

Authenticate using LDAP

If you're using LDAP, authenticate to UAAC as an admin before creating a client.

To authenticate to UAAC:

1. Target your UAA server.

```
uaac target https://YOUR-OPSMANAGER-FQDN/uaa
```

Where **YOUR-OPSMANAGER-FQDN** is the fully qualified domain name of your Tanzu Operations Manager installation.

2. Log in as an admin.

```
uaac token owner get
```

3. When prompted, type the Client ID, your user name, and your password. Leave the client secret blank.

```
Client ID: opsman
Client secret:
User name: admin
Password: *****
```

4. A message appears confirming that UAAC has “Successfully fetched token.”

Create a client

After you authenticate to UAAC, create a client to manage automated components and tasks.

To create a client:

1. Create a client with role-based permissions, an ID, and an authentication secret using UAAC:

```
uaac client add CLIENT-ID --authorized_grant_types client_credentials --authorities opsman.admin --secret CLIENT-SECRET
```

Where:

- **CLIENT-ID** is the name of your client.
- **CLIENT-SECRET** is the secret you use to authenticate to your client.

The **opsman.admin** authority referenced in the previous example grants specific permissions to the client. You can choose which permissions you want the client to have by assigning a different role.

For more information about Tanzu Operations Manager roles and what they do, see [Understanding Roles in Tanzu Operations Manager](#).

Add pre-created client

You can add a pre-created client to Tanzu Operations Manager using the Tanzu Operations Manager API in either of the following ways:

- [Add a Pre-Created Client Before Initial Deployment](#)
- [Add or Modify a Pre-Created Client for an Existing Deployment](#)

Add a pre-created client before initial deployment

This section describes how to add a pre-created client before you deploy Tanzu Operations Manager for the first time.

If you do not already have the UAAC installed, run **gem install cf-uaac** on the command line.

To add a pre-created client before you deploy Tanzu Operations Manager:

1. Authenticate to and access the Tanzu Operations Manager API by following the steps in [Using the Tanzu Operations Manager API](#).
2. Pass the following command:

```
POST /api/v0/setup
```

Include any configuration details your deployment requires. For more information about configuring Tanzu Operations Manager with the API, see [Setting up with SAML](#) in the Tanzu Operations Manager API documentation.

3. In the configuration details, specify this parameter:

```
"precreated_client_secret": "YOUR-SECRET"
```

Where **YOUR-SECRET** is the authentication secret you use to access the pre-created client.

The authentication secret for the pre-created client must be between 1 and 255 ASCII characters.

When **precreated_client_secret** is passed, a UAA client called **precreated-client** is created. To use this client, see [Authenticate as the Pre-Created Client](#).

Add or modify a pre-created client for an existing deployment

This section describes how to add a pre-created client after you have already deployed Tanzu Operations Manager. You can have only one pre-created client at a time. If you have an existing pre-created client, you might also use this procedure to change its authentication secret.

To add a pre-created client to a Tanzu Operations Manager that has already been deployed, or change the secret of an existing pre-created client:

1. Authenticate to and access the Tanzu Operations Manager API by following the steps in [Using the Tanzu Operations Manager API](#).
2. Pass the following command:

```
PUT /api/v0/uaa/precreated_client
```

```
"precreated_client_secret": "NEW-SECRET"
```

Where **NEW-SECRET** is the authentication secret you use to access the pre-created client. This secret must be different from the previous secret you used. The secret cannot be blank.

The authentication secret for the pre-created client must be between 1 and 255 ASCII characters.

To use this client, see [Authenticate as the pre-created client](#).

Authenticate as the pre-created client

To authenticate to UAAC as the pre-created client:

1. Target your UAA server.

```
uaac target https://YOUR-OPSMANAGER-FQDN/uaa
```

Where **YOUR-OPSMANAGER-FQDN** is the fully qualified domain name of your Tanzu Operations Manager installation.

2. Log in as a client.

```
uaac token client get precreated-client -s "PRECREATED-CLIENT-SECRET"
```

3. A message appears confirming that UAAC has “Successfully fetched token.”

You have authenticated to UAAC as the pre-created client. Use this client to perform any automation tasks.

For more information about the Tanzu Operations Manager API, see [Using the Tanzu Operations Manager API](#).

Configuring Role-Based Access Control in Tanzu Operations Manager

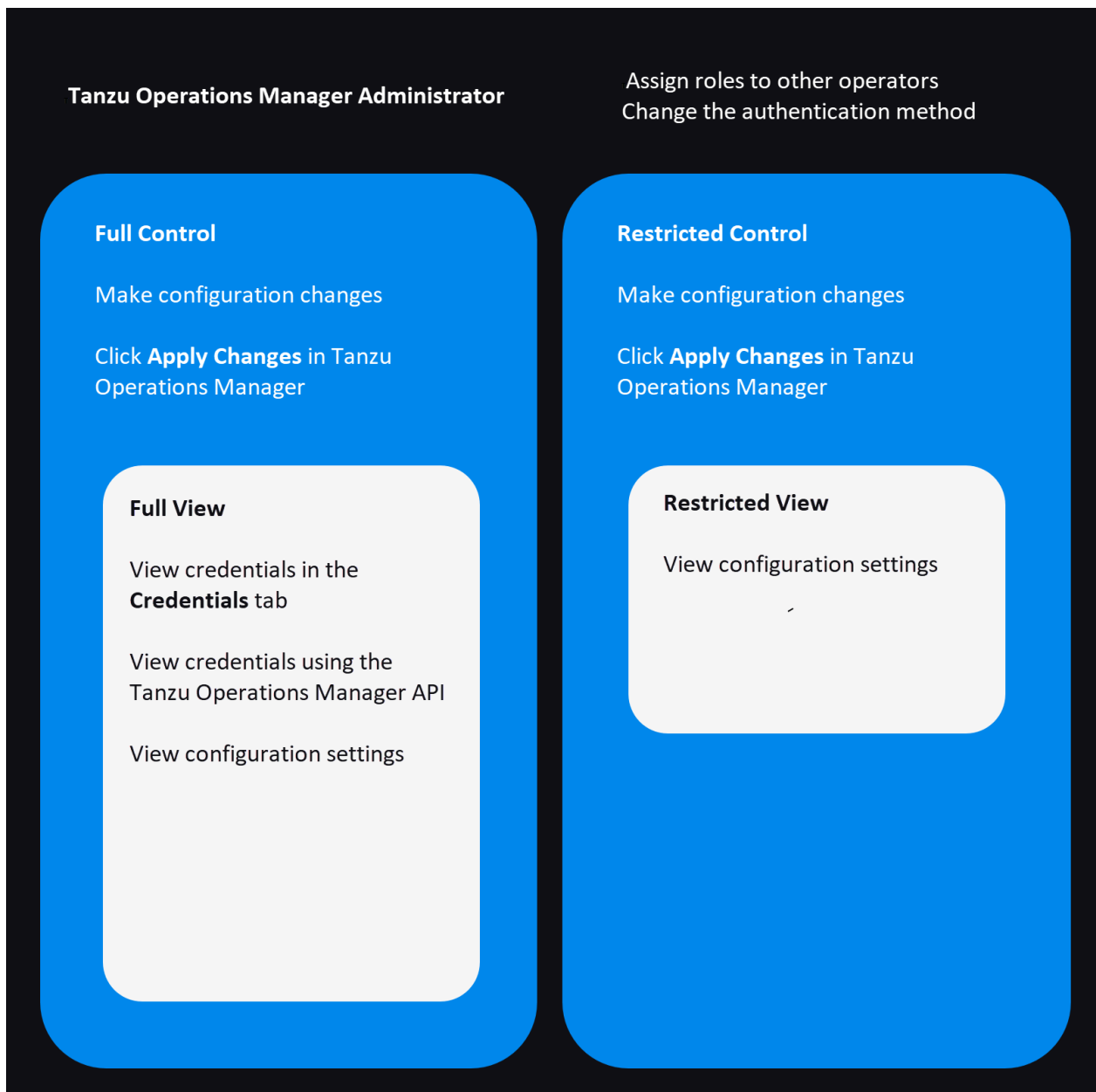
This topic describes how you can customize role-based access control (RBAC) in Tanzu Operations Manager to manage which operators in your organization can make deployment changes, view credentials, and manage user roles.

For information about configuring Tanzu Operations Manager to use internal authentication or SAML authentication, see the BOSH Manager configuration topic for your IaaS:

- [Configuring BOSH Director on AWS](#)
- [Configuring BOSH Director on Azure](#)
- [Configuring BOSH Director on GCP](#)
- [Configuring BOSH Director on vSphere](#)

Roles in Tanzu Operations Manager

The following graphic illustrates the roles you can assign to determine which operators in your organization make deployment changes, view credentials, and manage user roles in Tanzu Operations Manager. Full details are also shown in the following table.



Tanzu Operations Manager admins can use these roles to meet the security needs of their organization. The roles provide a range of privileges that are appropriate for different types of users. For example, assign either Restricted Control or Restricted View to an operator to prevent access to all Tanzu Operations Manager credentials.

For more information about each role, see the following table:

| Tanzu Operations Manager Role | Role Definition | UAA Scope |
|--|---|---------------------------|
| Tanzu Operations Manager Administrator | Admins can make configuration changes, click Review Pending Changes and Apply Changes in Tanzu Operations Manager, view credentials in the Credentials tab and Tanzu Operations Manager API endpoints, change the authentication method, and assign roles to other operators. | opsman.admin |
| Full Control | Operators can make configuration changes, click Review Pending Changes and Apply Changes in Tanzu Operations Manager, and view credentials in the Credentials tab and Tanzu Operations Manager API endpoints. | opsman.full_control |
| Restricted Control | Operators can make configuration changes and click Review Pending Changes and Apply Changes in Tanzu Operations Manager. They cannot view credentials in the Credentials tab or Tanzu Operations Manager API endpoints. | opsman.restricted_control |

| Tanzu Operations Manager Role | Role Definition | UAA Scope |
|-------------------------------|---|-------------------------------------|
| Full View | Operators can view Tanzu Operations Manager configuration settings and view credentials in the Credentials tab and Tanzu Operations Manager API endpoints. They cannot make configuration changes or click Apply Changes. | <code>opsman.full_view</code> |
| Restricted View | Operators can view Tanzu Operations Manager configuration settings. They cannot make configuration changes or view credentials in the Credentials tab or Tanzu Operations Manager API endpoints. | <code>opsman.restricted_view</code> |

To assign one of the preceding roles to an operator, see [Manage roles with internal authentication](#) or [Manage roles with SAML authentication](#).

When you install a new Tanzu Operations Manager instance, all existing users have the Tanzu Operations Manager Administrator role by default.

Simultaneous Tanzu Operations Manager admins

Tanzu Operations Manager allows multiple admins to log in to Tanzu Operations Manager simultaneously and make changes.

The interface does not provide visibility to other admins who are logged in. VMware recommends that admins communicate with each other and coordinate their changes.

Precedence for Apply Changes

If two admins are working at the same time, the admin who first clicks Apply Changes takes precedence. Tanzu Operations Manager overwrites all configurations made by other admins during deployment.

VMware recommends coordinating changes between admins to avoid overwriting configurations.

If you are having deployment issues or changes to your Tanzu Operations Manager are not persisting correctly, confirm that your work is not conflicting with an automated admin.

Enable RBAC in Tanzu Operations Manager after upgrade

When you install a new instance of Tanzu Operations Manager, RBAC is permanently enabled by default.

If your organization has operators who are devoted to managing certain services like Tanzu for MySQL, you can use RBAC to assign those services operators a more restricted role.

If you upgrade from an older Tanzu Operations Manager instance, you must enable RBAC and assign roles to users before they can access Tanzu Operations Manager. If you do not assign any roles to a user, they cannot log in to Tanzu Operations Manager.

Do not assign roles before you enable RBAC. Changing UAA authentication settings causes deletion of the roles.

Enable RBAC with internal authentication

If you upgrade from an earlier version of Tanzu Operations Manager and use internal authentication:

1. Log in to the Tanzu Operations Manager Installation Dashboard.
2. From the user account menu, click Settings.
3. Click Advanced.

4. Click Enable RBAC. When the confirmation dialog box appears, click Confirm and Logout.

Enabling RBAC is permanent. You cannot undo this action. When you upgrade Tanzu Operations Manager, your RBAC settings remain configured. This dialog does not appear if RBAC is already configured. With new instances of Tanzu Operations Manager, RBAC is permanently configured by default.

Enable RBAC with SAML authentication

If you upgrade from an earlier version of Tanzu Operations Manager and are using SAML authentication, follow the procedures in these sections to enable RBAC. To enable RBAC in Tanzu Operations Manager when using SAML authentication, you must configure groups in SAML for admins and non-admins, and then map the admin group to Tanzu Operations Manager.

Step 1: Configure SAML groups

To gather information from your SAML dashboard:

1. Log in to your SAML provider dashboard.
2. Create or identify the name of the SAML group that contains Tanzu Operations Manager admin users.
3. Identify the groups attribute tag you configured for your SAML server.

Step 2: Enable RBAC in Tanzu Operations Manager

Follow the procedure in [Enable RBAC with internal authentication](#) to configure Tanzu Operations Manager to recognize your SAML admin user group.

When RBAC is enabled, only users with the Tanzu Operations Manager Administrator role can edit the SAML configuration.

Create user accounts in Tanzu Operations Manager

To assign RBAC roles to operators, you must first create user accounts for them.

In addition to user accounts, you can create a client account to add to Tanzu Operations Manager. Client accounts manage automation tasks, such as upgrade scripts, log management, and other behaviors that might be negatively impacted if managed by a user account. You can add a client account either before initial deployment or to an existing deployment.

For more information about user and client accounts, see [Creating and managing Tanzu Operations Manager user and client accounts](#).

Manage RBAC roles in Tanzu Operations Manager

You can assign the roles defined in [Roles in Tanzu Operations Manager](#) to determine which operators in your organization can make deployment changes, view credentials, and manage user roles in Tanzu Operations Manager.

Manage roles with internal authentication

If you configured Tanzu Operations Manager to use internal authentication, you can configure roles using the UAA Command Line Interface (UAAC). For more information, see [Creating and managing users with the UAA CLI \(UAAC\)](#).

To use the UAAC to configure roles:

1. Target your UAA server and log in as an admin by running:

```
uaac target https://OPS-MANAGER-DOMAIN/uaa
uaac token owner get
```

Where **OPS-MANAGER-DOMAIN** is the domain of your Tanzu Operations Manager deployment.

2. When prompted, enter these credentials, leaving Client secret blank:

```
Client ID: opsman
Client secret:
User name: USERNAME
Password: PASSWORD
```

Where:

- **USERNAME** is your user name.
- **PASSWORD** is your password.

3. To assign a role to a user, run one of these commands:

- Tanzu Operations Manager Administrator:

```
uaac member add opsman.admin USERNAME
```

Where **USERNAME** is the user to which you want to assign the role.

- Full Control:

```
uaac member add opsman.full_control USERNAME
```

Where **USERNAME** is the user to which you want to assign the role.

- Restricted Control:

```
uaac member add opsman.restricted_control USERNAME
```

Where **USERNAME** is the user to which you want to assign the role.

- Full View:

```
uaac member add opsman.full_view USERNAME
```

Where **USERNAME** is the user to which you want to assign the role.

- Restricted View:

```
uaac member add opsman.restricted_view USERNAME
```

Where **USERNAME** is the user to which you want to assign the role.

Manage roles with SAML authentication

If you configured Tanzu Operations Manager with SAML authentication, you can assign non-admin user roles using UAAC. To assign non-admin user roles:

1. Target your UAA server and log in as an admin by running:

```
uaac target https://OPS-MANAGER-DOMAIN/uaa
uaac token sso get
```

Where **OPS-MANAGER-DOMAIN** is the domain of your Tanzu Operations Manager deployment.

2. When prompted, enter Client ID and Passcode, leaving Client secret blank:

```
Client ID: opsmán
Client secret:
Passcode: UAA-PASSCODE
```

Where **UAA-PASSCODE** is the passcode you retrieved in the previous step.

3. Run:

```
uaac group map SAML-GROUP --name 'OPS-MANAGER-SCOPE' --origin 'saml'
```

Where:

- **SAML-GROUP** is the name of the SAML group to which the user belongs.
 - **OPS-MANAGER-SCOPE** is the Tanzu Operations Manager UAA scope you want to assign to the user. To determine which UAA scope to use, see the table in [Roles in Tanzu Operations Manager](#).
4. Add new and existing users to the appropriate SAML groups in the SAML provider dashboard. Users must log out of both Tanzu Operations Manager and the SAML provider for role changes to take effect.

Manage roles with LDAP authentication

If you configured Tanzu Operations Manager with LDAP authentication, you can assign non admin user roles using UAAC. To assign non-admin user roles:

1. Target your UAA server and log in as an admin by running:

```
uaac target https://OPS-MANAGER-DOMAIN/uaa
uaac token sso get
```

Where **OPS-MANAGER-DOMAIN** is the domain of your Tanzu Operations Manager deployment.

2. When prompted, enter Client ID and Passcode, leaving Client secret blank:

```
Client ID: opsmán
Client secret:
Passcode: UAA-PASSCODE
```

Where **UAA-PASSCODE** is the passcode you retrieved in the previous step.

3. Run:

```
uaac group map LDAP-GROUP --name 'OPS-MANAGER-SCOPE'
```

Where:

- **LDAP-GROUP** is the name of the LDAP group to which the user belongs.
 - **OPS-MANAGER-SCOPE** is the Tanzu Operations Manager UAA scope you want to assign to the user. To determine which UAA scope to use, see the table in [Roles in Tanzu Operations Manager](#).
4. Add new and existing users to the appropriate LDAP groups in the LDAP provider dashboard. You must log out of both Tanzu Operations Manager and the LDAP provider for role changes to take effect.

Certificates and credentials in Tanzu Operations Manager

The following topics describe how you can work with the certificates and credentials used in your Tanzu Operations Manager deployment:

- [Certificate types used by Tanzu Operations Manager](#)
- [Managing certificates with the Tanzu Operations Manager API](#)
- [Overview of certificate rotation in Tanzu Operations Manager](#)
- [Checking Tanzu Operations Manager certificate expiration dates and rotation procedures](#)
- [Overriding duration for Tanzu Operations Manager and CredHub certificates](#)
- [Rotating CAs and leaf certificates using the Tanzu Operations Manager API](#)
- [Rotating non-configurable leaf certificates in Tanzu Operations Manager](#)
- [Rotating configurable Tanzu Operations Manager leaf certificates](#)
- [Rotating identity provider SAML certificates in Tanzu Operations Manager](#)
- [Using custom certificate authorities in Tanzu Operations Manager](#)
- [Retrieving credentials from your Tanzu Operations Manager deployment](#)
- [Using short-lived NATS bootstrap credentials in Tanzu Operations Manager](#)
- [Certificates and credentials managed by CredHub](#)
 - [CredHub for centralized Tanzu Operations Manager credential management](#)
 - [Credential types supported by CredHub](#)
 - [Resetting manually-set certificates in BOSH CredHub](#)
 - [Getting started with CredHub Maestro](#)
 - [Advanced certificate rotation with CredHub Maestro](#)
 - [Tanzu Operations Manager tiles that are compatible with CredHub Maestro](#)
 - [Troubleshooting CredHub Maestro safety violations](#)

Certificate types used by Tanzu Operations Manager

This topic describes the types of certificates used in Tanzu Operations Manager that require planned rotation because they expire.

Tanzu Operations Manager uses certificate authorities (CAs) and various leaf certificates. CAs are self-signed certificates that issue leaf certificates. CAs can be custom or generated by Tanzu Operations Manager.

Leaf certificates are signed by a CA and are used to identify resources in Tanzu Operations Manager. Both root CAs and leaf certificates require planned rotation in Tanzu Operations Manager.

CAs and leaf certificates are stored in Tanzu Operations Manager or CredHub. You can manage both Tanzu Operations Manager and CredHub certificates with the Tanzu Operations Manager API. You can also manage CredHub certificates with CredHub Maestro.

In addition to the types of certificates listed in this topic, some Tanzu Operations Manager products issue their own tile certificates that are not managed by or visible to the Tanzu Operations Manager API. These tile certificates do not require planned rotation because they rotate automatically with product upgrades.

Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI) both use tile certificates in addition to their Tanzu Operations Manager certificates.

Certificate types

The following types of Tanzu Operations Manager certificates require planned rotation and can be viewed and managed with the Tanzu Operations Manager API:

- **Tanzu Operations Manager root CA:** The Tanzu Operations Manager root CA issues other certificates that Tanzu Operations Manager uses. The root CA can be a Tanzu Operations Manager-generated CA or your own custom CA. The Tanzu Operations Manager root CA expires four years after creation.

For more information about viewing the root CAs for Tanzu Operations Manager, see [Listing the Root Certificate Authorities](#) in the Tanzu Operations Manager API documentation.

- **Other internal CAs:** The following CAs are used primarily for internal purposes:
 - **BOSH NATS CA:** The BOSH NATS CA is rotated automatically when you rotate the Tanzu Operations Manager root CA.
 - **BOSH DNS CAs:** The BOSH DNS CAs are managed by CredHub and are applied automatically.
- **Other CredHub-managed certificates:** You can also manage CAs and leaf certificates stored in CredHub; for example, the Diego root CA. In Container Orchestration Engine v1.9 and later, these include the cluster specific `kubo_ca_2018` and `etcd_ca_2018` CAs and their leaf certificates.
- **Non-configurable certificates:** Non-configurable certificates are leaf certificates either created by a CA stored in Tanzu Operations Manager, or created and stored by CredHub and managed by Tanzu Operations Manager calls to the CredHub API. Non-configurable certificates are issued directly by the Tanzu Operations Manager root CA, or by intermediate CAs in a chain of trust originated by the root CA. Non-configurable certificates expire after two years.

For more information about viewing non-configurable leaf certificates, see [Getting information about certificates from products](#) in the Tanzu Operations Manager API documentation.

For more information about generating non-configurable leaf certificates, see [Generating new certificates](#) in the Tanzu Operations Manager API documentation.

- **Configurable certificates:** Configurable certificates are leaf certificates supplied by the user and pasted into configuration fields in Tanzu Operations Manager. Some configuration panes include a Generate RSA Certificate button that supplies valid certificates, but users can obtain configurable certificates from elsewhere, too.
- **Non-rotatable certificates:** Non-rotatable certificates are leaf certificates that, like non-configurable certificates, are issued by the root CA. Unlike non-configurable certificates, non-rotatable certificates cannot be rotated by the Tanzu Operations Manager API.

Managing certificates with Tanzu Operations Manager API

This topic describes how you can use Tanzu Operations Manager API endpoints to manage and retrieve information about certificates in your Tanzu Operations Manager deployment. For more information about the Tanzu Operations Manager API endpoints for managing certificates, see the [Tanzu Operations Manager API documentation](#).

Prerequisites

To use the Tanzu Operations Manager API, you must generate an access token by authenticating with the Tanzu Operations Manager user account and authentication (UAA) server.

For more information about authenticating with UAA, see [Using the Tanzu Operations Manager API](#).

Generate a single RSA certificate

To generate and retrieve a new RSA certificate signed by the root certificate authority (CA):

1. Run:

```
curl "https://OPS-MANAGER-FQDN/api/v0/certificates/generate" \  
-X POST \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \  
-i
```

Where:

- **OPS-MANAGER-FQDN** is the fully-qualified domain name (FQDN) of your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your Tanzu Operations Manager access token without any newline characters such as `\n`.

Retrieve the Tanzu Operations Manager root CA

You can view the Tanzu Operations Manager root CA as a file or in JSON format.

Retrieve the Tanzu Operations Manager root CA as a file through the Tanzu Operations Manager UI

To retrieve the Tanzu Operations Manager root CA as a file through the Tanzu Operations Manager UI:

1. Go to the Tanzu Operations Manager Installation Dashboard.
2. From the account drop-down menu, select Settings.
3. Click Advanced Options.
4. Click Download Root CA Cert. Your browser downloads the Tanzu Operations Manager root CA certificate file.

Retrieve the Tanzu Operations Manager root CA as a file through the Tanzu Operations Manager API

To retrieve the Tanzu Operations Manager root CA as a file through the Tanzu Operations Manager API:

1. Run:

```
curl "https://OPS-MANAGER-FQDN/download_root_ca_cert" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \  
-i
```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your Tanzu Operations Manager access token without any newline characters such as `\n`.

Retrieve the Tanzu Operations Manager root CA in JSON format

To retrieve the Tanzu Operations Manager root CA in JSON format:

1. Run:

```
curl "https://OPS-MANAGER-FQDN/api/v0/security/root_ca_certificate" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-i
```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your Tanzu Operations Manager access token without any newline characters such as `\n`.

List all RSA certificates

To view metadata from all deployed RSA certificates visible to Tanzu Operations Manager, including the root CA and certificates that Tanzu Operations Manager stores in CredHub:

1. Run:

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/certificates" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-i
```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your Tanzu Operations Manager access token without any newline characters such as `\n`.

Overview of certificate rotation in Tanzu Operations Manager

This topic provides an overview of how you can rotate certificate authority (CA) and leaf certificates in Tanzu Operations Manager.

The Tanzu Operations Manager API manages and lists internal CAs and leaf certificates that enable Tanzu Operations Manager components to communicate with each other securely using TLS. It can also list certificates used externally, such as SAML certificates that authenticate to an external identity provider (IDP).

For more information about the CAs and leaf certificates visible to the Tanzu Operations Manager API, see [Certificate types](#).

Rotate CAs and leaf certificates before they expire to avoid downtime for your deployment. To rotate certificates in Tanzu Operations Manager, first check the expiration dates of all certificates. Then, follow the certificate rotation procedure recommended for the expiring certificates to regenerate them, and redeploy any BOSH deployments using those certificates to use the new versions.

When possible, the Tanzu Operations Manager API invokes CredHub Maestro to rotate certificates. You might encounter CredHub Maestro safety violation errors if you attempt to perform an unsafe operation, such as running a rotation step out of order. For more information about troubleshooting these errors, see [Troubleshooting CredHub Maestro Safety Violations During Certificate Rotation](#).

You can set a value to override the duration for certificates in the BOSH Director tile. VMware recommends that you set the value before starting a certificate rotation. For more information, see [Overriding duration for certificates](#).

Check certificate expiration dates and rotation procedures

Before determining which certificate rotation procedure to follow, you must determine:

- Which CAs and leaf certificates are due to expire soon
- Which procedure is recommended for the expiring certificates
- Whether customer provided CAs exist in your deployment

To check the expiration dates and rotation procedures of your certificates, see [Checking Expiration Dates and Rotation Procedures](#).

If customer provided CAs exist in your deployment, you must do an additional procedure to successfully rotate the certificates. To see if you have customer-provided CAs and to rotate them successfully, see [Rotating Certificates with Customer-Provided CAs](#).

Certificate rotation procedures

The topics listed in this section explain how to rotate certificates in Tanzu Operations Manager, including the Tanzu Operations Manager root CA, BOSH NATS CA, CAs stored in CredHub, and leaf certificates. There are different rotation procedures for each type of certificate that requires rotation.

The rotation procedures described in the following topics do not work if the certificates have already expired. If the certificates have expired, contact [Broadcom Support](#) for guidance.

To rotate certificates, follow one of these procedures:

- To rotate CAs stored in Tanzu Operations Manager and CredHub, see [Rotating CAs and Leaf Certificates](#). This procedure also rotates all leaf certificates associated with each CA.
- To rotate non-configurable leaf certificates, see [Rotating non-configurable Leaf Certificates](#).
- To rotate configurable leaf certificates, see [Rotating configurable Leaf Certificates](#).
- To rotate non-rotatable certificates, contact [Broadcom Support](#).

Rotating other certificates

Certificates that the Tanzu Operations Manager API does not rotate include:

- The Services TLS CA. To rotate this CA, see [Advanced certificate rotation with CredHub Maestro](#).
- IDP SAML certificates. To rotate IDP SAML certificates, see [Rotating identity provider SAML Certificates](#).
- IPsec certificates. To rotate IPsec certificates, see [Rotating IPsec certificates](#) in the IPsec Encryption documentation.
- Certificates for any products that you deploy manually with BOSH. To rotate these certificates, you must redeploy the BOSH release and re-create its VMs using the BOSH CLI.

- CredHub-managed certificates for the Elastic Application Runtime versions before v2.9

Checking Tanzu Operations Manager certificate expiration dates and rotation procedures

This topic describes how you can use the Tanzu Operations Manager UI or API to manually check the expiration dates and rotation procedures of the certificate authorities (CAs) and leaf certificates managed by the Tanzu Operations Manager. After identifying the certificates that expire soon, you can determine which certificate rotation procedure to follow.

To configure Concourse to automatically monitor expiring certificates, you can use Platform Automation. For more information, see the [Platform Automation documentation](#).

Check expiration dates and rotation procedures using the Tanzu Operations Manager UI

The Certificates page in the Tanzu Operations Manager UI lists the certificates your deployment uses.

To check certificate expiration dates and rotation procedures using the Tanzu Operations Manager UI:

1. Go to Tanzu Operations Manager Installation Dashboard.
2. In the top menu, click Certificates.
3. Look at the Expiration Date column for the expiration dates of each certificate and the number of days before expiration.
4. For any certificates expiring soon, identify the procedure listed in the Rotation Procedure column. This will link to the documentation page describing how to rotate that certificate.

Note that most procedures operate on multiple certificates at once. For example, all CA certificates listed as using the Standard CA Procedure are rotated together; there is no need to repeat the procedure for each of these certificates. It is also recommended to rotate CA certificates first, as rotating CAs will also rotate any leaf certificates signed by those CAs.

Check expiration dates and certificate types using the Tanzu Operations Manager API

To check certificate expiration dates and types using the Tanzu Operations Manager API:

1. Follow the procedure in [Using Tanzu Operations Manager API](#) to target and authenticate with the Tanzu Operations Manager User Account and Authentication (UAA) server. Record your Tanzu Operations Manager access token, and use it for **UAA-ACCESS-TOKEN** in the following steps. When you record your Tanzu Operations Manager access token, remove any newline characters such as `\n`.
2. To retrieve the certificates, call the `/deployed/certificates` endpoint of the Tanzu Operations Manager API by running:

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/certificates" \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \  
-i
```

Where:

- **OPS-MANAGER-FQDN** is the fully-qualified domain name (FQDN) of your Tanzu Operations Manager deployment.

- `UAA-ACCESS-TOKEN` is the `access_token` value you recorded in the previous step.

More options:

- To limit command output to certificates that expire within a given time interval, append `?expires_within=TIME` to the endpoint, replacing `TIME` with an integer-letter code.
 - Valid letter codes are `d` for days, `w` for weeks, `m` for months, and `y` for years. For example, querying to `https://OPS-MANAGER-FQDN/api/v0/deployed/certificates?expires_within=6m` searches for certificates expiring within six months.
 - To make the JSON output more readable, you can pipe your `curl` command to `jq` or another text editor with JSON formatting.
3. The `deployed/certificates` output lists all CAs and leaf certificates visible to the Tanzu Operations Manager API, whether they are stored in Tanzu Operations Manager directly or they are stored in CredHub. To determine the expiration date and rotation procedure for each certificate listed:
1. Determine the expiration date from the `valid_until` value. For example, the root CA listed in the following example expires on August 12, 2020:

```
{
  "configurable": true,
  "is_ca": true,
  "property_reference": ".properties.root_ca.fb10484dd5541a273c9d",
  "property_type": "rsa_cert_credentials",
  "product_guid": "ops_manager",
  "location": "ops_manager",
  "variable_path": null,
  "issuer": "/CN=ToolsmithsCA",
  "valid_from": "2019-08-13T15:30:22Z",
  "valid_until": "2020-08-12T15:30:22Z",
  "rotation_procedure_name": "Standard CA Procedure",
  "rotation_procedure_url": "https://techdocs.broadcom.com/us/en/vmware-tanzu/platform/tanzu-operations-manager/3-0/tanzu-ops-manager/security-pcf-infrastructure-rotate-cas-and-leaf-certs.html"
}
```

2. For any certificates expiring soon, identify the procedure listed in the `rotation_procedure_name` field. This is the name of the rotation procedure that will be used to rotate the certificate. The URL for the documentation for this procedure is listed in the `rotation_procedure_url` field.

Note that most procedures operate on multiple certificates at once. For example, all CA certificates listed as using the Standard CA Procedure are rotated together; there is no need to repeat the procedure for each of these certificates. VMware also recommends rotating CA certificates first, because rotating CAs also rotates any leaf certificates signed by those CAs.

Overriding duration for Tanzu Operations Manager and CredHub certificates

You can set values to override the duration for certificate authority (CA) and leaf certificates managed by Tanzu Operations Manager and CredHub. By default, certificates that Tanzu Operations Manager and CredHub generate use the duration set by the tile and certificate authors, which can vary from certificate to certificate.

When you apply an override, you can increase certificate duration and reduce the frequency of required certificate rotations. The following table describes what happens when you enable this feature:

| If you... | then... |
|---|--|
| configure a duration that is shorter than the default | the certificate is generated using the default duration. |
| configure a duration that is longer than the default | the certificate is generated using the duration you set. |

For example, Tanzu Platform for Cloud Foundry might create a leaf certificate with a duration of 365 days. If you configure the minimum leaf duration to be 730 days, then when Elastic Application Runtime is first deployed or regenerated, the leaf certificates are created using a duration of 730 days instead. A Elastic Application Runtime certificate that is normally created with a duration of 1460 days continues to be created with a duration of 1460 days.

VMware recommends that you do not change your scheduled certificate rotations to use the duration override feature. Instead, you can enable a duration override and use it when the next scheduled rotation takes place.

Enable duration overrides

To enable the duration override feature:

1. Click the BOSH Director tile.
2. Click Director Config.
3. Scroll down to the Certificate Duration Overrides section.
4. Select the On option.
5. Enter the desired duration for CA certificates in the CA Certificate Duration field.
6. Enter the desired duration for leaf certificates in the Leaf Certificate Duration field.
7. Click Save.

After you enable the duration overrides feature, you must take additional steps to apply the setting to existing certificates, and any new certificates generated by CredHub.

Depending on your deployment, continue to one of the following sections.

Apply a duration override on an existing Tanzu Operations Manager

This procedure describes how to apply a duration override on an existing Tanzu Operations Manager.

To apply a duration override:

1. Click Apply Changes on the BOSH Director. This ensures that future certificates generated by CredHub use the new duration that you set.
2. To apply a duration override to leaf certificates, you must do a leaf certificate rotation:
 1. To rotate non-configurable leaf certificates, see [Rotating non-configurable leaf certificates](#).
 2. To rotate certificates generated using Generate RSA Certificate, see [Rotating configurable leaf certificates](#).
3. To apply a duration override to both CA and leaf certificates, you must do a full CA rotation. To rotate CAs and leaf certificates, see [Rotating CAs and Leaf Certificates](#).
4. By default, the Services TLS CA is excluded by the Tanzu Operations Manager certificate rotation API. You can rotate this certificate separately to apply the configured duration. To rotate the Services TLS CA, see [Rotate the Services TLS CA and Its Leaf Certificates](#).

Apply a duration override on a new Tanzu Operations Manager

This procedure describes how to apply a duration override on a new Tanzu Operations Manager.

If you have already run Apply Changes on the BOSH Director tile, you must use the procedure in [Apply a duration override on an existing Tanzu Operations Manager](#).

To apply a duration override:

1. To apply duration overrides to leaf certificates, no action is required. Leaf certificates are created using the configured duration the first time you run Apply Changes.
2. To apply duration overrides to CA certificates, you must regenerate the CA certificates. Follow the procedure in [Rotating CAs and Leaf Certificates](#) with the following modifications:
 1. You do not need to enable the Recreate VMs deployed by the BOSH Director check box.
 2. You do not need to run Apply Changes.

Excluded certificates

This section describes the certificates that are excluded from the duration override setting. The duration override setting applies only to certificates that are rotated by the Tanzu Operations Manager certificate rotation procedures.

The following certificates do not use the duration override setting:

- User-provided certificates: Products installed using Tanzu Operations Manager can contain fields that request operator-provided certificates. These certificates are not generated by Tanzu Operations Manager or CredHub.

You can use Generate RSA Certificate to generate a certificate instead of requiring an operator-provided certificate. Certificates generated with Generate RSA Certificate use the duration overrides you set.
- Tanzu Operations Manager SAML Certificate: If you configure Tanzu Operations Manager to use SAML, UAA uses the Tanzu Operations Manager SAML Certificate. This certificate is valid for 730 days. To rotate this certificate, see [How to check and rotate Tanzu Operations Manager SAML Certificate before it expires](#).
- Tanzu Operations Manager SSL Certificate: The Tanzu Operations Manager SSL Certificate is used to secure communication between the operator's browser and the Tanzu Operations Manager app. By default, this is a self-signed certificate created when the Tanzu Operations Manager VM is first launched. This certificate is valid for 365 days, and is automatically regenerated when upgrading Tanzu Operations Manager.
- NATS client certificates: Each BOSH deployed VM has a unique NATS client certificate used to communicate with the BOSH Director. This certificate is regenerated each time the VM is re-created.
- Diego identity certificates: Each app instance deployed with Elastic Application Runtime has a unique identity certificate used to communicate with Diego. The Diego Cell rep supplies a new certificate and private key pair to the app instance before the end of the validity period. For more information, see [Using instance identity credentials](#) in the Cloud Foundry documentation.
- All other certificates that have a default duration that is longer than the duration configuration.

Certificates and TLS in your Tanzu Operations Manager deployment

See the following topics for information about certificate and Transport Layer Security (TLS) infrastructure in Tanzu Operations Manager.

- [TLS connections in Elastic Application Runtime](#): Describes TLS used in Tanzu Operations Manager deployments, including supported TLS cipher suites.
- [Securing traffic into Elastic Application Runtime](#): Describes how to configure TLS termination for HTTP traffic into Tanzu Platform for Cloud Foundry.
- [Providing a certificate for your TLS termination point](#): Describes how to configure TLS certificates for Elastic Application Runtime.
- [Managing certificates with the Tanzu Operations Manager API](#): Describes how to manage internal certificate authorities (CAs) and certificates in Tanzu Operations Manager that are visible to the Tanzu Operations Manager API.
- [Overriding duration for Tanzu Operations Manager and CredHub certificates](#): Describes how to configure duration overrides for CA and leaf certificates.
- [Overview of certificate rotation in Tanzu Operations Manager](#): Describes how to rotate internal CAs and certificates in Tanzu Operations Manager.
- [Using custom certificate authorities in Tanzu Operations Manager](#): Describes how to add a custom CA to issue digital certificates in a Tanzu Operations Manager deployment.
- [Trusted system certificates](#): Describes where apps deployed to Elastic Application Runtime can find trusted system certificates.

Rotating CAs and leaf certificates using Tanzu Operations Manager API

This topic describes how you can rotate certificate authorities (CAs) and leaf certificates using the Tanzu Operations Manager API. To rotate only leaf certificates, see one of the leaf certificate rotation topics listed in [Certificate Rotation Procedures](#) in *Overview of Certificate Rotation*.

This procedure uses the Tanzu Operations Manager API to rotate all CA and leaf certificates generated by Tanzu Operations Manager and CredHub.

This procedure includes three redeployments. The first deployment regenerates the new CA certificates used by Tanzu Operations Manager components. The second deploy replaces the old leaf certificates with the new leaf certificates. The third deploy removes the old CA certificates, leaving only the new CA certificates.

Certificates rotated by the procedure

This procedure rotates all certificates listed on the Certificates page in Tanzu Operations Manager, except those with specialized rotation procedures.

The following certificates types will be rotated:

- CA certificates
- Non-configurable leaf certificates
- Configurable leaf certificates stored in Tanzu Operations Manager

Some certificates require a different procedure to rotate. For example, the Services TLS CA, [/services/tls_ca](#), cannot be rotated using this procedure. For more information, see [Certificates with specialized procedures](#).

The Tanzu Operations Manager API also excludes from rotation the CAs for certain versions of the Tanzu for MySQL, Tanzu for Valkey, Tanzu RabbitMQ, Tanzu GemFire tiles, and IPsec Encryption. For information about rotating these CAs, see the documentation for those products.

Prerequisites

Before you rotate certificates:

- You must have Jammy stemcells at version 1.8 or later, Xenial stemcells at version 621.171 or later, and be running Windows 2019.41 or later.

If you have any VMs deployed with any of the following stemcells, you will have to recreate the VMs at every Apply Changes step:

- Xenial stemcells earlier than release 621.171
- Windows stemcells earlier than release 2019.41
- Jammy stemcells earlier than release 1.8

If you are still using these stemcell versions and you cannot immediately upgrade, you must enable the Recreate all service instances errand, if the service tile provides it.

Running this errand pushes BOSH Agent certificate updates to service instances.

If the service tile does not include the Recreate all service instances errand, see the procedure [Running with old Xenial or Windows stemcells](#) on *Rotating CAs and leaf certificates using the Tanzu Operations Manager API* for instructions.

If you use VMware Tanzu Container Orchestration Engine (formerly TKGI), rotate your certificates using the instructions in the [Container Orchestration Engine documentation](#). For certificates that are not rotated by the Container Orchestration Engine procedures, see [Advanced certificate rotation with CredHub Maestro](#).

Procedure

You must complete these steps in the exact order specified. Otherwise, you can damage your deployment.

You can set a value to override the duration for certificates in the BOSH Director tile. VMware recommends that you set the value before starting a certificate rotation. For more information, see [Overriding duration for certificates](#).

To rotate CAs, configurable leaf certificates, and non-configurable leaf certificates:

1. Regenerate CAs. This step regenerates CAs for all CAs visible to the Tanzu Operations Manager API. For more information, see [Regenerate CAs](#).
2. Activate the new CAs. This step activates the new CAs regenerated in the previous step. See [Activate the CAs](#).

3. Regenerate configurable leaf certificates. This step regenerates configurable certs to use the new CAs. See [Regenerate configurable Leaf Certificates](#).
4. Regenerate non-configurable leaf certificates. This step regenerates non-configurable certs to use the new CAs. See [Regenerate non-configurable Leaf Certificates](#).
5. Delete the old CAs. This step deletes the old unused CAs, leaving only the regenerated new CAs. See [Delete the old CAs](#).

Step 1: Regenerate CAs

This section describes how to regenerate the CA certificates used by Tanzu Operations Manager, tiles, and service instances.

You can provide your own custom CA for the Tanzu Operations Manager root CA instead of using one generated by Tanzu Operations Manager, if you want to. The custom CA will be used to sign leaf certificates that are managed by Tanzu Operations Manager. Leaf certificates managed by CredHub are not signed using the Tanzu Operations Manager root CA and will not be signed by the provided custom CA.

To regenerate CAs:

1. If you have not already done so, follow the procedure in [Using Tanzu Operations Manager API](#) to target and authenticate with the Tanzu Operations Manager UAA server. Record your Tanzu Operations Manager access token, and use it for [UAA-ACCESS-TOKEN](#) in the following steps. When you record your Tanzu Operations Manager access token, remove any newline characters such as `\n`.
2. Use Tanzu Operations Manager to generate a new CA, or add your own custom CA:
 - To use your own custom CA: Call the Tanzu Operations Manager API [certificate_authorities](#) endpoint by running:

```
curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities" \
-X POST \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-H "Content-Type: application/json" \
-d '{"cert_pem": "-----BEGIN CERTIFICATE-----CUSTOM-CERTIFICATE", "private_key_pem": "-----BEGIN RSA PRIVATE KEY-----RSA-KEY"}' \
-i
```

Where:

- [OPS-MANAGER-FQDN](#) is the fully-qualified domain name (FQDN) of your Tanzu Operations Manager deployment.
- [CUSTOM-CERTIFICATE](#) is your custom CA.
- [RSA-KEY](#) is your RSA key. Elliptic Curve Digital Signature Algorithm (ECDSA) certificates are not supported as the root CA in Tanzu Operations Manager.
- [UAA-ACCESS-TOKEN](#) is your UAA access token.

If the command succeeds, the API returns a response that includes the new CA. For example:

```

HTTP/1.1 200 OK
{
  "certificate_authorities": [
    {
      "guid": "f7bc18f34f2a7a9403c3",
      "issuer": "CUSTOM-CERTIFICATE",
      "created_on": "2017-01-09",
      "expires_on": "2021-01-09",
      "active": true,
      "cert_pem": "-----BEGIN CERTIFICATE-----\nMIIC+zCCAeOgAwIBAg
I....etc"
    }
  ]
}

```

- To use a Tanzu Operations Manager-generated CA: Call the Tanzu Operations Manager API **generate** endpoint by running:

```

curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities/generate" \
-X POST \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-H "Content-Type: application/json" \
-d '{}' \
-i

```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your UAA access token.

If the command succeeds, the API returns a response that includes the new CA. For example:

```

HTTP/1.1 200 OK
{
  "guid": "f7bc18f34f2a7a9403c3",
  "issuer": "VMware",
  "created_on": "2017-01-19",
  "expires_on": "2021-01-19",
  "active": false,
  "cert_pem": "-----BEGIN EXAMPLE CERTIFICATE-----
MIIC+zCCAeOgAwIBAgIBADANBgkqhkiG9w0BAQsFADAfMQswCQYDVQQGEwJVUzEQ
EXAMPLEoCgWHUGl2b3RhbDAeFw0xNDarthgyMTQyMjVaFw0yMTAxMTkyMTQyMjVa
EXAMPLEoBgNVBAYTA1VTMRAdDgYDVVaderdQaXZvdGFsMIIBIjANBgkqhkiG9w0B
EXAMPLEoAQ8AMIIBCgKCAQEAYV40hPIIZTEym90cdcNVip9Ev0ijPPLo9WPLUMzT
EXAMPLEo/TgD+DP09mwVXfqwB1Jmoj9DqRED1x/6bc0Ki/BAFo/P4MmOKm3QnDCt
EXAMPLEo0qgA++2HYrNTKWJ5fsXmERs8lK9AXXT7RKXhkyWwU3oNGf7zo0e3YKp
EXAMPLEoh1NwIbNcGT1AurIDSxy0Zy1HVzBLTisMyDogJmSCLsOw3qUDQjatjXKw
EXAMPLEojG3nv2hvD4/aTOiHuKM3+AGbnaS2MdIOvFOh/7Y79tUp89csK0gs6uOd
EXAMPLEohe4DcKw5CzUTfHKNXgHyeoVOBPCVQTP4lJp1iQIDAQABo0IwQDAdBgNV
EXAMPLEoyH4y7VEuImLStXM0CKR8uVqxX/gwDwYDVR0TAQH/BAUwAwEB/zA0BgNV
EXAMPLEoBAMCAQYwDQYJKoZIhvcNAQELBQADggEBALmHOPxdyBGnuR0HgR9V4TwJ
EXAMPLEoGLKVT7am5z6G20q5cwACFHWAfFrPG4W9Jm577QtewiY/Rad/PbkY0YSY
EXAMPLEokrfNjxjxI0H2sr7qLBFjJ0wBZHhVmDs06A9PkfAPu4eJvqRMuL/xGmSQ
EXAMPLEoCynMNz7FgHyFbd9D9X5YW8fWGSeVBPPikcONdRv9aEeAtbGEh8eZCP
EXAMPLEob33RuR+CTNqThXY9k8d7/7ba4KVdd4gP8ynFgwvnDQ0jcJZ6Go5QY5HA
EXAMPLEoPFw8pAYcvWrXKR0rE8fL5o9qgTyjm0+5yyyvWIYrKPqgIUIvMCdNr84=
-----END EXAMPLE CERTIFICATE-----"
}

```

3. Confirm that the new CA is added by listing all the CAs for Tanzu Operations Manager. Run:

```

curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-i

```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your UAA access token.

The API call returns the GUID of the new CA. For example:

```

HTTP/1.1 200 OK
{
  "certificate_authorities": [
    {
      "guid": "f7bc18f34f2a7a9403c3",
      "issuer": "VMware",
      "created_on": "2017-01-09",
      "expires_on": "2021-01-09",
      "active": true,
      "cert_pem": "-----BEGIN CERTIFICATE-----\nMIIC+zCCAeOgAwIBAgI....et
c"
    }
    {
      "guid": "a8ee01e33e3e3e3303e3",
      "issuer": "VMware",
      "created_on": "2017-04-09",
      "expires_on": "2021-04-09",
      "active": false,
      "cert_pem": "-----BEGIN CERTIFICATE-----\nzbBBC+eAAAE1gAwAAAEZ....et
c"
    }
  ]
}

```

4. Identify the new CA, which has **active** set to **false**. Record its GUID.
5. Return to Tanzu Operations Manager Installation Dashboard.
6. If you have service tiles installed, for each service tile, do the following:

The name of the Upgrade all service instances errand might be slightly different between services. In the TKGi tile, the errand is called Upgrade all clusters.

1. Click the tile.
2. Click the Errands tab.
3. Enable the Upgrade all service instances errand. Running this errand is necessary to push certificate updates to each service instance.
7. Click Review Pending Changes.
8. Click Apply Changes.

If you are running a deployment with old Xenial or Windows stemcells, you must recreate all VMs. See [Running with old versions of Xenial or Windows stemcells](#).

When the deployment finishes and you have completed all necessary tasks to update service instances, continue to the next section, [Activate the CAs](#).

Step 2: Activate the CAs

To activate the new CAs created in [Regenerate CAs](#):

1. Call the Tanzu Operations Manager API **activate** endpoint by running:


```
curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities/CERTIFICATE-GUID/activate" \
-X POST \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-H "Content-Type: application/json" \
-d '{}' \
-i
```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **CERTIFICATE-GUID** is the GUID of your CA that you retrieved in the previous section.
- **UAA-ACCESS-TOKEN** is your UAA access token.

The API returns a successful response:

```
HTTP/1.1 200 OK
```

2. List your Tanzu Operations Manager root CAs to confirm that the new root CA is active. Run:

```
curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-i
```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your UAA access token.

3. Examine the response to ensure that your new root CA has the **active** property set to **true**.

Step 3: Regenerate configurable leaf certificates

Any configurable leaf certificates generated from the Tanzu Operations Manager root CA must also be regenerated. These are the leaf certificates that you generated using the Generate RSA Certificate button in the Tanzu Operations Manager UI. For example, see [Providing a Certificate for Your TLS Termination Point](#) in the Elastic Application Runtime documentation.

To regenerate each configurable leaf certificate that expires soon:

1. Find the text field where the leaf certificate is configured in the Tanzu Operations Manager UI. You might need to look through multiple configuration panes to identify the location of the certificate configuration text field. Use the following fields to identify the location of the certificate configuration text field:
 - The **product_guid** field in the Tanzu Operations Manager API output can help identify the tile in which the certificate is configured. For example, the prefix **p-bosh-** refers to the BOSH Director tile, and the prefix **cf-** refers to the Elastic Application Runtime tile.
 - The **property_reference** field in the Tanzu Operations Manager API output can help identify the configuration pane in which the certificate is configured. For example, the **uaa.service_provider_key_credentials** property is configured in the UAA pane of the Elastic Application Runtime tile.
2. Copy a new value for the leaf certificate into the text field or generate a new leaf certificate using the Generate RSA Certificate button.
3. Click Save on the pane in which you added new leaf certificates.

Step 4: Regenerate non-configurable leaf certificates

After you activate the new CAs, you must regenerate the non-configurable leaf certificates to ensure they are signed by the new CAs.

To regenerate the non-configurable leaf certificates:

1. Call the Tanzu Operations Manager API `regenerate` endpoint by running:

```
curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities/active/regenerate" \  
-X POST \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \  
-H "Content-Type: application/json" \  
-d '{}' \  
-i
```

Where:

- `OPS-MANAGER-FQDN` is the FQDN of your Tanzu Operations Manager deployment.
- `UAA-ACCESS-TOKEN` is your UAA access token.

The API returns a successful response:

```
HTTP/1.1 200 OK
```

2. Return to Tanzu Operations Manager Installation Dashboard.
3. If you have service tiles installed, for each service tile do the following:

The name of the Upgrade all service instances errand might be slightly different between services. In the TKGi tile, the errand is called Upgrade all clusters.

1. Click the tile.
 2. Click the Errands tab.
 3. Enable the Upgrade all service instances errand. Running this errand is necessary to push certificate updates to each service instance.
4. Click Review Pending Changes.
 5. Click Apply Changes.

If you are running a deployment with old Xenial or Windows stemcells, you must recreate all VMs. See [Running with old versions of Xenial or Windows stemcells](#).

6. When the deployment finishes and you have completed all necessary tasks to update service instances, continue to the next section, [Delete the old CAs](#).

Step 5: Delete the old CAs

After activating new CAs and regenerating leaf certificates from the new CAs, you must delete the old CAs. Deleting the old CAs ensures that each leaf certificate is associated only with the new CA that signs it, and that the old CAs are no longer trusted.

To delete the old CAs:

1. List your root CAs and retrieve the GUID of the old, inactive Tanzu Operations Manager root CA.
Run:

```
curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-i
```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your UAA access token.

2. Delete the old Tanzu Operations Manager root CA by running:

```
curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities/OLD-CERTIFICATE-GUID" \
-X DELETE \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-i
```

Where:

- **OPS-MANAGER-FQDN** is the FQDN of your Tanzu Operations Manager deployment.
- **OLD-CERTIFICATE-GUID** is the GUID of the old Tanzu Operations Manager root CA.
- **UAA-ACCESS-TOKEN** is your UAA access token.

The API returns a successful response:

```
HTTP/1.1 200 OK
```

3. If you have service tiles installed, for each service tile, do the following:

The name of the Upgrade all service instances errand might be slightly different between services. In the TKGi tile, the errand is called Upgrade all clusters.

1. Click the tile.
2. Click the Errands tab.
3. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.
4. Click Review Pending Changes.
5. Click Apply Changes.

If you are running a deployment with old Xenial or Windows stemcells, you must recreate all VMs. See [Running with old versions of Xenial or Windows stemcells](#).

When the deployment finishes and you have completed all necessary tasks to update service instances, the rotation procedure is complete.

Certificates with specialized procedures

Some certificates are not rotated when using the standard rotation procedure. The following table lists certificates that use a separate, specialized rotation procedure and provides links to documentation and help for rotating these certificates.

| Certificate | Tile | Procedure |
|----------------------------------|------|--|
| /services/tls_ca | n/a | Rotate the Services TLS CA and its Leaf Certificates |

Troubleshooting

Safety Violation Errors

The Tanzu Operations Manager API runs CredHub Maestro when rotating certificates. If a certificate rotation API command is unsafe, CredHub Maestro stops the command and returns one or more safety violations.

For example, CredHub Maestro stops a certificate rotation API command if you try to follow certificate rotation steps in the wrong order. Because performing these steps in the wrong order can make your deployment unstable or cause downtime, CredHub Maestro stops the command and returns an error message.

For information about how to troubleshoot safety violation errors that are returned when rotating certificates, see [Troubleshooting CredHub Maestro safety violations during certificate rotation] (troubleshoot-cert-errors.html).

CredHub Garbage Collection

This procedure does not delete leaf certificates stored in CredHub. It marks CredHub leaf certificates as transitional, which indicates that the leaf certificate is inactive. To delete CredHub leaf certificates, use the CredHub Maestro garbage collection feature. For information about CredHub Maestro certificate garbage collection, see [Delete Inactive Certificate Versions](#) in *Advanced Certificate Rotation with CredHub Maestro*.

Running with old versions of Xenial or Windows stemcells

If you are running old versions of the Xenial and/or Jammy stemcells, you should upgrade the stemcells. If there is some reason you cannot upgrade the stemcells, then you must recreate all VMs by enabling the service tile Recreate all service instances errand. If you are running with Jammy stemcells, you don't need to recreate the VMs.

If the tile does not include this errand, use the procedure here. This procedure propagates the new CA to the BOSH Agent on each VM deployed by the BOSH Director, which enables the new leaf certificates to trust the new CA.

- Go to <https://OPS-MANAGER-FQDN> in a browser, where [OPS-MANAGER-FQDN](#) is the FQDN of your Tanzu Operations Manager deployment.
- Log in to the Tanzu Operations Manager Installation Dashboard.
- Click the BOSH Director tile, then click Director Config.
- Select the Recreate VMs deployed by the BOSH Director check box.

Or you can run:

```
bosh -d SERVICE-INSTANCE-DEPLOYMENT recreate
```

Where [SERVICE-INSTANCE-DEPLOYMENT](#) is the BOSH deployment name for the service instance.

For details, see the Prerequisites section earlier in this topic.

Rotating non-configurable leaf certificates in Tanzu Operations Manager

You can rotate non-configurable Tanzu Operations Manager leaf certificates that are visible to the Tanzu Operations Manager API, whether they are managed and stored by Tanzu Operations Manager directly, or by CredHub at Tanzu Operations Manager request. To rotate all certificates in your Tanzu Operations Manager deployment, see [Rotating CAs and Leaf Certificates](#).

This procedure does not rotate the Tanzu Operations Manager root certificate authority (CA) or other CAs in your deployment. To rotate CAs and leaf certificates, see [Rotating CAs and Leaf Certificates](#).

Prerequisites

This procedure always rotates non-configurable leaf certs with location “ops_manager”. When VMware Tanzu Container Orchestration Engine (formerly TKGI) is not installed, this procedure also rotates all non-configurable leaf certs with location “credhub”.

If you use Container Orchestration Engine, rotate your certificates using the instructions in the [Container Orchestration Engine documentation](#). For certificates that are not rotated by the Container Orchestration Engine procedures, see [Advanced certificate rotation with CredHub Maestro](#).

Procedure

You can set a value to override the duration for certificates in the BOSH Director tile. VMware recommends that you set the value before starting a certificate rotation. For more information, see [Overriding duration for certificates](#).

To rotate non-configurable leaf certificates:

1. Call the Tanzu Operations Manager API [regenerate](#) endpoint by running:

```
curl "https://OPS-MANAGER-FQDN/api/v0/certificate_authorities/active/regenerate" \
-X POST \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-H "Content-Type: application/json" \
-d '{}' \
-i
```

Where:

- [OPS-MANAGER-FQDN](#) is the fully-qualified domain name (FQDN) of your Tanzu Operations Manager deployment.
- [UAA-ACCESS-TOKEN](#) is your UAA access token.

The API returns a successful response:

```
HTTP/1.1 200 OK
```

2. Go to Tanzu Operations Manager Installation Dashboard.
3. If you have any on-demand service tiles installed, do the following for each one:
 1. Click the tile.
 2. Click the Errands tab.

3. Enable the Upgrade All Service Instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.
 4. Click Review Pending Changes.
 5. Click Apply Changes.
4. If you do not have any on demand service tiles installed:
1. Click Review Pending Changes.
 2. Click Apply Changes.

Troubleshooting

The Tanzu Operations Manager API runs CredHub Maestro when rotating certificates. If a certificate rotation API command is unsafe, CredHub Maestro stops the command and returns one or more safety violations.

For example, CredHub Maestro stops a certificate rotation API command if you try to follow certificate rotation steps in the wrong order. Because performing these steps in the wrong order can make your deployment unstable or cause downtime, CredHub Maestro stops the command and returns an error message.

For information about how to troubleshoot safety violation errors that are returned when rotating certificates, see [Troubleshooting CredHub Maestro safety violations during certificate rotation] (troubleshoot-cert-errors.html).

Rotating configurable Tanzu Operations Manager leaf certificates

This topic describes how you can rotate configurable leaf certificates for your Tanzu Operations Manager deployment. To rotate all certificates in your Tanzu Operations Manager deployment, see [Rotating CAs and Leaf Certificates](#).

You can generate configurable certificates and paste them into Tanzu Operations Manager configuration panes where ever needed. Examples include certificates that stop SSL traffic into Tanzu Platform for Cloud Foundry, or authenticate a Single Sign-On service plan to an external SAML server.

To rotate SAML certificates for both Elastic Application Runtime and the Single Sign-On service, see [Rotating identity provider SAML certificates](#).

This procedure does not rotate the Tanzu Operations Manager root certificate authority (CA) or other CAs in your deployment. To rotate CAs and leaf certificates, see [Rotating CAs and Leaf Certificates](#).

Procedure

You can set a value to override the duration for certificates in the BOSH Director tile. VMware recommends that you set the value before starting a certificate rotation. For more information, see [Overriding duration for certificates](#).

To rotate configurable leaf certificates:

1. Go to Tanzu Operations Manager Installation Dashboard.
2. For each certificate you want to rotate:

1. Find the text box where the leaf certificate is configured in the Tanzu Operations Manager UI. You might need to look through multiple configuration panes to identify the location of the certificate configuration text field, and in some cases, the tile might not expose the field in the UI. In that case the value must be updated using the Tanzu Operations Manager API. Use the following fields to identify the location of the certificate configuration text field:
 - The `product_guid` text box in the Tanzu Operations Manager API output can help identify the tile in which the certificate is configured. For example, the prefix `p-bosh-` refers to the BOSH Director tile, and the prefix `cf-` refers to the Elastic Application Runtime tile.
 - The `property_reference` text box in the Tanzu Operations Manager API output can help identify the configuration pane in which the certificate is configured. For example, the `uaa.service_provider_key_credentials` property is configured in the UAA pane of the Elastic Application Runtime tile.
2. Copy a new value for the leaf certificate into the text field or generate a new leaf certificate using the Generate RSA Certificate button.
3. Click Save for each new leaf certificates value you add.
3. Return to Tanzu Operations Manager Installation Dashboard.
4. Click Review Pending Changes.
5. Click Apply Changes.

Certificates with specialized procedures

Some certificates are not rotated when using the standard rotation procedure. The following table lists certificates that use separate, specialized rotation procedures and provides links to documentation and Knowledge Base articles to help you rotate these certificates.

| Certificate | Tile | Procedure |
|---|-----------------------------|---|
| <code>.properties.saml_service_provider_cert</code> | BOSH Director | How to check and rotate Tanzu Operations Manager SAML Certificate before it expires |
| <code>.uaa.service_provider_key_credentials</code> | Elastic Application Runtime | Rotating Identity Provider SAML Certificates |

Troubleshooting

The Tanzu Operations Manager API runs CredHub Maestro when rotating certificates. If a certificate rotation API command is unsafe, CredHub Maestro stops the command and returns one or more safety violations.

For example, CredHub Maestro stops a certificate rotation API command if you try to follow certificate rotation steps in the wrong order. Because performing these steps in the wrong order can make your deployment unstable or cause downtime, CredHub Maestro stops the command and returns an error message.

For information about how to troubleshoot safety violation errors that are returned when rotating certificates, see [Troubleshooting CredHub Maestro safety violations during certificate rotation] (troubleshoot-cert-errors.html).

Rotating identity provider SAML certificates in Elastic Application Runtime

You rotate SAML service provider (SP) credentials in your Tanzu Operations Manager deployment.

SAML SP credentials are one example of configurable certificates in Tanzu Platform for Cloud Foundry. When Elastic Application Runtime is configured to use SAML as an IDP, it uses a configurable certificate authority (CA) certificate to authenticate to an external SAML server. The CA generates ephemeral certificates that Elastic Application Runtime includes in its outbound request message headers. This CA has a two-year expiration period.

In addition, the Single Sign-On service shares the use of Elastic Application Runtime SAML certificates for every SAML external IDP integration, such as trust, partnership, or federation. You must rotate these in lockstep with Elastic Application Runtime. For more information about Single Sign-On, see the [Single Sign-On documentation](#).

This topic also provides an example of how to rotate certificates for each IDP, including temporarily deactivating certificate validation on the IDP side during the rotation.

For more information about rotating SAML certificates in Tanzu Operations Manager, see [How to check and rotate Tanzu Operations Manager SAML Certificate before it expires](#).

Prerequisites

SAML SP credentials are required for your Elastic Application Runtime deployment only if all of these conditions are met:

- You are using Single Sign-On in production for log in to Elastic Application Runtime or you are using the Single Sign-On service for login to apps.
- You are using SAML IDPs for Elastic Application Runtime or Single Sign-On service plans.
- You had Tanzu Operations Manager generate a certificate for you by clicking the Generate RSA Certificate button.
- You are validating the signature of SAML authentication request with your IDP.

Rotate SAML certificates

To regenerate and rotate SAML SP certificates without disrupting Elastic Application Runtime or your apps using the Single Sign-On service:

1. Deactivate certificate validation in your IDP.
2. For Elastic Application Runtime, follow the procedure in the following table that corresponds to your use case. This includes downloading and importing a new certificate and updated SAML metadata in your IDP.

| Solution Name | Procedure |
|--|--|
| CA Single Sign-On aka Symantec SiteMinder | Configure a SAML 2.0 Identity Provider |
| PingFederate | Configuring PingFederate as an Identity Provider |
| Active Directory Federation Services (AD FS) | Configuring AD FS as an Identity Provider |

3. For the Single Sign-On service, follow the procedure in the following table that corresponds to your use case. This includes downloading the SAML SP metadata for each SAML IDP integration.

For example, trust, partnership, or federation, and importing the updated SAML SP metadata in your IDP.

| Solution Name | Procedure |
|--------------------------|--|
| AD FS | Configuring a Single Sign-On Service Provider |
| CA Single Sign-On | Configuring a Single Sign-On Service Provider |
| Okta | Configure Okta as an Identity Provider |
| PingFederate | Configure PingFederate as an Identity Provider |
| Additional Documentation | Integration Guides |

4. Enable certificate validation again in your IDP.

Rotating identity provider SAML certificates for Tanzu Operations Manager

You can rotate SAML service provider (SP) credentials for your Tanzu Operations Manager. This procedure requires Tanzu Operations Manager v3.0.30+LTS-T or higher.

SAML SP credentials are one example of configurable certificates in Tanzu Operations Manager. When Tanzu Operations Manager is configured to use SAML as an IDP, it uses a configurable certificate authority (CA) certificate to authenticate to an external SAML server for both Tanzu Operations Manager and the BOSH Director. This CA has a two-year expiration period.

Some IDPs require you to provide the SAML certificate before allowing authentication. Collaborate with your IDP regarding the new SAML certificate before following this procedure. Failure to do so may result in all SSO users being locked out of the platform.

Follow this procedure:

1. Go to Settings, then SAML Settings.
2. Under the SAML service provider certificate and private key text box, click Change.
3. You can provide your own certificate and private key or you can click Generate RSA Key.
 1. In the Generate RSA Key dialog box, use the FQDN of Tanzu Operations Manager as your domain name.
4. Click Save.
5. Changes to Tanzu Operations Manager's configuration take effect immediately.
6. Apply the changes by deploying the BOSH Director.

Using custom certificate authorities in Tanzu Operations Manager

This topic provides an overview of how you can use custom certificate authorities (CAs) in your Tanzu Operations Manager deployment.

To secure traffic in your Tanzu Operations Manager deployment, you must provide a CA to issue digital certificates. This can be either a Tanzu Operations Manager-generated or custom CA. When you add and activate a new CA, a digital certificate is issued to BOSH Director. BOSH Director then passes the certificate to other components in your Tanzu Operations Manager deployment.

VMware recommends that you supply a CA from a trusted provider when using a production environment. While you can create your own custom CAs, if necessary, a trusted CA is more secure because it has been authenticated by the trusted entities permitted to issue them.

Elliptic Curve Digital Signature Algorithm (ECDSA) certificates are not supported as the root CA in Tanzu Operations Manager.

Add a custom CA

You can add a new custom CA as part of the procedure for rotating CAs and other certificate types in Tanzu Operations Manager. To add and activate a new custom CA in Tanzu Operations Manager, see [Rotating CAs and Leaf Certificates](#).

Retrieving credentials from your Tanzu Operations Manager deployment

This topic describes how your Tanzu Operations Manager deployment stores its credentials, and how you can access them.

- What credentials does Tanzu Operations Manager store?
 - Many Tanzu Operations Manager components use credentials to authenticate connections, and Tanzu Operations Manager installations often have hundreds of active credentials. This includes certificates, virtual machine (VM) credentials, and credentials for jobs running on the VMs.
- Where does Tanzu Operations Manager store these credentials?
 - Tanzu Operations Manager stores credentials in either the Tanzu Operations Manager database or in [BOSH CredHub](#).
- When do I need to access these credentials?
 - You may need to access credentials for Tanzu Operations Manager, Elastic Application Runtime, and service tiles as part of regular administrative tasks in Tanzu Operations Manager, including troubleshooting. Many procedures in this documentation require you to retrieve credentials.
- How can I retrieve credentials?
 - The workflow for retrieving credentials depends on where they are stored. For more information, see the following procedures.

Retrieve credentials stored in BOSH CredHub

To retrieve credentials from CredHub using the Tanzu Operations Manager API:

1. Follow the procedures in [Using the Tanzu Operations Manager API](#) to authenticate and access the Tanzu Operations Manager API.
2. Use the Tanzu Operations Manager API to retrieve a list of deployed products:

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/products" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where **UAA-ACCESS-TOKEN** is the access token recorded in the previous step.

3. In the response to the preceding request, locate the **guid** for the product from which you want to retrieve credentials. For example, if you want to retrieve Elastic Application Runtime credentials, find the **installation_name** starting with **cf-** and copy its **guid**.
4. Run the following **curl** command to list the names of the credentials stored in CredHub for the product you selected. If you already know the name of the credential, you can skip this step.

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/products/PRODUCT-GUID/variables" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where **PRODUCT-GUID** is the value of **guid** from the previous step.

5. Run the following command to view the credential:

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/products/PRODUCT-GUID/variables?name=VARIABLE-NAME" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where **VARIABLE-NAME** is the name of the credential you want to retrieve.

Retrieve credentials stored in the Tanzu Operations Manager database

To retrieve credentials stored in the Tanzu Operations Manager database and not CredHub, use the Tanzu Operations Manager UI or API as outlined in the following procedures.

Retrieve credentials using the Tanzu Operations Manager UI

1. From Tanzu Operations Manager, select the product tile for which you want to retrieve credentials.
2. Click the Credentials tab.
3. Locate the credential that you need and click Link to Credential.

Retrieve credentials using the Tanzu Operations Manager API

1. Follow the procedures in [Using the Tanzu Operations Manager API](#) to authenticate and access the Tanzu Operations Manager API.
2. Use the Tanzu Operations Manager API to retrieve a list of deployed products:

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/products" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where **UAA-ACCESS-TOKEN** is the access token recorded in the previous step.

3. In the response to the preceding request, locate the **guid** for the product from which you want to retrieve credentials. For example, if you want to retrieve Elastic Application Runtime credentials, find the **installation_name** starting with **cf-** and copy its **guid**.
4. Run the following command to list references for the credentials stored in Tanzu Operations Manager for the product you selected. If you already know the reference for the credential, you can skip this step.

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/products/PRODUCT-GUID/credentials" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where **PRODUCT-GUID** is the value of **guid** from the previous step.

5. Run the following command to view the credential:

```
curl "https://OPS-MANAGER-FQDN/api/v0/deployed/products/PRODUCT-GUID/credentials/CREDENTIAL-REFERENCE" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where **CREDENTIAL-REFERENCE** is the name of the credential you want to retrieve.

Changing Tanzu Operations Manager credentials

Tanzu Operations Manager password

1. In a web browser, go to <https://OPS-MANAGER-FQDN> and log in, where **OPS-MANAGER-FQDN** is the Fully Qualified Domain Name of your Tanzu Operations Manager.
2. In Tanzu Operations Manager, go to My Account in the drop-down menu. You can access this at <https://OPS-MANAGER-FQDN/uaa/profile>.
3. Go to Change Password. You can access this at https://OPS-MANAGER-FQDN/uaa/change_password.
4. Enter your current password and a new password.

Tanzu Operations Manager decryption passphrase

You must have the existing passphrase to update the decryption passphrase.

1. In a web browser, go to <https://OPS-MANAGER-FQDN> and log in, where **OPS-MANAGER-FQDN** is the Fully Qualified Domain Name of your Tanzu Operations Manager.
2. In Tanzu Operations Manager, go to Settings in the drop-down menu. You can access this at https://OPS-MANAGER-FQDN/encryption_passphrase/edit.
3. In the **Decryption Passphrase** pane, enter your current decryption passphrase and a new decryption passphrase.
4. Click Update.

S3 compatible blobstore credentials

If you use an S3 compatible blobstore, you can rotate your blobstore credentials from the Director Config pane of the BOSH Director tile. After you enter your new credentials and click Review Pending Changes, then click Apply Changes, BOSH recreates the VMs in your deployment to apply the new credentials.

1. In a web browser, go to <https://OPS-MANAGER-FQDN> and log in, where **OPS-MANAGER-FQDN** is the Fully Qualified Domain Name of your Tanzu Operations Manager.
2. In the Installation Dashboard, click the BOSH Director tile.
3. In the Director Config pane, select Recreate VMs deployed by the BOSH Director check box.
4. Under Blobstore Location, click S3 Compatible Blobstore
5. For Access Key, enter a new access key.

6. For Secret Key, enter a new secret key.
7. Click Review Pending Changes, then click Apply Changes.
8. Deselect the Recreate VMs deployed by the BOSH Director check box.

Using short-lived NATS bootstrap credentials in Tanzu Operations Manager

This topic describes how you can use short-lived NATS bootstrap credentials in your Tanzu Operations Manager deployment.

BOSH NATS is a service that allows the BOSH Director and BOSH Agents in deployed VMs to communicate with each other. The BOSH Director uses BOSH NATS to send commands and information to the BOSH Agent, so that the BOSH Agent can execute tasks or return information about the current state of the VM.

BOSH NATS uses TLS certificates to authenticate communication between the BOSH Director and the BOSH Agent. When VMs are created during a new deployment of Tanzu Operations Manager, BOSH gives each VM a TLS certificate through the metadata service for your IaaS, such as AWS instance user data.

However, any user with non-root access to a VM or process running in a VM can access VM metadata within your IaaS. This could allow a malicious user or process to obtain the TLS certificate for the VM and use it to communicate with the BOSH Director through BOSH NATS.

To increase the security of your Tanzu Operations Manager deployment, you can configure your VMs to use short-lived bootstrap credentials when communicating through BOSH NATS. When you configure your VMs to use short-lived bootstrap credentials, the TLS certificate that BOSH gives to a VM through the metadata service for your IaaS contains credentials that are only to be used during the bootstrap process. Before the bootstrap process finishes, the BOSH Director sends the BOSH Agent a new TLS certificate. From that point forward, BOSH NATS rejects the previous TLS certificate that is stored in the metadata service. Because no user or process can access VMs during the bootstrap process, and apps are installed after the bootstrap process is complete, using short-lived bootstrap credentials ensures that any malicious user or process that obtains the TLS certificate stored in the metadata service cannot use it to access your VMs or apps.

To learn about the stemcells that are compatible with short-lived bootstrap credentials, see [Stemcell compatibility](#).

To configure your VMs to use short-lived bootstrap credentials when communicating through BOSH NATS, see [Configure Short-Lived NATS Bootstrap credentials](#).

Stemcell compatibility

You can configure short-lived bootstrap credentials for VMs using the following stemcells:

- Microsoft Windows 2019.41 and later.
- Xenial 621.171 and later.
- Bionic 1.36 and later.
- Jammy 1.95 and later.

All Jammy stemcell versions are compatible with short-lived bootstrap credentials. However, if you use Jammy 1.94 or earlier, using short-lived bootstrap credentials causes VMs with re-sized disk

allocations to fail when they are re-created. To fix this issue, upgrade your stemcell to Jammy 1.95 or later.

If you configure short-lived bootstrap credentials for VMs using unsupported stemcells, the VMs become unresponsive.

Configure short-lived NATS bootstrap credentials

To configure your VMs to use short-lived bootstrap credentials when communicating through BOSH NATS:

1. In a browser window, go to Tanzu Operations Manager Installation Dashboard.
2. Click the BOSH Director tile.
3. Click Director Config.
4. Select the Enable Short Lived NATS Bootstrap Credentials check box.
5. Click Save.
6. Return to Tanzu Operations Manager Installation Dashboard.
7. Click Review Pending Changes.
8. Click Apply Changes.

This configuration affects all new deployments of Tanzu Operations Manager. If you want to configure VMs in existing Tanzu Operations Manager deployments to use short-lived bootstrap credentials, you must recreate them.

Certificates and credentials managed by CredHub

The following topics describe how you can work with certificates and credentials that are managed by CredHub:

- [CredHub for centralized Tanzu Operations Manager credential management](#)
- [Credential types supported by CredHub](#)
- [Accessing BOSH CredHub with the CredHub CLI](#)
- [Resetting manually-set certificates in BOSH CredHub](#)
- [Getting started with CredHub Maestro](#)
- [Advanced certificate rotation with CredHub Maestro](#)
- [Tanzu Operations Manager tiles that are compatible with CredHub Maestro](#)
- [Troubleshooting CredHub Maestro safety violations](#)

CredHub for centralized Tanzu Operations Manager credential management

For information about the CredHub component, which is designed for centralized credential management in Tanzu Operations Manager, see the [CredHub documentation](#).

Credential types supported by CredHub

For information about the different credential types that CredHub supports, see the [Tanzu Platform for Cloud Foundry documentation](#).

Accessing BOSH CredHub with the CredHub CLI

Tanzu Operations Manager stores some of its internal CA (certificate authority) and non-CA certificates in the BOSH CredHub credentials store. This topic describes how you can use the CredHub CLI to access BOSH CredHub.

For information about BOSH CredHub, see the [CredHub documentation](#).

Procedure

To access the BOSH CredHub credentials store, you must retrieve credentials from the BOSH Director and then use the credentials to log in to CredHub from the Tanzu Operations Manager VM.

1. In the Tanzu Operations Manager Installation Dashboard, click the BOSH Director tile.
2. Click the Credentials tab.
3. In the BOSH Director section, click the BOSH Commandline Credentials link.
4. Record the values for **BOSH_CLIENT** and **BOSH_CLIENT_SECRET**.

For example:

```
{"credential":"BOSH_CLIENT=ops_manager
BOSH_CLIENT_SECRET=abCdE1FgHijKL2m3n-3PqrsT4EUvwXy5
BOSH_CA_CERT=/var/tempest/workspaces/default/root_ca_certificate
BOSH_ENVIRONMENT=10.0.0.5 bosh "}
```

The **BOSH_CLIENT** is the BOSH CredHub client name and the **BOSH_CLIENT_SECRET** is the BOSH CredHub client secret.

5. Follow the procedure in [Gather credential and IP Address information](#) to obtain the information needed to log in to the BOSH Director VM. Record the IP address for the BOSH Director and the Director Credentials.
6. Log in to the Tanzu Operations Manager VM by following the procedure in [Log in to the Tanzu Operations Manager VM with SSH](#).
7. From the Tanzu Operations Manager VM, set the API target of the CredHub CLI to your BOSH CredHub server by running the following command:

```
credhub api \
https://BOSH-DIRECTOR-IP:8844 \
--ca-cert=/var/tempest/workspaces/default/root_ca_certificate
```

Where **BOSH-DIRECTOR-IP** is the IP address of the BOSH Director VM you previously recorded.

For example:

```
$ credhub api \
https://10.0.0.5:8844 \
--ca-cert=/var/tempest/workspaces/default/root_ca_certificate
```

8. Log in to CredHub by running the following command:

```
credhub login \  
--client-name=CREDHUB-CLIENT-NAME \  
--client-secret=CREDHUB-CLIENT-SECRET
```

Where:

- **CREDHUB-CLIENT-NAME** is the value you recorded for **BOSH_CLIENT** earlier in this procedure.
- **CREDHUB-CLIENT-SECRET** is the value you recorded for **BOSH_CLIENT_SECRET** earlier in this procedure.

For example:

```
$ credhub login \  
--client-name=credhub \  
--client-secret=abcdefghijklm123456789
```

Where:

- **CREDHUB-CLIENT-NAME** is the value you recorded for **BOSH_CLIENT** earlier in this procedure.
- **CREDHUB-CLIENT-SECRET** is the value you recorded for **BOSH_CLIENT_SECRET** earlier in this procedure.

Alternatively, to improve security, you can export the client name and secret from environment variables. Using a space before the **export** command prevents the clear text credentials from being saved in the bash history.

```
export CREDHUB_CLIENT=ops_manager CREDHUB_SECRET=abCdE1FgHjKl2m3n-3PqrsT4EUvWxY5  
  
credhub login
```

Resetting manually-set certificates in BOSH CredHub

This topic describes how you can identify and update certificates that were set manually: certificates that an operator added directly to BOSH CredHub through the CredHub CLI instead of using a CredHub-generated certificate. For example, an operator might have provided an existing certificate to BOSH CredHub as described in the [MySQL for Elastic Application Runtime documentation](#).

To ensure that Tanzu Operations Manager does not rotate manually-set certificates alongside CredHub-generated certificates, you must ensure that the **generated** field of all manually-set certificates is set to **false** in CredHub.

In this procedure, use the CredHub CLI to review existing certificates and then check for manually-set certificates. If you find any manually-set certificates that require updates, you can use the CredHub CLI to reset the certificate.

Find BOSH client credentials and BOSH Director IP address

You need the BOSH client name and client secret to use the CredHub CLI.

To find the BOSH CredHub client name and client secret:

1. Go to Tanzu Operations Manager Installation Dashboard.

2. Click the BOSH Director tile.
3. Click the Credentials tab.
4. In the BOSH Director section, go to the BOSH Commandline Credentials row, and click the corresponding link, Link to Credential.

| JOB | NAME | CREDENTIALS |
|---------------|--|------------------------------------|
| BOSH Director | Vm Credentials | Link to Credential |
| | Agent Credentials | Link to Credential |
| | Director Credentials | Link to Credential |
| | Postgres Credentials | Link to Credential |
| | Blobstore Credentials | Link to Credential |
| | Health Monitor Credentials | Link to Credential |
| | Nats Sync Credentials | Link to Credential |
| | Count Cores Credentials | Link to Credential |
| | Uaa Admin User Credentials | Link to Credential |
| | Uaa Login Client Credentials | Link to Credential |
| | Uaa Jwt Key | Link to Credential |
| | Bbr Ssh Credentials | Link to Credential |
| | Uaa Bbr Client Credentials | Link to Credential |
| | Bosh Commandline Credentials | Link to Credential |
| | Blobstore Certificate | Link to Credential |
| | Director Metrics Server Client Certificate | Link to Credential |

5. Record the values for **BOSH_CLIENT** and **BOSH_CLIENT_SECRET**. Here is an example of the credentials page:

```
{
  "credential": {
    "BOSH_CLIENT": "ops_manager",
    "BOSH_CLIENT_SECRET": "abcdefghijklmnopqrstuvwxyz123456789",
    "BOSH_CA_CERT": "/var/tempest/workspaces/default/root_ca_certificate",
    "BOSH_ENVIRONMENT": "10.0.0.5 bosh"
  }
}
```

6. Click the Status tab.
7. In the BOSH Director row, locate and record the IP address of the BOSH Director. For example, **10.0.0.5**.

Target and log in to the BOSH CredHub API server

You can run CredHub CLI commands on the Tanzu Operations Manager VM.

To log in to and access the CredHub API:

1. SSH into the Tanzu Operations Manager VM. For IaaS-specific instructions on how to SSH, see [Log in to the Tanzu Operations Manager VM with SSH](#).
2. Set the CredHub API target of the CLI by running:

```
credhub api https://BOSH-DIRECTOR:8844 --ca-cert=/var/tempest/workspaces/default/root_ca_certificate
```

Where **BOSH-DIRECTOR** is the IP address of the BOSH Director VM.

3. Log in to CredHub by running:

```
credhub login --client-name=CREDHUB-CLIENT-NAME --client-secret=CREDHUB-CLIENT-SECRET
```

Where:

- **CREDHUB-CLIENT-NAME** is the value you recorded for **BOSH_CLIENT** in [Find BOSH Client Credentials and BOSH Director IP Address](#).
- **CREDHUB-CLIENT-SECRET** is the value you recorded for **BOSH_CLIENT_SECRET** in [Find BOSH Client Credentials and BOSH Director IP Address](#).

Check certificates in CredHub

After you log in to CredHub, you can query the CredHub API on the command line.

To retrieve a list of CredHub certificates:

1. Retrieve a list of all the certificates stored in CredHub by running:

```
credhub curl -p "/api/v1/certificates" -X GET | jq .
```

2. In the JSON output, check the value of the **generated** field of each certificate. For example:

```
{
  "certificates": [
    {
      "name": "/services/tls_ca",
      "id": "1b378f9c-d3dc-4519-9cde-20397f9d8d6b",
      "versions": [
        {
          "id": "3e41bba8-1373-4b51-b774-d65d9a0b87b4",
          "expiry_date": "2020-09-19T14:36:36Z",
          "transitional": false,
          "certificate_authority": true,
          "self_signed": true,
          "generated": null
        }
      ]
    }
  ]
}
```

3. For each certificate where **generated** value is **null**, follow the procedure in [Reset a certificate in CredHub](#).

Reset a certificate in CredHub

To reset a certificate in CredHub:

1. Retrieve a manually-set certificate by running:

```
credhub get -n SET-CERTIFICATE -k certificate > /tmp/certificate.yml
credhub get -n SET-CERTIFICATE -k ca > /tmp/ca.yml
credhub get -n SET-CERTIFICATE -k private_key > /tmp/private_key.yml
```

Where **SET-CERTIFICATE** is the value of the **name** field in the manually-set certificate you want to retrieve.

2. Reset the certificate depending on the certificate type.

- If the retrieved certificate is a root CA certificate, run:

```
credhub set -n SET-CERTIFICATE -t certificate -r /tmp/ca.yml -c /tmp/certificate.yml -p /tmp/private_key.yml
```

Where **SET-CERTIFICATE** is the name of the manually-set certificate you want to reset.

- If the retrieved certificate is an intermediate or leaf certificate, run:

```
credhub set -n SET-CERTIFICATE -t certificate -c /tmp/certificate.yml -p /tmp/private_key.yml -m SIGNING-CA
```

Where:

- **SET-CERTIFICATE** is the name of the manually-set certificate you want to reset.
- **SIGNING-CA** is the name of the signing root CA certificate.

Getting started with CredHub Maestro

This topic describes how you can set up and use CredHub Maestro, a command-line interface (CLI) that rotates certificates in CredHub.

Using CredHub Maestro, you can:

- Determine if any of your CredHub certificates are expiring soon.
- Rotate CredHub certificates.
- Clean up inactive certificate versions so that CredHub does not run out of disk space.

Set up environment variables

This section describes setting up the environment variables you need to use CredHub Maestro.

Required environment variables

To set the CredHub environment variables you need for CredHub Maestro, follow the guidance in the following table:

To set BOSH environment variables, see [Advanced troubleshooting with the BOSH CLI](#).

| Environment Variable | Description |
|----------------------|--|
| BOSH_ENVIRONMENT | BOSH Director URL or IP address |
| BOSH_CLIENT | Name of BOSH client |
| BOSH_CLIENT_SECRET | BOSH client secret |
| BOSH_CA_CERT | Path or value of BOSH Director trusted certificate authority (CA) |
| CREDHUB_SERVER | URL of BOSH Director CredHub server. This must be BOSH_ENVIRONMENT:8844 . |

| Environment Variable | Description |
|----------------------|--|
| CREDHUB_CLIENT | Name of CredHub client. This is the same as <code>BOSH_CLIENT</code> . |
| CREDHUB_SECRET | CredHub client secret. This is the same as <code>BOSH_CLIENT_SECRET</code> . |
| CREDHUB_CA_CERT | Path or value of CredHub trusted CA certificate. This is the same as <code>BOSH_CA_CERT</code> . |

Using CredHub Maestro from outside your foundation

To use CredHub Maestro from outside your foundation, you must also set the following variables in addition to the required environment variables:

| Environment Variable | Description |
|----------------------|---|
| BOSH_ALL_PROXY | SOCKS5 proxy address of BOSH Director jumpbox |
| CREDHUB_PROXY | SOCKS5 proxy address of CredHub jumpbox |

Optional environment variables

You can also set the following optional environment variables:

| Environment Variable | Description |
|----------------------|--|
| MAESTRO_DEBUG | Enable debug logging |
| MAESTRO_OUTPUT_JSON | Print all CredHub Maestro command output as JSON |

Commands

This section describes the commands you can use in CredHub Maestro.

To review a list of commands in CredHub Maestro, run `maestro --help`.

maestro list

`maestro list` displays expiry information for all actively-deployed certificates in CredHub.

You can use the following flags with `maestro list`:

- `expires-within`: Filter certificates by expiry window. Valid units are `d` for days, `w` for weeks, `m` for months, and `y` for years.
- `deployment-name`: Filter certificates by deployment name.
- `name`: Show metadata for a single certificate.
- `include-all`: Include inactive certificates in response.
- `ca-only`: Filter certificates to include only CAs.
- `leaf-only`: Filter to include only leaf certificates.
- `generated-only`: Include only certificates that have been generated by CredHub.

The `deployment-name` and `include-all` flags are mutually exclusive.

maestro topology

`maestro topology` displays expiry information for all actively-deployed certificates in CredHub.

You can use the following flags with `maestro topology`:

- **name:** Display topology for a single certificate.
- **include-all:** Display topology for both active and inactive certificates.

maestro regenerate

maestro regenerate regenerates CredHub-generated certificates. By default, certificates that have been manually-set in CredHub are not regenerated.

maestro regenerate ca

maestro regenerate ca regenerates actively-deployed CAs and adds transitional flags to these new CA versions. The transitional status indicates that these new CA versions will not yet sign any leaf certificates, though they will be trusted by clients.

You can use the following flags with **maestro regenerate ca**:

- **all:** Regenerate all actively-deployed CAs.
- **name:** Regenerate a single CA by name.
- **dry-run:** List CAs to be regenerated.
- **exclude:** Comma-separated list of CAs to exclude from being regenerated. This includes all leaf certificates of excluded CAs.
- **force:** Regenerate both CredHub-generated and manually-set CAs.
- **show-excluded:** Show all certificates excluded from regeneration.

maestro regenerate leaf

maestro regenerate leaf regenerates actively-deployed leaf certificates.

You can use the following flags with **maestro regenerate leaf**:

- **all:** Regenerate all actively-deployed leaf certificates.
- **name:** Regenerate single leaf by name.
- **signed-by:** Regenerate all actively-deployed leaf certificates signed by a specific CA.
- **dry-run:** List leaf certificates to be regenerated.
- **exclude-signed-by:** Exclude all leaf certificates signed by a list of CAs from being regenerated.
- **force:** Regenerate both CredHub-generated and manually-set leaf certificates.
- **show-excluded:** Show all certificates excluded from regeneration.

maestro update-transitional

maestro update-transitional updates the transitional flag for CAs.

maestro update-transitional signing

maestro update-transitional signing updates the transitional flag for the version of the actively-deployed CA that signed deployed leaf certificates.

You can use the following flags with **maestro update-transitional signing**:

- **all:** Update the transitional flag for all actively-deployed CAs.
 - **name:** Update the transitional flag for a single CA.
 - **dry-run:** List CAs to be updated.
 - **exclude:** Exclude a list of CAs and their leaf certificates from being updated.
 - **show-excluded:** Show all certificates excluded from update.
-

maestro update-transitional remove removes the transitional flag for all versions of the actively-deployed CA.

You can use the following flags with **maestro update-transitional remove**:

- **all**: Update the transitional flag for all actively-deployed CAs.
 - **name**: Update the transitional flag for a single CA.
 - **dry-run**: List CAs to be updated.
 - **exclude**: Exclude a list of CAs and their leaf certificates from being updated.
 - **show-excluded**: Show all certificates excluded from update.
-

maestro garbage-collect

maestro garbage-collect deletes inactive certificate versions.

maestro garbage-collect leaf

maestro garbage-collect leaf deletes inactive leaf certificate versions that are older than the active certificate version.

You can use the following flags with **maestro garbage-collect leaf**:

- **all**: Remove all inactive versions of deployed leaf certificates.
 - **name**: Remove all inactive versions of a single leaf certificate.
 - **dry-run**: List leaf certificate versions to be removed.
 - **force**: Remove all inactive leaf certificates.
-

maestro garbage-collect ca

maestro garbage-collect ca deletes inactive CA versions that are older than the active CA version.

You can use the following flags with **maestro garbage-collect ca**:

- **all**: Remove all inactive versions of deployed CAs.
 - **name**: Remove all inactive versions of a single CA.
 - **dry-run**: List CA versions to be removed.
 - **force**: Remove all inactive CAs. Note that running garbage collection with the force flag on a CA that has an actively deployed transitional version will result in the inability to regenerate that CA until the transitional flag is removed.
-

Safety checks

CredHub Maestro performs basic safety checks when rotating certificates to prevent unsafe operations.

To skip CredHub Maestro safety checks, you can pass the **--skip-safety-check** flag to CredHub Maestro certificate rotation commands.

For more information, see [Troubleshooting CredHub Maestro Safety Violations During Certificate Rotation](#).

Advanced certificate rotation with CredHub Maestro

You can rotate certificates and certificate authorities (CAs) in BOSH CredHub using CredHub Maestro. This topic only covers rotating certificates that are managed by CredHub.

For more information about setting up and using CredHub Maestro, see [Getting started with CredHub Maestro](#).

When to use CredHub Maestro for certificate rotation

VMware recommends that you use the Tanzu Operations Manager API to perform bulk rotation of certificates. For more information, see [Overview of Certificate rotation](#).

However, there are some cases when you cannot use the Tanzu Operations Manager API to do bulk certificate rotation. In these cases, you can use CredHub Maestro to rotate certificates.

The following describes when to use CredHub Maestro instead of Tanzu Operations Manager API for certificate rotation:

- You need to rotate one or more specific CredHub managed certificates as part of a certificate rotation repair process due to errors returned by the standard procedure.
- If you use VMware Tanzu Container Orchestration Engine (formerly TKGI), rotate your certificates using the instructions in the [Container Orchestration Engine documentation](#). Use the following instructions for only the certificates that are not rotated by the Container Orchestration Engine procedures.

Prerequisites

Before you rotate CredHub-managed certificates with CredHub Maestro:

- To ensure that there is no app availability downtime during a rotation, VMware recommends that you scale up all instance groups and use an external blobstore.
- Check which tiles in your installation support certificate rotation with CredHub Maestro. Not every tile currently supports certificate rotation, so there is a potential for downtime. For more information about which tiles support certificate rotation, see [CredHub Maestro tile compatibility](#).

Check expiration dates and certificate types

This section describes how to manually check the expiration dates of the CAs and leaf certificates managed by CredHub.

After you identify the types of certificates that expire soon, you can determine which certificate rotation procedure to follow.

To check certificate expiration dates and types:

1. Retrieve a list of certificates expiring within a given timeframe by running:

```
maestro list --expires-within TIME-PERIOD
```

Where **TIME-PERIOD** is the unit for which you want to check the expiry window. Valid units are **d** for days, **w** for weeks, **m** for months, and **y** for years.

2. After you identify the list of certificates that expire soon, follow the procedure in [Rotate certificates](#).

Rotate certificates

This section explains how to rotate leaf certificates and CAs stored in CredHub. There are different rotation procedures for each type of certificate that requires rotation.

You can determine which rotation procedure to follow based on the types of certificates in your foundation.

To rotate certificates, use one of the following procedures:

- If multiple CAs are expiring soon, see [Rotate all CAs and Leaf Certificates](#).
- If only leaf certificates are expiring, see [Rotate all Leaf Certificates](#).
- To rotate only a subset of CAs, see [Rotate a single CA and the Leaf Certificates](#).
- To rotate the Services TLS CA, see [Rotate the Services TLS CA and the Leaf Certificates](#).
- To rotate CredHub-set CAs, see [Rotate CredHub-set CAs](#).
- To rotate CredHub-set leaf certificates, see [Rotate CredHub-set Leaf Certificates](#).

All CredHub Maestro rotation commands have a dry-run flag you can use to preview the certificates to be modified.

Safety checks

CredHub Maestro performs basic safety checks when rotating certificates to prevent unsafe operations.

To skip CredHub Maestro safety checks, you can pass the `--skip-safety-check` flag to CredHub Maestro certificate rotation commands.

For more information, see [Troubleshooting CredHub Maestro safety violations during certificate rotation](#).

Rotate all CAs and leaf certificates

This section describes the procedure for bulk rotating all CAs and leaf certificates using CredHub Maestro.

VMware recommends that you use the Tanzu Operations Manager API to perform bulk rotation of certificates unless you meet the conditions described in [When to use CredHub Maestro for Certificate Rotation](#when-to-use). If your deployment includes VMware Tanzu Container Orchestration Engine (formerly TKGI), using the `--all` option can damage your deployment. You can rotate individual CA certificates and their leaf certs instead. For TKGI-specific certificates, follow the certificate rotation instructions at [Rotating Certificates for Container Orchestration Engine](#).

If any of your deployments use the Services TLS CA:

- Run all CredHub Maestro commands except `maestro regenerate leaf` with `--exclude /services/tls_ca`.
- Run `maestro regenerate leaf` with `--exclude-signed-by /services/tls_ca`.

The process for rotating the Services TLS CA requires additional steps to ensure that there is no downtime. To rotate the Services TLS CA, see [Rotate the Services TLS CA and its Leaf Certificates](#).

If you have enabled the [Automatic rotation of BOSH DNS certificates](#) feature:

- Run all CredHub Maestro commands except `maestro regenerate leaf` with `--exclude /opsmgr/bosh_dns/tls_ca`.

To rotate all CAs and leaf certificates in CredHub:

1. Regenerate all CAs. This creates a new version of every CA and marks these new CA versions as transitional. The transitional status indicates that these new CA versions do not yet sign any leaf certificates, although they are trusted by clients. Run:


```
maestro regenerate ca --all
```

2. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes.
3. On the Review Pending Changes pane, select the Select All Products check box.
4. For each service tile you have deployed:
 1. Verify that the Recreate all service instances errand is unselected. You do not need to run this errand because, when run by itself, CredHub Maestro does not update BOSH Agent certificates.
 2. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.

The names of the Recreate all service instances and the Upgrade all service instances errands might be slightly different between services.

5. Click Apply Changes.

3. Mark the signing version of each CA as transitional. Run:

```
maestro update-transitional signing --all
```

4. Regenerate all leaf certificates. Run:

```
maestro regenerate leaf --all
```

5. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes.
3. On the Review Pending Changes pane, select the Select All Products check box.
4. For each service tile you have deployed:
 1. Verify that the Recreate all service instances errand is unselected. You do not need to run this errand because when run by itself, CredHub Maestro does not update BOSH Agent certificates.
 2. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.

The names of the Recreate all service instances and the Upgrade all service instances errands might be slightly different between services.

5. Click Apply Changes.

6. Remove the transitional flag on all certificate versions. This action removes the old, inactive CA versions on the next deploy. Run:

```
maestro update-transitional remove --all
```

7. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes.
3. On the Review Pending Changes pane, select the Select All Products check box.
4. For each service tile you have deployed:
 1. Verify that the Recreate all service instances errand is unselected. You do not need to run this errand because when run by itself, CredHub Maestro does not update BOSH Agent certificates.
 2. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.

The names of the Recreate all service instances and the Upgrade all service instances errands might be slightly different between services.

5. Click Apply Changes.

Rotating all leaf certificates

This section describes the procedure for bulk rotating all leaf certificates using CredHub Maestro.

VMware recommends that you use the Tanzu Operations Manager API to perform bulk rotation of certificates if you don't meet the conditions described in [When to use CredHub Maestro for Certificate Rotation](#).

If your deployment includes Container Orchestration Engine, using the `--all` option can damage your deployment. You can rotate leaf certs signed by individual CAs instead. For TKGi-specific certificates, follow the certificate rotation instructions at [Rotating Certificates for Container Orchestration Engine](#).

To rotate all leaf certificates in CredHub:

1. Regenerate all leaf certificates. Run:

```
maestro regenerate leaf --all
```

2. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes.
3. On the Review Pending Changes pane, select the Select All Products check box.
4. For each service tile you have deployed:
 1. Verify that the Recreate all service instances errand is unselected. You do not need to run this errand because, when run by itself, CredHub Maestro does not update BOSH Agent certificates.
 2. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.

The names of the Recreate all service instances and the Upgrade all service instances errands might be slightly different between services.

5. Click Apply Changes.

Rotate a single CA and its leaf certificates

This section describes the procedure for rotating a single CA and its leaf certificates.

To rotate a single CA and its leaf certificates:

1. Regenerate the CA you want to rotate. This creates a new version of this CA. Run:

```
maestro regenerate ca --name "CA-NAME"
```

Where **CA-NAME** is the name of the CA you want to rotate. The CA name you provide must be the full path to the credential.

2. Redeploy all deployments that use the CA.
3. Mark the signing version of the CA as transitional. Run:

```
maestro update-transitional signing --name "CA-NAME"
```

Where **CA-NAME** is the name of the CA you want to mark as transitional.

4. Regenerate all leaf certificates signed by the CA. Run:

```
maestro regenerate leaf --signed-by "CA-NAME"
```

Where **CA-NAME** is the name of the CA for which you want to regenerate leaf certificates.

5. Redeploy all deployments that use the CA.
6. Remove the transitional flag from the CA. This removes the old, inactive CA version on the next deploy. Run:

```
maestro update-transitional remove --name "CA-NAME"
```

Where **CA-NAME** is the name of the CA you want to remove.

7. Redeploy all deployments that use the CA.

Rotate the Services TLS CA and its leaf certificates

You can set a value to override the duration for certificates in the BOSH Director tile. VMware recommends that you set the value before starting a certificate rotation. For more information, see [Overriding duration for certificates](#).

This section describes the procedure for rotating the Services TLS CA and its leaf certificates.

Before you begin this rotation procedure, ensure that you have the following:

- Only one Tanzu Operations Manager foundation and not a multi-site or Wide Area Network (WAN) setup. For a multi-site or WAN setup, you must perform the Apply Changes steps across all your sites or WAN.
- All certificates must include Subject Alternate Name (SAN) and Common Name (CN) entries.

- Your services are configured for high availability to minimize potential app impact during CA rotation. Rotating the Services TLS CA requires upgrading all on-demand service instances, which can cause service downtime.
- You must know whether your deployment contains non-Java MySQL apps that rely on `VCAP_SERVICES` for TLS. You must rebind and restage these apps during the procedure, which can require additional time and coordination for the rotation. You do not need to rebind and restage MySQL apps that use `jdbcURL` or consume CA certificates from `/etc/ssl/certs/ca-certificate.crt`. For more information about determining which apps are affected, see [Rotating /services/tls_ca certificate using CredHub transitional certificate rotation feature](#).

To rotate the Services TLS CA and its leaf certificates for Tanzu Gemfire, WAN configuration only

Follow the procedure in [Rotating Certificates \(Tanzu Gemfire WAN Setups only\)](#).

To rotate the Services TLS CA and its leaf certificates (all other configurations)

1. Check the [CredHub Maestro tile compatibility](#) document to ensure that all service tile versions currently deployed are compatible with Maestro.
If you have any service tiles that are not compatible with CredHub Maestro, follow their rotation procedure. Do not follow this procedure. If you do, you might experience extended downtime or data loss.
2. Regenerate the Services TLS CA and mark the latest version of the CA as transitional. This creates a new version of the Services TLS CA and indicates that the latest version is inactive, so that all leaf certificates trust the new CA.

1. On the Services TLS CA, see if your Services TLS CA was CredHub-set. Run:

```
maestro list
```

After you run this command, you see one Services TLS CA listed per deployment, but it is the same certificate.

2. If the Services TLS CA was CredHub-set, run:

```
maestro regenerate ca --name "/services/tls_ca" --force
```

3. For deployments that use Tanzu for MySQL, see the [Tanzu for MySQL documentation](#) to add the new Services TLS CA to the Trusted Certificates of the BOSH Director.

3. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. On the Review Pending Changes page, select Select All Products check box.
3. For each service tile, enable the Upgrade all service instances errand.

The name of the **Upgrade all service instances** errand might be slightly different between services.

4. Click Apply Changes.

4. Rebind and restage:

- For Tanzu GemFire apps, rebind and restage the apps.
- For deployments that use Tanzu for MySQL, rebind and restage all non-Java MySQL apps that rely on `VCAP_SERVICES` for TLS.

The bg-restage plug-in enables zero-downtime blue-green restaging without access to the app source code. For more information, see [bg-restage](#) on GitHub.

5. Mark the signing version of the Services TLS CA as transitional. Run:

```
maestro update-transitional signing --name "/services/tls_ca"
```

6. Regenerate all service instance certificates signed by the Services TLS CA. Run:

```
maestro regenerate leaf --signed-by "/services/tls_ca"
```

7. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. On the Review Pending Changes page, select the Select All Products check box.
3. For each service tile, enable the Upgrade all service instances errand.

The name of the Upgrade all service instances errand might be slightly different between services.

4. Click Apply Changes.
8. Rebind and restage the following apps:
 - For Tanzu GemFire apps, rebind and restage the apps.
 - For deployments that use Tanzu for MySQL, rebind and restage all non Java MySQL apps that rely on `VCAP_SERVICES` for TLS.
9. Remove the transitional flag from the Services TLS CA. This removes the old, inactive version of the Services TLS CA on the next deployment. Run:

```
maestro update-transitional remove --name "/services/tls_ca"
```

10. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. On the Review Pending Changes page, select the Select All Products check box.
3. For each service tile, enable the Upgrade all service instances errand.
The name of the Upgrade all service instances errand might be slightly different between services.
4. Click Apply Changes.

11. For the following apps:

- For Tanzu GemFire apps, rebind and restage the apps.
- For deployments that use Tanzu for MySQL, rebind and restage all non Java MySQL apps that rely on `VCAP_SERVICES` for TLS.

Rotate CredHub-set CAs

This section describes the procedure for rotating CAs that are manually set in CredHub.

To rotate a CredHub-set CA:

1. Create a new CA using the same parameters you used for the original CA.
2. Use the CredHub CLI or CredHub API to set the new CA at the same path as the original. To download and install the CredHub CLI, see the [CredHub CLI repository](#) on GitHub. For more information about the CredHub API, see the [CredHub API documentation](#).
3. Find out which tiles or service instances are using the rotated certificate.
 1. Get the list of deployment names using one of the following methods:
 - To get a list of deployments using the CA or its leaf certificates, run the following command:


```
maestro --json ls | jq -r '.metadata[] | select(.name == "/PATH/TO/CA/IN/CREDHUB" or .signed_by == "/PATH/TO/CA/IN/CREDHUB") | .deployment_name' | sort | uniq
```
 - To verify the deployments manually:
 1. Use Maestro to get a list of all certificates. Run:


```
maestro ls
```
 2. Locate the output entries whose name or `signed_by` field matches the CredHub path of the CA that was rotated. Certificates might be listed multiple times if they are associated with multiple deployments.
 3. For each output entry, look at the `deployment_name` key. This corresponds to a tile that you must redeploy.
 2. Entries with a deployment name like `service-instance-*` are certificates associated with a service instance created by a service tile. You must redeploy the corresponding service tile and enable its Upgrade all service instances errand.
4. Redeploy:
 1. Return to the Tanzu Operations Manager Installation Dashboard.
 2. Click Review Pending Changes.
 3. On the Review Pending Changes page, select the check box for each of the products using the rotated certificates. If you are unsure, select the Select All Products check box.
 4. For each service tile you are deploying:
 1. Verify that the Recreate all service instances errand is unselected. You do not need to run this errand because, when run by itself, CredHub Maestro does not update BOSH Agent certificates.
 2. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.

The names of the Recreate all service instances and the Upgrade all service instances errands might be slightly different between services.
 5. Click Apply Changes.
5. Mark the signing version of the single CA as transitional. Run:


```
maestro update-transitional signing --name "CA-NAME"
```
6. Regenerate all leaf certificates that are signed by the CA you are rotating. Run:

```
maestro regenerate leaf --signed-by "CA-NAME"
```

You might need to use the `--force` flag when regenerating leaf certificates if they were set by CredHub. Otherwise, you must provide a new leaf certificate at the same path.

7. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes.
3. On the Review Pending Changes page, select the check box for all products using the rotated certificates. If you are unsure, select the Select All Products check box.
4. For each service tile you are deploying:
 1. Verify that the Recreate all service instances errand is unselected. You do not need to run this errand because, when run by itself, CredHub Maestro does not update BOSH Agent certificates.
 2. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.

The names of the Recreate all service instances and the Upgrade all service instances errands might be slightly different between services.

5. Click Apply Changes.
8. Remove the transitional flag from the single CA. This removes the old, inactive CA version on the next deploy. Run:

```
maestro update-transitional remove --name "CA-NAME"
```

9. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes.
3. On the Review Pending Changes page, select the check box for all products using the rotated certificates. If you are unsure, select the Select All Products check box.
4. For each service tile you are deploying:
 1. Verify that the Recreate all service instances errand is unselected. You do not need to run this errand because, when run by itself, CredHub Maestro does not update BOSH Agent certificates.
 2. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.

The names of the Recreate all service instances and the Upgrade all service instances errands might be slightly different between services.

5. Click Apply Changes.

Rotate CredHub-set leaf certificates

This section describes the procedure for rotating leaf certificates that are manually set in CredHub.

To rotate a CredHub-set leaf certificate:

1. Create a new leaf certificate using the same parameters you used for the original leaf certificate.
2. Use the CredHub CLI or CredHub API to set the new leaf certificate at the same path as the original. To download and install the CredHub CLI, see the [CredHub CLI](#) repository on GitHub. For more information about the CredHub API, see the [CredHub API documentation](#).
3. Verify which tiles or service instances are using the rotated certificate.

1. Get the list of deployment names using one of the following methods:

- To get a list of deployments using the CA or its leaf certificates, run:

```
maestro --json ls | jq -r '.metadata[] | select(.name == "/PATH/TO/CERTIFICATE/IN/CREDHUB") | . deployment_name' | sort | uniq
```

- To verify the deployments manually:

1. Use Maestro to get a list of all certificates by running:

```
maestro ls
```

2. Locate the output entries whose name matches the CredHub path of the certificate that was rotated. Note that a certificate might be listed multiple times if it is associated with multiple deployments.

2. For each output entry, look at the `deployment_name` key. This corresponds to a tile that you must redeploy. Entries with a deployment name like `service-instance-*` are certificates associated with a service instance created by a service tile. You must redeploy the corresponding service tile and enable its Upgrade all service instances errand.

4. Redeploy:

1. Return to the Tanzu Operations Manager Installation Dashboard.
2. Click Review Pending Changes.
3. On the Review Pending Changes page, select the check box for all products using the rotated certificate. If you are unsure, select the Select All Products check box.
4. For each service tile you are deploying:
 1. Verify that the Recreate all service instances errand is unselected. You do not need to run this errand because when run by itself, CredHub Maestro does not update BOSH Agent certificates.
 2. Enable the Upgrade all service instances errand. Running this errand is necessary to push CredHub certificate updates to each service instance.

The names of the Recreate all service instances and the Upgrade all service instances errands might be slightly different between services.

5. Click Apply Changes.

Delete inactive certificate versions

Over the course of many rotations, it is possible for the CredHub database to fill up. In order to prevent this, you can periodically use CredHub Maestro's garbage collection functionality to delete old, unused certificate versions.

You can run garbage collect on leaf certificates and CAs. Generally, deleting leaf certificates is enough to free up space in the database.

To see if you need to run garbage collect, do one of the following:

- Manually verify the BOSH Director's VM vitals through your IaaS provider.
- SSH into the BOSH Director and run `df -h /var/vcap/store`.

To enhance the safety of the garbage collect command, CredHub Maestro only deletes versions of a certificate older than the currently active version. To delete all inactive versions of CAs or leaf certificates, add the `-force` flag to the garbage collect command. VMware does not recommend using the `-force` flag during a rotation.

Delete all leaf certificates

To delete all leaf certificates:

1. Determine which certificate versions you want to delete by running:

```
maestro garbage-collect leaf --all --dry-run
```

2. Delete the certificate versions by running:

```
maestro garbage-collect leaf --all
```

Delete all CAs

To delete all CAs:

1. Determine which CA versions you want to delete by running:

```
maestro garbage-collect ca --all --dry-run
```

2. Delete the CA versions by running:

```
maestro garbage-collect ca --all
```

Delete a single leaf certificate

To delete a single leaf certificate:

1. Determine which certificate versions you want to delete by running:

```
maestro garbage-collect leaf --name "LEAF-CERTIFICATE-NAME" --dry-run
```

2. Delete the certificate versions by running:

```
maestro garbage-collect leaf --name "LEAF-CERTIFICATE-NAME"
```

Delete a single CA

To delete a single CA:

1. Determine which CA you want to delete by running:

```
maestro garbage-collect ca --name "CA-NAME" --dry-run
```

Where **CA-NAME** is the name of the CA you want to delete.

2. Delete the CA by running:

```
maestro garbage-collect ca --name "CA-NAME"
```

Where **CA-NAME** is the name of the CA you want to delete.

Tanzu Operations Manager tiles that are compatible with CredHub Maestro

This topic describes which Tanzu Operations Manager tiles are compatible with CredHub Maestro. A compatible tile has certificates that you can rotate using CredHub Maestro.

For more information about CredHub Maestro, see [Getting started with CredHub Maestro](#) and [Advanced certificate rotation with CredHub Maestro](#).

If you see out-of-date or incorrect information on this page, please contact the CredHub team or log a GitHub issue.

| Product name | Product slug | Minimum required version | Compatible | Notes |
|--|---------------------------|--------------------------|------------|---|
| VMware Tanzu Container Orchestration Engine (formerly TKGI) v1.9 and later | pivotal-container-service | n/a | Yes | <p>If a multi-runtime Tanzu Operations Manager deployment includes Container Orchestration Engine v1.9 or later, you must use CredHub Maestro to rotate the following certificates:</p> <ul style="list-style-type: none">• The Container Orchestration Engine certificate authorities (CAs) <code>kubo_ca_2018</code> and <code>etcd_ca_201</code>, as well as their leaf certificates. For more information, see Rotate Cluster-Specific Certificates (Non-NSX-T) in <i>Rotating Cluster Certificates</i> in the Container Orchestration Engine documentation.• Cluster-specific <code>tls-nsx-t</code> certificates. To rotate these certificates, contact Support.• CredHub-managed certificates for all other runtimes. For more information, see When to use CredHub Maestro for certificate rotation in <i>Advanced certificate rotation with CredHub Maestro</i>. |
| Container Orchestration Engine v1.8 and earlier | pivotal-container-service | n/a | No | <p>You cannot use CredHub Maestro to rotate certificates for Container Orchestration Engine v1.8 and earlier. However, if a multi-runtime Tanzu Operations Manager deployment includes Container Orchestration Engine v1.8 or earlier, you must use CredHub Maestro to rotate CredHub-managed certificates for all other runtimes. For more information, see When to Use CredHub Maestro for Certificate Rotation in <i>Advanced Certificate Rotation with CredHub Maestro</i>.</p> |
| File Integrity Monitoring | p-fim | v2.0.0 | Yes | |

| Product name | Product slug | Minimum required version | Compatible | Notes |
|---|---------------------------------|---|------------|--|
| IPsec Encryption | ipsec | n/a | No | IPsec uses its own rotation method. For more information, see Rotating Active IPsec Certificates in the IPsec documentation. |
| Metric Store | p-metric-store | v1.3.1 | Yes | |
| Metrics Forwarder | p-metrics-forwarder | v1.11.4-build.20 | Yes | |
| App Metrics | apmPostgress | v1.6.2-build.1 | Yes | |
| Monitoring indicator protocol | p-monitoring-indicator-protocol | v1.0.0-beta4.1 | Yes | |
| Tanzu for MySQL | pivotal-mysql | v2.8 | Yes | |
| Event alerts | p-event-alerts | v2.9.1-build.1 | Yes | |
| Anti-Virus Scanning | p-antivirus | v2.2.2 | Yes | |
| Anti-Virus Mirror | p-antivirus-mirror | v2.2.2 | Yes | |
| Tanzu Platform for Cloud Foundry | cf | v2.7.21 and later are compatible, except for v2.8.0 and v2.8.1. | Yes | |
| Elastic Application Runtime Windows add-on | pas-windows | v2.7.17 and later are compatible, except for v2.8.0 and v2.8.1. | Yes | |
| Isolation Segment | p-isolation-segment | v2.7.21 and later are compatible, except for v2.8.0 and v2.8.1. | Yes | |
| Small Footprint Elastic Application Runtime | cf | v2.7.21 and later are compatible, except for v2.8.0 and v2.8.1. | Yes | |
| Tanzu Gemfire | p-cloudcache | v1.11 | Yes | |

| Product name | Product slug | Minimum required version | Compatible | Notes |
|---|---------------------------------|--------------------------|------------|-------|
| Compliance Scanning | p-compliance-scanner | v1.2.16 | Yes | |
| HealthWatch | p-healthwatch | v1.8.0-build.66 | Yes | |
| Push notification | p-push-notifications | v1.10.6-build.6 | Yes | |
| Telemetry | pivotal-telemetry-om | v1.0.1 | Yes | |
| Tanzu RabbitMQ | p-rabbitmq | v1.20 | Yes | |
| Tanzu for Valkey | p-redis | v2.4 | Yes | |
| Reliability View exporter for Elastic Application Runtime (Beta) | p-reliability-view-pas-exporter | v0.4.0-build.124 | Yes | |
| Reliability view | p-reliability-view | v0.4.0-build.124 | Yes | |
| Reliability view exporter for Container Orchestration Engine (Beta) | p-reliability-view-pks-exporter | v0.4.0-build.124 | Yes | |
| Scheduler | p-scheduler | v1.2.30-build.1 | Yes | |
| Tanzu Cloud Service Broker for AWS | aws-services | v1.4.14 | Yes | |
| Single Sign-On | pivotal_single_sign-on_service | v1.11.0 | Yes | |
| Spring Cloud Data Flow for Cloud Foundry | p-dataflow | v1.6.6 | Yes | |
| Spring Cloud Services | p_spring-cloud-services | v3.1.5 | Yes | |
| Spring Cloud Gateway | p_spring-cloud-gateway-service | v1.0.0 | Yes | |

| Product name | Product slug | Minimum required version | Compatible | Notes |
|--------------------------------|--------------|--------------------------|------------|-------|
| VMware NSX-T Container Plug-in | VMware-NSX-T | v2.5.1 | Yes | |

Troubleshooting CredHub Maestro safety violations

You can troubleshoot safety violation errors that you encounter when rotating certificates with either the Tanzu Operations Manager API or the CredHub Maestro CLI.

About CredHub Maestro safety checks

CredHub Maestro completes basic safety checks when rotating certificates to prevent unsafe operations.

The Tanzu Operations Manager API runs CredHub Maestro when rotating certificates. If a certificate rotation API command is unsafe, CredHub Maestro stops the command and returns one or more safety violations.

CredHub Maestro returns safety violations if a rotation step is run out of order. For example, if you generate a new root certificate authority (CA), and attempt to activate it before clicking Apply Changes. Performing steps out of the correct order during certificate rotations can make your deployment unstable or risk downtime.

For information about certificate rotation using the Tanzu Operations Manager API, see [Overview of Certificate Rotation](#).

For information about using the CredHub Maestro CLI, see [Getting started with CredHub Maestro](#).

Skip safety checks

To skip CredHub Maestro safety checks in Tanzu Operations Manager:

- In API calls - Pass `skip_safety_check=true` as a parameter to certificate rotation API endpoints.
- In the CredHub Maestro CLI - Specify the `--skip-safety-check` flag when running certificate rotation commands.

Skipping safety checks is strongly discouraged. Use the option only if absolutely necessary. For example, you might want to skip safety checks if your CA has already expired and your apps are experiencing downtime. Skipping safety checks can help bring your apps back online faster. If certificates have expired, contact [Broadcom Support](#) for guidance.

Troubleshoot certificate rotation safety violations

This section lists known safety violations that might occur when using the Tanzu Operations Manager API or the CredHub Maestro CLI to rotate certificates. Each table includes the safety violation message, describes the safety violation, and presents possible solutions to resolve the underlying cause.

Multiple signing versions of a CA

there is more than one signing version of a certificate authority

| | |
|------------------|---|
| Related Commands | <p>This safety violation might appear after calling the following Tanzu Operations Manager API endpoints:</p> <ul style="list-style-type: none"> • <code>POST /api/v0/certificate_authorities/generate</code> • <code>POST /api/v0/certificate_authorities/{certificate_authority_guid}/activate</code> • <code>POST /api/v0/certificate_authorities/active/regenerate</code> • <code>DELETE /api/v0/certificate_authorities/{certificate_authority_guid}</code> <p>Or after running one of the following commands:</p> <ul style="list-style-type: none"> • <code>maestro regenerate ca</code> • <code>maestro update-transitional signing</code> • <code>maestro regenerate leaf</code> • <code>maestro update-transitional remove</code> |
| Cause | CredHub Maestro expects that only one version of a CA is used to sign active leaf certificates. During the rotation process, Maestro moves forward from one CA version to another in bulk, but is not equipped to handle mixed-mode operations where multiple CA versions are actively in use. |
| Solution | <p>There are two possible problems, with different solutions. To identify and resolve the root issue:</p> <ol style="list-style-type: none"> 1. Run <code>maestro topology --name CERTIFICATE-NAME</code>, where <code>CERTIFICATE-NAME</code> is the name and full path of the certificate returned in the error message. 2. Check the command line output for the following: <ul style="list-style-type: none"> ◦ If the <code>signs</code> field contains a child certificate that has two versions set to <code>active: true</code> with different product tile deployment names: <ol style="list-style-type: none"> 1. Go to the Tanzu Operations Manager Installation Dashboard. 2. Click Review Pending Changes. 3. Select the product tile that is associated with the older certificate version. 4. Click Apply Changes. ◦ If there are multiple child certificates and their active versions have different <code>signed-by</code> versions: <ol style="list-style-type: none"> 1. Run <code>maestro regenerate leaf --signed-by PARENT-CA-NAME</code>, where <code>PARENT-CA-NAME</code> is the name of the signing CA. 2. Go to the Tanzu Operations Manager Installation Dashboard. 3. Click Review Pending Changes. 4. Select all products. 5. Click Apply Changes. |

Transitional version of CA detected

certificate authorities can not have a transitional version

| | |
|------------------|---|
| Related Commands | <p>This safety violation might appear after calling the <code>POST /api/v0/certificate_authorities/generate</code> Tanzu Operations Manager API endpoint or after running the <code>maestro regenerate ca</code> command.</p> |
| Cause | CredHub Maestro has detected an existing transitional CA version. This means that the certificate rotation process has already started. |

| | |
|----------|--|
| Solution | Continue with the certificate rotation procedure. In most cases, this means clicking Apply Changes in the Tanzu Operations Manager Installation Dashboard. Do not run earlier commands after the rotation process has started. |
|----------|--|

Multiple active child certificates

| | |
|------------------|---|
| Safety Violation | active child certificate version is not the latest non-transitional version or more than one active version exists |
| Related Commands | <p>This safety violation might appear after calling the following Tanzu Operations Manager API endpoints:</p> <ul style="list-style-type: none"> POST /api/v0/certificate_authorities/generate POST /api/v0/certificate_authorities/{certificate_authority_guid}/activate <p>Or after running the following commands:</p> <ul style="list-style-type: none"> maestro regenerate ca maestro update-transitional signing |
| Cause | This error usually occurs if you have not applied your changes to all the deployments that are required to propagate the latest version of the child certificate. A child certificate can refer to a leaf certificate or an intermediate certificate. |
| Solution | <p>To resolve this error:</p> <ol style="list-style-type: none"> Go to the Tanzu Operations Manager Installation Dashboard. Click Review Pending Changes. Select all products. Click Apply Changes. After the deployment finishes, confirm that the latest version is propagated. Run <code>maestro topology --name CERTIFICATE-NAME</code>, where <code>CERTIFICATE-NAME</code> is the name of the certificate returned in the error message. |

Latest CA is not transitional

| | |
|------------------|--|
| Safety Violation | certificate authorities latest version is not transitional |
| Related Commands | <p>This safety violation might appear after calling the <code>POST /api/v0/certificate_authorities/{certificate_authority_guid}/activate</code> Tanzu Operations Manager API endpoint or after running the <code>maestro update-transitional signing</code> command.</p> |
| Cause | The latest version of each CA is expected to be transitional before the latest CA version becomes the future signing version. This allows trust to be incrementally added to the latest version before it ever gets used. |

| | |
|----------|---|
| Solution | <p>Continue with the certificate rotation procedure. In most cases, this means clicking Apply Changes in the Tanzu Operations Manager Installation Dashboard. Do not run earlier commands after the rotation process has started.</p> <p>If you believe you are running the steps in the correct order, it is possible that a new CA has been created in CredHub after the rotation process has begun.</p> <p>To work around this situation, take one of the following actions:</p> <ul style="list-style-type: none"> • If the CA is new, exclude this CA by adding the <code>--exclude=CA-NAME</code> option for future commands during this rotation. For example, <code>maestro update-transitional signing --exclude=CA-NAME</code>. • If you need to rotate the CA now, re-run the earlier CredHub Maestro rotation steps <code>maestro regenerate ca CA-NAME</code>, where <code>CA-NAME</code> is the name of a CA that was not originally present. Repeat this step for all CAs not originally present. This gets all targeted CAs into the same state. Then, run the <code>maestro update-transitional signing</code> step again. |
|----------|---|

Signing version must be transitional

| | |
|------------------|--|
| Safety Violation | signing version has to be transitional if there is a transitional certificate authority |
| Related Commands | This safety violation might appear after calling the <code>POST /api/v0/certificate_authorities/active/regenerate</code> Tanzu Operations Manager API endpoint or after running the <code>maestro regenerate leaf</code> command. |
| Cause | Before regenerating new leaf certificates, all the existing leaf certificates must only be signed by older versions of their appropriate CAs. These older CA versions are marked as transitional, which indicates that they are inactive but still need to be trusted by clients. This means the new leaf certificates are signed by the newer CA version and clients trust both the old and new CA versions. |
| Solution | <p>Continue with the certificate rotation procedure. In most cases, this means clicking Apply Changes in the Tanzu Operations Manager Installation Dashboard. Do not run earlier commands after the rotation process has started.</p> <p>If you believe you are running the steps in the correct order, it is possible that a new leaf was issued while the rotation process was in progress. You might also see the <code>there is more than one signing version</code> error described in Multiple Signing Versions of a CA. If this safety violation error appears, follow the mitigation steps for that error.</p> |

Active child must trust signing CA

| | |
|------------------|---|
| Safety Violation | active child certificate does not trust the certificate authority that will regenerate it |
| Related Commands | This safety violation might appear after calling the <code>POST /api/v0/certificate_authorities/active/regenerate</code> Tanzu Operations Manager API endpoint or after running the <code>maestro regenerate leaf</code> command. |
| Cause | <p>This safety violation occurs when the active leaf certificate is not signed by the CA version that would sign new leaf certificates, and also the leaf does not include that CA in the list of trusted CAs. This situation can occur after running <code>maestro regenerate ca</code> or <code>maestro update-transitional</code> while skipping safety checks, or using the <code>credhub</code> CLI or API to manually generate a new version of the CA.</p> <p>During a CA rotation, if the active version of the leaf certificate does not trust the signing version of the CA, then trust will be broken when deploying a regenerated leaf certificate.</p> |
| Solution | The repair procedure for this safety violation can vary depending on the exact state of the CA and leaf certificates. Contact Support for guidance. |

CA must have a non-transitional version

Safety Violation

certificate authorities must have a non-transitional version

| | |
|------------------|--|
| Related Commands | This safety violation might appear after calling the POST /api/v0/certificate_authorities/active/regenerate Tanzu Operations Manager API endpoint or after running the maestro regenerate leaf command. |
| Cause | <p>This safety violation occurs when the CA that would sign the leaf certificate has a single version, and that version is marked as transitional. This situation can occur after running maestro garbage-collect ca --force, or using the credhub CLI or API to delete CA versions.</p> <p>When regenerating leaf certificates, if there is only a transitional version of the CA, then regenerating leaf certificates will fail.</p> |
| Solution | <p>Remove the transitional flag from the CA certificate using the Update Transitional Version CredHub API endpoint and setting the version to <code>null</code>.</p> <pre>credhub curl -X PUT -p /api/v1/certificates/CA_CERTIFICATE_ID/update_transitional_version -d '{"version": null}'</pre> |

Signed by non-latest CA version

Safety Violation

active child certificate is signed by a certificate authority that is not the latest version

| | |
|------------------|--|
| Related Commands | This safety violation might appear after calling the POST /api/v0/certificate_authorities/active/regenerate Tanzu Operations Manager API endpoint or after running the maestro regenerate leaf command. |
| Cause | <p>This safety violation occurs when the existing leaf certificates are not signed by the CA version that will sign the new leaf certificate versions, and they are also not signed by the current CA version that is marked as transitional (if one exists). This situation can occur after running maestro regenerate ca or maestro update-transitional while skipping safety checks, or using the CredHub CLI or API to generate new CA versions or change the transitional flag of existing CAs.</p> <p>During a CA rotation, if the leaf certificate does not trust both the old and new versions of the CA, then trust will be broken when deploying a regenerated leaf certificate.</p> |
| Solution | <p>If the active leaf certificate is signed by one of the two most recent versions of the CA, the leaf certificate trusts both CA versions, and neither CA version is marked transitional, then use the Update Transitional Version CredHub API endpoint to set the older CA version as transitional.</p> <pre>credhub curl -X PUT -p /api/v1/certificates/CA_CERTIFICATE_ID/update_transitional_version -d '{"version": "OLDER_CA_VERSION_ID"}'</pre> <p>Then continue the CA rotation starting on the maestro regenerate leaf step of the rotation procedure.</p> <p>For other scenarios, please reach out to Support for guidance.</p> |

Active certificates not signed by latest CA detected

Safety Violation

active child certificate version is signed by a certificate authority that is not the latest version

| | |
|------------------|---|
| Related Commands | This safety violation might appear after calling the DELETE /api/v0/certificate_authorities/{certificate_authority_guid} Tanzu Operations Manager API endpoint or after running the maestro update-transitional remove command. |
| Cause | There are still active intermediate CAs or leaf certificates that are not signed by the latest CA version. |

| | |
|----------|--|
| Solution | Continue with the certificate rotation procedure. In most cases, this means clicking Apply Changes in the Tanzu Operations Manager Installation Dashboard. Do not run earlier commands after the rotation process has started. |
|----------|--|

Stale leaf certificates detected

| | |
|------------------|--|
| Safety Violation | active certificate version is not the latest non transitional version |
| Related Commands | This safety violation might appear after calling the DELETE /api/v0/certificate_authorities/{certificate_authority_guid} Tanzu Operations Manager API endpoint or after running the <code>maestro update-transitional remove</code> command. |
| Cause | There are still stale leaf certificates in use with new versions ready to be deployed. |
| Solution | Continue with the certificate rotation procedure. In most cases, this means clicking Apply Changes in the Tanzu Operations Manager Installation Dashboard. Do not run earlier commands after the rotation process has started. |

Transitional CA detected as latest CA

| | |
|------------------|--|
| Safety Violation | transitional certificate authority is the latest version |
| Related Commands | This safety violation might appear after calling the DELETE /api/v0/certificate_authorities/{certificate_authority_guid} Tanzu Operations Manager API endpoint or after running the <code>maestro update-transitional remove</code> command. |
| Cause | The latest version of one or more CAs is transitional, when CredHub Maestro is expecting the transitional version to be an older version. |
| Solution | <p>Continue with the certificate rotation procedure. In most cases, this means clicking Apply Changes in the Tanzu Operations Manager Installation Dashboard. Do not run earlier commands after the rotation process has started.</p> <p>It is possible that you are attempting to halt a rotation in the early stages, likely after new transitional CAs have been issued but not yet used. Continue with the rotation at this point. If continuing with the rotation is impossible due to other deployment needs, reach out to Support for guidance.</p> |

Backing up and restoring Tanzu Operations Manager

The following topics describe how you can back up your Tanzu Operations Manager deployment, restore a deployment, and recover from incidents:

- [Backing up and restoring your Tanzu Operations Manager deployment](#)
- [Incident response guides for Tanzu Operations Manager](#)
 - [Restoring lost BOSH Director persistent disk](#)
 - [Restoring lost Elastic Application Runtime data](#)
 - [Recovering from Tanzu Operations Manager and Elastic Application Runtime upgrade failure](#)
- [Configuring AD FS as your identity provider in Tanzu Operations Manager](#)
- [Disaster Recovery](#)
- [Backing up your Tanzu Operations Manager deployments with BBR](#)
- [Tanzu Operations Manager file storage backup levels](#)
- [Elastic Application Runtime component availability during backup process](#)
- [Restoring Tanzu Operations Manager deployments from backup with BBR](#)
- [Enabling external blobstore backups](#)
- [Troubleshooting BOSH backup and restore](#)
- [TLS connections in Elastic Application Runtime](#)

Backing up and restoring your Tanzu Operations Manager deployment

This topic describes how to back up and restore your Tanzu Operations Manager deployment.

Before you back up

When backing up data in your deployment, consider:

- If your deployment uses external databases, check that it is a supported database configuration. For more information, see [External storage support across Tanzu Operations Manager Versions in Backing up Deployments with BBR](#).
- If your deployment uses internal databases, follow the backup and restore instructions for the MySQL server in the BBR documentation. For more information, see [Backup and restore with BBR](#).

General Data Protection Regulation (GDPR)

The General Data Protection Regulation (GDPR) came into effect on May 25, 2018 and applies to any company processing the data of the citizens or residents of the European Union (EU), even if the company is not EU-based. The GDPR sets forth how companies should handle privacy issues, securely store data, and respond to security breaches.

Backup artifacts might contain personal data covered by GDPR. For example, a backup of a Elastic Application Runtime instance could contain a user email address. For more information about personal data that can be stored in your deployment, see [General Data Protection Regulation and Elastic Application Runtime](#).

Back up and restore with BBR

BOSH Backup and Restore (BBR) is a command-line tool for backing up and restoring BOSH deployments.

- To perform a backup of your deployment with BBR, see [Backing up deployments with BBR](#).
- To restore your deployment with BBR, see [Restoring deployments from backup with BBR](#).
- To troubleshoot problems with BBR, see [Troubleshooting BBR](#).

BBR restore scenarios

There are many different restore cases that can arise with BOSH Directors and Elastic Application Runtime deployments. The following links provide some guidance on how to use BBR in common restore scenarios:

- [Restoring lost BOSH Director persistent disk](#)
- [Restoring lost Elastic Application Runtime data](#)
- [Recovering from Tanzu Operations Manager and Elastic Application Runtime upgrade failure](#)

These guides do not cover all possible restore scenarios. If you encounter another restore case, some information in these guides might be applicable. For guidance, contact [Support](#).

Incident response guides for Tanzu Operations Manager

The following guides describe how you can use BOSH Backup and Restore (BBR) to respond to upgrade failures and data loss in your Tanzu Operations Manager deployment:

- [Restoring lost BOSH Director persistent disk](#)
- [Restoring lost Elastic Application Runtime data](#)
- [Recovering from Tanzu Operations Manager and Elastic Application Runtime upgrade failure](#)

Restoring lost BOSH Director persistent disk

You can use BOSH Backup and Restore (BBR) to restore a BOSH Director that has lost its persistent disk.

Prerequisite

Restoring a BOSH Director persistent disk with BBR is only possible if you have previously taken a backup using the `bbr director backup` command.

This command produces a backup artifact in the format `DIRECTOR_IP_TIMESTAMP/`.

BOSH Director backups require identical references to all BOSH-deployed VMs. If BOSH re-creates your deployment VMs for any reason, such as changes to stemcells, networks, or availability zones (AZs), your BBR backup artifact is no longer compatible for performing a restore.

Prepare your environment

Your BOSH Director must be in a healthy state before you can perform a BOSH Director restore. The BOSH Director is in a healthy state when all jobs are running.

If your BOSH Director has lost its persistent disk, you must create a new disk to return your BOSH Director to a healthy state.

Create a new BOSH Director persistent disk

To create a new persistent disk for the BOSH Director:

1. Create a new persistent disk using your IaaS console.
2. Log in to Tanzu Operations Manager VM. For more information, see [Log Into the Tanzu Operations Manager VM with SSH](#).
3. Open the `bosh-state.json` file in a text editor.
4. Edit the `disks.cid` value to match the new persistent disk you created in the first step.
5. Update the persistent disk size in Tanzu Operations Manager and then click Apply Changes. This creates a new BOSH Director VM and attaches the new persistent disk to it.

Perform the restore

BBR restore is a destructive process which removes any current data in your deployment. Performing a restore overwrites all new data since you created your most recent backup artifact.

Perform a BOSH Director restore by following the BBR instructions in [Step 8: Restore the BOSH Director in Restoring Deployments From Backup with BBR](#).

Consolidate your environment

After you successfully restore your BOSH Director, the BOSH Director uses the VM references that were stored in the backup artifact. If you clicked Apply Changes in Tanzu Operations Manager between when the backup and the restore took place, check if any BOSH-deployed VMs were deleted or added.

If changes have occurred, ensure that the BOSH Director database is consistent with the current state of your IaaS after the restore. If VMs were deleted, the BOSH Director still has references to these deleted VMs after the restore. If VMs were added, the BOSH Director has no knowledge of the new VMs.

To ensure that the BOSH Director database matches your IaaS console:

1. Find and delete outdated VM references for an environment by running:

```
bosh --deployment DEPLOYMENT-NAME cloud-check
```

Where `DEPLOYMENT-NAME` is the name of your BOSH deployment.

2. Log in to your IaaS console and delete any orphaned VMs that your BOSH Director does not reference.

Your BOSH Director is now in a healthy state with the new persistent disk attached. Your BOSH deployment VMs are aligned with the current state of your IaaS.

Restoring lost Elastic Application Runtime data

You can use BOSH Backup and Restore (BBR) to restore a lost piece of Tanzu Platform for Cloud Foundry data, such as a user or space.

Prerequisite

Restoring Elastic Application Runtime data with BBR is only possible if you have previously taken a backup using the `bbr deployment backup` command.

This command produces a backup artifact in the format `TAS_DEPLOYMENT_TIMESTAMP/`.

Validate your backup artifact

Before restoring, verify that the backup artifact you have is suitable for your restore need. You can `untar` and explore the Elastic Application Runtime backup artifact to verify that it contains the data that you need to restore.

Use any of the checks in this section, depending on the data you need to restore.

| Verify contents of backup artifact | Run: | You can restore this backup if output shows artifact you need to restore |
|--|--|---|
| To verify that your backup artifact contains the org you need to restore | <pre>tar -xvf backup_restore-0-bbr-cloudcontrollerdb.tar cat cloudcontroller-artifact-file grep "ORG-TO-RESTORE"</pre> Where <code>ORG-TO-RESTORE</code> is the org you need to restore. | <div>INSERT INTO `organizations` VALUES (... , 'ORG-TO-RESTORE', ...);</div> <div>Where <code>ORG-TO-RESTORE</code> is the org you need to restore.</div> |
| To verify that your backup artifact contains the space you need to restore | <pre>tar -xvf backup_restore-0-bbr-cloudcontrollerdb.tar cat cloudcontroller-artifact-file grep "SPACE-TO-RESTORE"</pre> Where <code>SPACE-TO-RESTORE</code> is the space you need to restore. | <div>INSERT INTO `spaces` VALUES (... , 'SPACE-TO-RESTORE', ...);</div> <div>Where <code>SPACE-TO-RESTORE</code> is the space you need to restore.</div> |
| To verify that your backup artifact contains the user you need to restore | <pre>tar -xvf backup_restore-0-bbr-uaadb.tar cat uaadb-artifact-file grep "USER-TO-RESTORE"</pre> Where <code>USER-TO-RESTORE</code> is the user you need to restore. | <div>INSERT INTO `users` VALUES (... , 'USER-TO-RESTORE', ...);</div> <div>Where <code>USER-TO-RESTORE</code> is the user you need to restore.</div> |
| To verify that your backup artifact contains the route you need to restore | <pre>tar -xvf backup_restore-0-bbr-cloudcontrollerdb.tar cat cloudcontroller-artifact-file grep "ROUTE-TO-RESTORE"</pre> Where <code>ROUTE-TO-RESTORE</code> is the route you need to restore. | <div>INSERT INTO `routes` VALUES (... , 'ROUTE-TO-RESTORE', ...);</div> <div>Where <code>ROUTE-TO-RESTORE</code> is the route you need to restore.</div> |

To verify that your backup artifact contains the domain you need to restore

```
tar -xvf backup_restore-0-bbr-  
cloudcontrollerdb.tar  
cat cloudcontroller-artifact-file |  
grep "DOMAIN-TO-RESTORE"  
Where DOMAIN-TO-RESTORE  
is the domain you need to  
restore.
```

```
INSERT INTO `domains` VALUES (... , 'DOMAIN-TO-R  
ESTORE', ...);
```

Where DOMAIN-TO-RESTORE is the domain you need to restore.

To verify that your backup artifact contains the app you need to restore

```
tar -xvf backup_restore-0-bbr-  
cloudcontrollerdb.tar  
cat cloudcontroller-artifact-file |  
grep "APP-TO-RESTORE"  
Where APP-TO-RESTORE is  
the app you need to restore.
```

Run:

```
tar -xvf blobstore-backup.tar  
tree blobstore-backup | grep "APP-DROPLET"
```

Where APP-DROPLET is the droplet for the app that you need to restore.

Look for the following in the output:

```
INSERT INTO `apps` VALUES (... , 'APP-TO-RESTOR  
E', 'APP-DROPLET', ...);
```

Where APP-TO-RESTORE is the app you need to restore, and APP-DROPLET is the droplet for the app that you need to restore.

NOTE: If you use an external blobstore, go to your IaaS console and verify that the backup bucket for the droplet is present.

Perform the restore

BBR restore is a destructive process that removes any current data in your deployment. Performing a restore overwrites all new data since you created your most recent backup artifact.

VMware recommends that you verify the compatibility of your backup artifact by attempting the restore on an environment that is identical to the target environment. If this is not possible, see [Compatibility of Restore](#) in *Restoring Deployments from Backup with BBR* to confirm that your backup is compatible with your restore environment.

To run a Elastic Application Runtime restore, follow the BBR instructions in [Step 14: Restore Elastic Application Runtime](#) in *Restoring Deployments from Backup with BBR*.

After a restore, Elastic Application Runtime rolls back to its state at the time when the backup was taken, and contains the data that you had previously lost.

Consolidate your environment

If you have service instances that were not present when you performed the backup, then the BOSH service deployments become orphaned. Elastic Application Runtime has no reference to these service deployments, but they continue to exist. You can locate orphaned service deployments using the BOSH command-line interface (CLI).

You can locate orphaned on-demand service instances by running the `orphan-deployments` errand. For more information, see [Orphan deployments](#) in *Broker and Service Management*.

Recovering from Operations Manager and Elastic Application Runtime upgrade failure

You can use BOSH Backup and Restore (BBR) to restore Tanzu Operations Manager and Tanzu Platform for Cloud Foundry to a healthy state after a failed upgrade between Elastic Application Runtime minor versions.

BBR can only restore across the same Elastic Application Runtime minor versions. To verify which version of Elastic Application Runtime was used to create the backup artifact, see `metadata/cf-VERSION.yml` in the backup artifact.

Evaluate your deployment

You can restore Tanzu Operations Manager and Elastic Application Runtime to a healthy state after a failed upgrade between Elastic Application Runtime minor versions if your backups, BOSH Director, and Elastic Application Runtime deployment meet certain conditions. Follow the procedures in this section to confirm that your deployment can be restored by BBR.

Review your backup

Before you perform a restore, you must:

1. Run the following `bbr` commands from your Tanzu Operations Manager VM:

- Back up your deployments by running:

```
bbr deployment --deployment DEPLOYMENT-NAME backup
```

Where `DEPLOYMENT-NAME` is the name of your Elastic Application Runtime deployment.

- Back up your BOSH Director by running:

```
bbr director backup
```

The preceding `bbr` commands produce the following backup artifacts:

- `DEPLOYMENT_NAME_TIMESTAMP/`
- `DIRECTOR_IP_TIMESTAMP/`
- `installation.zip`

2. Export your Tanzu Operations Manager settings. For more information, see [Settings page](#) in *Using the Tanzu Operations Manager Interface*.

Validate your environment for restore

Before you can restore after a failed upgrade between Elastic Application Runtime minor versions, you must confirm that your BOSH Director and Elastic Application Runtime deployments are healthy.

To confirm that your BOSH Director is healthy:

To verify that the BOSH Director is in a healthy state for a `bbr restore`, SSH to the Tanzu Operations Manager VM and ensure that all jobs are running.

To ensure that all jobs are running:

1. SSH into your BOSH Director. For more information, see [Log Into the BOSH Director VM with SSH](#) in *Advanced Troubleshooting with the BOSH CLI*.
2. View all deployment job instances by running:

```
monit summary
```

Monit should report that all jobs are running.

3. Ensure that the BOSH API is responsive by running the following command:

```
bosh configs
```

If this command runs successfully, then the BOSH Director API is in a healthy and responsive state and can be restored by BBR.

If your BOSH Director is unhealthy, see [Restoring an Unhealthy BOSH Director](#).

To confirm that your Elastic Application Runtime deployment is healthy:

To confirm that Elastic Application Runtime is in a healthy state:

1. Run:

```
bosh -d DEPLOYMENT-NAME instances
```

Where **DEPLOYMENT-NAME** is the name of your Elastic Application Runtime deployment.

If your Elastic Application Runtime deployment is healthy, then the output states that all of your instances are in the **running** state. If all instances are running, Elastic Application Runtime can be restored by BBR.

If your Elastic Application Runtime deployment is unhealthy, see [Restoring an Unhealthy Elastic Application Runtime Deployment](#).

Restore your deployment

After you confirm that your BOSH Director is healthy with its data intact, and all instances in your Elastic Application Runtime deployment are responsive, you can restore Elastic Application Runtime with BBR.

Restoring Elastic Application Runtime

If your BOSH Director is healthy, you can restore across the same Elastic Application Runtime minor version by following the procedures in [Step 14: Restore Elastic Application Runtime](#) in *Restoring Deployments from Backup with BBR*.

Recover to a healthy state

If either the BOSH Director or Elastic Application Runtime are not in a healthy state, they must be recovered before you can run **bbt restore**. If you need guidance on the following subsections, contact [Support](#).

Restoring an unhealthy BOSH Director

If the BOSH Director is not in a healthy state, VMware recommends following the full restore workflow to regain your Tanzu Operations Manager, BOSH Director, and BOSH deployments. For more information,

see [Restore Workflow](#) in *Restoring Deployments from Backup with BBR*.

Restoring an unhealthy Elastic Application Runtime deployment

If your BOSH Director is healthy, you can restore your previous BOSH Director state by following the procedure in [Step 8: Restore the BOSH Director](#) in *Restoring Deployments from Backup with BBR*.

After your BOSH Director is restored, you can then restore your deployments by following the procedure in [Step 14: Restore Elastic Application Runtime](#) in *Restoring Deployments from Backup with BBR*.

Configuring AD FS as your identity provider in Tanzu Operations Manager

You can configure Active Directory Federation Services (AD FS) as your identity provider (IDP) in Tanzu Operations Manager and AD FS.

You can use AD FS as your SAML IDP for Tanzu Operations Manager and Tanzu Platform for Cloud Foundry.

- To use AD FS as your SAML IDP for Tanzu Operations Manager, follow these procedures:
 - [Configure SAML integration in Tanzu Operations Manager](#).
 - [Configure SAML integration in AD FS](#).
- To use AD FS as your SAML IDP for Elastic Application Runtime, follow these procedures:
 - [Configure SAML integration in Elastic Application Runtime](#).
 - [Configure SAML integration in AD FS](#).

Configure SAML integration in Tanzu Operations Manager

To configure Tanzu Operations Manager to use AD FS as your SAML IDP:

1. Download your IDP metadata from <https://AD-FS-HOSTNAME/federationmetadata/2007-06/federationmetadata.xml>, where **AD-FS-HOSTNAME** is the hostname of your AD FS deployment.
2. Follow the procedure in *Use an Identity Provider* in the BOSH Director configuration topic for your IaaS:
 - [Configuring BOSH Director on AWS](#)
 - [Configuring BOSH Director on Azure](#)
 - [Configuring BOSH Director on GCP](#)
 - [Configuring BOSH Director on vSphere](#)

You can set up SAML access for Tanzu Operations Manager during the initial Tanzu Operations Manager installation or later by navigating to Settings in the user menu in the Tanzu Operations Manager Installation Dashboard, configuring the Authentication Method pane, and then clicking Review Pending Changes and Apply Changes.

Configure SAML integration in Elastic Application Runtime

To configure Elastic Application Runtime to use AD FS as your SAML IDP:

1. Download your IDP metadata from <https://AD-FS-HOSTNAME/federationmetadata/2007-06/federationmetadata.xml>, where **AD-FS-HOSTNAME** is the hostname of your AD FS deployment.

2. Follow the procedure in [Configure Elastic Application Runtime as a Service Provider for SAML in Configuring authentication and enterprise SSO for Elastic Application Runtime](#).

Configure SAML integration in AD FS

To designate Tanzu Operations Manager as your SAML service provider (SP) in AD FS:

1. Download your SP metadata from <https://login.SYSTEM-DOMAIN/saml/metadata>, where **SYSTEM-DOMAIN** is the system domain of your Tanzu Operations Manager deployment.
2. Open your ADFS Management console.
3. To add a relying party trust:
 1. Select Actions, and click Add Relying Party Trust.
 2. On the Welcome step, click Start.
 3. Select Import data about the relying party from a file.
 4. Choose the downloaded SP metadata file, and click Next.
 5. Enter a Display name for the new relying party trust, and click Next.
 6. Leave the default multi-factor authentication selection, and click Next.
 7. Select Permit all users to access this relying party, and click Next.
 8. Review your settings.
 9. Click Next.
 10. Click Close to finish the wizard.
4. To modify your relying party trust:
 1. Double-click the new relying party trust.
 2. Select the Encryption tab.
 3. Click Remove to remove the encryption certificate you imported.
 4. Select the Advanced tab.
 5. For the Secure hash algorithm, select SHA256.
5. (Optional) If you are using a self-signed certificate and you want to deactivate CRL checks:
 1. Open Windows Powershell as an Administrator.
 2. Run:

```
set-ADFSRelyingPartyTrust -TargetName "RELYING-PARTY-TRUST" -SigningCertificateRevocationCheck  
None
```

Where **RELYING-PARTY-TRUST** is the relying party trust for which you want to deactivate CRL checks.
6. To add claim rules for your relying party trust, select your relying party trust and click Edit Claim Rules.
7. In the Issuance Transform Rules tab, create two claim rules:
 1. Click Add Rule.
 2. For Claim rule template, select Send LDAP Attributes as Claims.
 3. Click Next.

4. Enter a Claim rule name.
 5. For Attribute store, select Active Directory.
 6. For LDAP Attribute, select E-Mail-Addresses. If you do not have the email attribute configured for users, you can select User-Principle-Name.
 7. For Outgoing Claim Type, select E-Mail Address.
 8. Click Finish.
 9. Click Add Rule.
 10. For Claim rule template, select Transform an Incoming Claim.
 11. Click Next.
 12. Enter a Claim rule name.
 13. For Incoming claim type, select E-Mail Address.
 14. For Outgoing claim type, select Name ID.
 15. For Outgoing name ID format, select Email.
 16. Click Finish.
8. To permit access to users based on a security group:
1. Select the Issuance Authorization Rules tab.
 2. Click Add Rule.
 3. For Claim rule template, select Permit or Deny Users Based on an Incoming Claim.
 4. Click Next.
 5. Enter a Claim rule name.
 6. For Incoming claim type, select Group SID.
 7. Click Browse.
 8. Locate the security group in your domain of which Tanzu Operations Manager developers are a part.
 9. Click OK.
 10. Ensure Permit access to users with this incoming claim is selected.
 11. Click Finish.

Disaster Recovery

There are a range of approaches for ensuring that you can recover your deployment, apps, and data in case of a disaster.

The approaches fall into two categories:

- Using data from a backup to restore the data in the deployment. For more information, see [Back up and restore using BOSH Backup and Restore \(BBR\)](#).
- Recreating the data in the deployment by automating the creation of state. For more information, see [Disaster recovery by recreating the deployment](#).

Back up and restore using BOSH Backup and Restore (BBR)

BOSH Backup and Restore (BBR) is a CLI for orchestrating backing up and restoring BOSH deployments and BOSH Directors. BBR triggers the backup or restore process on the deployment or BOSH Director, and transfers the backup artifact to and from the deployment or BOSH Director.

Use BBR to reliably create backups of core components of your deployments and their data. These core components include CredHub, UAA, BOSH Director, and Tanzu Platform for Cloud Foundry.

Each component includes its own backup scripts. This decentralized structure helps keep scripts synchronized with the components. At the same time, *locking* features ensure data integrity and consistent, distributed backups across your deployment.

For more information about the BBR framework, see [BOSH Backup and Restore](#) in the open-source Cloud Foundry documentation.

Backing up your deployment

Backing up your deployment requires backing up the following components:

- Tanzu Operations Manager settings
- BOSH Director, including CredHub and UAA
- Elastic Application Runtime
- Data services

For more information, see [Backing up Deployments with BBR](#). With these backup artifacts, operators can re-create their deployment exactly as it was when the backup was taken.

Restoring your deployment

The restore process involves creating a new deployment starting with the Tanzu Operations Manager VM. For more information, see [Restoring Deployments from Backup with BBR](#).

The time required to restore the data is proportionate to the size of the data because the restore process includes copying data. For example, restoring a 1 TB blobstore takes 1,000 times as long as restoring a 1 GB blobstore.

Benefits

Unlike other backup solutions, using BBR to back up your deployment enables:

- **Completeness:** BBR supports backing up BOSH, including CredHub, UAA, and service instances created with an on-demand service broker.
- **Consistency:** BBR provides referential integrity between the database and the blobstore because a lock is held while both the database and blobstore are backed up.
- **Correctness:** Using the BBR restore flow addresses container-to-container networking and routing issues that can occur during restore.

API downtime during backups

Apps are not affected during backups, but certain APIs are unavailable. The downtime occurs only while the backup is being taken, not while the backup is being copied to the jumpbox.

In a consistent backup, the blobs in the blobstore match the blobs in the Cloud Controller database. To ensure a consistent backup, changes to the data are prevented during the backup. This means that the Cloud Foundry API (CAPI), Routing API, Usage service, Autoscaler, Notification Service, Network Policy

Server, and CredHub are unavailable while the backup is being taken. UAA is in read-only mode during the backup.

Backup timings

The first three phases of the backup are lock, backup, and unlock. During this time, the API is unavailable. The drain and checksum phase starts after the backup scripts finish. BBR downloads the backup artifacts from the instances to the BBR VM, and performs a checksum to ensure the artifacts are not corrupted. The size of the blobstore significantly influences backup time.

The following table provides an indication of the downtime that you can expect. Actual downtime varies based on hardware and configuration. These example timings were recorded with Elastic Application Runtime deployed on Google Cloud Platform (GCP) with all components scaled to one and only one app pushed.

| Disaster Recovery | | | | |
|-------------------|--------------------|---|---|---------------------------------|
| API state | Backup phase | Duration for external versioned S3-compatible blobstore | Duration for external unversioned S3-compatible blobstore | Duration for internal blobstore |
| API unavailable | lock | 15 seconds | 15 seconds | 15 seconds |
| | backup | <30 seconds | Proportional to blobstore size | 10 seconds |
| | unlock | 3 minutes | 3 minutes | 3 minutes |
| API available | drain and checksum | <10 seconds | <10 seconds | Proportional to blobstore size |

Blobstore backup and restore

Blobstores can be very large. To minimize downtime, BBR only takes blob metadata during the backup. For example, in the case of internal blobstores such as WebDav and NFS, BBR takes a list of hard links that point to the blobs. After the API becomes available, BBR makes copies of the blobs.

Unsupported products

- Data services: The following data services do not support BBR. Operators of these services can use the automatic backups feature of each tile, available within Tanzu Operations Manager.
 - Tanzu for MySQL
 - Tanzu Gemfire
 - Tanzu RabbitMQ [VMs]
 - Tanzu for Valkey
- External blobstores and databases: BBR support for backing up and restoring external databases and blobstores varies across Tanzu Operations Manager versions. For more information, see [Supported Components](#) and [External Storage Support Across Tanzu Operations Manager Versions](#) in *Backing up Deployments with BBR*.

Best practices

This section describes best practices for backing up your deployment.

| Best Practice | |
|-------------------------------|---|
| Frequency of backups | VMware recommends that you take backups in proportion to the rate of change of the data in your deployment to minimize the number of changes lost if a restore is required. VMware recommends starting with backing up every 24 hours. If app developers make frequent changes, you can increase the frequency of backups. |
| Retention of backup artifacts | <p>You must retain backup artifacts based on the timeframe you need to be able to restore to. For example, if backups are taken every 24 hours and the deployment must be able to be restored to three days prior, three sets of backup artifacts must be retained.</p> <p>Artifacts can be stored in two data centers other than the data center. When deciding the restore timeframe, you can take into account other factors, such as compliance and auditability.</p> |
| Security | VMware recommends that you encrypt artifacts and store them securely. |

Disaster Recovery by re-creating the deployment

An alternative strategy for recovering your deployment after a disaster is to have automation in place so that all the data can be re-created. This requires that every modification to settings and state is automated, typically through the use of a pipeline.

Recovery steps include creating a new deployment, re-creating orgs, spaces, users, services, service bindings and other state, and re-pushing apps.

For more information about this approach, see the Cloud Foundry Summit presentation [Multi-DC Cloud Foundry: What, Why and How?](#) on YouTube.

Disaster Recovery for different topologies

This section describes disaster recovery strategies for different deployment topologies.

Active-Active

To prevent app downtime, some customers run an active-active deployment, where they run two or more identical deployments in different data centers. If one deployment becomes unavailable, traffic is seamlessly routed to the other deployment. To achieve identical deployments, all operations are automated so they can be applied to both deployments in parallel.

Because all operations have been automated, the automation approach to disaster recovery is a viable option for an active-active deployment. Disaster recovery requires re-creating the deployment, then running all the automation to re-create state.

This option requires discipline to automate all changes to the deployment. Some of the operations that need to be automated are:

- App push, restage, scale
- Org, space, and user create, read, update, and delete (CRUD)
- Service instance CRUD
- Service bindings CRUD
- Routes CRUD
- Security groups CRUD
- Quota CRUD

Human-initiated changes always make their way into the system. These changes can include quotas being raised, new settings being enabled, and incident responses. For this reason, VMware recommends taking backups even when using an automated disaster recovery strategy.

The following table compares backing up and restoring using BBR to recreating a failed deployment in active-active:

| Disaster Recovery | | |
|--------------------------------|--|---|
| | Restore the data | Re-create the data |
| Preconditions | IaaS prepared for Tanzu Operations Manager and runtime install | |
| Steps | <div>1. Re-create the deployment</div> <div>2. Restore</div> <div>3. Apply changes to make the restored deployment match the other active deployment</div> | <div>1. Re-create the deployment</div> <div>2. Trigger automation to re-create orgs, spaces, etc.</div> <div>3. Notify app developers to re-push apps, re-create service instances and bindings</div> |
| RTO (recovery time objective) | | |
| Platform | Time to re-create the deployment | Time to re-create the deployment |
| Apps | Time to restore | Time until orgs, spaces, etc. have been re-created and apps have been re-pushed |
| RPO (recovery point objective) | | |
| Platform | Time of the last backup | Current time |
| Apps | Time of the last backup | Current time |

Active-Passive

Instead of having a true active-active deployment across all layers, some customers prefer to install a deployment on a backup site. The backup site resides on-premises, in a co-location facility, or the public cloud. The backup site includes an operational deployment, with only the most critical apps ready to accept traffic if a failure occurs in the primary data center. Disaster recovery in this scenario involves:

1. Switching traffic to the passive deployment, making it active.
2. Recovering the formerly-active deployment. Operators can choose to do this through automation, if that option is available, or by using BBR and the restore process.

The RTO and RPO for re-creating the active deployment are the same as outlined in the preceding table.

Reducing RTO

Both the restore and re-create data disaster recovery options require standing up a new deployment, which can take hours. If you require shorter RTO, several options involving a pre-created standby hardware are available. The following table outlines these options:

| | |
|-------------|--|
| Active-cold | Public cloud environment ready for installation, no Tanzu Operations Manager and runtime installed. This saves both IaaS costs and instance costs. For on-premise installations, this requires hardware on standby that is ready to install on, which might not be a realistic option. |
|-------------|--|

| | |
|-------------------------|--|
| Active-warm | The deployment installed on standby hardware and kept up to date, VMs scaled down to zero (that you spin up each time there is a deployment update), no apps installed, no orgs or spaces defined. |
| Active-inflate platform | Bare-minimum deployment installation, either with no apps, or a small number of each app in a stopped state. On recovery, push a small number of apps or start current apps, while simultaneously triggering automation to scale the deployment to the primary node size, or a smaller version if large percentages of loss are acceptable. This mode allows you to start sending some traffic immediately, while not paying for a full non-primary deployment. This method requires data to be seeded, but it is usually acceptable to complete data sync while the deployment is scaling up. |
| Active-inflate apps | Non-primary deployment scaled to the primary node size, or smaller version if large percentages of loss are acceptable, with a small number of Diego Cells (VMs). On fail-over, scale Diego Cells up to primary node counts. This mode allows you to start sending most traffic immediately, while not paying for all the AIs of a fully-fledged node. This method requires data to be there very quickly after failure. It does not require real-time sync, but near-real time. |

There is a trade-off between cost and RTO: the less the replacement deployment needs to be deployed and scaled, the faster the restore.

Automating backups

BBR generates the required backup artifacts, but does not handle scheduling, artifact management, or encryption. You can use the starter Concourse pipeline from the [BBR PCF Pipeline Tasks](#) repository on GitHub to automate backups with BBR.

You can also use Stark & Wayne's Shield as a front end management tool using the BBR plug-in.

Validating backups

To ensure that backup artifacts are valid, the BBR tool creates checksums of the generated backup artifacts, and ensures that the checksums match the artifacts on the jumpbox.

However, the only way to be sure that the backup artifact can be used to successfully re-create the deployment is to test it in the restore process. This is a cumbersome, dangerous process that must be done with care. For instructions, see [Step 11: \(Optional\) Validate your backup](#) in *Backing up Deployments with BBR*.

Backing up your Tanzu Operations Manager deployments with BBR

You can back up critical back-end components with BOSH Backup and Restore (BBR), a command-line tool for backing up and restoring BOSH deployments. For information about restoring your backup, see [Restoring Tanzu Operations Manager deployments from backup with BBR](#).

To create a consistent backup, BBR stops the Cloud Controller API and the Cloud Controller workers during backup. This affects API functionality, such as when pushing apps or using the Tanzu cf CLI. The deployed apps do not experience downtime.

- Security: Backup artifacts can contain secrets. Secure backup artifacts by using encryption or other means.
- Compatibility: A BBR backup can only be restored to environments matching its source environment. To ensure that a restore environment is compatible with a

backup, see [Compatibility of Restore](#).

- Failed Restore: Restore is a destructive operation. BBR is designed to restore deployments after a disaster. If a restore fails, the environment might be left in an unusable state and require re-provisioning. For more information, see [Restoring Deployments from Backup with BBR](#).
- Service Data: BBR does not back up any service data.
- API Outage: The Cloud Controller API does not send or receive calls during restoration.

Component availability during backup

If the Cloud Controller API and workers are stopped during a backup, any services that rely on the API and workers are not functional during that backup.

Most of the affected components are stopped and restarted after a successful backup. For more detail about how the components are affected, see [Elastic Application Runtime availability during backup process](#).

Here are some examples of the effects of component unavailability:

- Autoscaler is unable to scale apps during a backup.
- Any component within the Elastic Application Runtime deployment that contains lock or unlock scripts for backup is temporarily unavailable during backup process.
- UAA is in read-only mode for the duration of the backup.

Recommendations

VMware recommends:

- Follow the full procedure documented in this topic when creating a backup. This ensures that you always have a consistent backup to restore from.
- Back up frequently. This is especially important before making any changes to your deployment, such as the configuration of any tiles in Tanzu Operations Manager.
- You can prune the BOSH blobstore by running `bosh clean-up --all` before you run a backup of the BOSH Director. The command removes all unused resources, including packages compiled against older stemcell versions. If a lot of unused resources have accumulated over time, this helps create a smaller, faster backup of the BOSH Director. For more information see [Clean-Up](#) in the BOSH documentation.

Running the `bosh clean-up --all` command is a destructive operation and can remove resources that are unused but still needed. For example, if an On-Demand Service Broker is deployed and no service instances have been created, the releases needed to create a service instance are categorized as unused.

Supported components

BBR is a binary that can back up and restore BOSH deployments and BOSH Directors. BBR requires that the backup targets supply scripts that implement the backup and restore functions. BBR can communicate securely with external blobstores and databases, using TLS, if these are configured accordingly.

BBR can back up Tanzu Platform for Cloud Foundry and BOSH Director.

Elastic Application Runtime

BBR can back up and restore Elastic Application Runtime configured with:

- An internal MySQL database or a supported external database. For a list of supported external databases, see [Supported External Databases](#) in *Configuring Cloud Foundry for BOSH Backup and Restore* in the Cloud Foundry documentation.
- An internal WebDAV/NFS blobstore, an external Amazon S3 or S3-compatible blobstore, or an external Azure blobstore.

For guidance about backing up unsupported databases and blobstores, see [Unsupported External Blobstores and Databases in Elastic Application Runtime](#).

BOSH Director

BBR can back up and restore the BOSH Director configured with:

- An internal PostgreSQL database or a supported external database. As part of backing up the BOSH Director, BBR backs up the BOSH UAA database and the CredHub database. For a list of supported external databases, see [Supported External Databases](#) in *Configuring Cloud Foundry for BOSH Backup and Restore* in the Cloud Foundry documentation.
- An internal blobstore or an external versioned S3 blobstore.

BBR support for backing up and restoring external databases and blobstores varies across Tanzu Operations Manager versions. For more information, see [External Storage Support across Tanzu Operations Manager versions](#).

External storage support across Tanzu Operations Manager versions

You can use BBR to back up and restore the following types of external database and blobstore configurations:

| Deployment type | External storage type |
|-----------------------------|---|
| Elastic Application Runtime | <ul style="list-style-type: none">• External database• Amazon S3 blobstore *• S3-compatible blobstore• Azure blobstore• GCS blobstore |
| BOSH Director | <ul style="list-style-type: none">• External database• Amazon S3 blobstore *• S3-Compatible blobstore• GCS blobstore |

* Any S3 clone that supports the versioning API

Unsupported external blobstores and databases in Elastic Application Runtime

If you configured an unsupported external blobstore or an unsupported external database in Elastic Application Runtime, see the following guidelines to perform a backup successfully.

| Scenario | Action |
|--|--|
| You configured a supported database and an unsupported external blobstore in Elastic Application Runtime. | Follow the backup procedures and copy the external blobstore using the IaaS-specific tool. |
| You configured an unsupported external database and a supported blobstore in Elastic Application Runtime. | Follow the backup procedures and copy the external database using the IaaS-specific tool. |
| You configured both an unsupported external database and an unsupported external blobstore in Elastic Application Runtime. | Follow the backup procedures but skip Back up your Elastic Application Runtime Deployment . Copy the external database and blobstore using the IaaS-specific tool. |

You might encounter inconsistencies between the blobstore and database. If apps do not appear during a restore, re-push those apps.

Backing up services

BBR does not back up any service data.

When backing up services:

- BBR backs up and restores the service bindings, but not the service data.
- You can redeploy on-demand service instances manually during restore, but the data in the instance is not backed up.
- BBR does not back up managed services or their data.
- You can back up and restore brokered services with the procedures documented in this topic and in [Restoring deployments from backup with BBR](#).

Backup workflow

BBR examines the jobs in the BOSH deployment, and triggers the scripts in the following stages:

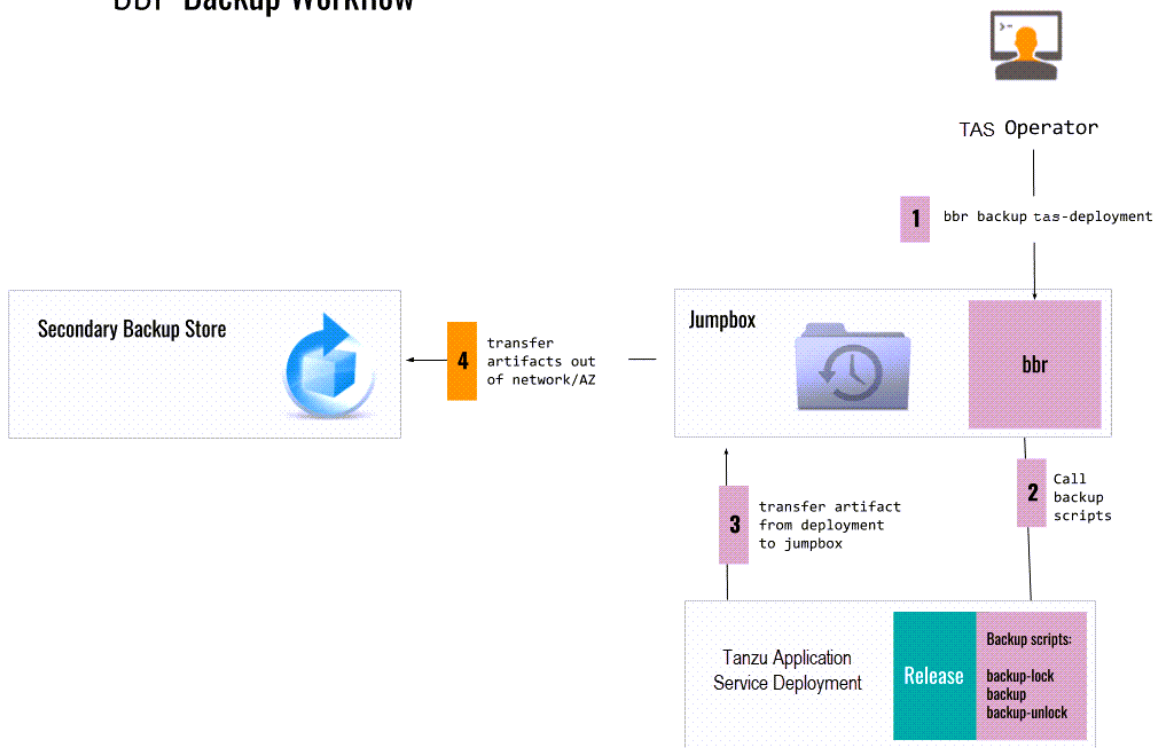
1. Pre-backup lock: The pre-backup lock scripts locks the job so that backups are consistent across the cluster.
2. Backup: The backup script backs up the release.
3. Post-backup unlock: The post-backup unlock script unlocks the job after the backup is complete.

Scripts in the same stage are all triggered together. For instance, BBR triggers all pre-backup lock scripts before any backup scripts. Scripts within a stage can be triggered in any order.

The backup artifacts are drained to the jumpbox, where the operator can transfer them to storage and use them to restore the deployment.

The following diagram shows a sample backup flow. A detailed description follows.

bbr Backup Workflow



1. You run `bbr backup pcf-deployment` on the jumpbox.
2. BBR calls the following backup scripts on the deployment:
 - `backup-lock`
 - `backup`
 - `backup-unlock`
3. BBR transfers the backup artifacts from the deployment to the jumpbox.
4. BBR transfers the backup artifacts from the jumpbox to a secondary backup store that is outside of the network or Availability Zone.

Preparing to create your backup

Step 1: Set up your jumpbox

You must have a jumpbox before you can install BBR to the jumpbox. A jumpbox is a separate, hardened server on your network that provides a controlled means of access to the VMs on computers on your network.

VMware recommends using the Tanzu Operations Manager VM as your jumpbox. If you use the Tanzu Operations Manager VM as your jumpbox, the path to the root certificate authority (CA) is:

```
/var/tempest/workspaces/default/root_ca_certificate
```

You must use this path to run BBR commands.

As an alternative to using the Tanzu Operations Manager VM as your jumpbox, you can:

- Use Concourse to back up a deployment using BBR. For more information, see the [BBR PCF Pipeline Tasks](#) repository on GitHub.

- Configure a custom jumpbox. For more information, see [Configuring a custom jumpbox for BBR](#).

Step 2: Record the Cloud Controller database encryption credentials

To retrieve the Cloud Controller database encryption credentials from the Elastic Application Runtime tile:

1. Go to Tanzu Operations Manager Installation Dashboard.
2. Click the Elastic Application Runtime tile.
3. Click Credentials.
4. Locate the Cloud Controller section.
5. Beside Db Encryption Credentials, click Link to Credential, and record the credentials. You must provide these credentials when you contact [Support](#) for help restoring your installation.

Step 3: Retrieve BOSH Director credentials

To use BBR, you must retrieve and record the following credentials:

- Bosh Director Credentials
- Bbr Ssh Credentials
- Uaa Bbr Client Credentials

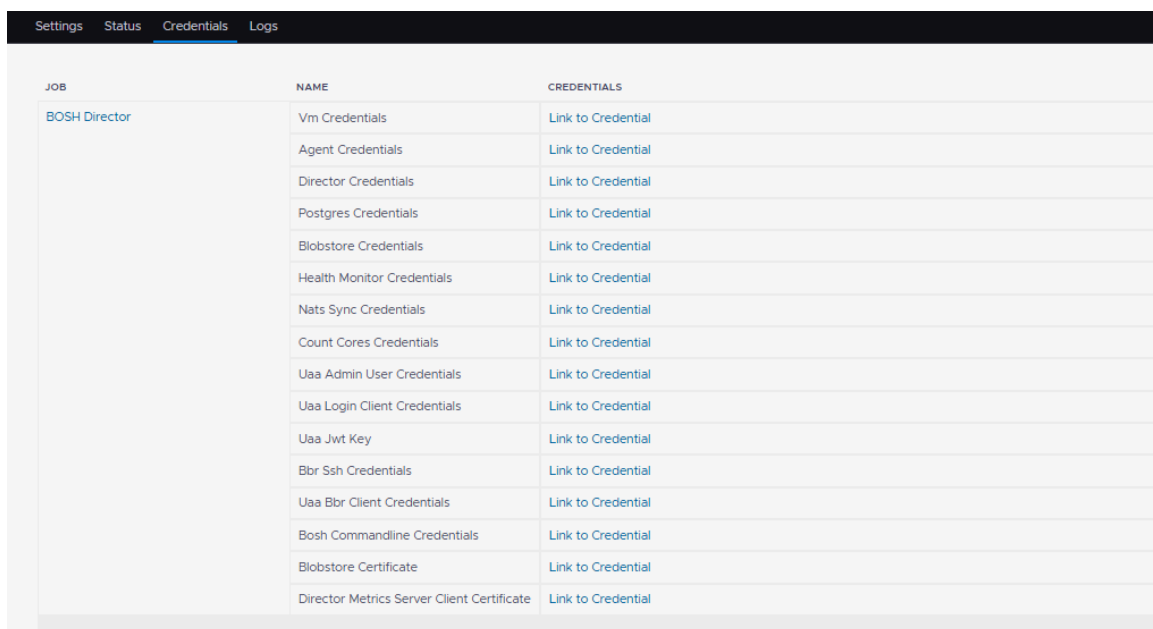
There are two ways to retrieve BOSH Director credentials:

- Option 1: Tanzu Operations Manager Installation Dashboard
- Option 2: Tanzu Operations Manager API

Option 1: Tanzu Operations Manager Installation Dashboard

To retrieve your BOSH Director credentials using the Tanzu Operations Manager Installation Dashboard:

1. Go to Tanzu Operations Manager Installation Dashboard.
2. Click the BOSH Director tile.
3. Click the Credentials tab.



| Settings Status Credentials Logs | | |
|----------------------------------|--|------------------------------------|
| JOB | NAME | CREDENTIALS |
| BOSH Director | Vm Credentials | Link to Credential |
| | Agent Credentials | Link to Credential |
| | Director Credentials | Link to Credential |
| | Postgres Credentials | Link to Credential |
| | Blobstore Credentials | Link to Credential |
| | Health Monitor Credentials | Link to Credential |
| | Nats Sync Credentials | Link to Credential |
| | Count Cores Credentials | Link to Credential |
| | Uaa Admin User Credentials | Link to Credential |
| | Uaa Login Client Credentials | Link to Credential |
| | Uaa Jwt Key | Link to Credential |
| | Bbr Ssh Credentials | Link to Credential |
| | Uaa Bbr Client Credentials | Link to Credential |
| | Bosh Commandline Credentials | Link to Credential |
| | Blobstore Certificate | Link to Credential |
| | Director Metrics Server Client Certificate | Link to Credential |

4. Locate Director Credentials.
 1. Click Link to Credential next to it.

2. Verify the value of the **identity** text box. It can be **director**.
 3. Copy and record the value of the **password** text box.
5. Locate Bbr Ssh Credentials.
1. Click Link to Credential next to it.
 2. Copy the value of the **private_key_pem** text box.
 3. Reformat the copied value and save it in the current directory, in a file named **PRIVATE-KEY-FILE**.
- Run:

```
printf -- "YOUR-PRIVATE-KEY" > PRIVATE-KEY-FILE
```

Where:

- **YOUR-PRIVATE-KEY** is the text of your private key.
- **PRIVATE-KEY-FILE** is the path to the private key file you are creating.

For example:

```
$ printf -- "-----BEGIN RSA PRIVATE KEY----- MIIEkeycontents -----END R
SA PRIVATE KEY-----" > bbr_key.pem
```

6. Locate Uaa Bbr Client Credentials.
1. Click Link to Credential next to it.
 2. Verify the value of the **identity** text box. It can be **bbr_client**.
 3. Record the value of the **password** text box.

Option 2: Tanzu Operations Manager API

To retrieve BOSH Director credentials using the Tanzu Operations Manager API:

1. Obtain your UAA access token. For more information, see [Access the API](#) in *Using the Tanzu Operations Manager API*.
2. Retrieve the Director Credentials.
 1. Run:

```
curl "https://OPS-MAN-FQDN/api/v0/deployed/director/credentials/director_credentials" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where:

- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
 - **UAA-ACCESS-TOKEN** is your UAA access token.
2. Verify the value of the **identity** text box. It can be **director**.
 3. Record the value of the **password** text box.
3. Retrieve the Bbr Ssh Credentials.
1. Run:

```
curl "https://OPS-MAN-FQDN/api/v0/deployed/director/credentials/bbr_ssh_credentials" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where:

- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your UAA access token.

2. Copy the value of the **private_key_pem** text box.
3. Reformat the copied value and save it in the current directory to a file named **PRIVATE-KEY-FILE**. Run:

```
printf -- "YOUR-PRIVATE-KEY" > PRIVATE-KEY-FILE
```

Where:

- **YOUR-PRIVATE-KEY** is the text of your private key.
- **PRIVATE-KEY-FILE** is the path to the private key file you are creating.

For example:

```
$ printf -- "-----BEGIN RSA PRIVATE KEY----- MIIEkeycontents -----END R
SA PRIVATE KEY-----" > bbr_key.pem
```

4. Retrieve the Uaa Bbr Client Credentials.

1. Run:

```
curl "https://OPS-MAN-FQDN/api/v0/deployed/director/credentials/uaa_bbr_client_credentials" \
-X GET \
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where:

- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
 - **UAA-ACCESS-TOKEN** is your UAA access token.
2. Verify the value of the **identity** text box. It can be **bbr_client**.
 3. Record the value of the **password** text box.

For more information about using the Tanzu Operations Manager API, see [Using the Tanzu Operations Manager API](#).

Step 4: Retrieve BOSH Director address

To retrieve the IP address of your BOSH Director from the BOSH Director tile:

1. If you are not using the Tanzu Operations Manager VM as your jumpbox, install the latest [BOSH CLI](#) on your jumpbox.
2. From the Installation Dashboard in Tanzu Operations Manager, click BOSH Director, then Status, and record the IP address listed for the Director. You access the BOSH Director using this IP address.

| <div>Tanzu Ops Manager Installation Dashboard Stemcell Library Change Log Certificates</div> <div>BOSH Director for GCP</div> <div>Settings <u>Status</u> Credentials Logs</div> | | | | |
|--|-------|-----------|---------------|---|
| JOB | INDEX | IPS | AZ | CID |
| BOSH Director | 0 | 10.0.0.23 | us-central1-a | vm-552a4e93-44a3-49e0-6adb-8f2f7c76568f |

- On the command line, log in to the BOSH Director using the IP address that you previously used. Run:

```
bosh -e DIRECTOR-IP \
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE log-in
```

Where:

- DIRECTOR-IP** is the BOSH Director IP address previously recorded.
 - PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the root Certificate Authority (CA) certificate as outlined in [Set up your Jumpbox](#).
- When prompted for Email (), specify **director**.
 - When prompted for Password (), enter the Director Credentials that you obtained in [Retrieve BOSH Director Credentials](#).

For example:

```
$ bosh -e 10.0.0.3 \
--ca-cert /var/tempest/workspaces/default/root_ca_certificate log-in
Email (): director
Password (): *****
Successfully authenticated with UAA
Succeeded
```

Step 5: Check your BOSH Director

This section describes the procedures you must perform to confirm that your BOSH Director is reachable and can be backed up.

You can establish a connection to your jumpbox in one of the following ways:

- [Option 1: Connect with SSH](#)
- [Option 2: Connect with BOSH_ALL_PROXY](#)

These procedures require that you have a jumpbox configured for SSH access. If you do not have a jumpbox, see [Enabling SSH Access](#) and [Tunneling](#) in the BOSH documentation.

Option 1: Connect with SSH

To connect to your jumpbox with SSH, perform one of these steps:

- If you are using the Tanzu Operations Manager VM as your jumpbox, log in to the Tanzu Operations Manager VM. For more information, see [Log Into the Tanzu Operations Manager VM with SSH](#).
- To connect to your jumpbox using the command line, run:

```
ssh -i PATH-TO-KEY JUMPBOX-USERNAME@JUMPBOX-ADDRESS
```

Where:

- **PATH-TO-KEY** is the local path to your private key file for the jumpbox host.
- **JUMPBOX-USERNAME** is your jumpbox username.
- **JUMPBOX-ADDRESS** is the IP address or hostname of the jumpbox.

If you connect to your jumpbox with SSH, you must run the BBR commands in this topic from within your jumpbox.

Option 2: Connect with BOSH_ALL_PROXY

You can use the environment variable **BOSH_ALL_PROXY** to open an SSH tunnel with SOCKS5 to the jumpbox. This tunnel enables you to forward requests from your local machine to the BOSH Director through the jumpbox. When **BOSH_ALL_PROXY** is set, the BBR CLI uses its value to forward requests to the BOSH Director.

For the following procedures to work, ensure the SOCKS port is not already in use by a different tunnel or process.

Using **BOSH_ALL_PROXY** can result in longer backup and restore times due to network performance degradation. All operations must pass through the proxy, so moving backup artifacts can be significantly slower.

To connect with **BOSH_ALL_PROXY**, perform one of the following procedures:

- Establish the tunnel separately from the BOSH CLI.
- Establish the tunnel using the BOSH CLI.

To establish the tunnel separately from the BOSH CLI:

1. Establish the tunnel and make it available on a local port by running:

```
ssh -4 -D SOCKS-PORT -fNC JUMPBOX-USERNAME@JUMPBOX-ADDRESS -i JUMPBOX-KEY-FILE -o ServerAliveInterval=60
```

Where:

- **SOCKS-PORT** is the local SOCKS port.
 - **JUMPBOX-USERNAME** is your jumpbox username.
 - **JUMPBOX-ADDRESS** is the IP address or hostname of the jumpbox.
 - **JUMPBOX-KEY-FILE** is the local SSH private key for accessing the jumpbox.
2. Provide the BOSH CLI with access to the tunnel using the **BOSH_ALL_PROXY** environment variable by running:

```
export BOSH_ALL_PROXY=socks5://localhost:SOCKS-PORT
```

Where is **SOCKS-PORT** is your local SOCKS port.

To establish the tunnel using the BOSH CLI:

1. Provide the BOSH CLI with the necessary SSH credentials to create the tunnel. Ensure that you are using BBR v1.5.1 or later, and run:

```
export BOSH_ALL_PROXY=ssh+socks5://JUMPBOX-USERNAME@JUMPBOX-ADDRESS:SOCKS-PORT?private_key=JUMPBOX-KEY-FILE
```

Where:

- **JUMPBOX-USERNAME** is your jumpbox username.
- **JUMPBOX-ADDRESS** is the IP address or hostname of the jumpbox.
- **SOCKS-PORT** is the local SOCKS port.
- **JUMPBOX-KEY-FILE** is the local SSH private key for accessing the jumpbox.

To check your BOSH Director:

1. Run:

```
bbr director \  
--private-key-path PRIVATE-KEY-FILE \  
--username bbr \  
--host HOST \  
pre-backup-check
```

Where:

- **PRIVATE-KEY-FILE** is the path to the private key file that you created from Bbr Ssh Credentials in [Step 3: Retrieve BOSH Director Credentials](#).
- **HOST** is the address of the BOSH Director. If the BOSH Director is public, **HOST** is a URL, such as <https://bosh.xxx.cf-app.com>. Otherwise, **HOST** is the **BOSH-DIRECTOR-IP** you retrieved in [Step 4: Retrieve BOSH Director Address](#).

To enable debug logs, use the optional **--debug** flag. For more information, see [Logging](#).

2. After the pre-backup verification succeeds, continue to [Step 6: Confirm backup Restore Node is deployed](#). If the pre-backup check fails, the BOSH Director might not have the correct backup scripts or the connection to the BOSH Director might have failed.

Step 6: Confirm backup restore node is deployed

When BBR backs up Elastic Application Runtime, it needs a Backup Restore node.

To confirm that the Backup Restore node is deployed:

1. Go to Tanzu Operations Manager Installation Dashboard.
2. Click the Elastic Application Runtime tile.
3. Click Resource Config.
4. Find the Backup Restore Node job and check that the Instances dropdown is set to **1**.

Resource Config

| JOB | INSTANCES | VM TYPE | PERSISTENT DISK TYPE |
|-----------------------|--------------|---|----------------------|
| > Database | Automatic: 1 | Automatic: e2-standard-2 disk (machine_type: e2-standard-2, cpu: 2, ram: 8 GB, disk: 10 GB) | Automatic: 100 GB |
| > File Storage | Automatic: 1 | Automatic: e2-medium (machine_type: e2-medium, cpu: 2, ram: 4 GB, disk: 8 GB) | Automatic: 100 GB |
| > Control | Automatic: 1 | Automatic: e2-highmem-2 (machine_type: e2-highmem-2, cpu: 2, ram: 16 GB, disk: 6 GB) | None |
| > Compute | Automatic: 1 | Automatic: e2-highmem-2 (machine_type: e2-highmem-2, cpu: 2, ram: 16 GB, disk: 6 GB) | None |
| > Backup Restore Node | Automatic: 1 | Automatic: e2-micro (machine_type: e2-micro, cpu: 1, ram: 1 GB, disk: 8 GB) | Automatic: 200 GB |
| > Router | Automatic: 1 | Automatic: e2-small (machine_type: e2-small, cpu: 2, ram: 2 GB, disk: 8 GB) | None |
| > TCP Router | 0 | Automatic: e2-micro (machine_type: e2-micro, cpu: 1, ram: 1 GB, disk: 8 GB) | None |
| > MySQL Monitor | Automatic: 1 | Automatic: e2-micro (machine_type: e2-micro, cpu: 1, ram: 1 GB, disk: 8 GB) | None |

SAVE

5. If the Instances drop-down is not set to 1:
 1. Set the Instances drop-down to 1.
 2. Click Save.
6. Go to Tanzu Operations Manager Installation Dashboard.
7. Click Review Pending Changes.
8. Review your changes. For more information, see [Reviewing pending product changes](#).
9. Click Apply Changes.

Step 7: Identify your deployment

To identify the name of the BOSH deployment:

1. Log in to your BOSH Director.
2. To identify the name of the BOSH deployment, run:

```
bosh -e BOSH-DIRECTOR-IP --ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE deployments
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Step 4: Retrieve BOSH Director Address](#).
- **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA for the BOSH Director.

Step 8: Check your deployment

To check that your BOSH Director is reachable and has a deployment that can be backed up:

1. Run:

```
bbr deployment \
--target BOSH-DIRECTOR-IP \
--username BOSH-CLIENT \
--password BOSH-PASSWORD \
--deployment DEPLOYMENT-NAME \
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \
pre-backup-check
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Step 4: Retrieve BOSH Director Address](#).
 - **BOSH-CLIENT**, **BOSH-PASSWORD** are the Uaa Bbr Client Credentials, identity and password, that you retrieved in [Step 3: Retrieve BOSH Director Credentials](#).
 - **DEPLOYMENT-NAME** is the deployment name you retrieved in [Step 6: Identify Your Deployment](#).
 - **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the root CA for the BOSH Director. If you are using the Tanzu Operations Manager VM as your jumpbox, the path is `/var/tempest/workspaces/default/root_ca_certificate`.
2. After the pre-backup verification succeeds, continue to [Create Your Backup](#). If the pre-backup check fails, the deployment you selected might not have the correct backup scripts, or the connection to the BOSH Director might have failed.

Create your backup

This section describes the procedures for creating your backup with BBR.

Step 9: Export installation settings

VMware recommends that you back up your Tanzu Operations Manager installation settings by exporting frequently.

When exporting your installation settings, keep in mind:

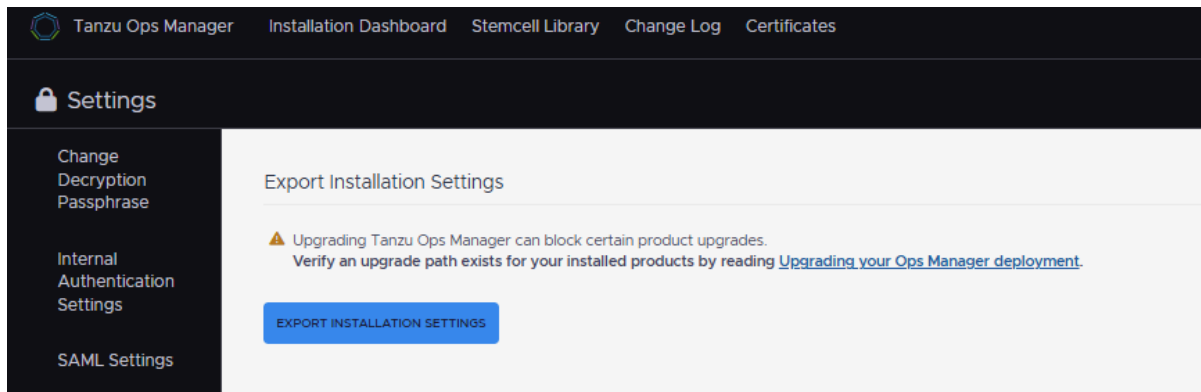
- This option is only available after you have deployed at least once.
- Always export your installation settings before following the procedure in [Deploy Tanzu Operations Manager and Import Installation Settings](#) in *Restoring Deployments from Backup with BBR*.
- Exporting your installation settings only backs up the settings you configured in Tanzu Operations Manager. It does not back up your VMs or any external MySQL databases.
- Your Tanzu Operations Manager settings are encrypted. Ensure you keep track of your Decryption Passphrase, because you need it to restore the Tanzu Operations Manager settings.

When you export installation settings for Tanzu Operations Manager v1.12 and later, releases are not included in the output file. Releases are now included in the BOSH Director backup, so the output file is much smaller than in previous Tanzu Operations Manager versions.

There are two ways to export Tanzu Operations Manager installation settings:

Option 1: Use the Tanzu Operations Manager UI

1. From the Tanzu Operations Manager Installation Dashboard, click your user name at the top right of the navigation.
2. Click Settings.
3. Click Export Installation Settings.
4. Click Export Installation Settings. Note the warning: "Upgrading Tanzu Ops Manager can block certain product upgrades. Verify an upgrade path exists for your installed products by reading [Upgrading your Operations Manager deployment](#)".



Option 2: Use the Tanzu Operations Manager API

Run:

```
curl https://OPS-MANAGER-FQDN/api/v0/installation_asset_collection -H "Authorization: Bearer UAA-ACCESS-TOKEN" > installation.zip
```

Where:

- **OPS-MANAGER-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your UAA access token. For more information, see [Access the API](#) in *Using the Tanzu Operations Manager API*.

Step 10: Back up BOSH Director

The time it takes to back up the BOSH Director varies depending on conditions such as network infrastructure and underlying hardware. For large foundations, this step can take a considerable amount of time because the BBR **backup** command can take a long time to complete. VMware recommends you run it independently of the SSH session so that the process can continue running even if your connection to the jumpbox fails. Use **nohup**, a **screen**, or a **tmux** session.

To back up BOSH Director:

1. Run:

```
bbr director \
--private-key-path PRIVATE-KEY-FILE \
--username bbr \
--host HOST \
backup
```

Where:

- **PRIVATE-KEY-FILE** is the path to the private key file that you created from Bbr Ssh Credentials in [Step 3: Retrieve BOSH Director Credentials](#).
- **HOST** is the address of the BOSH Director:
 - If the BOSH Director is public, **HOST** is a URL, such as <https://bosh.xxx.cf-app.com>.

- If the BOSH Director is not public, **HOST** is the **BOSH-DIRECTOR-IP** retrieved in [Step 4: Retrieve BOSH Director Address](#).

To enable debug logs, use the optional `--debug` flag. For more information, see [Logging](#).

If your backup ends early or fails, you must perform a cleanup.

To clean up:

1. Run:

```
bbr director \  
--private-key-path PRIVATE-KEY-FILE \  
--username bbr \  
--host HOST \  
backup-cleanup
```

Where:

- **PRIVATE-KEY-FILE** is the path to the private key file that you created from Bbr Ssh Credentials in [Step 3: Retrieve BOSH Director Credentials](#).
- **HOST** is the address of the BOSH Director:
 - If the BOSH Director is public, **HOST** is a URL, such as <https://bosh.xxx.cf-app.com>.
 - If the BOSH Director is not public, **HOST** is the **BOSH-DIRECTOR-IP** retrieved in [Step 4: Retrieve BOSH Director Address](#).

Step 11: Back up your Elastic Application Runtime deployment

Backing up Elastic Application Runtime can take a considerable amount of time, and varies depending on network and hardware infrastructure. In controlled conditions, backup durations have been observed ranging from a few minutes to several hours. To reduce your backup lock times, consider using selective backups. For more information, see [File storage backup level](#). Part of the backup process causes downtime for Cloud Controller, during which you cannot push, scale, or delete apps. Your apps are not affected. Because the BBR **backup** command can take a long time to complete, VMware recommends you run it independently of the SSH session, so that the process can continue running even if your connection to the jumpbox fails. The preceding command uses **nohup**, but you could also run the command in a **screen** or **tmux** session.

To back up your Elastic Application Runtime deployment:

1. (Optional) If you are not using an internal blobstore or a versioned S3-compatible external blobstore, create a copy of the blobstore with your IaaS-specific tool. Your blobstore backup might be slightly inconsistent with your Elastic Application Runtime backup depending on the duration of time between performing the backups.
2. Run:

```
bbr deployment \  
--target BOSH-DIRECTOR-IP \  
--username BOSH-CLIENT \  
--password BOSH-PASSWORD \  
--deployment DEPLOYMENT-NAME \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \  
backup
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Step 4: Retrieve BOSH Director Address](#).
 - **BOSH-CLIENT**, **BOSH-PASSWORD** are the Uaa Bbr Client Credentials, or the identity and password, that you retrieved in [Step 3: Retrieve BOSH Director Credentials](#).
 - **DEPLOYMENT-NAME** is the deployment name you retrieved in [Step 6: Identify Your Deployment](#).
 - **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA for the BOSH Director.
 - To enable debug logs, use the optional **--debug** flag. For more information, see [Logging](#).
 - To ensure that BBR downloads your current deployment manifest when backing up, use the optional **--with-manifest** flag. These manifests are included in the Tanzu Operations Manager export, but are useful for reference.
3. After the command successfully completes, continue to [Step 12: Transfer Backup Artifacts](#). If the command fails:
4. Run:

```
bbr deployment \  
--target BOSH-DIRECTOR-IP \  
--username BOSH-CLIENT \  
--password BOSH-PASSWORD \  
--deployment DEPLOYMENT-NAME \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \  
backup-cleanup
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Step 4: Retrieve BOSH Director Address](#).
 - **BOSH-CLIENT**, **BOSH-PASSWORD** are the Uaa Bbr Client Credentials, or the identity and password, that you retrieved in [Step 3: Retrieve BOSH Director Credentials](#).
 - **DEPLOYMENT-NAME** is the deployment name you retrieved in [Step 6: Identify Your Deployment](#).
 - **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA for the BOSH Director.
 - To enable debug logs, use the optional **--debug** flag. For more information, see [Logging](#).
 - To ensure that BBR downloads your current deployment manifest when backing up, use the optional **--with-manifest** flag. These manifests are included in the Tanzu Operations Manager export, but are useful for reference.
5. Run the BBR **backup** command again after ensuring that:
- All the parameters in the command are set.
 - The BOSH Director credentials are valid.
 - The specified deployment exists.

After taking your backup

This section describes the procedures you must perform after taking your backup.

Step 12: Transfer backup artifacts

After creating your backup successfully:

1. Move the backup artifacts off the jumpbox to your preferred storage space. The backup created by BBR consists of a directory with the backup artifacts and metadata files. VMware recommends compressing and encrypting the files.
2. Make redundant copies of your backup artifacts and store them in multiple locations to minimize the risk of losing backups in the event of a disaster.
3. Attempt to restore every backup to validate the artifacts. For more information, see [Step 13: Validate Your Backup](#).

Backup artifacts might contain data covered by European Union regulations. For more information, see [European Union's General Data Protection Regulation \(GDPR\)](#).

Step 13: (Optional) Validate your backup

After you back up Tanzu Operations Manager, consider validating your backup by restoring it and checking the apps. BBR is designed for disaster recovery and BBR backups are only compatible with environments matching their source environment's configuration.

The VMs and disks from the backed-up BOSH Director must not be visible to the new BOSH Director when validating your backup, so VMware recommends that you deploy the new BOSH Director to a different IaaS network and account for the VMs and disks of the backed-up BOSH Director.

Data services outside of Tanzu Operations Manager might produce unexpected side effects during restoration. Restored apps and services might attempt to connect to data services when you restore to a new environment.

For example, consider an app that processes mail queues and connects to an external database. When you validate your backup in a test environment, the app might start processing the queue, and this work might be lost.

Option 1: Validate your backup in a second environment

1. Export your Tanzu Operations Manager installation by performing the procedure in [Step 8: Export Installation Settings](#).
2. Create a new Tanzu Operations Manager VM in a different network to the original. Ensure that the Tanzu Operations Manager VM has enough persistent disk to accommodate the files exported in the previous step. For more information, see [IaaS-Specific Deployment Guidelines](#) in *Deploying BOSH and Tanzu Operations Manager*.
3. Ensure that the restore environment is compatible with the backup. For more information, see [Compatibility of Restore](#) in *Restoring Deployments from Backup with BBR*.
4. Follow the procedures in [Restoring Deployments from Backup with BBR](#).

Option 2: Validate your Elastic Application Runtime backup only

For a sandbox or other non-production environment, you can optionally perform an *in-place* restore of Elastic Application Runtime only. In this case, you restore the Elastic Application Runtime backup to the same Tanzu Operations Manager environment that the backup was created from. For more information, see [Step 12: Redeploy Elastic Application Runtime](#) in *Restoring Deployments from Backup with BBR*.

Logging

BBR outputs logs to stdout. By default, BBR logs:

- The backup and restore scripts that it finds

- When it starts or finishes a stage, such as pre-backup scripts or backup scripts
- When the process is complete
- When any error occurs

If you require additional logging, use the optional `--debug` flag to print:

- Logs about the API requests made to the BOSH server
- All commands executed on remote instances
- All commands executed on local environment
- Standard in and standard out streams for the backup and restore scripts when they are executed

Canceling a backup

To cancel a backup:

1. Enter `CTRL-C` to terminate the BBR process, then enter `yes` to confirm.
2. Run the BBR `backup-cleanup` command.
 - For a canceled director backup, run:

```
bbr director \
--private-key-path PRIVATE-KEY-FILE \
--username bbr \
--host HOST \
backup-cleanup
```

Where:

- `PRIVATE-KEY-FILE` is the path to the private key file that you created from Bbr Ssh Credentials in [Step 3: Retrieve BOSH Director Credentials](#).
- `HOST` is the address of the BOSH Director:
 - If the BOSH Director is public, `HOST` is a URL, such as `https://bosh.xxx.cf-app.com`.
 - If the BOSH Director is not public, `HOST` is the `BOSH-DIRECTOR-IP` retrieved in [Step 4: Retrieve BOSH Director Address](#).
- For a canceled deployment backup, run:

```
bbr deployment \
--target BOSH-DIRECTOR-IP \
--username BOSH-CLIENT \
--password BOSH-PASSWORD \
--deployment DEPLOYMENT-NAME \
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \
backup-cleanup
```

Where:

- `BOSH-DIRECTOR-IP` is the BOSH Director IP you retrieved in [Step 4: Retrieve BOSH Director Address](#).
- `BOSH-CLIENT`, `BOSH-PASSWORD` are the Uaa Bbr Client Credentials, or the identity and password, that you retrieved in [Step 3: Retrieve BOSH Director Credentials](#).

- `DEPLOYMENT-NAME` is the deployment name you retrieved in [Step 6: Identify Your Deployment](#).
- `PATH-TO-BOSH-SERVER-CERTIFICATE` is the path to the CA for the BOSH Director.
- To enable debug logs, use the optional `--debug` flag. For more information, see [Logging](#).
- To ensure that BBR downloads your current deployment manifest when backing up, use the optional `--with-manifest` flag. These manifests are included in the Tanzu Operations Manager export, but are useful for reference.

European Union General Data Protection Regulation

Backup artifacts might contain personal and other data covered by the European Union's *General Data Protection Regulation (GDPR)*.

For example, a Elastic Application Runtime backup could contain user email addresses.

As such, handle backup artifacts in accordance with organizational policies and applicable laws as they pertain to security, confidentiality, and privacy.

Tanzu Operations Manager file storage backup levels

You can configure Tanzu Platform for Cloud Foundry file storage backups to include all files, exclude droplets, or exclude droplets and packages. For information about configuring the Elastic Application Runtime blobstore to exclude droplets or to exclude droplets and packages, see the [Elastic Application Runtime documentation](#).

Considerations for configuring file storage backups

When considering configuration for your use case, the two main factors are backup artifact size and Recovery Time Objective (RTO).

If you want a smaller backup artifact, you can exclude certain blobs from the backup. This can enable you to take more frequent backups without using a large amount of storage space and causes shorter periods of app downtime. However, you will have to re-push or restage all apps, which results in a higher RTO.

If you want a lower RTO, you can do a complete file storage backup. However, this results in a larger backup artifact size.

The following table gives approximate values for the proportion of the blobstore used by each object:

| Name | Portion of blobstore |
|------------|----------------------|
| Buildpacks | 0–5% |
| Packages | 5–25% |
| Droplets | 70–80% |

Complete file storage backup

A complete file storage backup contains the complete blobstore, including droplets, packages, and buildpacks. This is the default behavior. This backup is the largest and slowest but has the lowest RTO.

When restoring from this backup, apps automatically return in a running state because Elastic Application Runtime is restored with the expected droplets and can reschedule the apps without any manual intervention.

File storage backups excluding droplets

If you configure Elastic Application Runtime to exclude droplets from the file storage backups, the backup is a smaller and faster, but has an increased RTO.

When restoring from this backup, apps return in a stopped state because the foundation does not have their corresponding droplets. Developers must restage each app to return them to a running state. Elastic Application Runtime must recompile the droplet using the packages in the backup then deploy the new droplet. This process increases the RTO.

For more information about configuring the Elastic Application Runtime blobstore to exclude droplets, see [Configure file storage](#) in *Configuring Elastic Application Runtime*.

File storage backups excluding droplets and packages

If you configure Elastic Application Runtime to exclude droplets and packages from the file storage backup, the backup is smaller and faster than the other two options, but with an even greater RTO.

When restoring from this backup, apps return in a stopped state because the foundation does not have their corresponding droplets or packages. Developers must re-push each app to return them to a running state. This uploads the app packages, compiles them, and deploys a new droplet. This process increases the RTO.

Elastic Application Runtime component availability during backup process

This topic describes component availability while you back up Tanzu Platform for Cloud Foundry using BOSH Backup and Restore (BBR). To ensure the correctness of backups, each component that requires a backup of its database has its own set of scripts.

Cloud Controller

The Cloud Controller is unavailable during backup. Apps and services continue to run as normal, but you cannot perform operations that require the Cloud Controller API. This includes:

- Pushing new apps or creating new services
- Modifying existing apps or services
- Using the clients of the Cloud Controller API, such as:
 - The Tanzu cf CLI
 - Apps Manager and its integrations
 - The [Java client](#) used by Spring apps

The following table describes the backup process for the Cloud Controller:

| Stage | Description |
|--------------------|---|
| 1: Pre-backup lock | The processes running on the Cloud Controller Worker, Cloud Controller, and Clock Global VMs are stopped. |

| Stage | Description |
|-----------------------|---|
| 2: Backup | The BBR SDK backup script runs to back up the Cloud Controller database (CCDB), which contains state information for apps on your deployment. |
| 3: Post-backup unlock | The processes start again on the Cloud Controller Worker, Cloud Controller, and Clock Global VMs. |

For more information, see the [Cloud Controller API](#) documentation.

UAA

UAA remains available in read-only mode during backup. This means that you cannot perform write operations for clients, users, groups, identity providers, or zone configuration. However, you can continue performing read operations, such as generating, validating, and revoking tokens. Additionally, UAA continues to authenticate users and authorize requests for users and clients.

The read-only behavior during backup applies to all of the following ways of accessing UAA: the UAA API, the UAA CLI, Tanzu cf CLI, login screens, and services such as the Single Sign-On Service tile.

The following table describes the backup process for UAA:

| Stage | Description |
|-----------------------|--|
| 1: Pre-backup lock | UAA enters read-only mode. |
| 2: Backup | The BBR SDK backup script runs to back up the UAA database, which contains Elastic Application Runtime user credentials. |
| 3: Post-backup unlock | UAA exits read-only mode. |

Routing API

The Routing API remains available during backup. However, you cannot perform write operations using the Routing API because the routing database is locked. All read operations offered by the Routing API remain available.

The BBR SDK backup script for the Routing API backs up its database, which contains router groups, routes, and internal implementation information.

For more information, see the [Routing API documentation](#) in the Routing API repository on GitHub.

Usage service

The Usage service is unavailable during backup. You cannot access the API as described in [Reporting app, task, and service instance usage](#). Additionally, you cannot view usage and accounting reports as described in [Reporting instance usage with Apps Manager](#).

The following table describes the backup process for Usage service:

| Stage | Description |
|-----------------------|---|
| 1: Pre-backup lock | The Usage service apps in the <code>system</code> org stop. This lock occurs before the Cloud Controller and UAA components lock. |
| 2: Backup | The BBR SDK backup script runs to back up the Usage service database. |
| 3: Post-backup unlock | The Usage service apps in the <code>system</code> org start again. This unlock occurs after the Cloud Controller and UAA components unlock. |

The Usage service runs as a set of Elastic Application Runtime apps in the [system](#) org.

App Autoscaler

The App Autoscaler service is unavailable during backup. You cannot access the UI or API. For any apps configured to use the App Autoscaler, the service does not scale these apps during backup.

The following table describes the backup process for App Autoscaler:

| Stage | Description |
|-----------------------|---|
| 1: Pre-backup lock | The Autoscaler apps in the system org stop. This lock occurs before the Cloud Controller and UAA components lock. |
| 2: Backup | The BBR SDK backup script runs to back up the App Autoscaler database. |
| 3: Post-backup unlock | The Autoscaler apps in the system org start again. This unlock occurs after the Cloud Controller and UAA components unlock. |

The App Autoscaler service runs as a set of Elastic Application Runtime apps in the [system](#) org.

For more information, see [Using App Autoscaler to scale apps in Elastic Application Runtime](#).

NFS volume service

The NFS Service Broker backup scripts rely on the locking of the Cloud Controller to stop traffic to its service. This is because the Cloud Controller is responsible for invoking the NFS Service Broker.

When the Cloud Controller locks during backup, you cannot create or delete new instances or bindings of a volume service. However, apps already bound to a volume service continue to operate normally during backups.

The NFS Service Broker backup script performs a backup of the database used to store service instances and service bindings for the NFS Service Broker.

Notification Service

The Notification Service is not available during backup with BBR due to its dependency on the Cloud Controller. Notifications cannot be sent while the Cloud Controller is unavailable.

Network Policy Server

The Network Policy Server is unavailable during backup. While existing policies are still enforced, you cannot use the Tanzu cf CLI to add or remove policies for container networking as described in [Configuring container-to-container networking](#).

CredHub

Runtime CredHub is unavailable during backup. If the binding credentials for an app are stored in CredHub, the app cannot fetch those credentials during backup. In some cases, apps might not start if they cannot fetch credentials for a service instance binding.

Restoring Tanzu Operations Manager deployments from backup with BBR

This topic describes how you can restore critical back end components using BOSH Backup and Restore (BBR), a command-line tool for backing up and restoring BOSH deployments. These procedures prepare your environment for Tanzu Operations Manager, deploy Tanzu Operations Manager, import your installation settings, and use BBR to restore your Tanzu Operations Manager components.

To perform the procedures in this topic, you must have backed up your deployment by following the procedure in [Backing up deployments with BBR](#).

Restoring a deployment with BBR is a destructive operation. If the restore fails, the new environment could be left in an unusable state and require re-provisioning. Only perform the procedures in this topic for the purpose of disaster recovery, such as recreating a deployment after a storage-area network (SAN) corruption.

When validating your backup, the VMs and disks from the backed-up BOSH Director must not be visible to the new BOSH Director. As a result, VMware recommends that you deploy the new BOSH Director to a different IaaS network and account than the VMs and disks of the backed-up BOSH Director.

If the BOSH Director you are restoring had any deployments that were deployed manually rather than through a Tanzu Operations Manager tile, you must restore them manually at the end of the process. For more information, see [Restore non-tile deployments](#).

Compatibility of restore

This section describes the requirements for a backup artifact to be restorable to another environment. This section is for guidance only, and VMware recommends that you validate your backups by using the backup artifacts in a restore.

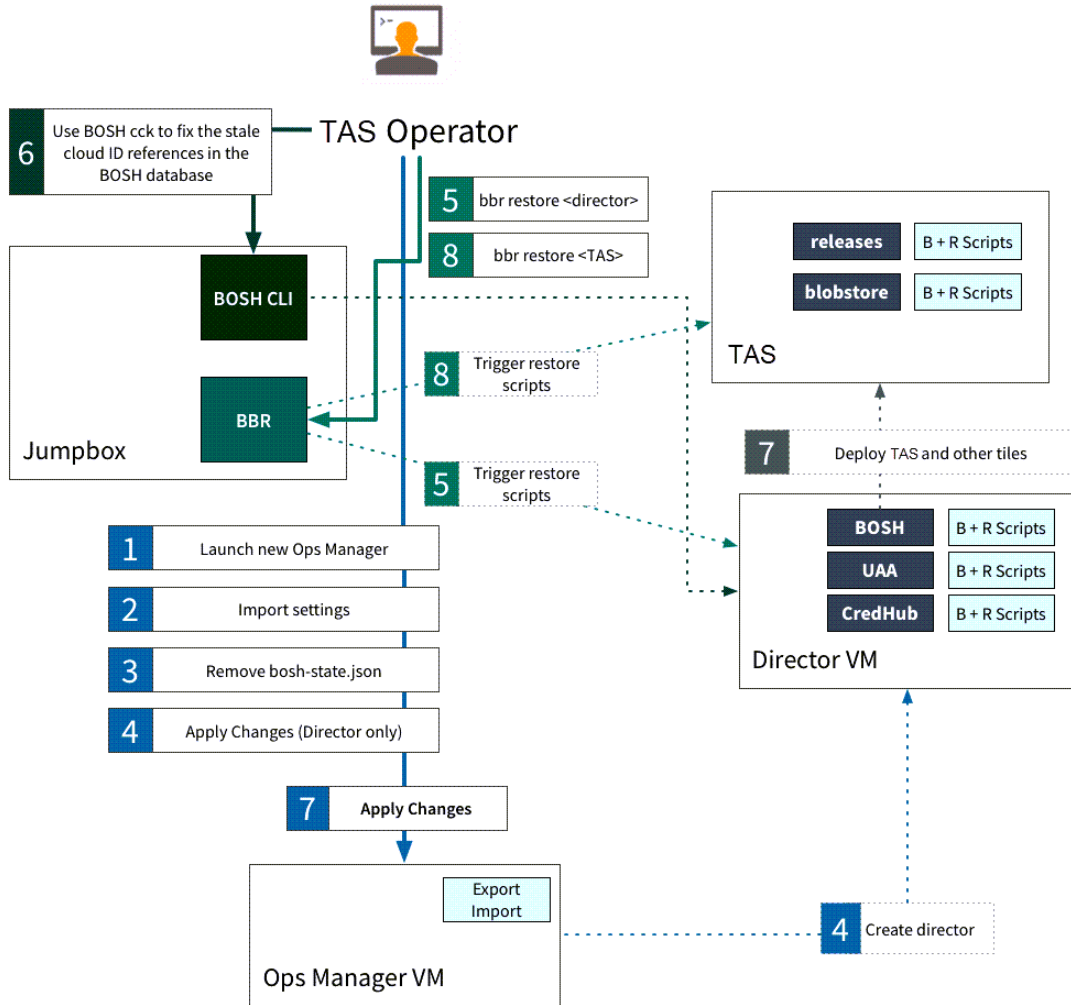
For a backup artifact to be restorable, these restrictions apply:

- CIDR ranges: BBR requires the IP address ranges to be the same in the restore environment as in the backup environment.
- Topology: BBR requires the BOSH topology of a deployment to be the same in the restore environment as it was in the backup environment.
- Naming of instance groups and jobs: For any deployment that implements the backup and restore scripts, the instance groups and jobs must have the same names.
- Number of instance groups and jobs: For instance groups and jobs that have backup and restore scripts, there must be the same number of instances.
- Limited validation: BBR puts the backed-up data into the corresponding instance groups and jobs in the restored environment, but cannot validate the restore beyond that. For example, if the MySQL encryption key is different in the restore environment, the BBR restore might succeed, but the restored MySQL database will be unusable.
- Tanzu Operations Manager version: BBR can restore to the same version of Tanzu Operations Manager that was backed up. BBR does not support restoring to other major, minor, or patch releases.

A change in VM size or underlying hardware does not affect the ability of BBR to restore data, as long as there is adequate storage space to restore the data.

Restore workflow

The following diagram illustrates the flow of the restore process in a series of steps performed by the operator. These steps are described in more detail throughout this topic.



[View a larger version of this diagram.](#)

As an Operator, complete the following steps:

1. Launch new Tanzu Operations Manager: Follow the procedures for your IaaS to deploy Tanzu Operations Manager. For more information, see step 1 of [Deploy Tanzu Operations Manager and Import Installation Settings](#).
2. Import settings: You can import settings either with the Tanzu Operations Manager UI or API. For more information, see step 2 of [Deploy Tanzu Operations Manager and Import Installation Settings](#).
3. Remove bosh-state.json: SSH into your Tanzu Operations Manager VM and delete the `bosh-state.json` file. For more information, see [Remove BOSH State File](#).
4. Apply Changes (BOSH Director only): Use the Tanzu Operations Manager API only to deploy the BOSH Director. For more information, see [Deploy the BOSH Director](#).
5. `bbr restore <director>`: Run the BBR restore command from your jumpbox to restore the BOSH Director. For more information, see [Restore the BOSH Director](#).
6. Use BOSH `cck` to fix the stale cloud ID references in the BOSH database: For each deployment in the BOSH Director, you must run a `bosh cloud-check` command. For more information, see [Remove Stale Cloud IDs for All Deployments](#).

7. Apply Changes: In the Tanzu Operations Manager Installation Dashboard, click Review Pending Changes, review your changes, and click Apply Changes. For more information, see [Reviewing Pending Product Changes](#).
8. `bbr restore <Elastic Application Runtime>`: Run the BBR restore command from your jumpbox to restore Elastic Application Runtime. For more information, see [Restore Elastic Application Runtime](#).

Prepare to restore

These sections describe the procedure you must perform before restoring your backup with BBR.

Step 1: Prepare your environment (optional)

In the event of a disaster, you can lose not only your VMs and disks, but your IaaS resources as well, such as networks and load balancers.

If you need to recreate your IaaS resources, prepare your environment by following the instructions specific to your IaaS in [Installation Overview](#).

If you recreate your IaaS resources, you must also add those resources to Tanzu Operations Manager by following the procedures in [\(Optional\) Configure Tanzu Operations Manager for New Resources](#).

Step 2: Deploy Tanzu Operations Manager and import installation settings

1. Follow the procedures for your IaaS to deploy Tanzu Operations Manager:
 - AWS: See [Deploying Tanzu Operations Manager on AWS](#).
 - GCP: See [Deploying Tanzu Operations Manager on GCP](#).
 - vSphere: See [Deploying Tanzu Operations Manager on vSphere](#).
2. Import your installation settings. This can be done in two ways:
 1. Using the Tanzu Operations Manager UI:
 1. Access your new Tanzu Operations Manager deployment by navigating to the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment in a browser.
 2. On the Welcome to Ops Manager page, click Import Existing Installation.
 3. In the import panel:
 - Enter the Decryption Passphrase in use when you exported the installation settings from Tanzu Operations Manager.
 - Click Browse and browse to the installation ZIP file that you exported in [Export Installation Settings](#) in *Backing Up Deployments with BBR*.
 4. Click Import.

Some browsers do not provide feedback on the status of the import process, and might appear to hang. The import process takes at least 10 minutes, and takes longer if there are more tiles present on the backed-up Tanzu Operations Manager.
 5. When the import is complete, a Successfully imported installation message appears.
 2. Using the Tanzu Operations Manager API:

1. Run:

```
curl -v "https://OPS-MANAGER-FQDN/api/v0/installation_asset_collection \
-X POST \
-H "Authorization: Bearer UAA-ACCESS-TOKEN" \
-F 'installation[file]=@installation.zip' \
-F 'passphrase=DECRYPTION-PASSPHRASE'
```

Where:

- **OPS-MANAGER-FQDN** is the FQDN for your Tanzu Operations Manager deployment.
- **UAA-ACCESS-TOKEN** is your UAA access token. For more information about how to retrieve this token, see [Using the Tanzu Operations Manager API](#).
- **DECRYPTION-PASSPHRASE** is the decryption passphrase in use when you exported the installation settings from Tanzu Operations Manager.

Do not click Apply Changes in Tanzu Operations Manager until you reach the instructions in [Redeploy Elastic Application Runtime](#).

(Optional) Step 3: Configure Tanzu Operations Manager for new resources

If you recreated IaaS resources such as networks and load balancers by following the procedure in [\(Optional\) Prepare your environment](#), you must update Tanzu Operations Manager with your new resources.

To update Tanzu Operations Manager:

1. Enable the advanced mode of Tanzu Operations Manager. For more information, see [Enabling Advanced Mode in Tanzu Operations Manager](#). In advanced mode, Tanzu Operations Manager allows you to make changes that are normally disallowed. You might see warning messages when you save changes.
2. Go to the Installation Dashboard.
3. Click the BOSH Director tile.
4. If you are using Google Cloud Platform (GCP), click Google Config and update:
 1. Project ID to reflect the GCP project ID.
 2. Default Deployment Tag to reflect the environment name.
 3. AuthJSON to reflect the service account.
5. Click Create Networks and update the network names to reflect the network names for the new environment.
6. If your BOSH Director had an external hostname, modify the Director Hostname field in the Director Config pane of the BOSH Director tile to ensure it does not conflict with the hostname of the backed-up Director.
7. Return to the Installation Dashboard.
8. Click the Elastic Application Runtime tile.
9. Click Resource Config.

10. If necessary for your IaaS, enter the name of your new load balancers in the Load Balancers column.
11. If necessary, select Networking and update the load balancer SSL certificate and private key under Certificates and private keys for the Gorouter.
12. If your environment has a new DNS address, update the old environment DNS entries to point to the new load balancer addresses. For more information, see the [Elastic Application Runtime documentation](#) and follow the instructions for your IaaS.
13. If your Elastic Application Runtime deployment uses an external blobstore, ensure that Elastic Application Runtime is configured to use a different blobstore. Otherwise, it attempts to connect to the blobstore that the backed-up Elastic Application Runtime deployment is using.
14. Click Domains.
15. Update your System domain and Apps domain to refer to the new environment's domains.
16. Ensure that there are no outstanding warning messages in the BOSH Director tile, then deactivate Tanzu Operations Manager advanced mode. For more information, see [Enabling Advanced Mode in Tanzu Operations Manager](#).

Step 4: Remove BOSH state file

Tanzu Operations Manager records the state of the BOSH Director in a BOSH state file. Tanzu Operations Manager uses this file when performing a BOSH operation or upgrading the BOSH Director. For more information, see the [BOSH documentation](#).

Tanzu Operations Manager fails to apply changes if it cannot contact the BOSH Director. In most cases, this is because the BOSH Director, as described by the BOSH state file, is missing or unresponsive.

If this failure would be dangerous, then it is recommended that you delete the BOSH state file. This forces Tanzu Operations Manager to create a new BOSH Director.

To remove the BOSH state file:

1. SSH into your Tanzu Operations Manager VM. For more information, see [Log Into the Tanzu Operations Manager VM with SSH](#).
2. To delete the `/var/tempest/workspaces/default/deployments/bosh-state.json` file, go to the Tanzu Operations Manager VM and run:

```
sudo rm /var/tempest/workspaces/default/deployments/bosh-state.json
```

3. In a browser, go to the FQDN of your Tanzu Operations Manager deployment.
4. Log in to Tanzu Operations Manager.

Step 5: Deploy the BOSH Director

Use the Tanzu Operations Manager API or the check box on the Review Pending Changes page to deploy the BOSH Director by itself.

Step 6: Transfer artifacts to jumpbox

In the [Back up your Elastic Application Runtime Deployment](#) step of *Backing Up Deployments with BBR*, you moved the TAR and metadata files of the backup artifacts off your jumpbox to your preferred storage space. Now you must transfer those files back to your jumpbox.

Restore your backup

These sections describe the procedure to restore your backup with BBR.

Step 7: Retrieve BOSH Director credentials

To use BBR, you must retrieve and record the following credentials:

- Bosh Director Credentials
- Bbr Ssh Credentials
- Uaa Bbr Client Credentials

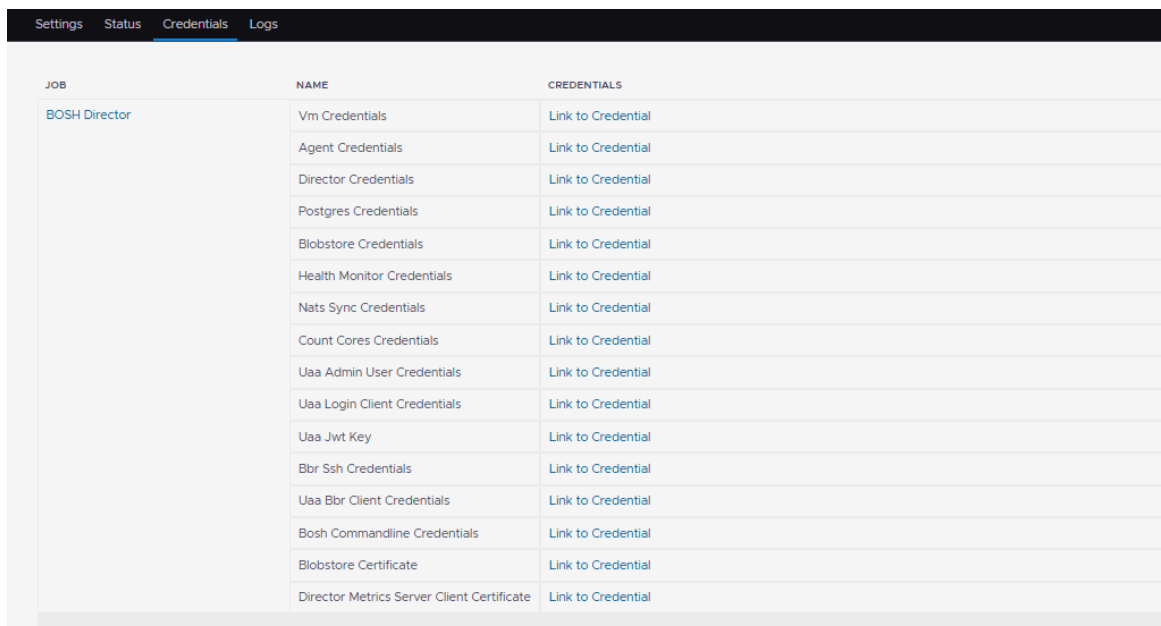
There are two ways to retrieve BOSH Director credentials:

- Option 1: Tanzu Operations Manager Installation Dashboard
- Option 2: Tanzu Operations Manager API

Option 1: Tanzu Operations Manager Installation Dashboard

To retrieve your BOSH Director credentials using the Tanzu Operations Manager Installation Dashboard:

1. Go to Tanzu Operations Manager Installation Dashboard.
2. Click the BOSH Director tile.
3. Click the Credentials tab.



The screenshot shows the 'Credentials' tab of the Tanzu Operations Manager Installation Dashboard. The table lists various credentials for the 'BOSH Director' job, each with a 'Link to Credential' button.

| JOB | NAME | CREDENTIALS |
|---------------|--|------------------------------------|
| BOSH Director | Vm Credentials | Link to Credential |
| | Agent Credentials | Link to Credential |
| | Director Credentials | Link to Credential |
| | Postgres Credentials | Link to Credential |
| | Blobstore Credentials | Link to Credential |
| | Health Monitor Credentials | Link to Credential |
| | Nats Sync Credentials | Link to Credential |
| | Count Cores Credentials | Link to Credential |
| | Uaa Admin User Credentials | Link to Credential |
| | Uaa Login Client Credentials | Link to Credential |
| | Uaa Jwt Key | Link to Credential |
| | Bbr Ssh Credentials | Link to Credential |
| | Uaa Bbr Client Credentials | Link to Credential |
| | Bosh Commandline Credentials | Link to Credential |
| | Blobstore Certificate | Link to Credential |
| | Director Metrics Server Client Certificate | Link to Credential |

4. Locate Director Credentials.
 1. Click Link to Credential next to it.
 2. Verify the value of the **identity** text box. It can be **director**.
 3. Copy and record the value of the **password** text box.
5. Locate Bbr Ssh Credentials.
 1. Click Link to Credential next to it.
 2. Copy the value of the **private_key_pem** text box.
 3. Reformat the copied value and save it in the current directory, in a file named **PRIVATE-KEY-FILE**.
Run:

```
printf -- "YOUR-PRIVATE-KEY" > PRIVATE-KEY-FILE
```

Where:

- **YOUR-PRIVATE-KEY** is the text of your private key.
- **PRIVATE-KEY-FILE** is the path to the private key file you are creating.

For example:

```
$ printf -- "-----BEGIN RSA PRIVATE KEY----- MIIEkeycontents -----END R  
SA PRIVATE KEY-----" > bbr_key.pem
```

6. Locate Uaa Bbr Client Credentials.

1. Click Link to Credential next to it.
2. Verify the value of the **identity** text box. It can be **bbr_client**.
3. Record the value of the **password** text box.

Option 2: Tanzu Operations Manager API

To retrieve BOSH Director credentials using the Tanzu Operations Manager API:

1. Obtain your UAA access token. For more information, see [Access the API](#) in *Using the Tanzu Operations Manager API*.
2. Retrieve the Director Credentials.
 1. Run:

```
curl "https://OPS-MAN-FQDN/api/v0/deployed/director/credentials/director_credentials" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where:

- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
 - **UAA-ACCESS-TOKEN** is your UAA access token.
2. Verify the value of the **identity** text box. It can be **director**.
 3. Record the value of the **password** text box.
3. Retrieve the Bbr Ssh Credentials.
 1. Run:

```
curl "https://OPS-MAN-FQDN/api/v0/deployed/director/credentials/bbr_ssh_credentials" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where:

- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
 - **UAA-ACCESS-TOKEN** is your UAA access token.
2. Copy the value of the **private_key_pem** text box.

3. Reformat the copied value and save it in the current directory to a file named **PRIVATE-KEY-FILE**. Run:

```
printf -- "YOUR-PRIVATE-KEY" > PRIVATE-KEY-FILE
```

Where:

- **YOUR-PRIVATE-KEY** is the text of your private key.
- **PRIVATE-KEY-FILE** is the path to the private key file you are creating.

For example:

```
$ printf -- "-----BEGIN RSA PRIVATE KEY----- MIIKeycontents -----END R  
SA PRIVATE KEY-----" > bbr_key.pem
```

4. Retrieve the Uaa Bbr Client Credentials.

1. Run:

```
curl "https://OPS-MAN-FQDN/api/v0/deployed/director/credentials/uaa_bbr_client_credentials" \  
-X GET \  
-H "Authorization: Bearer UAA-ACCESS-TOKEN"
```

Where:

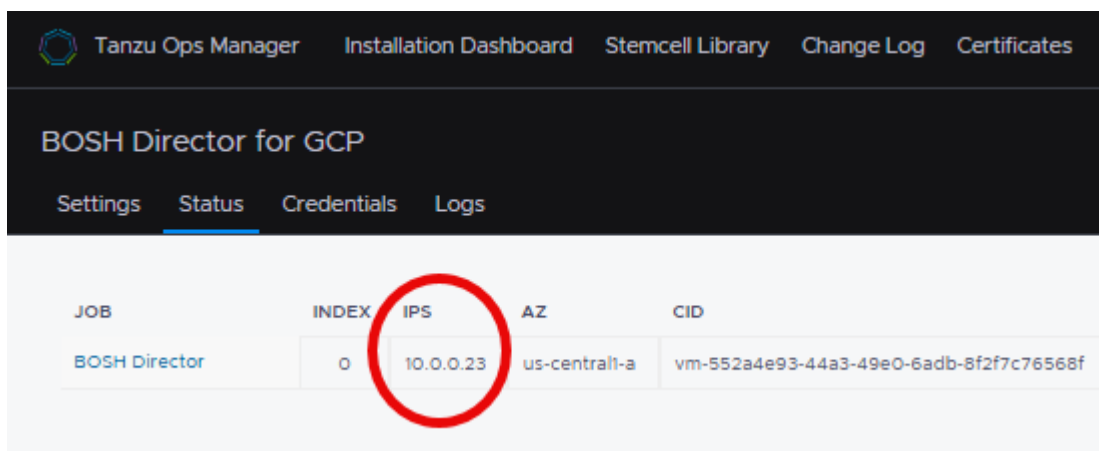
- **OPS-MAN-FQDN** is the fully-qualified domain name (FQDN) for your Tanzu Operations Manager deployment.
 - **UAA-ACCESS-TOKEN** is your UAA access token.
2. Verify the value of the **identity** text box. It can be **bbr_client**.
 3. Record the value of the **password** text box.

For more information about using the Tanzu Operations Manager API, see [Using the Tanzu Operations Manager API](#).

Step 8: Retrieve BOSH Director address

To retrieve the IP address of your BOSH Director from the BOSH Director tile:

1. If you are not using the Tanzu Operations Manager VM as your jumpbox, install the latest [BOSH CLI](#) on your jumpbox.
2. From the Installation Dashboard in Tanzu Operations Manager, click BOSH Director, then Status, and record the IP address listed for the Director. You access the BOSH Director using this IP address.



| Tanzu Ops Manager | | | | |
|------------------------|-------|-----------|---------------|---|
| Installation Dashboard | | | | |
| Stemcell Library | | | | |
| Change Log | | | | |
| Certificates | | | | |
| BOSH Director for GCP | | | | |
| Settings | | | | |
| Status | | | | |
| Credentials | | | | |
| Logs | | | | |
| JOB | INDEX | IPS | AZ | CID |
| BOSH Director | 0 | 10.0.0.23 | us-central1-a | vm-552a4e93-44a3-49e0-6adb-8f2f7c76568f |

3. On the command line, log in to the BOSH Director using the IP address that you previously used.
Run:

```
bosh -e DIRECTOR-IP \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE log-in
```

Where:

- **DIRECTOR-IP** is the BOSH Director IP address previously recorded.
 - **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the root Certificate Authority (CA) certificate as outlined in [Set up your Jumpbox](#).
4. When prompted for Email (), specify **director**.
 5. When prompted for Password (), enter the Director Credentials that you obtained in [Retrieve BOSH Director Credentials](#).

For example:

```
$ bosh -e 10.0.0.3 \  
--ca-cert /var/tempest/workspaces/default/root_ca_certificate log-in  
Email (): director  
Password (): *****  
Successfully authenticated with UAA  
Succeeded
```

Step 9: Restore BOSH Director

The BBR BOSH Director restore command can take at least 15 minutes to complete. VMware recommends that you run it independently of the SSH session, so that the process can continue running even if your connection to the jumpbox fails. The preceding command uses **nohup**, but you could also run the command in a **screen** or **tmux** session.

To restore BOSH Director:

1. SSH into your jumpbox. If you are using the Tanzu Operations Manager VM as your jumpbox, see [Log into the Tanzu Operations Manager VM with SSH](#) for how to use SSH to connect to the Tanzu Operations Manager VM.
2. Ensure the BOSH Director backup artifact is in the directory you from which you run BBR.
3. Run the BBR restore command from your jumpbox to restore the BOSH Director:

```
bbr director \  
--private-key-path PRIVATE-KEY-FILE \  
--username bbr \  
--host HOST \  
restore \  
--artifact-path PATH-TO-BOSH-DIRECTOR-BACKUP
```

Where:

- **PRIVATE-KEY-FILE** is the path to the private key file you created in [Retrieve BOSH Director Credentials](#).
- **HOST** is the address of the BOSH Director.
 - If the BOSH Director is public, **HOST** is a URL, such as <https://bosh-director.xxx.cf-app.com>.

- If the BOSH Director is not public, **HOST** is the **BOSH-DIRECTOR-IP** you retrieved in [Retrieve BOSH Director Address](#).
 - **PATH-TO-BOSH-DIRECTOR-BACKUP** is the path to the BOSH Director backup you want to restore.
 - (Optional) To enable debug logs, use the `--debug` flag. For more information, see [Logging in Backing Up Deployments with BBR](#).
4. After the command succeeds, continue to [Identify Your Deployment](#). If the command fails:

1. Run:

```
bbr director \
--private-key-path PRIVATE-KEY-FILE \
--username bbr \
--host HOST \
restore-cleanup
```

Where:

- **PRIVATE-KEY-FILE** is the path to the private key file you created in [Retrieve BOSH Credentials](#).
 - **HOST** is the address of the BOSH Director.
 - If the BOSH Director is public, **HOST** is a URL, such as <https://bosh-director.xxx.cf-app.com>.
 - If the BOSH Director is not public, **HOST** is the **BOSH-DIRECTOR-IP** you retrieved in [Retrieve BOSH Director Address](#).
2. Ensure that the following are true and then run the BBR restore command again:
 - All the parameters in the command are set.
 - The BOSH Director credentials are valid.
 - The specified deployment exists.
 - The source deployment is compatible with the target deployment.
 - That the jumpbox can reach the BOSH Director.

Step 10: Identify your deployment

To identify the name of the BOSH deployment that contains Tanzu Operations Manager:

1. Log in to your BOSH Director.
2. Run:

```
bosh -e BOSH-DIRECTOR-IP --ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE deployments
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
- **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the certificate authority (CA) for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.

Step 11: Remove stale cloud IDs for all deployments

To remove stale cloud IDs for all deployments:

1. Review the deployments listed when performing [Identify your deployment](#).
2. To reconcile the BOSH Director's internal state with the state in the IaaS, run:

```
bosh -e BOSH-DIRECTOR-IP \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \  
-d DEPLOYMENT-NAME -n cck \  
--resolution delete_disk_reference \  
--resolution delete_vm_reference
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
- **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.
- **DEPLOYMENT-NAME** is the deployment name you retrieved in [Identify Your Deployment](#).

Run this command for each deployment.

Step 12: Redeploy Elastic Application Runtime

These sections describe the steps to redeploy Elastic Application Runtime.

Determine the required stemcell

To determine which stemcell is used by Elastic Application Runtime, do one of the following:

- Review the Stemcell Library:
 1. Go to Stemcell Library.
 2. Record the Elastic Application Runtime stemcell release number from the Staged column.
- Review a stemcell list using BOSH CLI:
 1. To retrieve the stemcell release using the BOSH CLI, run:

```
bosh -e BOSH-DIRECTOR-IP deployments
```

Where **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Retrieve BOSH Director address](#).

For more information about stemcells in Tanzu Operations Manager, see [Importing and managing stemcells](#).

Upload the stemcell

To upload the required stemcell:

1. Download the [Xenial Stemcell](#) from Broadcom Support.
2. Upload the stemcell used by Elastic Application Runtime by running:

```
bosh -e BOSH-DIRECTOR-IP \  
-d DEPLOYMENT-NAME \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \  
upload-stemcell \  
--fix PATH-TO-STEMCELL
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
 - **DEPLOYMENT-NAME** is the deployment name you retrieved in [Identify Your Deployment](#).
 - **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA certificate for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.
 - **PATH-TO-STEMCELL** is the path to your tile's stemcell.
3. To ensure the stemcells for all of your other installed tiles have been uploaded, repeat the previous step, running `bosh upload-stemcell --fix PATH-TO-STEMCELL` for each stemcell that is different from the already uploaded Elastic Application Runtime stemcell.

Create external databases

If your foundation is configured to use an external MySQL database instance, recreate the databases on the external database instance, and then ensure that your external database is suitably configured.

For an example of how to create external databases, see the [Elastic Application Runtime documentation](#).

If you do not know whether your foundation uses an external or internal database, look in the Databases pane of the Elastic Application Runtime tile.

If your external database VM instance has also changed, for example, if the database instance has been lost, update the configuration settings in the Databases pane of the Elastic Application Runtime tile. For information about the Databases pane, see the [Elastic Application Runtime documentation](#).

Redeploy Elastic Application Runtime

To redeploy Elastic Application Runtime:

1. Go to the Installation Dashboard.
2. Click the Elastic Application Runtime tile.
3. Click Resource Config.
4. Ensure the number of instances for MySQL Server is set to **1**. Restore fails if there is not exactly one MySQL Server instance deployed.
5. (Optional) Increase the CPU and memory of the **backup_restore** VM. This decreases how long the restore process takes during [Restore Elastic Application Runtime](#).
6. Verify that all errands needed by your system are set to Run.
7. Return to the Installation Dashboard.
8. Click Review Pending Changes.
9. Review your changes and ensure that the Elastic Application Runtime tile is selected. Other tiles are optional. For more information, see [Reviewing Pending Product Changes](#).
10. Click Apply Changes to redeploy.

Step 13: Restore service data (optional)

BBR does not back up or restore any service data.

For this step, you must restore data to pre-provisioned service tiles.

The procedures for restoring service data might vary. For more information, see the documentation for your service tile.

For example, if you are using Tanzu for Valkey for Tanzu Operations Manager, see [Using BOSH Backup and Restore \(BBR\)](#).

Step 14: Restore Elastic Application Runtime

The BBR Elastic Application Runtime restore command can take at least 15 minutes to complete.

After the BBR Elastic Application Runtime restore completes, then Diego brings the apps back into a running state, unless Elastic Application Runtime file storage has been configured to selectively back up the blobstore. In that case, you need to follow [Restore apps to a running state](#).

VMware recommends that you run the BBR Elastic Application Runtime restore command independently of the SSH session, so that the process can continue running even if your connection to the jumpbox fails. The command uses `nohup`, but you could also run the command in a `screen` or `tmux` session.

To restore your Elastic Application Runtime backup:

1. For more information, see the table in [External Blobstore support](#) in *Enabling External Blobstore Backups*. If external blobstore support is not included in the version of Tanzu Operations Manager and Elastic Application Runtime you are using, restore the external blobstore with your IaaS-specific tools before restoring Elastic Application Runtime.
2. From your jumpbox, run:

```
bbbr deployment \
--target BOSH-DIRECTOR-IP \
--username BOSH-CLIENT \
--password BOSH-PASSWORD \
--deployment DEPLOYMENT-NAME \
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \
restore \
--artifact-path PATH-TO-Elastic Application Runtime-BACKUP
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
 - **BOSH-CLIENT** and **BOSH-PASSWORD** are the Uaa Bbr Client Credentials, identity and password that you retrieved in [Retrieve BOSH Director Credentials](#).
 - **DEPLOYMENT-NAME** is the deployment name you retrieved in [Identify Your Deployment](#).
 - **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.
 - **PATH-TO-Elastic Application Runtime-BACKUP** is the path to the Elastic Application Runtime backup you want to restore.
3. (Optional) To scale the MySQL Server job back up to its previous number of instances:
 1. Go to the Installation Dashboard
 2. Click the Elastic Application Runtime tile.
 3. Click Resource Config.
 4. Scale the MySQL Server job back up.
 5. Return to the Installation Dashboard.
 4. Click Review Pending Changes.
 5. Review your changes. For more information, see [Reviewing Pending Product Changes](#).

6. Click Apply Changes to deploy.
7. (Optional) Run `cf apps` to view the status of apps. By observing which apps are running and which are still pending, you can estimate how much longer the restore takes.

(Optional) Step 15: Restore apps to a running state

Restoring apps to a running state is required only if you configured a Elastic Application Runtime file storage backup that excludes droplets or both droplets and packages. For more information about the advantages and disadvantages of excluding droplets and packages, see [File Storage Backup Level](#).

To return apps to a running state:

- If you excluded droplets from your backup, restage each app by running:

```
cf restage APP-NAME
```

Where `APP-NAME` is the name of the app you want to restage.

- If you excluded both droplets and packages from your backup, re-push each app by running:

```
cf push APP-NAME OPTIONAL-FLAGS
```

Where:

- `APP-NAME` is the name of the app you want to re-push.
- `OPTIONAL-FLAGS` are any flags you want to use while re-pushing your app.

Step 16: Restore on-demand service instances (optional)

If you have on-demand service instances provisioned by an on-demand service broker, you can restore them after successfully restoring Elastic Application Runtime.

If the leader is lost and you use an on-demand MySQL service instance plan that replicates between foundations, purge the lost service instance to avoid accidentally restoring the former leader. Delete the former leader before running the Upgrade All Service Instances errand for MySQL.

To restore your on-demand service instances:

1. Go to the Installation Dashboard.
2. Click the on-demand service tile.
3. Click the Errands tab.
4. Verify that the Upgrade All Service Instances errand is On.
5. Repeat for all on-demand service tiles.
6. Return to the Installation Dashboard.
7. Click Review Pending Changes.
8. Click Apply Changes. This runs the Upgrade All Service Instances errand for the on-demand service, which redeploys the on-demand service instances.
9. (Optional) Restore service data to every on-demand service instance.
10. Any app on Elastic Application Runtime bound to an on-demand service instance might need to be restarted to start consuming the restored on-demand service instances.

(Optional) Step 17: Restore non-tile deployments

If you have any deployments that were deployed manually with the BOSH Director rather than through a Tanzu Operations Manager tile, you must restore their VMs.

To restore the VMs:

1. To obtain a list of all deployments on your BOSH Director, run:

```
bosh -e BOSH-DIRECTOR-IP \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \  
deployments
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
 - **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA certificate for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.
2. Identify the names of the deployments that you need to restore. Do not include the deployments from Tanzu Operations Manager tiles.
 3. For each deployment you need to restore, run:

```
bosh -n -e BOSH-DIRECTOR-IP \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \  
-d DEPLOYMENT-NAME \  
cck --resolution=recreate_vm
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
 - **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.
 - **DEPLOYMENT-NAME** is the deployment name you retrieved in [Identify Your Deployment](#).
4. To verify the status of the VMs in each deployment, run:

```
bosh -e BOSH-DIRECTOR-IP \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE \  
-d DEPLOYMENT-NAME \  
vms
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
- **PATH-TO-BOSH-SERVER-CERTIFICATE** is the path to the CA for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.
- **DEPLOYMENT-NAME** is the deployment name you retrieved in [Identify Your Deployment](#).

The process state for all VMs show as **running**.

After restoring your backup

This section describes the procedures you must perform after restoring your backup with BBR.

Step 18: Remove unused disks

This is a very destructive operation.

Disks from a previous deployment prevent recreated deployments from working.

Use BOSH to clean up disks

To clean up disk references:

1. Run:

```
bosh cloud-check
```

Manually clean up disks

If `bosh cloud-check` does not clean up all disk references, you must manually delete the remaining disks.

To delete the remaining disks, follow one of these procedures:

- Use the BOSH CLI to delete the disks:
 1. Target the redeployed BOSH Director using the BOSH CLI by following the procedures in [Retrieve BOSH Director Address](#).
 2. List the deployments by running:

```
bosh -e BOSH-DIRECTOR-IP \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE deployments
```

Where:

- `BOSH-DIRECTOR-IP` is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
- `PATH-TO-BOSH-SERVER-CERTIFICATE` is the path to the CA for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.

3. Delete each deployment by running:

```
bosh -d DEPLOYMENT-NAME delete-deployment
```

Where `DEPLOYMENT-NAME` is the deployment name you retrieved in [Identify Your Deployment](#).

- Log in to your IaaS account and delete the disks manually.
 1. To retrieve a list of disk IDs, run:

```
bosh -e BOSH-DIRECTOR-IP \  
--ca-cert PATH-TO-BOSH-SERVER-CERTIFICATE instances -i
```

Where:

- `BOSH-DIRECTOR-IP` is the BOSH Director IP you retrieved in [Retrieve BOSH Director Address](#).
- `PATH-TO-BOSH-SERVER-CERTIFICATE` is the path to the CA for the BOSH Director, if the certificate is not verifiable by the local machine's certificate chain.

After the disks are deleted, continue with [Remove Stale Cloud IDs for all deployments](#).

Enabling external blobstore backups

This topic tells you how to enable external blobstore backups in your Tanzu Platform for Cloud Foundry tile.

BOSH Backup and Restore (BBR) supports:

- Versioned S3 or S3-compatible blobstores
- Unversioned S3 or S3-compatible blobstores
- Azure blobstores

For more information, see [Backup and restore for external blobstores](#) in the Cloud Foundry documentation.

To enable external blobstore backups for Elastic Application Runtime, you must enable Backup Restore Node. For more information, see [Step 6: Confirm backup restore node is deployed](#) in *Backing up Deployments with BBR*.

External blobstore support

External blobstore backup support varies based on which version of Tanzu Operations Manager you are running and what type of blobstore you are backing up.

In some cases, external blobstore support is included in the version of Tanzu Operations Manager you are using. In other cases, installing Blobstore Add-On is required.

To determine if external blobstore support is included in the version of Tanzu Operations Manager you are using, see the following table:

| Blobstore type | Tanzu Operations Manager Version | | | | |
|----------------|----------------------------------|---------------|---------------|---------------|---------------|
| | 2.2 | 2.3 | 2.4 | 2.5 | 2.6 and later |
| Versioned | Included | Included | Included | Included | Included |
| Unversioned | Included | Included | Included | Included | Included |
| Azure | Add-On | Included | Included | Included | Included |
| GCS | Not supported | Not supported | Not supported | Not supported | Included |

S3 or S3-compatible blobstores

BBR supports backing up and restoring versioned and unversioned S3 or S3-compatible blobstores by default. For more information about configuring an S3 or S3-compatible blobstore, see [AWS](#) in *Configuring File Storage for Elastic Application Runtime*.

Backup artifacts of external, S3-compatible, versioned blobstores do not contain the physical blobs. BBR requires that the original buckets still exist to be restored. To protect yourself from losing a bucket, see [Enable Replication on Your Versioned S3-Compatible Blobstore](#) in *Backup and Restore for External Blobstores*.

Azure blobstores

To configure your Azure blobstore for backups, you must enable soft delete in your Azure Storage account. You can also enable soft delete for existing Azure storage accounts that were not initially created with soft delete enabled. For more information, see [Soft delete for Azure Storage blobs](#) in the Microsoft documentation.

After activating soft delete, you can enable Azure blobstore backups in Tanzu Operations Manager for new or existing blobstores. For more information, see [Azure](#) in *Configuring File Storage for Elastic Application Runtime*.

To save storage space and cost, VMware recommends that you configure a retention.

GCS blobstores

For more information about configuring a GCS blobstore, see [GCP](#) in *Configuring File Storage for Elastic Application Runtime*.

Troubleshooting BOSH Backup and Restore

This topic explains how to solve common problems that you might encounter when using BOSH Backup and Restore (BBR) to back up and restore your deployments.

Troubleshooting during a restore

This section provides solutions for errors that occur during a restore.

Problem: Restore fails with a MySQL Monit start timeout

| | |
|-------------|---|
| Symptom | While running the BBR restore command, restoring the job <code>mysql-restore</code> fails with the following error: <div>1 error occurred:

* restore script for job mysql-restore failed on mysql/0.
...
"Monit start failed: Timed out waiting for monit": 2m0s</div> |
| Explanation | This happens when the MySQL job fails to start within the timeout period. It ends up in an Execution Failed state and <code>monit</code> never tries to start it again. |

Ensure that your MySQL Server cluster has only one instance. If there is more than one instance of MySQL Server, the restore fails with a **monit start** timeout. Scale down to one instance and retry.

If your MySQL Server cluster is already scaled down to one node, it might have taken longer than normal to restart the cluster. To manually verify and retry:

1. List the VMs in your deployment by running:

```
bosh -e BOSH-DIRECTOR-IP --ca-cert /var/tempest/workspaces/default/root_ca_certificate \
-d DEPLOYMENT-NAME \
ssh
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP. To retrieve this IP address, see [Retrieve BOSH Director Address](#) in *Restoring Deployments from Backup with BBR*.
- **DEPLOYMENT-NAME** is the deployment name. To retrieve the deployment name, see [Identify your Deployment](#) in *Restoring Deployments from Backup with BBR*.

2. Locate the **mysql** VM and SSH into it.
3. To check that the MySQL job process is running, run:

```
ps aux | grep galera-init
```

For more information, see [Configuring Elastic Application Runtime](#).

4. Check that **monit** reports that the MySQL job process is in an Execution Failed state by running:

```
sudo monit summary
```

5. If so, deactivate monitoring by running:

```
monit unmonitor
```

6. Enable monitoring by running:

```
monit monitor
```

7. After a few minutes, run:

```
monit summary
```

The command reports that all the processes are running.

8. Attempt the restore with BBR again.

Problem: Deployment does not match the structure of the backup

| | |
|-------------|--|
| Symptom | <p>The following error is displayed:</p> <div>Deployment 'deployment-name' does not match the structure of the provided backup.</div> |
| Explanation | The instance groups with the restore scripts in the destination environment do not match the backup metadata. For example, they might have the wrong number of instances of a particular instance group, or the metadata names an instance group that does not exist in the destination environment. |
| Solution | BBR only supports restoring to an environment that matches your original environment. VMware recommends altering the destination environment to match the structure of the backup. |

General troubleshooting

This section provides solutions for general errors.

Problem: Connection error

| | |
|-------------|---|
| Symptom | BBR displays an error message containing "SSH Dial Error" or another connection error. |
| Explanation | The jumpbox and the VMs in the deployment are experiencing connection problems. |
| Solution | <p>To resolve connection problems between the jumpbox and the VMs in the deployment:</p> <ol style="list-style-type: none"> 1. Ensure your deployment is healthy by running: <div>bosh vms</div> 2. Clean up the data from the failed backup on the instances by running the following. This step is required. Otherwise, further BBR commands fail. <div>bbr deployment backup-cleanup</div> 3. Repeat the BBR operation. |

Problem: Error running metadata script

| | |
|-------------|--|
| Symptom | <p>BBR backup or restore fails with a metadata error:</p> <div> <p>1 error occurred:</p> <p>error 1:</p> <p>An error occurred while running metadata script for job redis-server on redis/0ce9f81f-1756-480b-8e3e-a4609b14b6a6: error from metadata</p> </div> |
| Explanation | There is a problem with your deployment install. |
| Solution | Contact Broadcom Support . |

TLS connections in Elastic Application Runtime

For information about communication paths used by external clients, internal components, app containers, and app services in Tanzu Platform for Cloud Foundry, and how the platform uses Transport Layer Security (TLS) protocols to secure these communications, see [TLS connections in Elastic Application Runtime](#).

Using logging and metrics in your Tanzu Operations Manager deployment

The following topics describe how you can use logging and metrics in your Tanzu Operations Manager deployment:

- [Accessing logs and metrics for your Tanzu Operations Manager deployment.](#)
- [Auditing user activity for your Tanzu Operations Manager deployment.](#)
- [Monitoring VM status in Tanzu Operations Manager.](#)

Accessing logs and metrics for your Tanzu Operations Manager deployment

You can access logs and metrics for VMs and app containers on your Tanzu Operations Manager deployment. It includes information about the available logs and metrics for the Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI) runtimes.

There are different methods for accessing logs and metrics for VMs and app containers depending on if you use the Elastic Application Runtime or Container Orchestration Engine app runtime for your deployment.

There are also different permissions required to access logs and metrics on Elastic Application Runtime and Container Orchestration Engine. The sections in this topic explain how to access logs and metrics on your Elastic Application Runtime or Container Orchestration Engine deployment based on your user role permissions.

Access Platform Director VM logs

Platform Director logs include logs from the Tanzu Operations Manager and BOSH Director VMs.

For both Elastic Application Runtime and Container Orchestration Engine, operators can access platform director VM logs by configuring a syslog drain in the Tanzu Operations Manager tile. For more information, see [Settings Page](#) in *Using the Tanzu Operations Manager Interface*.

The format for the BOSH Director VM logs is Syslog RFC 5424. For more information, see [RFC 5424 - The Syslog Protocol](#) in the Internet Engineering Task Force (IETF) documentation.

The format for the Tanzu Operations Manager VM logs is Syslog RFC 3164. For more information, see [The BSD Syslog Protocol](#) in the IETF documentation.

Access VM logs

This section describes how to access logs for Elastic Application Runtime and Container Orchestration Engine VMs.

The format for VM logs is Syslog RFC 5424. For more information, see [RFC 5424 - The Syslog Protocol](#) in the IETF documentation.

See one of the following sections:

- [Access VM Logs on Elastic Application Runtime.](#)
- [Access VM Logs on Container Orchestration Engine.](#)

Access VM logs on Elastic Application Runtime

Elastic Application Runtime VM logs include logs for runtime component VMs and service tile VMs.

The following table describes how to access VM logs on foundations with Elastic Application Runtime based on user role and permissions:

| Title | Elastic Application Runtime Role | Procedure | Notes |
|-----------|----------------------------------|--|--|
| Operator | Admin | Configure system logging in Elastic Application Runtime. For more information, see the Elastic Application Runtime documentation . | To access logs for service instances, you must also configure syslog in each service tile for which you want logs. For more information, see the documentation for the service tile. |
| Developer | Space developer | Not applicable. | |

Access VM logs on Container Orchestration Engine

The following table describes how to access VM logs on foundations with Container Orchestration Engine based on user role and permissions:

| Title | Role | Procedure | Notes |
|-------------------------|-----------------------|--|-------|
| Operator | Admin | <p>Enable system logging in Container Orchestration Engine. For more information, see <i>Host Monitoring</i> in the Container Orchestration Engine installation topic for your IaaS:</p> <ul style="list-style-type: none"> • Installing Container Orchestration Engine on AWS • Installing Container Orchestration Engine on Azure • Installing Container Orchestration Engine on GCP • Installing Container Orchestration Engine on vSphere • Installing Container Orchestration Engine on vSphere with NSX-T | |
| Cluster admin/developer | Not applicable. | Not applicable. | |

Access VM metrics

This section describes how to access metrics for Elastic Application Runtime VMs and Container Orchestration Engine VMs.

See one of the following sections:

- [Access VM Metrics on Elastic Application Runtime.](#)
- [Access VM Metrics on Container Orchestration Engine.](#)

Access VM metrics on Elastic Application Runtime

For Elastic Application Runtime deployments, VM metrics include Elastic Application Runtime component and service tile VM metrics.

The following table describes how to access VM metrics on foundations with Elastic Application Runtime based on user role and permissions:

| Title | Elastic Application Runtime Role | Procedure | Notes |
|-----------|----------------------------------|--|---|
| Operator | Admin | No additional configuration is required. Elastic Application Runtime emits VM metrics through the Loggregator Firehose. For more information, see Loggregator Architecture . | The Firehose emits metrics as a gauge or counter Loggregator envelope type. For more information, see the loggregator-api repository on GitHub. |
| Developer | Space developer | Not allowed. | |

Access VM metrics on Container Orchestration Engine

For Container Orchestration Engine deployments, VM metrics include metrics for the primary and worker nodes.

VM metrics are available on Container Orchestration Engine v1.5 and later.

The following table describes how to access VM metrics on foundations with Container Orchestration Engine based on user role and permissions.

| Title | Role | Procedure | Notes |
|---------------|---|--|--|
| Operator | Admin | Configure an output plug-in in the Container Orchestration Engine tile. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. | Component VM Metrics include the primary node only. |
| Cluster admin | The operator defines the role-based access policy for accessing VM metrics. | Configure a ClusterMetricSink in the cluster. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. | <ul style="list-style-type: none">• You must configure a ClusterMetricSink for each new cluster.• Component VM Metrics include the worker nodes only. |
| Developer | The operator defines the role-based access policy for accessing VM metrics. | | |

Access App Container logs

This section describes how to access app logs on Tanzu Operations Manager.

See one of the following sections:

- [Access App Container logs on Elastic Application Runtime.](#)
- [Access App Container logs on Container Orchestration Engine.](#)

Access App Container logs on Elastic Application Runtime

App container logs for Elastic Application Runtime deployments include logs for apps and service instances.

The following table describes how to access app logs on foundations with Elastic Application Runtime based on user role and permissions.

| Title | Elastic Application Runtime Role | Procedure | Notes |
|-----------|----------------------------------|---|---|
| Operator | Admin | <p>Access logs with one of the following methods:</p> <ul style="list-style-type: none"> • Loggregator Firehose: No configuration required. The Firehose emits logs as text payload. The developer defines the log format. For more information, see the loggregator-api repository on GitHub. • Syslog Drain: Requires configuration of Syslog Adapter Drain. For more information, see the Elastic Application Runtime documentation. • Aggregate Syslog Drain: A syslog drain that sends all app logs to a single destination. Configure the aggregate syslog drain in the Elastic Application Runtime tile. For more information, see the Elastic Application Runtime documentation. | <ul style="list-style-type: none"> • The Firehose does not emit logs for service instances. To access service instance logs, use the syslog drain or aggregate syslog drain method. • Syslog drains emit logs as Syslog RFC 5424. |
| Developer | Space developer | <p>Access logs with one of the following methods:</p> <ul style="list-style-type: none"> • Loggregator Firehose: The Firehose emits logs as text payload. The developer defines the log format. For more information, see the loggregator-api repository on GitHub. • Syslog Drain: Requires configuration of Syslog Adapter Drain. For more information, see the Elastic Application Runtime documentation. | <ul style="list-style-type: none"> • The Firehose does not emit logs for service instances. To access service instance logs, use the syslog drain method. • Syslog drains emit logs as Syslog RFC 5424. |

Access App Container logs on Container Orchestration Engine

The following table describes how to access app container logs on foundations with Container Orchestration Engine based on user role and permissions:

| Title | Role | Procedure | Notes |
|---------------|---|---|--|
| Operator | Admin | Configure a ClusterSink in the cluster. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. | <ul style="list-style-type: none"> You can send logs from a namespace within a cluster to a log destination by configuring a namespace log Sink. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. You must configure a ClusterSink for each new cluster. |
| Cluster admin | The operator defines the role-based access policy for accessing app logs. | Configure a ClusterSink in the cluster. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. | <ul style="list-style-type: none"> You can send logs from a namespace within a cluster to a log destination by configuring a namespace log Sink. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. You must configure a ClusterSink for each new cluster. |
| Developer | The operator defines the role-based access policy for accessing app logs. | Configure a Sink in the Container Orchestration Engine tile for a cluster namespace. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. | <ul style="list-style-type: none"> You can send logs from a namespace within a cluster to a log destination by configuring a namespace log Sink. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. You must configure a ClusterSink for each new cluster. You must provide your own access policy. Developers might not have permissions to configure a ClusterSink. Operators must define an access policy that allows developers to create a ClusterSink. |

Access App Container metrics

This section describes how to access app container metrics on Tanzu Operations Manager.

See one of the following sections:

- [Access App Container metrics on Elastic Application Runtime.](#)
- [Access App Container metrics on Container Orchestration Engine.](#)

Access App Container Metrics on Elastic Application Runtime

App container metrics on Elastic Application Runtime include app metrics and service instance metrics. App container metrics on Elastic Application Runtime do not include task metrics. This is because Diego does not emit task metrics.

The following table describes how to access app container metrics on foundations with Elastic Application Runtime based on user role and permissions:

| Title | Elastic Application Runtime role | Procedure | Notes |
|-----------|----------------------------------|---|--|
| Operator | Admin | Access metrics with one of the following methods: <ul style="list-style-type: none"> • Loggregator Firehose: No configuration required. • Syslog Drain: Requires configuration of Syslog Adapter Drain. For more information, see Step 2: Create and Bind a User-Provided Service Instance in <i>Streaming App Logs to Log Management Services</i>. | <ul style="list-style-type: none"> • The Firehose emits metrics as a gauge or counter Loggregator envelope type. For more information, see the loggregator-api repository on GitHub. • Syslog drains emit logs as Syslog RFC 5424. |
| Developer | Space developer | Access metrics with one of the following methods: <ul style="list-style-type: none"> • Loggregator Firehose: No configuration required. • Syslog Drain: Requires configuration of Syslog Adapter Drain. For more information, see Step 2: Create and Bind a User-Provided Service Instance in <i>Streaming App Logs to Log Management Services</i>. | |

Access App Container metrics on Container Orchestration Engine

The following table describes how to access app container metrics on foundations with Container Orchestration Engine based on user role and permissions:

| Title | Role | Procedure | Notes |
|----------|-----------------------|--|--|
| Operator | Admin | Configure a ClusterMetricSink in the cluster. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. | <ul style="list-style-type: none"> • You must configure a ClusterMetricSink for each new cluster. • You must provide your own access policy. |

| Title | Role | Procedure | Notes |
|---------------|--|---|--|
| Cluster admin | The operator defines the role-based access policy for accessing app metrics. | Configure a ClusterMetricSink in the cluster. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. | <ul style="list-style-type: none"> You must configure a ClusterMetricSink for each new cluster. ClusterMetricSink automatically scrapes an annotated /metrics Prometheus endpoint. |
| Developer | The operator defines the role-based access policy for accessing app metrics. | Configure a MetricSink in the Container Orchestration Engine tile for a cluster namespace. For more information, see Creating Sink Resources in the Container Orchestration Engine documentation. | <ul style="list-style-type: none"> You must configure a ClusterMetricSink for each new cluster. ClusterMetricSink automatically scrapes an annotated /metrics Prometheus endpoint. Developers might not have permissions to configure a ClusterMetricSink. Operators must define an access policy that allows developers to create a ClusterMetricSink. |

Auditing user activity for your Tanzu Operations Manager deployment

This topic defines the relevant logs for auditing Tanzu Operations Manager user activity, and describes how you can access and forward the logs.

You can audit Tanzu Operations Manager user activity to help improve security practices in your environment. Auditing user activity also helps you ensure that your environment is compliant with existing security standards. As an example, you can view the timestamp of any requests made in the Tanzu Operations Manager UI and the user name of the user that made the request.

You can find relevant information for auditing user activity in Tanzu Operations Manager in the following Tanzu Operations Manager logs:

- SSH login logs. For more information, see [Logs for auditing user activity](#).
- UAA login and failed UAA login logs. For more information, see [Logs for auditing user activity](#).
- The Tanzu Operations Manager [audit_log.txt](#) file. For more information, see [Audit logs text file](#).

Access logs on the Tanzu Operations Manager VM

You can access logs for auditing user activity by logging on to the Tanzu Operations Manager VM using SSH.

To access logs for auditing user activity:

- SSH onto the Tanzu Operations Manager VM using your private SSH key. For more information, see [Log in to the Tanzu Operations Manager VM](#).
- Go to the location of the logs and open or preview the log file. For example, to view the [audit_log.txt](#) file, run:

```
cat /var/log/opsmanager/audit_log.txt
```

Logs for auditing user activity

The following table describes the types of user actions that you can audit with logs, and the location of the logs.

For information about how to access the logs in this table, see [Access logs on the Tanzu Operations Manager VM](#).

| User action | Description | Location of logs | Example log trace |
|-------------------|--|--|---|
| SSH logins | User logs on to the Tanzu Operations Manager VM with SSH. | /var/log/auth.log | 2019-06-20T21:58:50.864147+00:00
HOSTNAME sshd[18028]: Accepted
publickey for USERNAME from
IPADDRESS port PORT ssh2: RSA
SHA256:SIGNATURE 2019-06-
20T21:58:50.865750+00:00
HOSTNAME sshd[18028]:
pam_unix(sshd:session): session |
| UAA logins | User logs in to Tanzu Operations Manager through UAA. | /home/tempest-web/uaa/
tomcat/logs/uaa.log | [2019-08-01 07:57:45.830] uaa - 5320
[http-nio-127.0.0.1-8080-exec-7]
INFO --- Audit:
IdentityProviderAuthenticationSuccess
('admin'): principal=1df75be4-7875-
4e7e-97fa-fe76dbff4a41, origin=
[remoteAddress=127.0.0.1, sessionId=],
identityZonId=[uaa],
authenticationType=[uaa] [2019-08-01
07:57:45.832] uaa - 5320 [http-nio-
127.0.0.1-8080-exec-7] INFO ---
Audit: UserAuthenticationSuccess
('admin'): principal=1df75be4-7875-
4e7e-97fa-fe76dbff4a41, origin=
[remoteAddress=127.0.0.1, sessionId=],
identityZonId=[uaa] ... |
| Failed UAA logins | User makes failed login attempt through UAA. | /home/tempest-web/uaa/
tomcat/logs/uaa.log | [2019-07-31 23:13:48.437] uaa - 1184
[http-nio-127.0.0.1-8080-exec-5]
INFO --- Audit:
IdentityProviderAuthenticationFailure
('admin'): principal=null, origin=
[remoteAddress=209.234.137.222,
sessionId=], identityZonId=[uaa],
authenticationType=[uaa] [2019-07-31
23:13:48.438] uaa - 1184 [http-nio-
127.0.0.1-8080-exec-5] INFO ---
Audit: UserAuthenticationFailure
('admin'): principal=2bdf7f2a-862d-47d7-
bf7f-ba92da6850c0, origin=
[remoteAddress=209.234.137.222,
sessionId=], identityZonId=[uaa] ... |
| User requests | User makes requests in the Tanzu Operations Manager app. The audit_log.txt file includes metadata about user requests in the Tanzu Operations Manager app. | /var/log/opsmanager/
audit_log.txt | For information about the properties in the audit_log.txt file, see Tanzu Operations Manager Audit Logs Text File . |

Audit logs text file

The Tanzu Operations Manager `audit_log.txt` file includes metadata about actions taken by users in the Tanzu Operations Manager app in JSON format.

The following example shows the content in the `audit_log.txt` file:

```
{
  "timestamp": "2019-07-13T00:43:08.783+00:00",
  "component": "Tanzu Operations Manager 2.7.0-build.87",
  "component_scheme": "https",
  "component_host": "pcf.example.opsmanager.cf-app.com",
  "component_port": 443,
  "result": 200,
  "target": "/proxy_settings",
  "event_type": "PUT",
  "origination": "123.456.789",
  "user_identification": "admin",
  "uaa_authentication_mechanism": "uaa",
  "request_body": {
    "authenticity_token": "[FILTERED]",
    "proxy_settings": {
      "http_proxy": "http://example.com",
      "https_proxy": "https://example.com",
      "no_proxy": "1.2.3.4"
    }
  }
}
```

The following table describes the properties in the `audit_log.txt` file and provides example values for each property:

| User action Pproperty | Description | Example value |
|---|---|--|
| <code>user_identification</code> | The user name or client ID that submitted a request. | <code>admin</code> |
| <code>uaa_authentication_mechanism</code> | The authentication method used to make the request.

(<code>unauthenticated</code>) for unauthenticated requests, <code>uaa</code> for a UAA user login, or <code>client_authenticated</code> for an API-only client. | <code>uaa</code> |
| <code>origination</code> | The IP address that made the request to the server. | <code>198.10.1.25</code> |
| <code>timestamp</code> | The time that the server received the request. | <code>2019-07-21 08:34:00</code> |
| <code>target</code> | The relative URL path associated with the request. | <code>/infrastructure/director/resource_config/edit</code> |
| <code>event_type</code> | The HTTP verb associated with the request. | <code>GET</code> |

| User action Pproperty | Description | Example value |
|-----------------------|--|---|
| result | The HTTP response code associated with the request.

Results beginning with 2 indicate a successful request. Results beginning with 4 indicate a failed request due to user error. Results beginning with 5 indicate a server error. | 200 |
| component | The version of Tanzu Operations Manager on which the request was made. | Tanzu Operations Manager 2.7.0-build.234 |
| component_scheme | The transport protocol used for the request. | HTTPS |
| component_host | The domain name used for the request. | pcf.ops-manager-domain.com |
| component_port | The port used for the request. | 443 |
| request_body | A JSON object that contains the contents of the query parameters, request body, and routing parameters for the request. Any credentials sent within the request are replaced with the value [FILTERED]. Any file uploads are excluded. | { "authenticity_token": "[FILTERED]", "director_configuration": { "ntp_servers_string": "ntp-server", "encryption": { "keys": "[FILTERED]" }, "blobstore_type": "local" } } |

Forwarding logs

You can forward some Tanzu Operations Manager logs using syslog forwarding. This allows you to forward logs to a syslog server or other third-party destination for storage and analysis.

These logs contain relevant information for auditing user activity and can be forwarded using syslog forwarding:

- `audit_log.txt`
- `uaa.log`
- `uaa_events.logs`

When syslog forwarding is enabled, Tanzu Operations Manager automatically forwards BOSH Director access events. These events allow you to audit BOSH Director access and the execution of BOSH CLI commands. For more information about BOSH Director access events, see [Logging API access](#) in the BOSH documentation.

For more information about syslog forwarding in Tanzu Operations Manager, see [Syslog](#) in *Using the Tanzu Operations Manager Interface*.

Monitoring VM status in Tanzu Operations Manager

You can check and monitor current VM status in Tanzu Operations Manager.

For a complete guide to monitoring Tanzu Platform for Cloud Foundry, see the [Elastic Application Runtime documentation](#).

Monitoring VMs using the Tanzu Operations Manager interface

To view monitoring information:

1. Go to the Tanzu Operations Manager Installation Dashboard.

2. Click any product tile.
3. Click the Status tab.

| | | | | | | | | | | | | | |
|---|-------|-----------|---------------|---|------------|------|--------|------|-------------|-------------|------------|------|--|
| <div>Tanzu Ops Manager</div> <div>Installation Dashboard</div> <div>Stemcell Library</div> <div>Change Log</div> <div>Certificates</div> <div>admin</div> | | | | | | | | | | | | | |
| Telemetry | | | | | | | | | | | | | |
| <div>Settings</div> <div>Status</div> <div>Credentials</div> <div>Logs</div> | | | | | | | | | | | | | |
| JOB | INDEX | IPS | AZ | CID | LOAD AVG15 | CPU | MEMORY | SWAP | SYSTEM DISK | EPHEM. DISK | PERS. DISK | LOGS | |
| Telemetry Centralizer | 0 | 10.0.0.24 | us-central1-a | vm-fb98aa3c-2ec8-48b1-4452-864d5e9c1743 | 0.00 | 0.1% | 40% | 1% | 61% | 5% | 0% | | |

The columns display the following information:

| VM data point | Details |
|---------------|--|
| Job | Each job represents a component running on one or more VMs that Tanzu Operations Manager deployed. |
| Index | For jobs that run across multiple VMs, the index value indicates the order in which the job VMs were deployed. For jobs that run on only one VM, the VM has an index value of 0. |
| IPs | IP address of the job VM. |
| AZ | The availability zone. |
| CID | Uniquely identifies the VM. |
| Load avg15 | CPU load average over 15 minutes. |
| CPU | Current CPU usage. |
| Memory | Current memory usage. |
| Swap | Swap file percentage. |
| System disk | System disk space usage. |
| Ephem. disk | Ephemeral disk space usage. |
| Pers. disk | Persistent disk space usage. |
| Logs | Download link for the most recent log files. |

As of Tanzu Operations Manager v3.0.3, you can directly download logs for instances and instance groups from the Status page in a product tile.

| | | | | | | | | | | | | | |
|---|-------|-----------|---------------|---|------------|------|--------|------|-------------|-------------|------------|------|--|
| <div>Tanzu Ops Manager</div> <div>Installation Dashboard</div> <div>Stemcell Library</div> <div>Change Log</div> <div>Certificates</div> <div>admin</div> | | | | | | | | | | | | | |
| Telemetry | | | | | | | | | | | | | |
| <div>Settings</div> <div>Status</div> <div>Credentials</div> <div>Logs</div> | | | | | | | | | | | | | |
| JOB | INDEX | IPS | AZ | CID | LOAD AVG15 | CPU | MEMORY | SWAP | SYSTEM DISK | EPHEM. DISK | PERS. DISK | LOGS | |
| Telemetry Centralizer | 0 | 10.0.0.24 | us-central1-a | vm-fb98aa3c-2ec8-48b1-4452-864d5e9c1743 | 0.00 | 0.1% | 40% | 1% | 61% | 5% | 0% | | |

The Logs column for each instance group contains two links:

- The first link retrieves logs for the instance.
- The second link retrieves logs for the entire instance group.

Clicking either link shows a loading icon as logs are queued for download. When the logs are ready for download, the icon changes to a download link.

As in previous Tanzu Operations Manager versions, you can also download log files from the Logs tab in a product tile.

Tanzu Operations Manager VM disk space

Tanzu Operations Manager stores its' logs on the Tanzu Operations Manager VM in the `/tmp` directory.

The logs collect over time and do not self-delete. To prevent the VM from running out of disk space, clear the log entries from `/tmp` by restarting the VM.

Security and compliance in Tanzu Operations Manager

The following topics cover security and compliance issues that you should consider for your Tanzu Operations Manager deployment:

- [Security processes used in developing Tanzu Operations Manager](#)
 - [Tanzu Operations Manager security overview and policy](#)
 - [Testing, release, and security lifecycle for Tanzu Operations Manager](#)
- [IaaS security considerations for Tanzu Operations Manager](#)
 - [Encrypting Tanzu Operations Manager VM disks](#)
 - [Security guidelines for your IaaS provider](#)
- [Stemcell security](#)
 - [Linux stemcell hardening](#)
 - [Windows stemcell hardening](#)
 - [Floating stemcells in Tanzu Operations Manager](#)
- [Tanzu Operations Manager assessment against NIST SP 800-53\(r4\) controls](#)
- [Security-related Tanzu Operations Manager tiles and add-ons](#)

Security processes used in developing Tanzu Operations Manager

The following topics explain how VMware responds to security vulnerabilities and tests and updates stemcells, the versioned operating systems that VMware products run on.

- [Tanzu Operations Manager security overview and policy](#): Covers VMware's responsible disclosure and vulnerability response procedures for Tanzu Operations Manager.
- [Testing, release, and security lifecycle for Tanzu Operations Manager](#): Explains how VMware's practices, tools, and organizational structures work together to create and support stable releases of Tanzu Operations Manager.
- [Container Orchestration Engine Security Disclosure and Release Process](#): Describes the processes for disclosing security issues and releasing related fixes for VMware Tanzu Container Orchestration Engine (formerly TKGI), Kubernetes, Cloud Foundry Container Runtime (CFCR), VMware NSX, and VMware Harbor.

Tanzu Operations Manager security overview and policy

This topic is for operators who are deploying Tanzu Operations Manager, and outlines VMware's security policy. You can find a comprehensive overview of the security architecture of each Tanzu Platform for Cloud Foundry component in the [Elastic Application Runtime documentation](#).

How VMware monitors for security vulnerabilities

VMware receives private reports on vulnerabilities from customers and from field personnel through VMware's secure disclosure process. VMware also monitors public repositories of software security vulnerabilities to identify newly discovered vulnerabilities that might affect one or more of VMware's products.

How to report a vulnerability

VMware encourages users who become aware of a security vulnerability in VMware's products to contact VMware with details of the vulnerability. Please send descriptions of any vulnerabilities found to the Tanzu Security Team using the procedure described in the [VMware Security Response Policy](#). Include details about the software and hardware configuration of your system so that VMware can reproduce the issue.

VMware encourages use of encrypted email. For VMware's public PGP key, see [Security Response: VMware PGP Public Key for Security Alerts](#) in the Broadcom Knowledge Base.

Notification policy

Tanzu Operations Manager has many customer stakeholders who need to know about security updates. When there is a possible security vulnerability identified for a Tanzu Operations Manager component, VMware does the following:

1. Assess the impact to Tanzu Operations Manager.
2. If the vulnerability affects a Tanzu Operations Manager component, VMware schedules an update for the impacted components.
3. Update the affected components and perform system tests.
4. Announce the fix publicly through the following channels:
 1. Automated notification to end users who have downloaded or subscribed to a Tanzu Operations Manager product on [Broadcom Support portal](#) when a new, fixed version is available.
 2. Adding a new post to the [VMware Security page](#).

Classes of vulnerabilities

Attackers can exploit vulnerabilities to compromise user data and processing resources. This can affect data confidentiality, integrity, and availability to different degrees. For vulnerabilities related to Ubuntu provided packages, VMware follows Canonical's priority levels. For other vulnerabilities, VMware follows Common Vulnerability Scoring System v3.0 standards when assessing severity. For more information about Canonical's priority levels, see [CVE Priority](#) in Ubuntu CVE Tracker. For more information about Common Vulnerability Scoring System v3.0 standards, see [Common Vulnerability Scoring System version 3.1: Specification Document](#) on the FIRST website.

VMware uses Canonical's Ubuntu distribution of Linux for Tanzu Operations Manager Ubuntu stemcells and rootfs. Canonical provides VMware with support services allowing VMware to escalate CVEs that VMware determine might affect Tanzu Operations Manager. In general, VMware does not escalate to upstream open source software components or vendors for Medium or Low CVEs that are not yet patched. Tanzu Operations Manager can escalate on behalf of a customer for High or Critical CVEs. Tanzu Operations Manager customers who are interested in addressing CVEs in Ubuntu that are not yet patched can establish their own support relationship with Canonical through the [Support](#) page on the Ubuntu website.

VMware reports the severity of vulnerabilities using the following severity classes:

High

High severity vulnerabilities are those that can be exploited by an unauthenticated or authenticated attacker, from the Internet or those that break the guest/host Operating System isolation. The exploitation could result in the complete compromise of confidentiality, integrity, and availability of user data and/or processing resources without user interaction. Exploitation could be leveraged to propagate an Internet worm or execute arbitrary code between Virtual Machines (VMs) and/or the Host Operating System. This rating also applies to those vulnerabilities that could lead to the complete compromise of availability when the exploitation is by a remote unauthenticated attacker from the Internet or through a breach of VM isolation.

Medium

Medium vulnerabilities are those in which the ability to exploit is mitigated to a significant degree by configuration or difficulty of exploitation, but in certain deployment scenarios could still lead to the compromise of confidentiality, integrity, or availability of user data and/or processing resources.

Low

Low vulnerabilities are all other issues that have a security impact. These include vulnerabilities for which exploitation is believed to be extremely difficult, or for which successful exploitation would have minimal impact.

Release policy

VMware schedules regular monthly releases of software to address Low and Medium severity vulnerability exploits. When High severity vulnerability exploits are identified, VMware releases fixes to software as quickly as possible.

Alerts and actions archive

To review an archive of Tanzu Operations Manager vulnerability reports, see the VMware Product Vulnerability Reports on the [Tanzu security page](#).

Testing, release, and security lifecycle for Tanzu Operations Manager

This topic explains how VMware's development practices, automated build tools, and organizational structures work together to create and support stable releases of Tanzu Operations Manager.

- Tanzu Operations Manager teams building system components receive frequent feedback, which helps to secure code from exposure to vulnerability.
- Every Tanzu Operations Manager release follows a strict workflow and passes through numerous quality and compliance checks before distribution.
- Teams build tests into the product consistently and run them automatically with any code change.

Release mechanics

VMware releases, patches, and supports multiple versions of Tanzu Operations Manager simultaneously. This section explains the versioning and support conventions VMware follows.

Versioning

VMware numbers Tanzu Operations Manager releases following a semantic versioning format, X.Y.Z, where X and Y indicate major and minor releases, and Z designates patch releases. Major and minor releases change Tanzu Operations Manager functionality. Patch releases are backward-compatible security patches or bug fixes. For more information about semantic versioning, see the [Semantic Versioning 2.0.0](#) website.

Support

VMware supports each major and minor platform release according to the following criteria:

- VMware supports the release for at least nine months following its first publication date.
- VMware supports the last three major or minor releases, even if this extends coverage beyond nine months.

Support includes maintenance updates and upgrades, bug and security fixes, and technical help. For more details about support standards, technical guidance, and publication phases, see [Product Lifecycle Policy](#) on the VMware Tanzu website. For a definition of VMware support in legal terms, see [Broadcom Support Services Policies](#).

Patch releases

Patch releases are more frequent and less predictable than major and minor releases.

VMware identifies security issues using standard nomenclature from [Common Vulnerabilities and Exposures \(CVE\)](#), [Ubuntu Security Notices \(USN\)](#), and other third-party sources. For more information about security fixes in core Tanzu Operations Manager code or packaged dependencies, see [Tanzu Operations Manager v3.2 Release Notes](#) and [Elastic Application Runtime v10.3 Release Notes](#).

The Tanzu security team maintains a running list of security fixes in Tanzu Operations Manager and Tanzu Operations Manager dependencies. To see their most recent findings, see [Security Advisories - Tanzu](#).

Upgrading

All Tanzu Operations Manager releases pass through extensive test suites that include automated unit, integration, and acceptance tests on multiple IaaSes. Regardless of this extensive testing, VMware recommends that you test major and minor releases in a non-production environment before implementing them across your deployment. Upgrade your production environment as soon as possible after you validate the new release on your test environment.

Release testing, integration, and validation

This section describes VMware's software development processes and explains compliance and regulatory standards to which VMware software adheres.

Test-Driven Development

The VMware Tanzu development process relies on a strict workflow with continuous automated testing. VMware R&D does not separate engineering and testing teams. Rather, every member of each engineering team is responsible for ensuring the quality of their code. They write tests for all of the software components that they develop, often before writing the software itself.

With every software change, automated build pipelines trigger these tests for the new software component and for everything it touches. If a new code check-in does not pass its tests or causes a failure elsewhere, it pauses the build pipeline for the entire team, or sometimes all of VMware R&D. The transparency of this process encourages developers to work together to address code issues quickly.

VMware applies the following automated testing approaches, scenarios, and frameworks to Tanzu Operations Manager components and to the release as a whole:

- Unit tests: Development teams write unit tests to express and validate desired functional behavior of product components. Typical frameworks used are [RSpec](#) and [Ginkgo](#). These tests run continuously throughout the development cycle.
- OSS integration tests: The Release Integration team exercises a full deployment of open-source Cloud Foundry to validate all end-user features. They maintain the [Cloud Foundry Acceptance Test](#) (CATs) suite alongside the OSS cf-release. Cloud Foundry component teams also contribute acceptance test suites at the OSS Integration Test level. These tests exercise and validate their components' functional, performance, and integration health.
- Tanzu Operations Manager integration tests: The Tanzu Operations Manager Release Engineering (RelEng) team validates the quality and cross-product integration health of the commercial Tanzu Operations Manager release. RelEng runs OSS Acceptance Tests against all supported releases. These tests run on full Tanzu Operations Manager instances configured to represent diverse real-world customer scenarios on various IaaSes and using both internal and external load balancer, database, blobstore, and user store solutions.

Additional pre-release gates: Internal, PWS, and Compliance

In addition to its automated unit and integration testing, VMware deploys all upgrades slated for upcoming Tanzu Operations Manager releases on at-scale test environments. Prior to each major or minor commercial release, VMware runs the entire Tanzu Operations Manager suite of services on several internally-managed large integration environments that run customer-like data and workloads.

All Tanzu Operations Manager product teams participate in go-to-market steps for each release, as is often required for shipping a legally compliant product. Examples include Open Source License File attribution and an Export Compliance classification.

Patch releases: security and bug fixes

VMware uses established processes to track, disclose, and remediate vulnerabilities in Tanzu Operations Manager and related dependent components. This section explains how VMware identifies vulnerabilities and implements fixes for them.

Identifying security vulnerabilities

VMware has an established process to track and patch vulnerabilities in software dependencies and Tanzu Operations Manager software. Additionally, the [Product Security Center](#) page describes a responsible disclosure process for reporting vulnerabilities identified in VMware software by third parties.

VMware uses multiple methods to identify security vulnerabilities in VMware software and dependencies internally, including:

- Security notifications from [Canonical](#) for their Ubuntu operating system, provided through VMware's commercial relationship with Canonical
- Software component scans that use third-party security software to update continuously from external security vulnerability sources.
- Dependency analysis software that identifies and catalogs software dependencies
- Security vulnerability notifications from known software dependencies

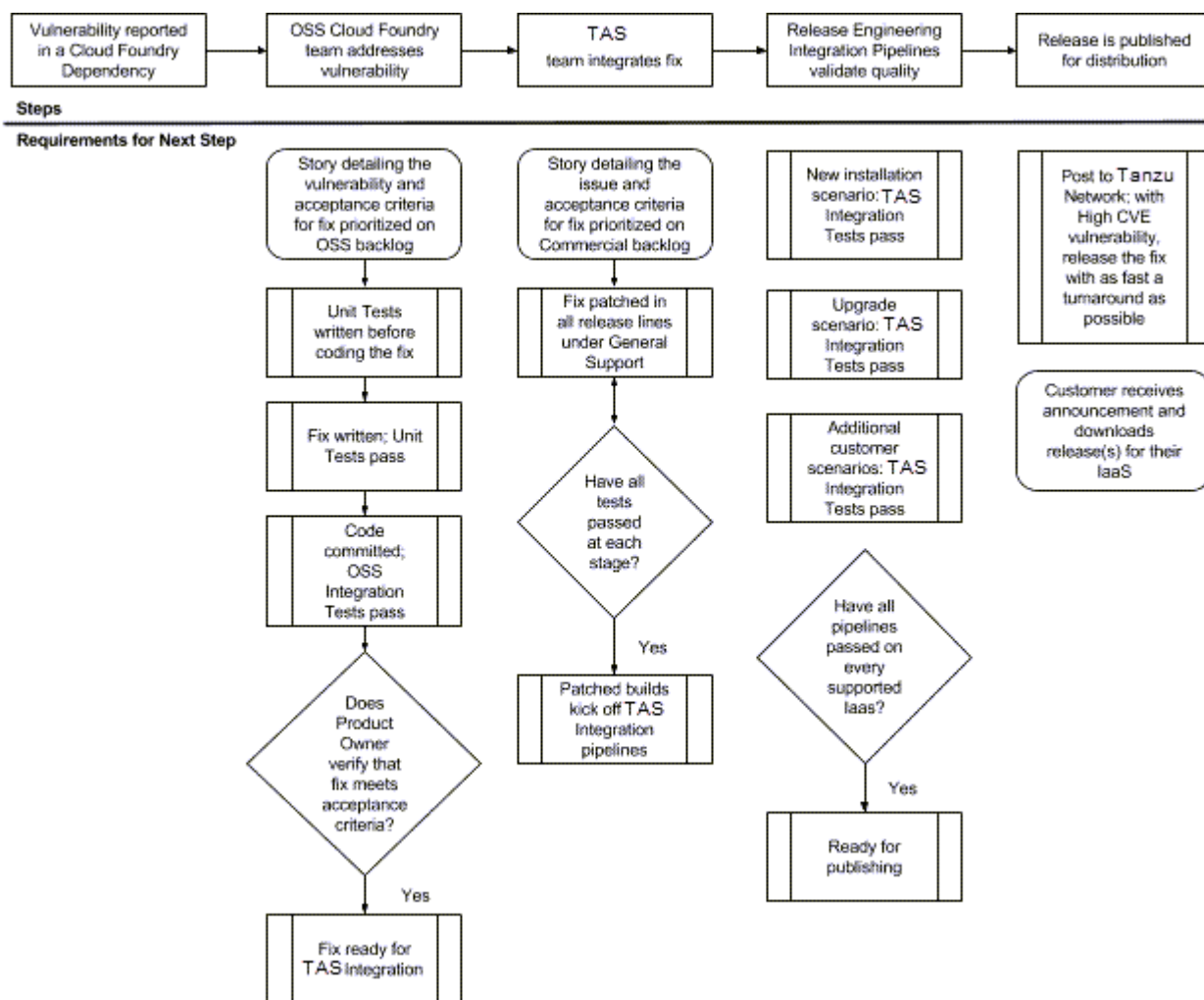
VMware also monitors externally-reported vulnerabilities from many sources, including:

- Third-party security analysis requested by VMware
- Cloud Foundry Foundation security notifications from member companies
- Customer, prospect and other third-party security reports

When VMware discovers a potential security vulnerability in Tanzu Operations Manager, the security team opens an issue to assess it. If it confirms the vulnerability exists, VMware identifies and updates affected components with plans to backport the fix to stable releases. Fixes are implemented on a target timeline based on the severity level of the vulnerability. For more information, see [Tanzu Operations Manager Security Overview and Policy](#).

Fix, test, and release lifecycle

The following flowchart details the steps that VMware performs on a typical high-priority CVE, to publish a patch release fix on the [Broadcom Support portal](#). A detailed description of the steps follows.



The triage flowchart shows the following steps:

1. Vulnerability reported in a Cloud Foundry Dependency.
2. OSS Cloud Foundry team addresses vulnerability.
 1. Story detailing the vulnerability and acceptance criteria for fix prioritized on OSS backlog.
 2. Unit Tests written before coding the fix.
 3. Fix written; Unit Tests pass.
 4. Code committed; OSS Integration Tests pass.
 5. Check: Does Product Owner verify that fix meets acceptance criteria?

6. If yes, fix is ready for integration.
3. VMware Tanzu team integrates fix.
 1. Story detailing the issue and acceptance criteria for fix prioritized on Commercial backlog.
 2. Fix patched in all release lines under General Support.
 3. Check: Have all tests passed each stage?
 4. If yes, patched builds kick off integration pipeline.
4. Release Engineering Integration Pipelines validate quality.
 1. Tests pass by scenario:
 - New installation scenario: Integration Tests pass.
 - Upgrade scenario: Integration Tests pass.
 - Additional customer scenarios: Integration Tests pass.
 2. Check: Have all pipelines passed on every supported IaaS?
 3. If yes, ready for Publishing.
5. Release is published for distribution.
 1. Post to [Broadcom Support Portal](#). With High CVE vulnerability, release the fix as quickly as possible.
 2. Customer receives announcements and downloads release for their IaaS.

IaaS security considerations for Tanzu Operations Manager

The following topics cover security considerations for your IaaS and your Tanzu Operations Manager deployment:

- [Encrypting Tanzu Operations Manager VM disks](#)
- [Security guidelines for your IaaS provider](#)

Encrypting Tanzu Operations Manager VM disks

This topic describes how you can secure Tanzu Operations Manager VMs by encrypting their disks or rotating their disk encryption keys. Disk encryption protects data integrity if computing resources are physically stolen.

Disk encryption for VMs works at the IaaS level. An IaaS encrypts disks when it first creates them, or re-encrypts them when it rotates encryption keys.

To encrypt disks in Tanzu Operations Manager, you must:

1. Configure the IaaS to encrypt disks when it creates or recreates them.
2. Trigger BOSH to recreate the existing VMs that use the disks, and create encrypted disks from now on for new VMs.

The procedures on this page describe how to do this for each IaaS.

The disks you can encrypt on a Tanzu Operations Manager VM are:

- The root file system for the VM. For BOSH-created VMs, this comes from the stemcell.
- Ephemeral disk for the VM

- Persistent disk for the VM

For each IaaS, there are two disk encryption procedures, which encrypt different VMs:

- The BOSH Director procedure encrypts the disks used by the BOSH Director VM when you first create a Tanzu Operations Manager environment.
- The BOSH-deployed VM procedure encrypts disks for the VMs the BOSH Director creates, after BOSH has been deployed.

Encrypt disks or rotate keys

You can use the same procedure to encrypt disks for the first time or to rotate encryption keys.

For BOSH-deployed VMs, some IaaS let you associate a policy with the BOSH process that automatically encrypts all disks BOSH creates. On AWS, BOSH must explicitly tell the IaaS to encrypt each disk that it creates, and passes in an encryption key. The following table summarizes these differences:

| IaaS | How configured | How encrypted | User can supply key | BOSH stores key ID |
|----------------------------|--|---|---------------------|--------------------|
| AWS | User pastes key Amazon Resource Name (ARN) into Tanzu Operations Manager | BOSH tells IaaS to encrypt disks it creates | Yes | Yes |
| Azure (with managed disks) | User configures IaaS to associate encrypt policy with BOSH | IaaS automatically encrypts disks it creates for BOSH | No | No |
| Azure (with Azure Storage) | User configures IaaS to associate encrypt policy with BOSH | IaaS automatically encrypts disks it creates for BOSH | Yes | No |
| vSphere | User configures IaaS to associate encrypt policy with BOSH | IaaS automatically encrypts disks it creates for BOSH | Yes | No |

Azure

Azure offers a managed disk service, which allocates disk space flexibly on demand.

For more information about managed disks, see [Introduction to Azure managed disks](#) in the Microsoft Azure documentation.

Encrypt Azure disks

For disk encryption, VMware recommends using managed disks. By default, managed disks use encryption keys that are managed by the IaaS, so you do not supply your own keys. You also do not need to re-create VMs after encrypting disks or rotating encryption keys, because the IaaS propagates the change to all VMs automatically.

To initiate or rotate disk encryption for BOSH-deployed VMs on Azure using platform-managed keys:

1. Log in to Azure Portal.
2. Encrypt new and existing Tanzu Operations Manager VMs by following the procedure in [Azure Disk Encryption for virtual machines and virtual machine scale sets](#) in the Microsoft Azure documentation.
3. For unmanaged Storage Account disks, see [Recreate BOSH-Deployed Disks](#) (at the bottom of this page) to propagate the change to existing VMs. If you are using managed disks, you can skip this step.

If you want to use customer-managed encryption keys, set the Custom Disk Encryption Set Name field on the **Azure Config* tab of the BOSH Director tile. You must redeploy all VMs if you change this setting for it to take effect.

For more information about how BOSH integrates with IaaS-level disk encryption on Azure, see [Encryption](#) in *Microsoft Azure* in the BOSH documentation.

vSphere

vSphere v6.5 and later support disk encryption for encrypted VMs.

If you are configuring volume encryption for VMware Tanzu Container Orchestration Engine (formerly TKGI), you cannot use Tanzu Operations Manager or the vSphere BOSH CPI to encrypt persistent disks.

To initiate or rotate disk encryption for BOSH-deployed VMs on vSphere v6.5 or later:

1. Log in to vCenter.
2. Follow the procedure in [Encrypt an existing virtual machine or virtual disk](#) in the VMware documentation. If you are configuring volume encryption for Container Orchestration Engine, change the Default Storage Policy to an existing encryption policy.
3. Follow the procedure in [Recreate BOSH-deployed disks](#) procedure to propagate the change to existing VMs.
4. If you configure volume encryption for Container Orchestration Engine:
 1. Delete the BOSH Director stemcell used for the Container Orchestration Engine foundation from the disk using vSphere WebClient. For more information on how to delete the BOSH Director stemcell, see [How to recreate BOSH Director VM when the Stemcell is deleted from vSphere](#).
 2. Re-upload the stemcell used by Container Orchestration Engine:

```
sudo cp /var/tempest/stemcells/STEMCELL-NAME.tgz /home/ubuntu/.
sudo chown ubuntu:ubuntu STEMCELL-NAME.tgz
bosh upload-stemcell STEMCELL-NAME.tgz --fix
```

Where **STEMCELL-NAME** is the name of the stemcell you want to reupload.

For example:

```
sudo cp /var/tempest/stemcells/bosh-stemcell-621.84-vsphere-esxi-ubuntu-xe
nial-go_agent.tgz /home/ubuntu/.
sudo chown ubuntu:ubuntu bosh-stemcell-621.84-vsphere-esxi-ubuntu-xenial-g
o_agent.tgz
bosh upload-stemcell bosh-stemcell-621.84-vsphere-esxi-ubuntu-xenial-go_ag
ent.tgz --fix
```

1. In the Container Orchestration Engine tile, select the Upgrade All Clusters errand.
2. In Tanzu Operations Manager, click Apply Changes.

For more information about how BOSH integrates with IaaS-level disk encryption on vSphere, see [Encryption](#) in *vSphere* in the BOSH documentation.

AWS

On AWS in Tanzu Operations Manager v2.0 and later, you can use your Amazon account key to encrypt Linux EBS volumes or you can supply your own key.

To encrypt BOSH-deployed VMs and the Tanzu Operations Manager VM on AWS, see [Configuring Amazon EBS Encryption](#).

For more information about how BOSH integrates with IaaS-level disk encryption on AWS, see [Encryption in Amazon Web Services](#) in the BOSH documentation.

Recreate BOSH-deployed disks

Unless you are using Azure managed disks, you must manually recreate disks on BOSH-deployed VMs after you have added or rotated disk encryption keys. To manually recreate disks:

1. Configure Tanzu Operations Manager to encrypt VM root, ephemeral disk, and persistent disk on next deploy:
 - Root File System: To recreate the root file system for VMs, you must upload a new stemcell. If you are already running the latest stemcell, you can:
 - Wait until a new stemcell comes out. This typically takes less than two weeks.
 - If propagating disk encryption is urgent, contact [Support](#).
 - Ephemeral Disks: In the Director Config pane of the Tanzu Operations Manager tile, enable the Recreate VMs deployed by the BOSH Director check box.
 - Persistent Disks: In the Director Config pane of the Tanzu Operations Manager tile, enable the Recreate All Persistent Disks check box.
2. Click Review Pending Changes.
3. Click Apply Changes.

Security guidelines for your IaaS provider

This topic provides security guidelines for the Infrastructure as a Service (IaaS) providers supported by Tanzu Operations Manager.

Tanzu Operations Manager supports a variety of IaaS providers. Different IaaS providers require different configuration steps to secure user data, identity information, and credentials.

Security requirements can vary broadly based on the unique configuration and infrastructure of each organization. Rather than provide specific guidance that might not apply to all use cases, VMware has collected links to IaaS providers' security and identity management documentation. The following documents can help you understand how your IaaS' security requirements impact your Tanzu Operations Manager deployment.

VMware does not endorse these documents for accuracy or guarantee that their contents apply to all Tanzu Operations Manager installations.

Find your IaaS provider in the following list. The documentation items linked for each IaaS can help you configure and secure your installation infrastructure.

Amazon Web Services (AWS)

These topics in the AWS documentation provide a general reference for AWS Identity and Access Management (IAM) features:

- [AWS Identity and Access Management Guide](#): If you are new to AWS, start here.
- [Identities \(Users, Groups, and Roles\)](#) in the AWS Identity and Access Management Guide

- [Temporary Security Credentials](#) in the AWS Identity and Access Management Guide: This documentation provides a general definition of IAM terms and provide best practices to help you manage IaaS users and permissions.
- [Use IMDSv2](#): This documentation provides a description of IMDSv2 and the security benefits it provides for the AWS metadata endpoints. IMDSv2 support must be enabled within Tanzu Operations Manager.

Enable IMDSv2 in Tanzu Operations Manager

Tanzu Operations Manager includes a feature that allows you to require the usage of IMDSv2. This security feature requires users to send a signed token header with any request to the instance metadata endpoint provided on AWS VMs. By default, IMDSv2 is not required when using the AWS metadata endpoint on a BOSH-deployed VM. You can configure Tanzu Operations Manager to require IMDSv2 using the Tanzu Operations Manager API. After enabling IMDSv2, you must re-create all VMs for the new configuration to take effect.

Tanzu Platform for Cloud Foundry v2.13.0 and later are compatible with IMDSv2. For all versions of VMware Tanzu Container Orchestration Engine (formerly TKGI), it might be necessary to set the hop limit using the `metadata_options` key. For more information, see [VM Types / VM Extensions](#) in the BOSH documentation.

All tiles and service instances must be using stemcell Xenial v621.183 or later in order to require IMDSv2. Requiring IMDSv2 while using an older stemcell causes Apply Changes to fail due to an unresponsive BOSH agent.

To require IMDSv2 on all BOSH-deployed VMs:

1. Ensure all tiles are using stemcell Xenial v621.183 or later.
2. Use the Tanzu Operations Manager API to set the `require_imds_v2` property to `true`. For more information, see:
 - [Using the Tanzu Operations Manager API](#)
 - [Tanzu Tanzu Operations Manager API Reference](#).
3. Configure the BOSH Director to recreate VMs:
 1. Go to the Tanzu Operations Manager Installation Dashboard.
 2. Click the BOSH Director tile.
 3. Click Director Config.
 4. Enable the Recreate VMs deployed by the BOSH Director check box.
 5. Return to the Tanzu Operations Manager Installation Dashboard.
4. If you have service tiles installed, for each service tile:

The names of the Upgrade all service instances and Recreate all service instances errands might be slightly different between services.

1. Click the tile.
2. Click the Errands tab.
3. Enable the Upgrade all service instances errand. You must run this errand to push CredHub certificate updates to each service instance.

4. If the service tile has the Recreate all service instances errand, enable the Recreate all service instances errand. Running this errand pushes BOSH Agent certificate updates to service instances.
5. Click Review Pending Changes.
6. Click Apply Changes.

Google Cloud Platform (GCP)

This topic in the GCP documentation describes general authentication guidelines for GCP:

- [Authentication overview](#)

Microsoft Azure

This topic in the Microsoft Azure documentation describes managing IaaS users and credentials:

- [Azure documentation](#)

VMware vSphere

This topic in the VMware vSphere documentation describes best practices for securing and managing a vSphere installation:

- [vSphere Security guide \(PDF\)](#)

Stemcell security

The following topics describe how VMware implements security in the stemcell layer of Tanzu Operations Manager.

- [Linux Stemcell Hardening](#): How VMware secures Linux stemcells through regular testing and minimizing their surface of vulnerability
- [Windows stemcell hardening](#): How VMware secures Windows stemcells through regular testing and minimizing their surface of vulnerability
- [Floating stemcells in Tanzu Operations Manager](#): How Tanzu Operations Manager automatically upgrades all compatible products when a new stemcell is available

For more information about stemcell availability and related security fixes, see [Stemcell Release Notes](#).

Linux Stemcell Hardening

This document applies to currently supported stemcells.

“Hardening” is the process by which VMware secures Tanzu Operations Manager; this is done by reducing its vulnerability surface from outside access. This document provides responses to some commonly-asked questions about the security configuration enhancements and hardening tests that are applied to the Cloud Foundry (CF) stemcell. This information will be helpful to:

- Customer accreditation teams who are responsible for running configuration scans of a Cloud Foundry deployment
- Auditors who need a documentation artifact to feed into the customers’ existing security assessment processes

- What is a stemcell? A stemcell is a versioned Operating System (“OS”) image wrapped with IaaS specific packaging. A typical stemcell contains a bare minimum OS skeleton with a few common utilities pre-installed, a BOSH Agent, and a few configuration files to securely configure the OS by default. Stemcells do not contain any specific information about any software that will be installed once that stemcell becomes a specialized machine in the cluster; nor do they contain any sensitive information which would make them unable to be shared with other BOSH users. This clear separation between base OS and later-installed software is what makes stemcells a powerful concept. In addition to being generic, a stemcell release contains nearly identical packages for the different infrastructures. This property of stemcells allows BOSH users to quickly and reliably switch between different infrastructures without worrying about the differences between OS images. The CF Foundation is responsible for producing and maintaining an official set of stemcells.
- What is stemcell hardening? Stemcell hardening is the process of securing a stemcell by reducing its surface of vulnerability, which is larger when a system performs more functions; in principle a single-function system is more secure than a multipurpose one. There are various methods of hardening Linux systems. Common techniques include reducing available methods of attack by implementing more restrictive and/or conservative configurations of the OS kernel and system services, changing default passwords, the removal of unnecessary software, unnecessary usernames and logins, and the disabling or removal of unnecessary services.
- What is our general approach to stemcell hardening? The CF stemcell is essentially a distinct Linux distribution. As such, industry-standard benchmarks are not entirely appropriate when assessing the security posture of the stemcell, but Cloud Foundry has considered and incorporated hardening guidance from various sources both commercial and government. Some parts of the existing recommended industry-standard hardening configurations will certainly apply, but some other parts do not apply. In addition, because each stemcell is a unique Linux distribution, existing industry-standard benchmarks are silent on some important aspects of hardening the stemcell configurations. The following paragraphs describe the different categories of stemcell hardening configurations.
 1. Baseline Passing: common hardening tests that pass without any changes to the stemcell or to test procedures.
 2. Test Amended: Stemcells are optimized for cloud deployment and some configuration settings are not stored in traditionally-expected locations. The industry standard test was changed to conform with stemcell design to accurately check the recommended setting. This new test reflects the changes to the industry standard test but the stemcell adheres to commonly accepted guidance.
 3. Additional Hardening: Configuration hardening improvements that have been made to the stemcell. As with most software, a stemcell’s security improves over time and every stemcell release is tested to ensure that it is suitable for use with its associated CF release. Later releases of a stemcell can include additional security features that were not present in earlier releases.
 4. New CF-specific Tests: New tests that have been added to check CF stemcell-specific configurations. These tests are not yet part of any industry standard Ubuntu benchmark. This category of tests is still under development and additional tests will be added over time.
- What are the major focus areas for our stemcell hardening approach?
 1. Maintenance, Updates, and Patching

1. Regular patches and feature enhancements are delivered through routine BOSH deployments of updated stemcells (obviates apt-get upgrade).
2. File System Hardening
 1. The temporary storage directories such as /tmp and /var/tmp are mounted on a separate partition, and configured with appropriately limiting options such as nodev, nosuid, and noexec.
 2. Each of the following directories is in a separate mount, with mount options managed through BOSH agent:
 - /var/log
 - /home
 - /tmp
 3. File system mount options for users' home directories are limited through appropriate mount options including nodev.
 4. Removable media might not be mounted as character or block special device.
 5. Executable programs might not run from removable media.
 6. setuid and setgid are not allowed on removable media.
 7. Users cannot create special devices in shared memory partitions.
 8. Users cannot put privileged programs onto shared memory partitions.
 9. Users cannot execute programs from shared memory partitions.
 10. Users cannot delete or rename files in world-writable directories such as /tmp that are owned by other users.
 11. Supplementary and exotic Linux file systems that are unused in CF have been deactivated.
 12. Automount of USB drives or disks is not permitted.
3. Boot Security
 1. The owner and group for the bootloader config (/boot/grub/grub.cfg) is set to root. Only root has read and write access to this file.
 2. Boot loader has been configured so that a password is required to reboot the system.
 3. Unauthorized users cannot reboot the system into single user mode.
4. Process Security
 1. Users cannot override the soft limit for core dumps.
 2. Randomized virtual memory region placement is enabled.
 3. Prelinking of shared libraries is deactivated.
5. Minimization of Attack Surface
 1. The Network Information Service ("NIS") is not used in CF and is not installed.
 2. The Berkeley rsh-server package is not used in CF and is deactivated.
 3. Classic rsh-related tools are not used in CF and are not installed.
 4. The following servers are not used on CF stemcells and are deactivated:

- talk server
- telnet server
- tftp-server
- Avahi
- print
- DHCP
- DNS
- FTP
- IMAP
- POP
- HTTP
- SNMP

5. The talk client is not used in CF and is not installed.
6. The eXtended InterNET Daemon (xinetd) is not used in CF and is deactivated.
7. The following network services are not used in CF and are deactivated:
 - chargen
 - daytime
 - echo
 - discard
 - time
8. The X Window system is not used in CF and is not installed.
9. NTP time setting is synchronized on the stemcell through the chrony utility.
10. The Samba daemon is not used in CF and is deactivated.
11. The Mail Transfer Agents (MTA) process only local mail.
12. The rsync service is not used in CF and is deactivated.
13. The biosdevname tool is deactivated.

6. Network Security

1. IPv4 networking is configured such that IP forwarding is deactivated.
2. The IPv4 networking has been configured such that the host cannot send ICMP redirects.
3. IPv4 networking has been configured such that the system does not accept source routed packets.
4. IPv4 networking is configured such that ICMP redirects are not accepted.
5. ICMP echo and timestamp requests with broadcast or multicast destinations will be ignored.
6. The stemcell will ignore malformed ICMP error responses.
7. IPv4 networking is configured for source route validation.

8. TCP SYN cookies are enabled.
9. Stemcells are set to refuse IPv6 router advertisements.
10. The /etc/hosts.allow file exists and is empty.
11. The /etc/hosts.allow and /etc/hosts.deny files are protected from unauthorized write access.
12. The /etc/hosts.deny file exists and is empty.
13. The following protocols are not used in CF and are deactivated:
 - SCTP
 - DCCP
 - TIPC
 - LDAP
 - RDS
14. Wireless interfaces are deactivated.
15. IPv6 is not used in CF deployments and the IPv6 protocol is deactivated.

7. Auditing

1. Audit log file size is configured for a manageable maximum size of 6 MB.
2. The system auditd logs have been configured such that the system is resilient in the event of a denial of service attack on the auditd daemon.
3. Auditd daemon is configured such that all auditd logs are kept after rotation.
4. The auditd service is enabled.
5. Auditing of successful and failed login/logout events is enabled.
6. The Linux auditing subsystem has been configured in accordance with best practice industry guidance to capture all security-relevant events. The /etc/audit/audit.rules configuration now contains more than 50 monitoring rules.
7. Audit records are created for loading and unloading of kernel modules and for system calls.
8. File Integrity Monitoring can be done on the stemcell (through a BOSH Add-on).

8. Authentication and Authorization

1. The cron daemon is enabled.
2. Access to the /etc/crontab file is limited to root.
3. Access to the cron utility configuration through the hourly, daily, weekly, and monthly directories is limited.
4. User authorization to schedule cron jobs is limited.
5. Only the vcap user is allowlisted to use the cron and at utilities.
6. Password requirements follow industry best practice guidance and enforce a minimum length of 14 characters, with at least one each of: digit, uppercase, lowercase and special characters.
7. Password reuse: users cannot reuse their twenty most recent passwords.
8. SSH protocol version is configured for SSH-2.

9. Logging level for SSH event is INFO.
10. Minimum permissions are set on `/etc/ssh/sshd_config`.
11. SSH X11 forwarding is deactivated.
12. The `MaxAuthTries` parameter for SSH is set to 3 attempts per connection.
13. SSH is configured to require passwords and ignore host-based authentication.
14. Root logins are not allowed over SSH.
15. Users cannot set environment variables through the SSH daemon.
16. SSH has been configured to use strong ciphers:
 - `aes128-ctr`
 - `aes192-ctr`
 - `aes256-ctr`
17. Idle SSH sessions are terminated after 15 minutes, and no client “keep alive” messages are sent.
18. The SSH login banner might be configured to display site-specific text before user authentication is permitted (through BOSH Add-on).
19. Root login is only permitted through console, not through tty devices.
20. Only the `vcap` user is authorized in the `sudo` group.
21. Only users in the root group (a.k.a. `wheel`) are authorized to run the `su` command.

9. Compliance

1. Contents of `/etc/issue` and `/etc/issue.net` have been configured to the phrase: “Unauthorized use is strictly prohibited. All access and activity is subject to logging and monitoring.” This might be amended if and as necessary through a BOSH Add-on.
2. The Message of the Day file `/etc/motd` is not used, but might be populated through a BOSH Add-on if needed.
3. Identification of the OS and/or version information about the OS does not appear in any login banners.

10. File System Permissions

1. The `/etc/passwd`, `/etc/shadow`, and `/etc/group` files are protected from unauthorized write access.
2. Use and/or presence of any world-writable files has been audited, and minimized to the extent possible for CF.
3. By default, all stemcell files are owned by a known user and group, and might not belong to a non-existent user or group.
4. Use of SUID and GUID is restricted, and only the `/usr/bin/sudo` and `/bin/su` programs are authorized as SUID and/or GUID programs.

11. User Account Management

1. Users cannot change their password more than once a day.
2. Users are notified 7 days before their passwords expire.
3. Interactive logins are deactivated for system accounts.

4. The GID for the root account is 0.
5. User accounts might not have empty passwords.
6. NIS is not used in CF, and integration of OS security configuration with legacy NIS permissioning is not enabled (e.g., for `/etc/passwd`, `shadow`, and `group`).
7. By default, the only UID 0 account present is `root`.
8. By default, the root `PATH` does not include any risky directory such as the current working (`.`) or any writable directory.
9. Minimum privileges are applied to all users' hidden configuration ("dot") files.
10. The `.netrc` and `.rhosts` and `.forward` files are not used in CF and are not present in any user home directory.
11. Any group present in the `/etc/passwd` file must also exist in the `/etc/group` file.
12. Users defined in `/etc/passwd` must have a valid home directory.
13. Users must own their home directories.
14. All references to user and group names, as well as UID and/or GID identifiers, are self consistent, with no duplicates or orphans allowed.
15. By default, the `shadow` group is not used in CF and must be empty.

Windows stemcell hardening

This topic provides information about the security hardening of Microsoft Windows stemcells.

A stemcell is a versioned OS image that is customized based on IaaS. A typical stemcell contains the OS image with common utilities, a BOSH agent, and configuration files to securely configure the OS.

Stemcell hardening is the process of securing a stemcell by reducing its surface of vulnerability. The surface of vulnerability for a stemcell is smaller when a system performs fewer functions. For example, a single-function system is more secure than a multipurpose one.

Microsoft baseline security standard

Windows Stemcells for both Tanzu Platform for Cloud Foundry and VMware Tanzu Container Orchestration Engine (formerly TKGI) follow the Microsoft Baseline Security Standard.

Windows stemcells do not yet align completely with the Microsoft Baseline Security Standard. For details about the ways in which Windows stemcell hardening differs from the Microsoft Baseline Security Standard, contact [Tanzu Support](#).

For more information about Microsoft Baseline Security Standard and to download security configuration baselines for Windows, see [Microsoft Security Compliance Toolkit](#) on the Microsoft website.

Audit policies

Audit policies for Windows Server 2019 stemcells are based on Microsoft Baseline Security Standard. Audit policies allow you to better audit security vulnerabilities in your environment.

The following list includes some of the key audit policies applied to Windows Server 2019 stemcells:

- Log success and failure audit events of user logins and logouts for Windows VMs
- Log audit events related to object access on Windows VMs
- Log audit events related to policy changes on Windows VMs

Firewall settings

Windows Server 2019 stemcells align with the firewall behavior recommended by the Microsoft Baseline Security Standard. However, they are not fully compliant with the Microsoft Baseline Security Standard.

The Windows stemcells block all inbound requests and permit all outbound requests. Specific ports are open for communication between Tanzu Operations Manager components and the Windows VM.

For more information about the firewall rules for the Windows Server 2019 stemcells, contact [Tanzu Support](#).

Floating stemcells in Tanzu Operations Manager

This topic explains how floating stemcells work in Tanzu Operations Manager, and how they affect your Tanzu Operations Manager product tile upgrades.

Floating stemcells: Benefits and limitations

Floating stemcells have the following benefits:

- Tanzu Operations Manager can quickly propagate a patched stemcell to all virtual machines in the deployment when a vulnerability is discovered.
- You can do certain deployment-wide updates, such as CVEs, by uploading a new stemcell instead of uploading `.pivotal` files for each tile, which reduces the time spent waiting for files to upload.

Stemcells have the following upgrade limitations:

Floating stemcells allow upgrade to the minor versions of stemcells but not to the major versions.

For example, a stemcell can float from `1234.56` to `1234.99` but not from `1234.991` to `1235.0`.

Product tile upgrades

When you upgrade a product tile, Tanzu Operations Manager completes the following steps:

1. Tanzu Operations Manager checks to see whether a new minor, or patch, version of the stemcell exists.
2. If a minor stemcell is available from the current major line, Tanzu Operations Manager installs the upgraded tile and all compatible product tiles in the deployment on the new stemcell.

Uploading and managing stemcells

You can upload new stemcells to your product tiles by using the Tanzu Operations Manager API or by using the Stemcell Library in the Tanzu Operations Manager Installation Dashboard. For more information about the Stemcell Library, see [Importing and Managing Stemcells](#).

Tanzu Operations Manager assessment against NIST SP 800-53(r4) controls

For an assessment of Tanzu Operations Manager against the NIST SP 800-53(r4) controls, and for guidance on how you can achieve compliance when using a shared responsibility model, see [Assessment of Tanzu Platform for Cloud Foundry against NIST SP 800-53\(r4\) Controls](#).

Security-related Tanzu Operations Manager tiles and add-ons

This topic provides links to other security-related services available for your Tanzu Operations Manager deployments. For information about other security-related tiles provided by VMware partners, see the [Broadcom Support portal](#) and the [Tanzu documentation](#).

Tiles

This section provides links to security related tiles available for Tanzu Operations Manager deployments.

- [Single Sign-On \(SSO\)](#): Provides support for native authentication, federated single sign-on, and authorization
- [CredHub Secrets Management \(formerly CredHub Service Broker\)](#): Allows apps running on Tanzu Platform for Cloud Foundry to access secure credentials in CredHub
- [Compliance Scanning](#): Enables operators and auditors to conduct compliance scanning of BOSH-managed Xenial VMs against configuration hardening guidelines

Add-ons

This section provides links to security-related BOSH add-ons available for Tanzu Operations Manager deployments.

- [File Integrity Monitoring](#): Provides File Integrity Monitoring (FIM) protection in the Tanzu Operations Manager environment. This monitoring is sometimes required for compliance purposes.
- [IPsec Encryption](#): Adds security to the network layer for each BOSH-deployed virtual machine (VM), increasing internal protection if a malicious actor breaches your firewall
 - [Securing Data in Transit with IPsec](#)
 - [Rotating IPsec Credentials](#)
 - [Installing IPsec Encryption](#)
- [Anti-Virus Scanning](#): Provides antivirus protection within the Tanzu Operations Manager environment. This protection is sometimes required for compliance purposes.

Troubleshooting and diagnostics for Tanzu Operations Manager

The following topics show you how to troubleshoot and diagnose issues you might encounter when installing or using Tanzu Operations Manager:

- [Troubleshooting problems in your Tanzu Operations Manager deployment](#)
- [Logging in to Tanzu Operations Manager VMs with SSH](#)
- [Advanced Tanzu Operations Manager troubleshooting with the BOSH CLI](#)
- [Troubleshooting BOSH Backup and Restore](#)
- [Troubleshooting Tanzu Operations Manager on Azure](#)
- [Troubleshooting Tanzu Operations Manager for VMware vSphere](#)

Troubleshooting problems in your Tanzu Operations Manager deployment

You can use the following procedures if you encounter issues while deploying products with Tanzu Operations Manager.

UAA token expires during product upload

During a product upload, the UAA tokens may expire if the upload takes longer than the token expiry time.

As a workaround, update the token expiry time to be longer than the product upload time.

To see or update the token expiry time, make a request to the [/api/v0/uaa/tokens_expiration](#) API endpoint.

See [Using the Tanzu Operations Manager API](#) for how to interact with the Tanzu Operations Manager API.

Retry the deployment

An installation or update can fail for many reasons, but the system is self healing, and can often automatically correct or work around hardware or network faults.

Click Install or Review Pending Changes, then click Apply Changes again, and the system might resolve a problem on its own.

Some failures only produce generic errors like [Exited with 1](#). In cases like this, where a failure is not accompanied by useful information, click Install or Review Pending Changes, then click Apply Changes to retry.

When the system does provide informative evidence, see [Common issues](#) to see if your problem is covered.

Common issues

Compare evidence that you have gathered to the following descriptions. If your issue is covered, try the recommended remediation procedures.

BOSH does not reinstall

You might want to reinstall BOSH for troubleshooting purposes. However, if Tanzu Operations Manager does not detect any changes, BOSH does not reinstall. To force a reinstall of BOSH, select BOSH Director, then Resource Sizes and change a resource value. For example, you can increase the amount of RAM by 4 MB. Making a small change like this allows BOSH to reinstall.

Creating bound missing VMs times out

This task happens immediately following package compilation, but before job assignment to agents. For example:

```
cloud_controller/0: Timed out pinging to f690db09-876c-475e-865f-2cece06aba79 after 600 seconds (00:10:24)
```

This is usually caused by a NATS issue with the affected VM. To identify a NATS issue, inspect the agent log for the VM. Since the BOSH Director is unable to reach the BOSH agent, you must access the VM using another method. You might be unable to access the VM using TCP. In this case, access the VM using your virtualization console.

To diagnose:

1. Access the VM using your virtualization console and log in.
2. Go to the Credentials tab of the Elastic Application Runtime tile and locate the VM to find the VM credentials.
3. Become root.
4. Run `cd /var/vcap/bosh/log`.
5. Open the file named `current`.
6. Determine whether the BOSH agent and director have successfully completed a handshake, represented in the logs as a “ping-pong:”

```
2013-10-03_14:35:48.58456 #[608] INFO: Message: {"method"=>"ping", "arguments"=>[], "reply_to"=>"director.f4b7df14-cb8f.19719508-e0dd-4f53-b755-58b6336058ab"}
```

```
2013-10-03_14:35:48.60182 #[608] INFO: reply_to: director.f4b7df14-cb8f.19719508-e0dd-4f53-b755-58b6336058ab: payload: {:value=>"pong"}
```

This handshake must finish successfully for the agent to receive instructions from the director.

7. If you do not see the handshake, look for another line near the beginning of the file, prefixed `INFO: loaded new infrastructure settings`. For example:

```
2013-10-03_14:35:21.83222 #[608] INFO: loaded new infrastructure settings:
{"vm"=>{"name"=>"vm-4d80ede4-b0a5-4992-aea6a0386e18e", "id"=>"vm-360"},
"agent_id"=>"56aea4ef-6aa9-4c39-8019-7024ccfdde4",
"networks"=>{"default"=>{"ip"=>"192.0.2.19",
"netmask"=>"255.255.255.0", "cloud_properties"=>{"name"=>"VMNetwork"},
"default"=>["dns", "gateway"]},
"dns"=>["192.0.2.2", "192.0.2.17"], "gateway"=>"192.0.2.2",
"dns_record_name"=>"0.nats.default.cf-d729343071061.microbosh",
"mac"=>"00:50:56:9b:71:67"}}, "disks"=>{"system"=>0, "ephemeral"=>1,
"persistent"=>{}}, "ntp"=>[], "blobstore"=>{"provider"=>"dav",
"options"=>{"endpoint"=>"http://192.0.2.17:25250",
"user"=>"agent", "password"=>"agent"}},
"mbus"=>"nats://nats:nats@192.0.2.17:4222",
"env"=>{"bosh"=>{"password"=>"$6$40ftQ9K4rvvC/8ADZHW0"}}}
```

This is a JSON blob of key-value pairs representing the expected infrastructure for the BOSH agent.

For this issue, the following section is the most important:

```
"mbus"=>"nats://nats:nats@192.0.2.17:4222"
```

This key-value pair represents where the agent expects the NATS server to be. Try pinging this NATS IP address from the VM to determine whether you are experiencing routing issues.

Install exits with a creates/updates/deletes app failure or with a 403 error

Scenario 1: Your Tanzu Operations Manager install exits with the following 403 error when you attempt to log in to the Apps Manager:

```
{"type": "step_finished", "id": "apps-manager.deploy"}

/home/tempest-web/tempest/web/vendor/bundle/ruby/1.9.1/gems/mechanize-2.7.2/lib/mechanize/http/agent.rb:306:in
`fetch': 403 => Net::HTTPForbidden for https://login.api.example.net/oauth/authorizeresponse_type=code&client_id=portal&redir
ect_uri=https%3...
-- unhandled response (Mechanize::ResponseCodeError)
```

Scenario 2: Your Tanzu Operations Manager install exits with a **creates/updates/deletes an app (FAILED - 1)** error message with the following stack trace:

```
1) App CRUD creates/updates/deletes an app
Failure/Error: Unable to find matching line from backtrace
CFoundry::TargetRefused:
Connection refused - connect(2)
```

In both of these scenarios, ensure that you have correctly entered your domains in wildcard format:

1. Browse to the fully qualified domain name (FQDN) of Tanzu Operations Manager.
2. Click the Elastic Application Runtime tile.
3. Click HAProxy, then Generate Self-Signed RSA Certificate.
4. Enter your system and app domains in wildcard format, as well as optionally any custom domains, and click Save. Refer to Elastic Application Runtime Cloud Controller for explanations of the domain values.

Install fails when Gateway instances exceed zero

If you configure the number of Gateway instances to be greater than zero for a given product, you create a dependency on Elastic Application Runtime for that product installation. If you attempt to install a product tile with a Elastic Application Runtime dependency before installing Elastic Application Runtime, the install fails.

To change the number of Gateway instances, click the product tile, then select Settings, then Resource sizes, followed by INSTANCES and change the value next to the product Gateway job.

To remove the Elastic Application Runtime dependency, change the value of this field to 0.

Out of disk space error

Tanzu Operations Manager displays an **Out of Disk Space** error if log files expand to fill all available disk space. If this happens, rebooting the Tanzu Operations Manager installation VM clears the tmp directory of these log files and resolves the error.

If you receive **Out of Disk Space** errors when trying to push apps, this can mean that Diego cells might be running out of disk space capacity.

To perform a detailed analysis of disk usage by containers and host VMs in your Tanzu Operations Manager deployment, see [Examining GrootFS disk usage](#).

Installing BOSH Director fails in vSphere

If the DNS information for the Tanzu Operations Manager VM is incorrectly specified when deploying the Tanzu Operations Manager .ova file, installing BOSH Director fails at the “Installing MicroBOSH” step.

To resolve this issue, correct the DNS settings in the Tanzu Operations Manager Virtual Machine properties and try again.

Deleting a deployment fails

Tanzu Operations Manager displays an error message when it cannot delete your installation. This scenario might happen if the BOSH Director cannot access the VMs or is experiencing other issues. To manually delete your installation and all VMs, do the following:

1. Use your IaaS dashboard to manually delete the VMs for all installed products, with the exception of the Tanzu Operations Manager VM.
2. SSH into your Tanzu Operations Manager VM and remove the **installation.yml** file from **/var/tempest/workspaces/default/**.

Deleting the **installation.yml** file does not prevent you from reinstalling a deployment. For future deploys, Tanzu Operations Manager regenerates this file when you click Save on any page in the BOSH Director tile.

Your installation is now deleted.

Installing Elastic Application Runtime fails

The following sections describe errors that may occur when installing Elastic Application Runtime and how to resolve them.

1. Elastic Application Runtime Cannot verify app push

If the DNS information for the Tanzu Operations Manager VM becomes incorrect after BOSH Director has been installed, installing Elastic Application Runtime fails at the “Verifying app push” step.

To resolve this issue, correct the DNS settings in the Tanzu Operations Manager Virtual Machine properties.

2. MySQL Monitor not running after update

When MySQL cannot communicate with UAA, Tanzu Operations Manager shows the following error:

```
Error: 'mysql_monitor/12a3b456-cd7e-8fgh-9012-345678b90ijk (0)' is not running after update. Review logs for failed jobs: replication-canary.
```

If you see this error, create firewall rules that allow MySQL to reach UAA, using [MySQL Network Communications](#) as a reference.

Tanzu Operations Manager hangs during MicroBOSH install or HAProxy states “IP Address Already Taken”

During a Elastic Application Runtime installation, you might see the following errors:

- The Tanzu Operations Manager GUI shows that the installation stops at the “Setting MicroBOSH deployment manifest” task.
- When you set the IP address for the HAProxy, the “IP Address Already Taken” message appears.

When you install the BOSH Director, you assign it an IP address. Tanzu Operations Manager then takes the next two consecutive IP addresses, assigns the first to MicroBOSH, and reserves the second. For example:

```
203.0.113.1 - Tanzu Operations Manager (User assigned)
203.0.113.2 - MicroBOSH (Tanzu Operations Manager assigned)
203.0.113.3 - Reserved (Tanzu Operations Manager reserved)
```

To resolve this issue, ensure that the next two subsequent IP addresses from the manually assigned address are unassigned.

Poor performance

If you notice poor network performance by your deployment and your deployment uses a Network Address Translation (NAT) gateway, your NAT gateway may be under-resourced.

1. Troubleshoot: To troubleshoot the issue, set a custom firewall rule in your IaaS console to route traffic originating from your private network directly to an S3-compatible object store. If you see decreased average latency and improved network performance, perform the solution below to scale up your NAT gateway.

1. Scale up your NAT gateway: Complete the following steps to scale up your NAT gateway:

1. Go to your IaaS console.
2. Spin up a new NAT gateway of a larger VM size than your previous NAT gateway.
3. Change the routes to direct traffic through the new NAT gateway.
4. Spin down the old NAT gateway.

The specific procedures vary depending on your IaaS. Consult your IaaS documentation for more information.

Common issues caused by firewalls

This section describes various issues you might encounter when installing Tanzu Operations Manager in an environment that uses a strong firewall.

DNS resolution fails

When you install Tanzu Operations Manager in an environment that uses a strong firewall, the firewall might block DNS resolution. To resolve this issue, see [Verify Tanzu Operations Manager resolves DNS Entries behind a firewall](#) in *Preparing Your Firewall*.

Logging into Tanzu Operations Manager VMs with SSH

To troubleshoot your Tanzu Operations Manager deployments, you might need to SSH into the Tanzu Operations Manager VM or BOSH Director VM. You can use SSH to connect to the Tanzu Operations Manager VM or the BOSH Director VM.

After you successfully SSH into the Tanzu Operations Manager VM, you can perform troubleshooting operations. For more information about troubleshooting, see [Advanced troubleshooting with the BOSH CLI](#).

Log in to Tanzu Operations Manager VM with SSH

Use SSH to connect to the Tanzu Operations Manager VM. To log in to the Tanzu Operations Manager VM, go to the procedure for your IaaS:

- [AWS](#)
- [Azure](#)
- [GCP](#)
- [vSphere](#)

AWS

To log in to the Tanzu Operations Manager VM with SSH in AWS, you need the key pair you used when you created the Tanzu Operations Manager VM. To see the name of the key pair, click the Tanzu Operations Manager VM and locate the **key pair name** in properties.

To log in to the Tanzu Operations Manager VM with SSH in AWS:

1. Locate the Tanzu Operations Manager FQDN on the AWS EC2 instances page.
2. Run `chmod 600 ops_mgr.pem` to change the permissions on the `.pem` file to be more restrictive. For example:

```
chmod 600 ops_mgr.pem
```

3. Log in to the Tanzu Operations Manager VM with SSH.

```
ssh -i ops_mgr.pem ubuntu@FQDN
```

Where **FQDN** is the fully qualified domain name of Tanzu Operations Manager.

For example:

```
$ ssh -i ops_mgr.pem ubuntu@my-opsmanager-fqdn.example.com
```


Azure

To log in to the Tanzu Operations Manager VM with SSH in Azure, you need the key pair you used when creating the Tanzu Operations Manager VM. If you need to reset the SSH key, locate the Tanzu Operations Manager VM in the Azure portal and click Reset Password.

To log in to the Tanzu Operations Manager VM with SSH in Azure:

1. From the Azure portal, locate the Tanzu Operations Manager FQDN by selecting the VM.
2. Change the permissions for your SSH private key by running the following command:

```
chmod 600 PRIVATE-KEY
```

Where **PRIVATE-KEY** is the name of your SSH private key.

3. SSH into the Tanzu Operations Manager VM and run:

```
ssh -i PRIVATE-KEY ubuntu@FQDN
```

Where:

- **FQDN** is the FQDN for your Tanzu Operations Manager deployment.
- **PRIVATE-KEY** is the name of your SSH private key.

GCP

To log in to the Tanzu Operations Manager VM with SSH in GCP:

1. Confirm that you have installed the [Google Cloud SDK and CLI](#). For more information, see the [Google Cloud Platform documentation](#).
2. Initialize Google Cloud CLI, using a user account with Owner, Editor, or Viewer permissions to access the project. Ensure that the Google Cloud CLI can log in to the project by running the command `gcloud auth login`.
3. From the GCP web console, go to Compute Engine.
4. Locate the Tanzu Operations Manager VM in the VM Instances list.
5. Under Remote access, click the SSH drop-down menu and click View gcloud command.
6. Copy the SSH command that appears in the pop-up window.
7. Paste the command into your terminal window to SSH to the VM. For example:

```
$ gcloud compute ssh "YOUR-VM" --zone "YOUR-ZONE-ID"
```

8. Run `sudo su - ubuntu` to switch to the `ubuntu` user.

vSphere

To log in to the Tanzu Operations Manager VM with SSH in vSphere, you must have the public SSH key that imports the Tanzu Operations Manager `.ova` or `.ovf` file into your virtualization system.

You set the public SSH key in the Public SSH Key text box of the Customize template screen when you deployed Tanzu Operations Manager. For more information, see [Deploy Tanzu Operations Manager in Deploying Tanzu Operations Manager on vSphere](#).

If you lose your SSH key, you must shut down the Tanzu Operations Manager VM in the vSphere UI and then reset the public SSH key. For more information, see [Edit vApp Settings](#) in the vSphere

documentation.

To log in to the Tanzu Operations Manager VM with SSH in vSphere:

1. Run the following command:

```
ssh ubuntu@FQDN
```

Where **FQDN** is the fully qualified domain name of Tanzu Operations Manager.

For example:

```
$ ssh ubuntu@my-fqdn.example.com
```

2. When you are prompted, enter the public SSH key.

Log in to the BOSH Director VM with SSH

Do the following steps to log in to the BOSH Director VM with SSH:

1. In Tanzu Operations Manager, open the BOSH Director tile.
2. Click the Credentials tab.
3. Next to Bbr Ssh Credentials, click Link to Credential. A tab opens containing a JSON credential structure.
4. Copy the **RSA PRIVATE KEY** and paste it into a file named **bbr.pem**. Include **-----BEGIN RSA PRIVATE KEY-----** and **-----END RSA PRIVATE KEY-----**.

Keep the key secure. The key provides full access to the entire Tanzu Operations Manager environment.

5. Replace all **\n** characters in **bbr.pem** with a line break.
6. Copy **bbr.pem** to the **~/.ssh/** directory on your computer.
7. Run **chmod 600 ~/.ssh/bbr.pem** to modify the permissions of the file.
8. Log in to the BOSH Director VM with SSH from your machine.

```
ssh bbr@BOSH-DIRECTOR-IP -i ~/.ssh/bbr.pem
```

If you use GCP, ensure SSH port **22** is open for the BOSH Director VM in your GCP console. If the SSH port is not open, open it by creating a firewall rule.

9. Run **sudo -i** to get the root privilege.

Advanced Tanzu Operations Manager troubleshooting with the BOSH CLI

You can use the BOSH CLI to help diagnose and resolve issues with your Tanzu Operations Manager deployment. Before using the information and techniques in this topic, review [Diagnosing Elastic Application Runtime deployment problems](#).

To follow the steps in this topic, you must log in to the BOSH Director VM. The BOSH Director runs on the virtual machine (VM) that Tanzu Operations Manager deploys on the first install of the BOSH Director tile.

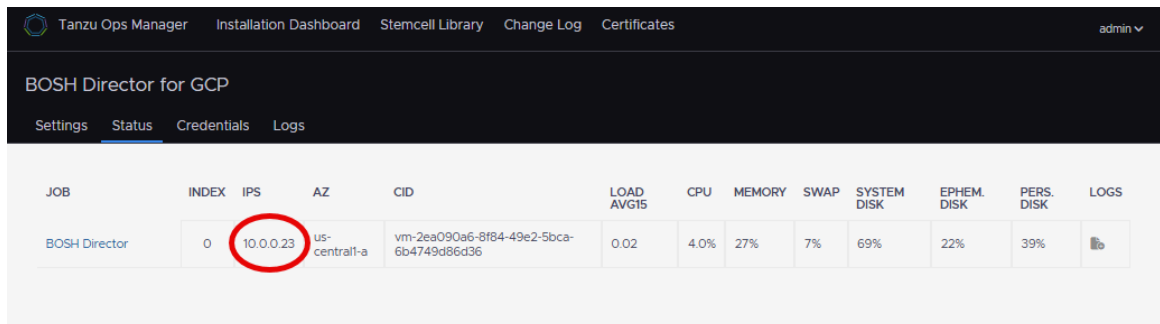
After you authenticate in to the BOSH Director, you can run specific commands using the BOSH Command Line Interface (BOSH CLI). BOSH Director diagnostic commands have access to information about your entire Tanzu Operations Manager installation.

Before you run any BOSH CLI commands, verify that no BOSH Director tasks are running on the Tanzu Operations Manager VM. For more information, see [Tasks](#) in *BOSH CLI commands* in the BOSH documentation.

Gather credential and IP address information

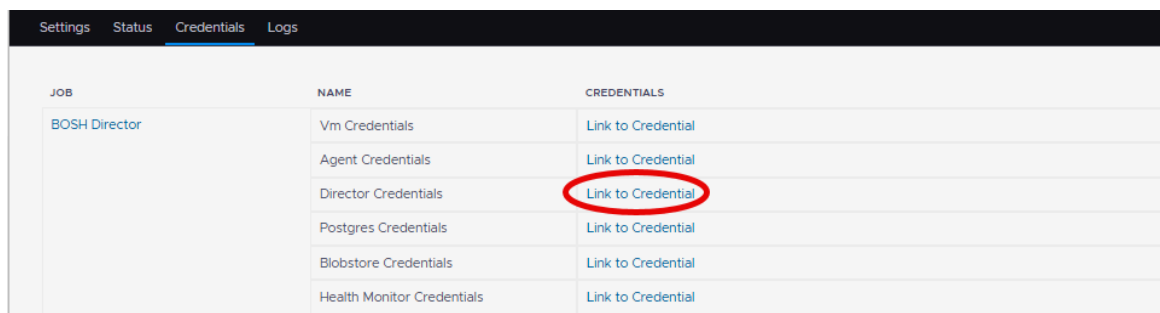
Before you begin troubleshooting with the BOSH CLI, follow these instructions to collect the information you need from the Tanzu Operations Manager interface:

1. Open the Tanzu Operations Manager interface by going to the Tanzu Operations Manager fully qualified domain name (FQDN) in a web browser.
2. Click the BOSH Director tile and click the Status tab.
3. Record the IP address for the Director job. This is the IP address of the VM where the BOSH Director runs.



| Tanzu Ops Manager Installation Dashboard Stemcell Library Change Log Certificates admin | | | | | | | | | | | | | | |
|---|-------|-----------|---------------|---|------------|------|--------|------|-------------|-------------|------------|------|--|--|
| BOSH Director for GCP | | | | | | | | | | | | | | |
| Settings Status Credentials Logs | | | | | | | | | | | | | | |
| JOB | INDEX | IPS | AZ | CID | LOAD AVG15 | CPU | MEMORY | SWAP | SYSTEM DISK | EPHEM. DISK | PERS. DISK | LOGS | | |
| BOSH Director | 0 | 10.0.0.23 | us-central1-a | vm-2ea090a6-8f84-49e2-5bca-6b4749d86d36 | 0.02 | 4.0% | 27% | 7% | 69% | 22% | 39% | | | |

4. Click the Credentials tab.
5. Go to Director Credentials and click Link to Credential. Record these credentials.



| Settings Status Credentials Logs | | |
|----------------------------------|----------------------------|------------------------------------|
| JOB | NAME | CREDENTIALS |
| BOSH Director | Vm Credentials | Link to Credential |
| | Agent Credentials | Link to Credential |
| | Director Credentials | Link to Credential |
| | Postgres Credentials | Link to Credential |
| | Blobstore Credentials | Link to Credential |
| | Health Monitor Credentials | Link to Credential |

6. Return to the Tanzu Operations Manager Installation Dashboard.
7. (Optional) To prepare to troubleshoot the job VM for any other product, click the product tile and repeat the previous procedure to record the IP address and VM credentials for that job VM.
8. Log out of Tanzu Operations Manager.

Ensure that there are no Tanzu Operations Manager installations or updates in progress while using the BOSH CLI.

Log in to the Tanzu Operations Manager VM with SSH

Use SSH to connect to the Tanzu Operations Manager VM. To log in to the Tanzu Operations Manager VM, go to the procedure for your IaaS:

- [AWS](#)
- [Azure](#)
- [GCP](#)
- [vSphere](#)

AWS

To log in to the Tanzu Operations Manager VM with SSH in AWS, you need the key pair you used when you created the Tanzu Operations Manager VM. To see the name of the key pair, click the Tanzu Operations Manager VM and locate the **key pair name** in properties.

To log in to the Tanzu Operations Manager VM with SSH in AWS:

1. Locate the Tanzu Operations Manager FQDN on the AWS EC2 instances page.
2. Run `chmod 600 ops_mgr.pem` to change the permissions on the `.pem` file to be more restrictive. For example:

```
chmod 600 ops_mgr.pem
```

3. Log in to the Tanzu Operations Manager VM with SSH.

```
ssh -i ops_mgr.pem ubuntu@FQDN
```

Where **FQDN** is the fully qualified domain name of Tanzu Operations Manager.

For example:

```
$ ssh -i ops_mgr.pem ubuntu@my-opsmanager-fqdn.example.com
```

Azure

To log in to the Tanzu Operations Manager VM with SSH in Azure, you need the key pair you used when creating the Tanzu Operations Manager VM. If you need to reset the SSH key, locate the Tanzu Operations Manager VM in the Azure portal and click Reset Password.

To log in to the Tanzu Operations Manager VM with SSH in Azure:

1. From the Azure portal, locate the Tanzu Operations Manager FQDN by selecting the VM.
2. Change the permissions for your SSH private key by running the following command:

```
chmod 600 PRIVATE-KEY
```

Where **PRIVATE-KEY** is the name of your SSH private key.

3. SSH into the Tanzu Operations Manager VM and run:

```
ssh -i PRIVATE-KEY ubuntu@FQDN
```

Where:

- **FQDN** is the FQDN for your Tanzu Operations Manager deployment.
- **PRIVATE-KEY** is the name of your SSH private key.

GCP

To log in to the Tanzu Operations Manager VM with SSH in GCP:

1. Confirm that you have installed the [Google Cloud SDK and CLI](#). For more information, see the [Google Cloud Platform documentation](#).
2. Initialize Google Cloud CLI, using a user account with Owner, Editor, or Viewer permissions to access the project. Ensure that the Google Cloud CLI can log in to the project by running the command `gcloud auth login`.
3. From the GCP web console, go to Compute Engine.
4. Locate the Tanzu Operations Manager VM in the VM Instances list.
5. Under Remote access, click the SSH drop-down menu and click View gcloud command.
6. Copy the SSH command that appears in the pop-up window.
7. Paste the command into your terminal window to SSH to the VM. For example:

```
$ gcloud compute ssh "YOUR-VM" --zone "YOUR-ZONE-ID"
```

8. Run `sudo su - ubuntu` to switch to the `ubuntu` user.

vSphere

To log in to the Tanzu Operations Manager VM with SSH in vSphere, you must have the public SSH key that imports the Tanzu Operations Manager `.ova` or `.ovf` file into your virtualization system.

You set the public SSH key in the Public SSH Key text box of the Customize template screen when you deployed Tanzu Operations Manager. For more information, see [Deploy Tanzu Operations Manager in Deploying Tanzu Operations Manager on vSphere](#).

If you lose your SSH key, you must shut down the Tanzu Operations Manager VM in the vSphere UI and then reset the public SSH key. For more information, see [Edit vApp Settings](#) in the vSphere documentation.

To log in to the Tanzu Operations Manager VM with SSH in vSphere:

1. Run the following command:

```
ssh ubuntu@FQDN
```

Where `FQDN` is the fully qualified domain name of Tanzu Operations Manager.

For example:

```
$ ssh ubuntu@my-fqdn.example.com
```

2. When you are prompted, enter the public SSH key.

Authenticate with the BOSH Director VM

To authenticate with BOSH, use one of the following methods:

- [Set the BOSH Environment Variables on the Tanzu Operations Manager VM.](#)
- [Create a Local BOSH Director Alias.](#)

Set the BOSH environment variables on the Tanzu Operations Manager VM

If you have access to the Tanzu Operations Manager VM, SSH into the Tanzu Operations Manager VM and do the following:

1. Record the Bosh Commandline Credentials from the Credentials tab of the BOSH Director tile.
2. SSH into the Tanzu Operations Manager VM. For more information, see [Log in to the Tanzu Operations Manager VM with SSH](#).
3. Export all the environment variables by running the following command:

```
export YOUR-ENV-VARIABLES
```

Where **YOUR-ENV-VARIABLES** is the value for **credential** in the BOSH command line credentials that you recorded in a previous step.

For example:

```
$ export BOSH_CLIENT=ops_manager \
BOSH_CLIENT_SECRET=some_secret \
BOSH_CA_CERT=/var/tempest/workspaces/default/root_ca_certificate \
BOSH_ENVIRONMENT=10.0.0.5 bosh
```

4. Verify that BOSH access works by running the following command:

```
bosh deployments
```

Create a local BOSH Director alias

To create a BOSH Director alias and log in to the BOSH Director VM:

1. Run the following command to create a local alias for the BOSH Director using the BOSH CLI:

```
bosh alias-env MY-ENV -e DIRECTOR-IP-ADDRESS --ca-cert /var/tempest/workspaces/default/root_ca_certificate
```

Where:

- **MY-ENV**: Enter an alias for the BOSH Director, such as **gcp**.
- **DIRECTOR-IP-ADDRESS**: Enter the IP address of your BOSH Director VM.

For example:

```
$ bosh alias-env gcp -e 10.0.0.3 --ca-cert /var/tempest/workspaces/default/root_ca_certificate
```

2. Log in to the BOSH Director VM using one of the following options:
 - [Internal User Store Login through UAA](#): Log in to the BOSH Director VM using BOSH.
 - [External User Store Login through SAML](#): Use an external user store to log in to the BOSH Director VM.

Log in to the BOSH Director VM with UAA

1. Retrieve the Director user name and password in one of the following ways:
 - In Tanzu Operations Manager, click the BOSH Director, select the Credentials tab, and click the link to Director Credentials.

- Browse to https://FQDN/api/v0/deployed/director/credentials/director_credentials to obtain the password, where **FQDN** is the fully qualified domain name of Tanzu Operations Manager.

2. Log in to the BOSH Director VM:

```
bosh -e MY-ENV log-in
```

Where **MY-ENV** is the alias for your BOSH Director.

For example:

```
$ bosh -e gcp log-in
```

Follow the BOSH CLI prompts and enter the BOSH Director credentials to log in to the BOSH Director VM.

Log in to the BOSH Director VM with SAML

1. Log in to your identity provider and use the following information to configure SAML Service Provider Properties:

- Service Provider Entity ID: **bosh-uaa**
- ACS URL: **<https://DIRECTOR-IP-ADDRESS:8443/saml/SSO/alias/bosh-uaa>**
- Binding: HTTP Post
- SLO URL: **<https://DIRECTOR-IP-ADDRESS:8443/saml/SSO/alias/bosh-uaa>**
- Binding: HTTP Redirect
- Name ID: Email Address

2. Log in to the BOSH Director VM:

```
bosh -e MY-ENV log-in
```

Where **MY-ENV** is the alias for your BOSH Director.

For example:

```
$ bosh -e gcp log-in
```

Follow the BOSH CLI prompts and enter your SAML credentials to log in to the BOSH Director VM.

Your browser must be able to reach the BOSH Director to log in with SAML.

3. Click Log in with organization credentials (SAML). A temporary authentication code appears.
4. Copy the Temporary Authentication Code that appears in your browser.
5. You see a login confirmation. For example:

```
Logged in as admin@example.org
```

SSH into the BOSH Director VM

Do the following steps to log in to the BOSH Director VM with SSH:

1. In Tanzu Operations Manager, open the BOSH Director tile.

2. Click the Credentials tab.
3. Next to Bbr Ssh Credentials, click Link to Credential. A tab opens containing a JSON credential structure.
4. Copy the **RSA PRIVATE KEY** and paste it into a file named **bbr.pem**. Include **-----BEGIN RSA PRIVATE KEY-----** and **-----END RSA PRIVATE KEY-----**.

Keep the key secure. The key provides full access to the entire Tanzu Operations Manager environment.

5. Replace all **\n** characters in **bbr.pem** with a line break.
6. Copy **bbr.pem** to the **~/ssh/** directory on your computer.
7. Run **chmod 600 ~/ssh/bbr.pem** to modify the permissions of the file.
8. Log in to the BOSH Director VM with SSH from your machine.

```
ssh bbr@BOSH-DIRECTOR-IP -i ~/ssh/bbr.pem
```

If you use GCP, ensure SSH port **22** is open for the BOSH Director VM in your GCP console. If the SSH port is not open, open it by creating a firewall rule.

9. Run **sudo -i** to get the root privilege.

Use the BOSH CLI for troubleshooting

This section describes five BOSH CLI commands commonly used during troubleshooting.

- **VMs**: Lists the VMs in a deployment
- **Cloud Check**: Runs a cloud consistency check and interactive repair
- **Create Recovery Plan and Recover**: Interactively create a plan to use in repairing resources
- **SSH**: Starts an interactive session or executes commands with a VM

BOSH VMs

The **bosh vms** command provides an overview of the virtual machines that BOSH manages.

- To see an overview of all virtual machines managed by BOSH, run:

```
bosh -e MY-ENV vms
```

- To see only the virtual machines associated with a particular deployment, run:

```
bosh -e MY-ENV -d MY-DEPLOYMENT vms
```

Where **MY-ENV** is your environment, and, if using the **-d** flag, **MY-DEPLOYMENT** is the name of a deployment

When you troubleshoot an issue with your deployment, **bosh vms** might show a VM in an unknown state. Run **bosh cloud-check** on a VM in an unknown state to instruct BOSH to diagnose problems with the VM.

You can also run **bosh vms** to identify VMs in your deployment, then use the **bosh ssh** command to log in to an identified VM with SSH for further troubleshooting.

bosh vms supports the following arguments:

- `--dns`: Report includes the DNS A record for each VM.
- `--vitals`: Report also includes load, CPU, memory usage, swap usage, system disk usage, ephemeral disk usage, and persistent disk usage for each VM.

The Status tab of the Elastic Application Runtime tile shows information similar to the `bosh vms` output.

BOSH Cloud check

Run the `bosh cloud-check` command to instruct BOSH to detect differences between the VM state database maintained by the BOSH Director and the actual state of the VMs. For each difference detected, `bosh cloud-check` can offer the following repair options:

- **Reboot VM**: Instructs BOSH to reboot a VM. Rebooting can resolve many transient errors.
- **Ignore problem**: Instructs BOSH to do nothing. You might want to ignore a problem and instead run `bosh ssh` to try to troubleshoot directly on the machine.
- **Reassociate VM with corresponding instance**: Updates the BOSH Director state database. Use this option if you believe that the BOSH Director state database is in error and that a VM is correctly associated with a job.
- **Recreate VM using last known apply spec**: Instructs BOSH to destroy the server and recreate it from the deployment manifest that the installer provides. Use this option if a VM is corrupted.
- **Delete VM reference**: Instructs BOSH to delete a VM reference in the Director state database. If a VM reference exists in the state database, BOSH expects to find an agent running on the VM. Select this option only if you know that this reference is in error. After you delete the VM reference, BOSH can no longer control the VM.

To use this command, run `bosh -e MY-ENV -d MY-DEPLOYMENT cloud-check`, where `MY-ENV` is your environment, and `MY-DEPLOYMENT` is your deployment.

Example scenarios

Here are some example scenarios to help you decide what actions you can take to resolve the problem.

Unresponsive Agent

```
$ bosh -e example-env -d example-deployment cloud-check
```

ccdb/0 (vm-3e37133c-bc33-450e-98b1-f86d5b63502a) is not responding:

- Ignore problem
- Reboot VM
- Recreate VM using last known apply spec
- Delete VM reference (DANGEROUS!)

Missing VM

```
$ bosh -e example-env -d example-deployment cloud-check
```

VM with cloud ID `vm-3e37133c-bc33-450e-98b1-f86d5b63502a` is missing:

- Ignore problem
- Recreate VM using last known apply spec
- Delete VM reference (DANGEROUS!)

Unbound Instance VM

```
$ bosh -e example-env -d example-deployment cloud-check
```

VM `vm-3e37133c-bc33-450e-98b1-f86d5b63502a` reports itself as `ccdb/0` but does not have a bound instance:

- Ignore problem
- Delete VM (unless it has persistent disk)
- Reassociate VM with corresponding instance

Out of Sync VM

```
$ bosh -e example-env -d example-deployment cloud-check
```

VM `vm-3e37133c-bc33-450e-98b1-f86d5b63502a` is out of sync: expected `cf-d7293430724a2c421061: ccdb/0`, got `cf-d7293430724a2c421061: nats/0`:

- Ignore problem
- Delete VM (unless it has persistent disk)

BOSH Create Recovery Plan and Recover

If underlying infrastructure issues cause widespread problems with your deployment, run the `bosh create-recovery-plan` and `bosh recover` commands. The `bosh create-recovery-plan` command scans the deployment for problems and then prompts you to generate a recovery plan, which is saved to a file. The `bosh recover` command consumes the recovery plan file.

These commands together work similarly to `bosh cloud-check`, with the following differences:

- You select resolutions to problems by instance group and problem type. The `bosh cloud-check` command asks for a resolution for each particular problem.
- You can override `max_in_flight` per instance group to speed deployment recovery. The `bosh cloud-check` command uses the `max_in_flight` values in the deployment manifest.

To use these commands:

1. Run `bosh -e MY-ENV -d MY-DEPLOYMENT create-recovery-plan MY-RECOVERY-PLAN`, where:
 - `MY-ENV` is your environment
 - `MY-DEPLOYMENT` is your deployment
 - `MY-RECOVERY-PLAN` is a filename for the recovery plan

For example:

```
$ bosh -e example-env -d example-deployment create-recovery-plan example-recovery-plan.yml
```

2. Run `bosh -e MY-ENV -d MY-DEPLOYMENT recover MY-RECOVERY-PLAN`, where:
 - `MY-ENV` is your environment
 - `MY-DEPLOYMENT` is your deployment
 - `MY-RECOVERY-PLAN` is the filename you provided to the `bosh create-recovery-plan` command

For example:

```
$ bosh -e example-env -d example-deployment recover example-recovery-plan.yml
```

For more information about the `bosh create-recovery-plan` and `bosh-recover` commands, see the [BOSH documentation](#).

BOSH SSH

Use `bosh ssh` to log in to the VMs in your deployment with SSH.

To use `bosh ssh`:

1. Identify a VM to log in to with SSH. Run the following command to list the VMs in the given deployment:

```
bosh -e MY-ENV -d MY-DEPLOYMENT vms
```

Where:

- `MY-ENV` is your environment alias.
- `MY-DEPLOYMENT` is the deployment name.

2. Run:

```
bosh -e MY-ENV -d MY-DEPLOYMENT ssh VM-NAME/GUID
```

For example:

```
$ bosh -e example-env -d example-deployment ssh diego-cell/abcd0123-a012-b345-c678-9def01234567
```

Troubleshooting BOSH Backup and Restore

This topic explains how to solve common problems that you might encounter when using BOSH Backup and Restore (BBR) to back up and restore your deployments.

Troubleshooting during a restore

This section provides solutions for errors that occur during a restore.

Problem: Restore fails with a MySQL Monit start timeout

Symptom

While running the BBR restore command, restoring the job `mysql-restore` fails with the following error:

```
1 error occurred:

* restore script for job mysql-restore failed on mysql/0.
...
"Monit start failed: Timed out waiting for monit": 2m0s
```

Explanation

This happens when the MySQL job fails to start within the timeout period. It ends up in an Execution Failed state and `monit` never tries to start it again.

Ensure that your MySQL Server cluster has only one instance. If there is more than one instance of MySQL Server, the restore fails with a **monit start** timeout. Scale down to one instance and retry.

If your MySQL Server cluster is already scaled down to one node, it might have taken longer than normal to restart the cluster. To manually verify and retry:

1. List the VMs in your deployment by running:

```
bosh -e BOSH-DIRECTOR-IP --ca-cert /var/tempest/workspaces/default/root_ca_certificate \
-d DEPLOYMENT-NAME \
ssh
```

Where:

- **BOSH-DIRECTOR-IP** is the BOSH Director IP. To retrieve this IP address, see [Retrieve BOSH Director Address](#) in *Restoring Deployments from Backup with BBR*.
 - **DEPLOYMENT-NAME** is the deployment name. To retrieve the deployment name, see [Identify your Deployment](#) in *Restoring Deployments from Backup with BBR*.
2. Locate the **mysql** VM and SSH into it.
 3. To check that the MySQL job process is running, run:

```
ps aux | grep galera-init
```

For more information, see [Configuring Elastic Application Runtime](#).

4. Check that **monit** reports that the MySQL job process is in an Execution Failed state by running:

```
sudo monit summary
```

5. If so, deactivate monitoring by running:

```
monit unmonitor
```

6. Enable monitoring by running:

```
monit monitor
```

7. After a few minutes, run:

```
monit summary
```

The command reports that all the processes are running.

8. Attempt the restore with BBR again.

Problem: Deployment does not match the structure of the backup

| | |
|-------------|--|
| Symptom | The following error is displayed: |
| | Deployment 'deployment-name' does not match the structure of the provided backup. |
| Explanation | The instance groups with the restore scripts in the destination environment do not match the backup metadata. For example, they might have the wrong number of instances of a particular instance group, or the metadata names an instance group that does not exist in the destination environment. |
| Solution | BBR only supports restoring to an environment that matches your original environment. VMware recommends altering the destination environment to match the structure of the backup. |

General troubleshooting

This section provides solutions for general errors.

Problem: Connection error

| | |
|-------------|---|
| Symptom | BBR displays an error message containing "SSH Dial Error" or another connection error. |
| Explanation | The jumpbox and the VMs in the deployment are experiencing connection problems. |
| Solution | <p>To resolve connection problems between the jumpbox and the VMs in the deployment:</p> <ol style="list-style-type: none"> 1. Ensure your deployment is healthy by running: <div data-bbox="604 1028 702 1052" data-label="Text"> <pre>bosh vms</pre> </div> 2. Clean up the data from the failed backup on the instances by running the following. This step is required. Otherwise, further BBR commands fail. <div data-bbox="604 1196 916 1223" data-label="Text"> <pre>bbr deployment backup-cleanup</pre> </div> 3. Repeat the BBR operation. |

Problem: Error running metadata script

| | |
|-------------|---|
| Symptom | BBR backup or restore fails with a metadata error: |
| | <div data-bbox="509 1541 1394 1662" data-label="Text"> <pre>1 error occurred: error 1: An error occurred while running metadata script for job redis-server on redis/0ce9f81f-1756-480b-8e3e-a4609b14b6a6: error from metadata</pre> </div> |
| Explanation | There is a problem with your deployment install. |
| Solution | Contact Broadcom Support . |

Troubleshooting Tanzu Operations Manager on Azure

This topic describes how you can troubleshoot known issues when deploying or running Tanzu Operations Manager on Azure.

Installation issues

This section describes how to troubleshoot known installation issues.

Problem: Cannot copy the Tanzu Operations Manager image

| | |
|-------------|---|
| Symptom | You cannot copy the Tanzu Operations Manager image into your storage account as part of Step 4: Boot Tanzu Operations Manager in <i>Deploying Tanzu Operations Manager on Azure</i> . |
| Explanation | <p>You have an outdated version of the Azure CLI. You need Azure CLI v2.0.0 or later. To display your current Azure CLI version, run:</p> <pre>az --version</pre> |
| Solution | Install the Azure CLI v2.0 by following the instructions for your operating system in Install the Azure CLI in the Microsoft Azure documentation. |

Problem: Deployment fails at “create-env”

| | |
|-------------|--|
| Symptom | <p>After you click Review Pending Changes, then Apply Changes to install Tanzu Operations Manager and Tanzu Platform for Cloud Foundry, the deployment fails at <code>create-env</code> with an error message similar to the following:</p> <pre>Command 'deploy' failed: Deploying: Creating instance 'bosh/0': Waiting until instance is ready: Starting SSH tunnel: Parsing private key file '/tmp/bosh_ec2_private_key.pem': asn1: structure error: tags don't match (16 vs {class:3 tag:28 length:127 isCompound:false}) {optional:false explicit:false application:false defaultValue: tag: stringType:0 timeType:0 set:false omitEmpty:false} pkcs1PrivateKe y @2 ===== 2016-09-29 16:28:22 UTC Finished "bosh create-env" /var/tempest/workspaces/default/deployments/bosh.yml"; Duration: 328s; Exit Status: 1 Exited with 1.</pre> |
| Explanation | You provided a passphrase when you created your key pair in Step 4: Boot Tanzu Operations Manager in <i>Deploying Tanzu Operations Manager on Azure</i> . |
| Solution | Create a new key pair with no passphrase and do the installation again. For more information, see Step 4: Boot Tanzu Operations Manager in <i>Deploying Tanzu Operations Manager on Azure</i> . |

Insufficient external database permissions

Upgrade issues can arise when the external database user that was used for the network policy database is given insufficient permissions. To avoid this upgrade issue, ensure that the Network Policy Server database user has the `ALL PRIVILEGES` permission.

Operational issues

This section describes how to troubleshoot known operational issues.

Problem: Slow performance or timeouts

| | |
|-------------|---|
| Symptom | Developers suffer from slow performance or timeouts when pushing or managing apps, and end users suffer from slow performance or timeouts when accessing apps. |
| Explanation | The Azure Load Balancer (ALB) disconnects active TCP connections when they have been idle for more than four minutes. |
| Solution | <p>To mitigate slow performance or timeouts, the default value of the Back end request and idle timeout field is set to 900 seconds, but operators can tune this parameter to fit their infrastructure.</p> <p>To edit the Back end request and idle timeout field:</p> <ol style="list-style-type: none"> 1. Go to Tanzu Operations Manager Installation Dashboard. 2. Click the Elastic Application Runtime tile. 3. Click Networking. 4. Enter your desired time, in seconds, in the Back end request and idle timeout text box. 5. Click Save. |

Problem: Service instance creation times out

| | |
|-------------|---|
| Symptom | You are unable to provision a service instance of a Java or Go service. Running <code>cf create-service</code> fails with error <code>Failure provisioning service instance... Timed out after 8 minutes...</code> or similar. |
| Explanation | HTTP libraries for Java and Go, running with default settings, prune connections that have been idle for 240 seconds from their connection pool without sending a TCP reset message back to the client service broker. This removes the ability of the broker to provision new service instances. |
| Solution | In the Networking pane of the Elastic Application Runtime tile, enter a value of 240 or less in the Front end idle timeout field, so that Tanzu Operations Manager regenerates the front end connection before it times out. |

Troubleshooting Tanzu Operations Manager for VMware vSphere

You can diagnose and resolve issues that can occur in Tanzu Operations Manager deployments on VMware vSphere. For infrastructure-agnostic troubleshooting help, see [Diagnosing Elastic Application Runtime deployment problems](#).

The following sections list common issues that you might encounter with Tanzu Operations Manager and possible resolutions.

Tanzu Operations Manager installation fails

If you modify the vCenter Statistics Interval Duration setting from its default setting of five minutes, the Tanzu Operations Manager installation might fail at the MicroBOSH deployment stage, and the logs might contain the following error message:

```
The specified parameter is not correct, interval
```

This failure happens because Tanzu Operations Manager expects a default value of five minutes, and the call to this method fails when the retrieved value does not match the expected default value.

To resolve this issue, launch vCenter, go to Administration, then vCenter Server Settings, followed by Statistics, and reset the vCenter Statistics Interval Duration setting to five minutes.

BOSH automated installation fails

Before you start an Tanzu Platform for Cloud Foundry deployment, you must set up and configure a vSphere cluster.

If you enable vSphere DRS (Distributed Resource Scheduler) for the cluster, you must set the Automation level to Partially automated or Fully automated.

If you set the Automation level to Manual, the BOSH automated installation fails with a `power_on_vm` error when BOSH attempts to create virtual VMs.

Tanzu Operations Manager loses IP address after HA or reboot

Tanzu Operations Manager can lose its IP address and use DHCP due to an issue in the open source version of VMware Tools. For more information to troubleshoot this issue, see [IP is not Getting Assigned in vSphere after Tanzu Operations Manager Restart](#).

Cannot connect to the OVF in a browser

If you deployed the OVF file but cannot connect to it in a browser, check that the network settings you entered in the wizard are correct.

1. Access the Tanzu Operations Manager installation VM using the vSphere Console. If your network settings are configured incorrectly, you cannot SSH into the installation VM.
2. Log in using the private SSH key associated with the public SSH key that you provided when you imported the Tanzu Operations Manager `.ova` in vCenter.
3. Confirm that the network settings are correct by checking that the `ADDRESS`, `NETMASK`, `GATEWAY`, and `DNS-NAMESERVERS` entries are correct in `/etc/network/interfaces`.
4. If any of the settings are incorrect, run `sudo vi /etc/network/interfaces` and correct the incorrect entries.
5. In vSphere, go to the Summary tab for the VM and confirm that the network name is correct.
6. If the network name is incorrect, right click the VM, select Edit Settings, then Network adapter 1, and select the correct network.
7. Reboot the installation VM.

Installation fails with failed network connection

If you experience a communication error while installing Tanzu Operations Manager or MicroBOSH Director, check the following:

- Tanzu Operations Manager requires that the NSX firewall rules are not blocked. By default, all communication between Tanzu Operations Manager VMs and vCenter or ESXi hosts route through the NSX firewall are blocked. For more information on which ports to enable, see [VMware ports and protocols](#) for vSphere.
- Open port 443. Tanzu Operations Manager and MicroBOSH Director VMs require access to vCenter and all ESX through port 443.
- Allocate more IP addresses. BOSH requires that you allocate a sufficient number of additional dynamic IP addresses when configuring a reserved IP address range during installation. BOSH uses these IP addresses during installation to compile and deploy VMs, install Elastic Application Runtime, and connect to services. VMware recommends that you allocate at least 36 dynamic IP addresses when deploying Tanzu Operations Manager and Elastic Application Runtime.

Insufficient external database permissions

Upgrade issues can be caused when the external database user used for the network policy database is given insufficient permissions. To avoid this upgrade issue, ensure that the networkpolicyserver database user has the **ALL PRIVILEGES** permission.

Additional Resources

- [AWS AMIs Stemcells deprecate after 2 years](#)
- [Setting up a Load Balancer in front of Tanzu Operations Manager](#)

Documentation for other products related to Tanzu Operations Manager

See these links for product documentation that relates to Tanzu Operations Manager:

- [Tanzu Platform for Cloud Foundry Documentation.](#)
- [VMware Tanzu Container Orchestration Engine \(formerly TKGI\) Documentation.](#)
- [Tanzu Kubernetes Grid \(TKG\) Documentation.](#)

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